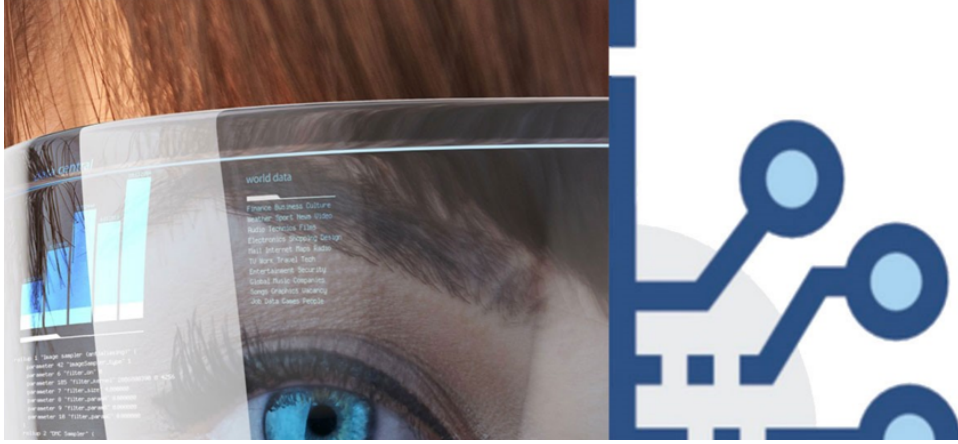




Multi-Asset Signal Generation

Principal Financial Group
Carnegie Mellon University

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1 Asset Classes

1.1 Summary of Asset Classes

Table 1: Commodity

	AssetClass	AC_ID	CSI_ID	Description	SOURCE	MOD_ID
14	Commodity	Commodity	CO_oil1	Commodities Oil 1	Principle	CO_oil1
15	Commodity	Commodity	CO_oil2	Commodities Oil 2	Principle	CO_oil2
16	Commodity	Commodity	CO_nat_gas	Commodities Natural Gas	Principle	CO_nat_gas
17	Commodity	Commodity	CO_gasoline	Commodities Gasoline	Principle	CO_gasoline
18	Commodity	Commodity	CO_gold	Commodities Gold	Principle	CO_gold
19	Commodity	Commodity	CO_silver	Commodities Silver	Principle	CO_silver
20	Commodity	Commodity	CO_grains	Commodities Grains	Principle	CO_grains

1.2 History: CO_{oil}

Current Version is $FI_{development.PriorVersionis}$.



1.3 History: CO_{il2}

Current Version is $FI_{development.PriorVersionis}$.



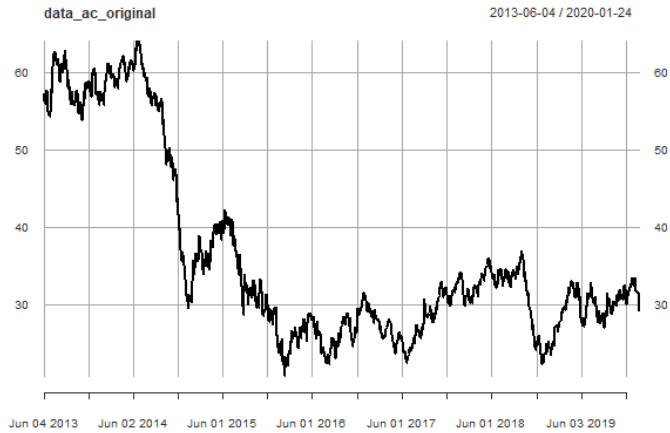
1.4 History: CO_{natgas}

Current Version is $FI_{development.PriorVersionis}$.



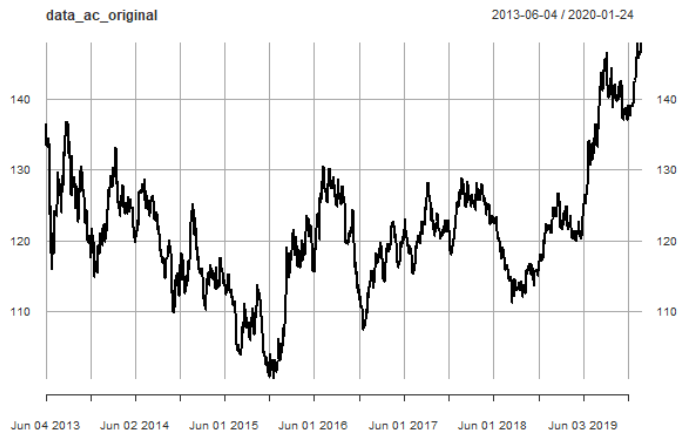
1.5 History: $CO_{gasoline}$

Current Version is $FI_{development.PriorVersionis}$.



1.6 History: CO_{gold}

Current Version is $FI_{development.PriorVersionis}$.



1.7 History: *CO_{silver}*

Current Version is *FI_{development.PriorVersionis}*.



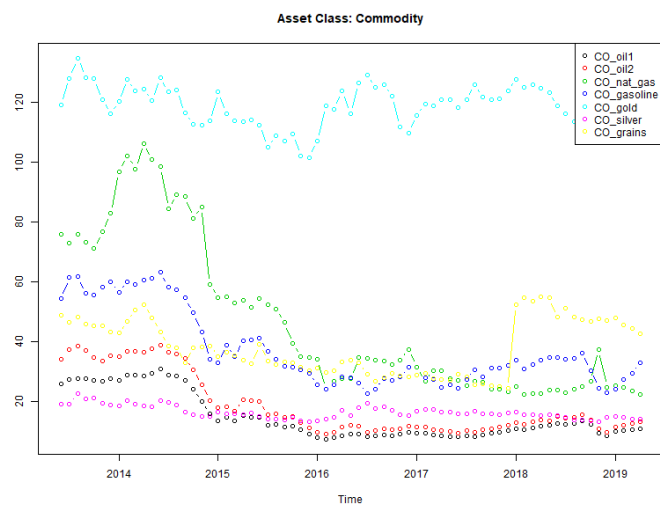
1.8 History: CO_{grains}

Current Version is $FI_{development.PriorVersionis}$.



2 Clusters Analysis of Asset Classes

2.1 Asset Class: *Commodity*



Commodity

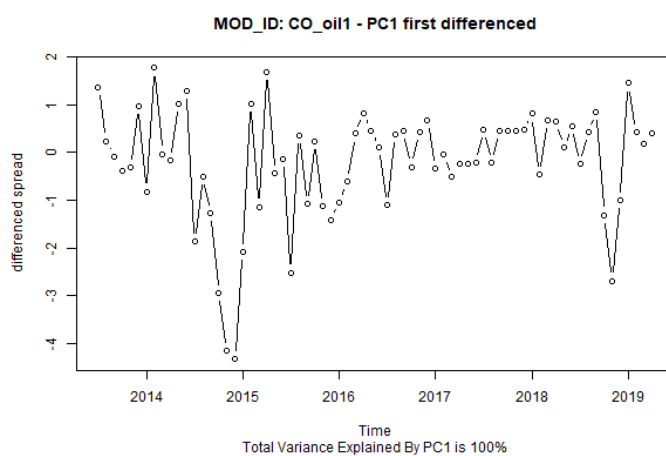
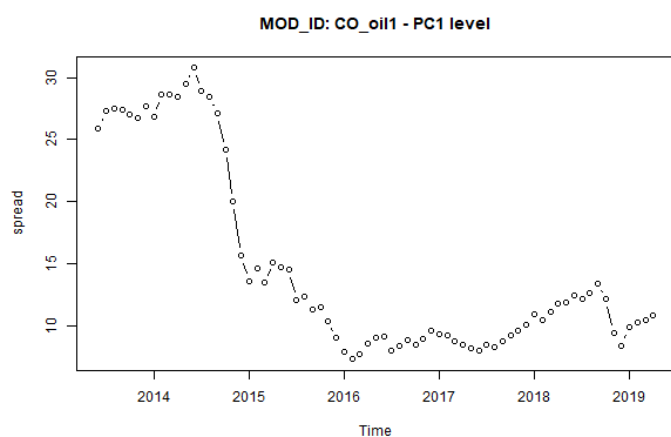
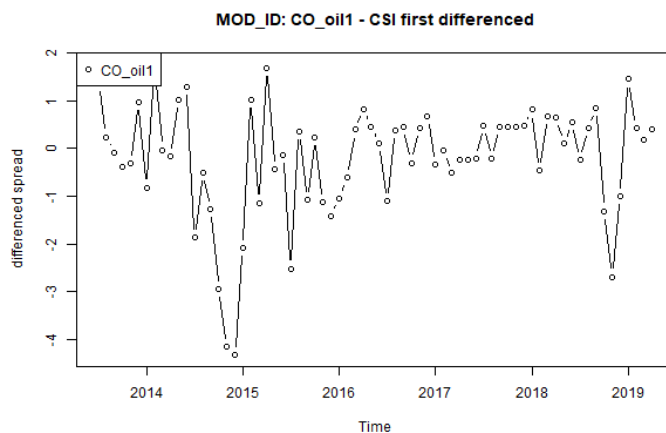
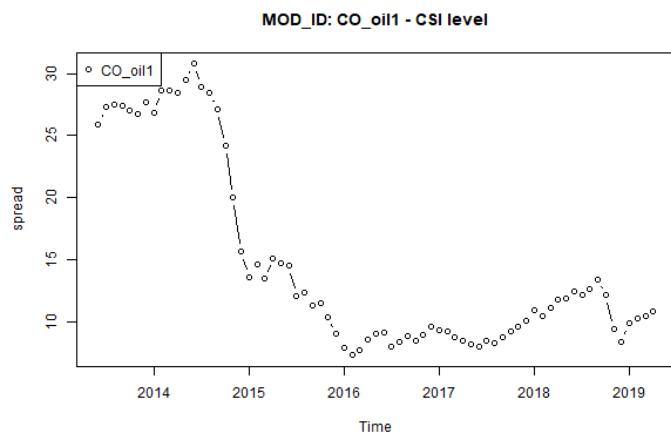
Variance%	CO_gasoline	CO_gold	CO_grains	CO_nat_gas	CO_oil1	CO_oil2	CO_silver	Spearman	Kendall
38.9%	1	1	1	1	1	1	1	-1%	-1%
53.7%	1	2	1	1	1	1	2	28%	20%
65.2%	1	3	2	2	1	1	3	31%	21%
91.6%	1	3	4	2	1	1	3	84%	65%
97.9%	3	4	5	2	1	1	4	84%	65%
97.9%	3	4	6	2	1	1	5	97%	87%
100%	4	5	7	3	1	2	6	100%	100%

Table 2: Cluster Analysis - Commodity

	1	2	3	4	5	6	7
CO_oil1	1	1	1	1	1	1	1
CO_oil2	1	1	1	1	1	1	2
CO_gasoline	1	1	1	1	3	3	4
CO_nat_gas	1	1	2	2	2	2	3
CO_grains	1	1	2	4	5	6	7
CO_gold	1	2	3	3	4	4	5
CO_silver	1	2	3	3	4	5	6
Variance% of PC1	38.9%	53.7%	65.2%	91.6%	97.9%	97.9%	100%
Retained Correlation	-1%	28%	31%	84%	84%	97%	100%

3 PCA for Clusters

3.1 Cluster: *CO_oil1*



	EIGENVECTOR	MULTIPLIER
CO_oil1	1.00	1.00

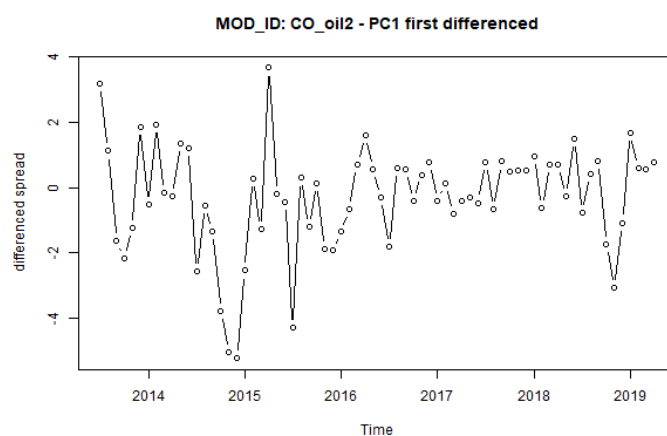
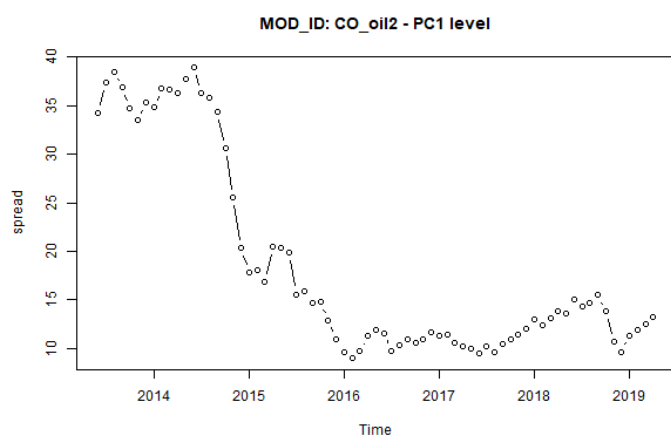
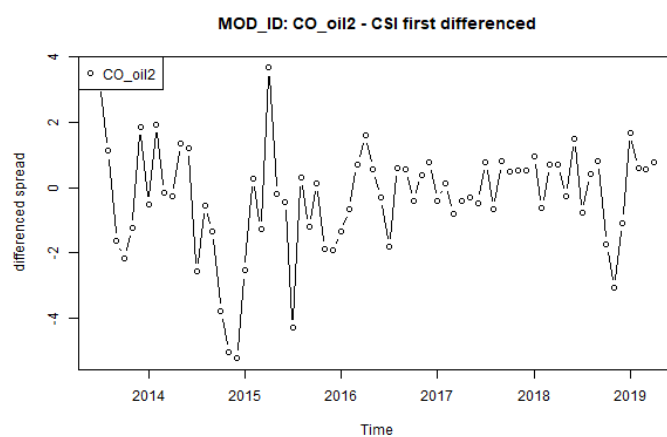
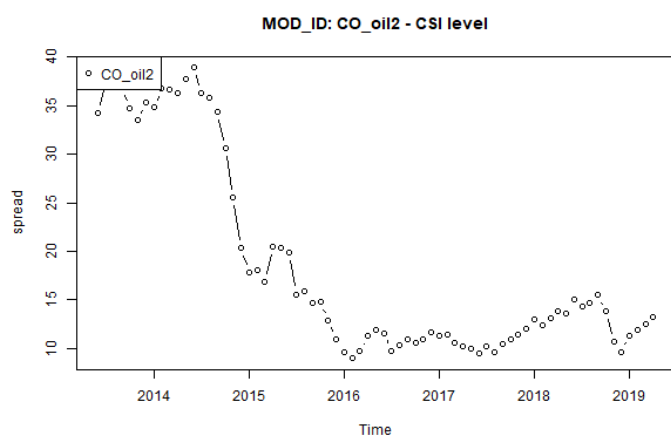
Kendall Correlation

	CO_oil1	pc1
CO_oil1	1	1
pc1	1	1

Spearman Correlation

	CO_oil1	pc1
CO_oil1	1	1
pc1	1	1

3.2 Cluster: *CO_{oil2}*



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
CO_oil2	1.00	1.00

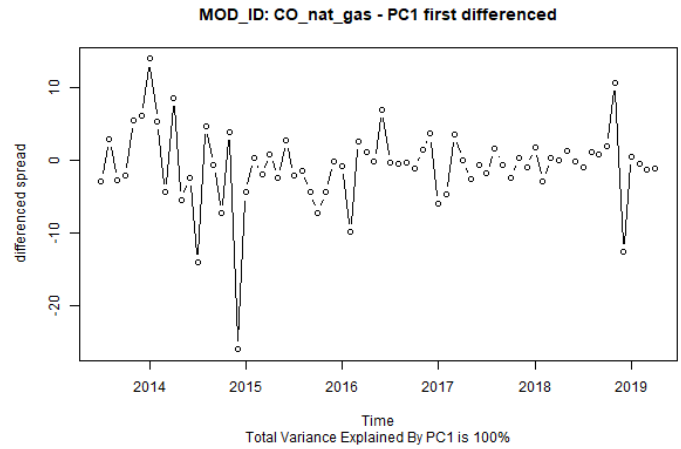
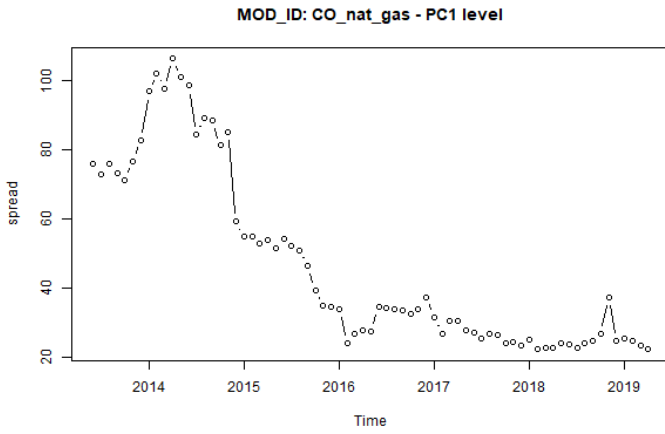
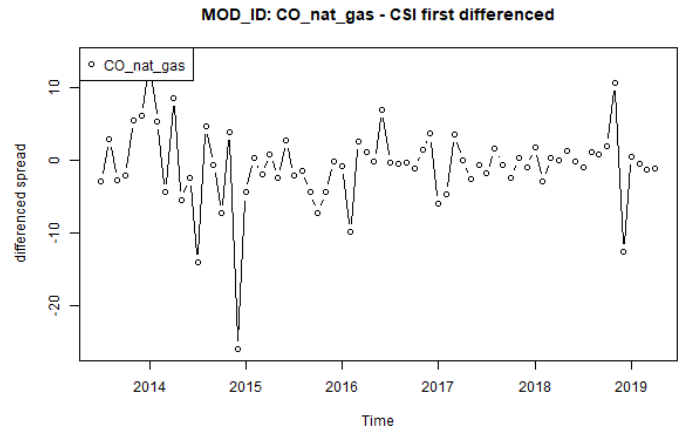
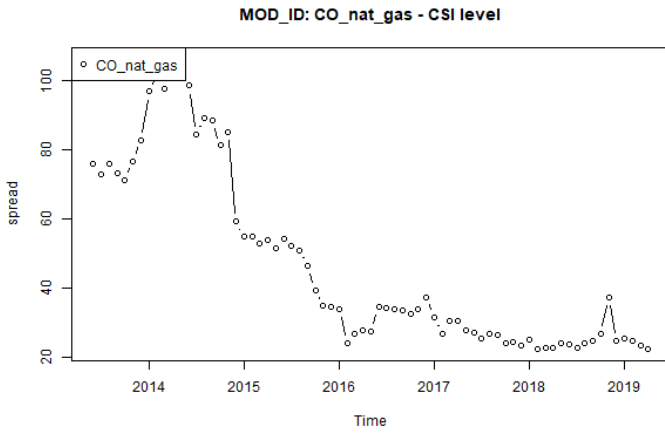
Kendall Correlation

	CO_oil2	pc1
CO_oil2	1	1
pc1	1	1

Spearman Correlation

	CO_oil2	pc1
CO_oil2	1	1
pc1	1	1

3.3 Cluster: CO_{nat_gas}



	EIGENVECTOR	MULTIPLIER
CO_nat_gas	1.00	1.00

Kendall Correlation

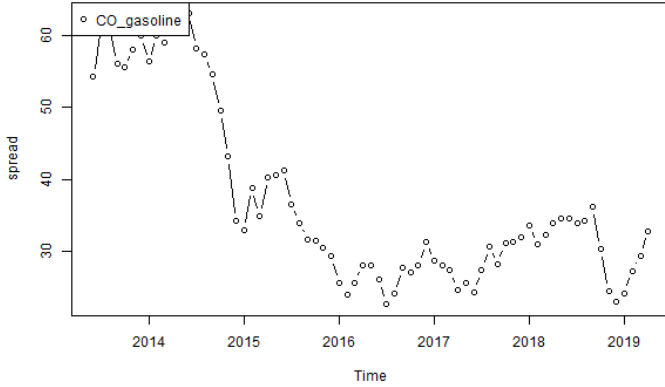
	CO_nat_gas	pc1
CO_nat_gas	1	1
pc1	1	1

Spearman Correlation

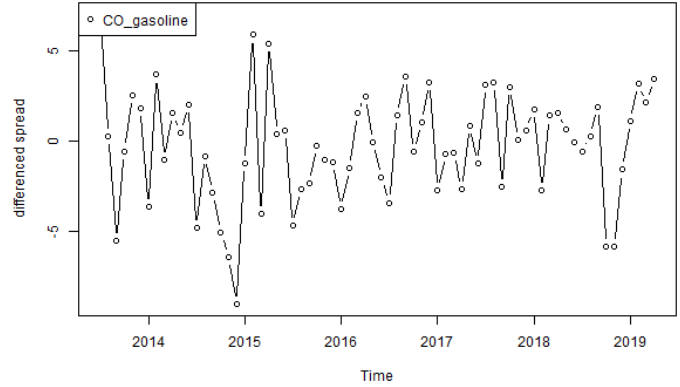
	CO_nat_gas	pc1
CO_nat_gas	1	1
pc1	1	1

3.4 Cluster: $CO_{gasoline}$

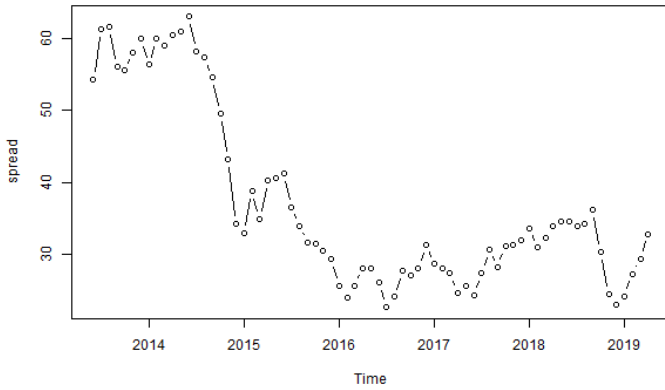
MOD_ID: CO_gasoline - CSI level



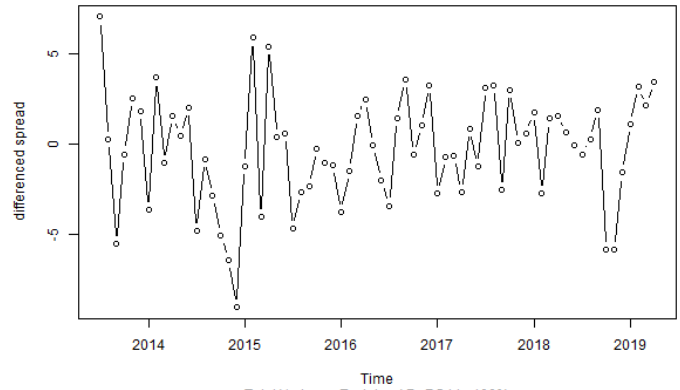
MOD_ID: CO_gasoline - CSI first differenced



MOD_ID: CO_gasoline - PC1 level



MOD_ID: CO_gasoline - PC1 first differenced



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
CO_gasoline	1.00	1.00

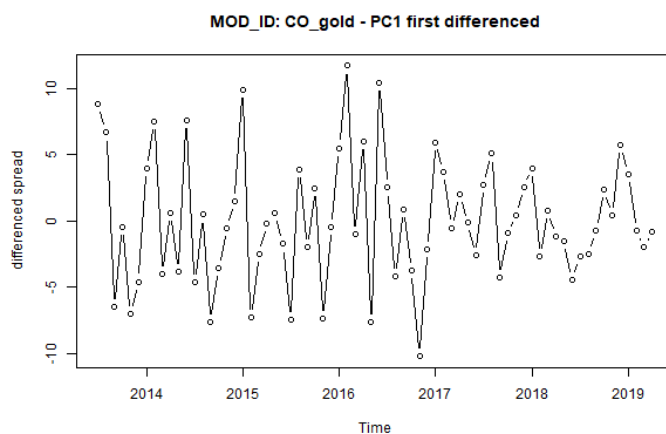
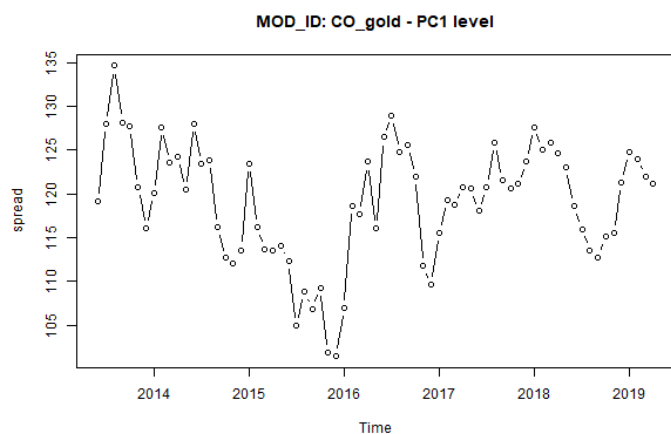
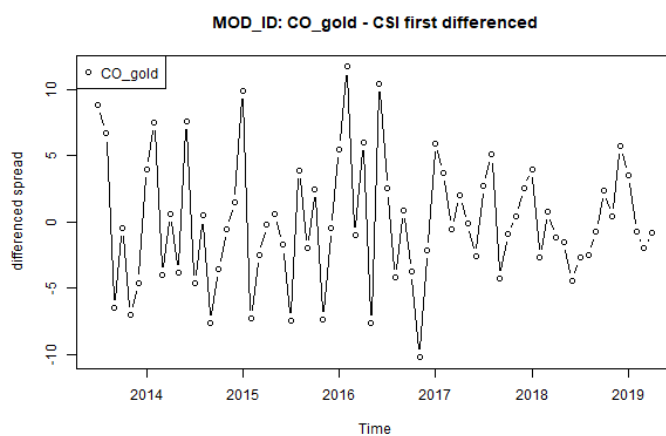
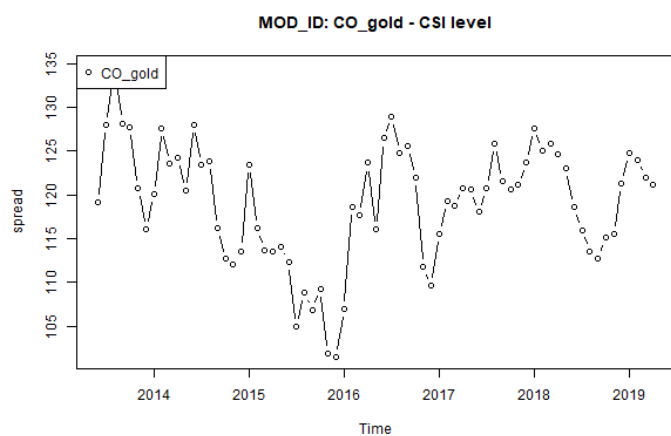
Kendall Correlation

	CO_gasoline	pc1
CO_gasoline	1	1
pc1	1	1

Spearman Correlation

	CO_gasoline	pc1
CO_gasoline	1	1
pc1	1	1

3.5 Cluster: CO_{gold}



Total Variance Explained By PC1 is 100%

	EIGENVECTOR	MULTIPLIER
CO_gold	1.00	1.00

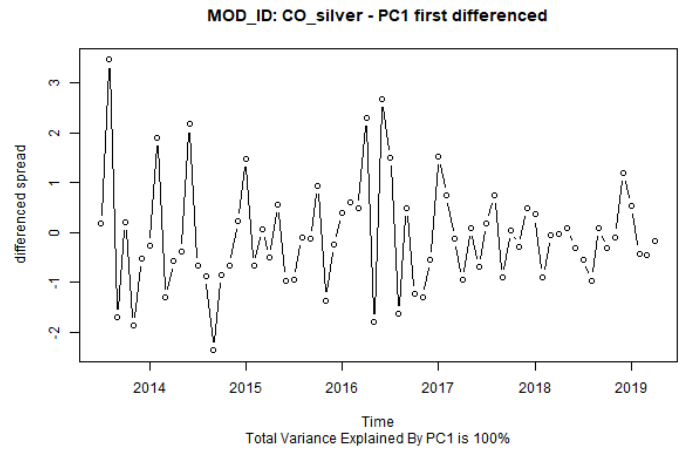
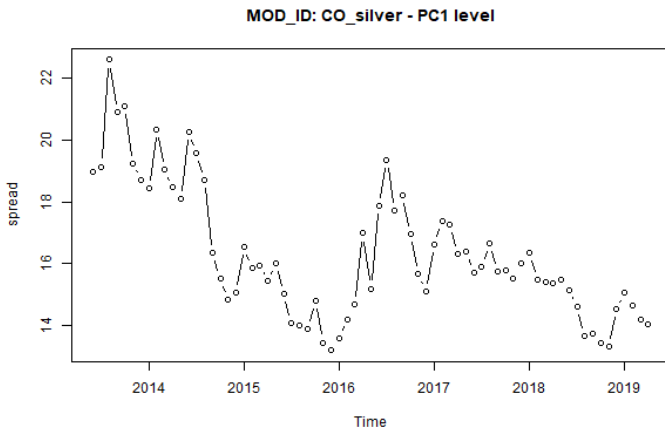
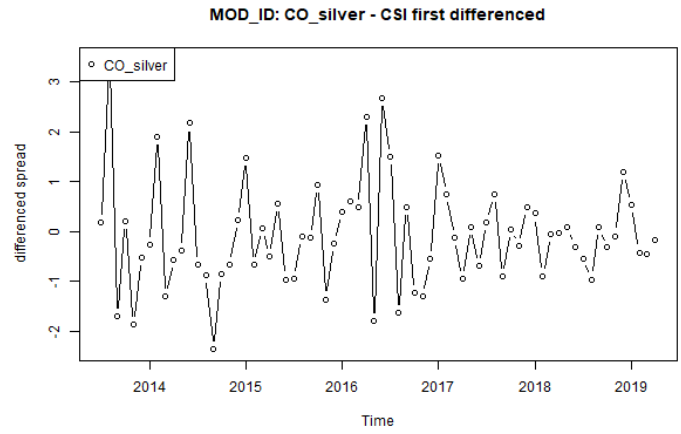
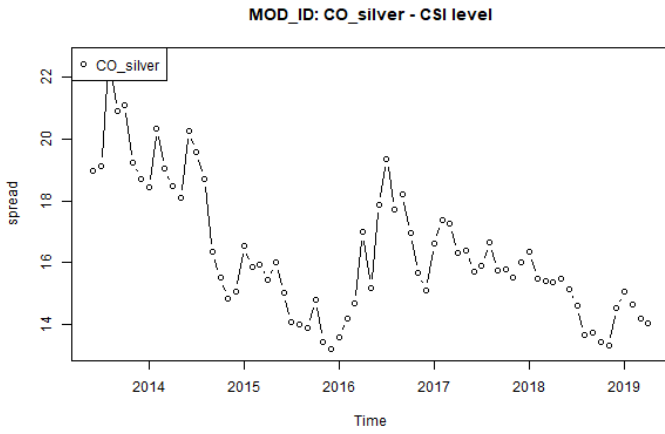
Kendall Correlation

	CO_gold	pc1
CO_gold	1	1
pc1	1	1

Spearman Correlation

	CO_gold	pc1
CO_gold	1	1
pc1	1	1

3.6 Cluster: *CO_{silver}*



	EIGENVECTOR	MULTIPLIER
CO_silver	1.00	1.00

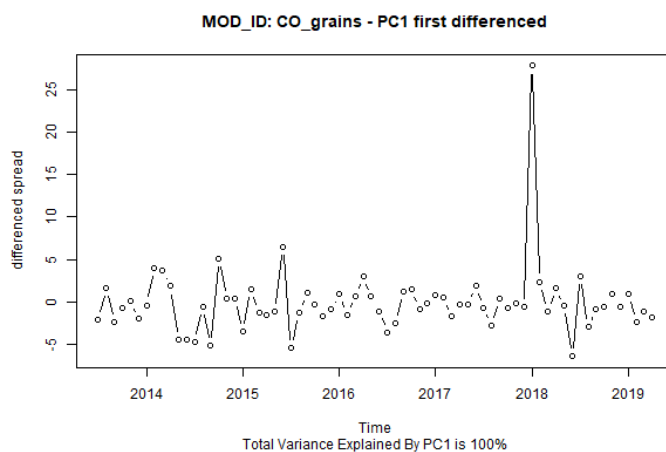
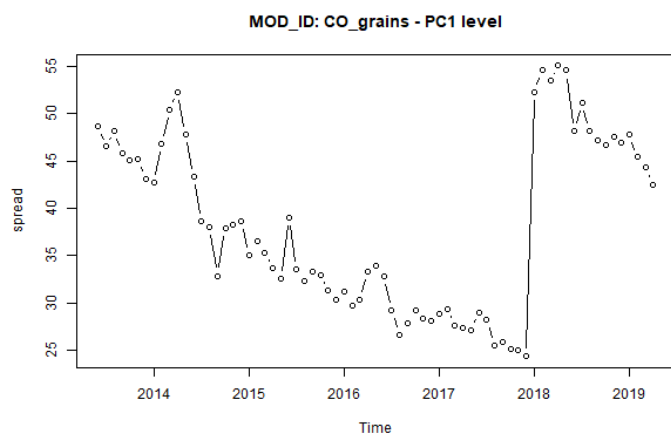
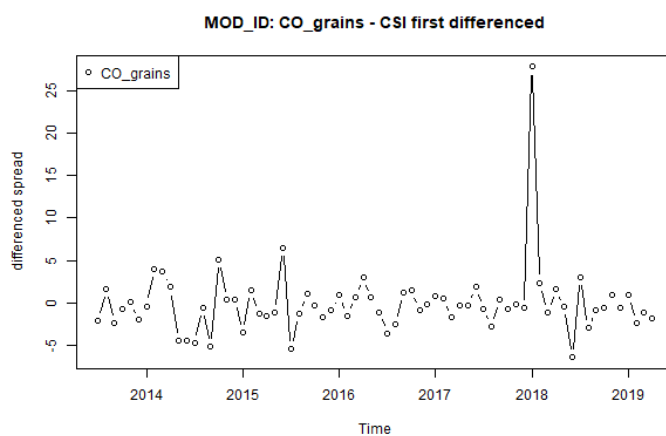
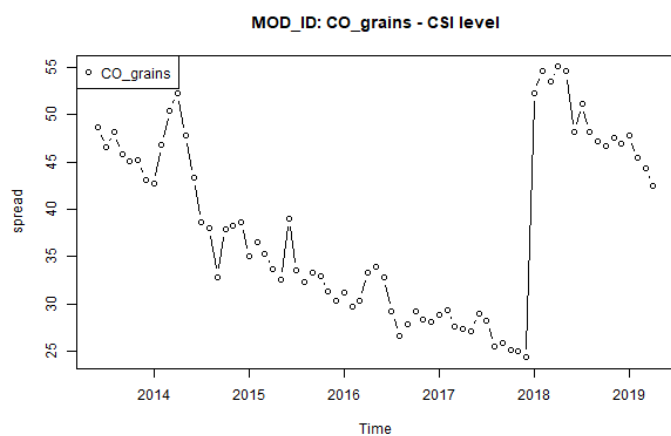
Kendall Correlation

	CO_silver	pc1
CO_silver	1	1
pc1	1	1

Spearman Correlation

	CO_silver	pc1
CO_silver	1	1
pc1	1	1

3.7 Cluster: CO_{grains}



	EIGENVECTOR	MULTIPLIER
CO_grains	1.00	1.00

Kendall Correlation

	CO_grains	pc1
CO_grains	1	1
pc1	1	1

Spearman Correlation

	CO_grains	pc1
CO_grains	1	1
pc1	1	1

4 Variable Selection

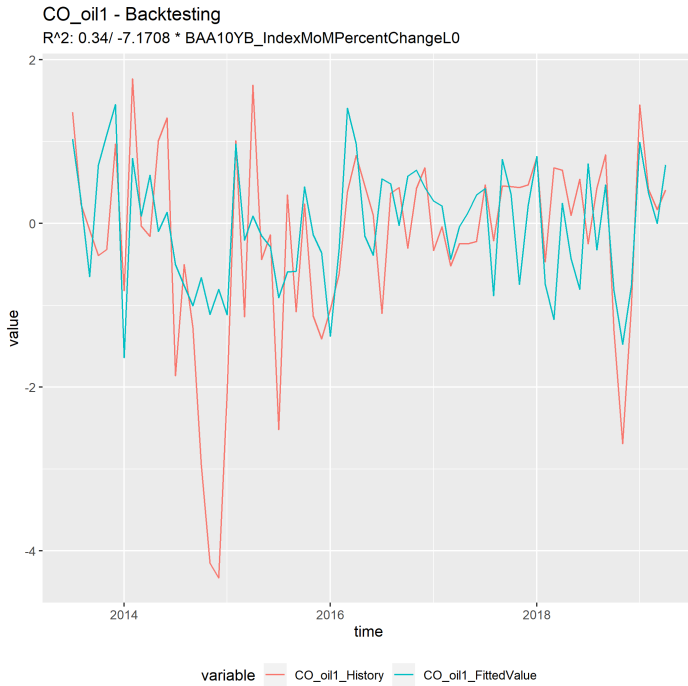
4.1 Single Factor Model for CO₂il1

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	BAA10YB_IndexMoMPercentChangeL0	0.34	198.58	-7.17	0.00	0.71			
2	BAA10YB_IndexMoMDifferenceL0	0.29	203.93	-0.05	0.00	0.61			
3	USGG30YR_IndexMoMDifferenceL0	0.16	216.06	2.96	0.00	0.83			
4	WPSID61MoMDifferenceL0	0.16	216.17	0.36	0.03	0.94			
5	USGG30YR_IndexMoMPercentChangeL0	0.15	216.86	8.14	0.00	0.82			
6	WPSFD49502MoMDifferenceL0	0.13	218.55	0.26	0.04	0.91			
7	ISRATIOMoMPercentChangeL0	0.13	218.81	-63.13	0.04	0.84			
8	WPSFD49207MoMDifferenceL0	0.12	218.84	0.36	0.04	0.86			
9	ISRATIOMoMDifferenceL0	0.12	219.11	-46.25	0.04	0.84			
10	USGG10YR_IndexMoMDifferenceL0	0.12	219.11	2.41	0.00	0.76			
11	ism_man_pricesMoMDifferenceL0	0.12	219.37	0.09	0.05	0.84			
12	USGG10YR_IndexMoMPercentChangeL0	0.11	220.13	4.99	0.00	0.75			

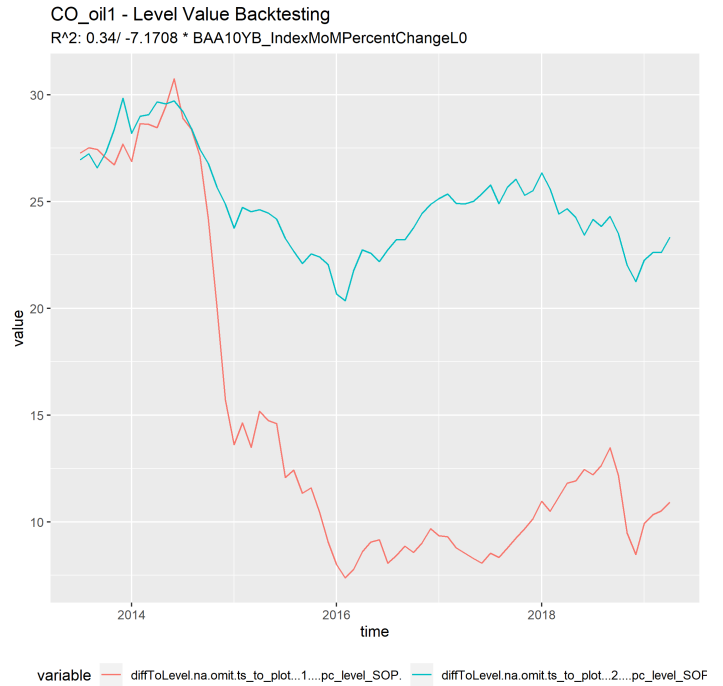
Table 3: Single Factor Analysis-CO₂il1

4.1.1 Model - 1

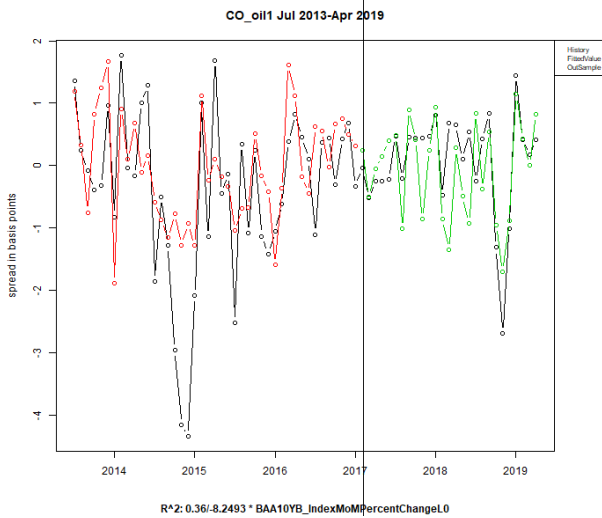
Backtesting on differenced values



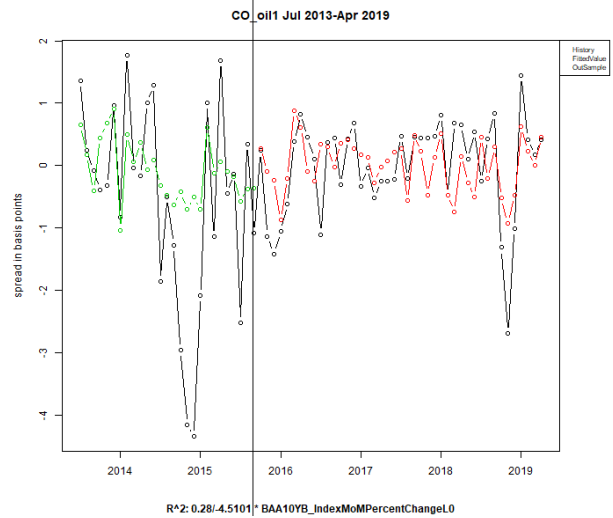
Backtesting on level values



Out-of-Sample testing

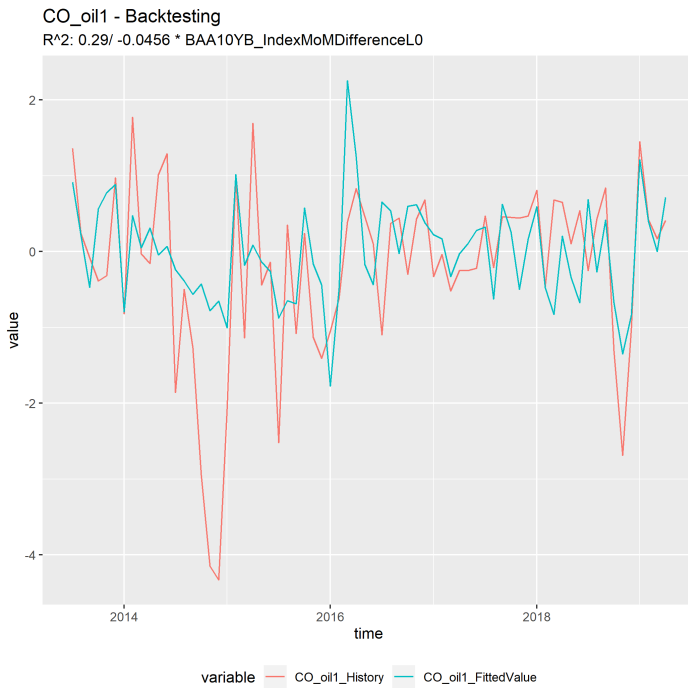


Out-of-Sample testing

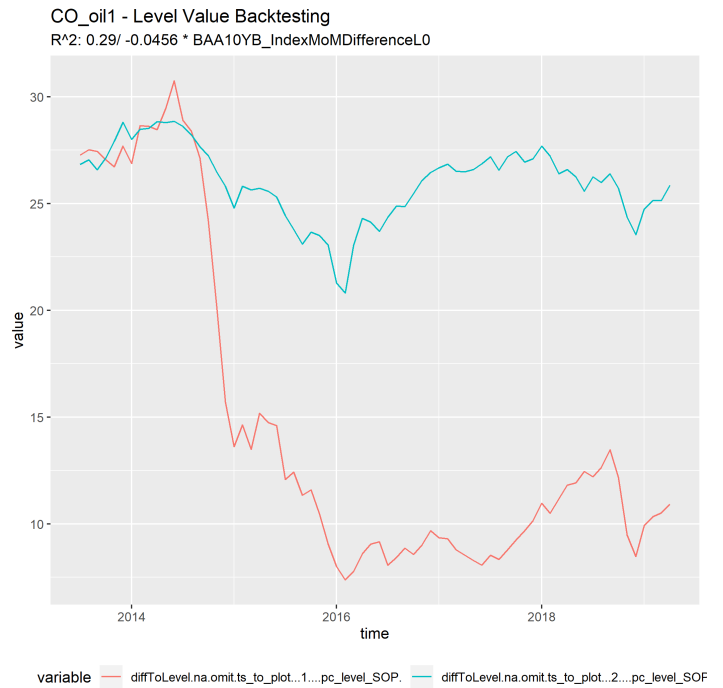


4.1.2 Model - 2

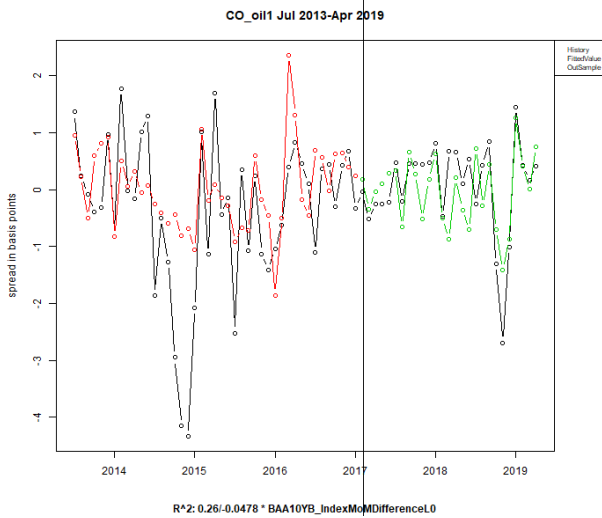
Backtesting on differenced values



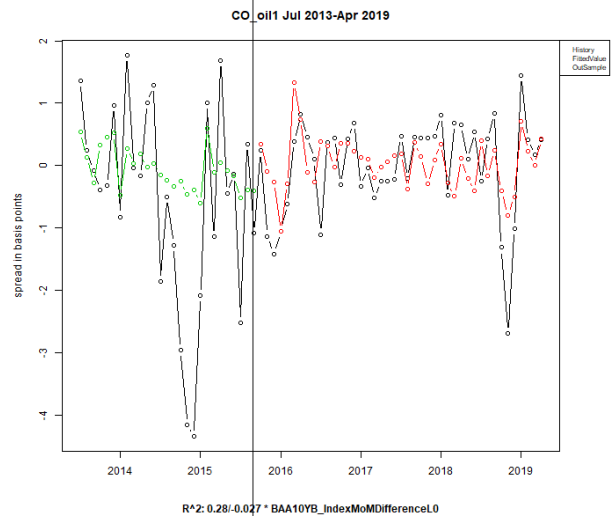
Backtesting on level values



Out-of-Sample testing

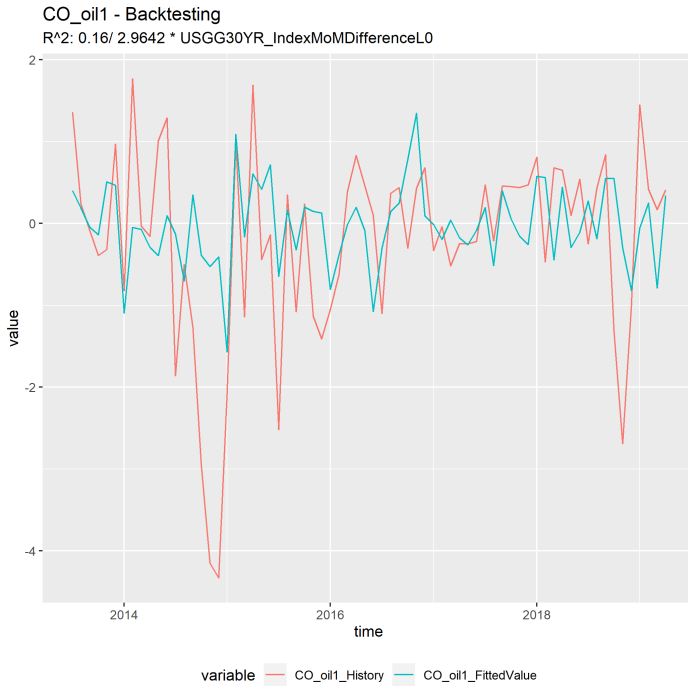


Out-of-Sample testing

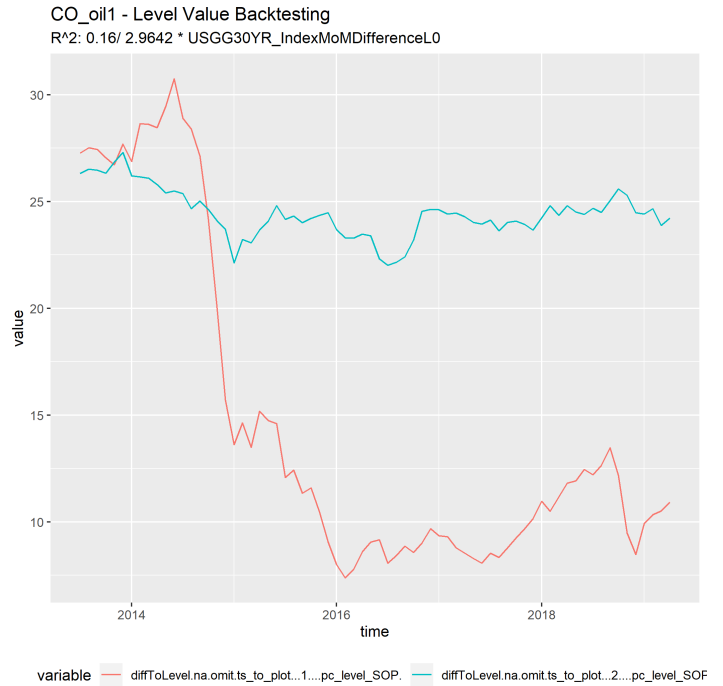


4.1.3 Model - 3

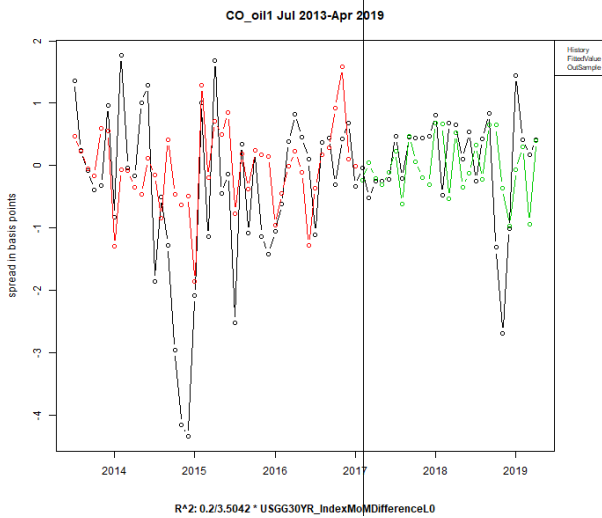
Backtesting on differenced values



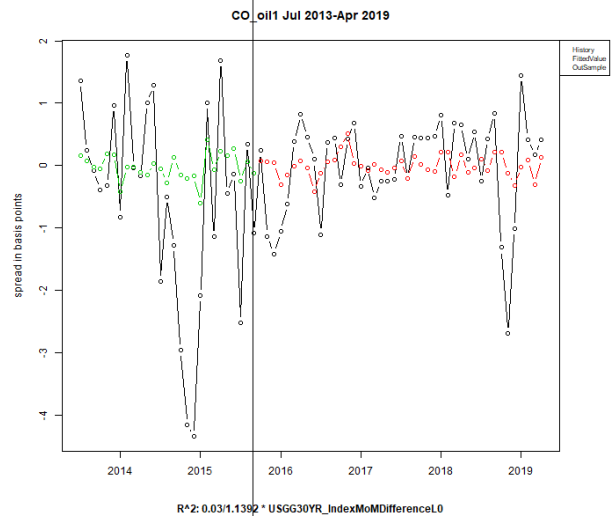
Backtesting on level values



Out-of-Sample testing

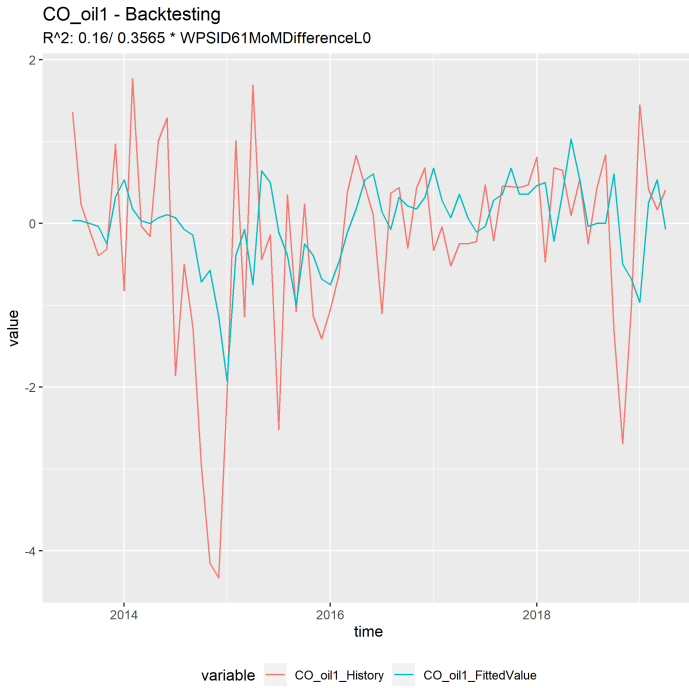


Out-of-Sample testing

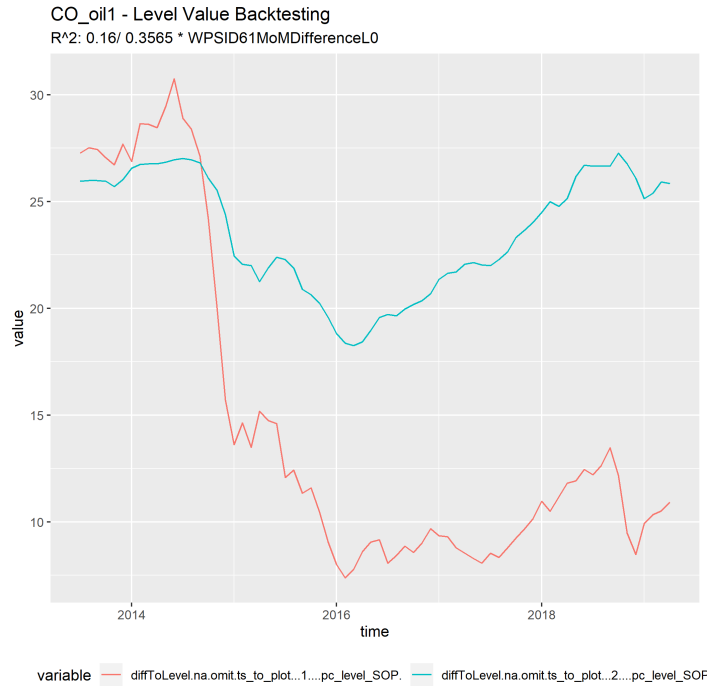


4.1.4 Model - 4

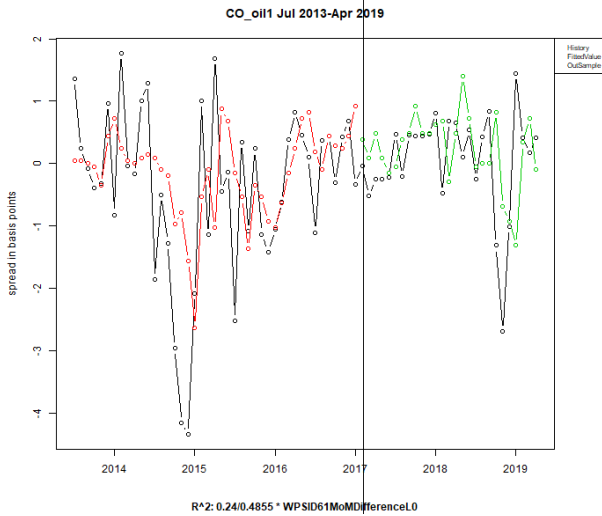
Backtesting on differenced values



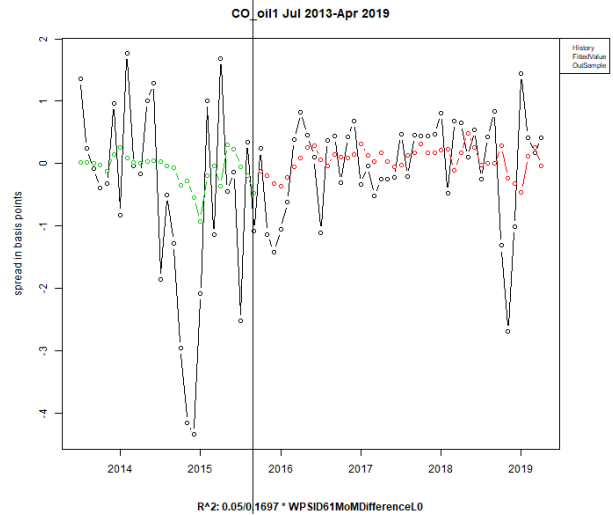
Backtesting on level values



Out-of-Sample testing

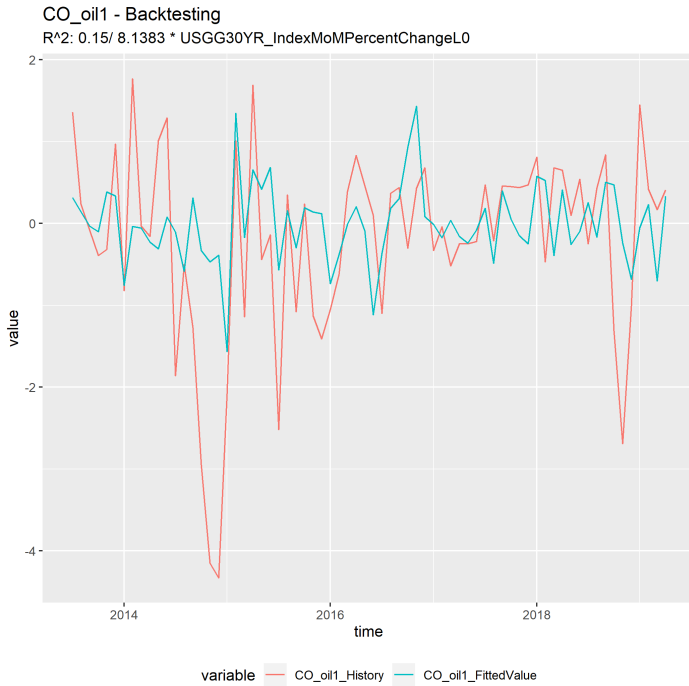


Out-of-Sample testing

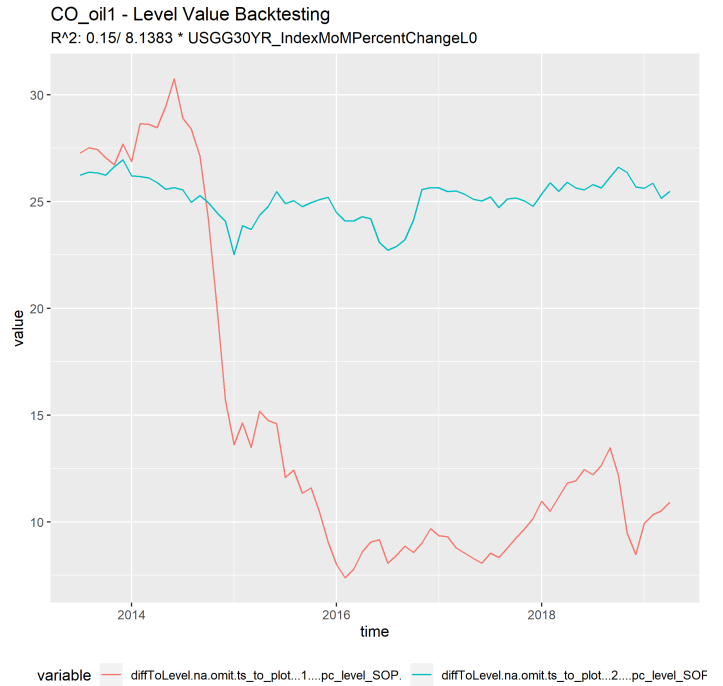


4.1.5 Model - 5

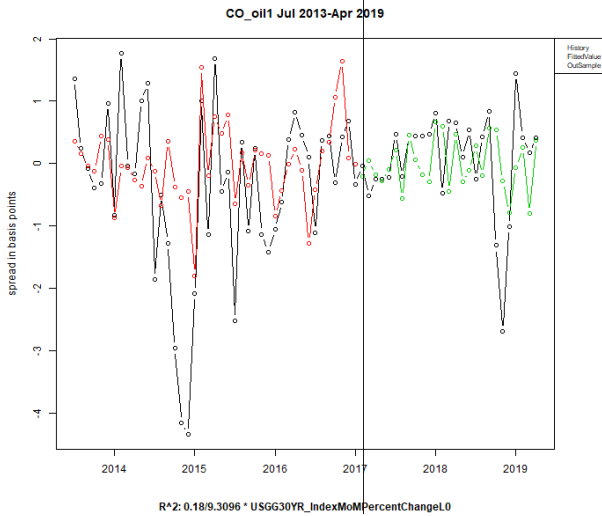
Backtesting on differenced values



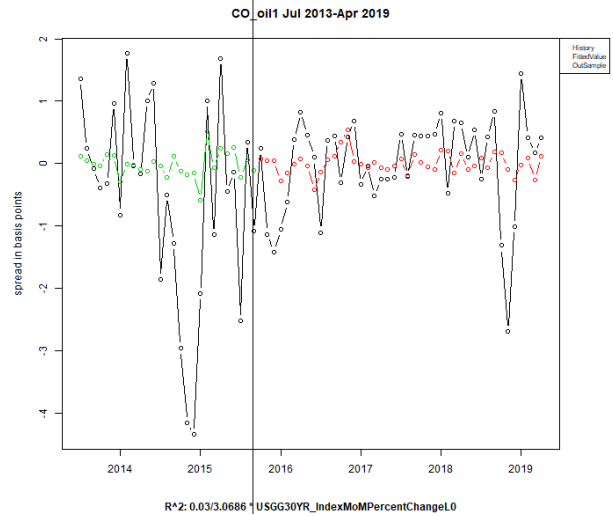
Backtesting on level values



Out-of-Sample testing

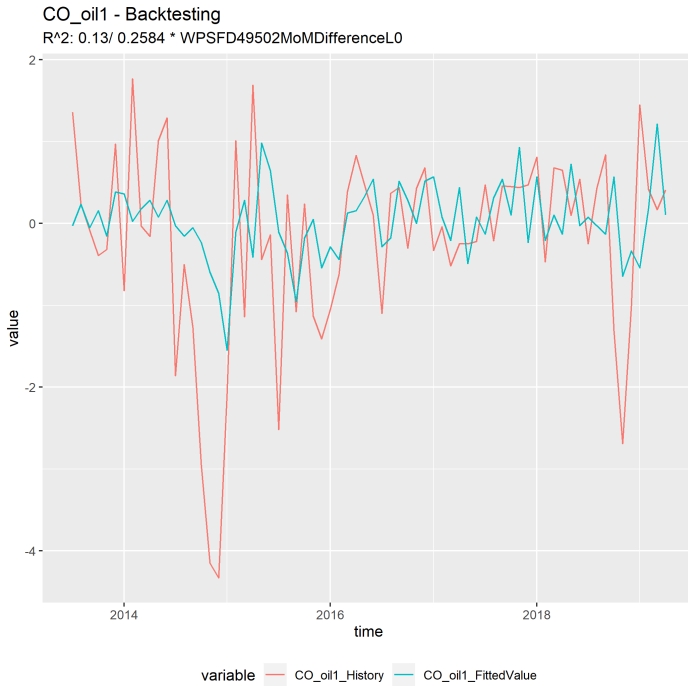


Out-of-Sample testing

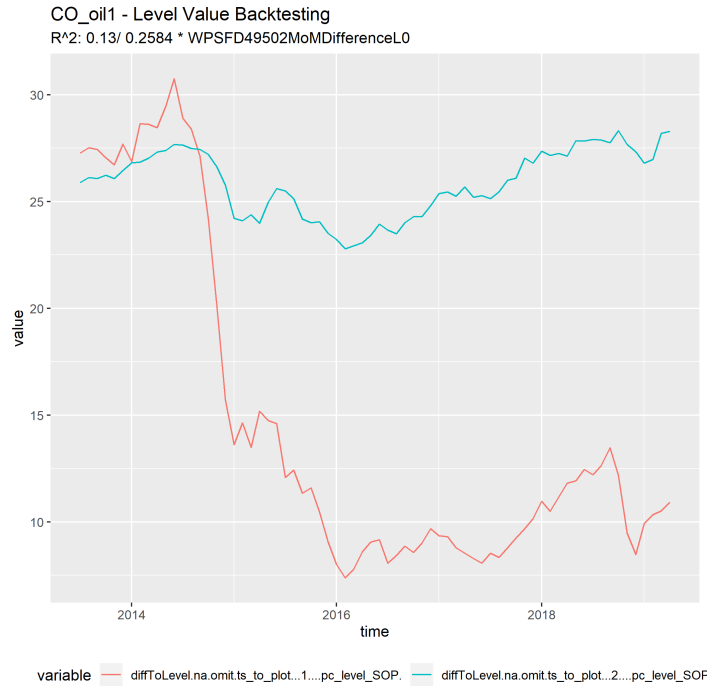


4.1.6 Model - 6

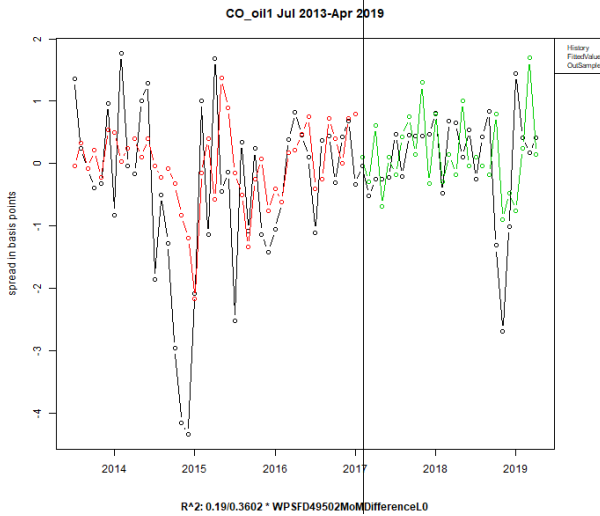
Backtesting on differenced values



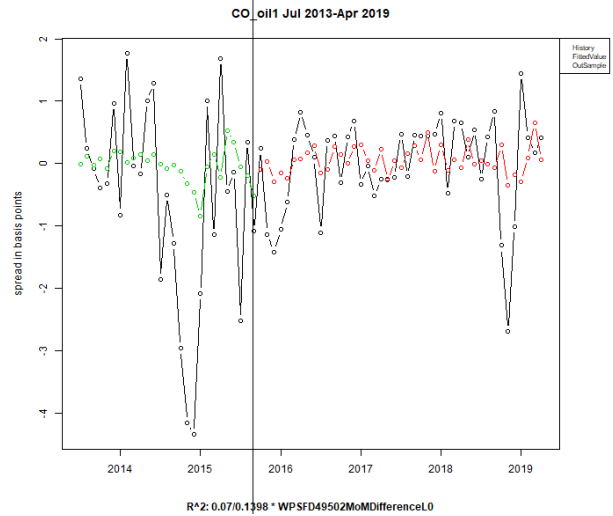
Backtesting on level values



Out-of-Sample testing

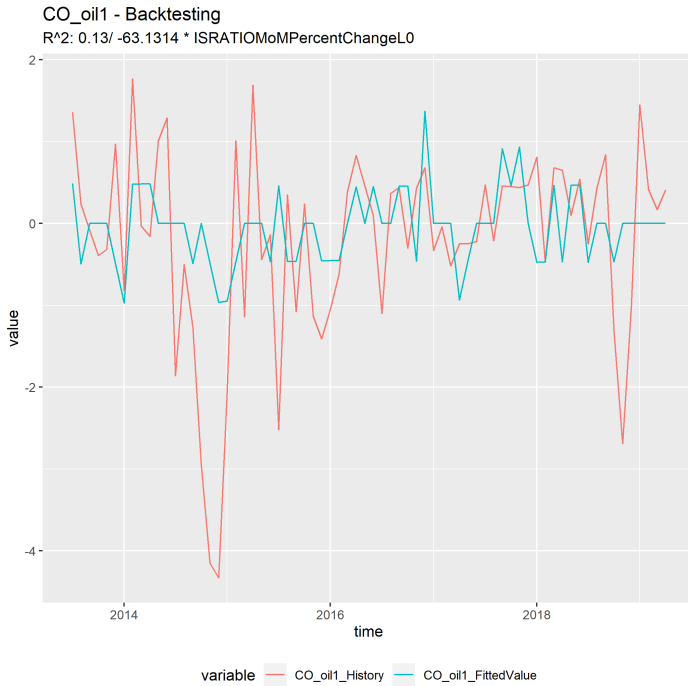


Out-of-Sample testing

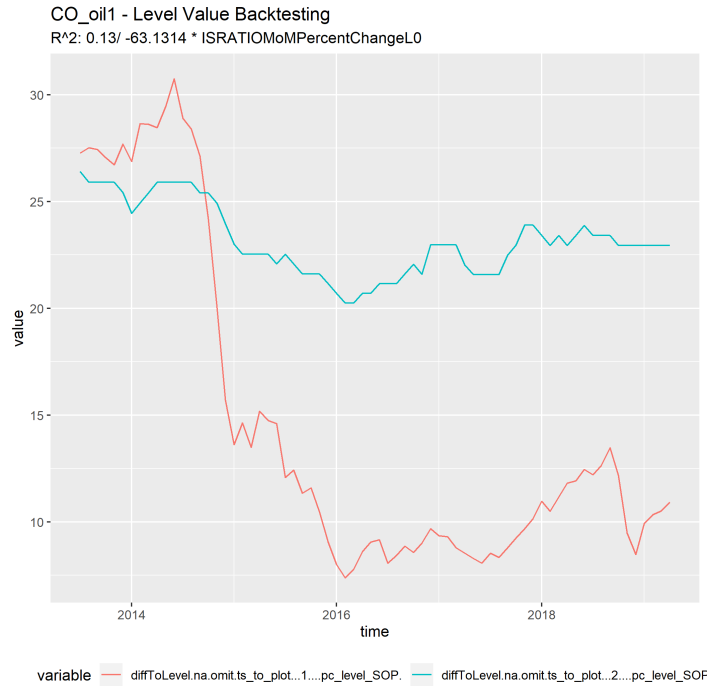


4.1.7 Model - 7

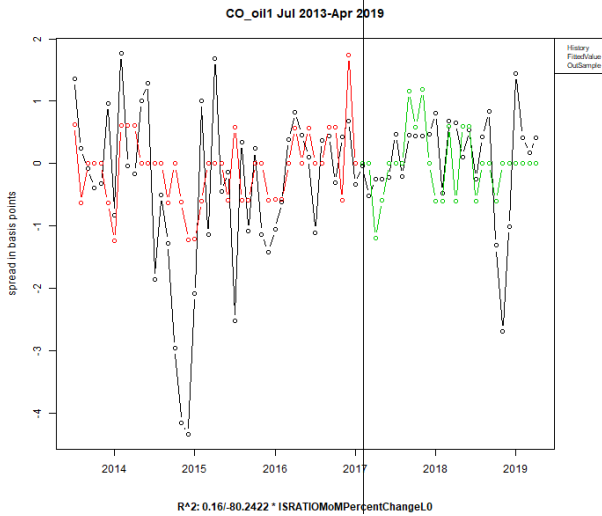
Backtesting on differenced values



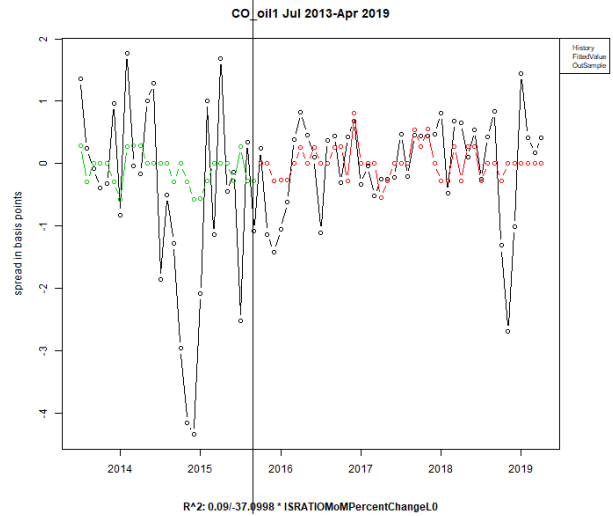
Backtesting on level values



Out-of-Sample testing

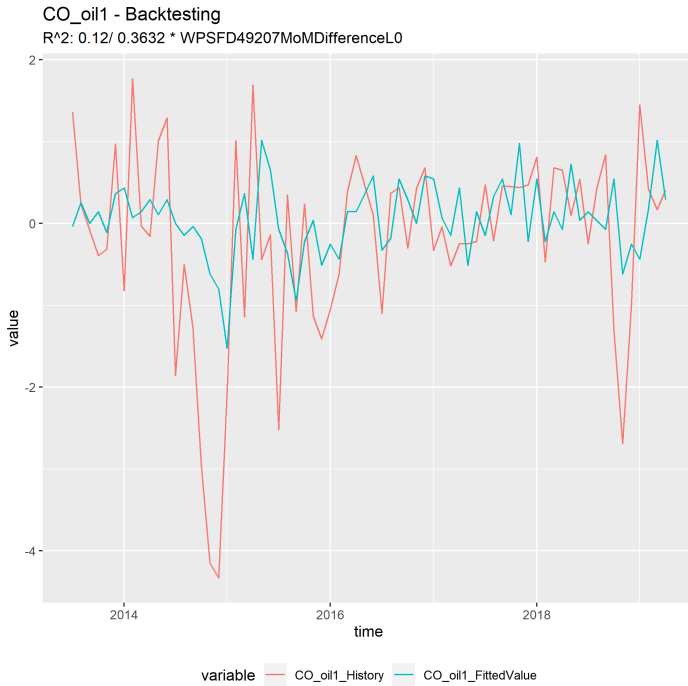


Out-of-Sample testing

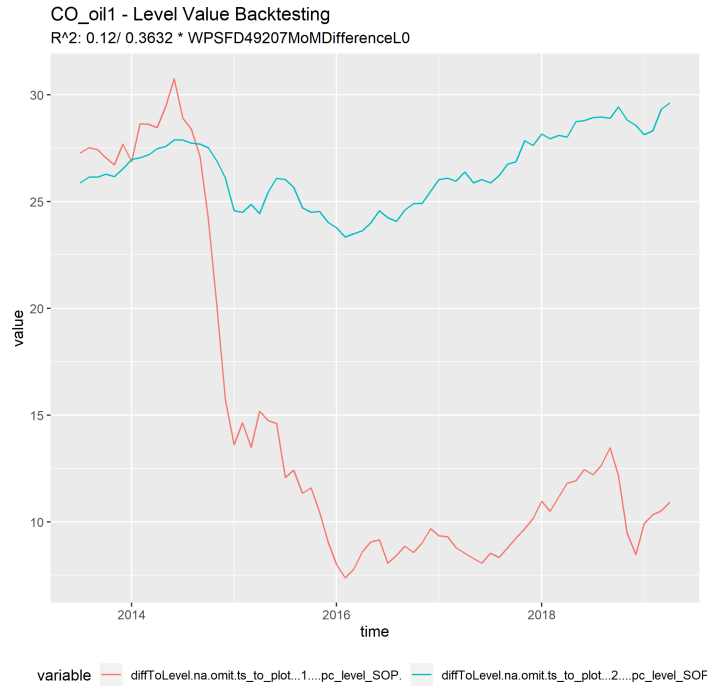


4.1.8 Model - 8

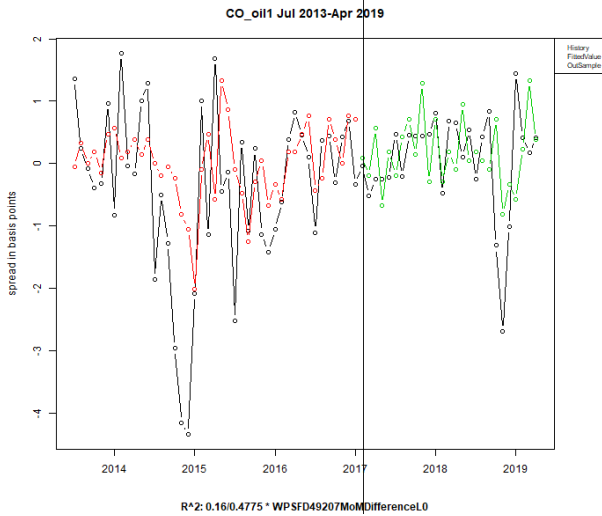
Backtesting on differenced values



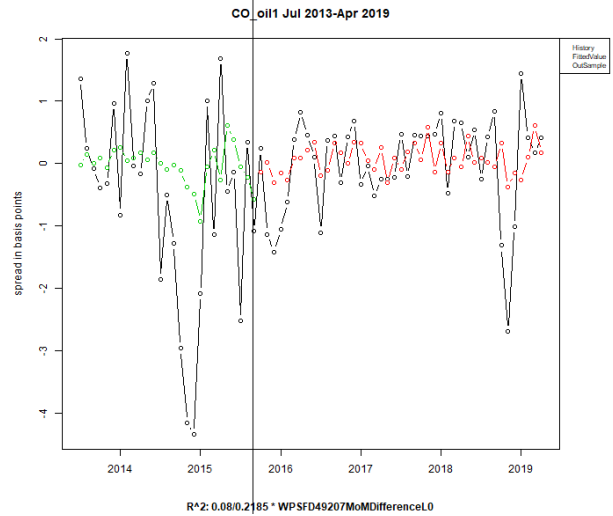
Backtesting on level values



Out-of-Sample testing

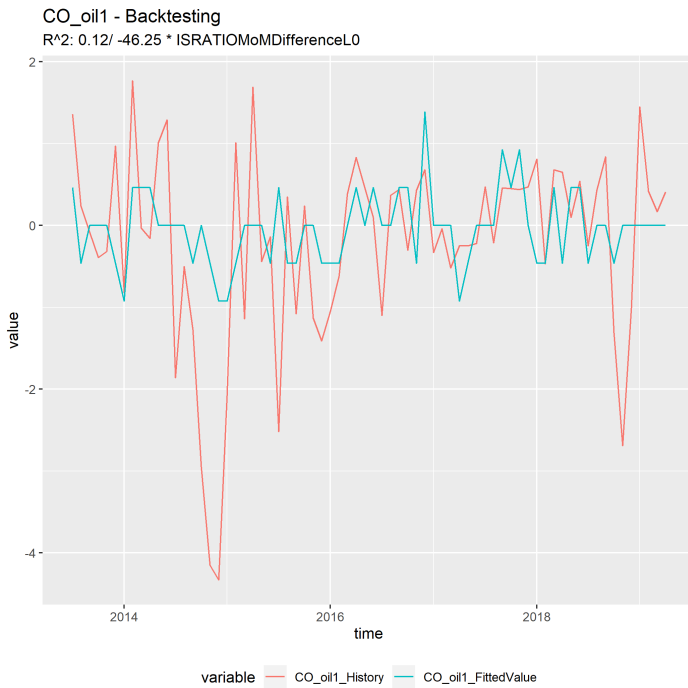


Out-of-Sample testing

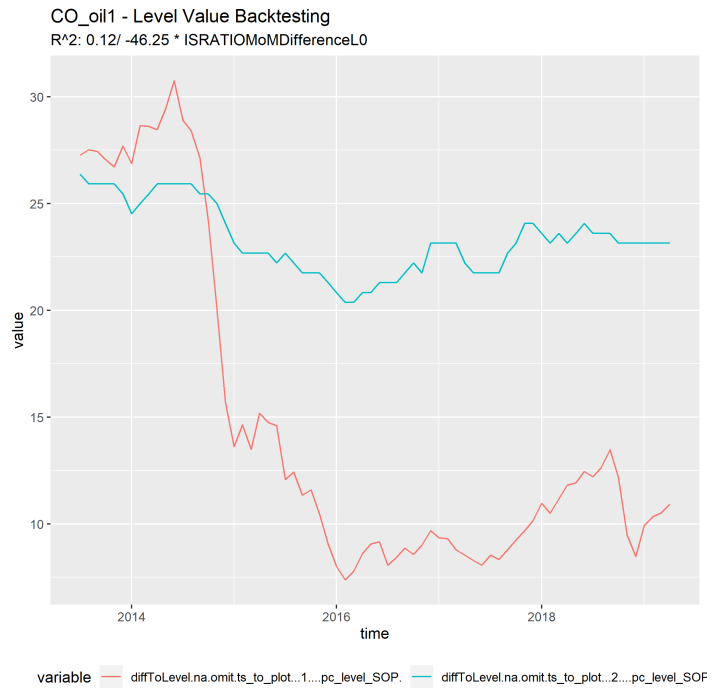


4.1.9 Model - 9

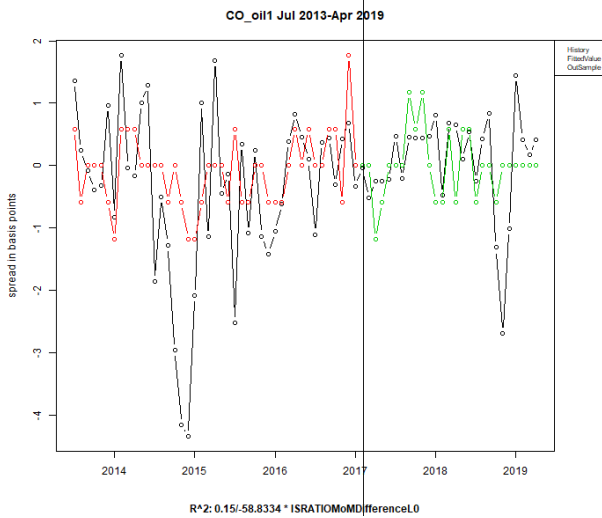
Backtesting on differenced values



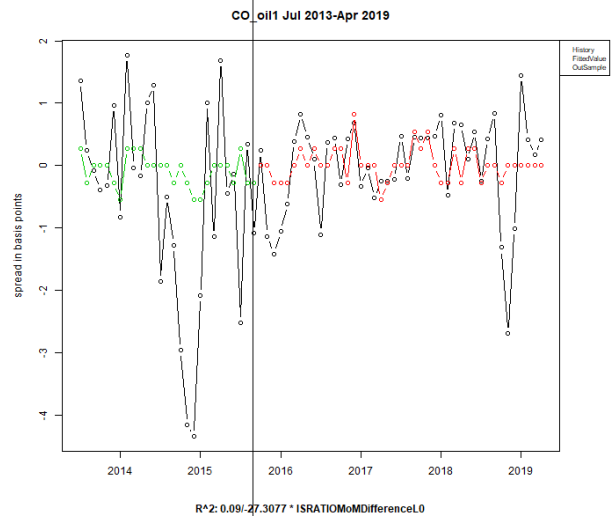
Backtesting on level values



Out-of-Sample testing

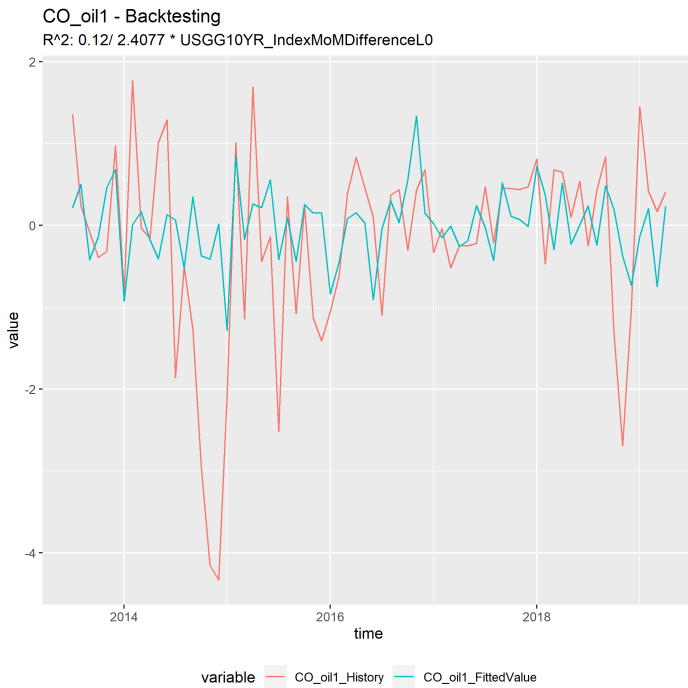


Out-of-Sample testing

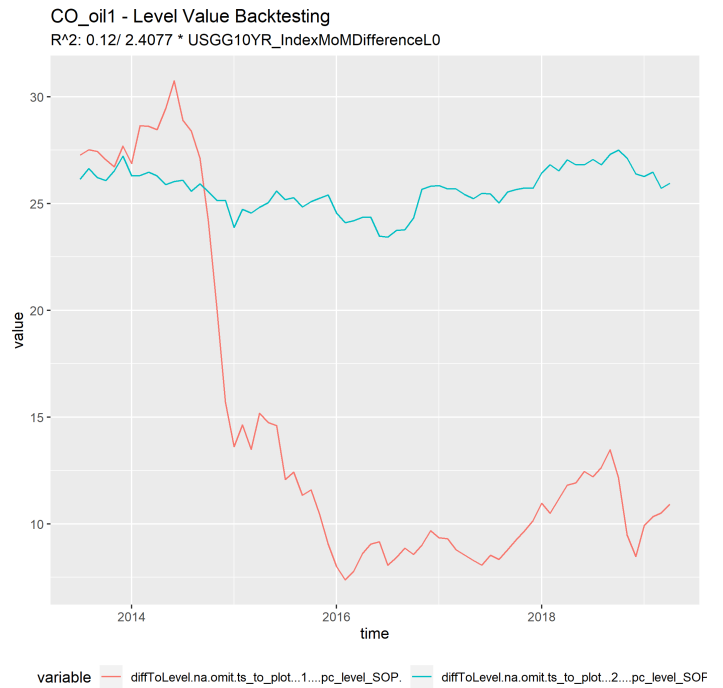


4.1.10 Model - 10

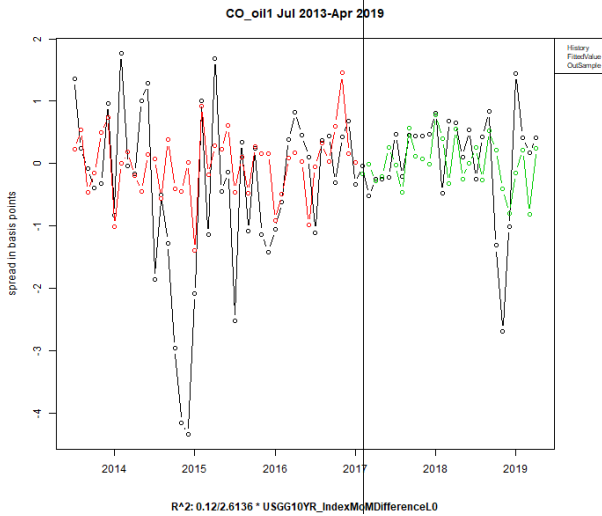
Backtesting on differenced values



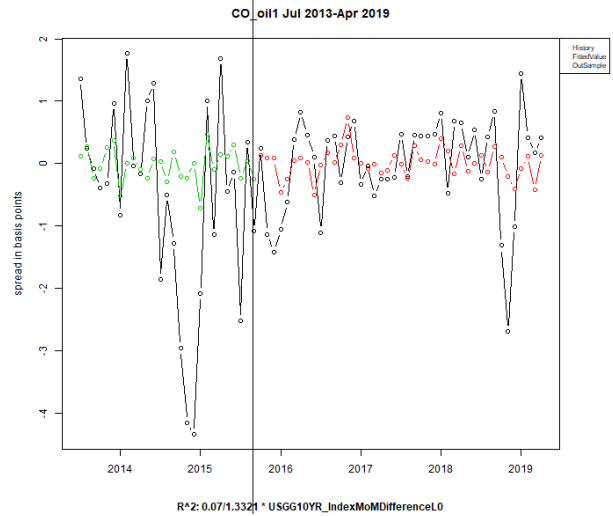
Backtesting on level values



Out-of-Sample testing

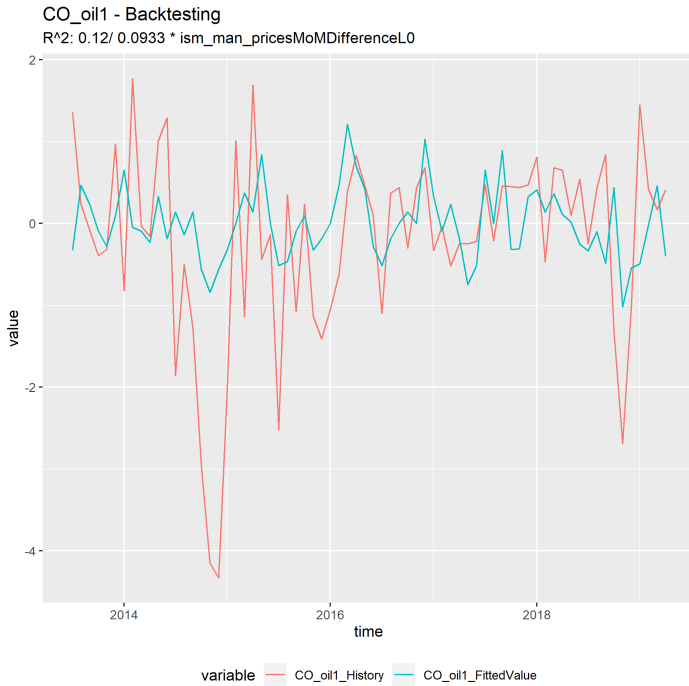


Out-of-Sample testing

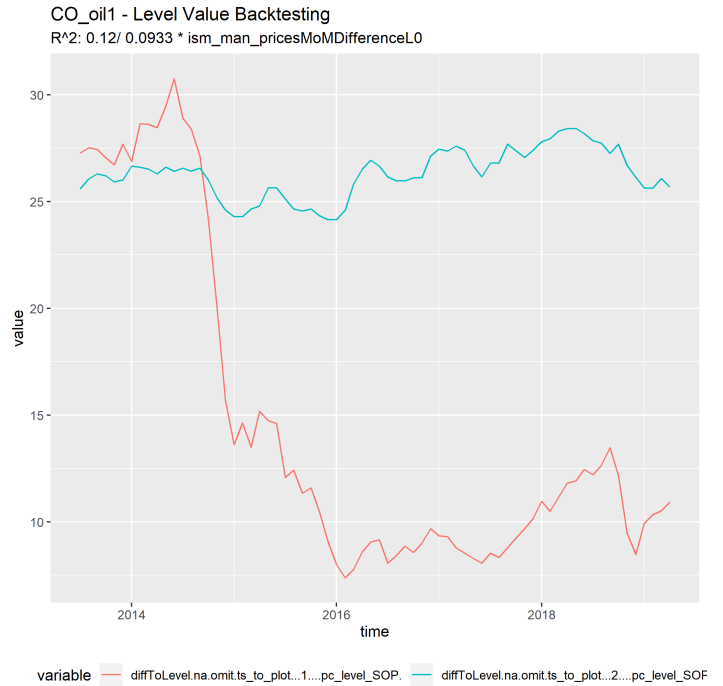


4.1.11 Model - 11

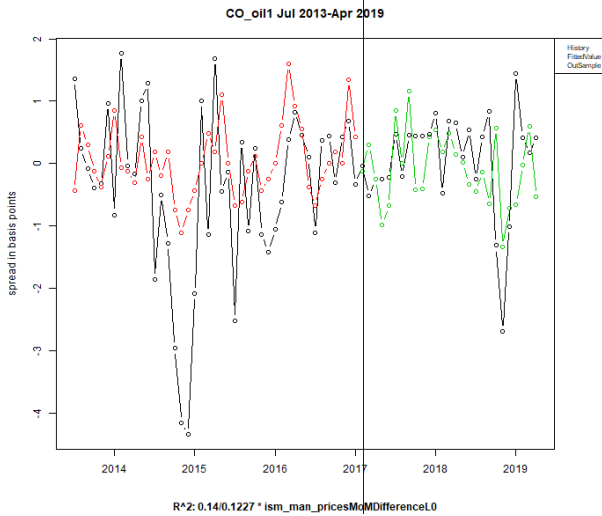
Backtesting on differenced values



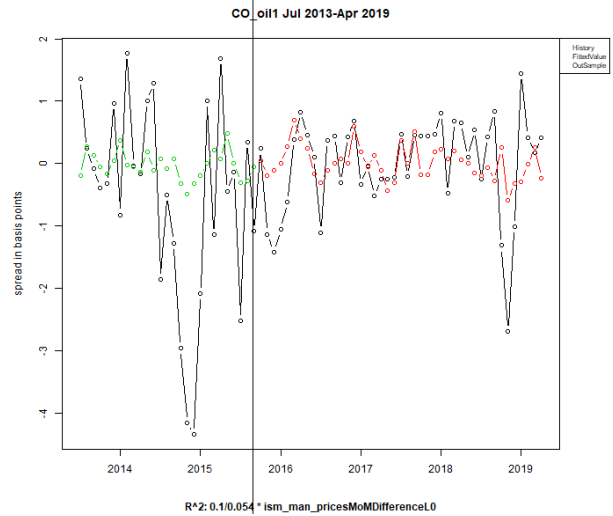
Backtesting on level values



Out-of-Sample testing

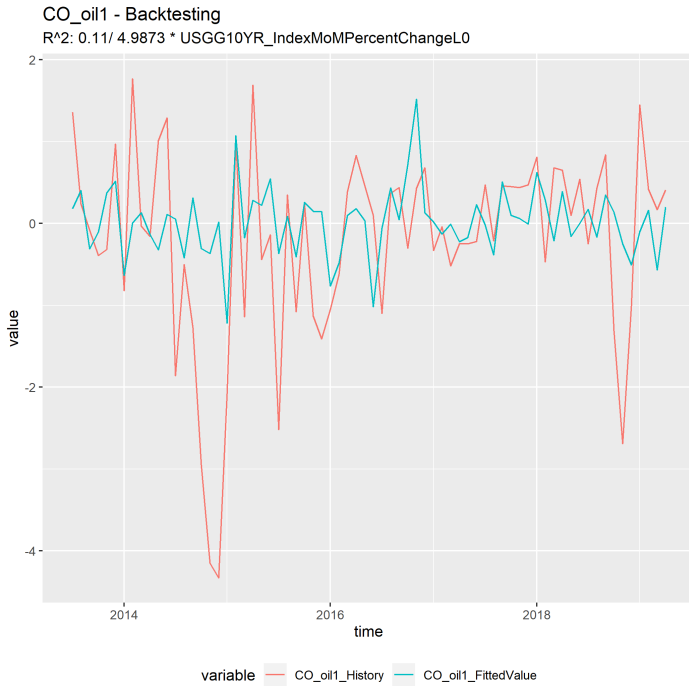


Out-of-Sample testing

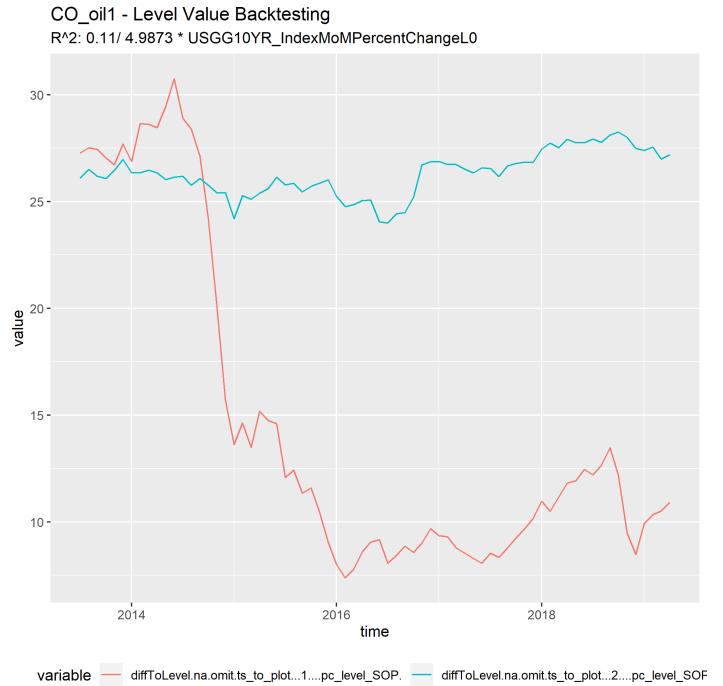


4.1.12 Model - 12

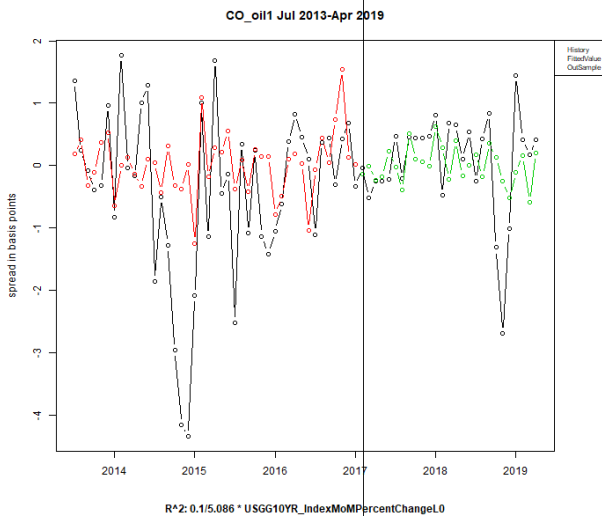
Backtesting on differenced values



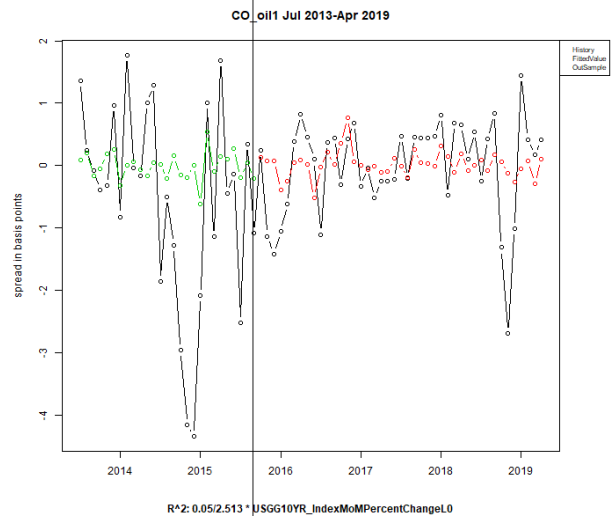
Backtesting on level values



Out-of-Sample testing



Out-of-Sample testing



4.2 Two Factor Model for CO₂il1

	V1	V2	r2adj	Var1	Var2	aic	RMSE	Criteria1	Criteria2	Criteria3
1	WPSID61MoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.43	0.26	-6.43	190.28	0.69			
2	WPSID61MoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.42	49.75	-6.44	190.81	0.68			
3	WPSFD49207MoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.40	0.25	-6.53	193.29	0.67			
4	WPSFD49502MoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.40	0.18	-6.50	193.32	0.68			
5	WPSFD49207MoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.40	49.49	-6.54	193.45	0.66			
6	WPSFD49502MoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.40	36.67	-6.51	193.55	0.68			
7	USTRADEMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.40	-120.61	-7.24	193.67	0.79			
8	INVESTMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.40	-42.43	-7.11	193.76	0.75			
9	USTRADEMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.40	-0.01	-7.22	193.81	0.79			
10	ism_man_pricesMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.40	0.06	-6.55	193.81	0.71		Fail	
11	ISRATIOMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-41.64	-6.48	194.17	0.60			
12	ISRATIOMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-30.39	-6.50	194.39	0.61			
13	INVESTMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-0.01	-7.13	194.40	0.77			
14	M2REALMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-55.39	-6.79	195.04	0.80			
15	ism_man_pricesMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.38	2.96	-6.56	195.25	0.68		Fail	
16	DPCERA3M086SBEAMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-36.59	-7.39	195.45	0.82			
17	M2REALMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-0.01	-6.78	195.48	0.82			
18	DPCERA3M086SBEAMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-0.30	-7.39	195.69	0.87			
19	IPDCONGDMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-9.58	-6.87	195.86	0.71			
20	ism_man_pricesMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.38	0.05	-6.82	195.96	0.75		Fail	
21	PAYEMSMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-146.04	-6.82	196.00	0.82			
22	HOUSTSMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-2.10	-7.10	196.02	0.72			
23	PAYEMSMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-0.00	-6.83	196.16	0.83			
24	WPSID61MoMDifferenceL0	BAA10YB_IndexMoMDifferenceL0	0.38	0.27	-0.04	196.19	0.66			
25	BUSLOANSMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-28.39	-7.05	196.24	0.79			
26	IPDCONGDMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.08	-6.86	196.30	0.71			
27	USFIREMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.37	-122.26	-6.94	196.30	0.81			
28	SRVPRDMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-145.87	-6.94	196.30	0.78			
29	SRVPRDMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.00	-6.95	196.42	0.79			
30	INVESTMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.37	-33.01	-7.15	196.48	0.67			
31	USFIREMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.01	-6.95	196.53	0.82			
32	ism_man_pricesMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.37	2.57	-6.75	196.55	0.70		Fail	
33	IPNMATMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-12.22	-6.95	196.57	0.70			
34	BUSLOANSMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.01	-7.02	196.58	0.82			
35	INVESTMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.01	-7.22	196.63	0.68			
36	WPSID61MoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.37	0.16	-7.30	196.72	0.70			
37	M2REALMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.37	-46.00	-7.25	196.73	0.77			
38	WPSID61MoMPercentChangeL0	BAA10YB_IndexMoMDifferenceL0	0.37	50.38	-0.04	196.73	0.65			
39	IPNMATMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.13	-6.96	196.75	0.70			
40	IPMANSICSMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-16.35	-6.79	196.87	0.71			
41	WPSID61MoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.37	30.39	-7.30	196.95	0.70			
42	WPSID62MoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.37	0.03	-6.73	197.09	1.01			
43	SRVPRDMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.37	-130.00	-7.08	197.09	0.88			
44	IPMANSICSMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.15	-6.81	197.16	0.72			
45	HOUSTSMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.00	-7.07	197.17	0.72			
46	M2REALMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.01	-7.26	197.20	0.79			
47	MANEMPMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-102.67	-6.44	197.20	0.89			
48	SRVPRDMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.00	-7.07	197.25	0.89			
49	MANEMPMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.37	-0.01	-6.45	197.33	0.90			
50	HOUSTWMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.36	-1.31	-7.22	197.56	0.71			

Table 4: Two Factor Analysis-CO₂il1

Table 5: Relevant Variables: CO₂i1

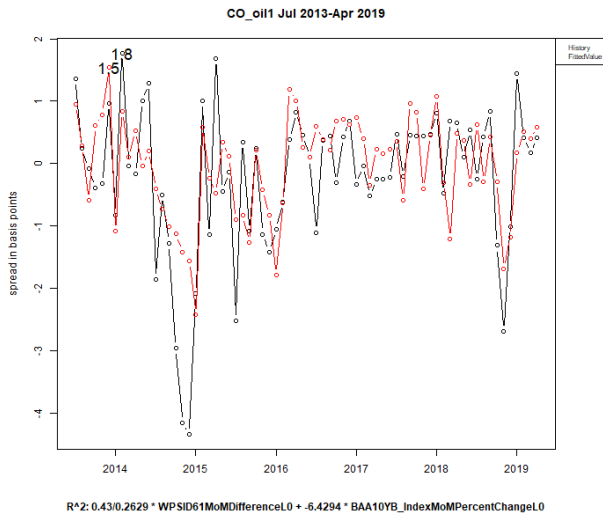
1	2	3
1 RPIMoMDifferenceL0	UEMP15T26MoMPercentChangeL0	BUSLOANSMoMDifferenceL0
2 RPIMoMDifferenceL1	UEMP15T26MoMPercentChangeL1	BUSLOANSMoMDifferenceL1
3 RPIMoMPercentChangeL0	UEMP27OVMoMDifferenceL0	BUSLOANSMoMPercentChangeL0
4 RPIMoMPercentChangeL1	UEMP27OVMoMDifferenceL1	BUSLOANSMoMPercentChangeL1
5 W875RX1MoMDifferenceL0	UEMP27OVMoMPercentChangeL0	REALLNMoMDifferenceL0
6 W875RX1MoMDifferenceL1	UEMP27OVMoMPercentChangeL1	REALLNMoMDifferenceL1
7 W875RX1MoMPercentChangeL0	PAYEMSMoMDifferenceL0	REALLNMoMPercentChangeL0
8 W875RX1MoMPercentChangeL1	PAYEMSMoMDifferenceL1	REALLNMoMPercentChangeL1
9 DPCERA3M086SBEAMoMDifferenceL0	PAYEMSMoMPercentChangeL0	WPSFD49207MoMDifferenceL0
10 DPCERA3M086SBEAMoMDifferenceL1	PAYEMSMoMPercentChangeL1	WPSFD49207MoMDifferenceL1
11 DPCERA3M086SBEAMoMPercentChangeL0	USGOODMoMDifferenceL0	WPSFD49207MoMPercentChangeL0
12 DPCERA3M086SBEAMoMPercentChangeL1	USGOODMoMDifferenceL1	WPSFD49207MoMPercentChangeL1
13 CMRMTSPLMoMDifferenceL0	USGOODMoMPercentChangeL0	WPSFD49502MoMDifferenceL0
14 CMRMTSPLMoMDifferenceL1	USGOODMoMPercentChangeL1	WPSFD49502MoMDifferenceL1
15 CMRMTSPLMoMPercentChangeL0	CES1021000001MoMDifferenceL0	WPSFD49502MoMPercentChangeL0
16 CMRMTSPLMoMPercentChangeL1	CES1021000001MoMDifferenceL1	WPSFD49502MoMPercentChangeL1
17 INDPROMoMDifferenceL0	CES1021000001MoMPercentChangeL0	WPSID61MoMDifferenceL0
18 INDPROMoMDifferenceL1	CES1021000001MoMPercentChangeL1	WPSID61MoMDifferenceL1
19 INDPROMoMPercentChangeL0	USCONSMoMDifferenceL0	WPSID61MoMPercentChangeL0
20 INDPROMoMPercentChangeL1	USCONSMoMDifferenceL1	WPSID61MoMPercentChangeL1
21 IPFPNSSMoMDifferenceL0	USCONSMoMPercentChangeL0	WPSID62MoMDifferenceL0
22 IPFPNSSMoMDifferenceL1	USCONSMoMPercentChangeL1	WPSID62MoMDifferenceL1
23 IPFPNSSMoMPercentChangeL0	MANEMPMoMDifferenceL0	WPSID62MoMPercentChangeL0
24 IPFPNSSMoMPercentChangeL1	MANEMPMoMDifferenceL1	WPSID62MoMPercentChangeL1
25 IPFINALMoMDifferenceL0	MANEMPMoMPercentChangeL0	MZMSLMoMDifferenceL0
26 IPFINALMoMDifferenceL1	MANEMPMoMPercentChangeL1	MZMSLMoMDifferenceL1
27 IPFINALMoMPercentChangeL0	DMANEMPMoMDifferenceL0	MZMSLMoMPercentChangeL0
28 IPFINALMoMPercentChangeL1	DMANEMPMoMDifferenceL1	MZMSLMoMPercentChangeL1
29 IPCONGDMoMDifferenceL0	DMANEMPMoMPercentChangeL0	DTCOLNVHFNMMDifferenceL0
30 IPCONGDMoMDifferenceL1	DMANEMPMoMPercentChangeL1	DTCOLNVHFNMMDifferenceL1
31 IPCONGDMoMPercentChangeL0	SRVPRDMoMDifferenceL0	DTCOLNVHFNMMDifferenceL0
32 IPCONGDMoMPercentChangeL1	SRVPRDMoMDifferenceL1	DTCOLNVHFNMMDifferenceL1
33 IPDCONGDMoMDifferenceL0	SRVPRDMoMPercentChangeL0	DTCTHFNMMDifferenceL0
34 IPDCONGDMoMDifferenceL1	SRVPRDMoMPercentChangeL1	DTCTHFNMMDifferenceL1
35 IPDCONGDMoMPercentChangeL0	USWTRADEMoMDifferenceL0	DTCTHFNMMDifferenceL0
36 IPDCONGDMoMPercentChangeL1	USWTRADEMoMDifferenceL1	DTCTHFNMMDifferenceL1
37 IPNCONGDMoMDifferenceL0	USWTRADEMoMPercentChangeL0	INVESTMoMDifferenceL0
38 IPNCONGDMoMDifferenceL1	USWTRADEMoMPercentChangeL1	INVESTMoMDifferenceL1
39 IPNCONGDMoMPercentChangeL0	USTRADEMoMDifferenceL0	INVESTMoMPercentChangeL0
40 IPNCONGDMoMPercentChangeL1	USTRADEMoMDifferenceL1	INVESTMoMPercentChangeL1
41 IPBUSEQMoMDifferenceL0	USTRADEMoMPercentChangeL0	ism_man_emplMoMDifferenceL0
42 IPBUSEQMoMDifferenceL1	USTRADEMoMPercentChangeL1	ism_man_emplMoMDifferenceL1
43 IPBUSEQMoMPercentChangeL0	USFIREMoMDifferenceL0	ism_man_emplMoMPercentChangeL0
44 IPBUSEQMoMPercentChangeL1	USFIREMoMDifferenceL1	ism_man_emplMoMPercentChangeL1
45 IPMATMoMDifferenceL0	USFIREMoMPercentChangeL0	ism_man_newordersMoMDifferenceL0
46 IPMATMoMDifferenceL1	USFIREMoMPercentChangeL1	ism_man_newordersMoMDifferenceL1
47 IPMATMoMPercentChangeL0	USGOVTMoMDifferenceL0	ism_man_newordersMoMPercentChangeL0
48 IPMATMoMPercentChangeL1	USGOVTMoMDifferenceL1	ism_man_newordersMoMPercentChangeL1
49 IPDMATMoMDifferenceL0	USGOVTMoMPercentChangeL0	ism_man_pmiMoMDifferenceL0
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51 IPDMATMoMPercentChangeL0	CES0600000007MoMDifferenceL0	ism_man_pmiMoMPercentChangeL0
52 IPDMATMoMPercentChangeL1	CES0600000007MoMDifferenceL1	ism_man_pmiMoMPercentChangeL1

53	IPNMATMoMDifferenceL0	CES0600000007MoMPercentChangeL0	ism_man_pricesMoMDifferenceL0
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56	IPNMATMoMPercentChangeL1	AWOTMANMoMDifferenceL1	ism_man_pricesMoMPercentChangeL1
57	IPMANSICSMoMDifferenceL0	AWOTMANMoMPercentChangeL0	USGG2YR_IndexMoMDifferenceL0
58	IPMANSICSMoMDifferenceL1	AWOTMANMoMPercentChangeL1	USGG2YR_IndexMoMDifferenceL1
59	IPMANSICSMoMPercentChangeL0	AWHMANMoMDifferenceL0	USGG2YR_IndexMoMPercentChangeL0
60	IPMANSICSMoMPercentChangeL1	AWHMANMoMDifferenceL1	USGG2YR_IndexMoMPercentChangeL1
61	IPB51222SMoMDifferenceL0	AWHMANMoMPercentChangeL0	USGG10YR_IndexMoMDifferenceL0
62	IPB51222SMoMDifferenceL1	AWHMANMoMPercentChangeL1	USGG10YR_IndexMoMDifferenceL1
63	IPB51222SMoMPercentChangeL0	HOUSTMoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0
64	IPB51222SMoMPercentChangeL1	HOUSTMoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL1
65	IPFUELSMoMDifferenceL0	HOUSTMoMPercentChangeL0	USGG30YR_IndexMoMDifferenceL0
66	IPFUELSMoMDifferenceL1	HOUSTMoMPercentChangeL1	USGG30YR_IndexMoMDifferenceL1
67	IPFUELSMoMPercentChangeL0	HOUSTNEMoMDifferenceL0	USGG30YR_IndexMoMPercentChangeL0
68	IPFUELSMoMPercentChangeL1	HOUSTNEMoMDifferenceL1	USGG30YR_IndexMoMPercentChangeL1
69	CUMFNSMoMDifferenceL0	HOUSTNEMoMPercentChangeL0	VIX_IndexMoMDifferenceL0
70	CUMFNSMoMDifferenceL1	HOUSTNEMoMPercentChangeL1	VIX_IndexMoMDifferenceL1
71	CUMFNSMoMPercentChangeL0	HOUSTMWMoMDifferenceL0	VIX_IndexMoMPercentChangeL0
72	CUMFNSMoMPercentChangeL1	HOUSTMWMoMDifferenceL1	VIX_IndexMoMPercentChangeL1
73	CLF16OVMoMDifferenceL0	HOUSTMWMoMPercentChangeL0	MOVE_IndexMoMDifferenceL0
74	CLF16OVMoMDifferenceL1	HOUSTMWMoMPercentChangeL1	MOVE_IndexMoMDifferenceL1
75	CLF16OVMoMPercentChangeL0	HOUSTSMoMDifferenceL0	MOVE_IndexMoMPercentChangeL0
76	CLF16OVMoMPercentChangeL1	HOUSTSMoMDifferenceL1	MOVE_IndexMoMPercentChangeL1
77	CE16OVMoMDifferenceL0	HOUSTSMoMPercentChangeL0	AAIIBULL_IndexMoMDifferenceL0
78	CE16OVMoMDifferenceL1	HOUSTSMoMPercentChangeL1	AAIIBULL_IndexMoMDifferenceL1
79	CE16OVMoMPercentChangeL0	HOUSTWMoMDifferenceL0	AAIIBULL_IndexMoMPercentChangeL0
80	CE16OVMoMPercentChangeL1	HOUSTWMoMDifferenceL1	AAIIBULL_IndexMoMPercentChangeL1
81	UNRATEMoMDifferenceL0	HOUSTWMoMPercentChangeL0	AAIIBEAR_IndexMoMDifferenceL0
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83	UNRATEMoMPercentChangeL0	ISRATIOMoMDifferenceL0	AAIIBEAR_IndexMoMPercentChangeL0
84	UNRATEMoMPercentChangeL1	ISRATIOMoMDifferenceL1	AAIIBEAR_IndexMoMPercentChangeL1
85	UEMPMEANMoMDifferenceL0	ISRATIOMoMPercentChangeL0	SPX_Index_14DRSIMoMDifferenceL0
86	UEMPMEANMoMDifferenceL1	ISRATIOMoMPercentChangeL1	SPX_Index_14DRSIMoMDifferenceL1
87	UEMPMEANMoMPercentChangeL0	M1SLMoMDifferenceL0	SPX_Index_14DRSIMoMPercentChangeL0
88	UEMPMEANMoMPercentChangeL1	M1SLMoMDifferenceL1	SPX_Index_14DRSIMoMPercentChangeL1
89	UEMPLT5MoMDifferenceL0	M1SLMoMPercentChangeL0	SPX_Index_30DRSIMoMDifferenceL0
90	UEMPLT5MoMDifferenceL1	M1SLMoMPercentChangeL1	SPX_Index_30DRSIMoMDifferenceL1
91	UEMPLT5MoMPercentChangeL0	M2SLMoMDifferenceL0	SPX_Index_30DRSIMoMPercentChangeL0
92	UEMPLT5MoMPercentChangeL1	M2SLMoMDifferenceL1	SPX_Index_30DRSIMoMPercentChangeL1
93	UEMP5TO14MoMDifferenceL0	M2SLMoMPercentChangeL0	TED_G_IndexMoMDifferenceL0
94	UEMP5TO14MoMDifferenceL1	M2SLMoMPercentChangeL1	TED_G_IndexMoMDifferenceL1
95	UEMP5TO14MoMPercentChangeL0	M2REALMoMDifferenceL0	TED_G_IndexMoMPercentChangeL0
96	UEMP5TO14MoMPercentChangeL1	M2REALMoMDifferenceL1	TED_G_IndexMoMPercentChangeL1
97	UEMP15OVMoMDifferenceL0	M2REALMoMPercentChangeL0	BAA10YB_IndexMoMDifferenceL0
98	UEMP15OVMoMDifferenceL1	M2REALMoMPercentChangeL1	BAA10YB_IndexMoMDifferenceL1
99	UEMP15OVMoMPercentChangeL0	AMBSLMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0
100	UEMP15OVMoMPercentChangeL1	AMBSLMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL1
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102	UEMP15T26MoMDifferenceL1	AMBSLMoMPercentChangeL1	

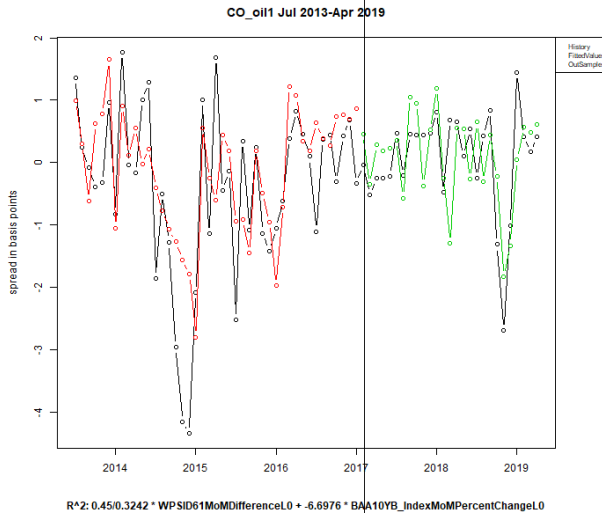
4.2.1 Model - 1

Backtesting on differenced values

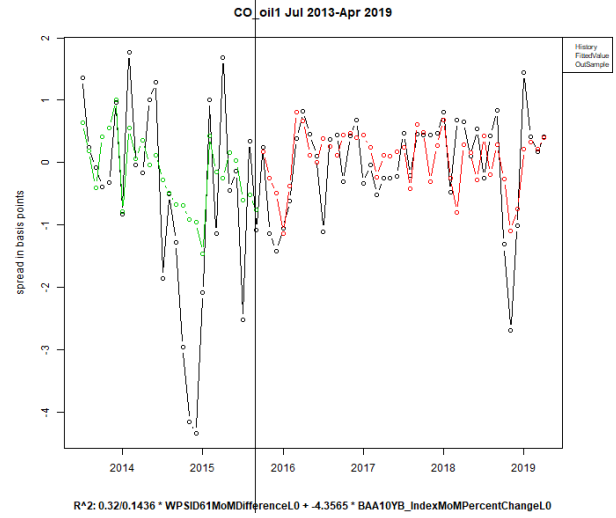
Backtesting on level values



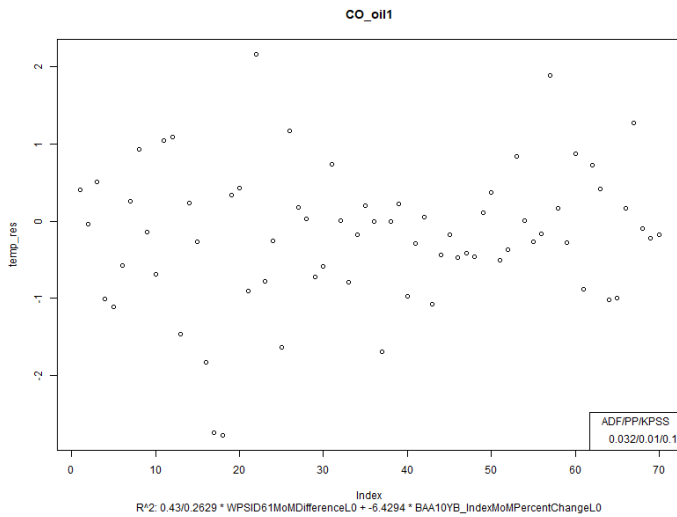
Out-of-Sample testing



Out-of-Sample testing



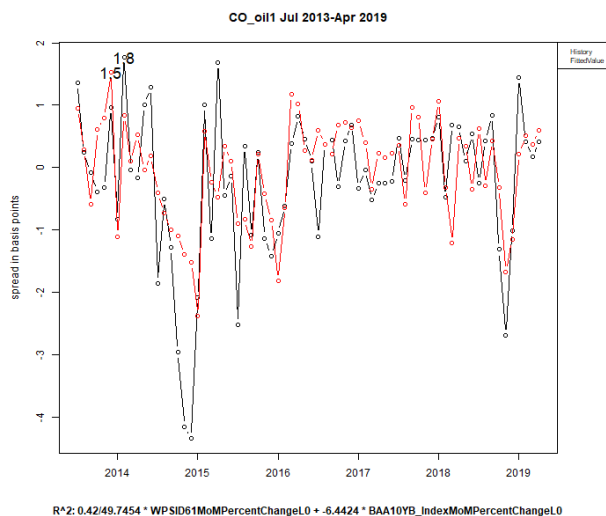
Stationarity of Residuals



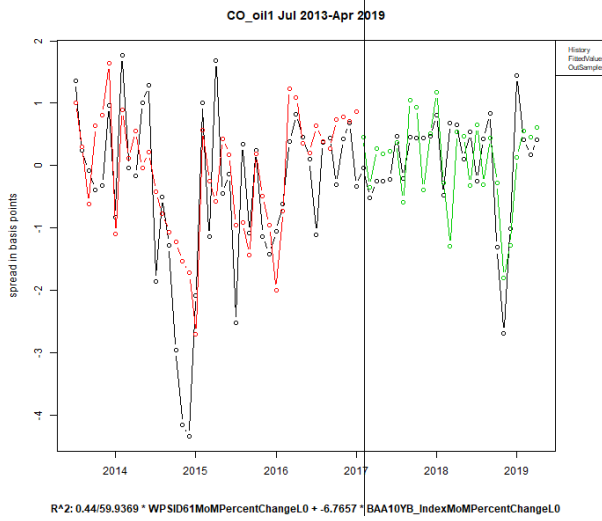
4.2.2 Model - 2

Backtesting on differenced values

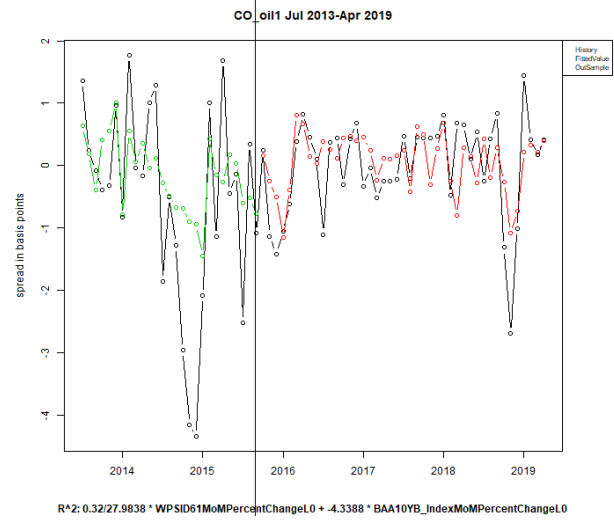
Backtesting on level values



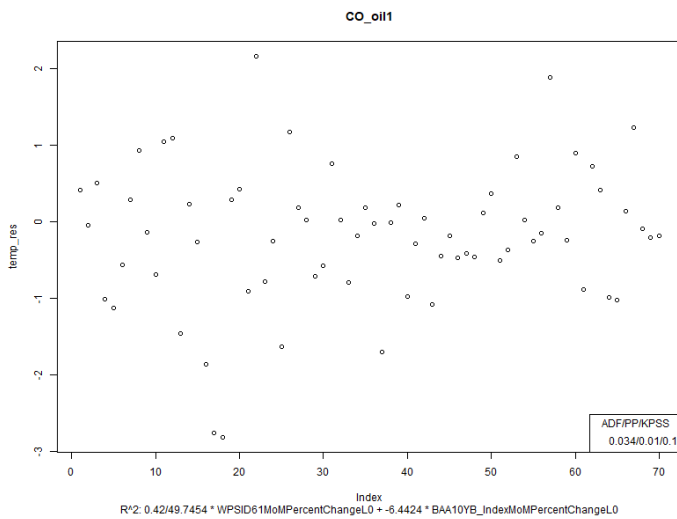
Out-of-Sample testing



Out-of-Sample testing



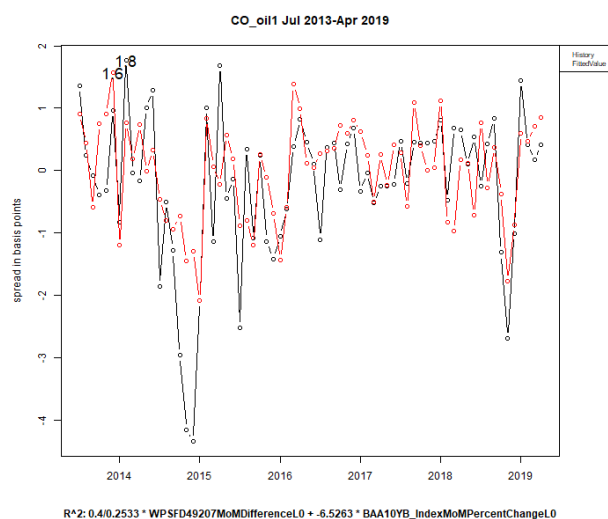
Stationarity of Residuals



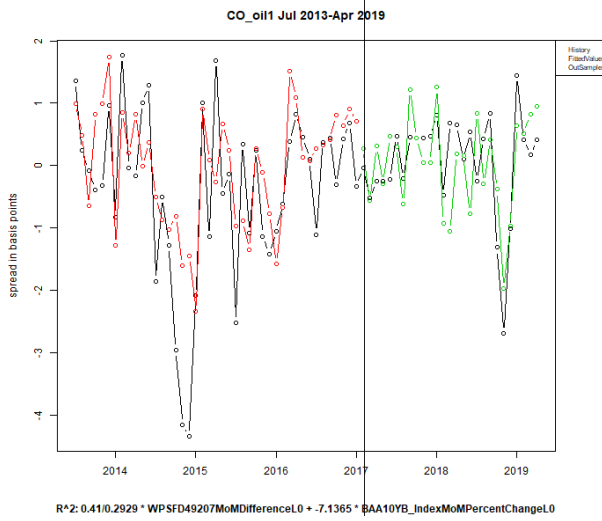
4.2.3 Model - 3

Backtesting on differenced values

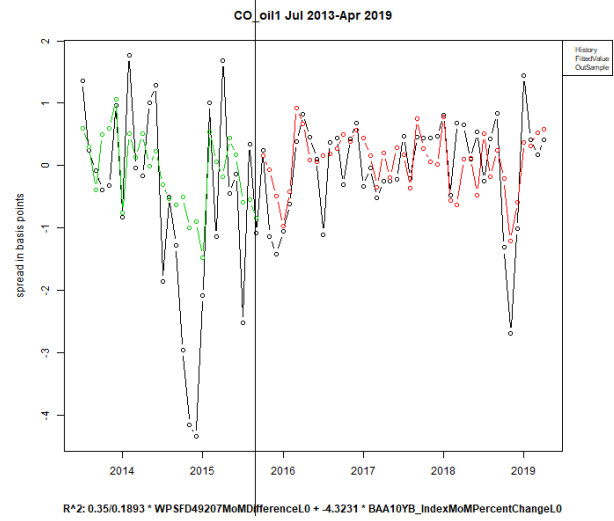
Backtesting on level values



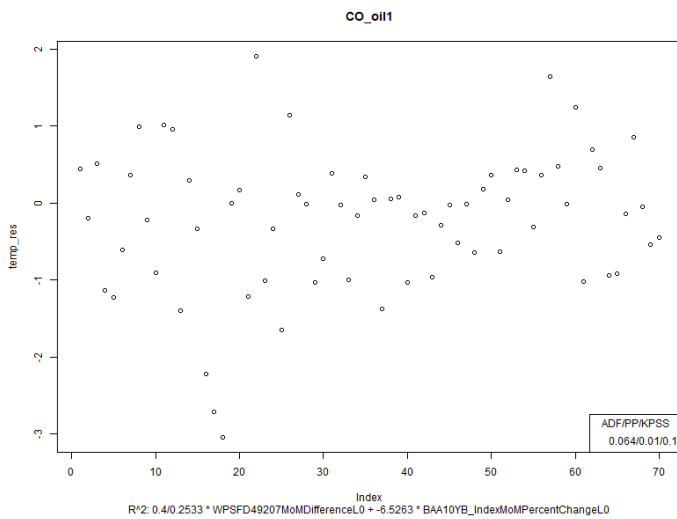
Out-of-Sample testing



Out-of-Sample testing



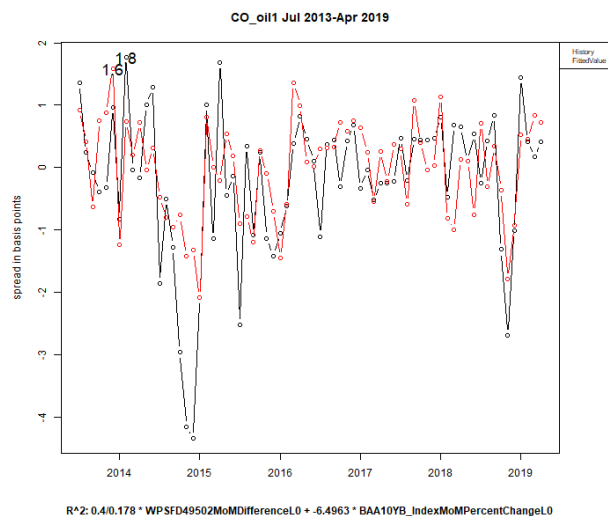
Stationarity of Residuals



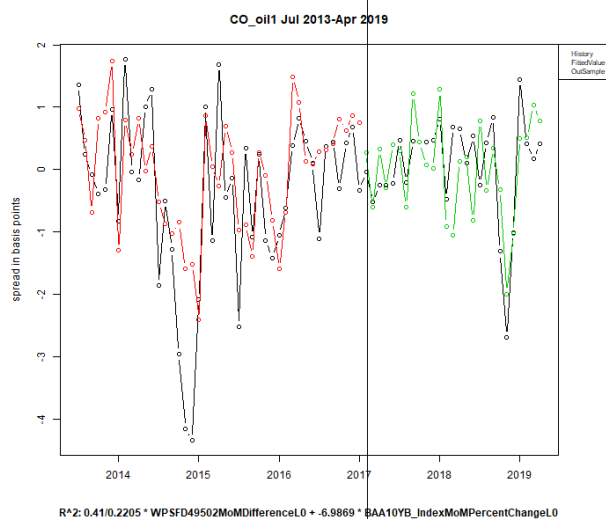
4.2.4 Model - 4

Backtesting on differenced values

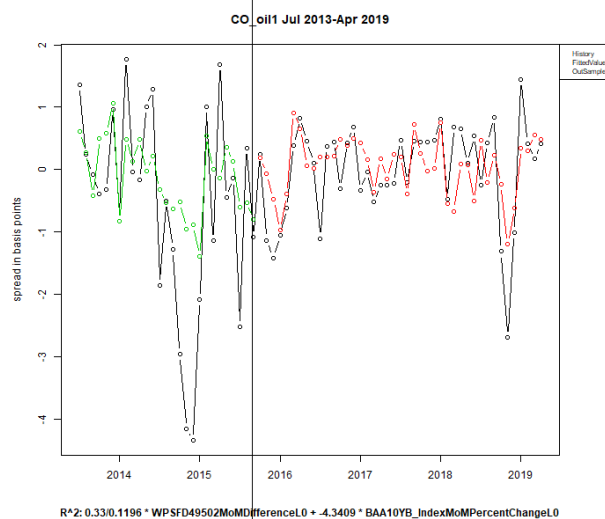
Backtesting on level values



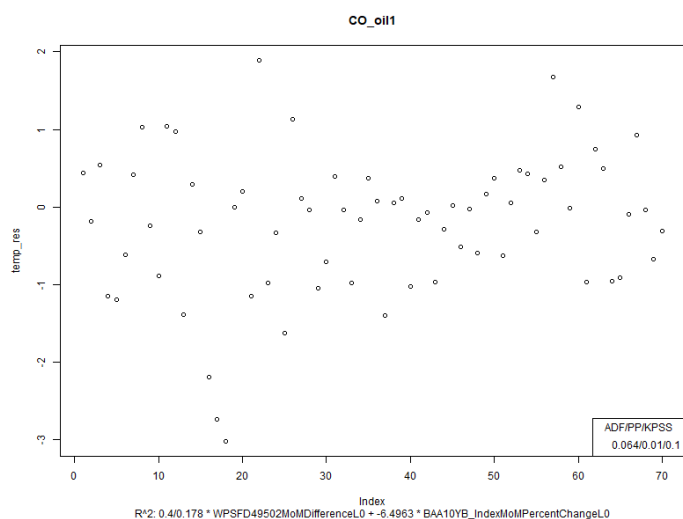
Out-of-Sample testing



Out-of-Sample testing



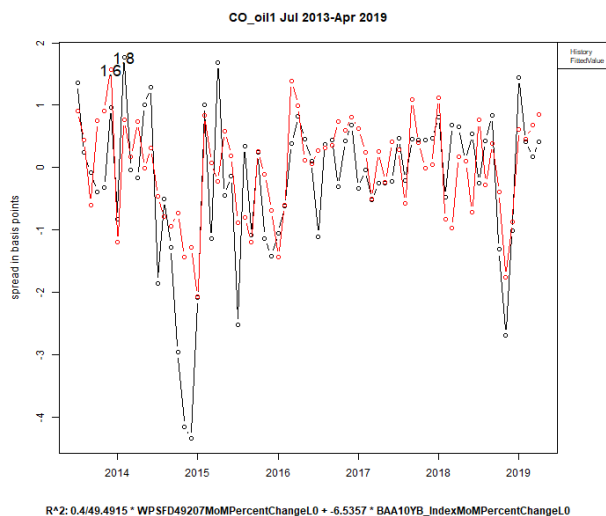
Stationarity of Residuals



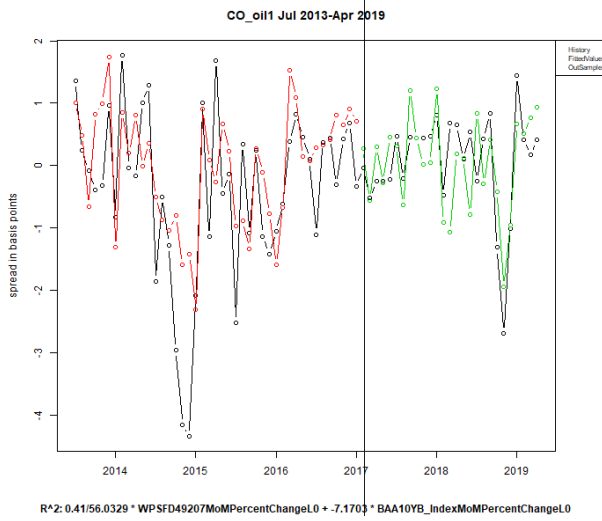
4.2.5 Model - 5

Backtesting on differenced values

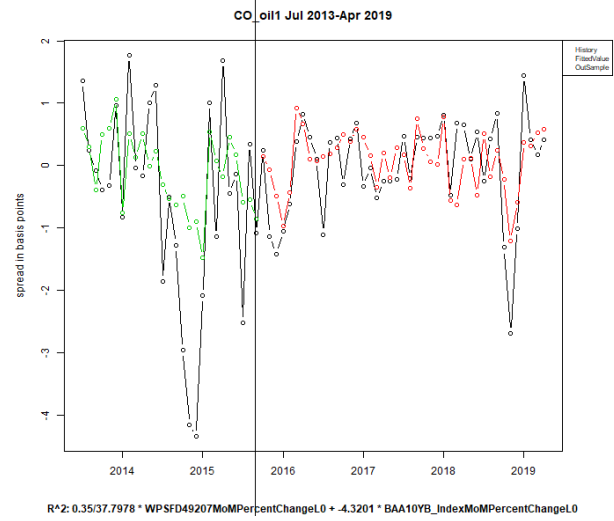
Backtesting on level values



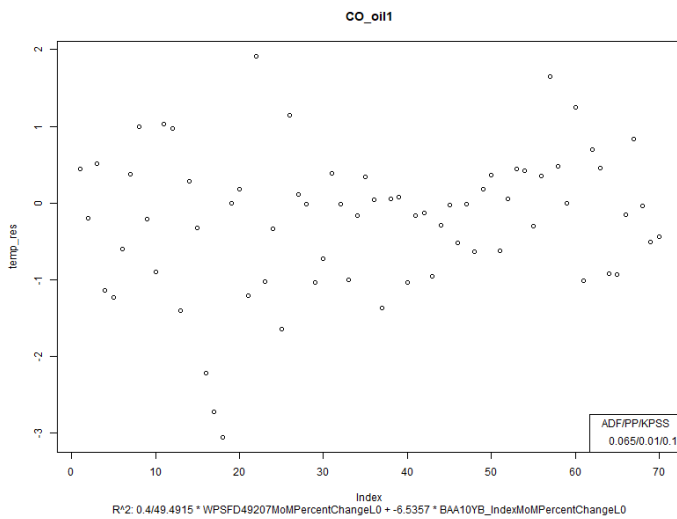
Out-of-Sample testing



Out-of-Sample testing



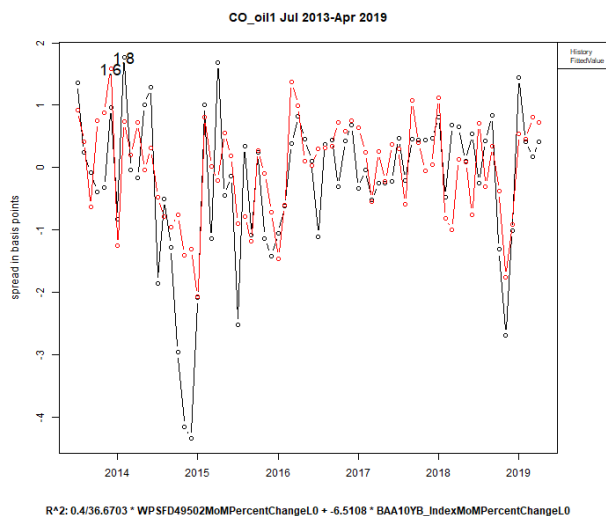
Stationarity of Residuals



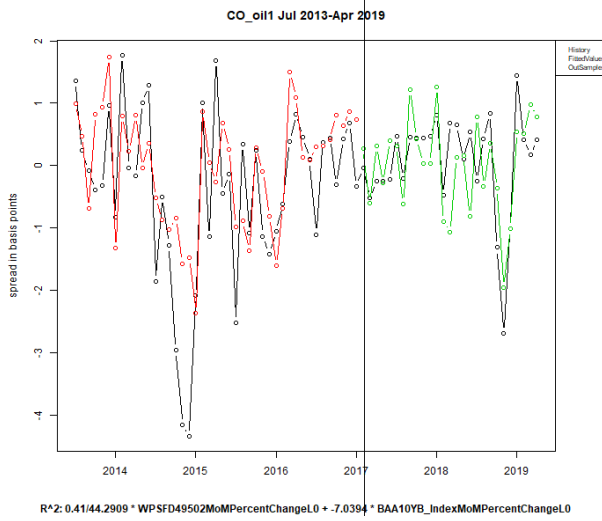
4.2.6 Model - 6

Backtesting on differenced values

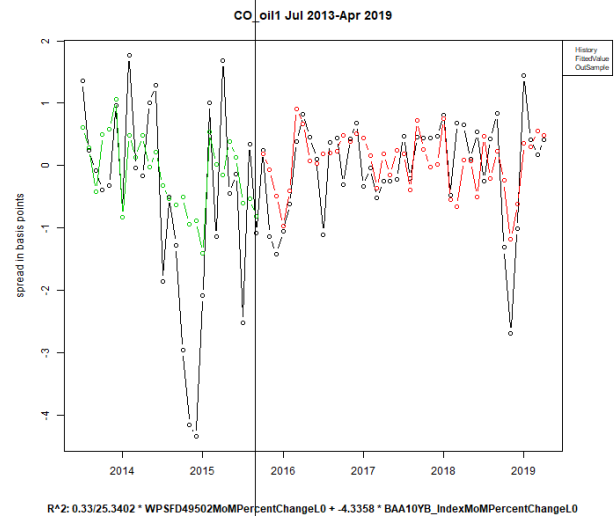
Backtesting on level values



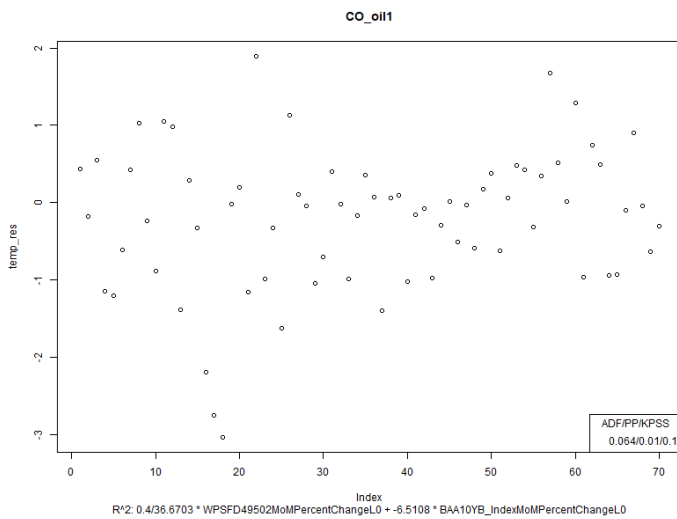
Out-of-Sample testing



Out-of-Sample testing



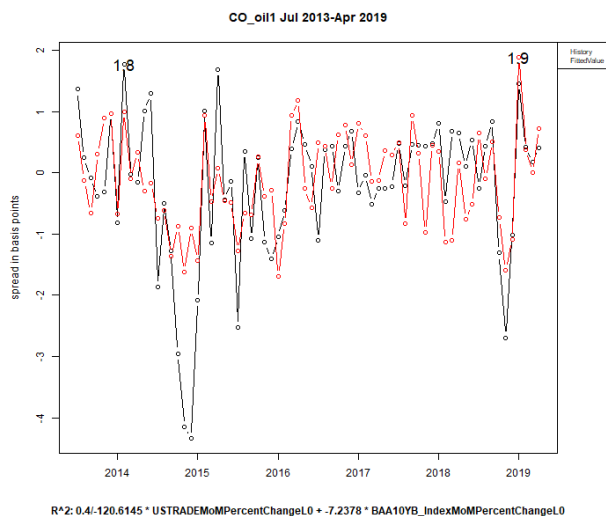
Stationarity of Residuals



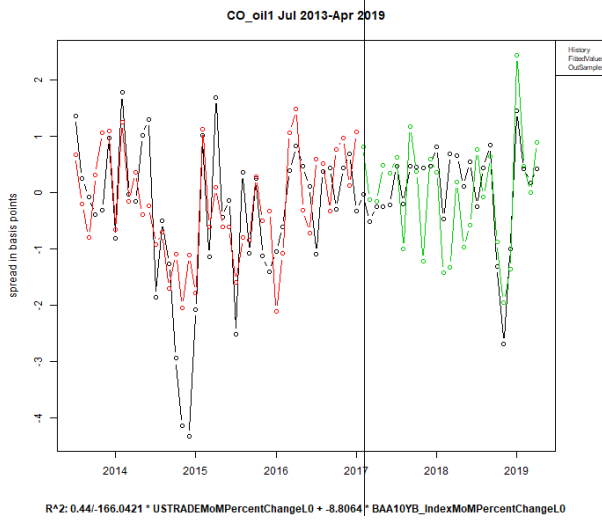
4.2.7 Model - 7

Backtesting on differenced values

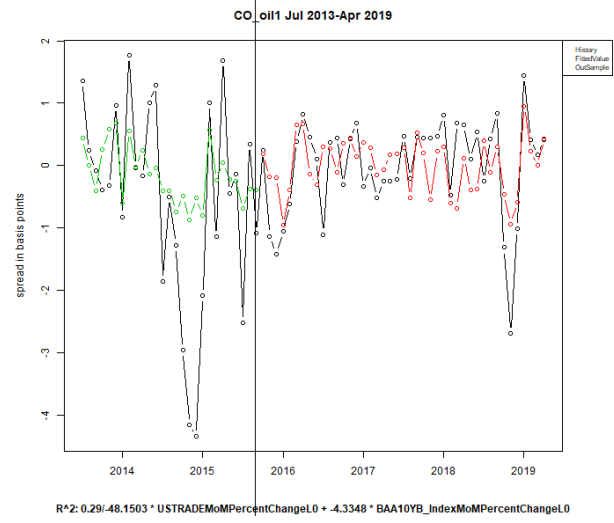
Backtesting on level values



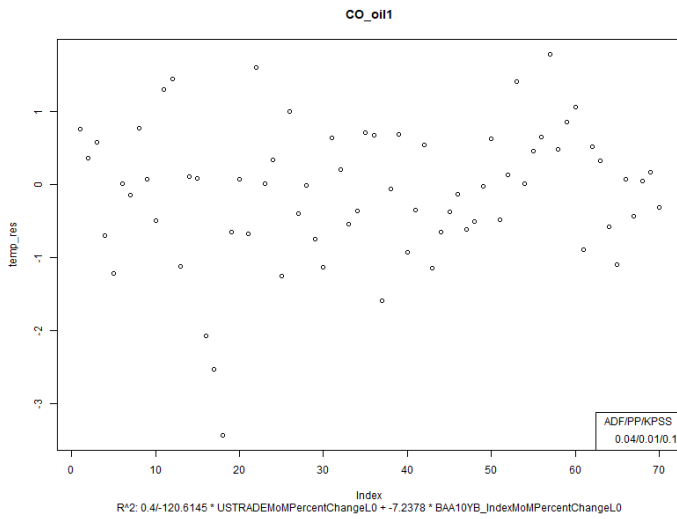
Out-of-Sample testing



Out-of-Sample testing



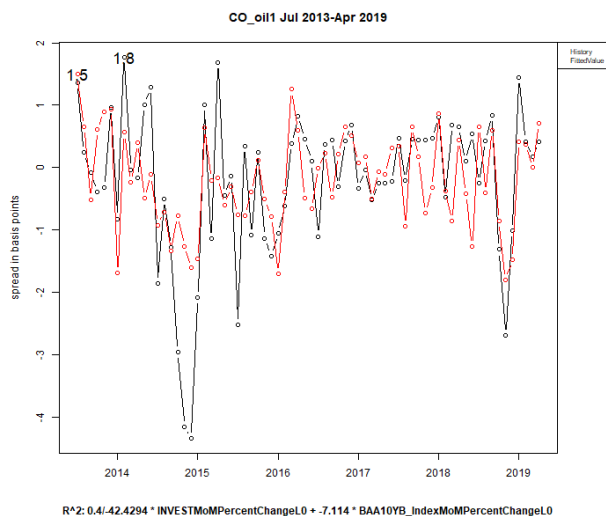
Stationarity of Residuals



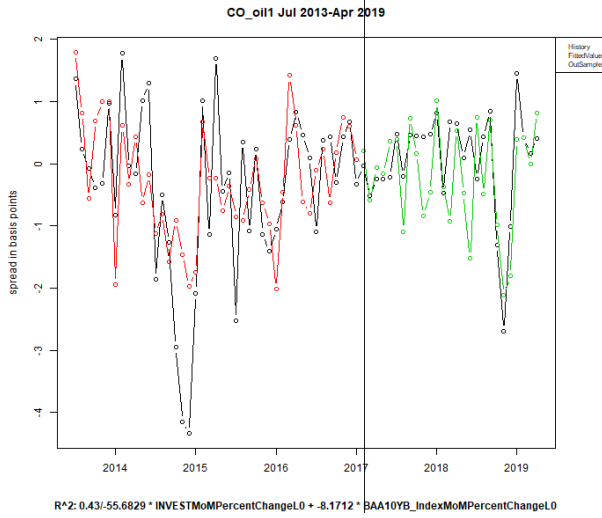
4.2.8 Model - 8

Backtesting on differenced values

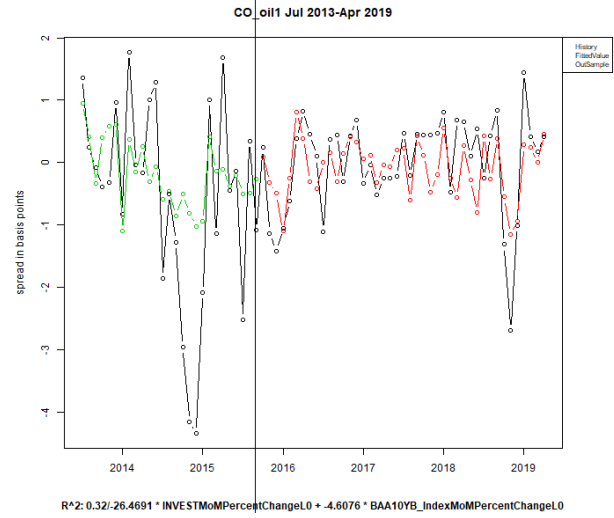
Backtesting on level values



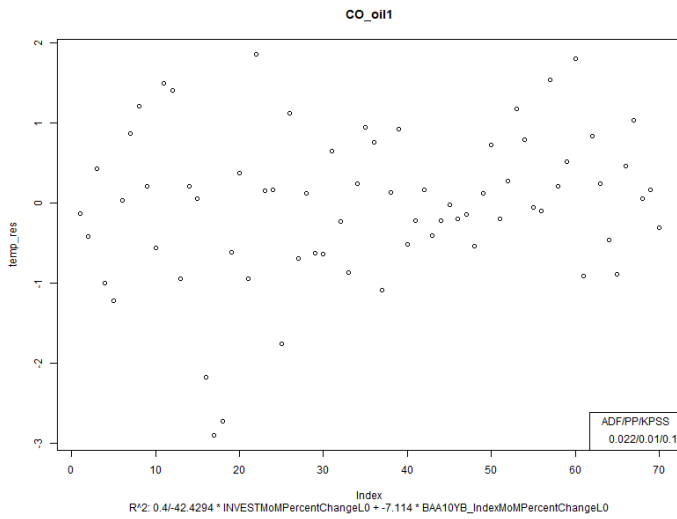
Out-of-Sample testing



Out-of-Sample testing



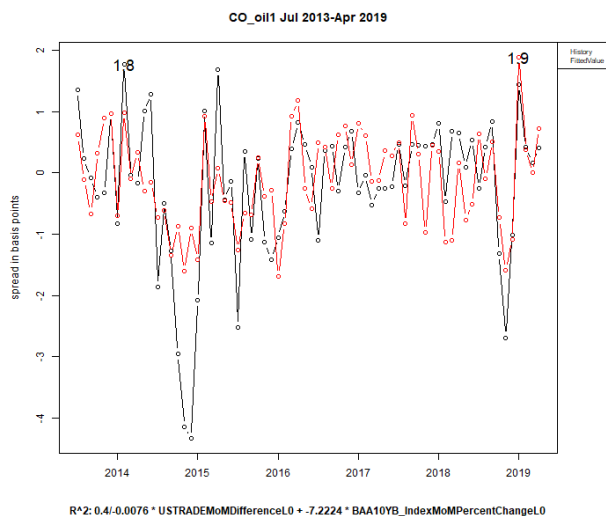
Stationarity of Residuals



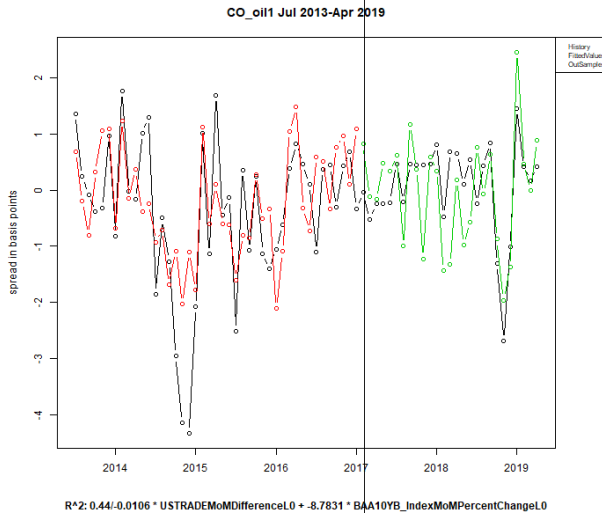
4.2.9 Model - 9

Backtesting on differenced values

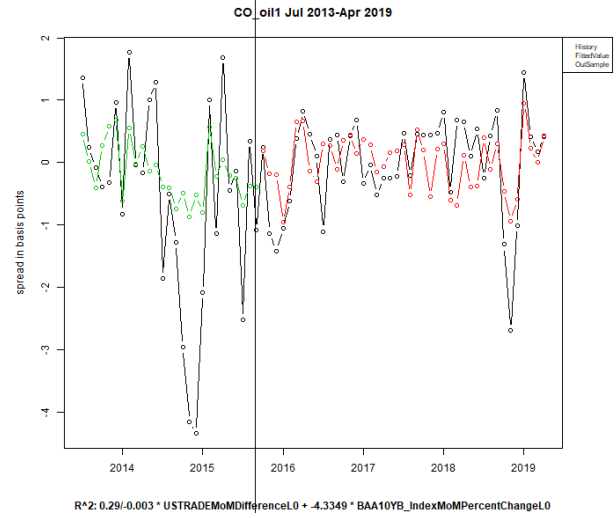
Backtesting on level values



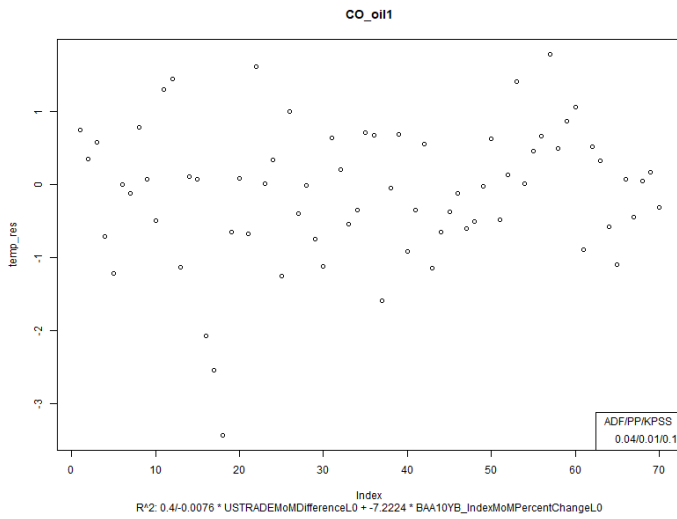
Out-of-Sample testing



Out-of-Sample testing



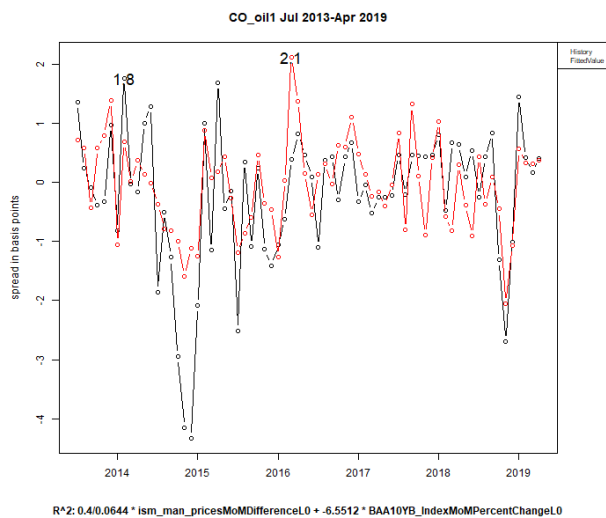
Stationarity of Residuals



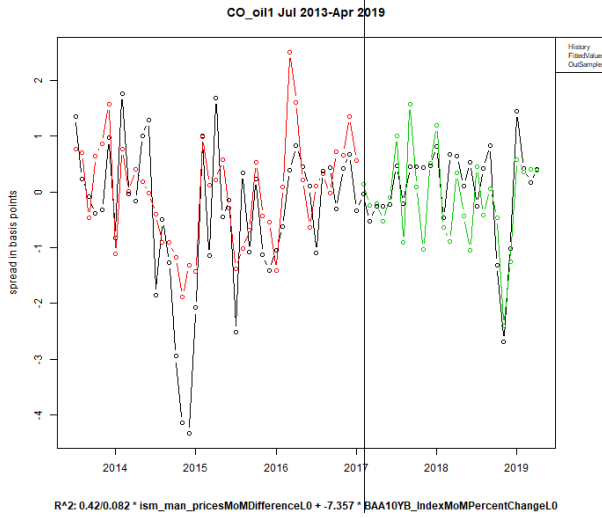
4.2.10 Model - 10

Backtesting on differenced values

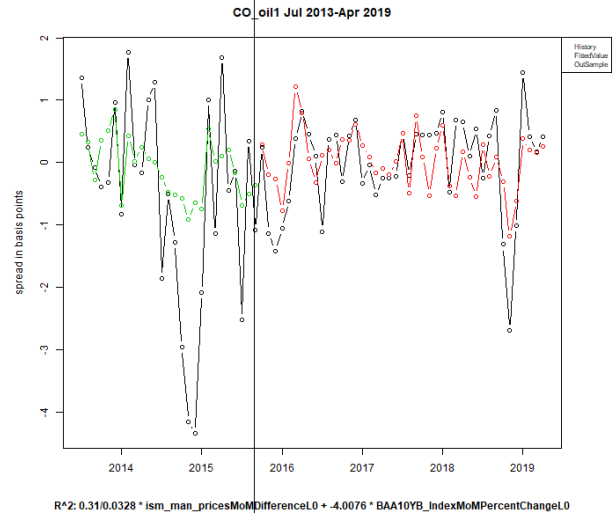
Backtesting on level values



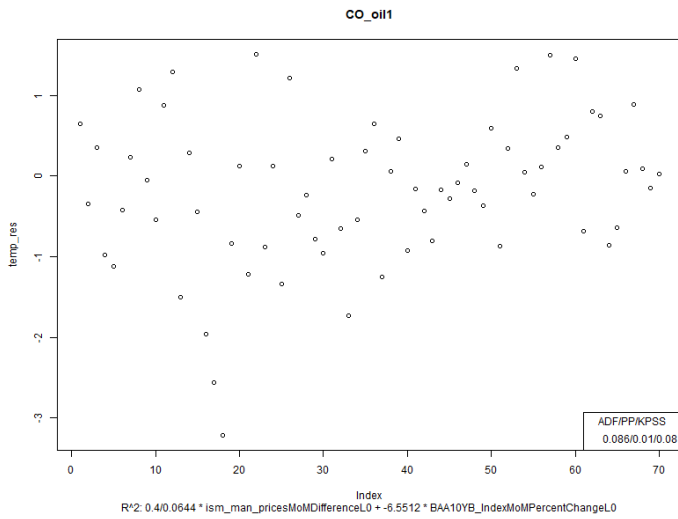
Out-of-Sample testing



Out-of-Sample testing



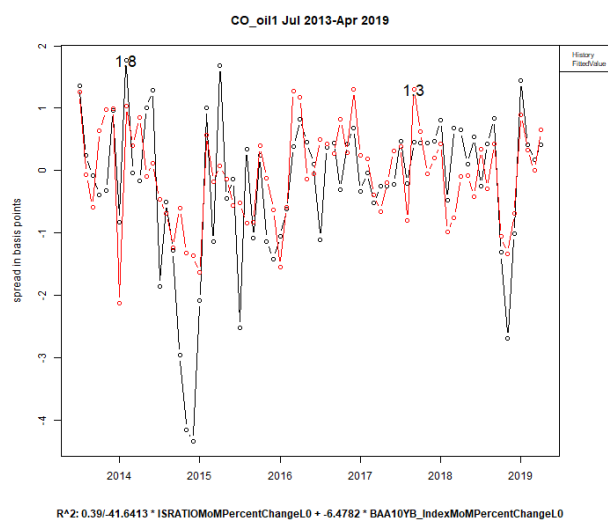
Stationarity of Residuals



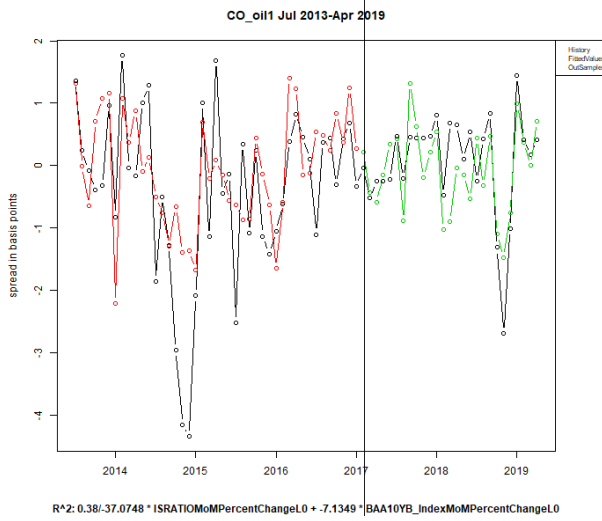
4.2.11 Model - 11

Backtesting on differenced values

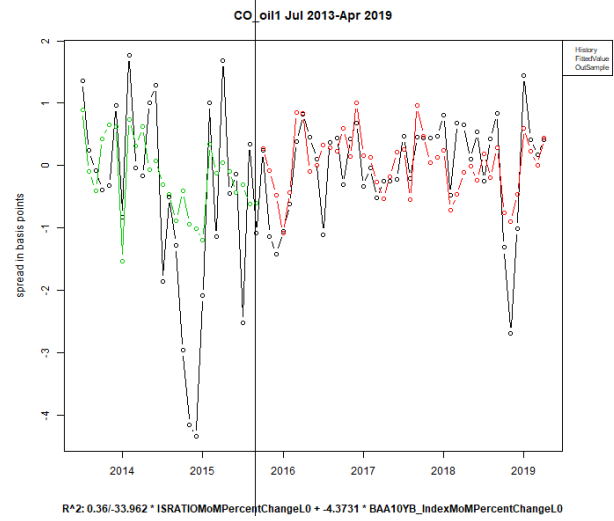
Backtesting on level values



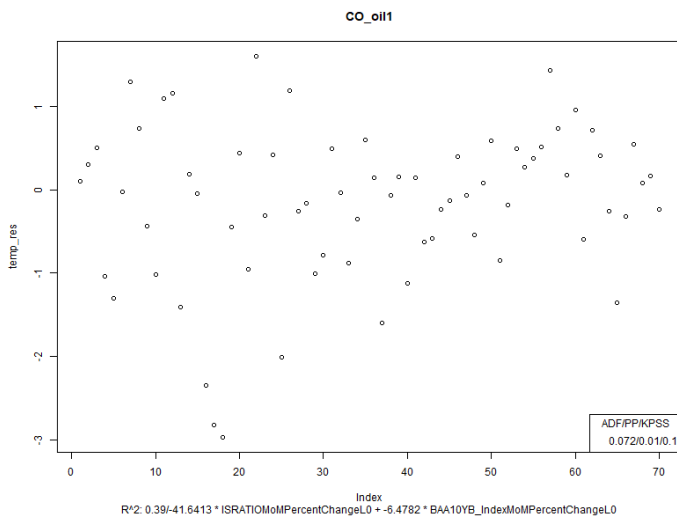
Out-of-Sample testing



Out-of-Sample testing



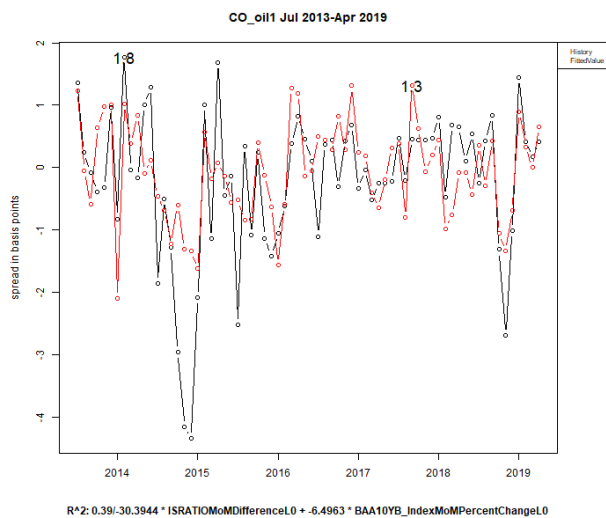
Stationarity of Residuals



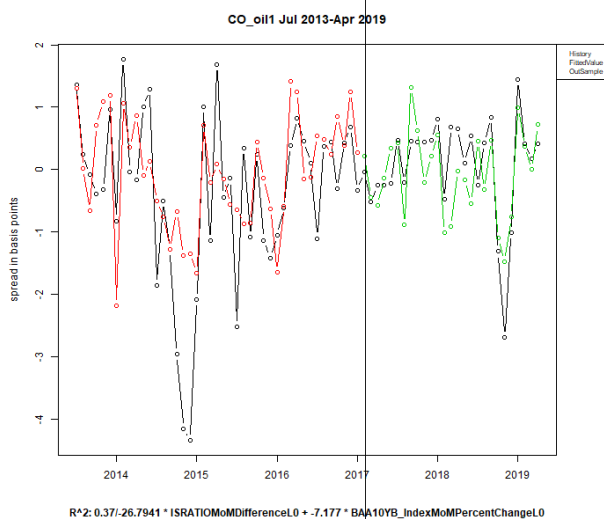
4.2.12 Model - 12

Backtesting on differenced values

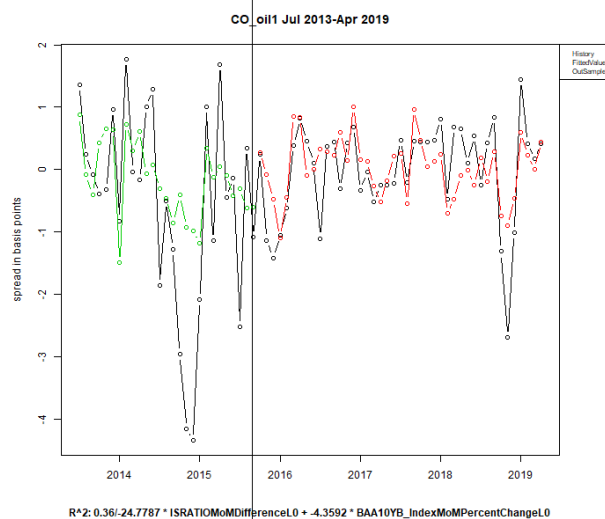
Backtesting on level values



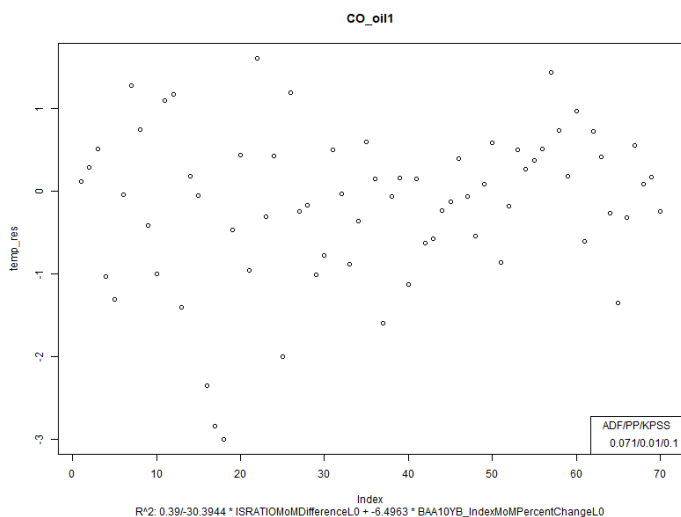
Out-of-Sample testing



Out-of-Sample testing



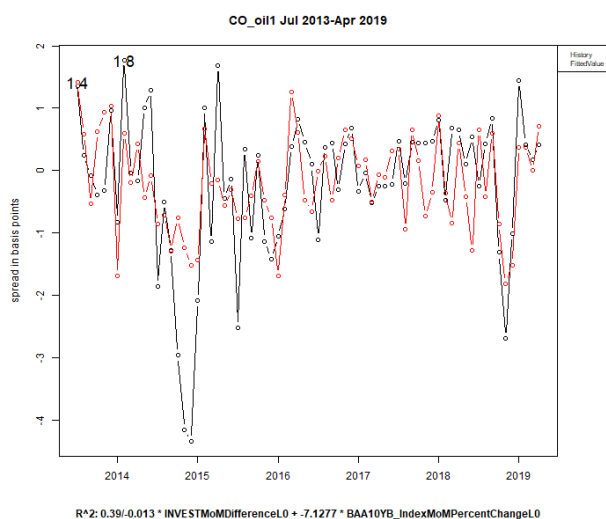
Stationarity of Residuals



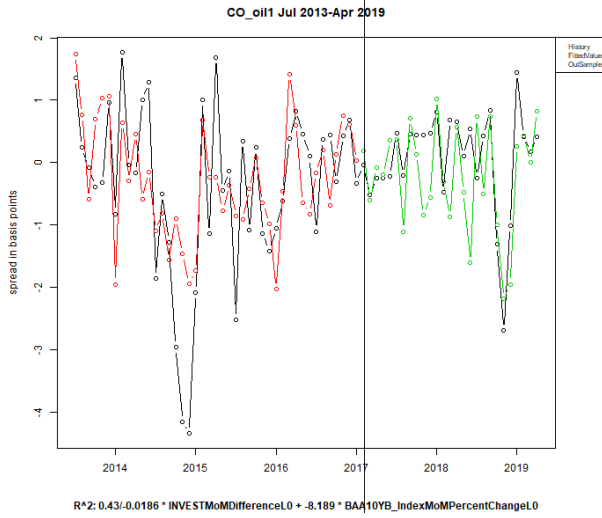
4.2.13 Model - 13

Backtesting on differenced values

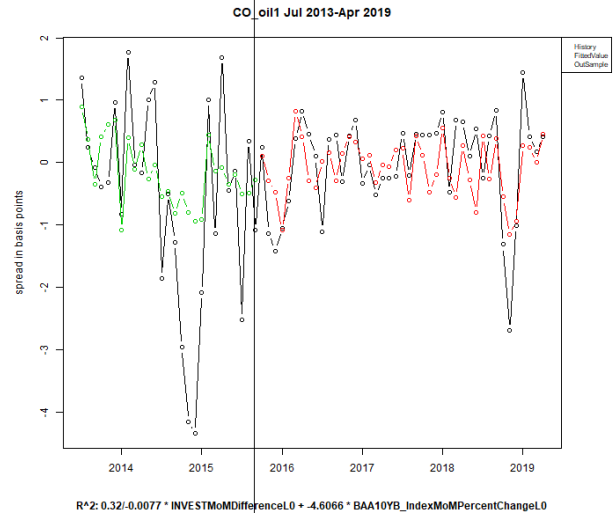
Backtesting on level values



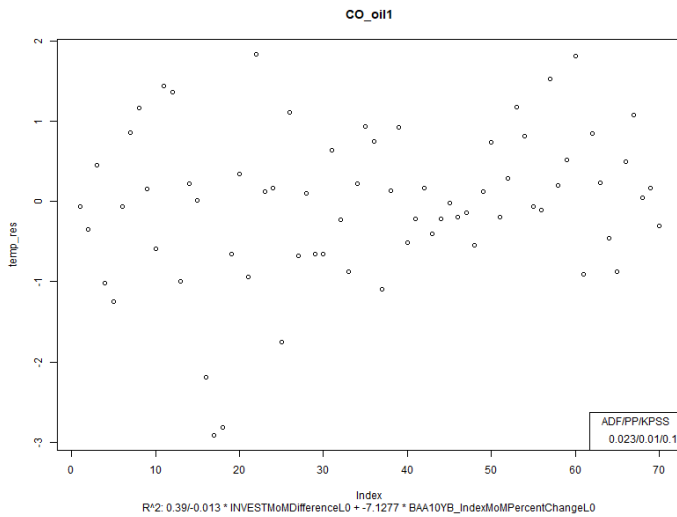
Out-of-Sample testing



Out-of-Sample testing



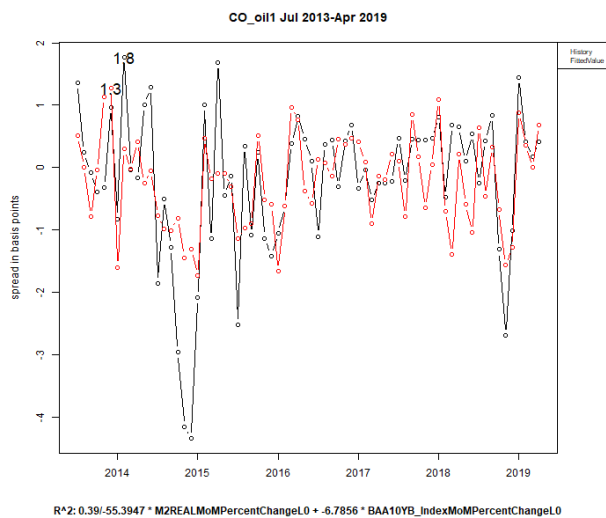
Stationarity of Residuals



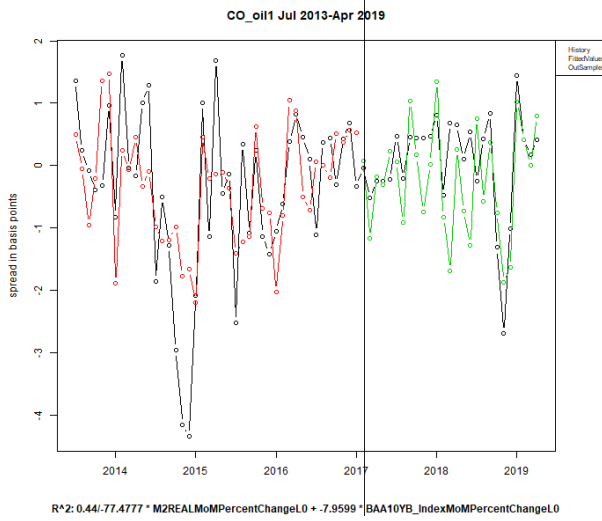
4.2.14 Model - 14

Backtesting on differenced values

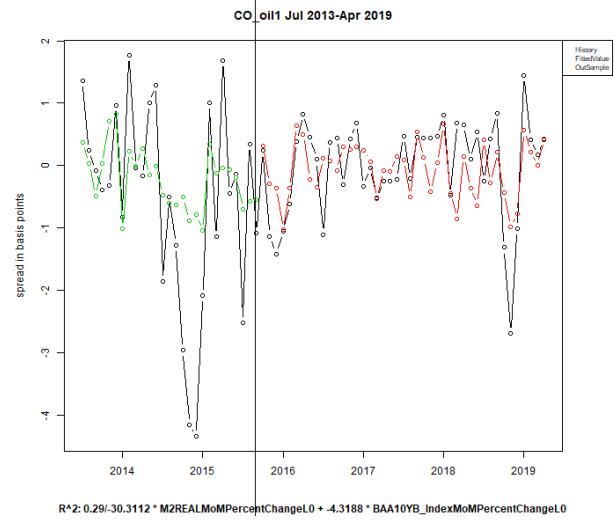
Backtesting on level values



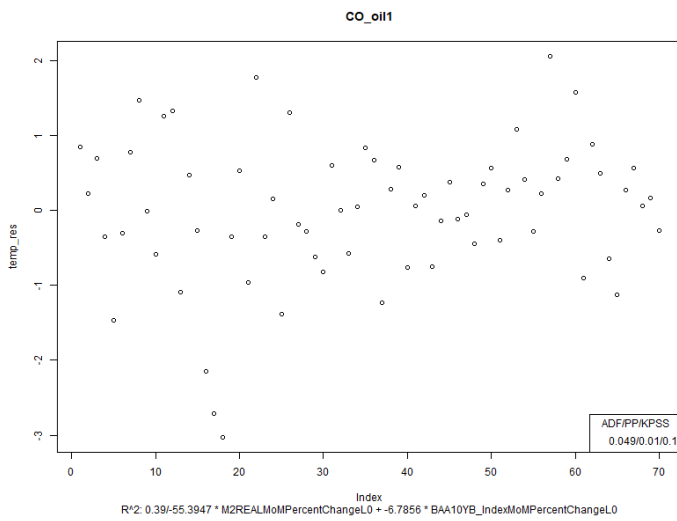
Out-of-Sample testing



Out-of-Sample testing



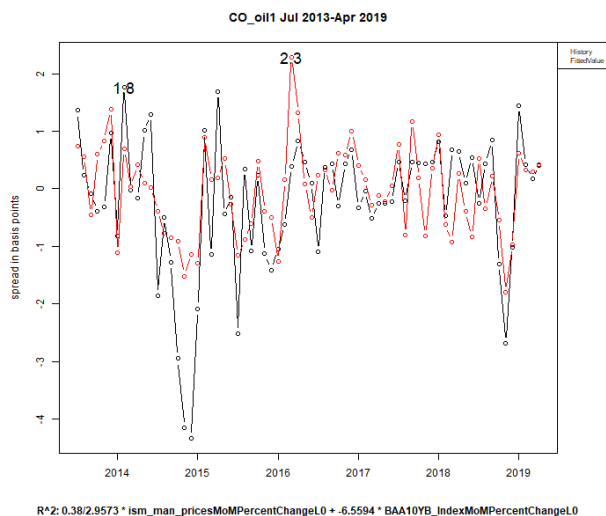
Stationarity of Residuals



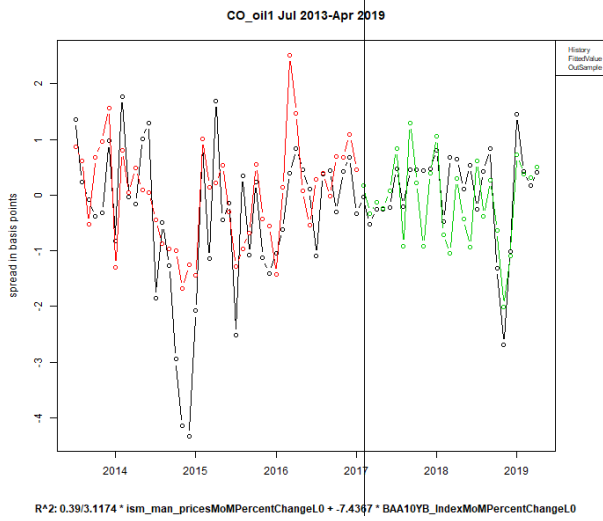
4.2.15 Model - 15

Backtesting on differenced values

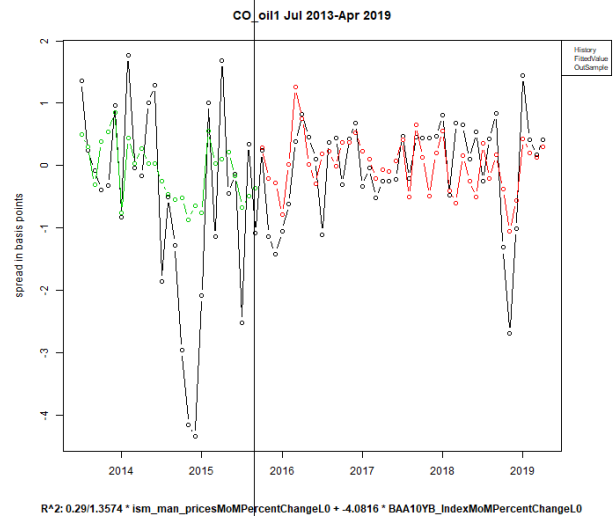
Backtesting on level values



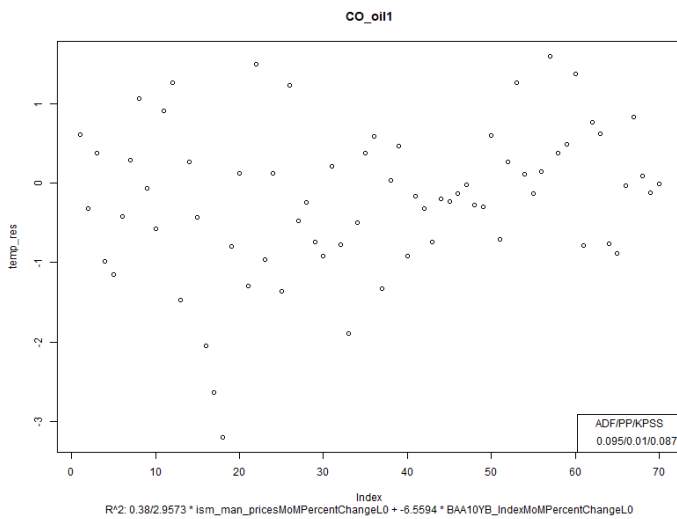
Out-of-Sample testing



Out-of-Sample testing



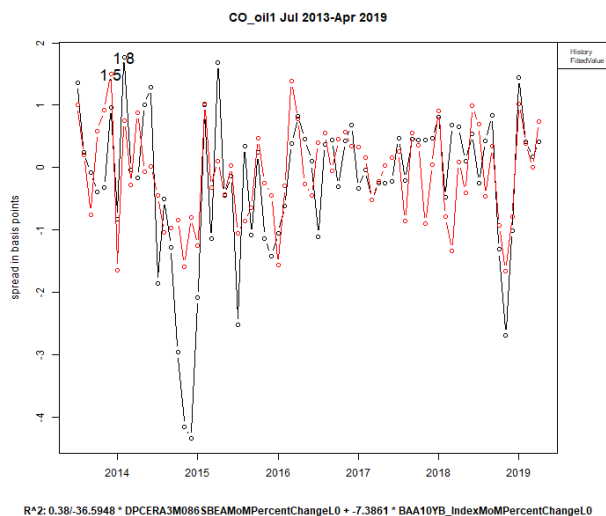
Stationarity of Residuals



4.2.16 Model - 16

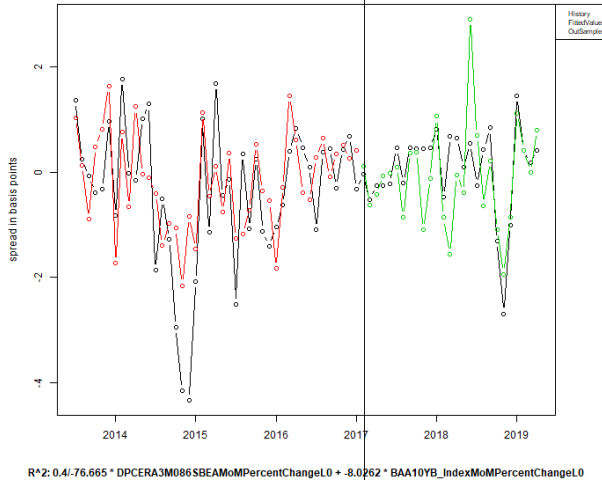
Backtesting on differenced values

Backtesting on level values



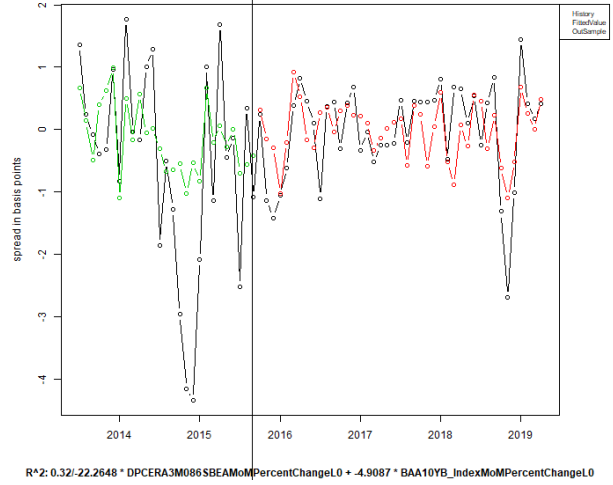
Out-of-Sample testing

CO_oil1 Jul 2013-Apr 2019



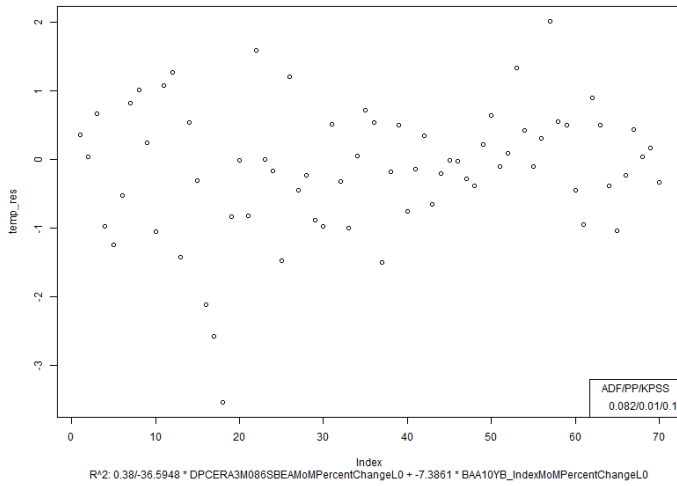
Out-of-Sample testing

CO_oil1 Jul 2013-Apr 2019



Stationarity of Residuals

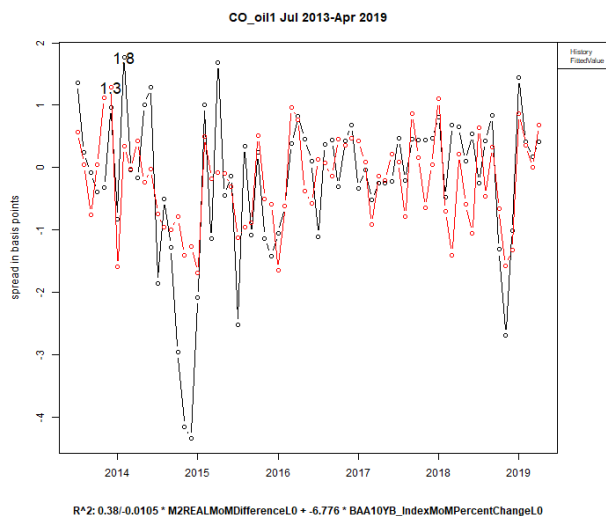
CO_oil1



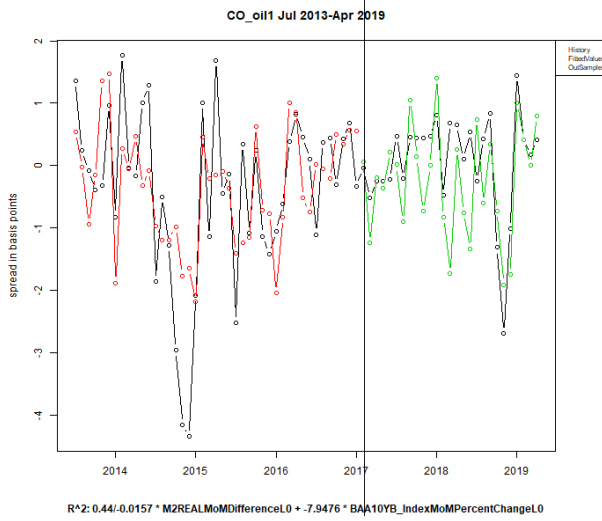
4.2.17 Model - 17

Backtesting on differenced values

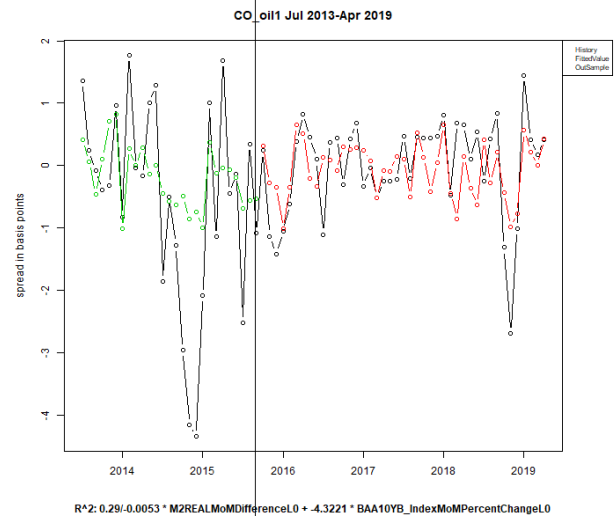
Backtesting on level values



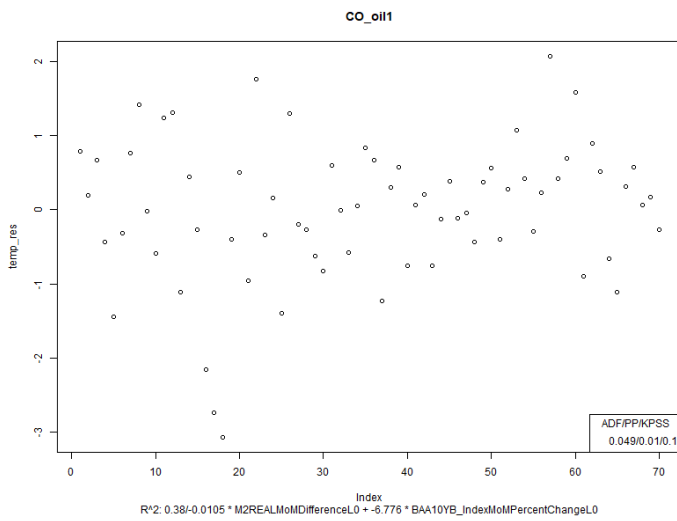
Out-of-Sample testing



Out-of-Sample testing



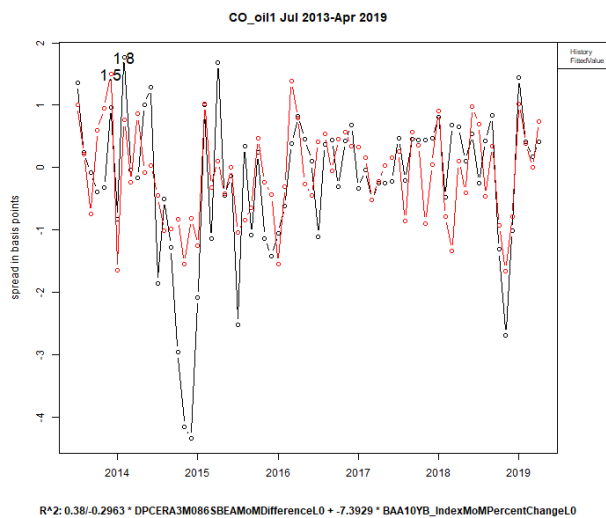
Stationarity of Residuals



4.2.18 Model - 18

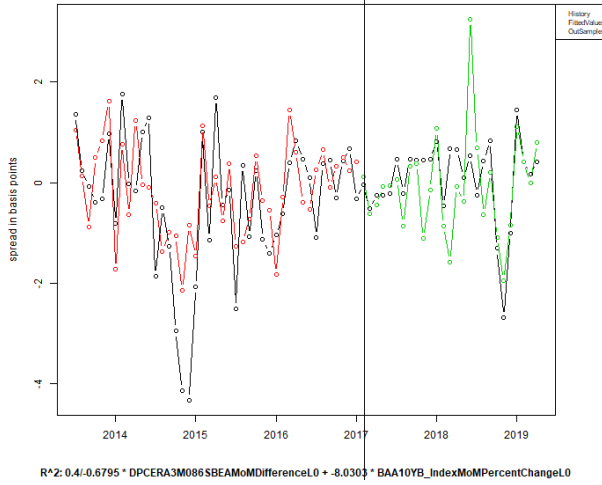
Backtesting on differenced values

Backtesting on level values



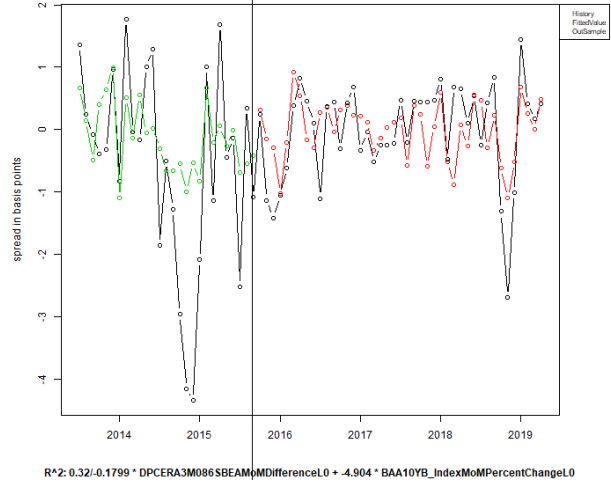
Out-of-Sample testing

CO_oil1 Jul 2013-Apr 2019



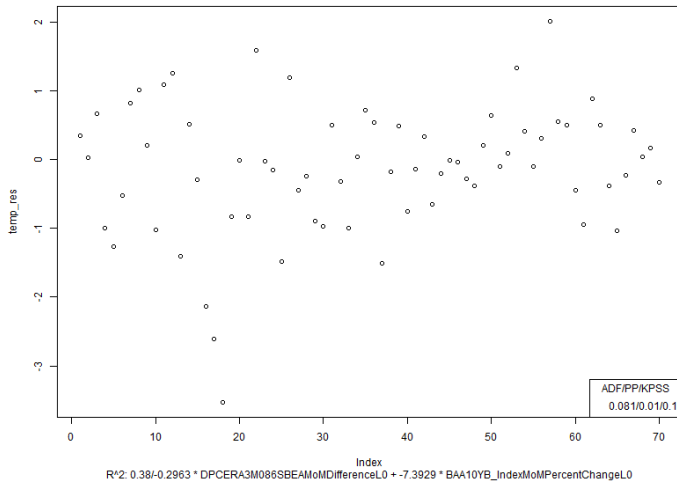
Out-of-Sample testing

CO_oil1 Jul 2013-Apr 2019



Stationarity of Residuals

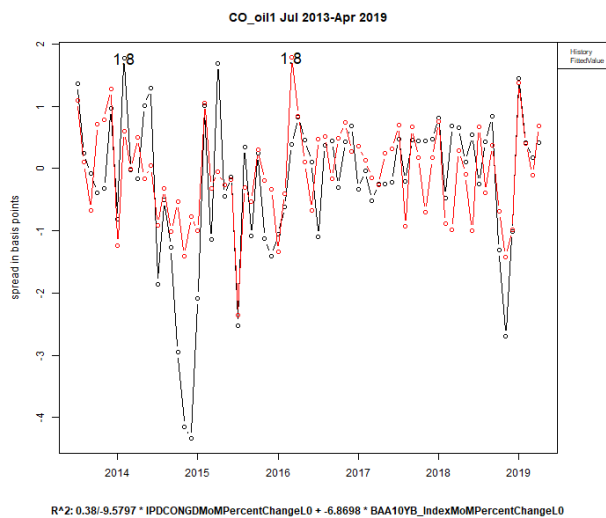
CO_oil1



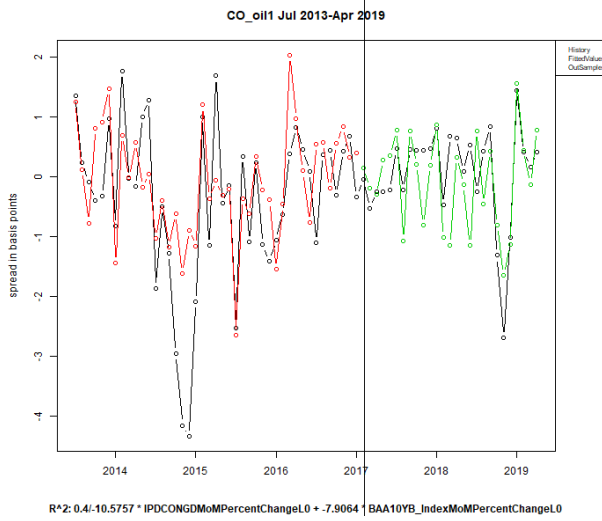
4.2.19 Model - 19

Backtesting on differenced values

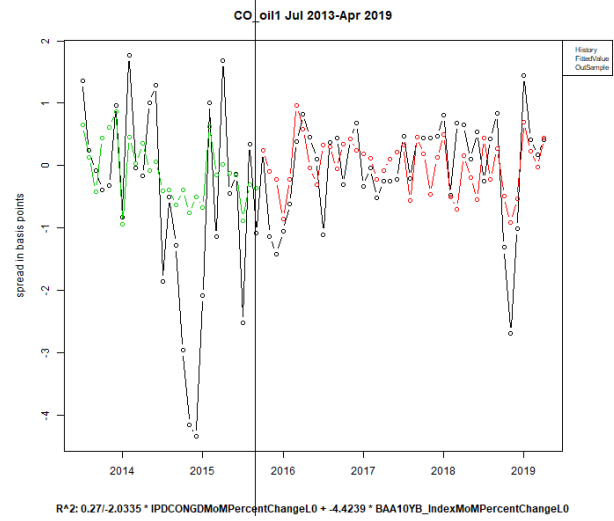
Backtesting on level values



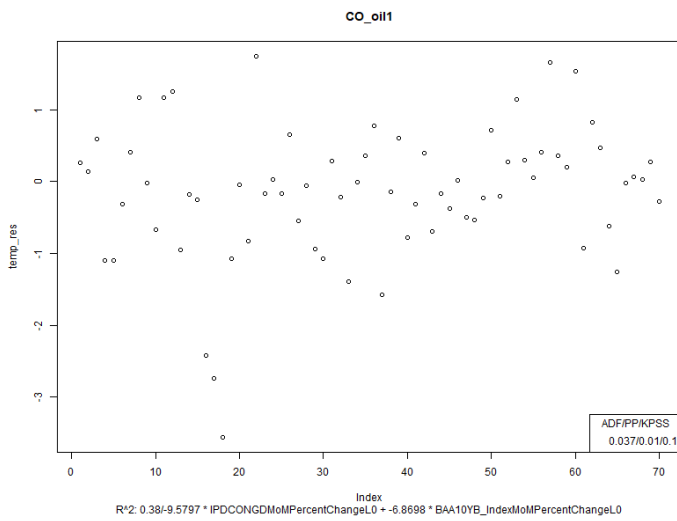
Out-of-Sample testing



Out-of-Sample testing



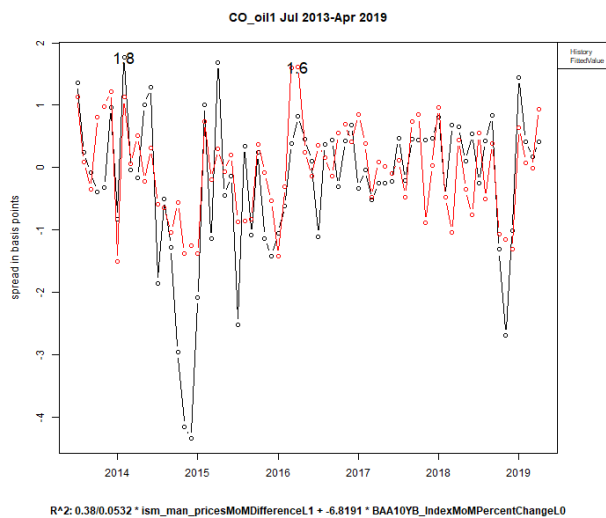
Stationarity of Residuals



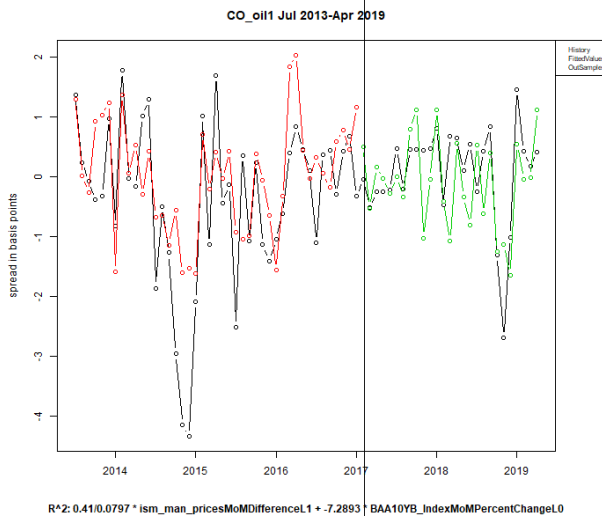
4.2.20 Model - 20

Backtesting on differenced values

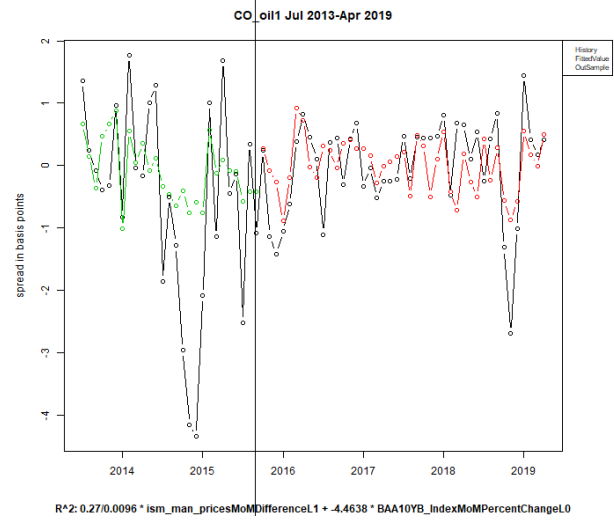
Backtesting on level values



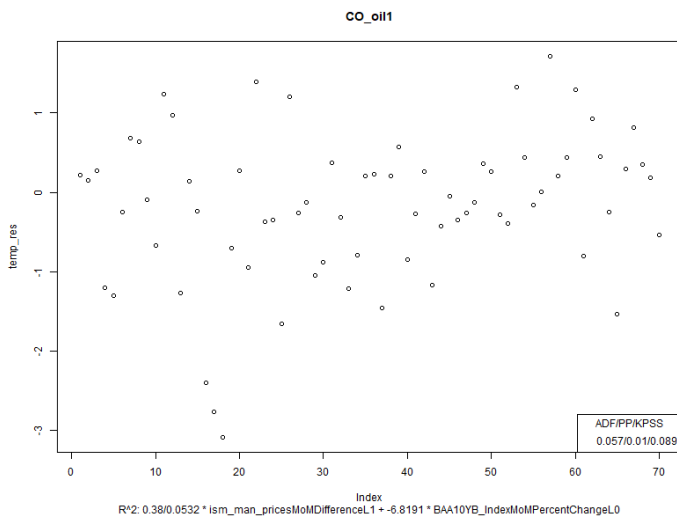
Out-of-Sample testing



Out-of-Sample testing



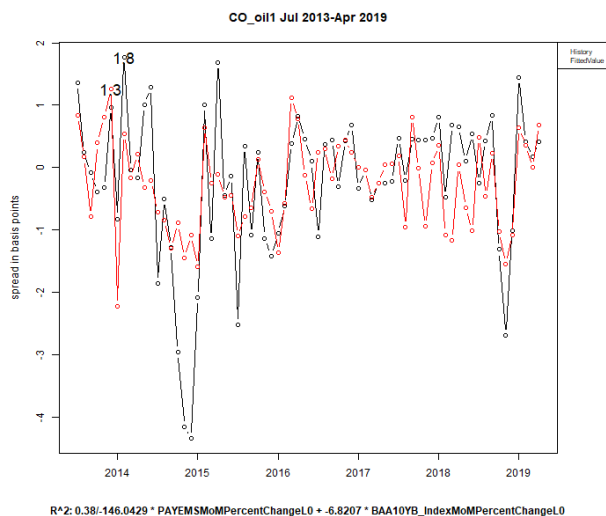
Stationarity of Residuals



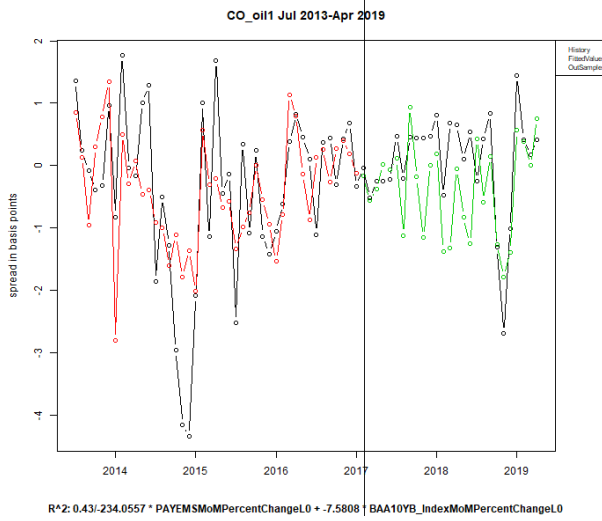
4.2.21 Model - 21

Backtesting on differenced values

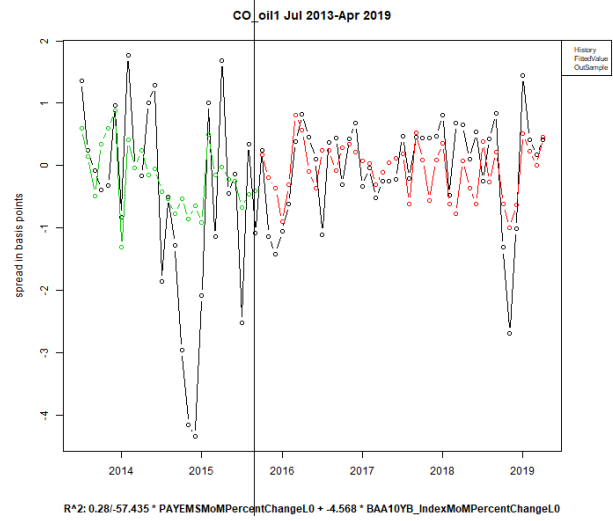
Backtesting on level values



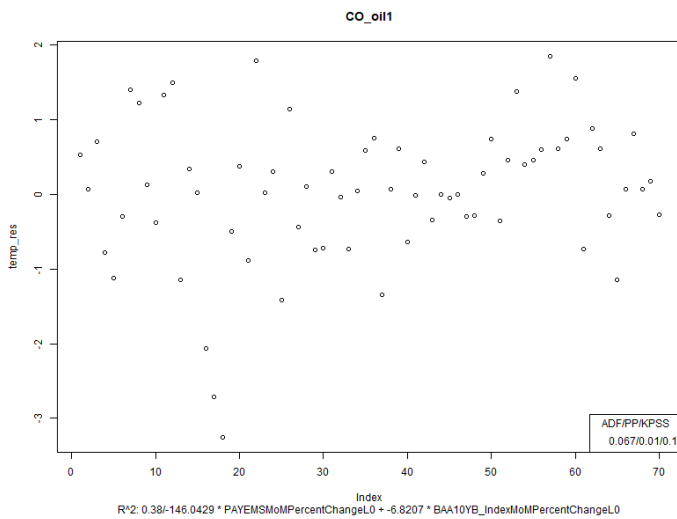
Out-of-Sample testing



Out-of-Sample testing



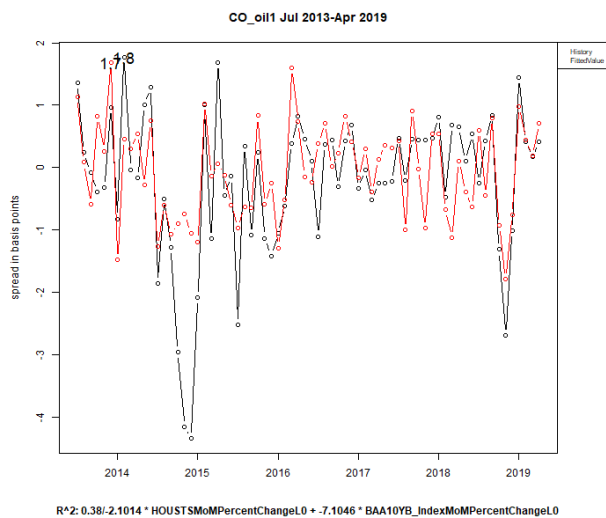
Stationarity of Residuals



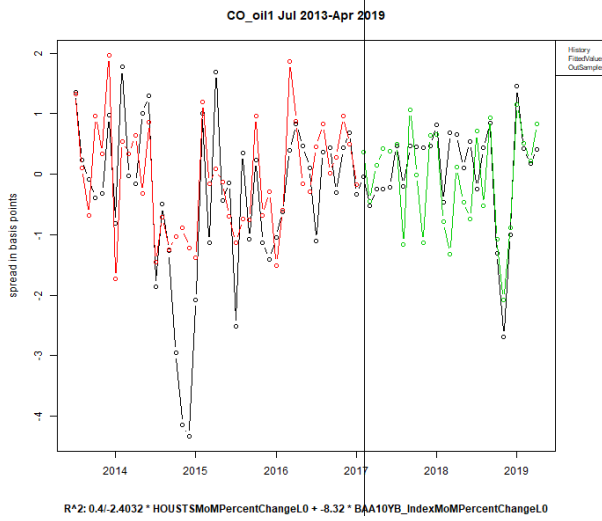
4.2.22 Model - 22

Backtesting on differenced values

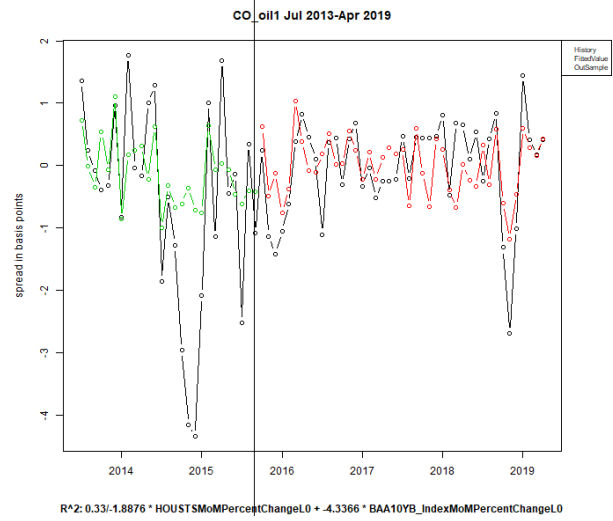
Backtesting on level values



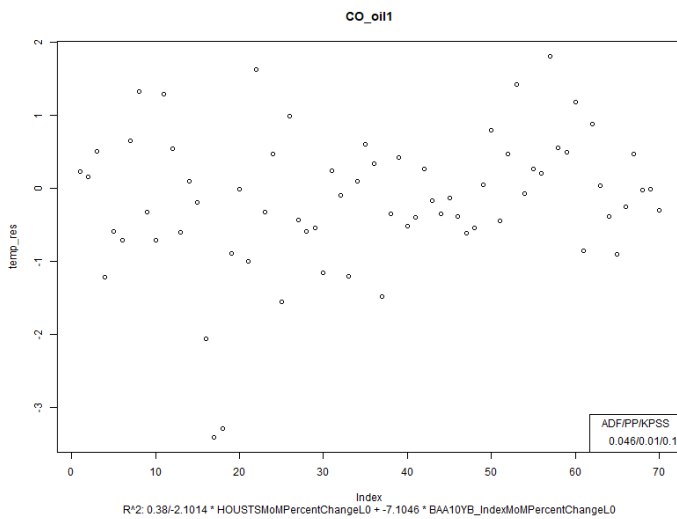
Out-of-Sample testing



Out-of-Sample testing



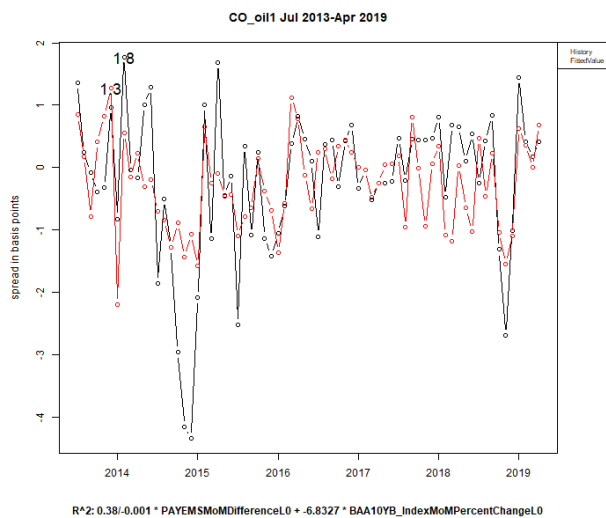
Stationarity of Residuals



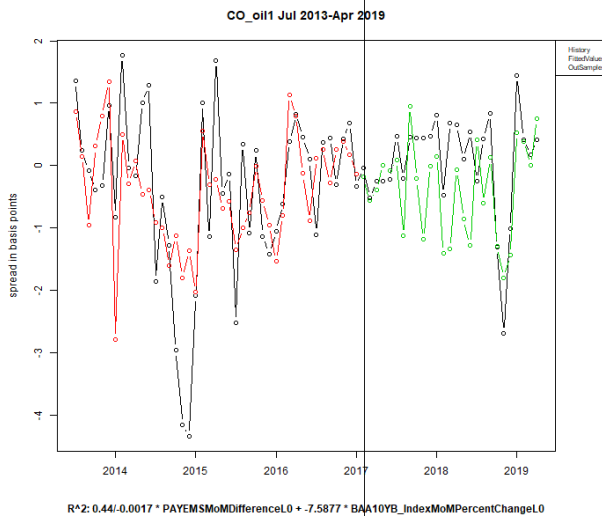
4.2.23 Model - 23

Backtesting on differenced values

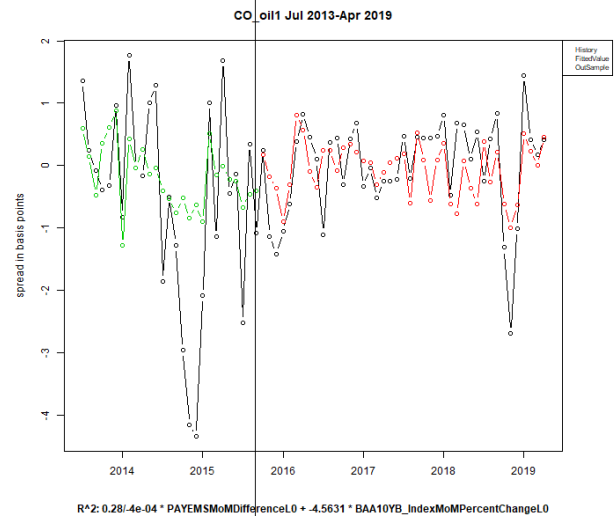
Backtesting on level values



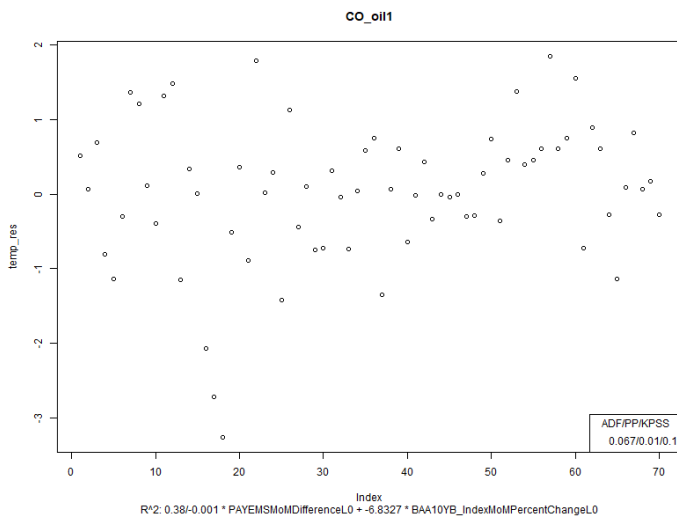
Out-of-Sample testing



Out-of-Sample testing



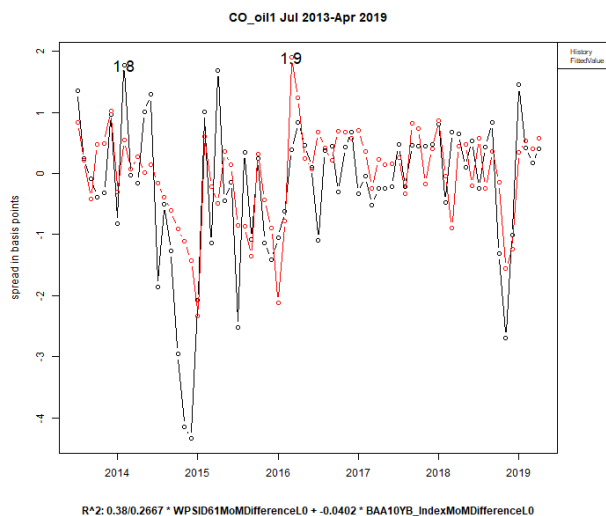
Stationarity of Residuals



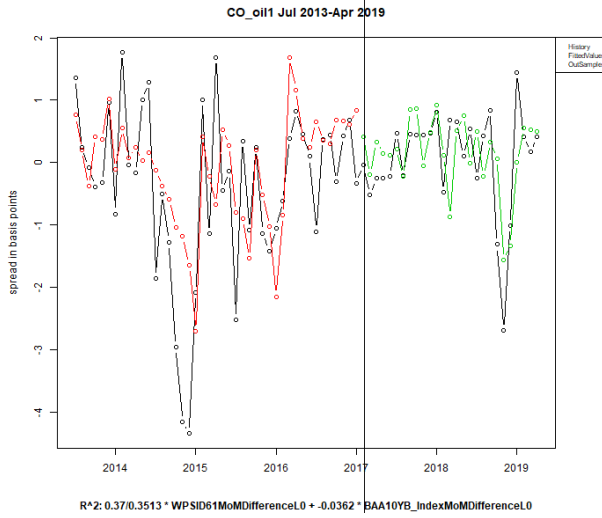
4.2.24 Model - 24

Backtesting on differenced values

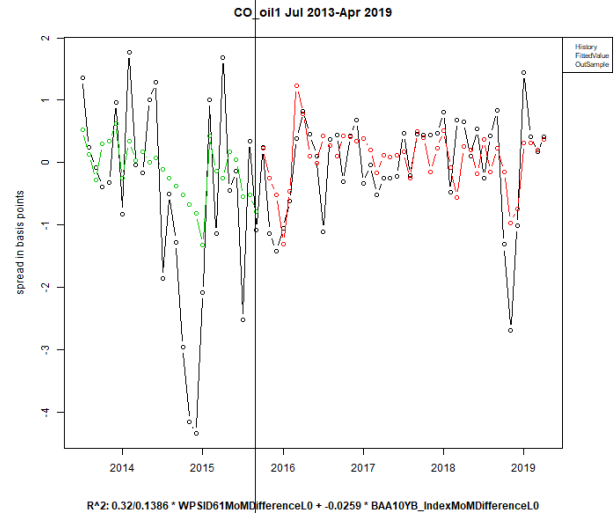
Backtesting on level values



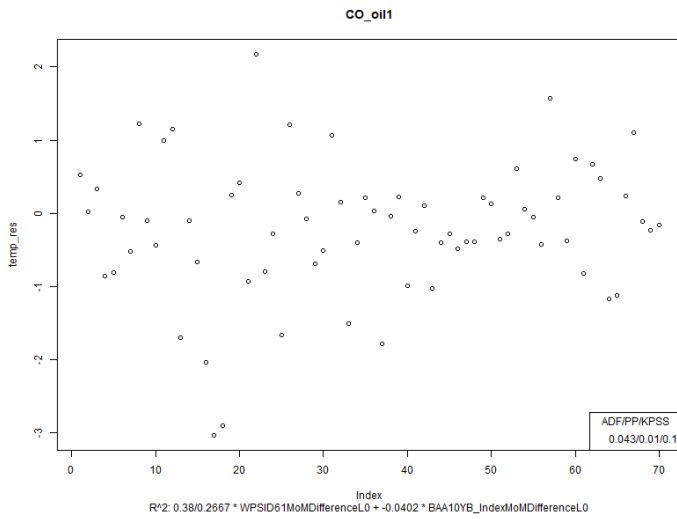
Out-of-Sample testing



Out-of-Sample testing



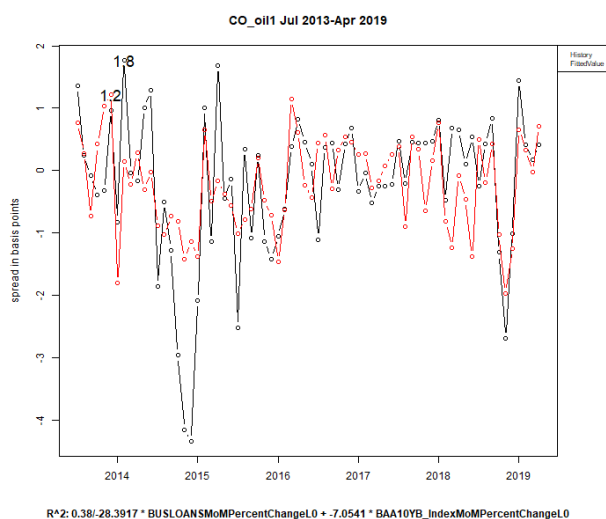
Stationarity of Residuals



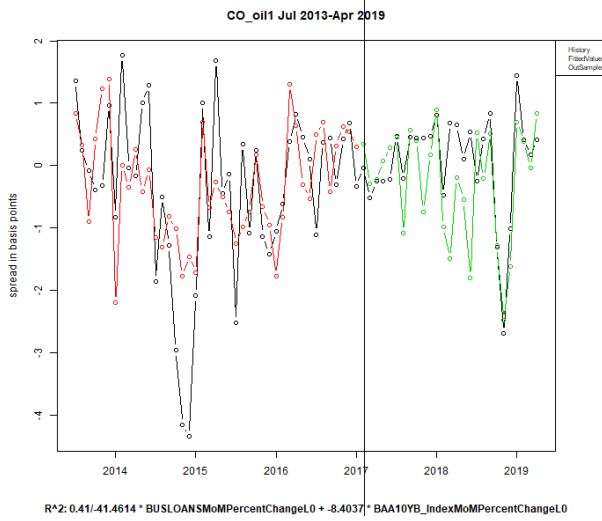
4.2.25 Model - 25

Backtesting on differenced values

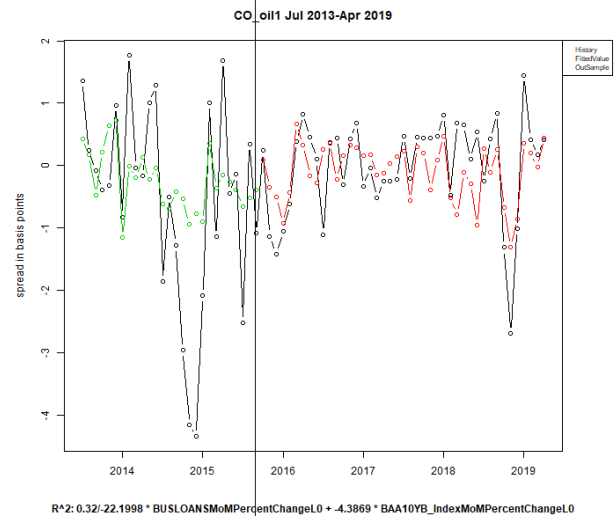
Backtesting on level values



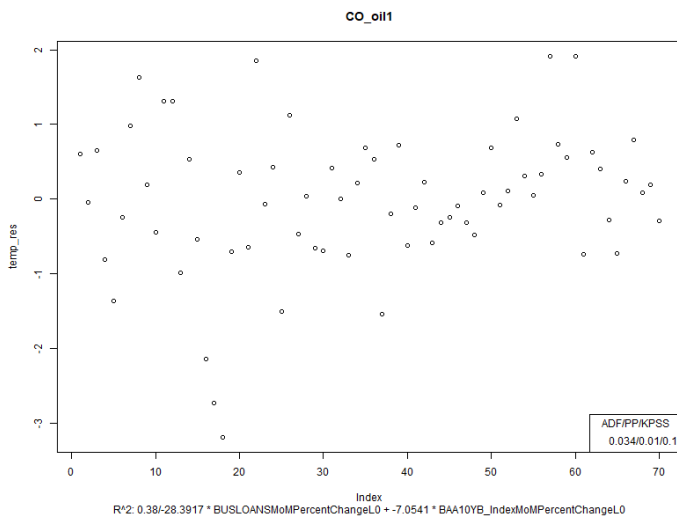
Out-of-Sample testing



Out-of-Sample testing



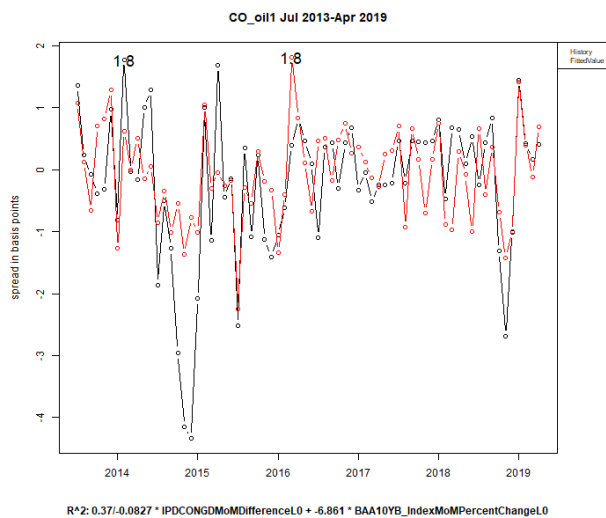
Stationarity of Residuals



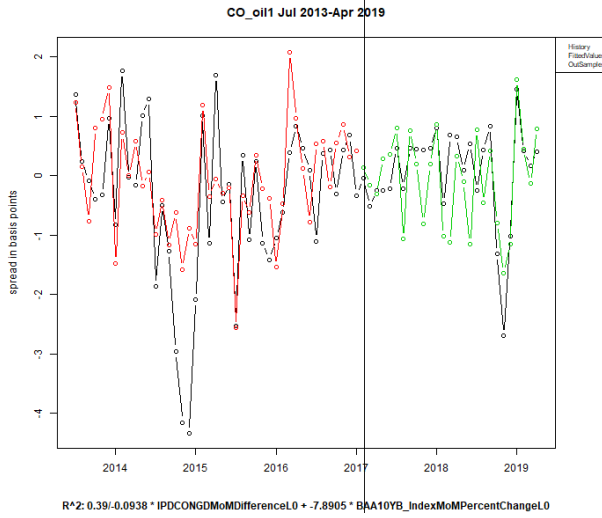
4.2.26 Model - 26

Backtesting on differenced values

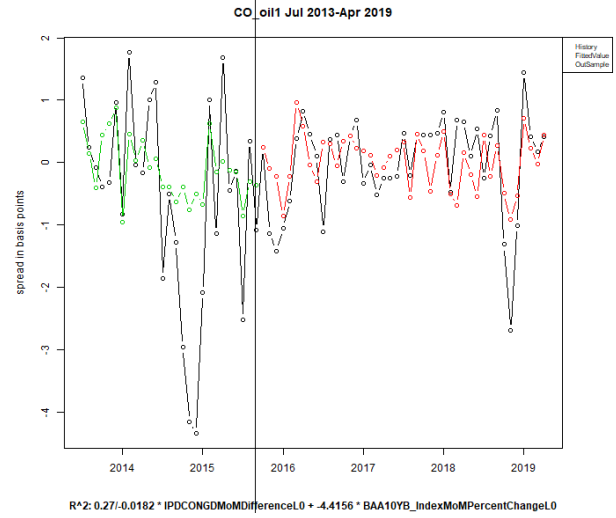
Backtesting on level values



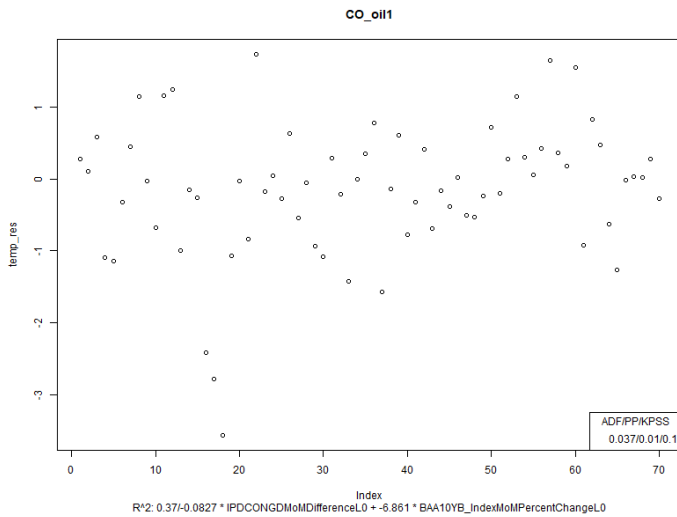
Out-of-Sample testing



Out-of-Sample testing



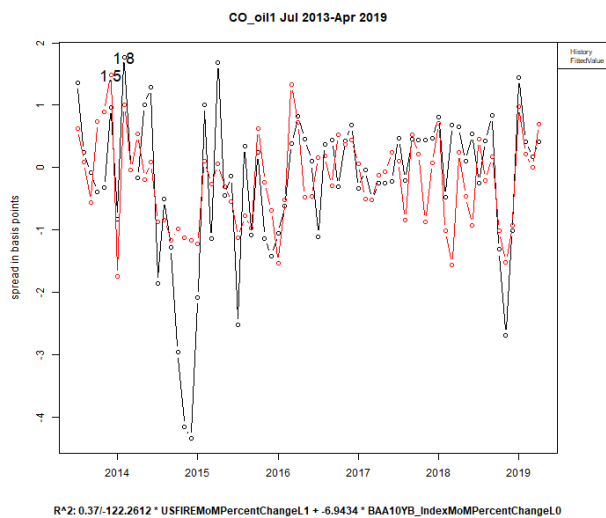
Stationarity of Residuals



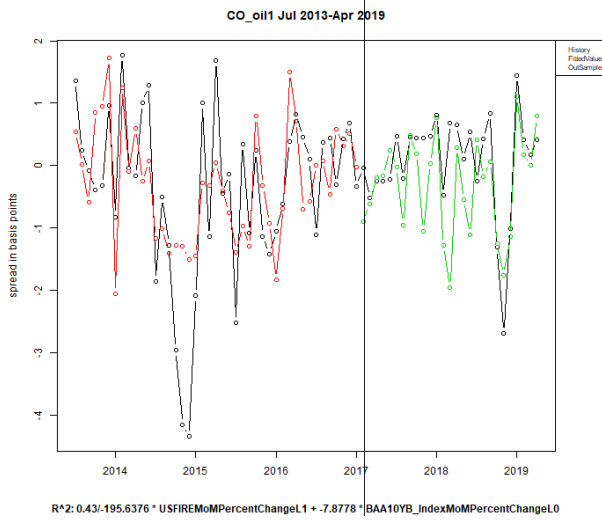
4.2.27 Model - 27

Backtesting on differenced values

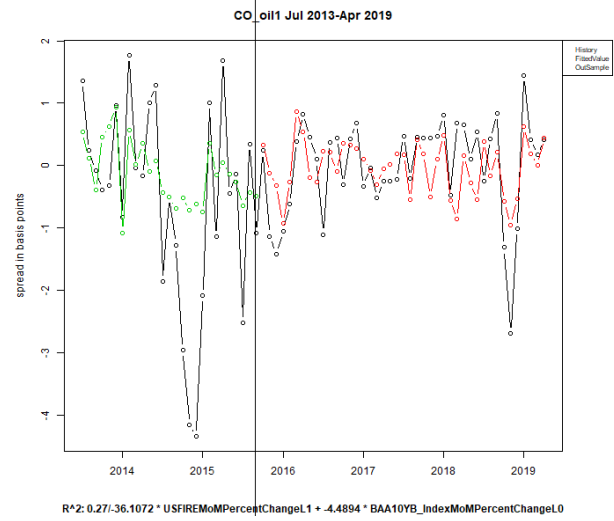
Backtesting on level values



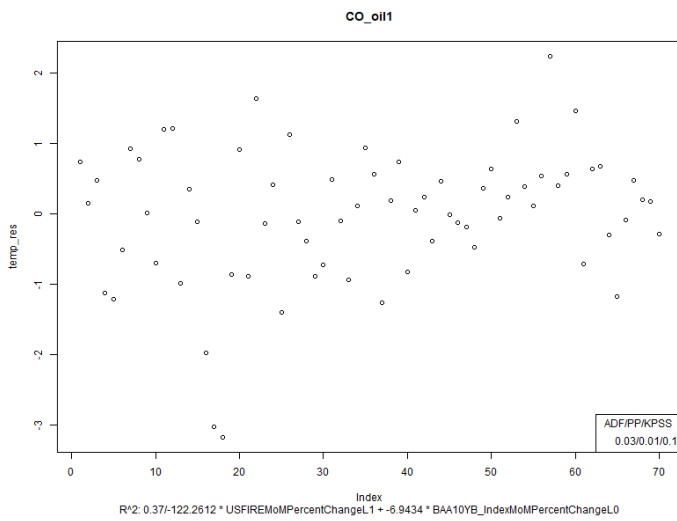
Out-of-Sample testing



Out-of-Sample testing



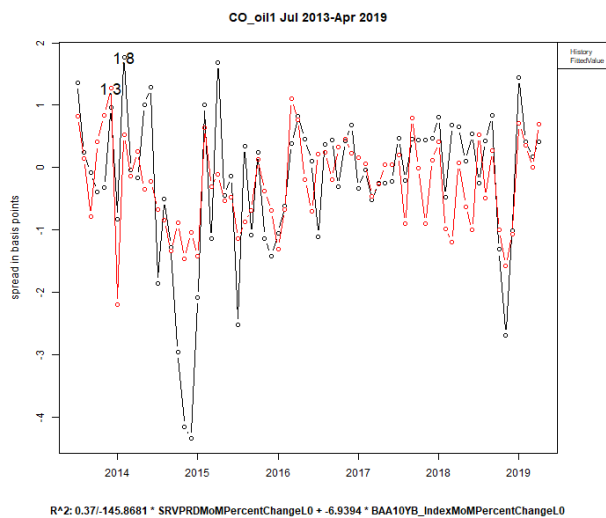
Stationarity of Residuals



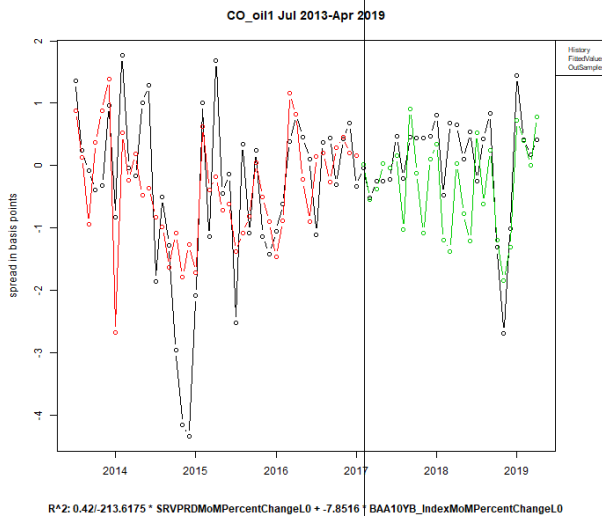
4.2.28 Model - 28

Backtesting on differenced values

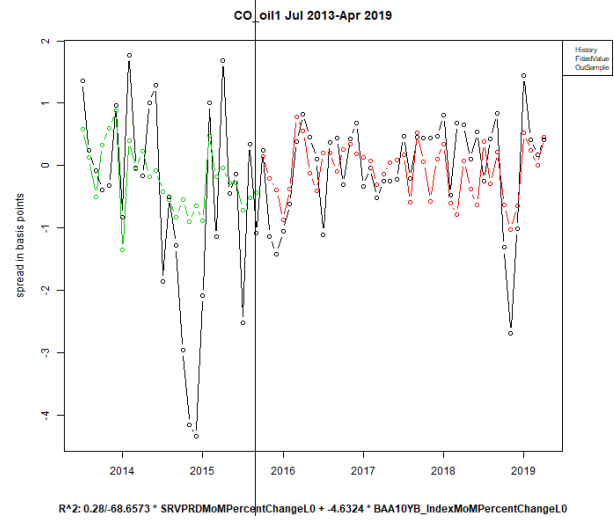
Backtesting on level values



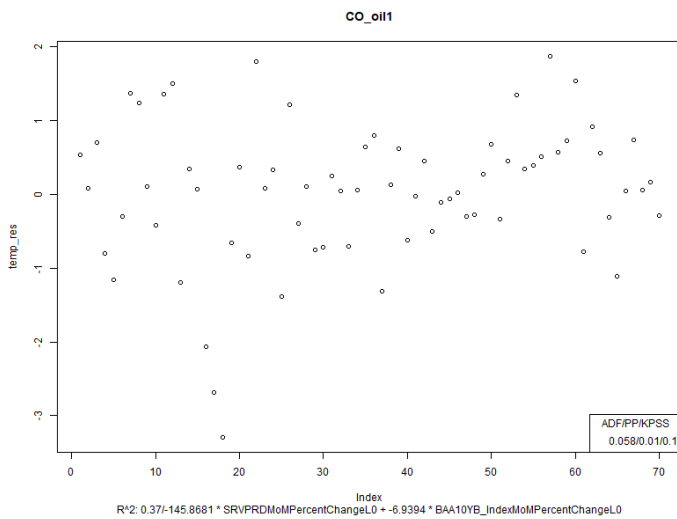
Out-of-Sample testing



Out-of-Sample testing



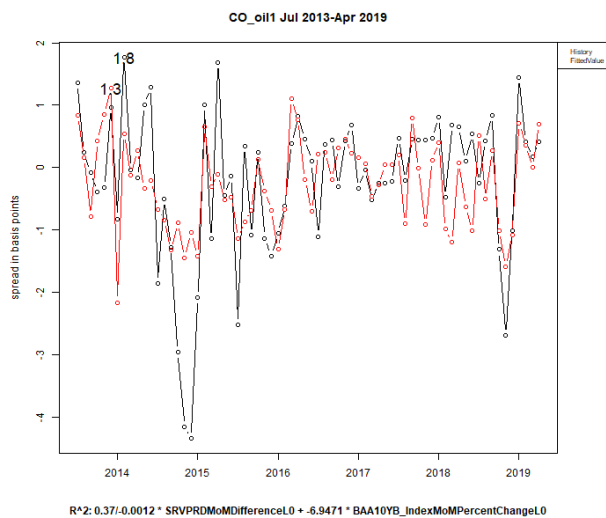
Stationarity of Residuals



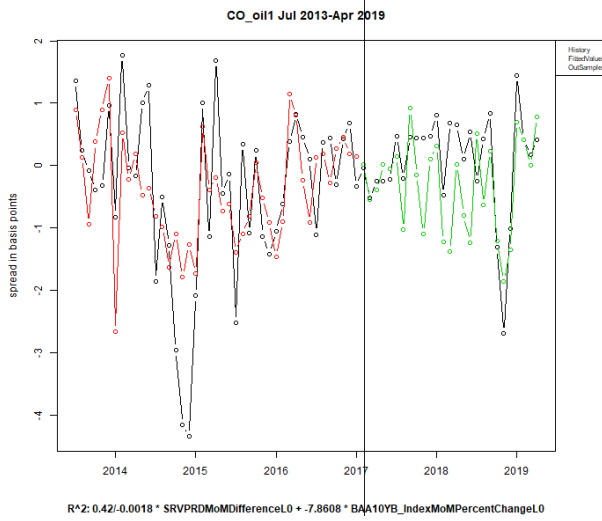
4.2.29 Model - 29

Backtesting on differenced values

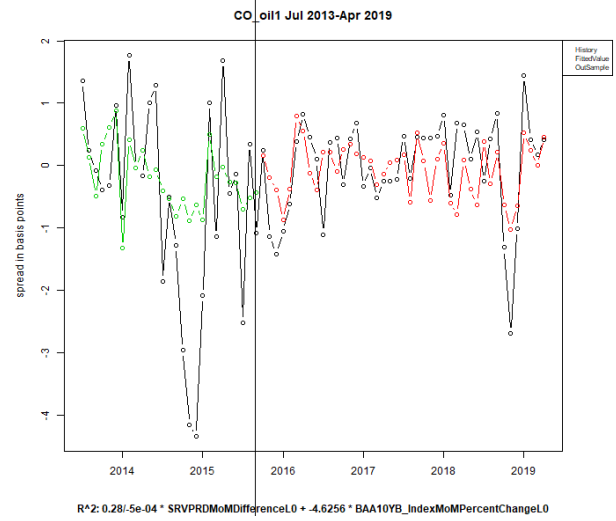
Backtesting on level values



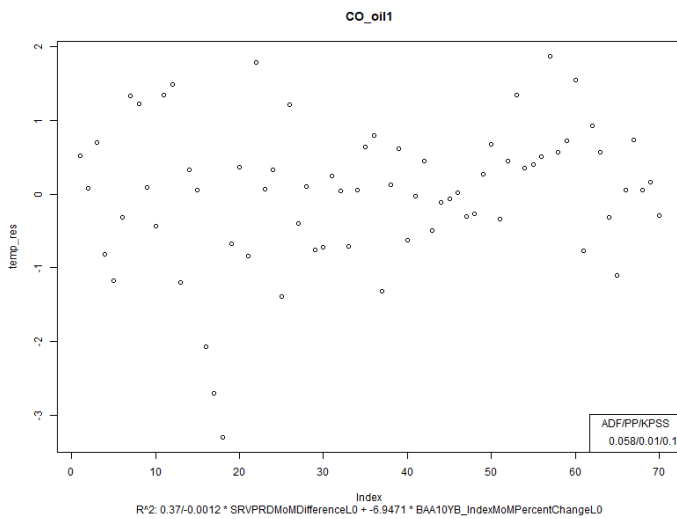
Out-of-Sample testing



Out-of-Sample testing



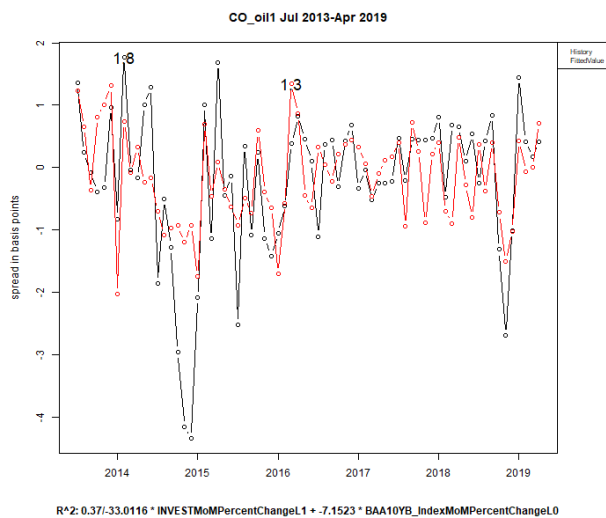
Stationarity of Residuals



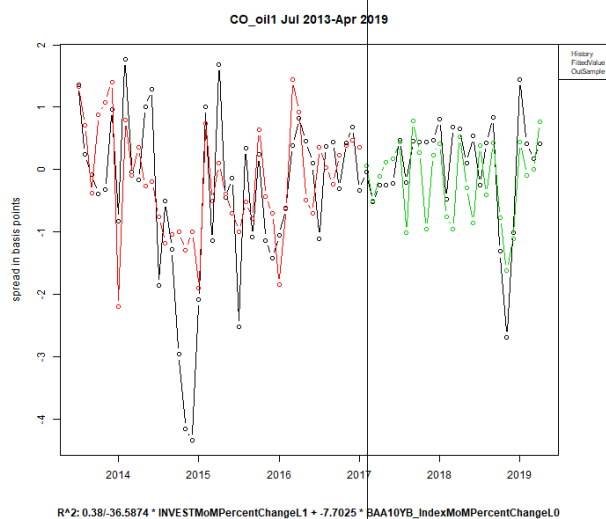
4.2.30 Model - 30

Backtesting on differenced values

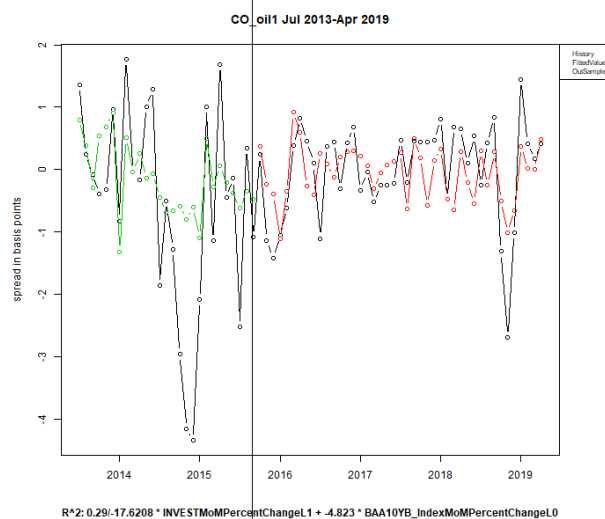
Backtesting on level values



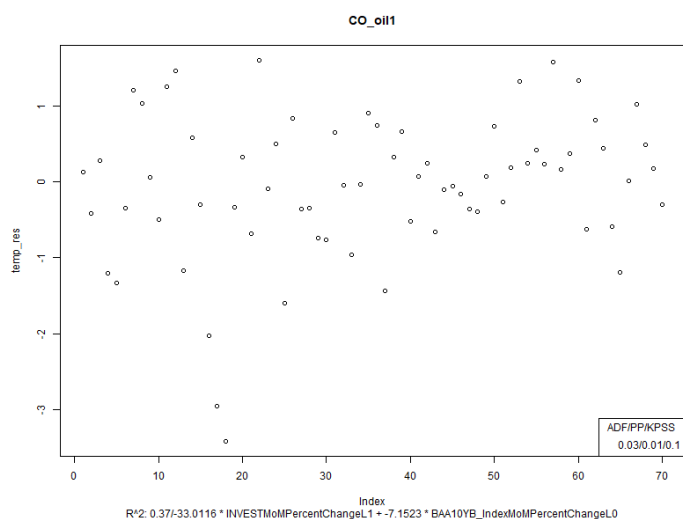
Out-of-Sample testing



Out-of-Sample testing



Stationarity of Residuals



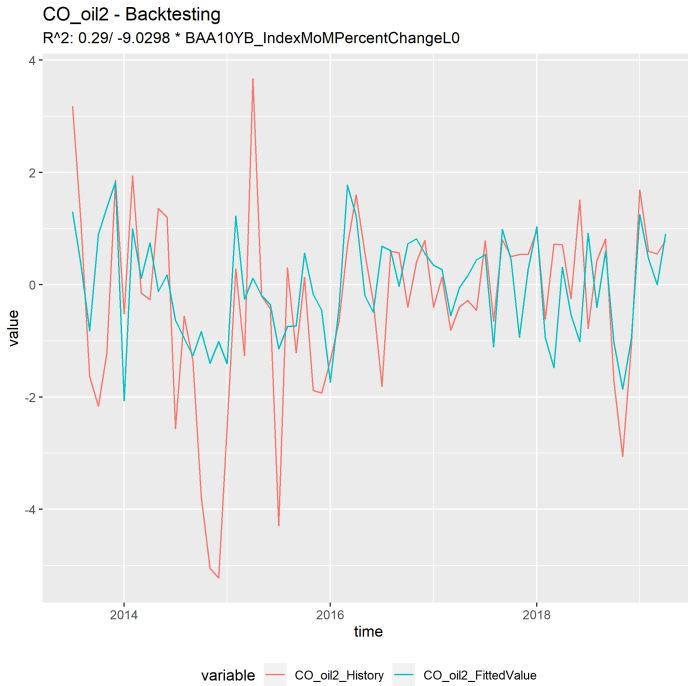
4.3 Single Factor Model for CO₂il2

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	BAA10YB_IndexMoMPercentChangeL0	0.29	249.47	-9.03	0.00	0.95			
2	BAA10YB_IndexMoMDifferenceL0	0.25	252.98	-0.06	0.00	0.83			
3	USGG30YR_IndexMoMDifferenceL0	0.14	262.69	3.82	0.00	1.09			
4	ism_man_pricesMoMDifferenceL0	0.13	263.14	0.13	0.02	1.13			
5	ism_man_pricesMoMPercentChangeL0	0.13	263.51	6.81	0.03	1.02			
6	USGG30YR_IndexMoMPercentChangeL0	0.13	263.52	10.42	0.00	1.06			
7	WPSID61MoMDifferenceL0	0.12	263.89	0.44	0.04	1.19			
8	MANEMPMoMPercentChangeL0	0.11	264.92	-267.39	0.03	1.26			
9	MANEMPMoMDifferenceL0	0.11	265.06	-0.02	0.03	1.27			
10	USGG10YR_IndexMoMDifferenceL0	0.11	265.11	3.13	0.00	0.99			

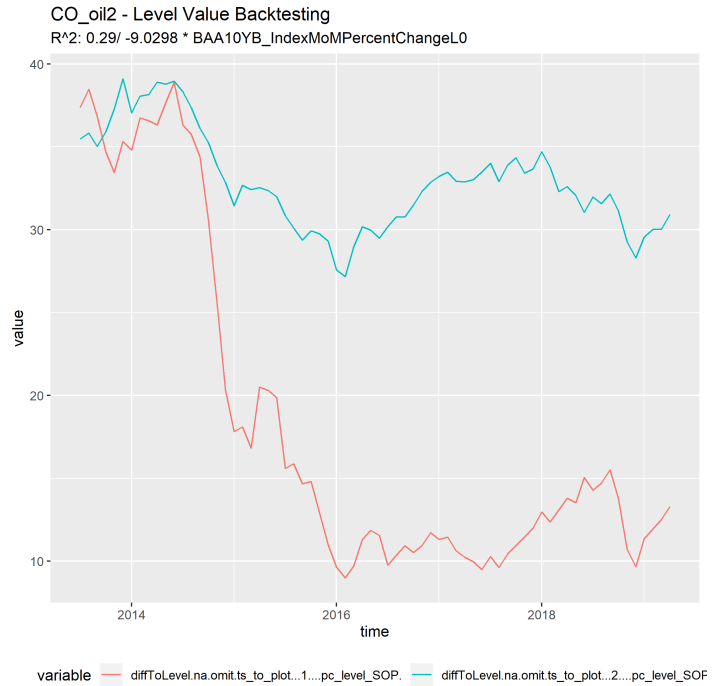
Table 6: Single Factor Analysis-CO₂il2

4.3.1 Model - 1

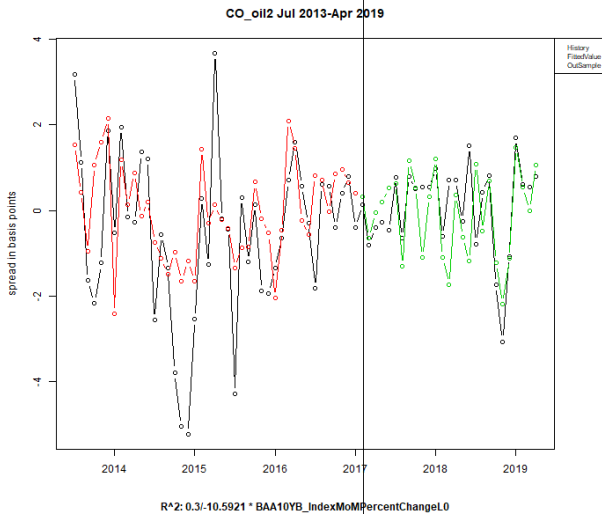
Backtesting on differenced values



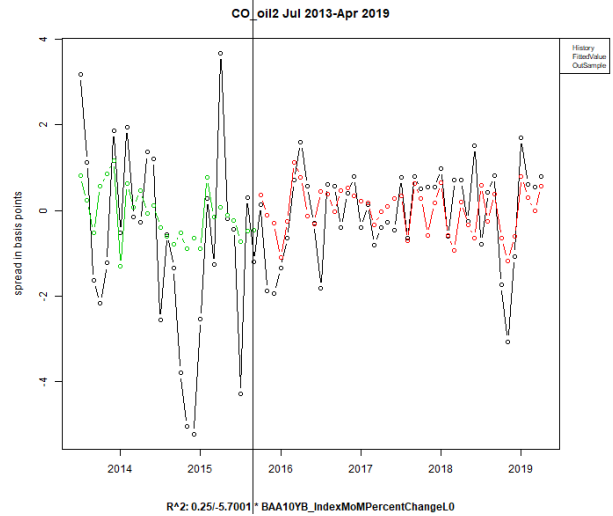
Backtesting on level values



Out-of-Sample testing

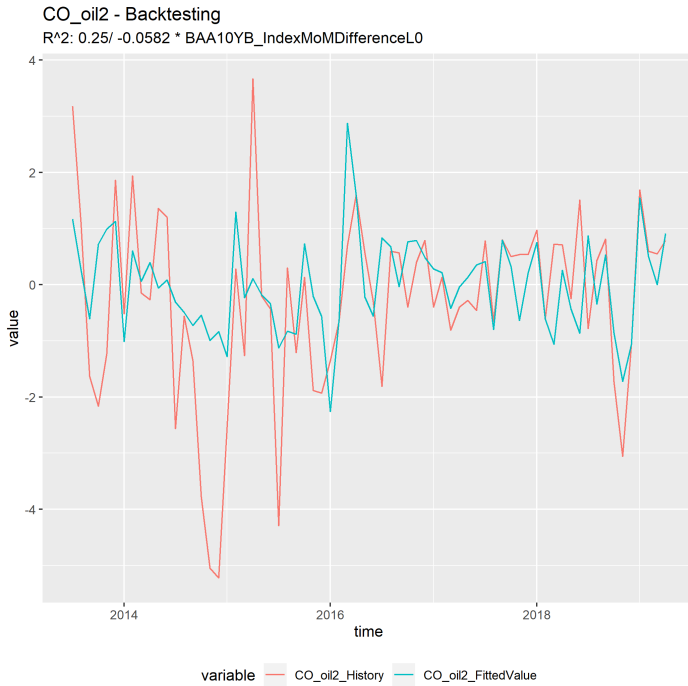


Out-of-Sample testing

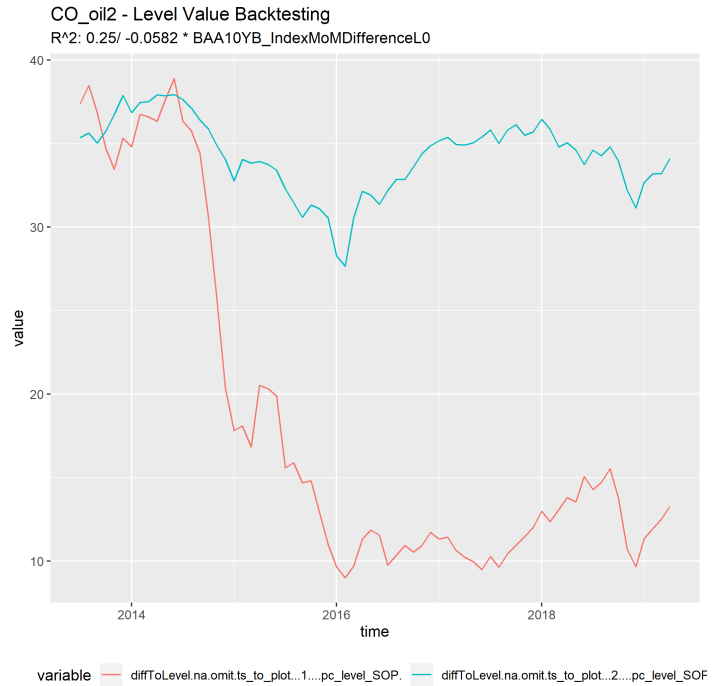


4.3.2 Model - 2

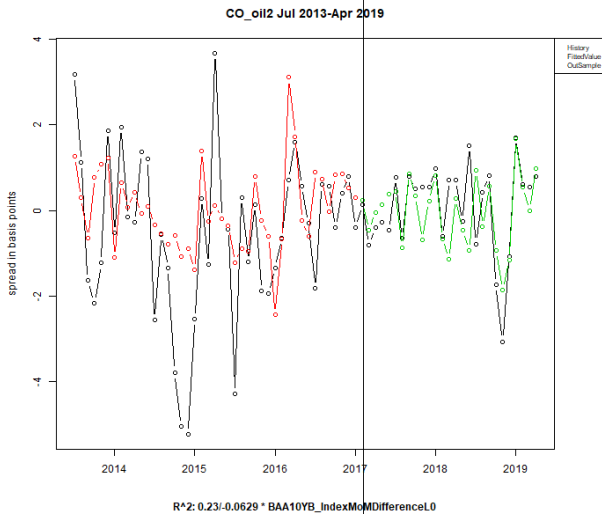
Backtesting on differenced values



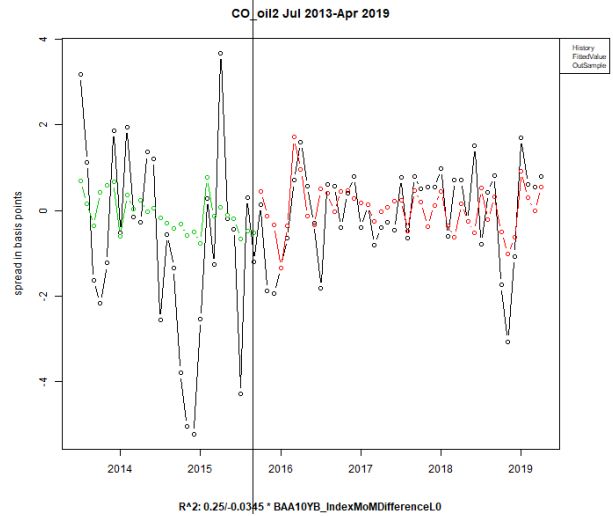
Backtesting on level values



Out-of-Sample testing

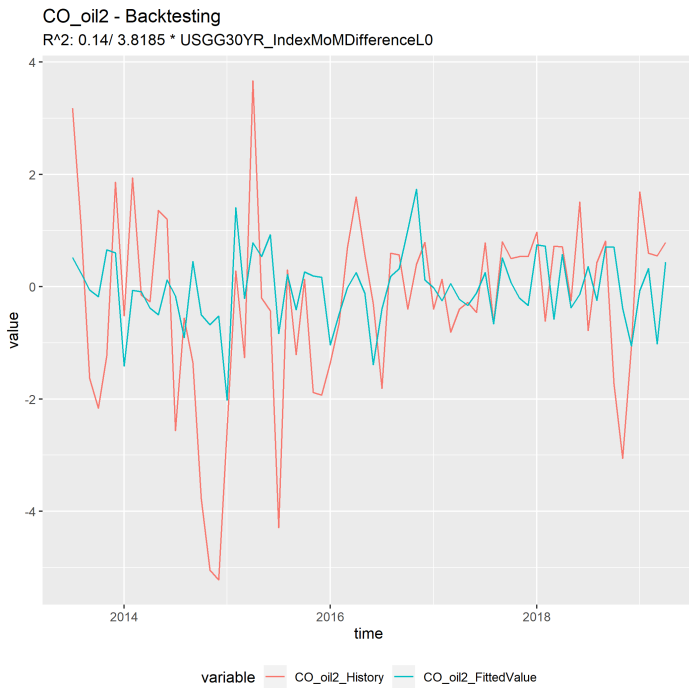


Out-of-Sample testing

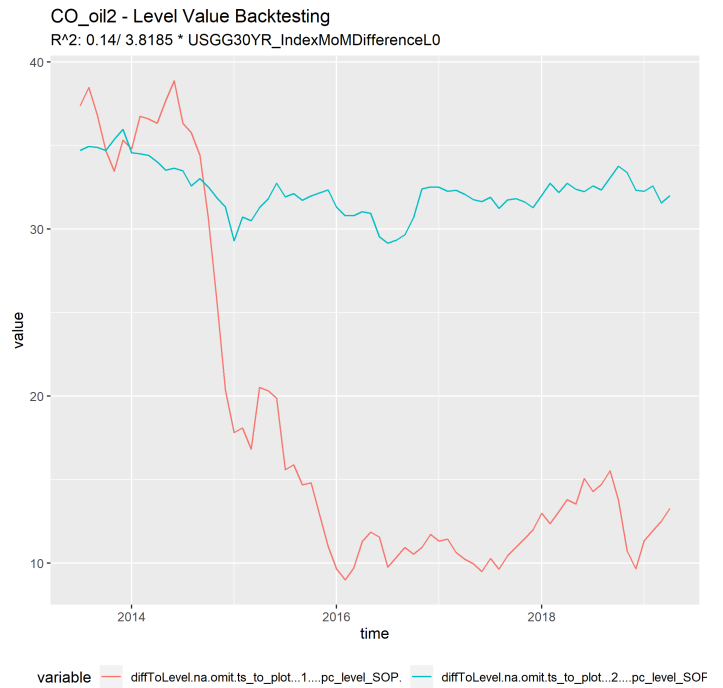


4.3.3 Model - 3

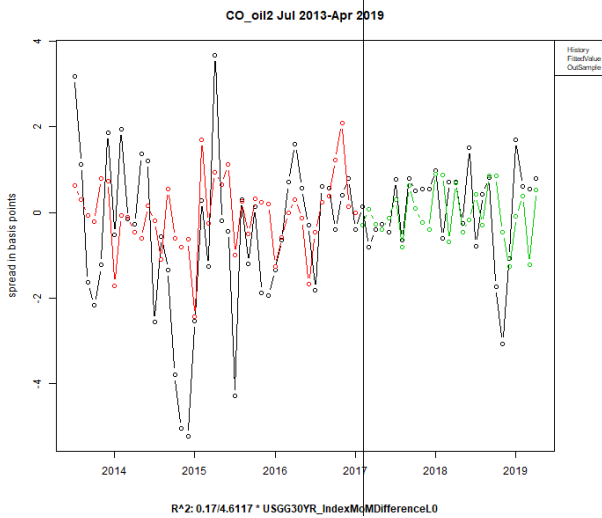
Backtesting on differenced values



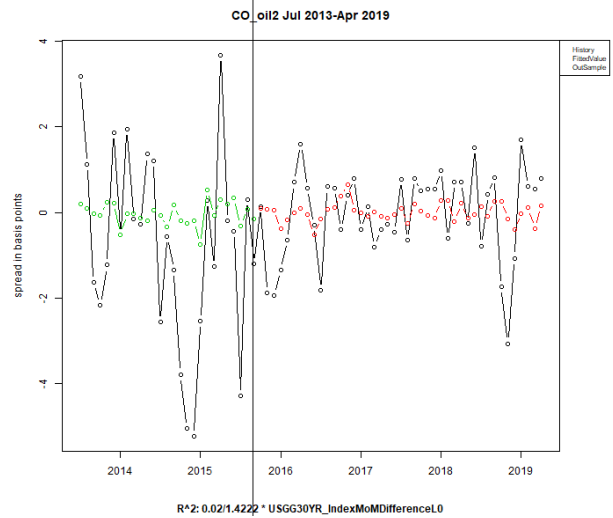
Backtesting on level values



Out-of-Sample testing

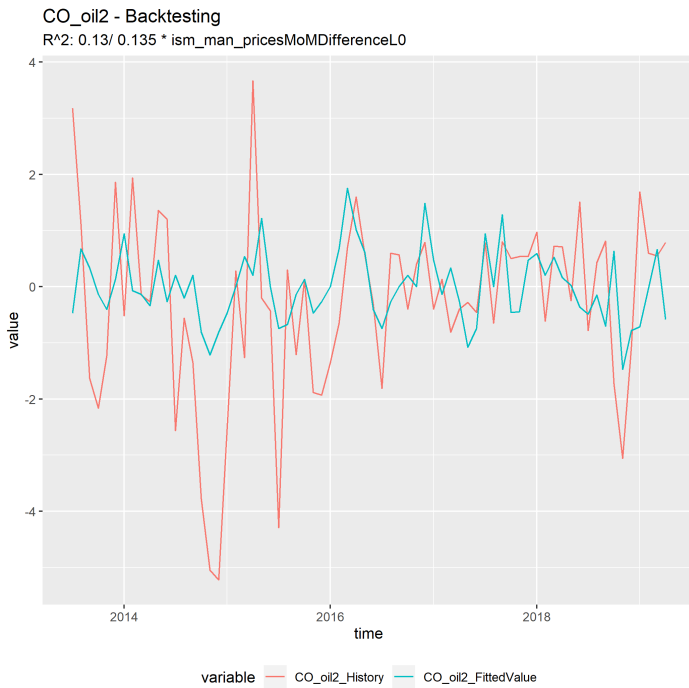


Out-of-Sample testing

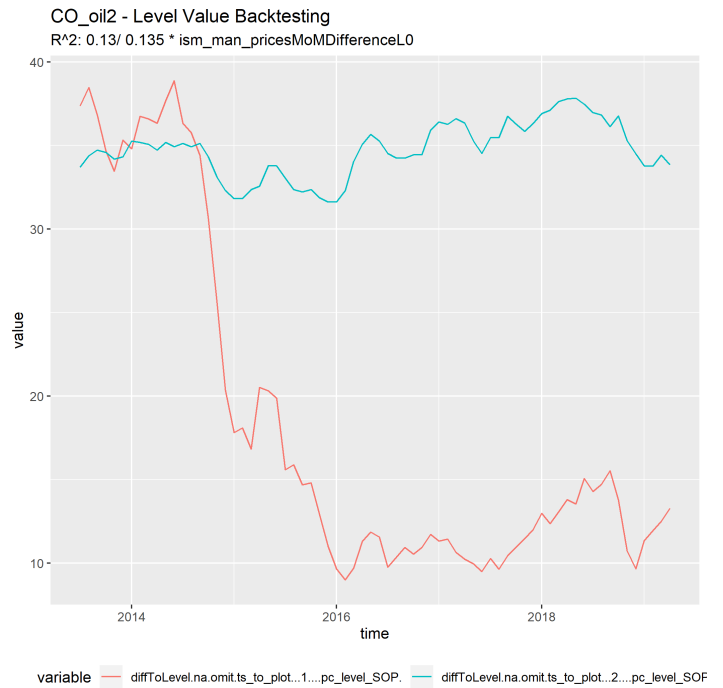


4.3.4 Model - 4

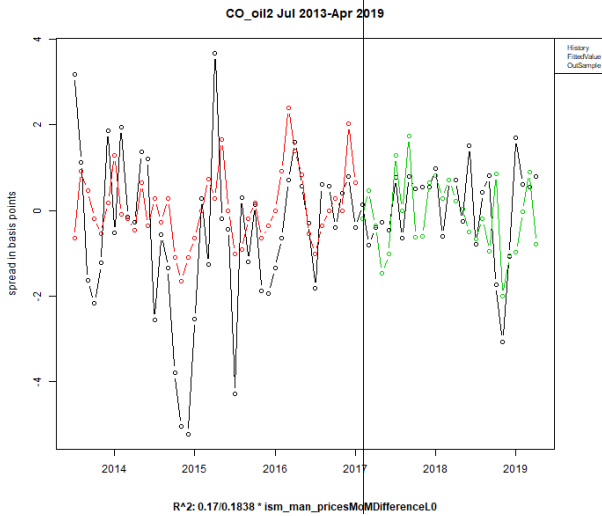
Backtesting on differenced values



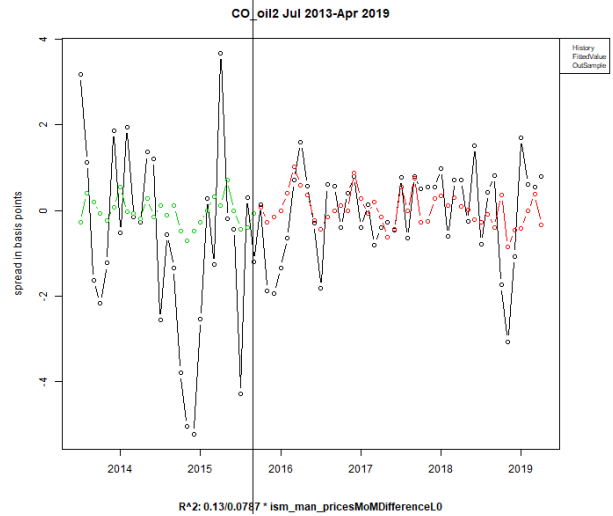
Backtesting on level values



Out-of-Sample testing

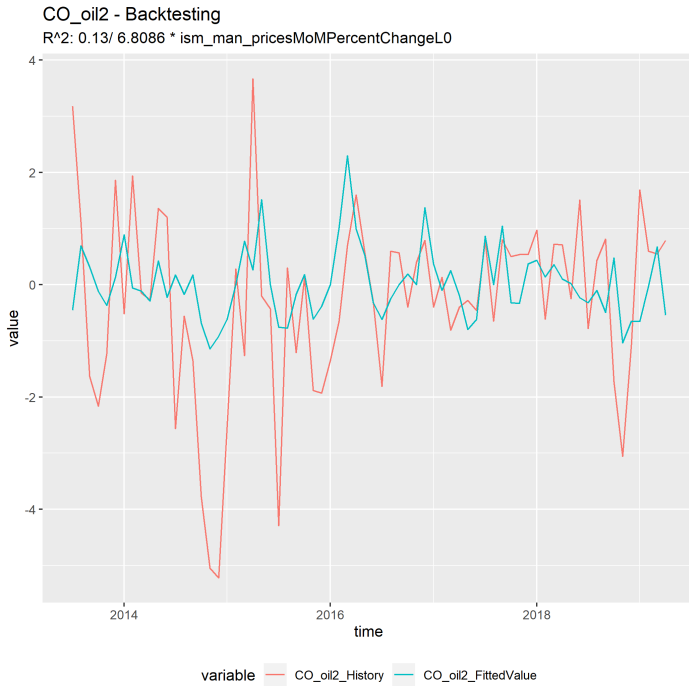


Out-of-Sample testing

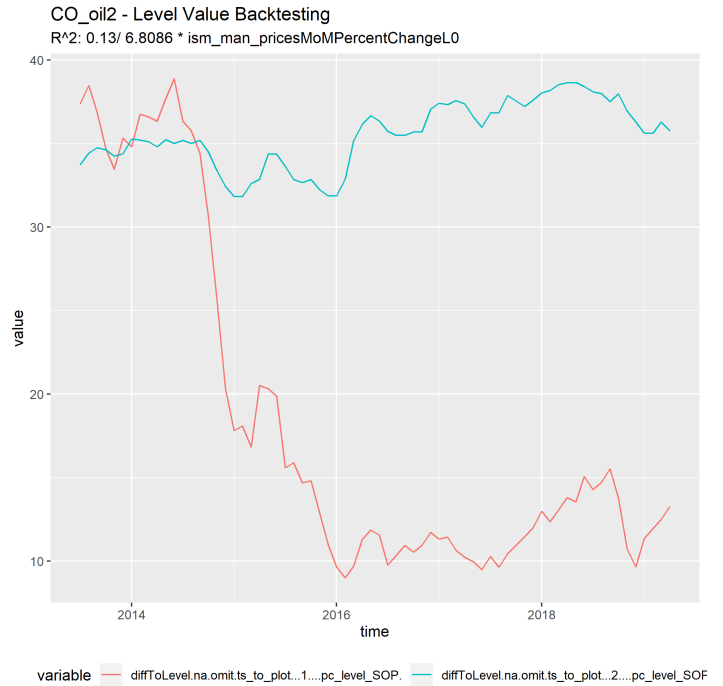


4.3.5 Model - 5

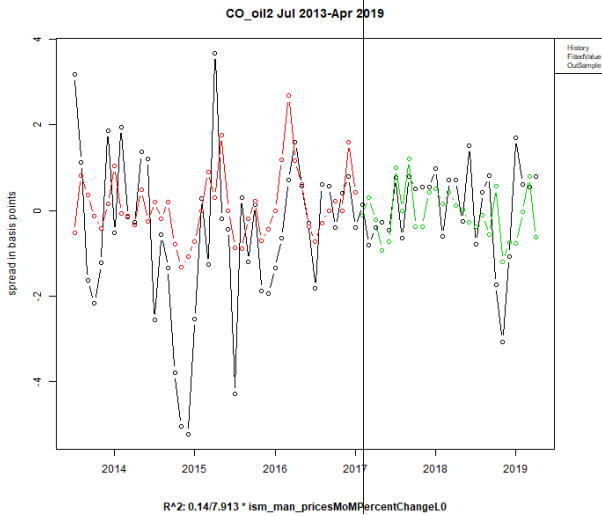
Backtesting on differenced values



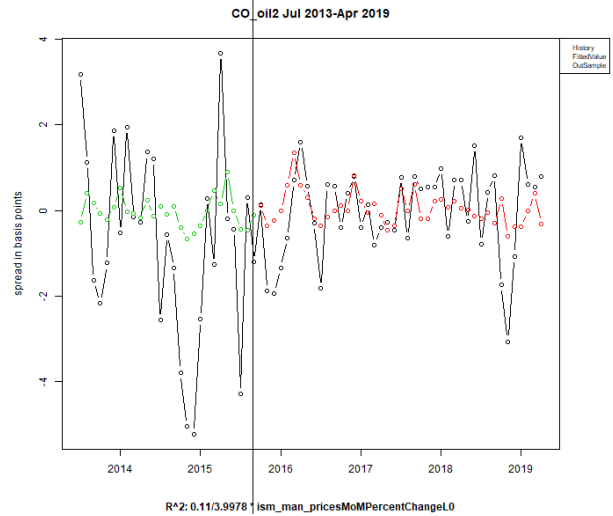
Backtesting on level values



Out-of-Sample testing

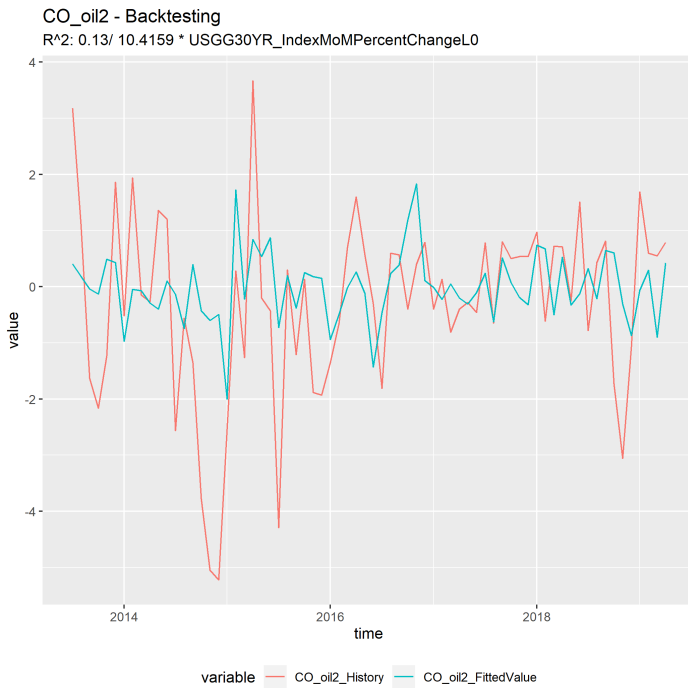


Out-of-Sample testing

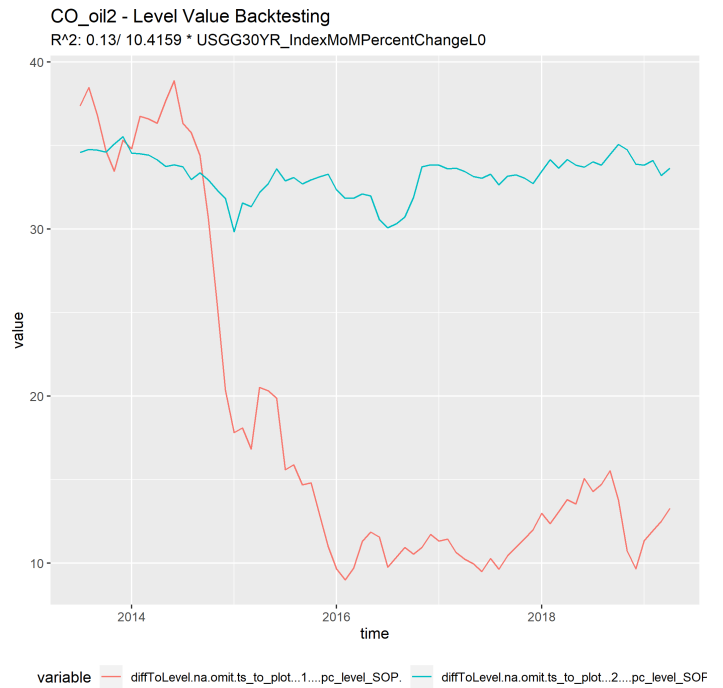


4.3.6 Model - 6

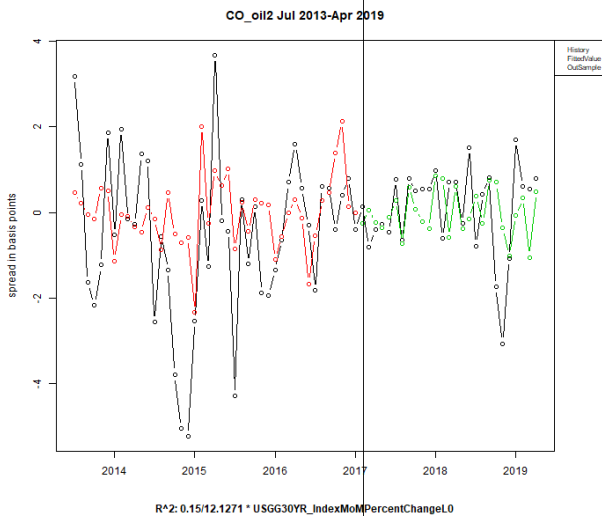
Backtesting on differenced values



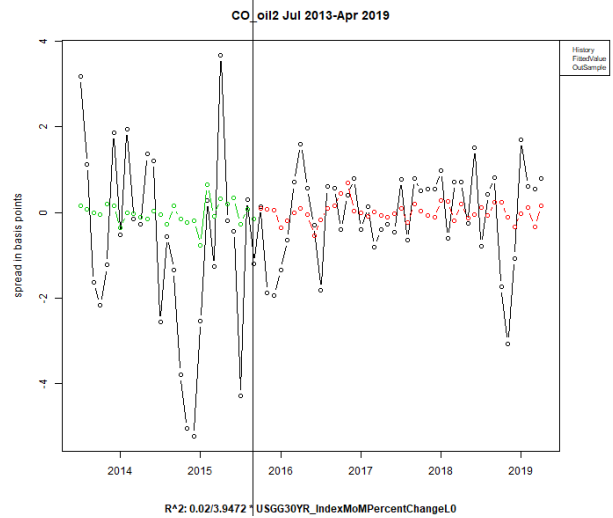
Backtesting on level values



Out-of-Sample testing

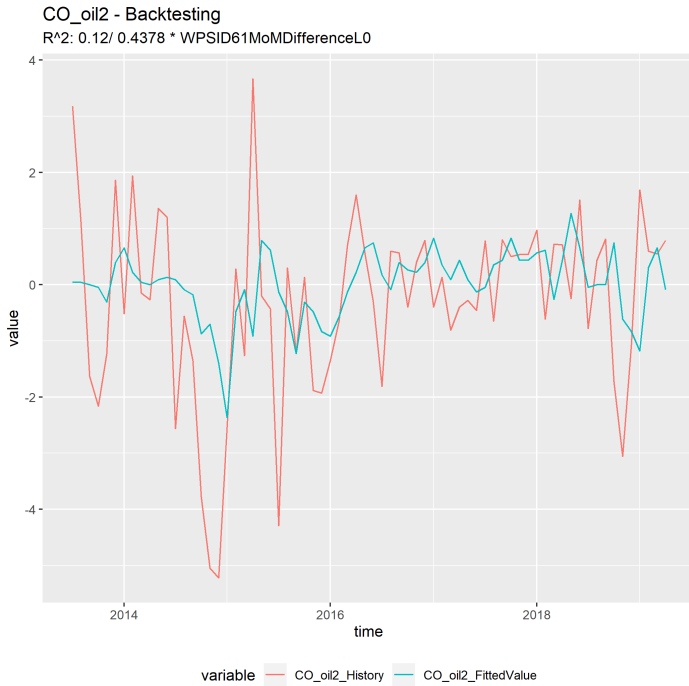


Out-of-Sample testing

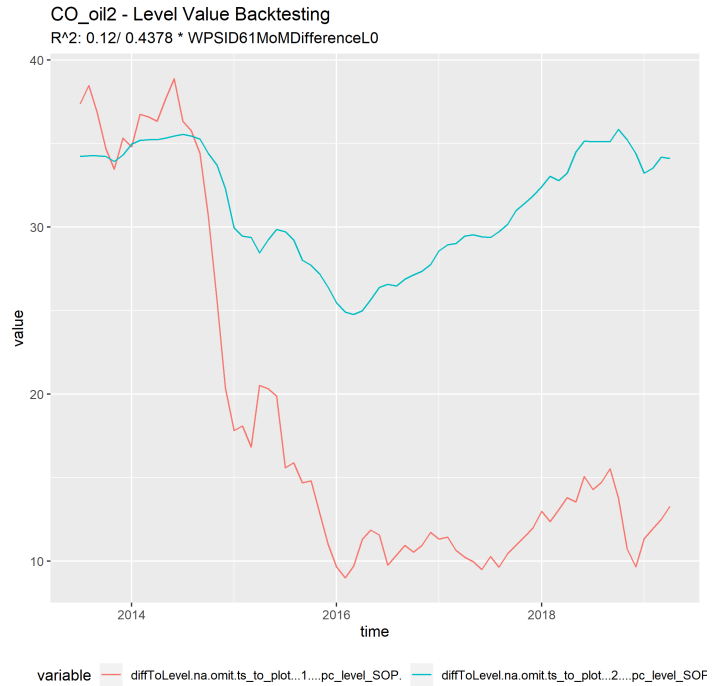


4.3.7 Model - 7

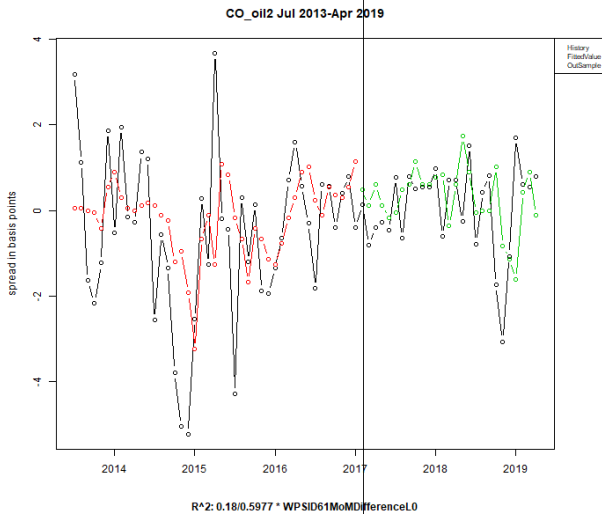
Backtesting on differenced values



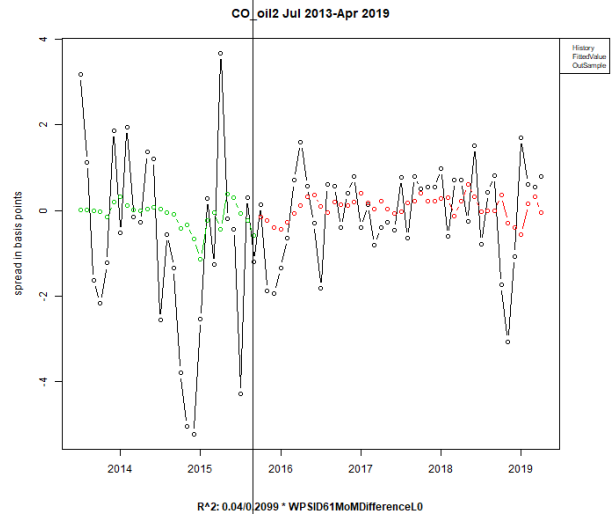
Backtesting on level values



Out-of-Sample testing

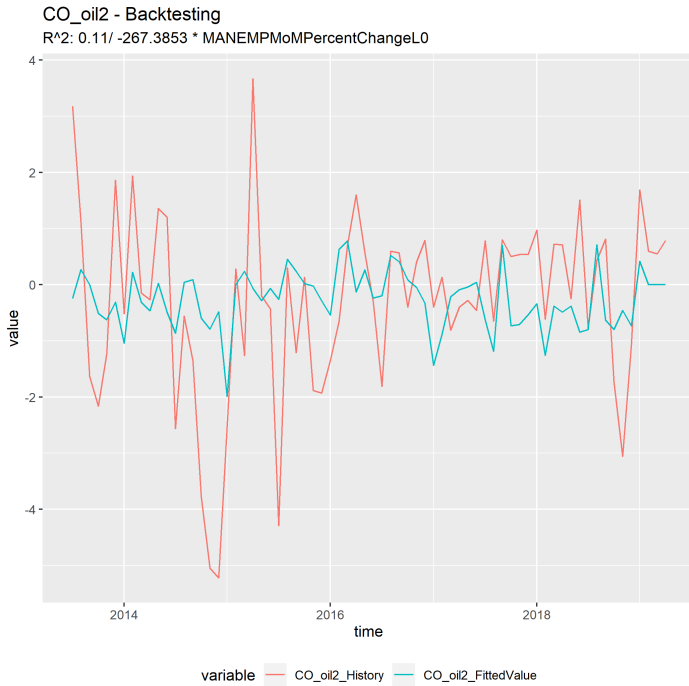


Out-of-Sample testing

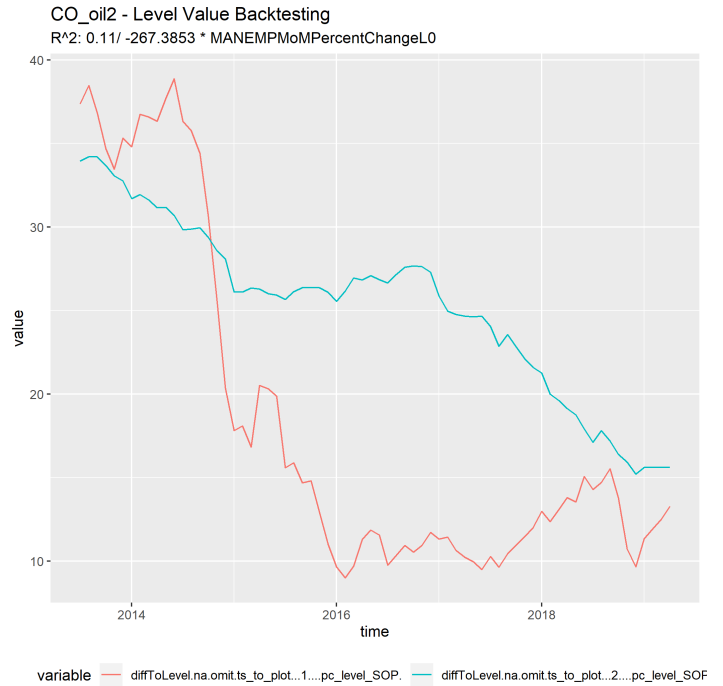


4.3.8 Model - 8

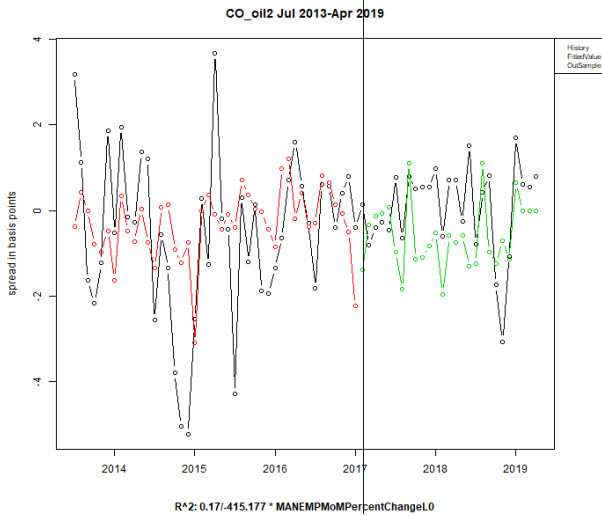
Backtesting on differenced values



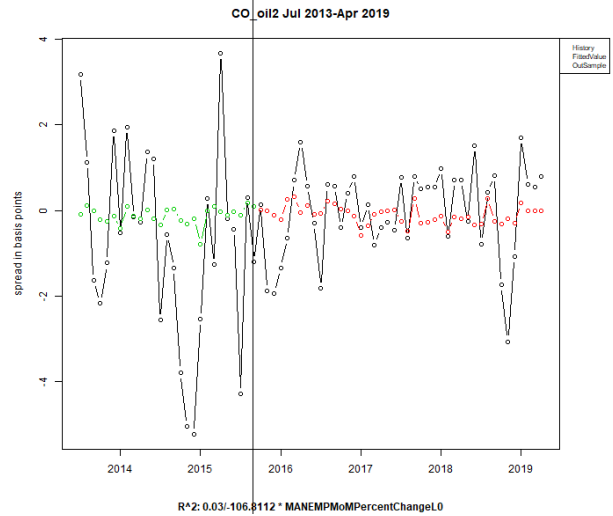
Backtesting on level values



Out-of-Sample testing

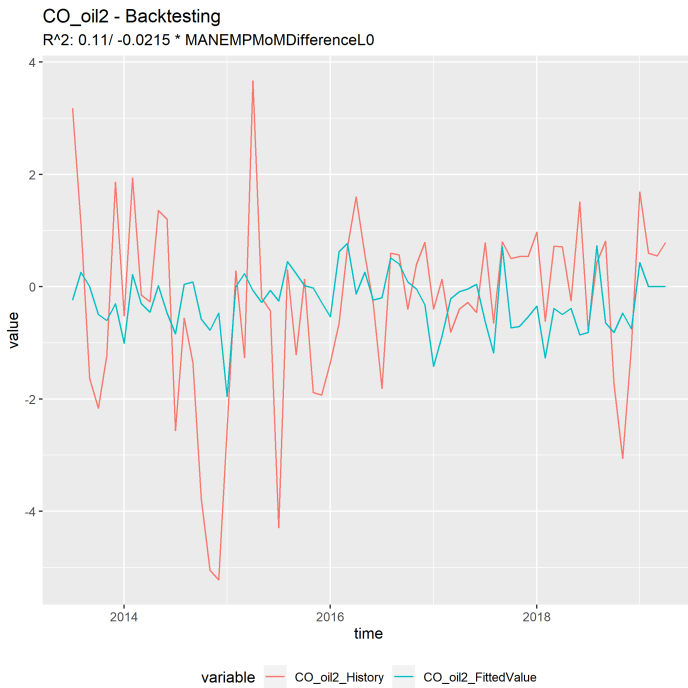


Out-of-Sample testing

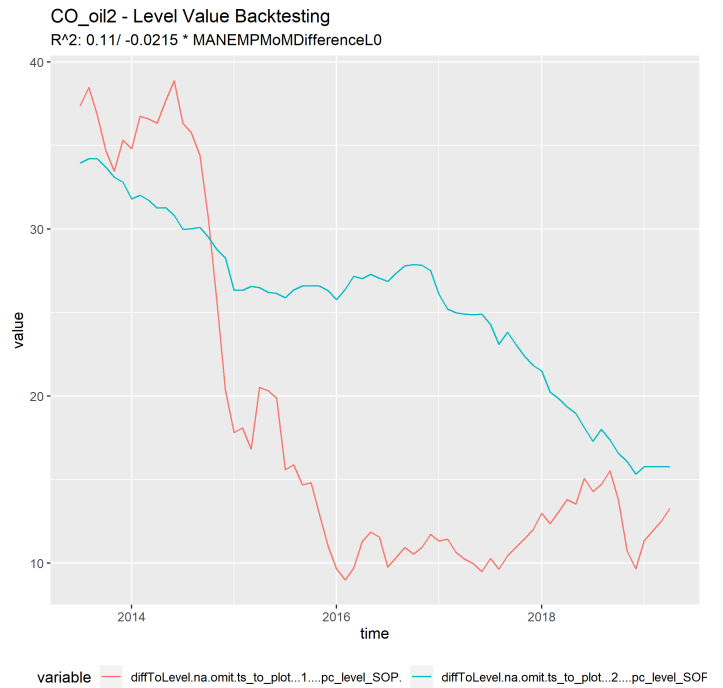


4.3.9 Model - 9

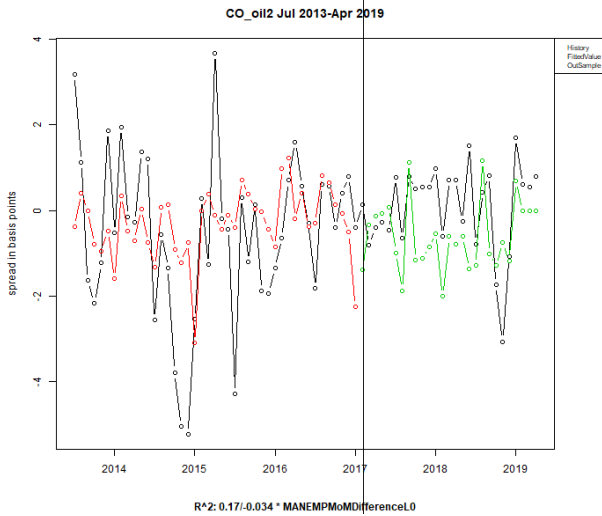
Backtesting on differenced values



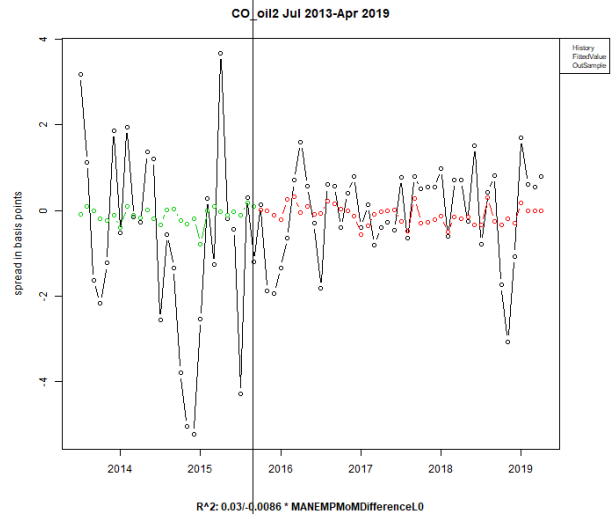
Backtesting on level values



Out-of-Sample testing

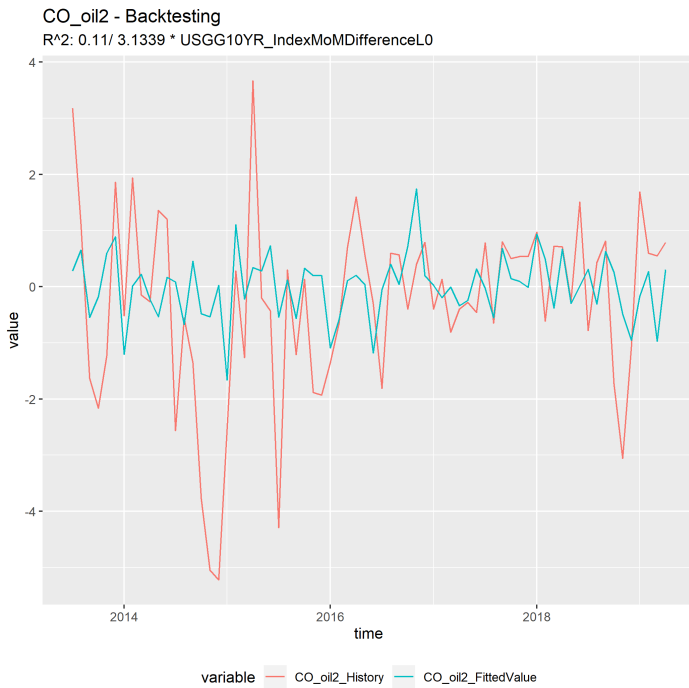


Out-of-Sample testing

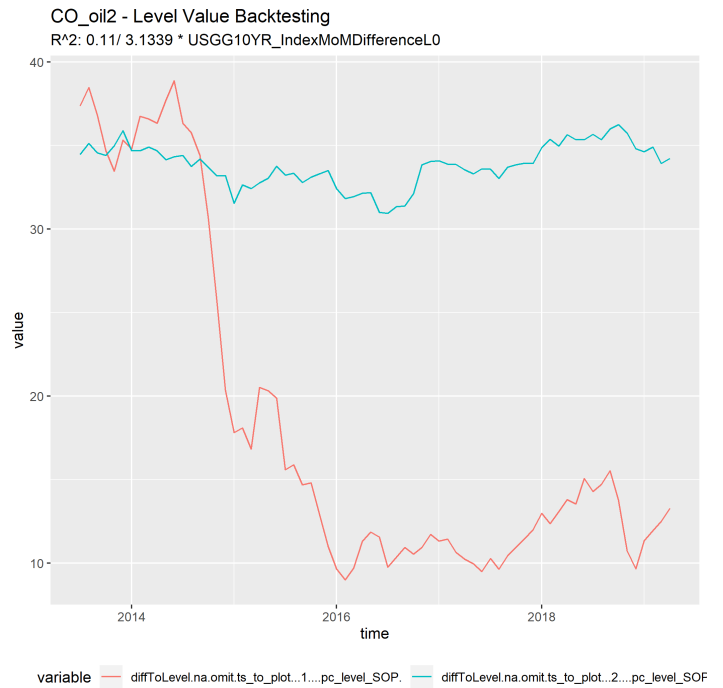


4.3.10 Model - 10

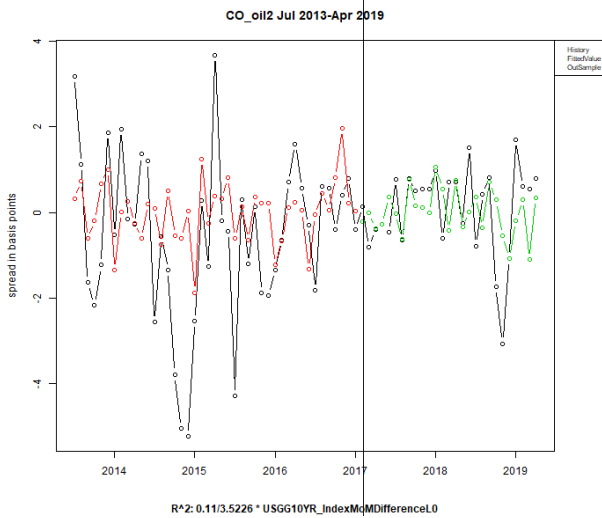
Backtesting on differenced values



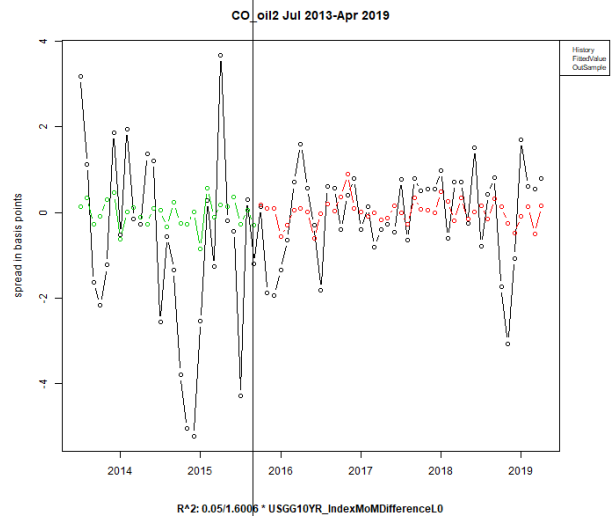
Backtesting on level values



Out-of-Sample testing



Out-of-Sample testing



4.4 Two Factor Model for CO₂il2

V1	V2	r2adj	Var1	Var2	aic	RMSE	Criteria1	Criteria2	Criteria3
1 ism_man_pricesMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.35	0.10	-8.07	243.64	0.99		Fail	
2 WPSID61MoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.35	0.32	-8.13	244.27	0.92			
3 DPCERA3M086SBEAMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.34	-61.02	-9.39	244.48	1.01			
4 ism_man_pricesMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.34	4.80	-8.04	244.60	0.91		Fail	
5 WPSID61MoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.34	60.26	-8.15	244.69	0.91			
6 DPCERA3M086SBEAMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.34	-0.50	-9.40	244.74	1.07			
7 USTRADEMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.34	-166.82	-9.12	245.10	1.05			
8 USTRADEMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.34	-0.01	-9.10	245.20	1.06			
9 IPDCONGDMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.33	-15.45	-8.54	245.49	0.99			
10 IPNMATMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.33	-21.02	-8.66	245.65	0.96			
11 M2REALMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.33	-80.02	-8.47	245.88	1.10			
12 IPNMATMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.33	-0.22	-8.66	245.92	0.97			
13 INVESTMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.33	-54.44	-8.96	246.09	1.06			
14 IPDCONGDMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.33	-0.13	-8.53	246.10	1.00			
15 M2REALMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.33	-76.97	-9.16	246.20	1.03			
16 M2REALMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.33	-0.02	-8.46	246.36	1.14			
17 WPSFD49502MoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.33	0.21	-8.22	246.46	0.91			
18 WPSFD49502MoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.32	44.18	-8.23	246.61	0.91			
19 WPSFD49207MoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.32	0.30	-8.27	246.62	0.90			
20 HOUSTSMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.32	-3.10	-8.93	246.65	0.96			
21 PAYEMSMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.32	-214.25	-8.52	246.70	1.09			
22 WPSFD49207MoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.32	58.53	-8.28	246.73	0.89			
23 INVESTMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.32	-0.02	-8.98	246.77	1.09			
24 M2REALMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.32	-0.01	-9.18	246.80	1.05			
25 PAYEMSMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.32	-0.00	-8.53	246.86	1.11			
26 IPB51222SMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.32	-0.07	-9.28	246.91	0.95			
27 ISRATIOMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.32	-50.55	-8.19	246.96	0.82			
28 IPB51222SMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.32	-6.61	-9.26	247.01	0.96			
29 SRVPRDMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.32	-212.72	-8.69	247.07	1.05			
30 ISRATIOMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.32	-37.00	-8.21	247.09	0.82			
31 WPSID62MoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.32	0.05	-8.33	247.18	1.42			
32 SRVPRDMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.32	-0.00	-8.70	247.18	1.06			
33 IPFPNSSMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.32	-34.35	-8.52	247.32	0.93			
34 IPCONGDMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.31	-27.52	-8.59	247.54	0.92			
35 IPMANSICSMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.31	-24.17	-8.46	247.55	1.12			
36 IPFPNSSMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.31	-0.33	-8.53	247.62	0.93			
37 INVESTMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.31	-45.38	-9.00	247.74	0.90			
38 INVESTMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.31	-0.01	-9.10	247.83	0.91			
39 IPMANSICSMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.31	-0.22	-8.50	247.86	1.15			
40 HOUSTSMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.31	-0.01	-8.88	247.88	0.96			
41 IPCONGDMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.31	-0.26	-8.60	247.90	0.92			
42 SRVPRDMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.31	-187.92	-8.90	247.94	1.17			
43 WPSID62MoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.31	9.15	-8.42	247.95	1.39			
44 WPSID61MoMDifferenceL0	BAA10YB_IndexMoMDifferenceL0	0.31	0.32	-0.05	248.05	0.86			
45 USFIREMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.31	-157.57	-8.74	248.05	1.09			
46 IPFINALMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.31	-35.37	-8.68	248.08	0.92			
47 SRVPRDMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.31	-0.00	-8.89	248.09	1.19			
48 MZMSLMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.31	-58.69	-9.22	248.09	0.99			
49 WPSID61MoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.31	0.22	-9.21	248.11	0.92			
50 ism_man_pricesMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.31	3.36	-8.47	248.14	0.94		Fail	

Table 7: Two Factor Analysis-CO₂il2

Table 8: Relevant Variables: CO₂i/2

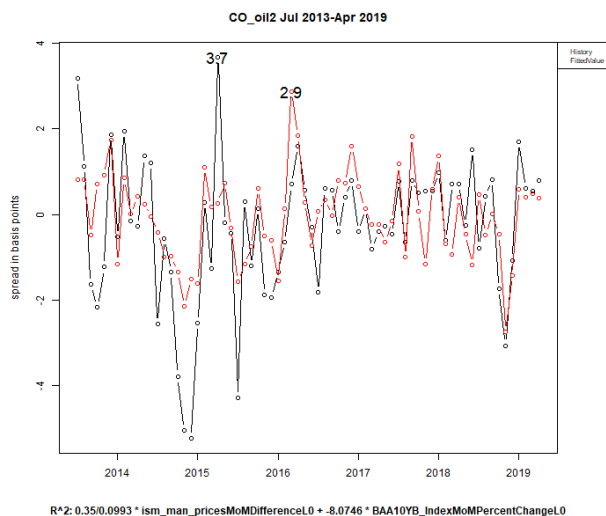
1	2	3
1 RPIMoMDifferenceL0	UEMP15T26MoMPercentChangeL0	BUSLOANSMoMDifferenceL0
2 RPIMoMDifferenceL1	UEMP15T26MoMPercentChangeL1	BUSLOANSMoMDifferenceL1
3 RPIMoMPercentChangeL0	UEMP27OVMoMDifferenceL0	BUSLOANSMoMPercentChangeL0
4 RPIMoMPercentChangeL1	UEMP27OVMoMDifferenceL1	BUSLOANSMoMPercentChangeL1
5 W875RX1MoMDifferenceL0	UEMP27OVMoMPercentChangeL0	REALLNMoMDifferenceL0
6 W875RX1MoMDifferenceL1	UEMP27OVMoMPercentChangeL1	REALLNMoMDifferenceL1
7 W875RX1MoMPercentChangeL0	PAYEMSMoMDifferenceL0	REALLNMoMPercentChangeL0
8 W875RX1MoMPercentChangeL1	PAYEMSMoMDifferenceL1	REALLNMoMPercentChangeL1
9 DPCERA3M086SBEAMoMDifferenceL0	PAYEMSMoMPercentChangeL0	WPSFD49207MoMDifferenceL0
10 DPCERA3M086SBEAMoMDifferenceL1	PAYEMSMoMPercentChangeL1	WPSFD49207MoMDifferenceL1
11 DPCERA3M086SBEAMoMPercentChangeL0	USGOODMoMDifferenceL0	WPSFD49207MoMPercentChangeL0
12 DPCERA3M086SBEAMoMPercentChangeL1	USGOODMoMDifferenceL1	WPSFD49207MoMPercentChangeL1
13 CMRMTSPLMoMDifferenceL0	USGOODMoMPercentChangeL0	WPSFD49502MoMDifferenceL0
14 CMRMTSPLMoMDifferenceL1	USGOODMoMPercentChangeL1	WPSFD49502MoMDifferenceL1
15 CMRMTSPLMoMPercentChangeL0	CES1021000001MoMDifferenceL0	WPSFD49502MoMPercentChangeL0
16 CMRMTSPLMoMPercentChangeL1	CES1021000001MoMDifferenceL1	WPSFD49502MoMPercentChangeL1
17 INDPROMoMDifferenceL0	CES1021000001MoMPercentChangeL0	WPSID61MoMDifferenceL0
18 INDPROMoMDifferenceL1	CES1021000001MoMPercentChangeL1	WPSID61MoMDifferenceL1
19 INDPROMoMPercentChangeL0	USCONSMoMDifferenceL0	WPSID61MoMPercentChangeL0
20 INDPROMoMPercentChangeL1	USCONSMoMDifferenceL1	WPSID61MoMPercentChangeL1
21 IPFPNSSMoMDifferenceL0	USCONSMoMPercentChangeL0	WPSID62MoMDifferenceL0
22 IPFPNSSMoMDifferenceL1	USCONSMoMPercentChangeL1	WPSID62MoMDifferenceL1
23 IPFPNSSMoMPercentChangeL0	MANEMPMoMDifferenceL0	WPSID62MoMPercentChangeL0
24 IPFPNSSMoMPercentChangeL1	MANEMPMoMDifferenceL1	WPSID62MoMPercentChangeL1
25 IPFINALMoMDifferenceL0	MANEMPMoMPercentChangeL0	MZMSLMoMDifferenceL0
26 IPFINALMoMDifferenceL1	MANEMPMoMPercentChangeL1	MZMSLMoMDifferenceL1
27 IPFINALMoMPercentChangeL0	DMANEMPMoMDifferenceL0	MZMSLMoMPercentChangeL0
28 IPFINALMoMPercentChangeL1	DMANEMPMoMDifferenceL1	MZMSLMoMPercentChangeL1
29 IPCONGDMoMDifferenceL0	DMANEMPMoMPercentChangeL0	DTCOLNVHFNMMDifferenceL0
30 IPCONGDMoMDifferenceL1	DMANEMPMoMPercentChangeL1	DTCOLNVHFNMMDifferenceL1
31 IPCONGDMoMPercentChangeL0	SRVPRDMoMDifferenceL0	DTCOLNVHFNMMDifferenceL0
32 IPCONGDMoMPercentChangeL1	SRVPRDMoMDifferenceL1	DTCOLNVHFNMMDifferenceL1
33 IPDCONGDMoMDifferenceL0	SRVPRDMoMPercentChangeL0	DTCTHFNMMoMDifferenceL0
34 IPDCONGDMoMDifferenceL1	SRVPRDMoMPercentChangeL1	DTCTHFNMMoMDifferenceL1
35 IPDCONGDMoMPercentChangeL0	USWTRADEMoMDifferenceL0	DTCTHFNMMoMPercentChangeL0
36 IPDCONGDMoMPercentChangeL1	USWTRADEMoMDifferenceL1	DTCTHFNMMoMPercentChangeL1
37 IPNCONGDMoMDifferenceL0	USWTRADEMoMPercentChangeL0	INVESTMoMDifferenceL0
38 IPNCONGDMoMDifferenceL1	USWTRADEMoMPercentChangeL1	INVESTMoMDifferenceL1
39 IPNCONGDMoMPercentChangeL0	USTRADEMoMDifferenceL0	INVESTMoMPercentChangeL0
40 IPNCONGDMoMPercentChangeL1	USTRADEMoMDifferenceL1	INVESTMoMPercentChangeL1
41 IPBUSEQMoMDifferenceL0	USTRADEMoMPercentChangeL0	ism_man_emplMoMDifferenceL0
42 IPBUSEQMoMDifferenceL1	USTRADEMoMPercentChangeL1	ism_man_emplMoMDifferenceL1
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48 IPMATMoMPercentChangeL1	USGOVTMoMDifferenceL1	ism_man_newordersMoMPercentChangeL1
49 IPDMATMoMDifferenceL0	USGOVTMoMPercentChangeL0	ism_man_pmiMoMDifferenceL0
50 IPDMATMoMDifferenceL1	USGOVTMoMPercentChangeL1	ism_man_pmiMoMDifferenceL1
51 IPDMATMoMPercentChangeL0	CES0600000007MoMDifferenceL0	ism_man_pmiMoMPercentChangeL0
52 IPDMATMoMPercentChangeL1	CES0600000007MoMDifferenceL1	ism_man_pmiMoMPercentChangeL1

53	IPNMATMoMDifferenceL0	CES0600000007MoMPercentChangeL0	ism_man_pricesMoMDifferenceL0
54	IPNMATMoMDifferenceL1	CES0600000007MoMPercentChangeL1	ism_man_pricesMoMDifferenceL1
55	IPNMATMoMPercentChangeL0	AWOTMANMoMDifferenceL0	ism_man_pricesMoMPercentChangeL0
56	IPNMATMoMPercentChangeL1	AWOTMANMoMDifferenceL1	ism_man_pricesMoMPercentChangeL1
57	IPMANSICSMoMDifferenceL0	AWOTMANMoMPercentChangeL0	USGG2YR_IndexMoMDifferenceL0
58	IPMANSICSMoMDifferenceL1	AWOTMANMoMPercentChangeL1	USGG2YR_IndexMoMDifferenceL1
59	IPMANSICSMoMPercentChangeL0	AWHMANMoMDifferenceL0	USGG2YR_IndexMoMPercentChangeL0
60	IPMANSICSMoMPercentChangeL1	AWHMANMoMDifferenceL1	USGG2YR_IndexMoMPercentChangeL1
61	IPB51222SMoMDifferenceL0	AWHMANMoMPercentChangeL0	USGG10YR_IndexMoMDifferenceL0
62	IPB51222SMoMDifferenceL1	AWHMANMoMPercentChangeL1	USGG10YR_IndexMoMDifferenceL1
63	IPB51222SMoMPercentChangeL0	HOUSTMoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0
64	IPB51222SMoMPercentChangeL1	HOUSTMoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL1
65	IPFUELSMoMDifferenceL0	HOUSTMoMPercentChangeL0	USGG30YR_IndexMoMDifferenceL0
66	IPFUELSMoMDifferenceL1	HOUSTMoMPercentChangeL1	USGG30YR_IndexMoMDifferenceL1
67	IPFUELSMoMPercentChangeL0	HOUSTNEMoMDifferenceL0	USGG30YR_IndexMoMPercentChangeL0
68	IPFUELSMoMPercentChangeL1	HOUSTNEMoMDifferenceL1	USGG30YR_IndexMoMPercentChangeL1
69	CUMFNSMoMDifferenceL0	HOUSTNEMoMPercentChangeL0	VIX_IndexMoMDifferenceL0
70	CUMFNSMoMDifferenceL1	HOUSTNEMoMPercentChangeL1	VIX_IndexMoMDifferenceL1
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72	CUMFNSMoMPercentChangeL1	HOUSTMWMoMDifferenceL1	VIX_IndexMoMPercentChangeL1
73	CLF16OVMoMDifferenceL0	HOUSTMWMoMPercentChangeL0	MOVE_IndexMoMDifferenceL0
74	CLF16OVMoMDifferenceL1	HOUSTMWMoMPercentChangeL1	MOVE_IndexMoMDifferenceL1
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76	CLF16OVMoMPercentChangeL1	HOUSTSMoMDifferenceL1	MOVE_IndexMoMPercentChangeL1
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80	CE16OVMoMPercentChangeL1	HOUSTWMoMDifferenceL1	AAIIBULL_IndexMoMPercentChangeL1
81	UNRATEMoMDifferenceL0	HOUSTWMoMPercentChangeL0	AAIIBEAR_IndexMoMDifferenceL0
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84	UNRATEMoMPercentChangeL1	ISRATIOMoMDifferenceL1	AAIIBEAR_IndexMoMPercentChangeL1
85	UEMPMEANMoMDifferenceL0	ISRATIOMoMPercentChangeL0	SPX_Index_14DRSIMoMDifferenceL0
86	UEMPMEANMoMDifferenceL1	ISRATIOMoMPercentChangeL1	SPX_Index_14DRSIMoMDifferenceL1
87	UEMPMEANMoMPercentChangeL0	M1SLMoMDifferenceL0	SPX_Index_14DRSIMoMPercentChangeL0
88	UEMPMEANMoMPercentChangeL1	M1SLMoMDifferenceL1	SPX_Index_14DRSIMoMPercentChangeL1
89	UEMPLT5MoMDifferenceL0	M1SLMoMPercentChangeL0	SPX_Index_30DRSIMoMDifferenceL0
90	UEMPLT5MoMDifferenceL1	M1SLMoMPercentChangeL1	SPX_Index_30DRSIMoMDifferenceL1
91	UEMPLT5MoMPercentChangeL0	M2SLMoMDifferenceL0	SPX_Index_30DRSIMoMPercentChangeL0
92	UEMPLT5MoMPercentChangeL1	M2SLMoMDifferenceL1	SPX_Index_30DRSIMoMPercentChangeL1
93	UEMP5TO14MoMDifferenceL0	M2SLMoMPercentChangeL0	TED_G_IndexMoMDifferenceL0
94	UEMP5TO14MoMDifferenceL1	M2SLMoMPercentChangeL1	TED_G_IndexMoMDifferenceL1
95	UEMP5TO14MoMPercentChangeL0	M2REALMoMDifferenceL0	TED_G_IndexMoMPercentChangeL0
96	UEMP5TO14MoMPercentChangeL1	M2REALMoMDifferenceL1	TED_G_IndexMoMPercentChangeL1
97	UEMP15OVMoMDifferenceL0	M2REALMoMPercentChangeL0	BAA10YB_IndexMoMDifferenceL0
98	UEMP15OVMoMDifferenceL1	M2REALMoMPercentChangeL1	BAA10YB_IndexMoMDifferenceL1
99	UEMP15OVMoMPercentChangeL0	AMBSLMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0
100	UEMP15OVMoMPercentChangeL1	AMBSLMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL1
101	UEMP15T26MoMDifferenceL0	AMBSLMoMPercentChangeL0	
102	UEMP15T26MoMDifferenceL1	AMBSLMoMPercentChangeL1	

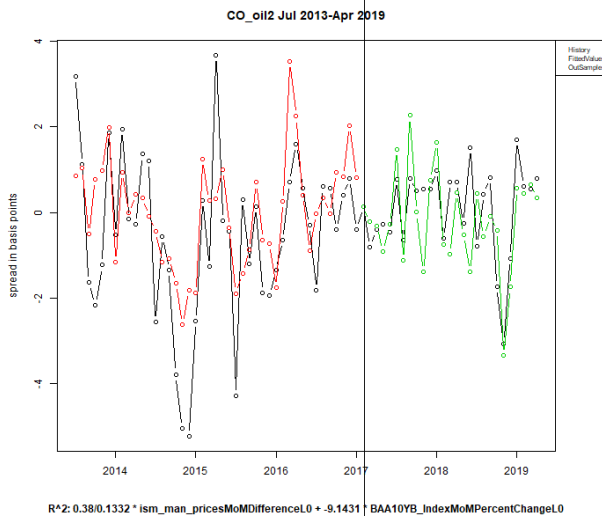
4.4.1 Model - 1

Backtesting on differenced values

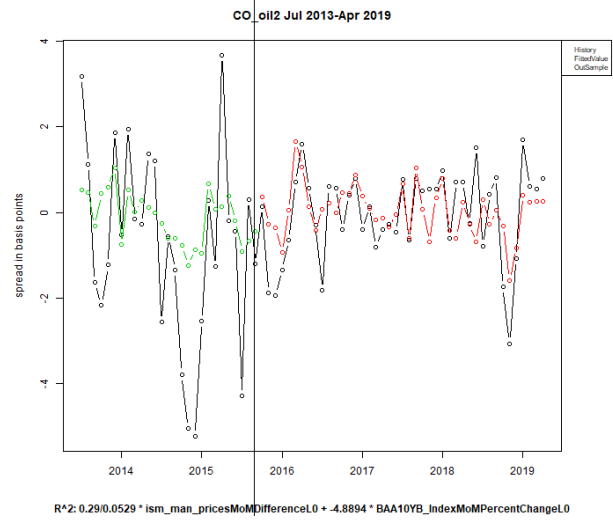
Backtesting on level values



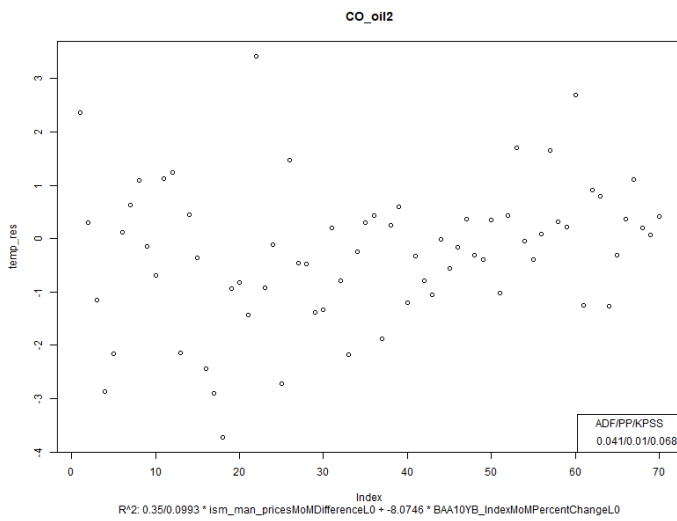
Out-of-Sample testing



Out-of-Sample testing



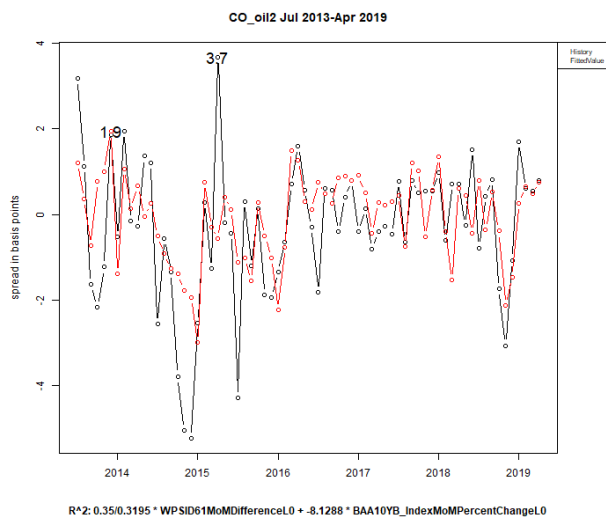
Stationarity of Residuals



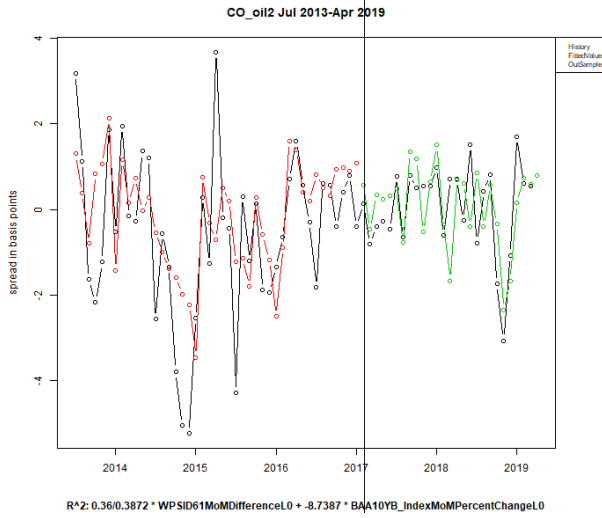
4.4.2 Model - 2

Backtesting on differenced values

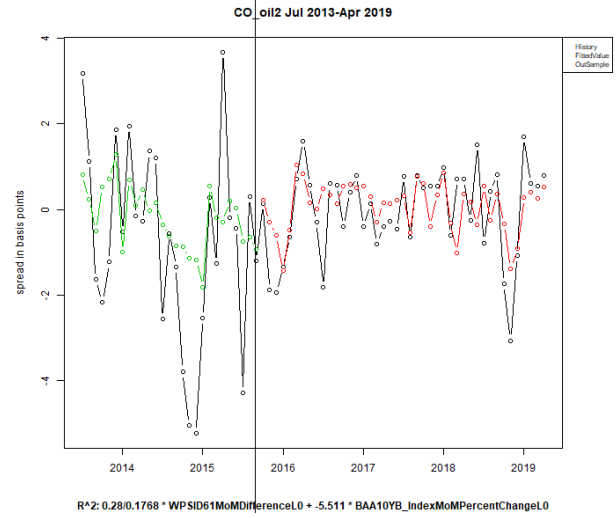
Backtesting on level values



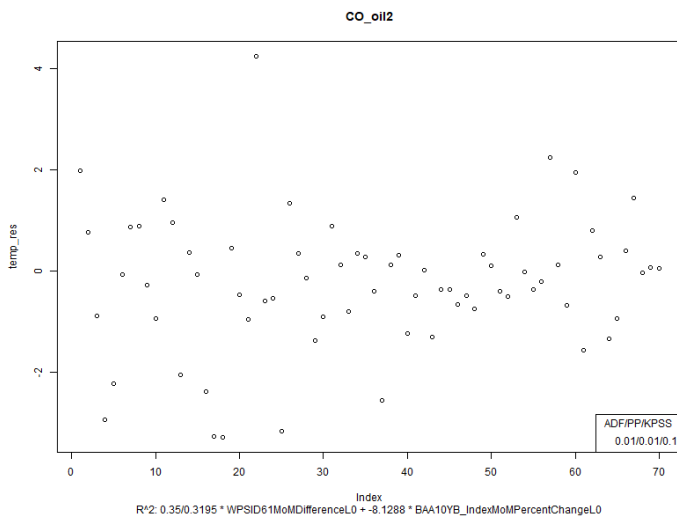
Out-of-Sample testing



Out-of-Sample testing



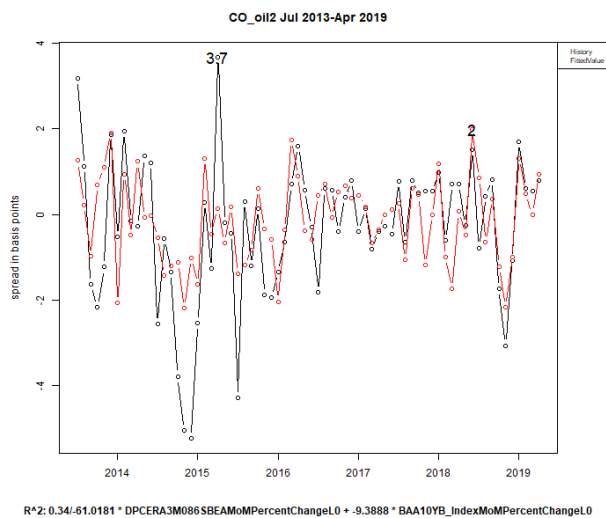
Stationarity of Residuals



4.4.3 Model - 3

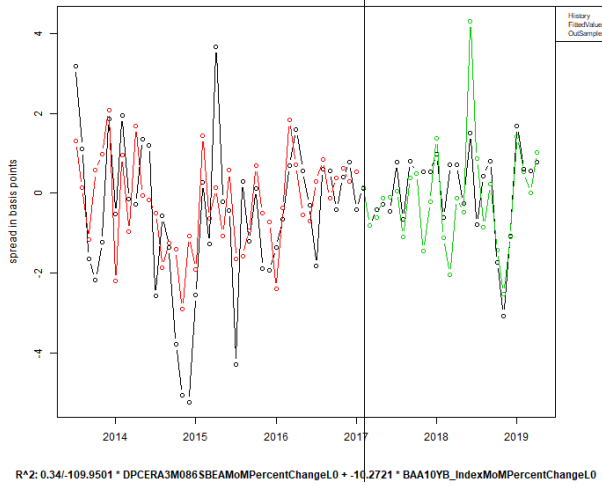
Backtesting on differenced values

Backtesting on level values



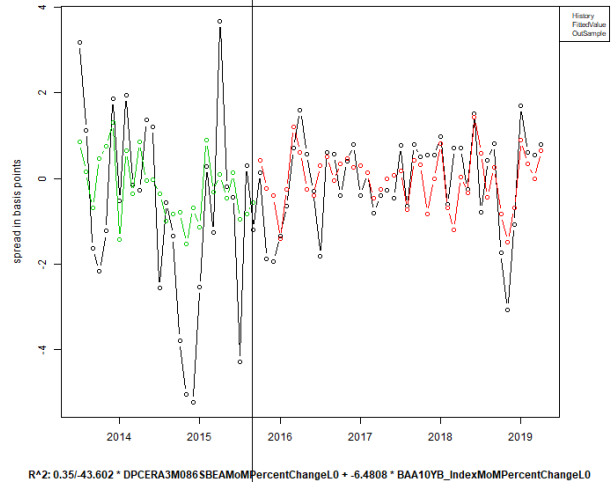
Out-of-Sample testing

CO_oil2 Jul 2013-Apr 2019



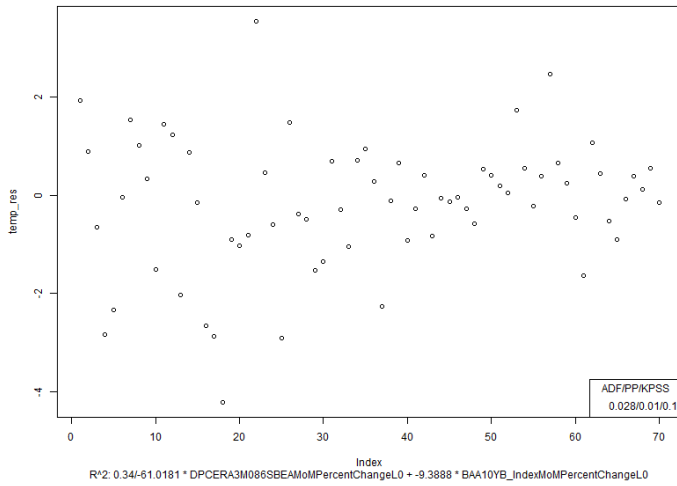
Out-of-Sample testing

CO_oil2 Jul 2013-Apr 2019



Stationarity of Residuals

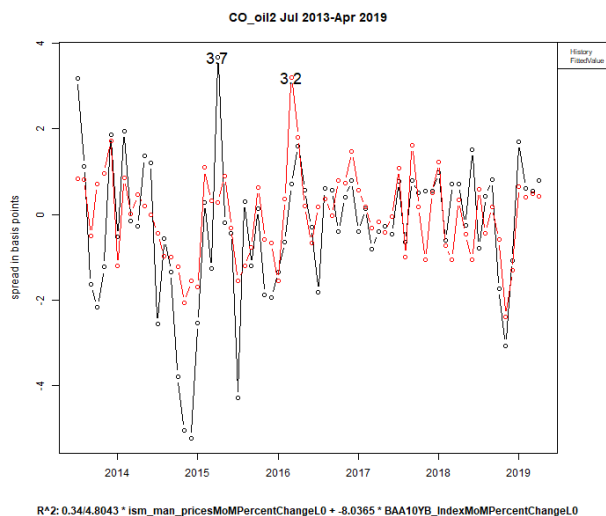
CO_oil2



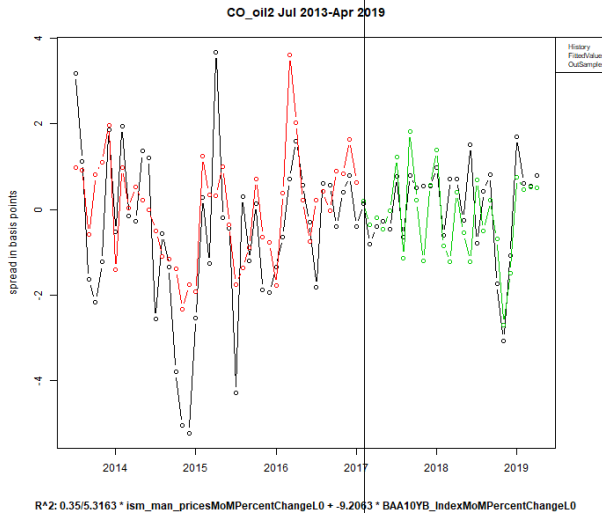
4.4.4 Model - 4

Backtesting on differenced values

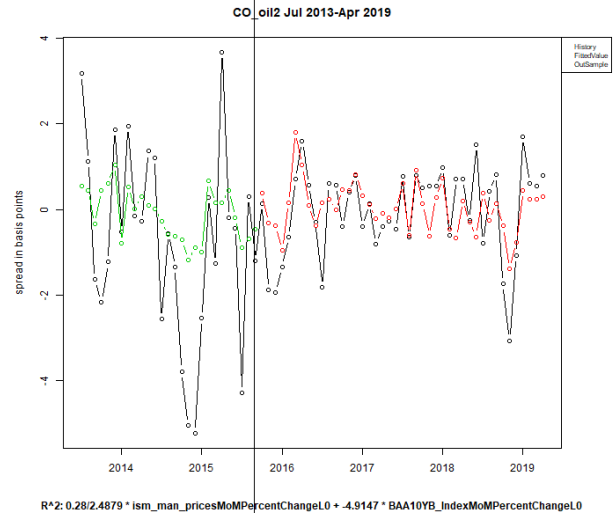
Backtesting on level values



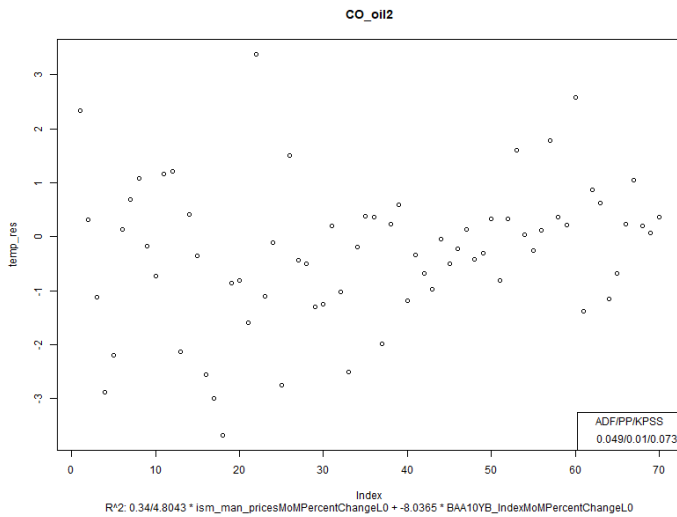
Out-of-Sample testing



Out-of-Sample testing



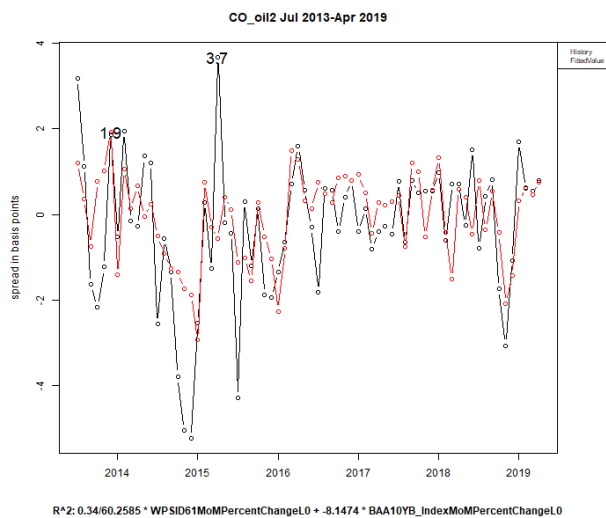
Stationarity of Residuals



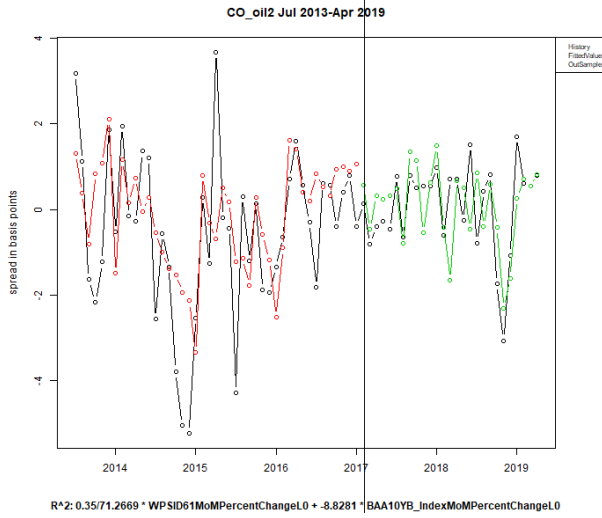
4.4.5 Model - 5

Backtesting on differenced values

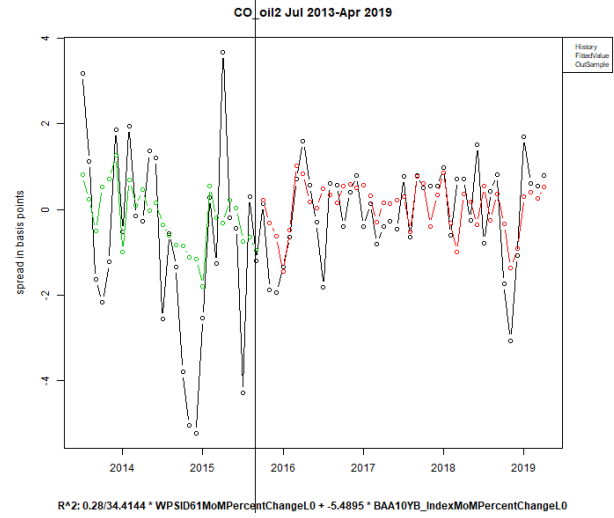
Backtesting on level values



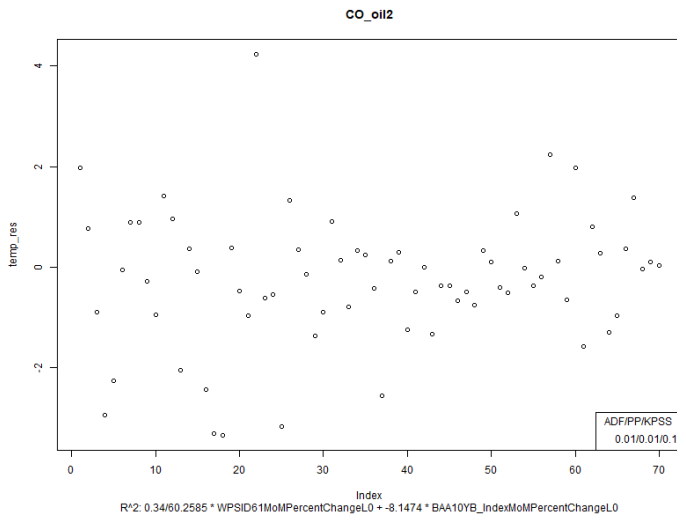
Out-of-Sample testing



Out-of-Sample testing



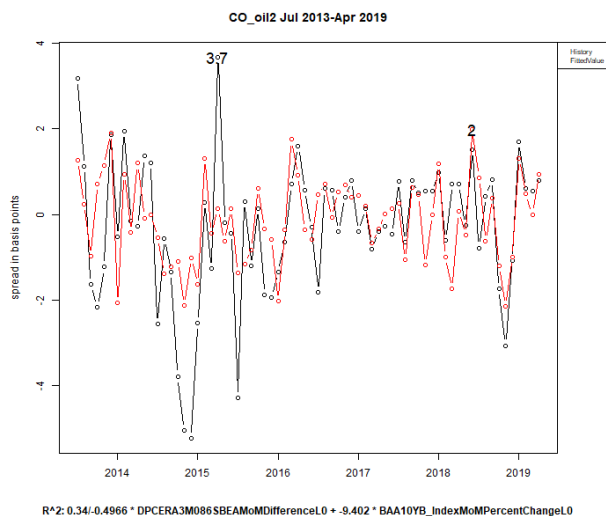
Stationarity of Residuals



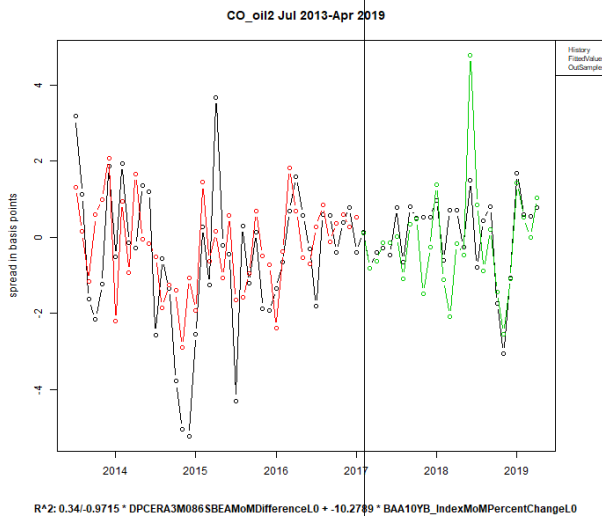
4.4.6 Model - 6

Backtesting on differenced values

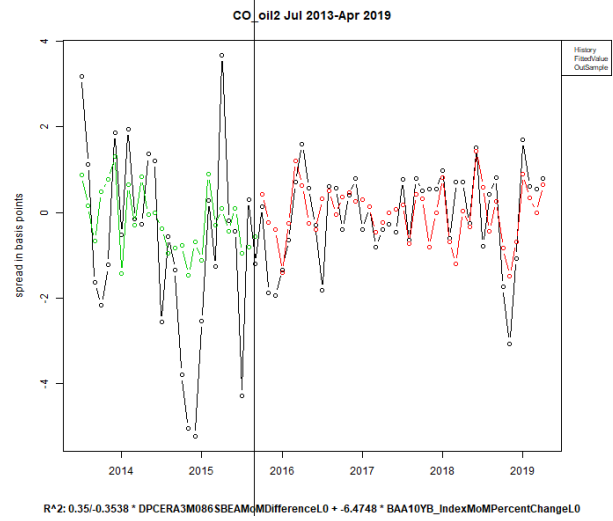
Backtesting on level values



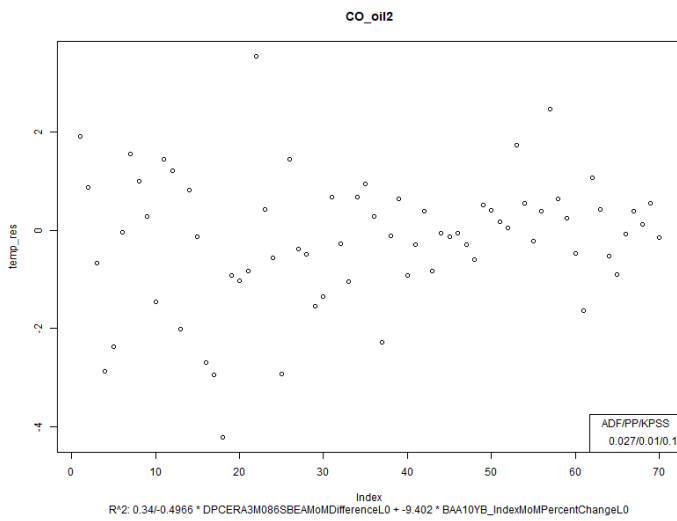
Out-of-Sample testing



Out-of-Sample testing



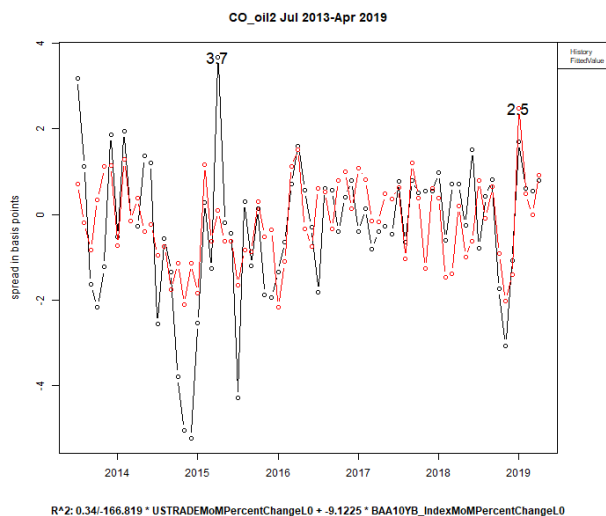
Stationarity of Residuals



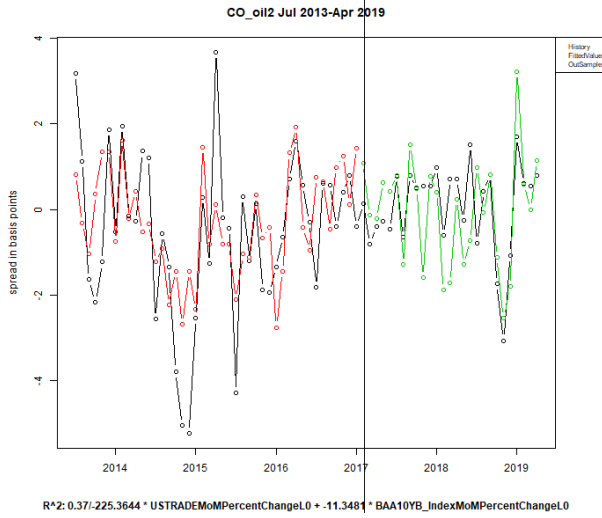
4.4.7 Model - 7

Backtesting on differenced values

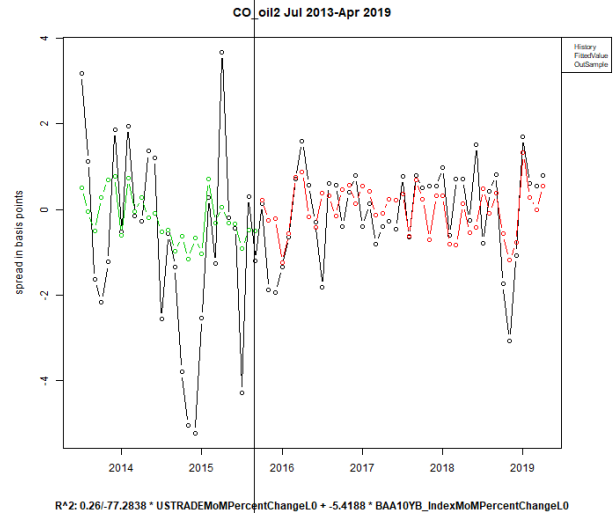
Backtesting on level values



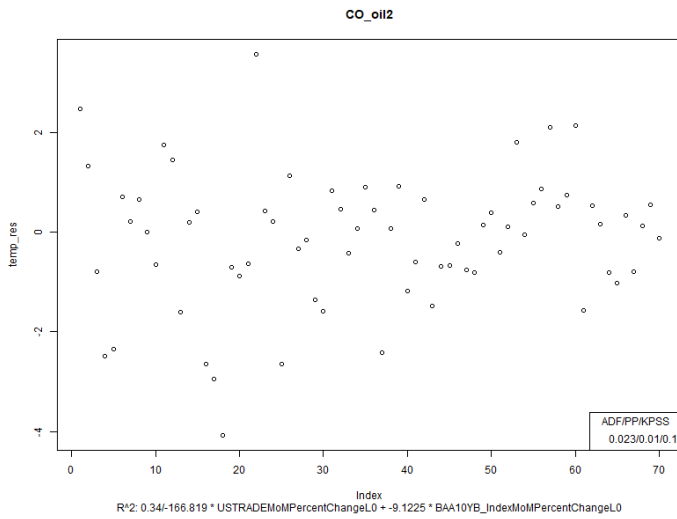
Out-of-Sample testing



Out-of-Sample testing



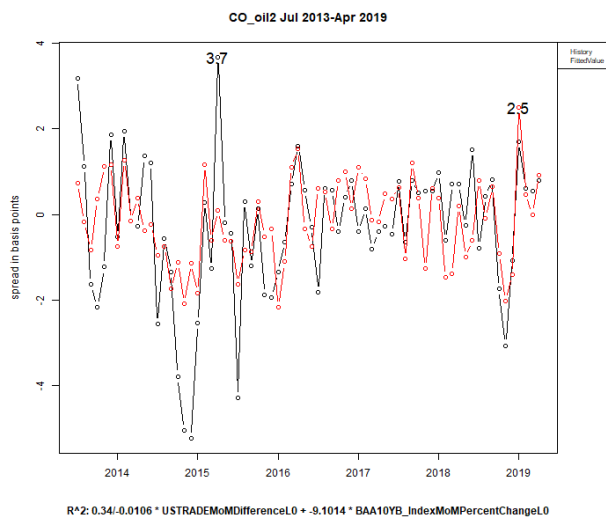
Stationarity of Residuals



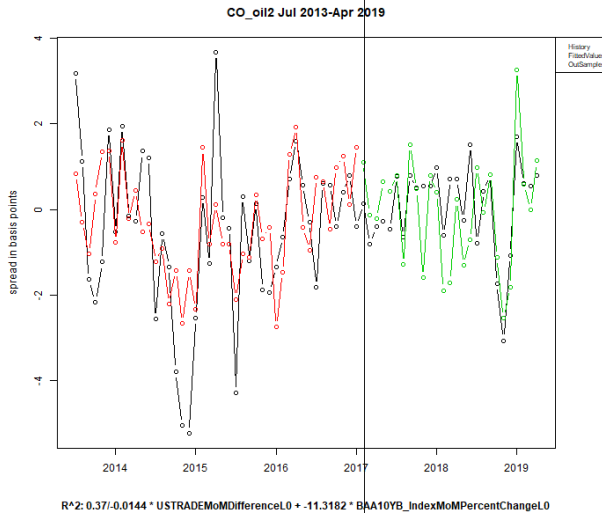
4.4.8 Model - 8

Backtesting on differenced values

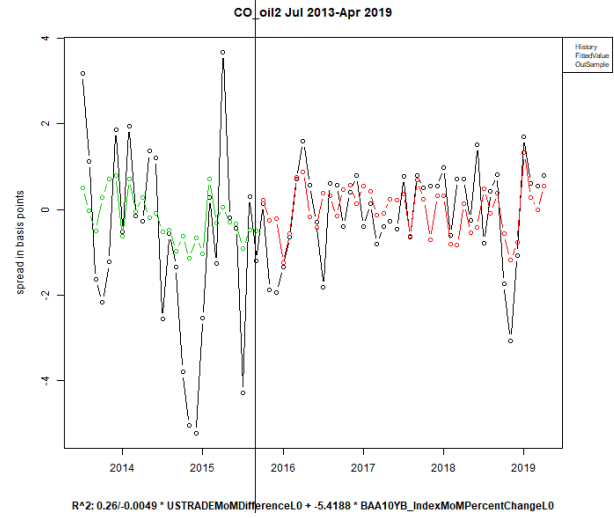
Backtesting on level values



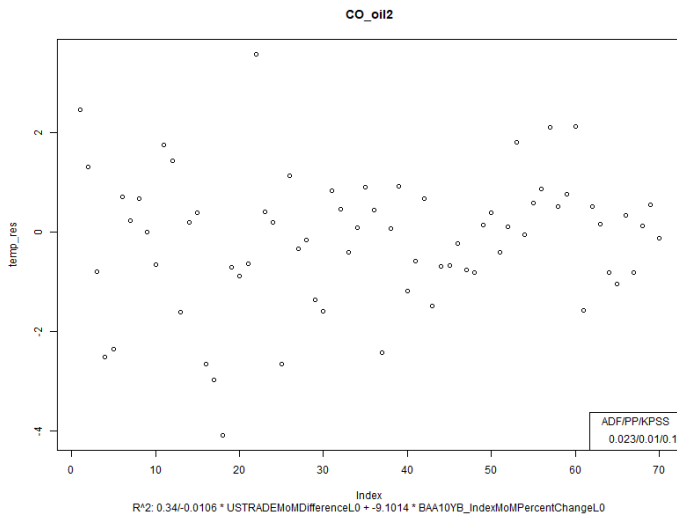
Out-of-Sample testing



Out-of-Sample testing



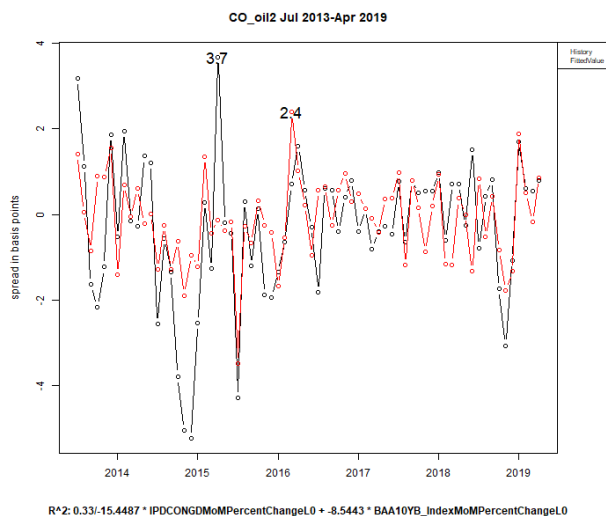
Stationarity of Residuals



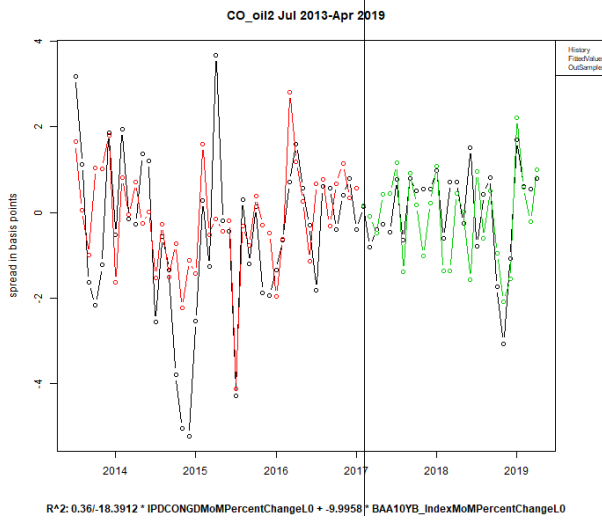
4.4.9 Model - 9

Backtesting on differenced values

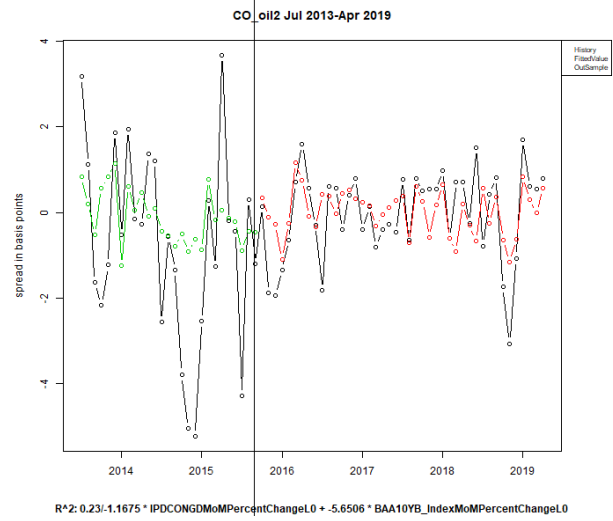
Backtesting on level values



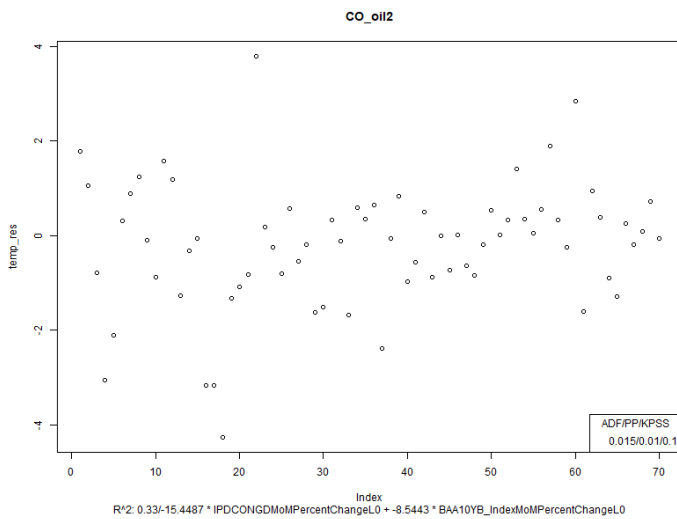
Out-of-Sample testing



Out-of-Sample testing



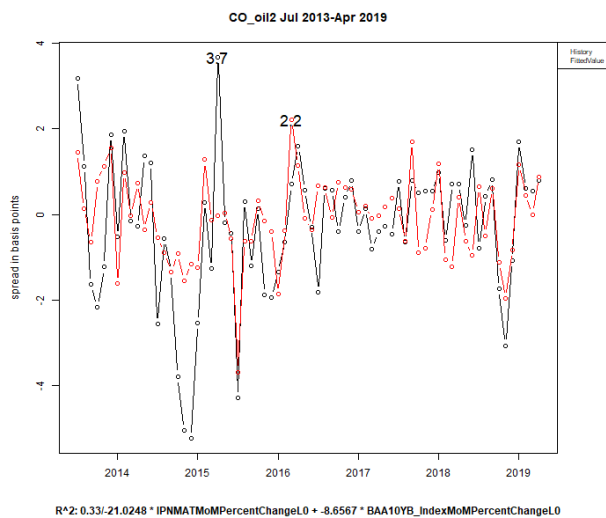
Stationarity of Residuals



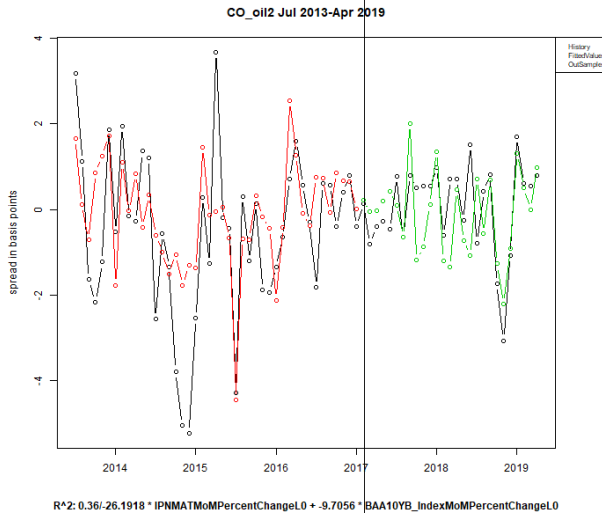
4.4.10 Model - 10

Backtesting on differenced values

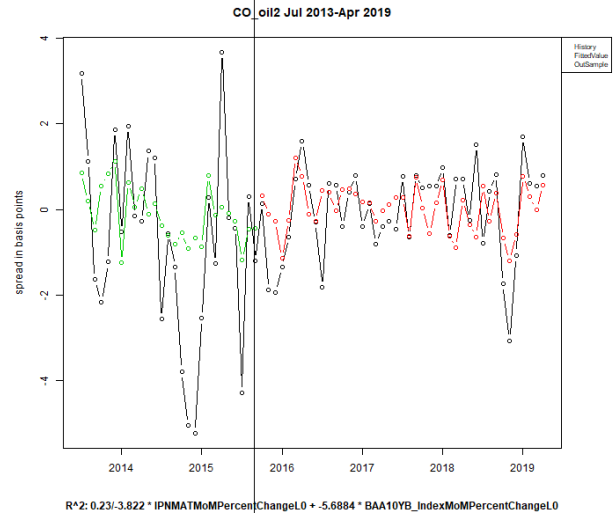
Backtesting on level values



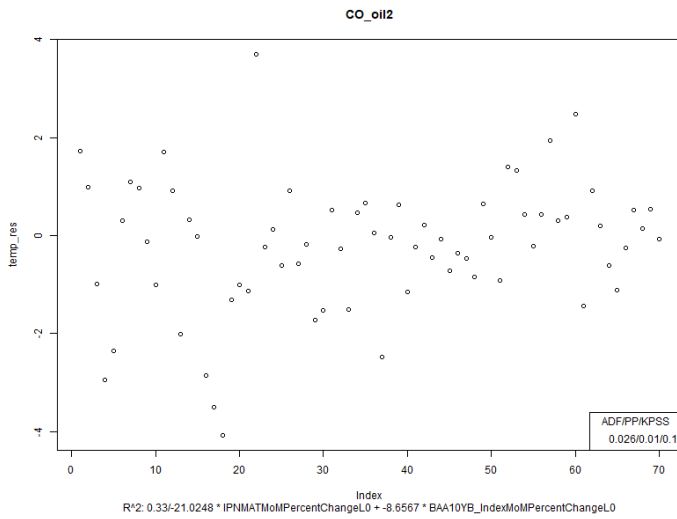
Out-of-Sample testing



Out-of-Sample testing



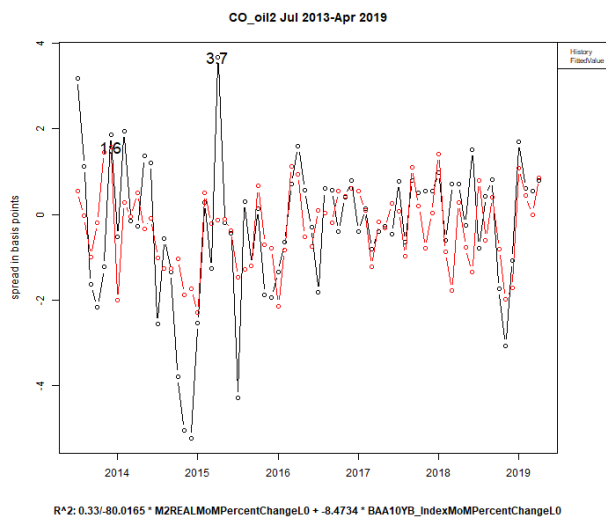
Stationarity of Residuals



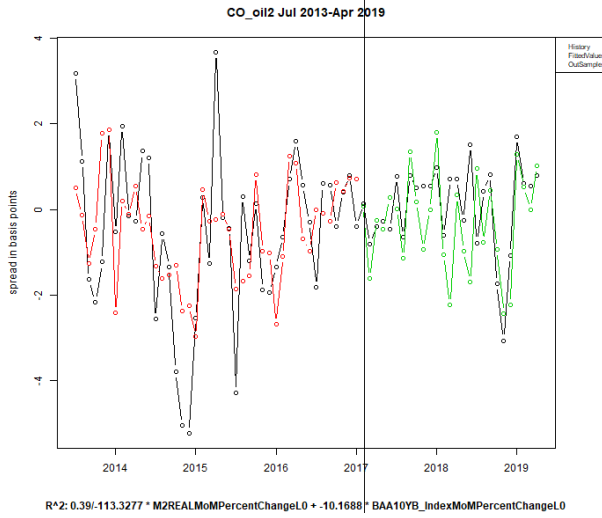
4.4.11 Model - 11

Backtesting on differenced values

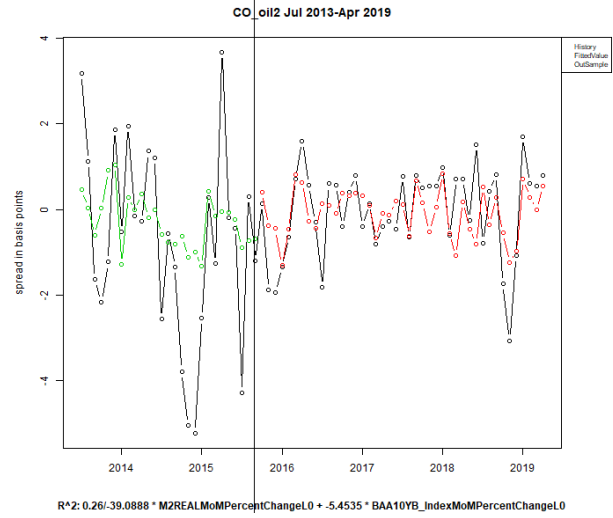
Backtesting on level values



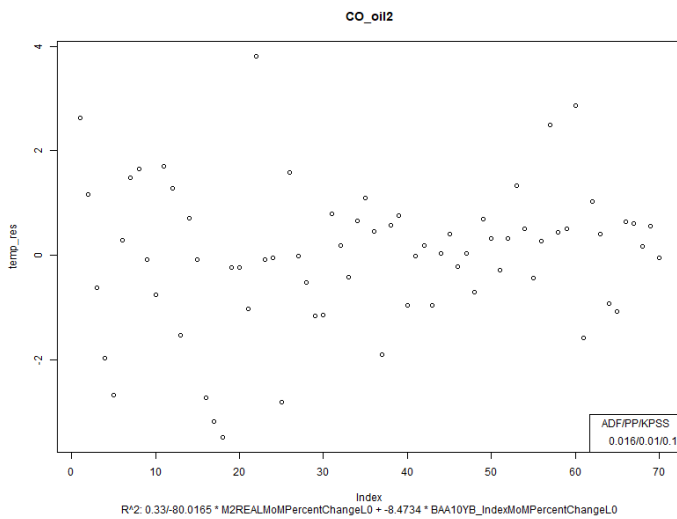
Out-of-Sample testing



Out-of-Sample testing



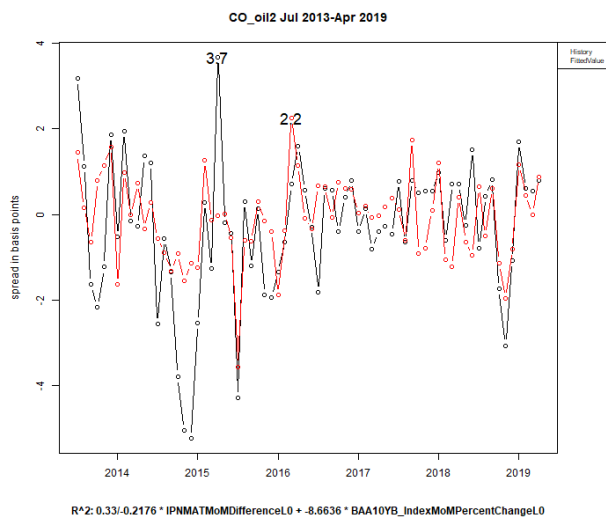
Stationarity of Residuals



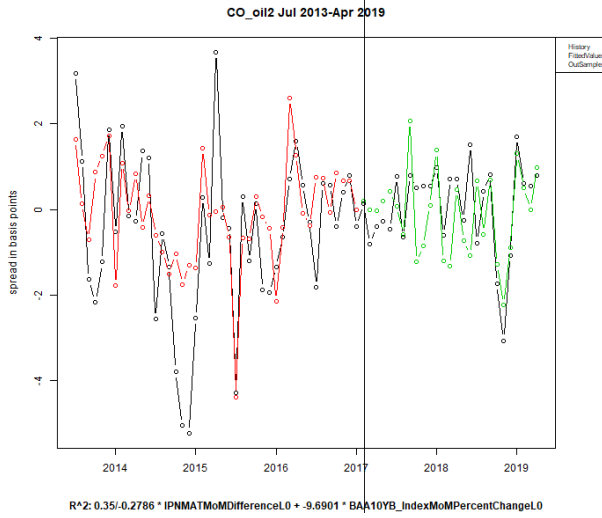
4.4.12 Model - 12

Backtesting on differenced values

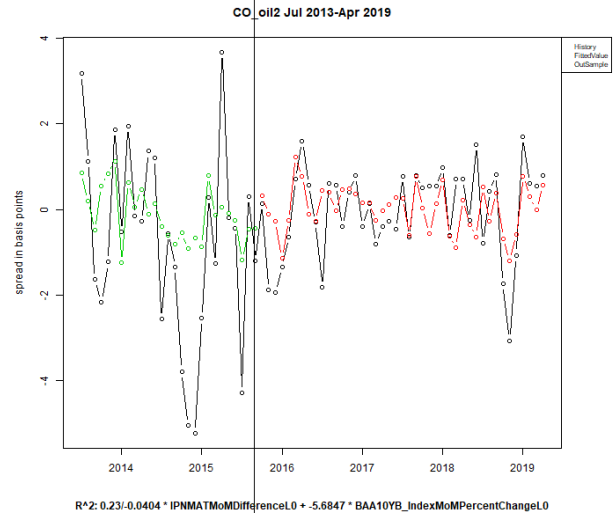
Backtesting on level values



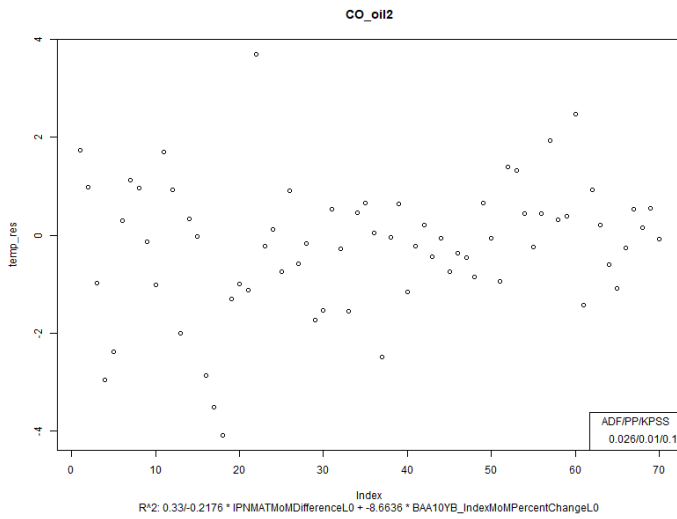
Out-of-Sample testing



Out-of-Sample testing



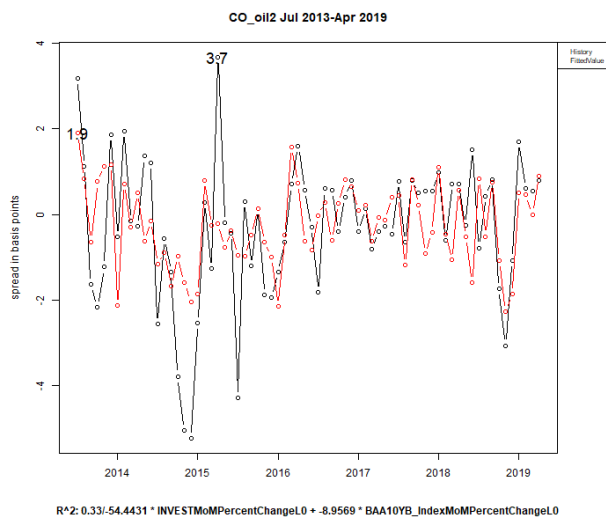
Stationarity of Residuals



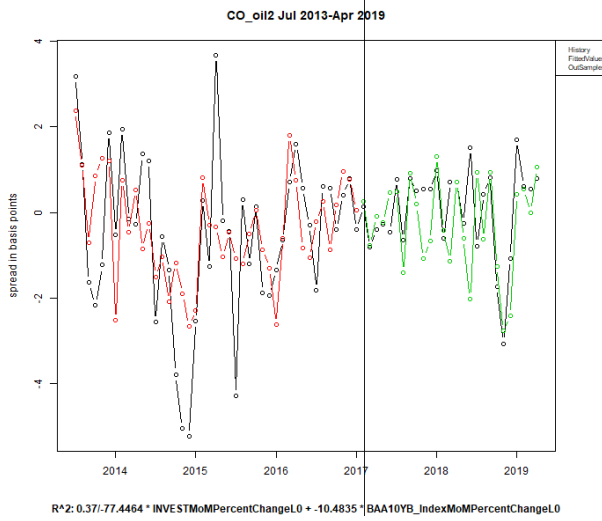
4.4.13 Model - 13

Backtesting on differenced values

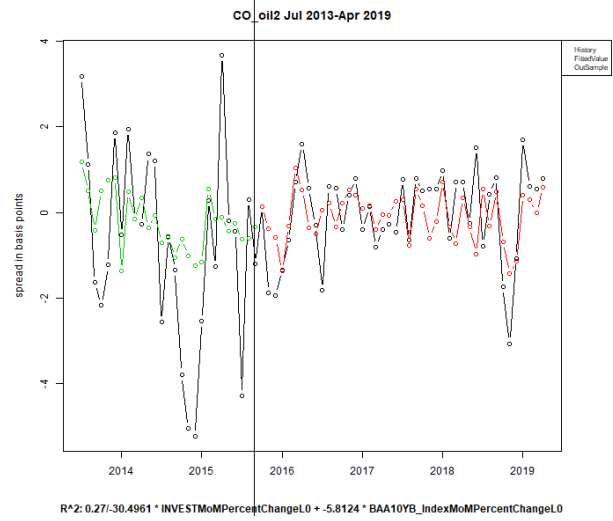
Backtesting on level values



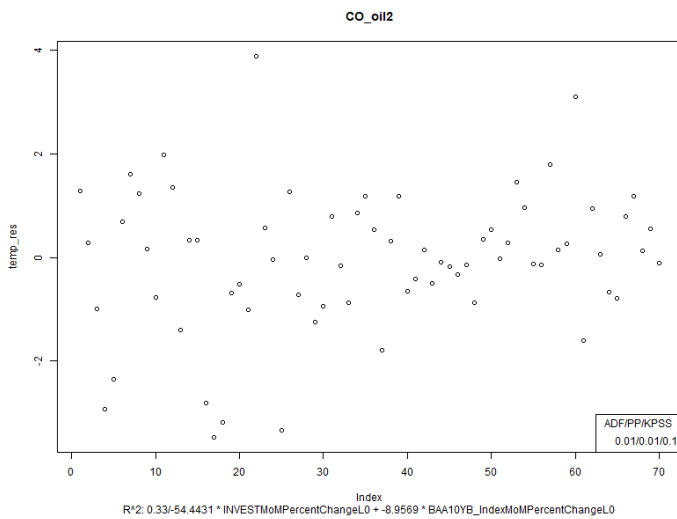
Out-of-Sample testing



Out-of-Sample testing



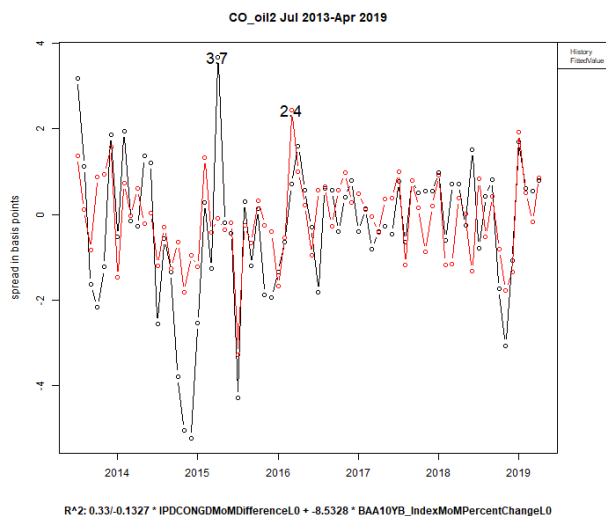
Stationarity of Residuals



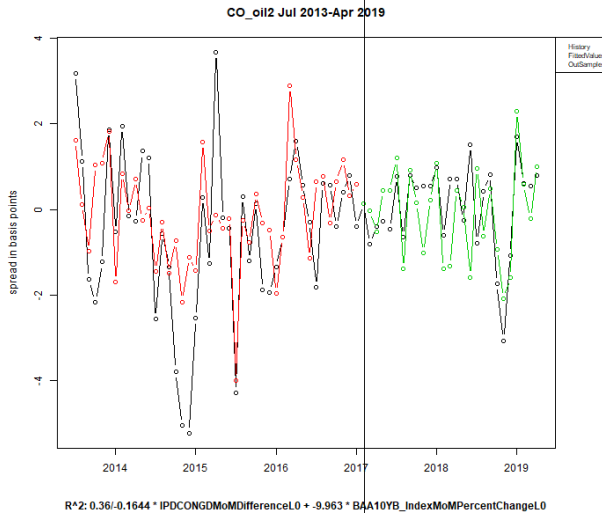
4.4.14 Model - 14

Backtesting on differenced values

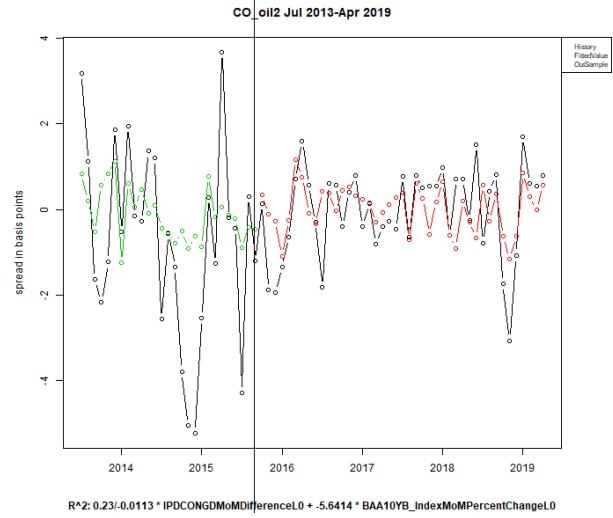
Backtesting on level values



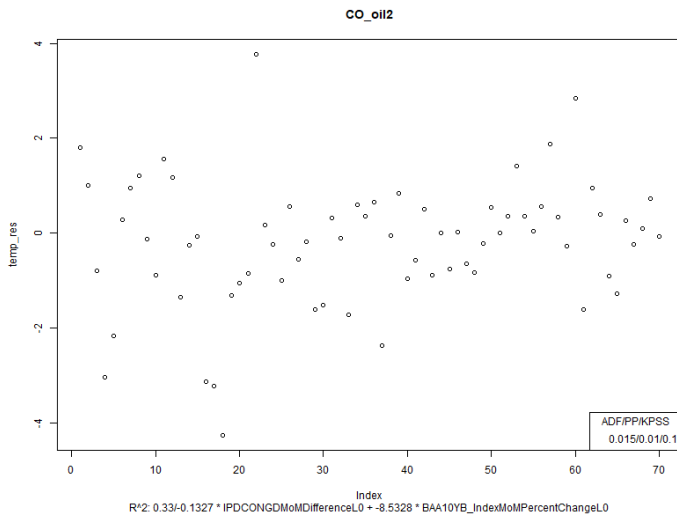
Out-of-Sample testing



Out-of-Sample testing



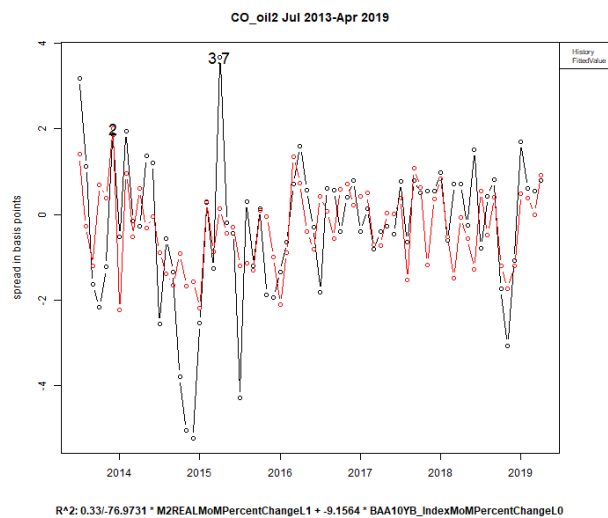
Stationarity of Residuals



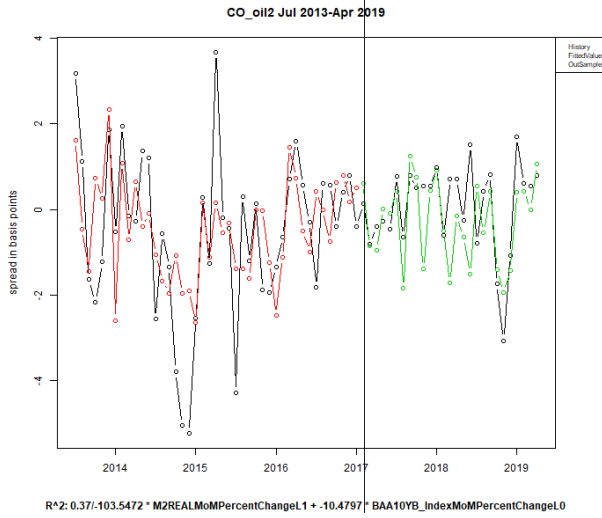
4.4.15 Model - 15

Backtesting on differenced values

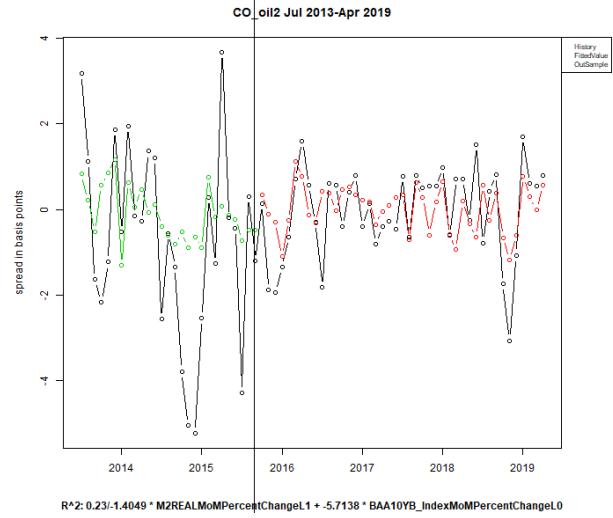
Backtesting on level values



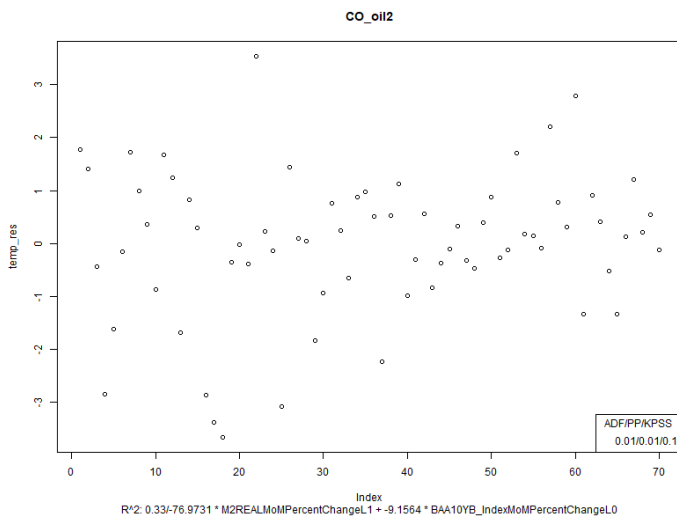
Out-of-Sample testing



Out-of-Sample testing



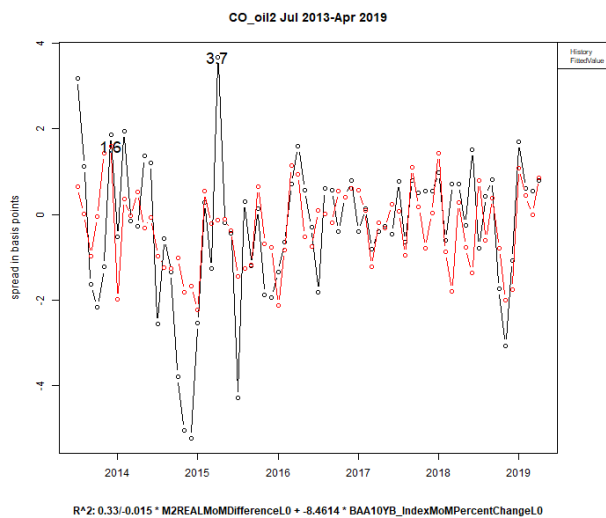
Stationarity of Residuals



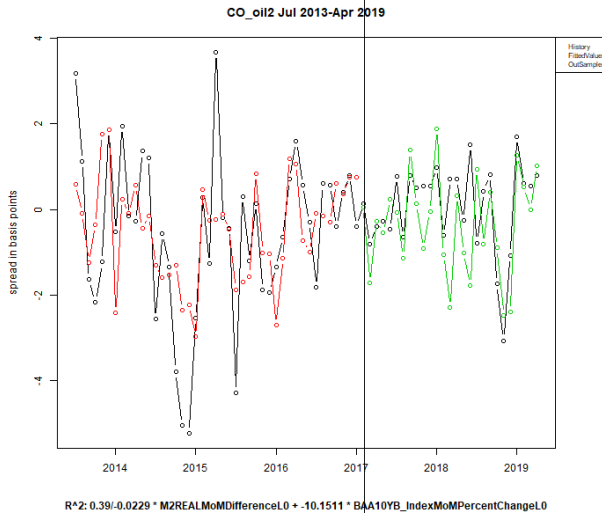
4.4.16 Model - 16

Backtesting on differenced values

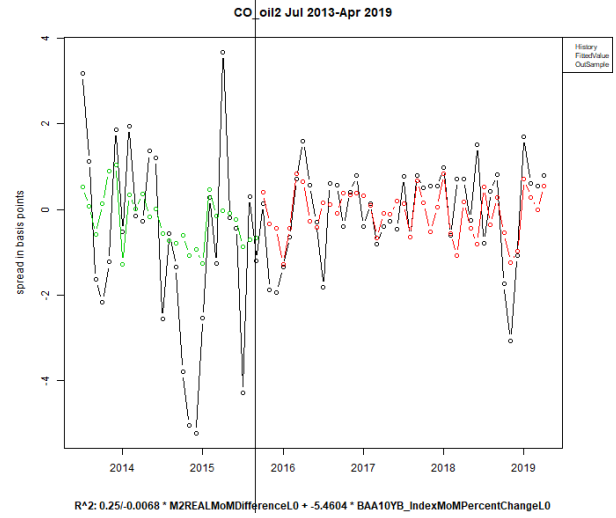
Backtesting on level values



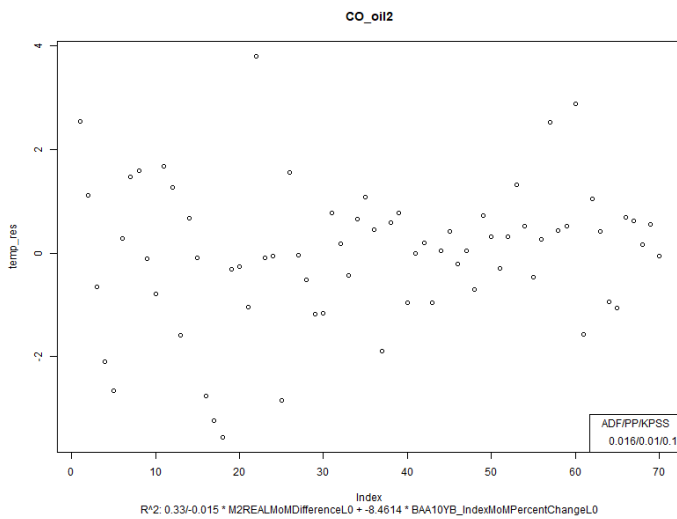
Out-of-Sample testing



Out-of-Sample testing



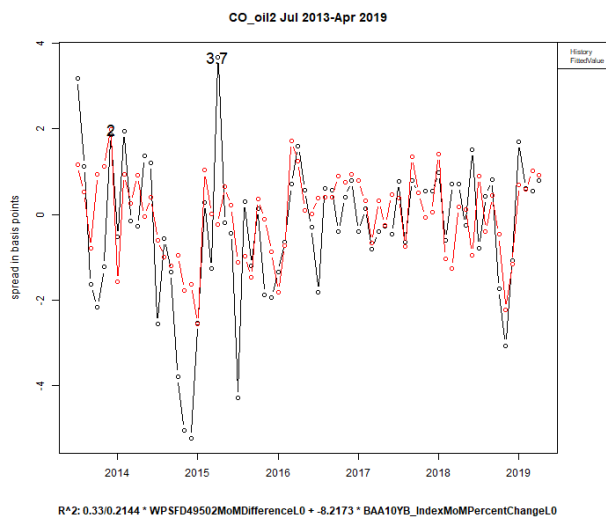
Stationarity of Residuals



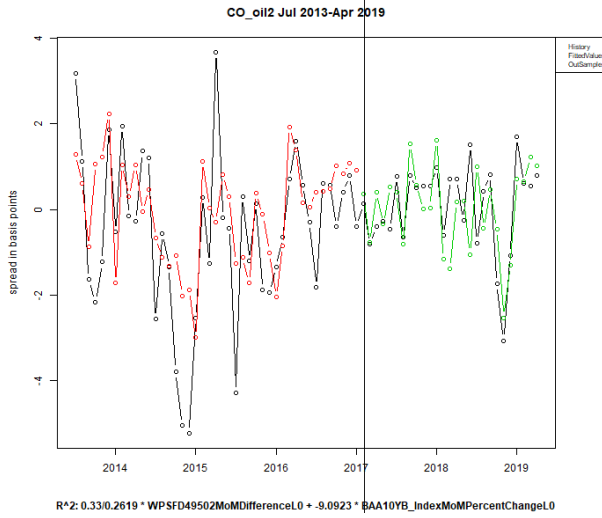
4.4.17 Model - 17

Backtesting on differenced values

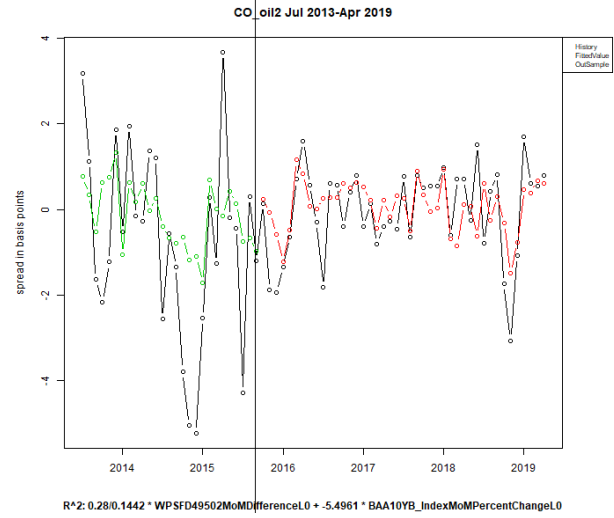
Backtesting on level values



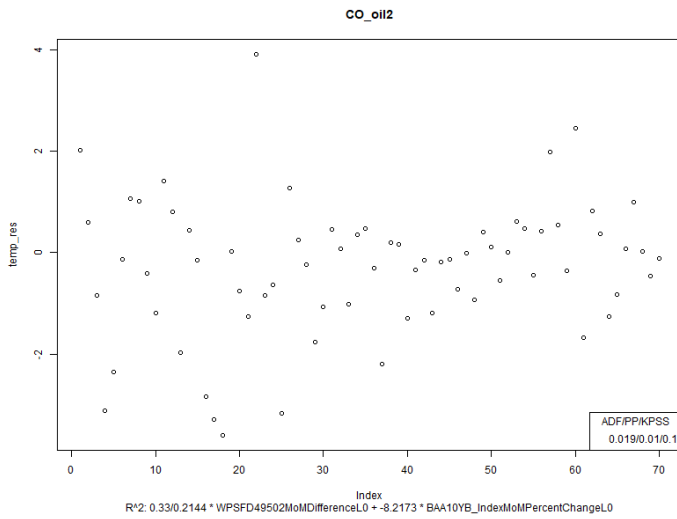
Out-of-Sample testing



Out-of-Sample testing



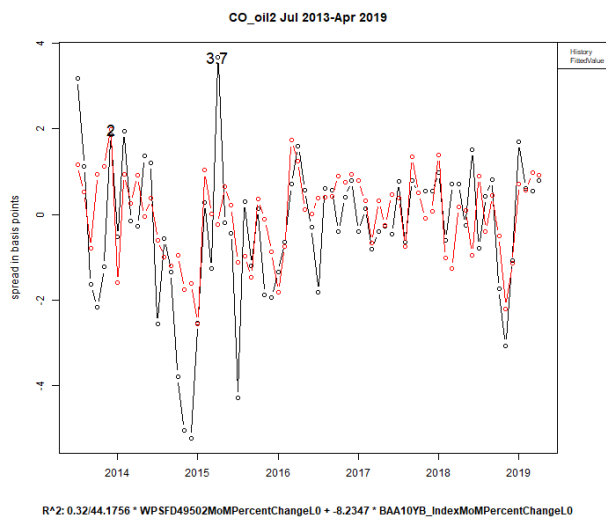
Stationarity of Residuals



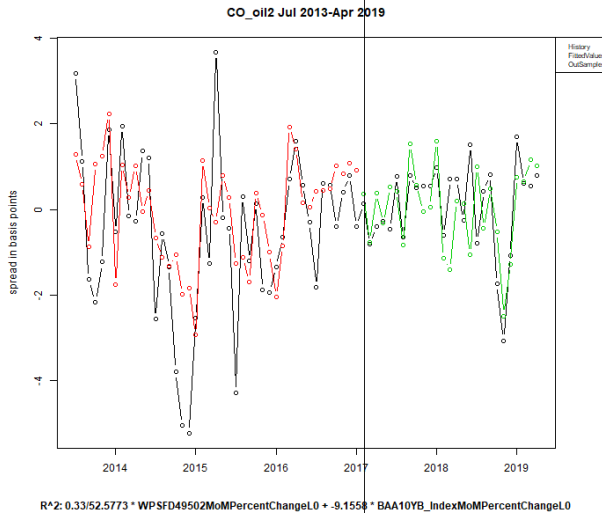
4.4.18 Model - 18

Backtesting on differenced values

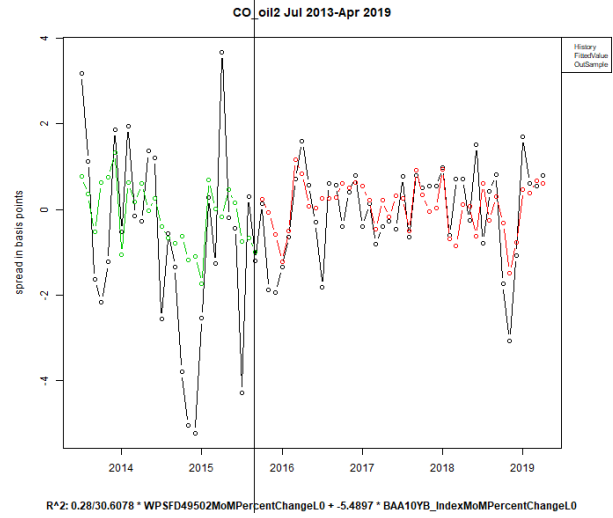
Backtesting on level values



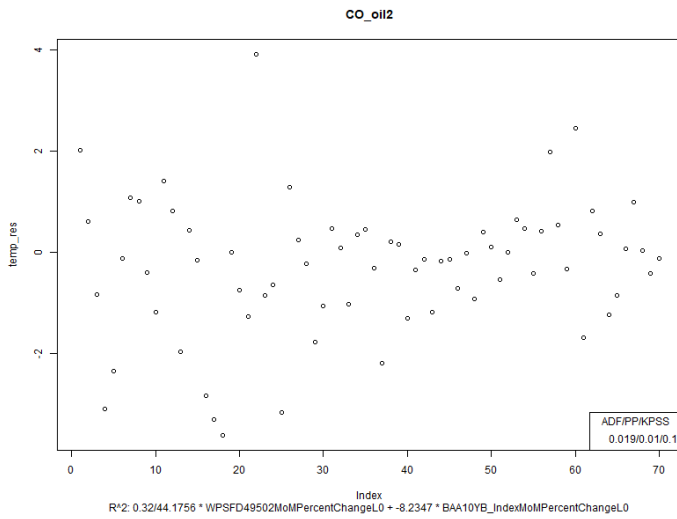
Out-of-Sample testing



Out-of-Sample testing



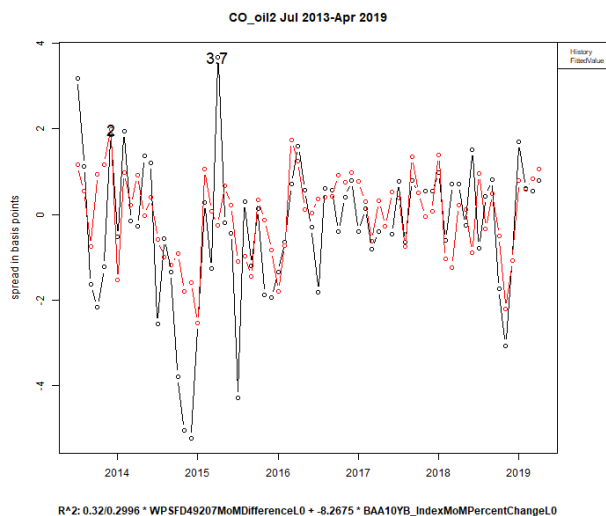
Stationarity of Residuals



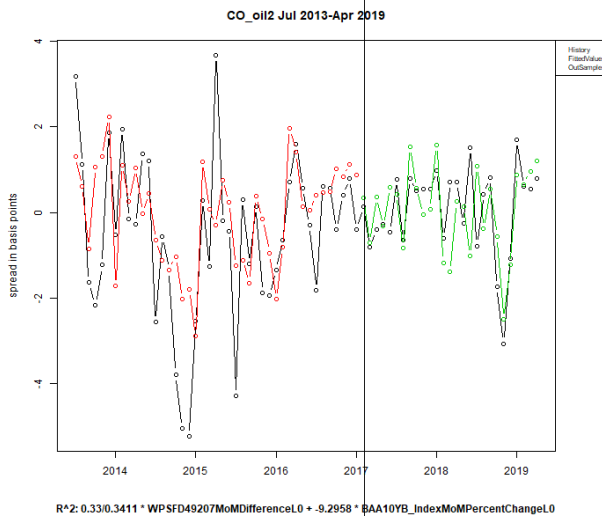
4.4.19 Model - 19

Backtesting on differenced values

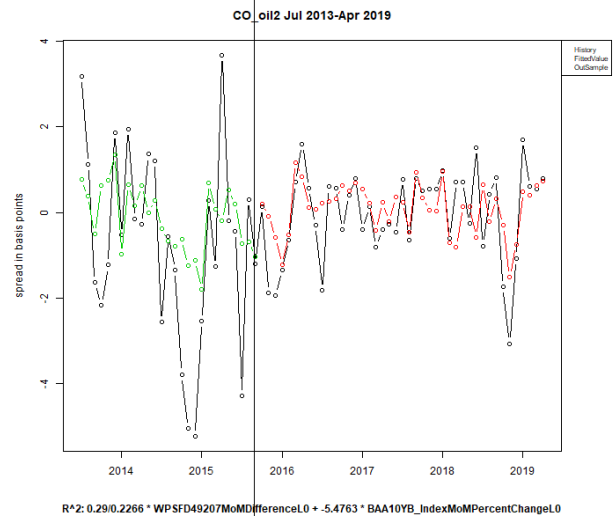
Backtesting on level values



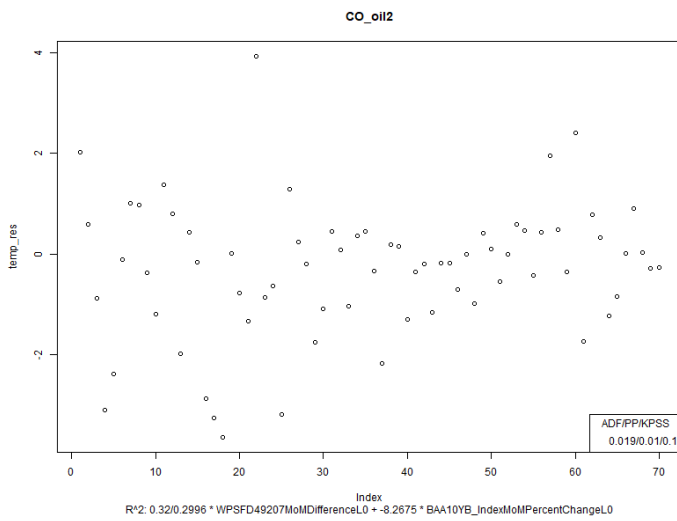
Out-of-Sample testing



Out-of-Sample testing



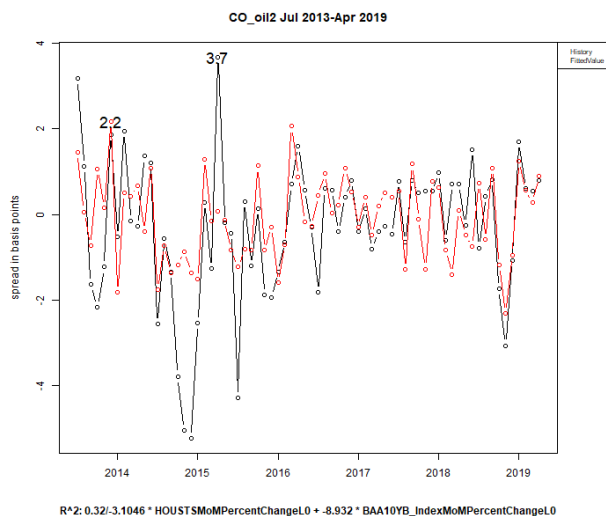
Stationarity of Residuals



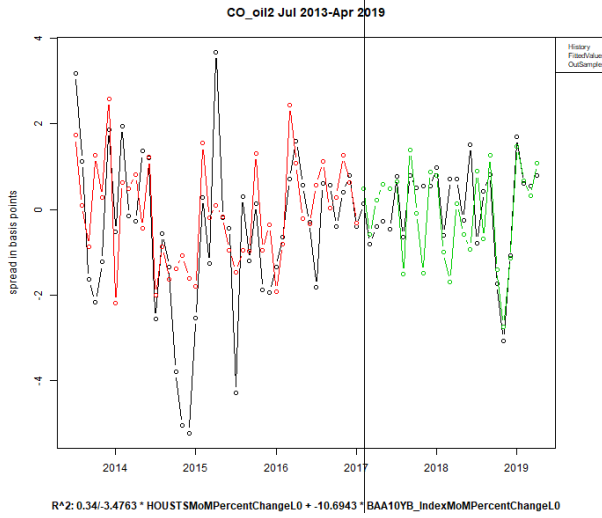
4.4.20 Model - 20

Backtesting on differenced values

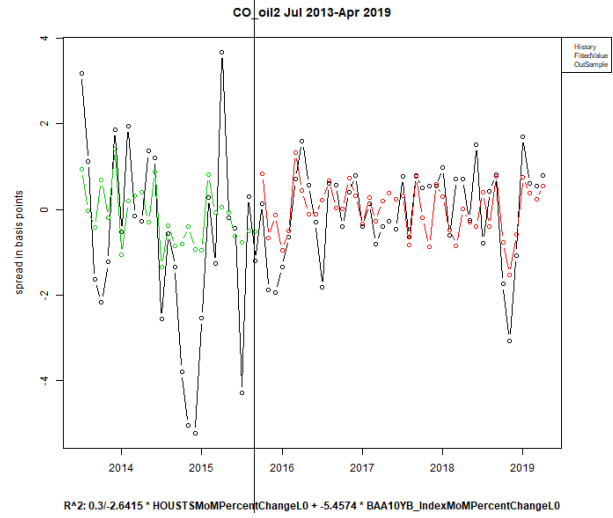
Backtesting on level values



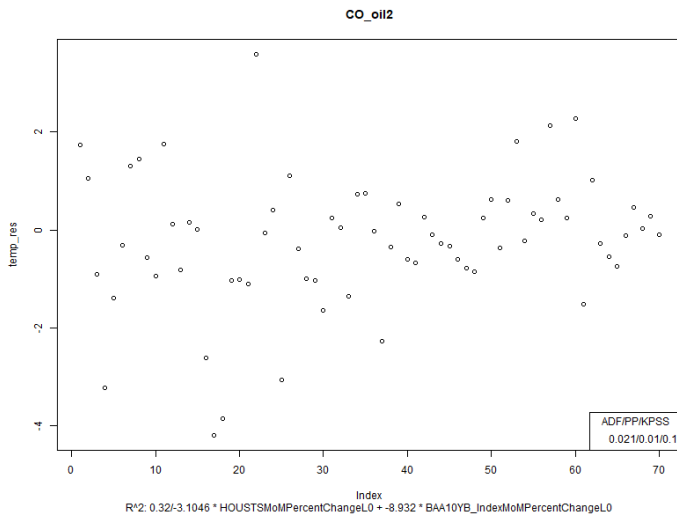
Out-of-Sample testing



Out-of-Sample testing



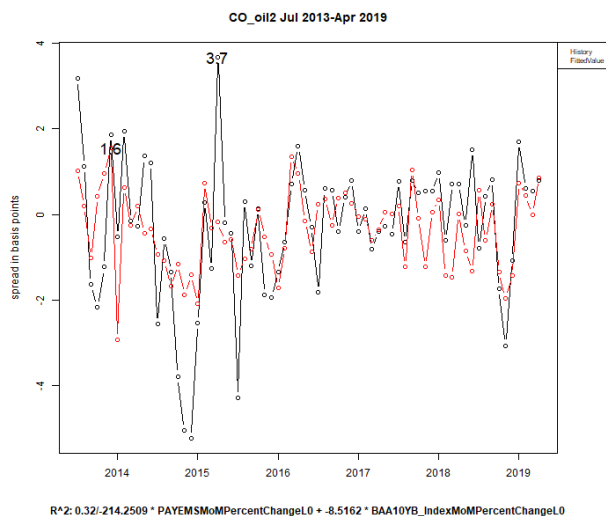
Stationarity of Residuals



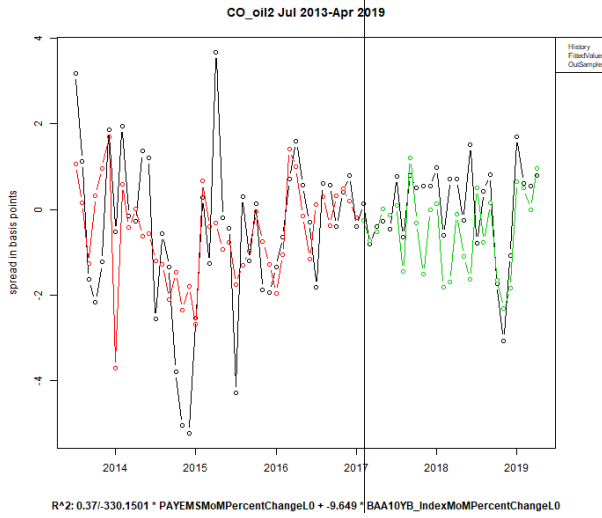
4.4.21 Model - 21

Backtesting on differenced values

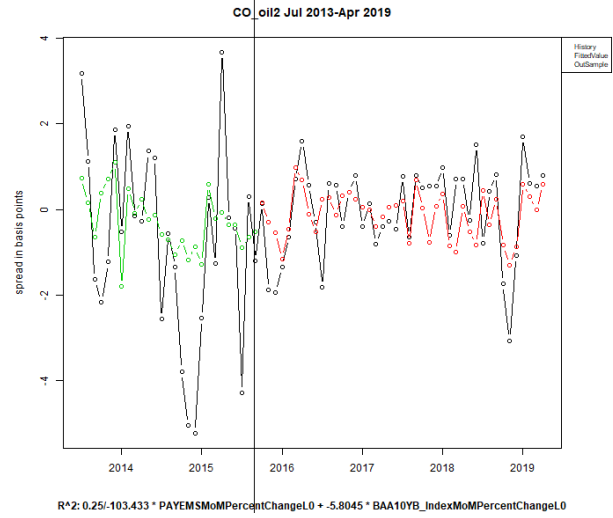
Backtesting on level values



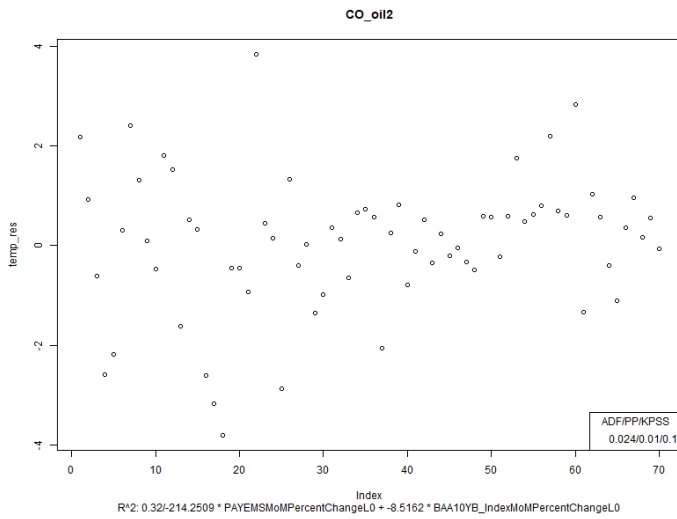
Out-of-Sample testing



Out-of-Sample testing



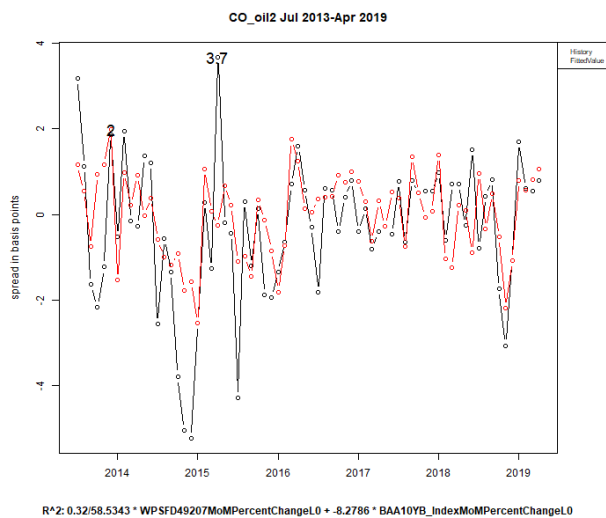
Stationarity of Residuals



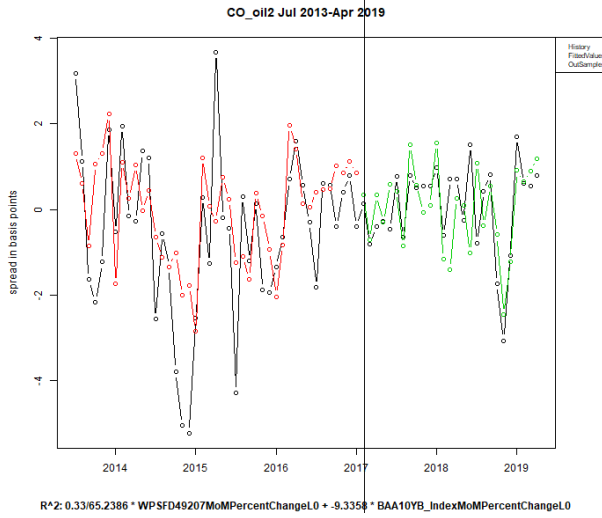
4.4.22 Model - 22

Backtesting on differenced values

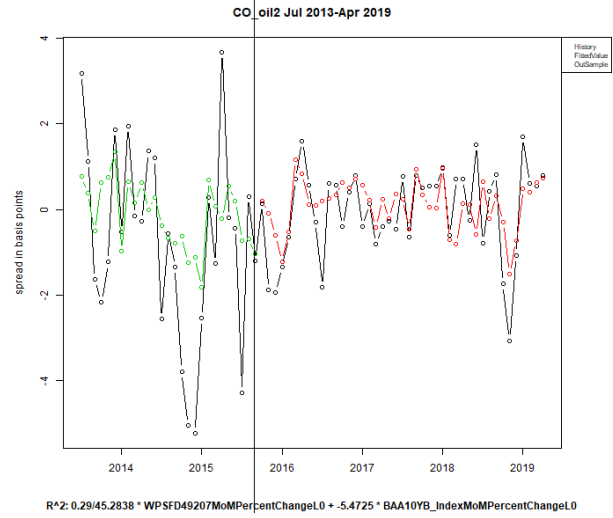
Backtesting on level values



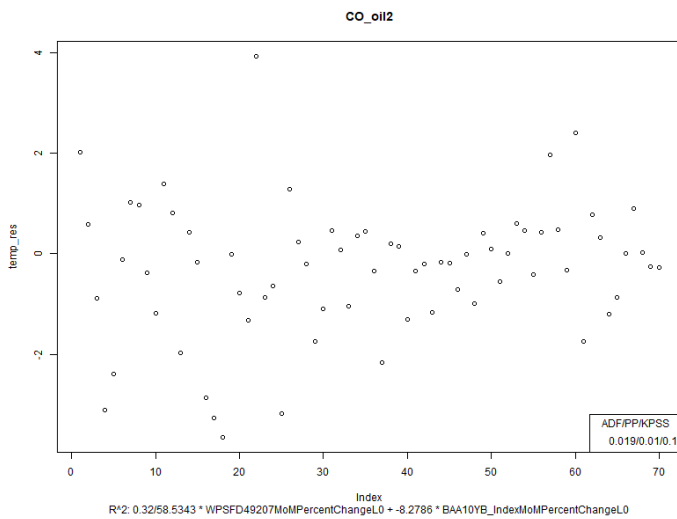
Out-of-Sample testing



Out-of-Sample testing



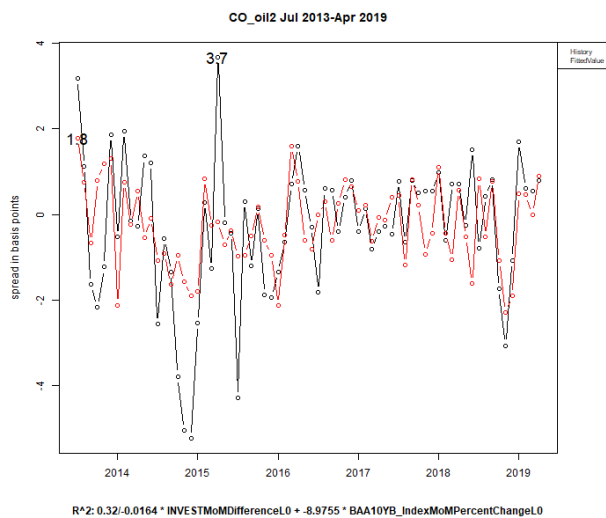
Stationarity of Residuals



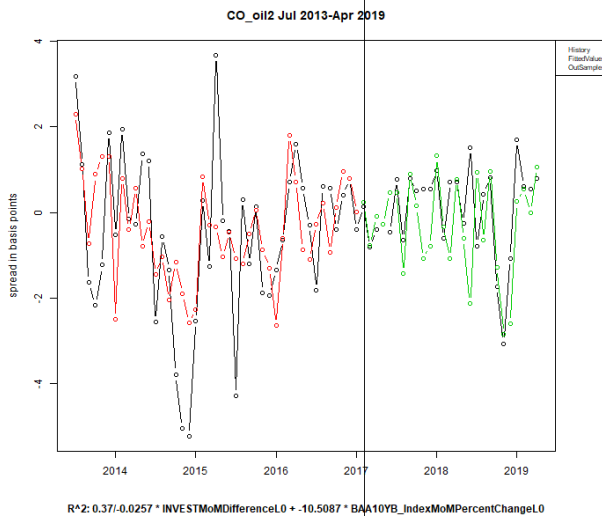
4.4.23 Model - 23

Backtesting on differenced values

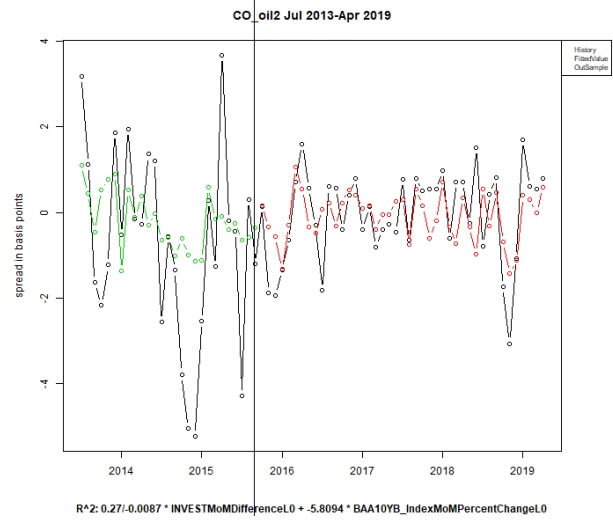
Backtesting on level values



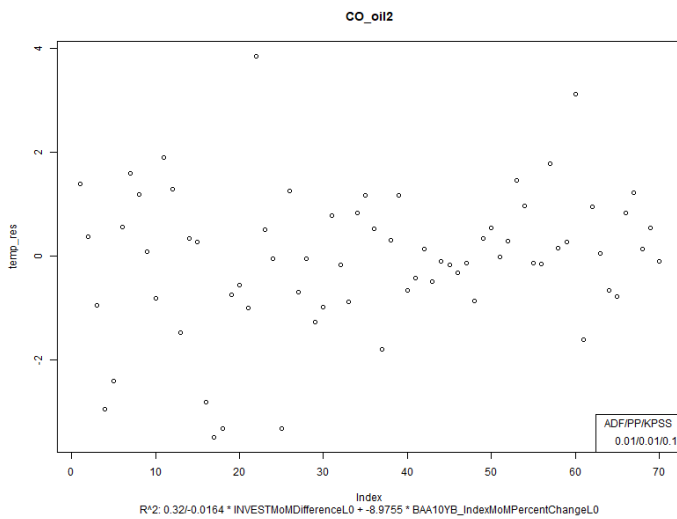
Out-of-Sample testing



Out-of-Sample testing



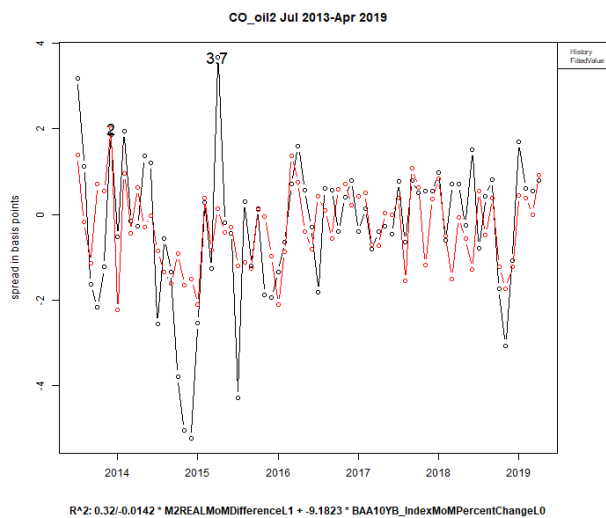
Stationarity of Residuals



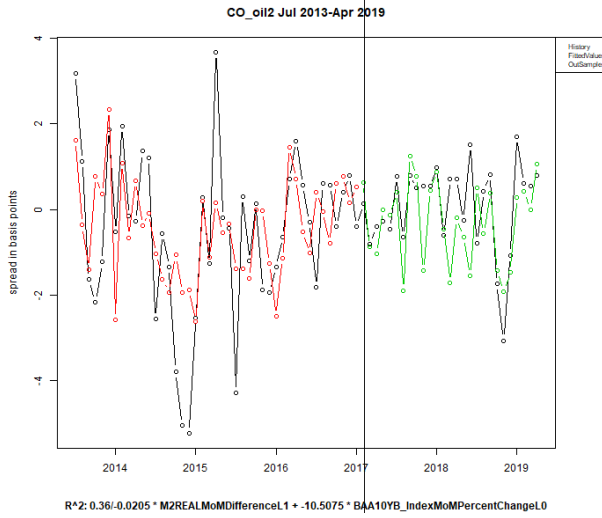
4.4.24 Model - 24

Backtesting on differenced values

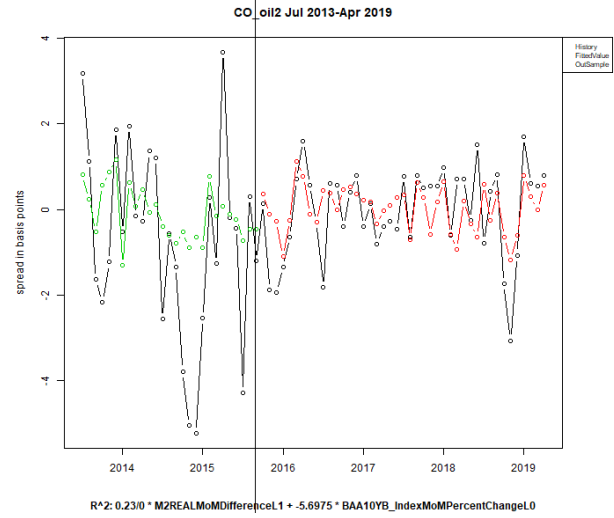
Backtesting on level values



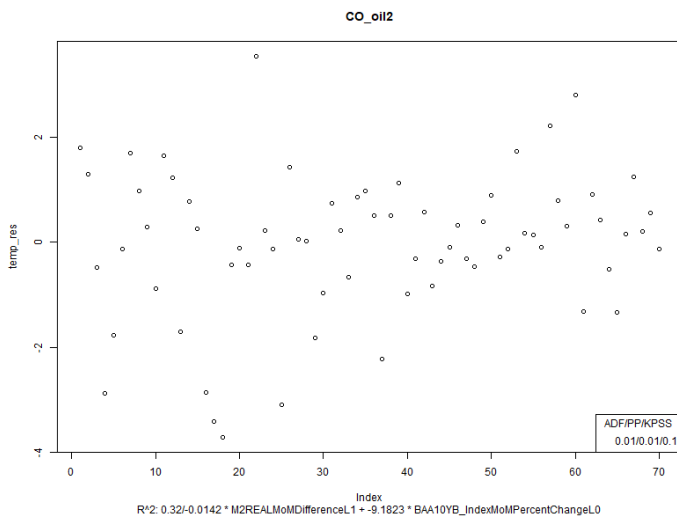
Out-of-Sample testing



Out-of-Sample testing



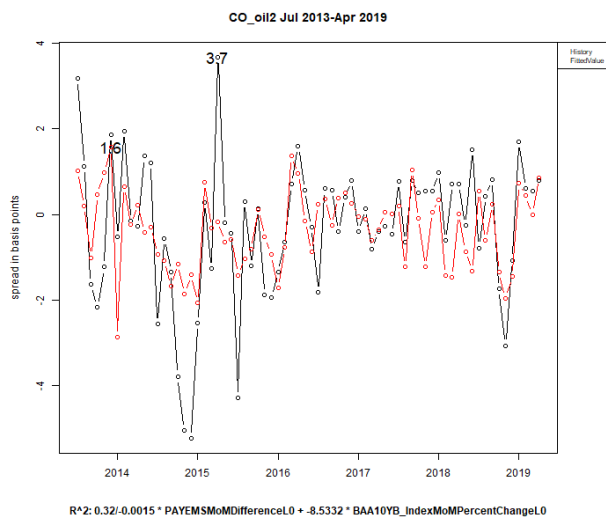
Stationarity of Residuals



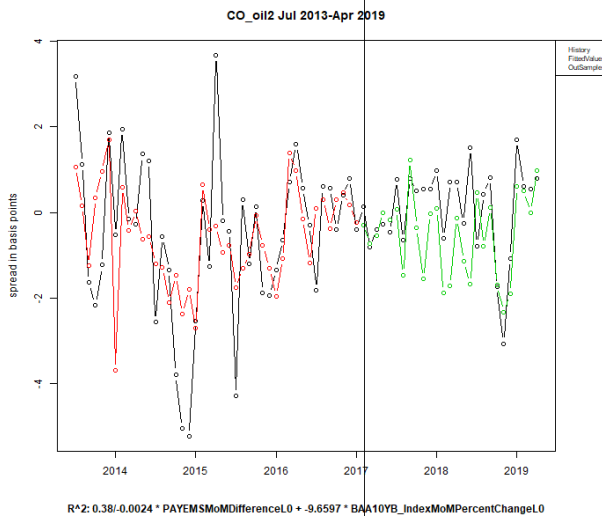
4.4.25 Model - 25

Backtesting on differenced values

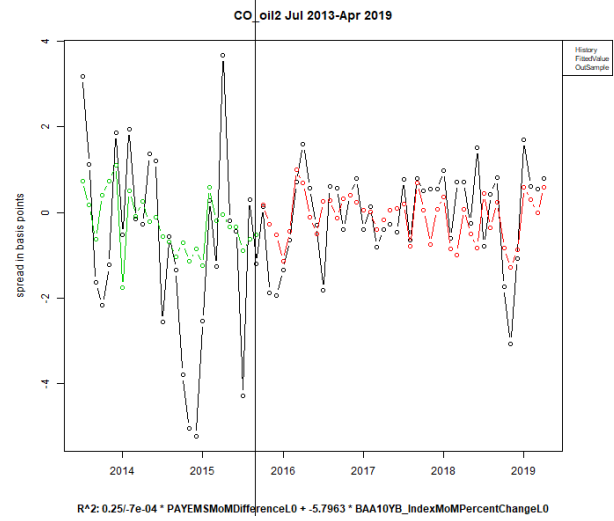
Backtesting on level values



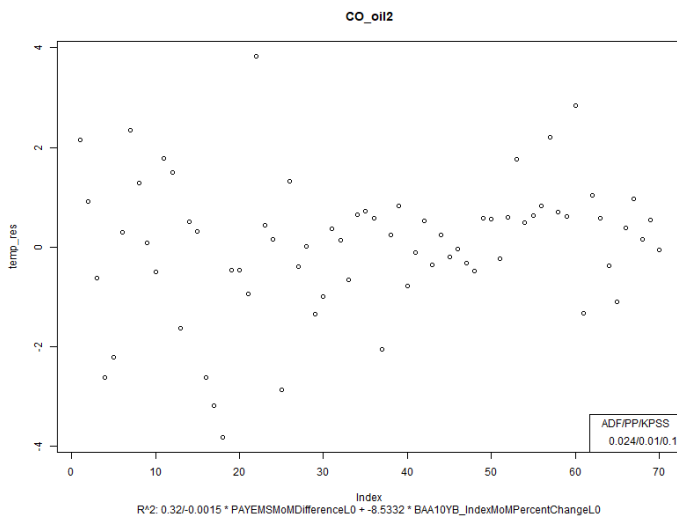
Out-of-Sample testing



Out-of-Sample testing



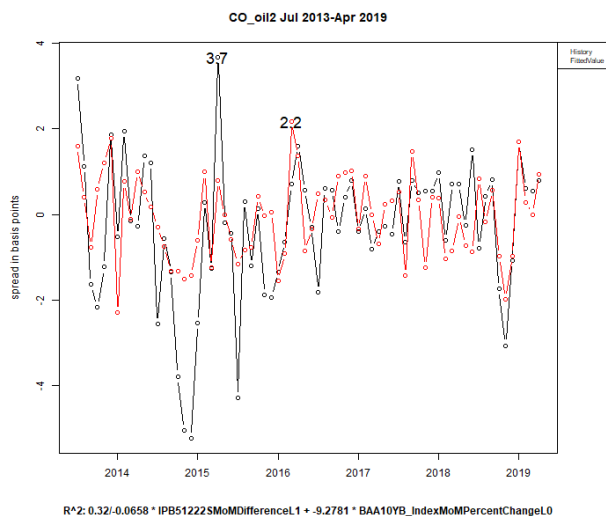
Stationarity of Residuals



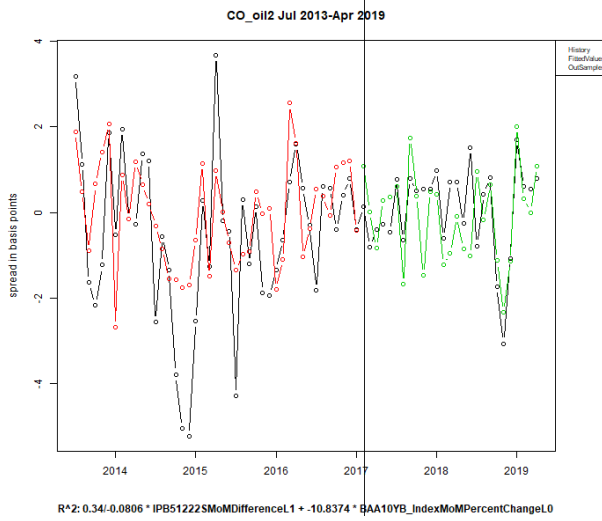
4.4.26 Model - 26

Backtesting on differenced values

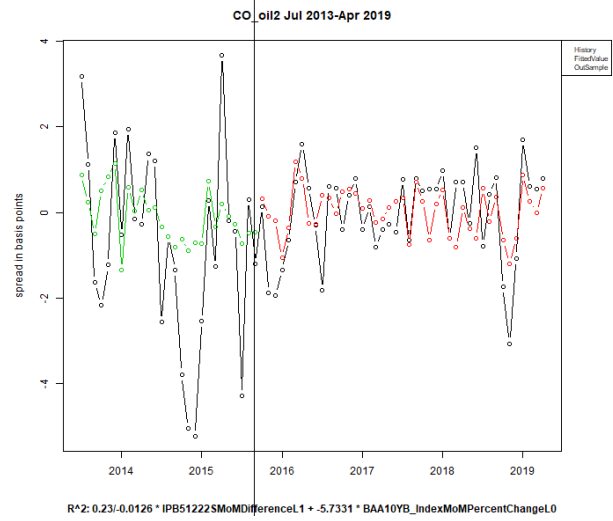
Backtesting on level values



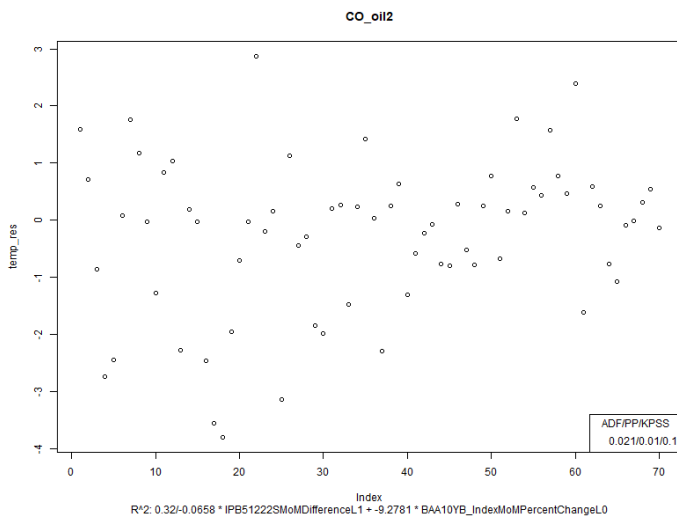
Out-of-Sample testing



Out-of-Sample testing



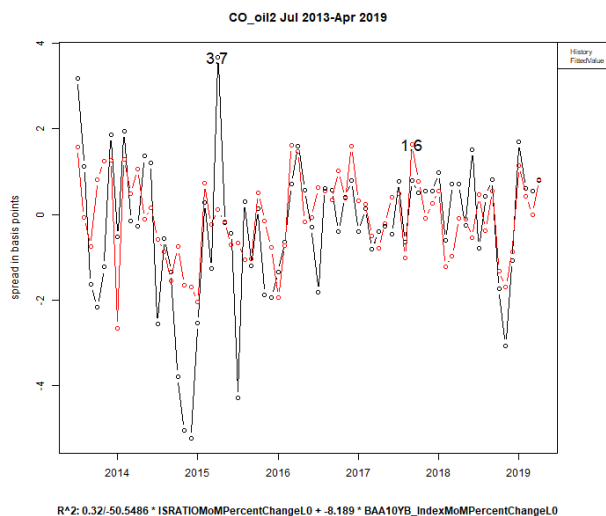
Stationarity of Residuals



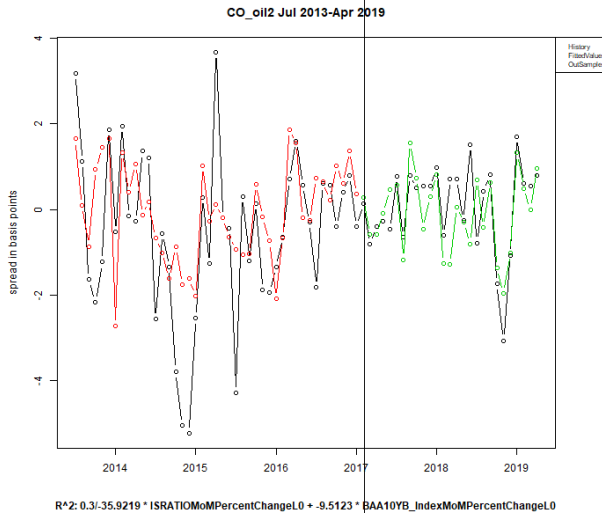
4.4.27 Model - 27

Backtesting on differenced values

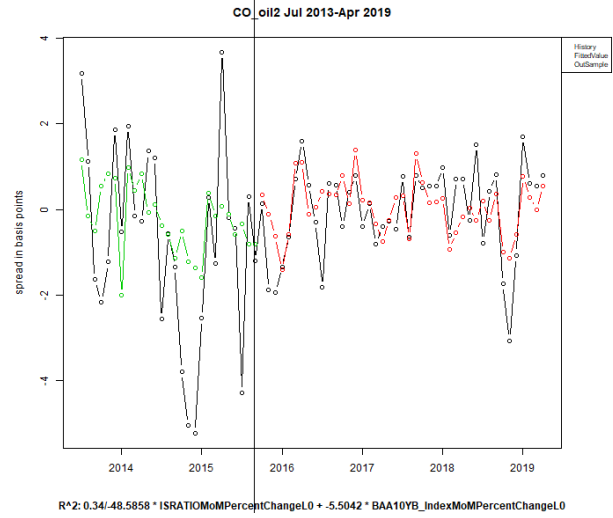
Backtesting on level values



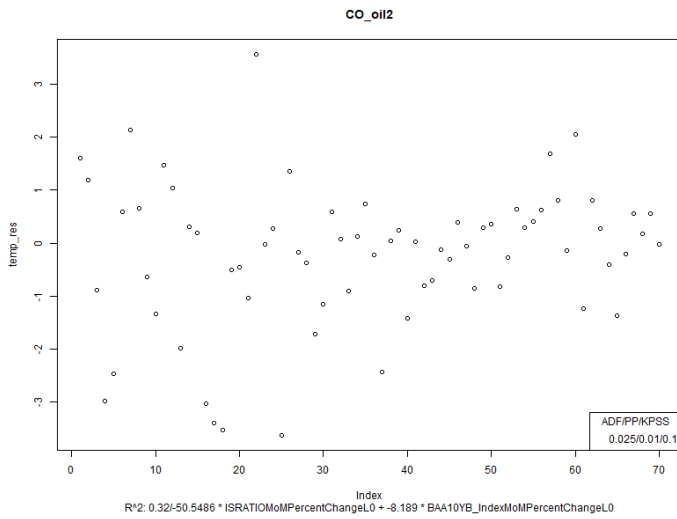
Out-of-Sample testing



Out-of-Sample testing



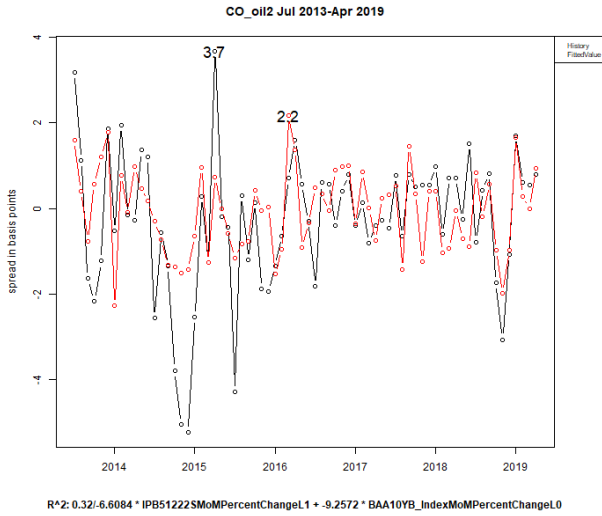
Stationarity of Residuals



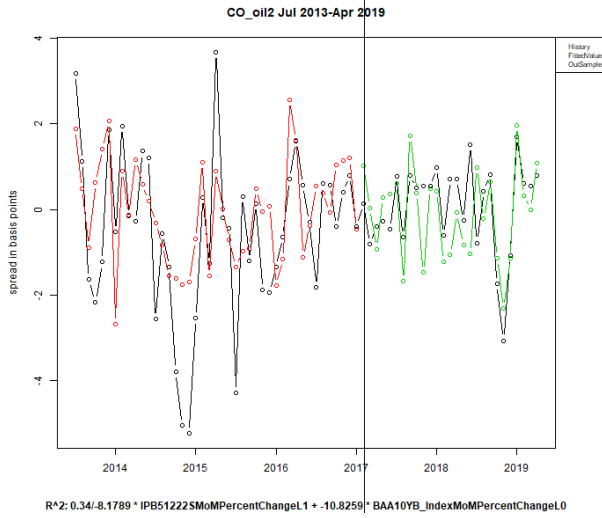
4.4.28 Model - 28

Backtesting on differenced values

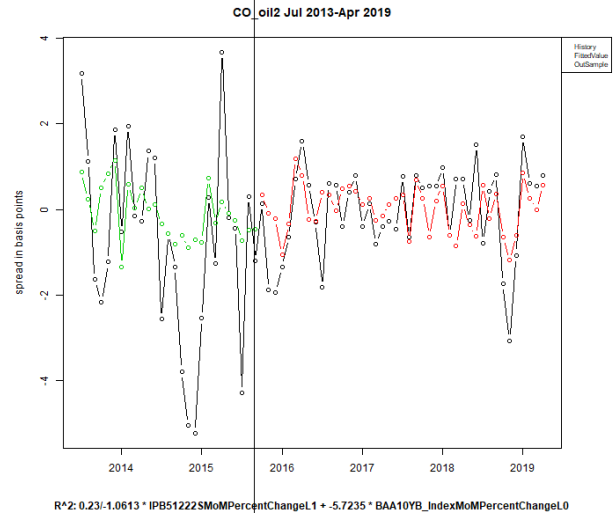
Backtesting on level values



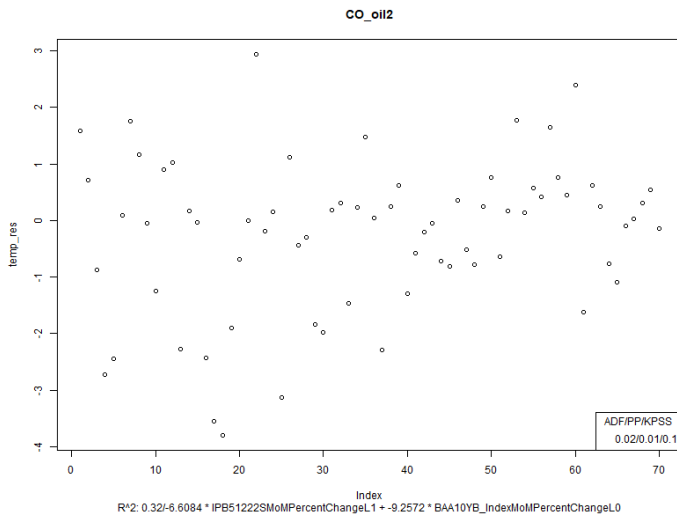
Out-of-Sample testing



Out-of-Sample testing



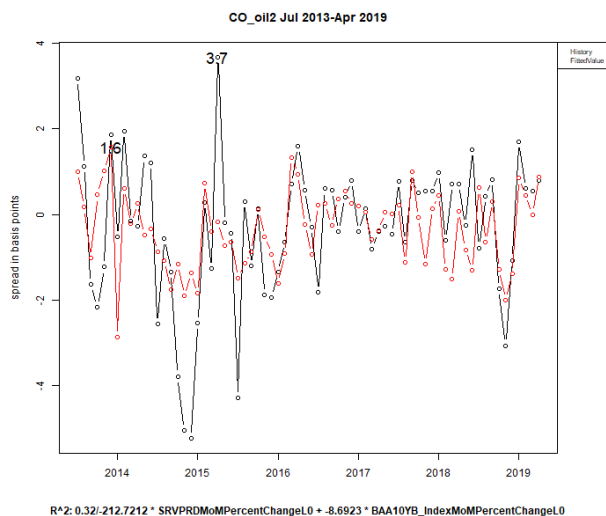
Stationarity of Residuals



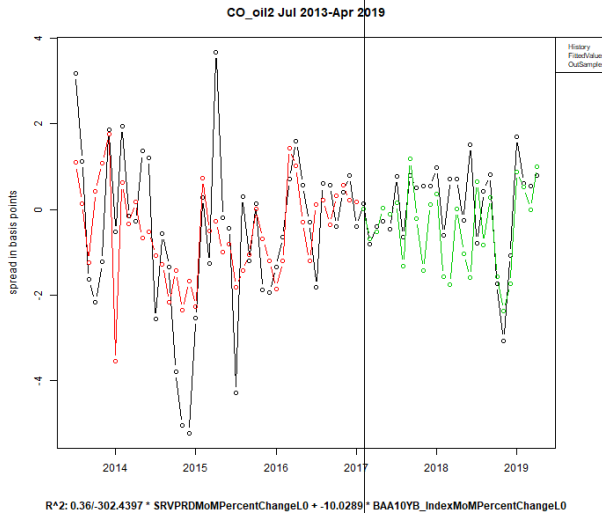
4.4.29 Model - 29

Backtesting on differenced values

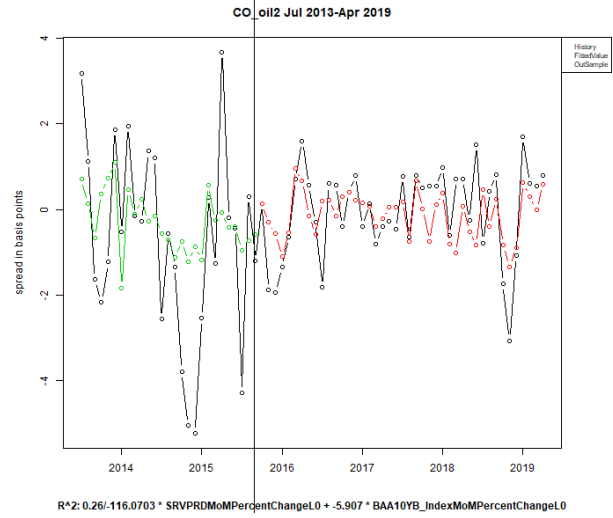
Backtesting on level values



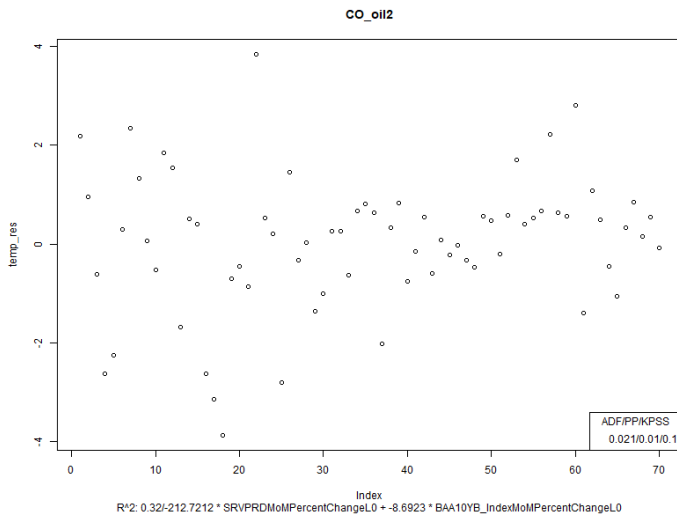
Out-of-Sample testing



Out-of-Sample testing



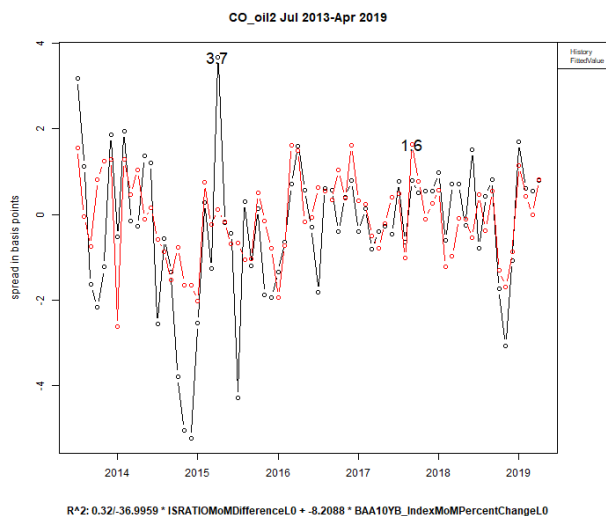
Stationarity of Residuals



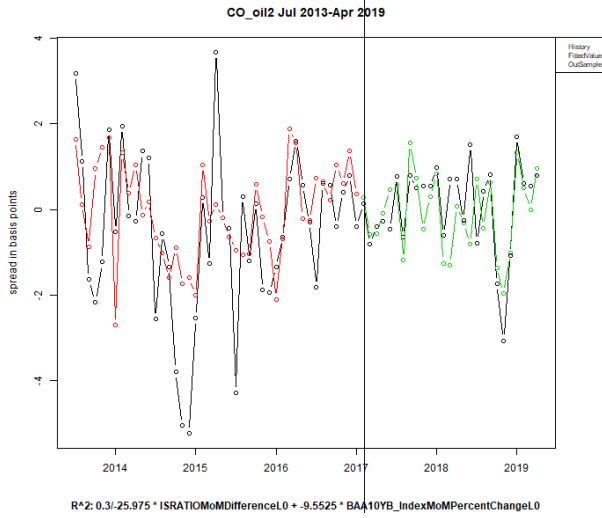
4.4.30 Model - 30

Backtesting on differenced values

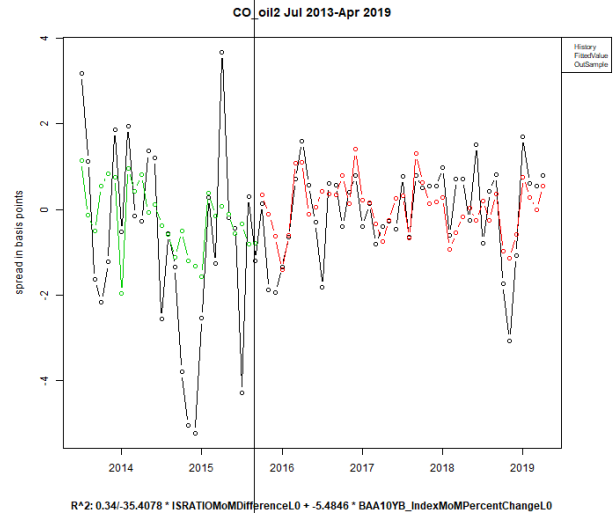
Backtesting on level values



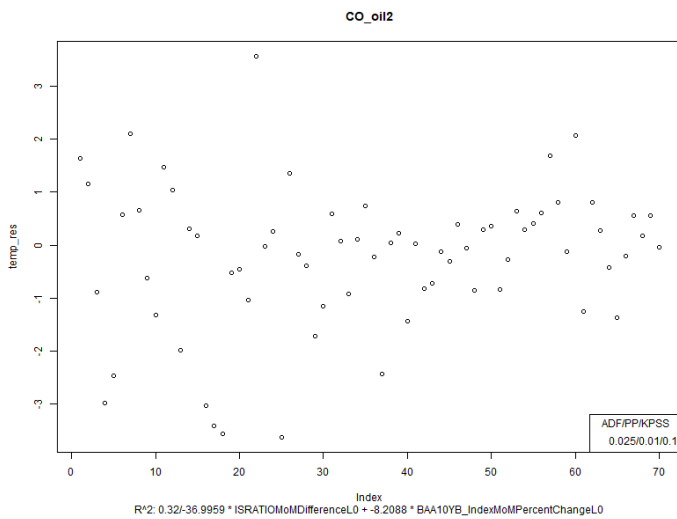
Out-of-Sample testing



Out-of-Sample testing



Stationarity of Residuals



4.5 Single Factor Model for CO_{natgas}

	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	IPB51222SMoMDifferenceL0	0.13	430.20	0.38	0.03	3.53			
2	IPB51222SMoMPercentChangeL0	0.12	430.79	37.79	0.03	3.51			
3	UEMP15T26MoMDifferenceL0	0.00	439.61	0.01	0.38	3.66			

Table 9: Single Factor Analysis- CO_{natgas}

4.5.1 Model - 1

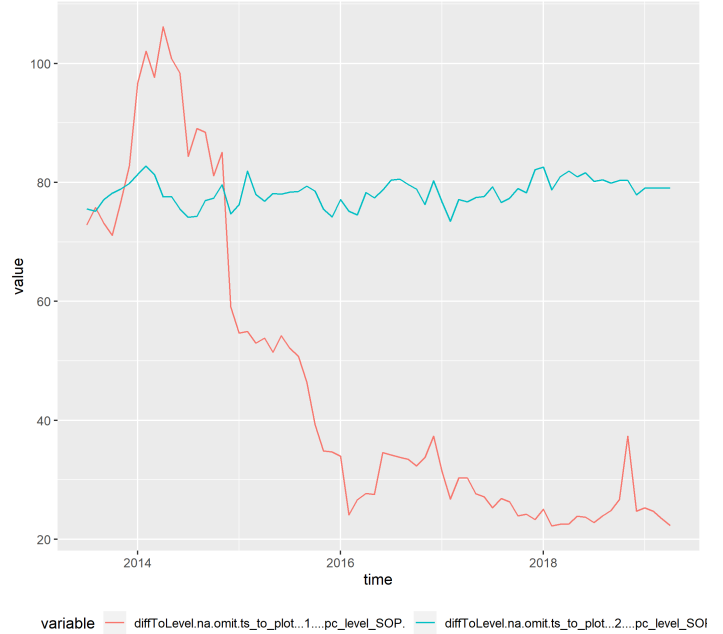
Backtesting on differenced values

CO_nat_gas - Backtesting
R²: 0.13/ 0.3824 * IPB51222SMoMDifferenceL0



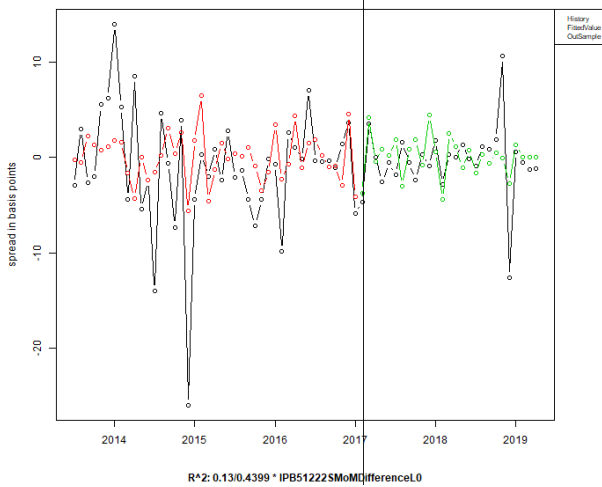
Backtesting on level values

CO_nat_gas - Level Value Backtesting
R²: 0.13/ 0.3824 * IPB51222SMoMDifferenceL0



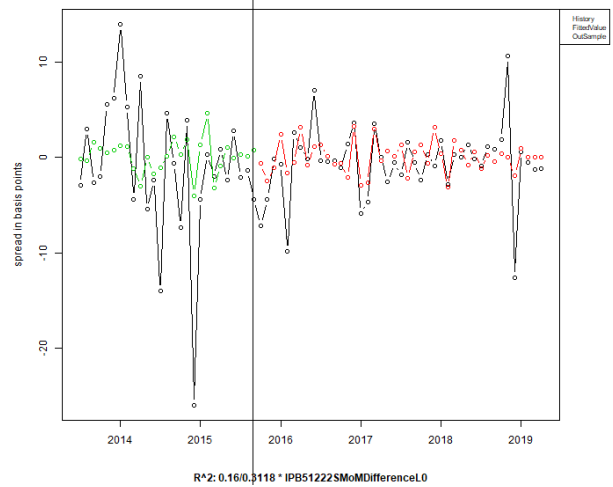
Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019



Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019

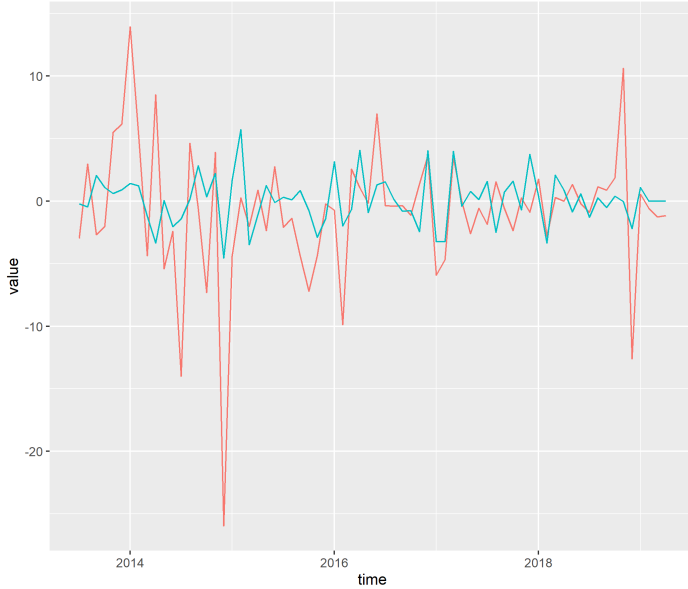


4.5.2 Model - 2

Backtesting on differenced values

CO_nat_gas - Backtesting

R²: 0.12/ 37.7925 * IPB51222SMoMPercentChangeL0

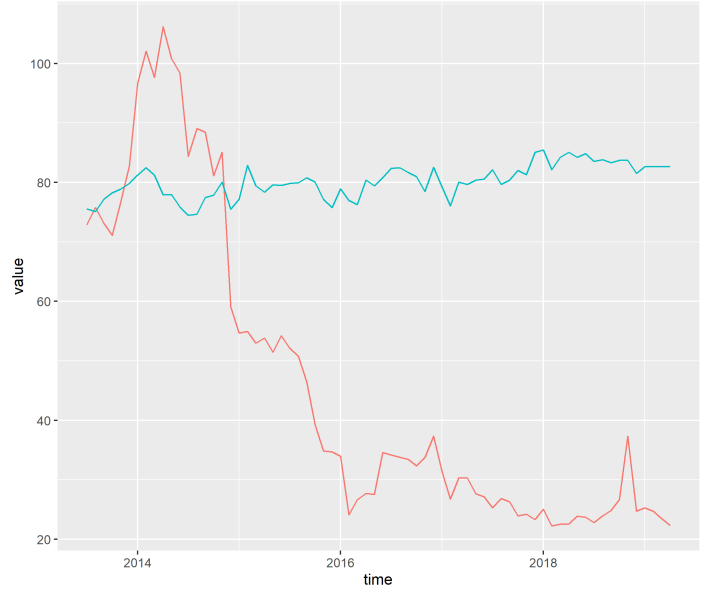


variable — CO_nat_gas_History — CO_nat_gas_FittedValue

Backtesting on level values

CO_nat_gas - Level Value Backtesting

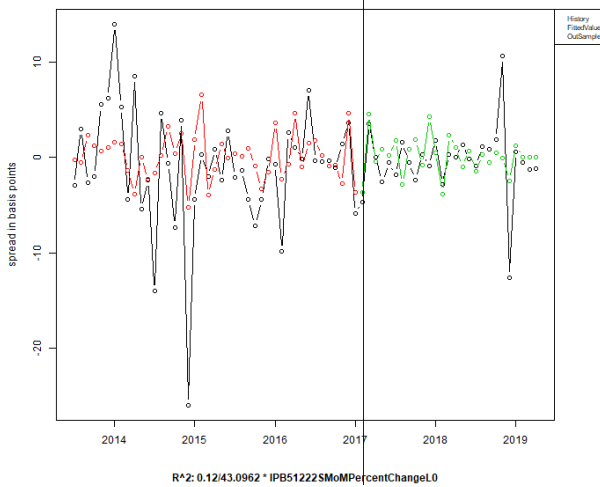
R²: 0.12/ 37.7925 * IPB51222SMoMPercentChangeL0



variable — diffToLevel.na.omit.ts_to_plot...1....pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2....pc_level_SOI

Out-of-Sample testing

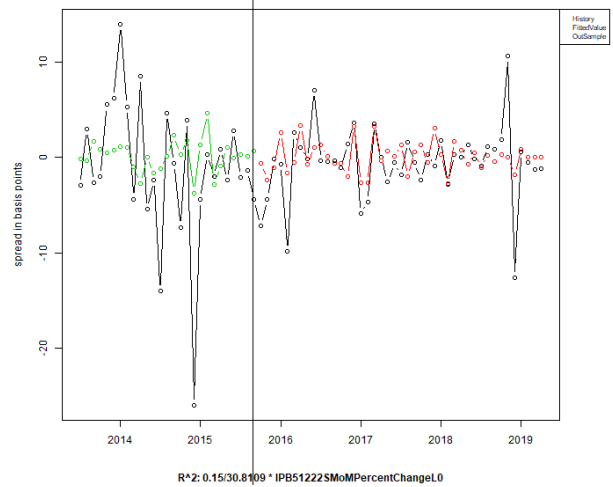
CO_nat_gas Jul 2013-Apr 2019



R²: 0.12/43.0962 * IPB51222SMoMPercentChangeL0

Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019

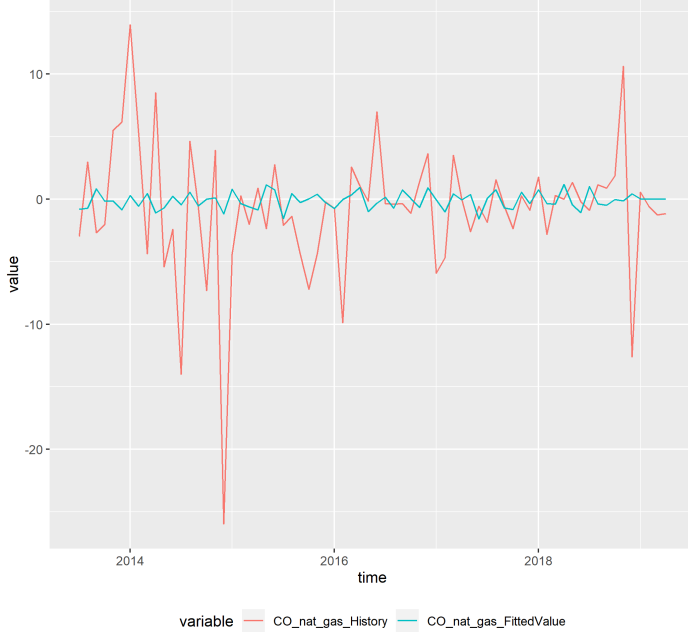


R²: 0.15/30.8109 * IPB51222SMoMPercentChangeL0

4.5.3 Model - 3

Backtesting on differenced values

CO_nat_gas - Backtesting
R²: 0/ 0.0075 * UEMP15T26MoMDifferenceL0



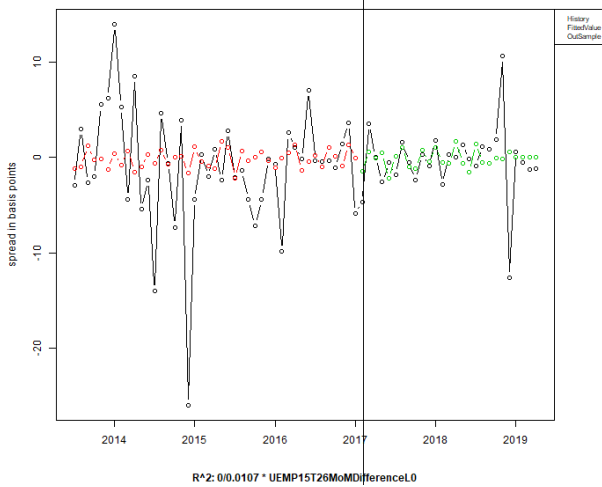
Backtesting on level values

CO_nat_gas - Level Value Backtesting
R²: 0/ 0.0075 * UEMP15T26MoMDifferenceL0



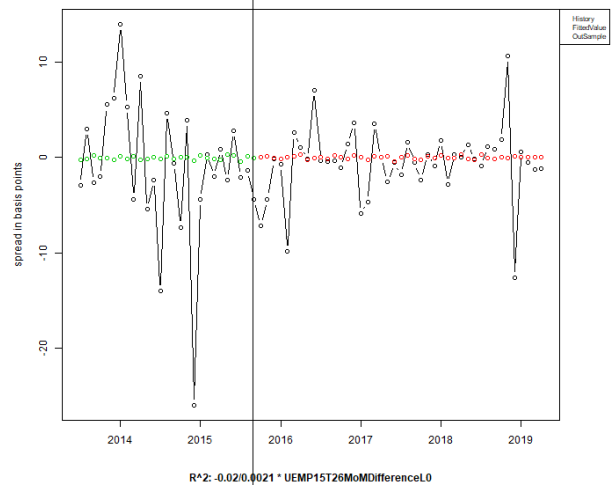
Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019



Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019



4.6 Two Factor Model for CO_{natgas}

	V1	V2	r2adj	Var1	Var2	aic	RMSE	Criteria1	Criteria2	Criteria3
1	IPB51222SMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL1	0.23	0.42	-18.49	422.28	3.76			
2	IPB51222SMoMDifferenceL0	WPSID61MoMDifferenceL0	0.23	0.38	1.31	422.48	4.16			
3	IPB51222SMoMDifferenceL0	CLF16OVMoMPercentChangeL0	0.23	0.35	-263.03	422.59	3.15			
4	IPB51222SMoMDifferenceL0	CLF16OVMoMDifferenceL0	0.23	0.35	-0.00	422.59	3.20			
5	INDPROMoMDifferenceL0	IPB51222SMoMDifferenceL0	0.23	-1.61	0.41	422.61	2.51			
6	INDPROMoMPercentChangeL0	IPB51222SMoMDifferenceL0	0.23	-170.11	0.41	422.65	2.52			
7	IPB51222SMoMDifferenceL0	WPSID61MoMPercentChangeL0	0.23	0.38	251.59	422.69	4.11			
8	IPB51222SMoMDifferenceL0	M2REALMoMPercentChangeL0	0.22	0.42	-370.44	422.81	3.20			
9	IPB51222SMoMDifferenceL0	M2REALMoMDifferenceL0	0.22	0.42	-0.07	422.94	3.19			
10	IPB51222SMoMPercentChangeL0	WPSID61MoMDifferenceL0	0.22	38.17	1.32	423.03	4.18			
11	IPB51222SMoMPercentChangeL0	CLF16OVMoMPercentChangeL0	0.22	34.57	-265.37	423.07	2.97			
12	IPB51222SMoMPercentChangeL0	CLF16OVMoMDifferenceL0	0.22	34.59	-0.00	423.07	3.01			
13	IPB51222SMoMPercentChangeL0	M2REALMoMPercentChangeL0	0.22	42.27	-378.73	423.08	3.16			
14	INDPROMoMDifferenceL0	IPB51222SMoMPercentChangeL0	0.22	-1.63	40.63	423.09	2.45			
15	INDPROMoMPercentChangeL0	IPB51222SMoMPercentChangeL0	0.22	-171.91	40.67	423.12	2.46			
16	IPB51222SMoMPercentChangeL0	M2REALMoMDifferenceL0	0.22	42.32	-0.07	423.21	3.15			
17	IPB51222SMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL1	0.22	41.45	-18.25	423.23	3.75			
18	IPB51222SMoMPercentChangeL0	WPSID61MoMPercentChangeL0	0.22	38.19	253.12	423.24	4.12			
19	IPB51222SMoMDifferenceL0	INVESTMoMPercentChangeL0	0.22	0.39	-251.04	423.34	3.31			
20	IPB51222SMoMDifferenceL0	WPSID61MoMDifferenceL1	0.22	0.42	1.25	423.64	4.00			
21	IPB51222SMoMPercentChangeL0	INVESTMoMPercentChangeL0	0.21	38.99	-255.76	423.66	3.32			
22	IPB51222SMoMDifferenceL0	INVESTMoMDifferenceL0	0.21	0.39	-0.08	423.73	3.36			
23	IPB51222SMoMDifferenceL0	WPSID61MoMPercentChangeL1	0.21	0.42	238.73	423.84	3.99			
24	IPB51222SMoMDifferenceL0	USFIREMoMPercentChangeL1	0.21	0.39	-872.16	423.94	3.70			
25	IPB51222SMoMPercentChangeL0	INVESTMoMDifferenceL0	0.21	39.23	-0.08	424.08	3.37			
26	IPB51222SMoMPercentChangeL0	WPSID61MoMDifferenceL1	0.21	42.54	1.26	424.10	4.04			
27	IPB51222SMoMDifferenceL0	USFIREMoMDifferenceL1	0.21	0.39	-0.10	424.19	3.73			
28	IPB51222SMoMPercentChangeL0	USFIREMoMPercentChangeL1	0.21	38.90	-892.82	424.19	3.70			
29	IPB51222SMoMPercentChangeL0	WPSID61MoMPercentChangeL1	0.21	42.38	241.83	424.30	4.02			
30	IPB51222SMoMDifferenceL0	BAA10YB_IndexMoMDifferenceL1	0.21	0.40	-0.11	424.42	3.60			
31	IPB51222SMoMPercentChangeL0	USFIREMoMDifferenceL1	0.21	38.80	-0.11	424.45	3.72			
32	CLF16OVMoMDifferenceL0	WPSID61MoMDifferenceL0	0.21	-0.00	1.20	424.48	4.55			
33	CLF16OVMoMPercentChangeL0	WPSID61MoMDifferenceL0	0.20	-270.12	1.20	424.55	4.55			
34	CLF16OVMoMDifferenceL0	WPSID61MoMPercentChangeL0	0.20	-0.00	230.21	424.69	4.45			
35	CLF16OVMoMPercentChangeL0	WPSID61MoMPercentChangeL0	0.20	-270.24	229.05	424.76	4.46			
36	IPB51222SMoMDifferenceL0	WPSFD49502MoMPercentChangeL0	0.20	0.40	192.72	424.77	4.11			
37	IPB51222SMoMDifferenceL0	WPSFD49502MoMDifferenceL0	0.20	0.40	0.92	424.78	4.18			
38	IPMATMoMPercentChangeL0	IPB51222SMoMDifferenceL0	0.20	-117.85	0.38	424.84	2.99			
39	IPMATMoMDifferenceL0	IPB51222SMoMDifferenceL0	0.20	-1.07	0.38	424.85	3.00			
40	IPB51222SMoMDifferenceL0	WPSFD49207MoMPercentChangeL0	0.20	0.40	256.79	424.88	4.06			
41	IPDMATMoMPercentChangeL0	WPSID61MoMDifferenceL0	0.20	-126.29	1.42	424.92	3.56			
42	IPB51222SMoMDifferenceL0	WPSFD49207MoMDifferenceL0	0.20	0.40	1.29	424.96	4.11			

Table 10: Two Factor Analysis- CO_{natgas}

Table 11: Relevant Variables: CO_{natgas}

	1	2	3
1	RPIMoMDifferenceL0	UEMP15T26MoMPercentChangeL0	BUSLOANSMoMDifferenceL0
2	RPIMoMDifferenceL1	UEMP15T26MoMPercentChangeL1	BUSLOANSMoMDifferenceL1

3 RPIMoMPercentChangeL0
4 RPIMoMPercentChangeL1
5 W875RX1MoMDifferenceL0
6 W875RX1MoMDifferenceL1
7 W875RX1MoMPercentChangeL0
8 W875RX1MoMPercentChangeL1
9 DPCERA3M086SBEAMoMDifferenceL0
10 DPCERA3M086SBEAMoMDifferenceL1
11 DPCERA3M086SBEAMoMPercentChangeL0
12 DPCERA3M086SBEAMoMPercentChangeL1
13 CMRMTSPLMoMDifferenceL0
14 CMRMTSPLMoMDifferenceL1
15 CMRMTSPLMoMPercentChangeL0
16 CMRMTSPLMoMPercentChangeL1
17 INDPROMoMDifferenceL0
18 INDPROMoMDifferenceL1
19 INDPROMoMPercentChangeL0
20 INDPROMoMPercentChangeL1
21 IPFPNSSMoMDifferenceL0
22 IPFPNSSMoMDifferenceL1
23 IPFPNSSMoMPercentChangeL0
24 IPFPNSSMoMPercentChangeL1
25 IPFINALMoMDifferenceL0
26 IPFINALMoMDifferenceL1
27 IPFINALMoMPercentChangeL0
28 IPFINALMoMPercentChangeL1
29 IPCONGDMoMDifferenceL0
30 IPCONGDMoMDifferenceL1
31 IPCONGDMoMPercentChangeL0
32 IPCONGDMoMPercentChangeL1
33 IPDCONGDMoMDifferenceL0
34 IPDCONGDMoMDifferenceL1
35 IPDCONGDMoMPercentChangeL0
36 IPDCONGDMoMPercentChangeL1
37 IPNCONGDMoMDifferenceL0
38 IPNCONGDMoMDifferenceL1
39 IPNCONGDMoMPercentChangeL0
40 IPNCONGDMoMPercentChangeL1
41 IPBUSEQMoMDifferenceL0
42 IPBUSEQMoMDifferenceL1
43 IPBUSEQMoMPercentChangeL0
44 IPBUSEQMoMPercentChangeL1
45 IPMATMoMDifferenceL0
46 IPMATMoMDifferenceL1
47 IPMATMoMPercentChangeL0
48 IPMATMoMPercentChangeL1
49 IPDMATMoMDifferenceL0
50 IPDMATMoMDifferenceL1
51 IPDMATMoMPercentChangeL0
52 IPDMATMoMPercentChangeL1
53 IPNMATMoMDifferenceL0
54 IPNMATMoMDifferenceL1
55 IPNMATMoMPercentChangeL0
56 IPNMATMoMPercentChangeL1
57 IPMANSICSMoMDifferenceL0

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UEMP27OVMoMPercentChangeL1
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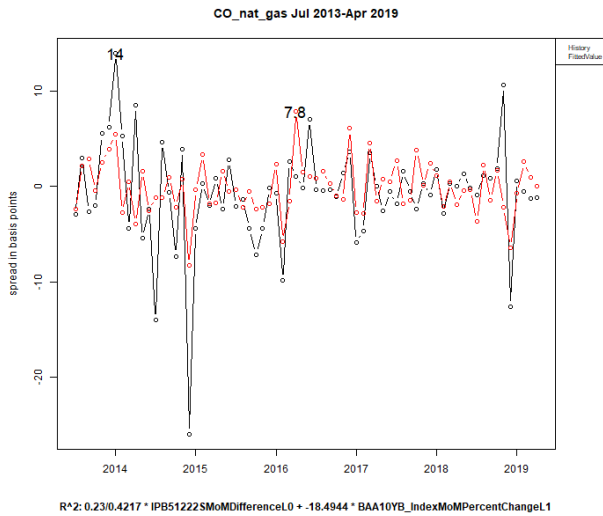
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USGG2YR_IndexMoMDifferenceL0

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68	IPFUELSMoMPercentChangeL1	HOUSTNEMoMDifferenceL1	USGG30YR_IndexMoMPercentChangeL1
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96	UEMP5TO14MoMPercentChangeL1	M2REALMoMDifferenceL1	TED_G_IndexMoMPercentChangeL1
97	UEMP15OVMoMDifferenceL0	M2REALMoMPercentChangeL0	BAA10YB_IndexMoMDifferenceL0
98	UEMP15OVMoMDifferenceL1	M2REALMoMPercentChangeL1	BAA10YB_IndexMoMDifferenceL1
99	UEMP15OVMoMPercentChangeL0	AMBSLMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0
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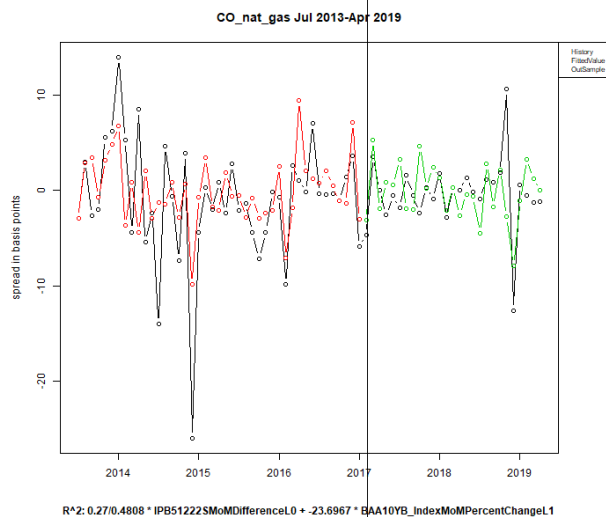
4.6.1 Model - 1

Backtesting on differenced values

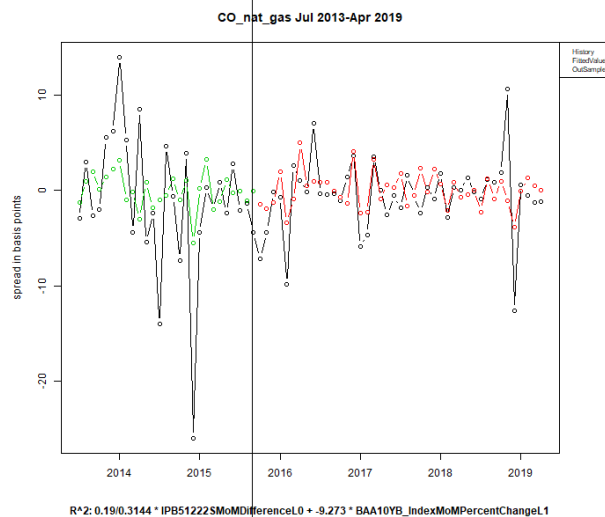
Backtesting on level values



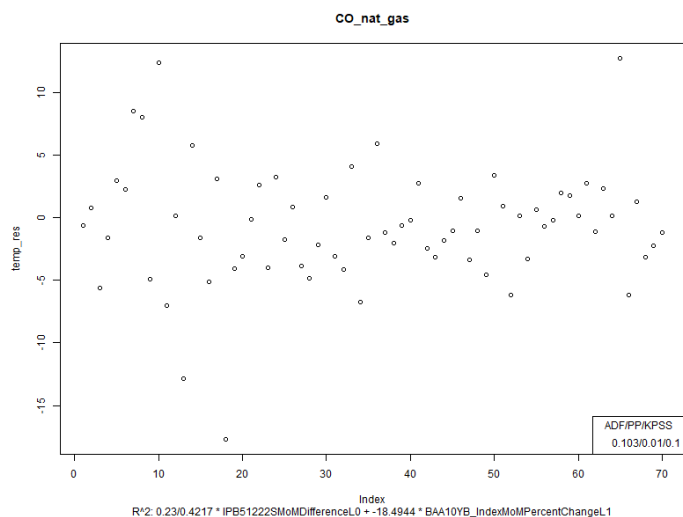
Out-of-Sample testing



Out-of-Sample testing



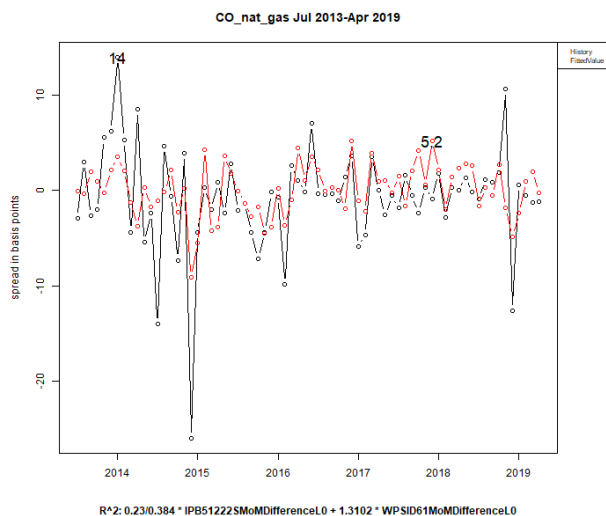
Stationarity of Residuals



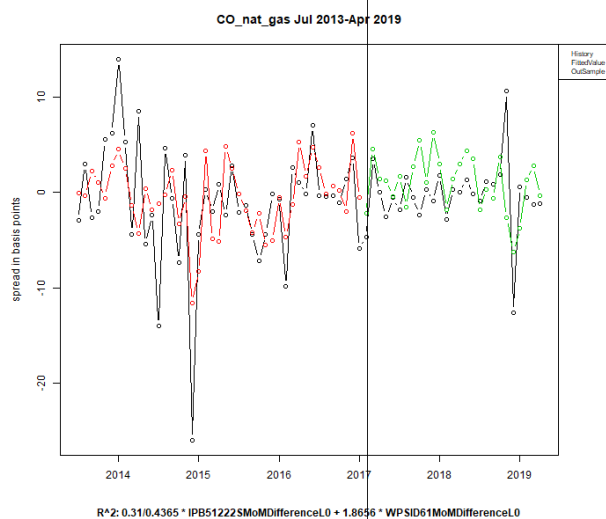
4.6.2 Model - 2

Backtesting on differenced values

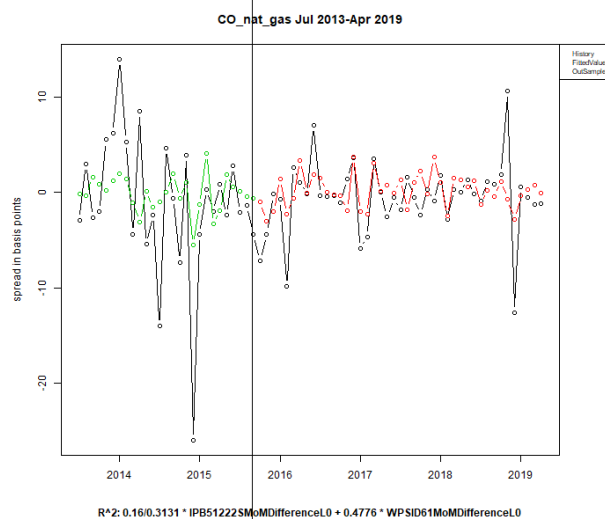
Backtesting on level values



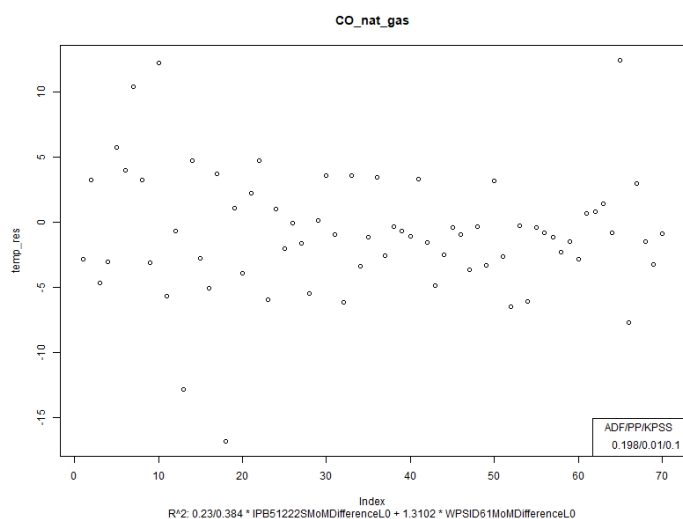
Out-of-Sample testing



Out-of-Sample testing



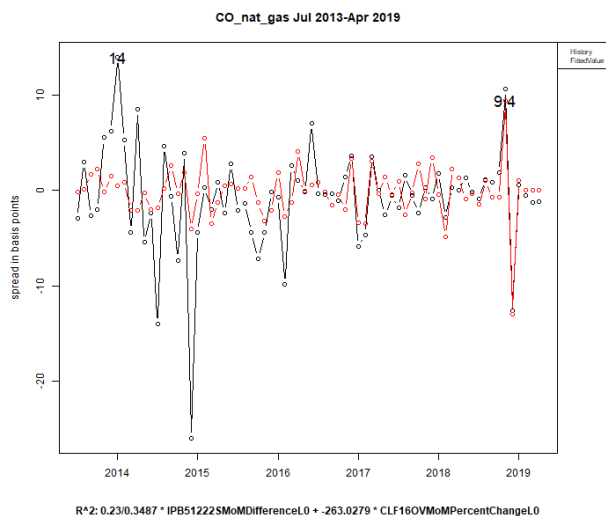
Stationarity of Residuals



4.6.3 Model - 3

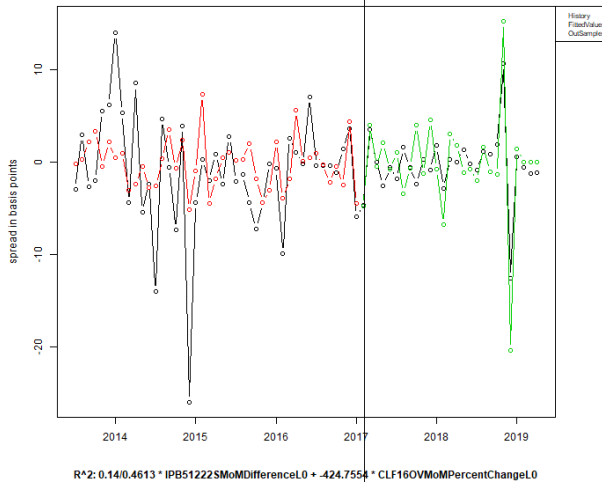
Backtesting on differenced values

Backtesting on level values



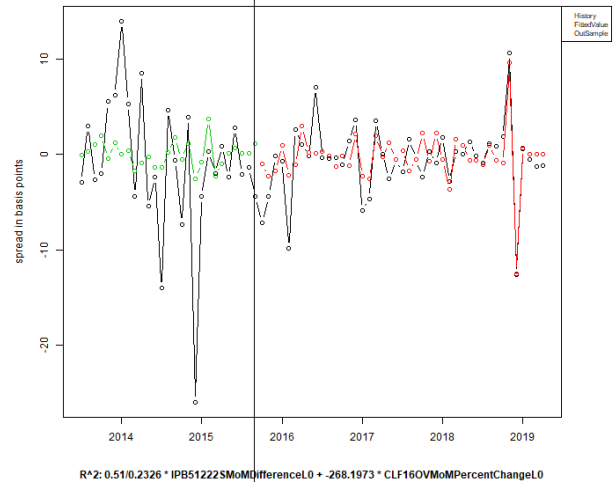
Out-of-Sample testing

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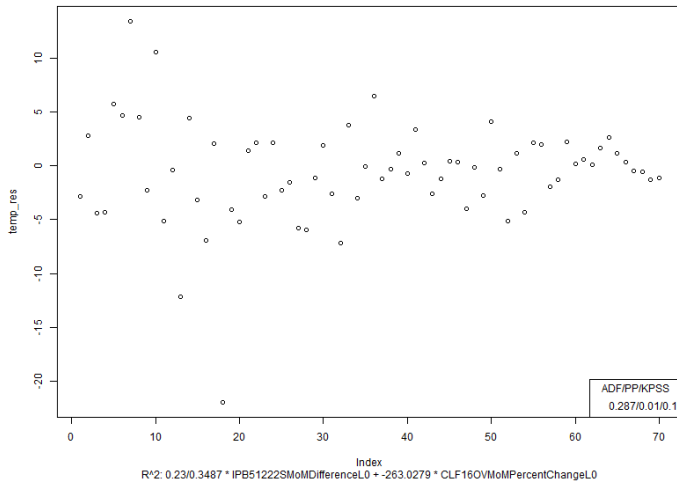
Out-of-Sample testing

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Stationarity of Residuals

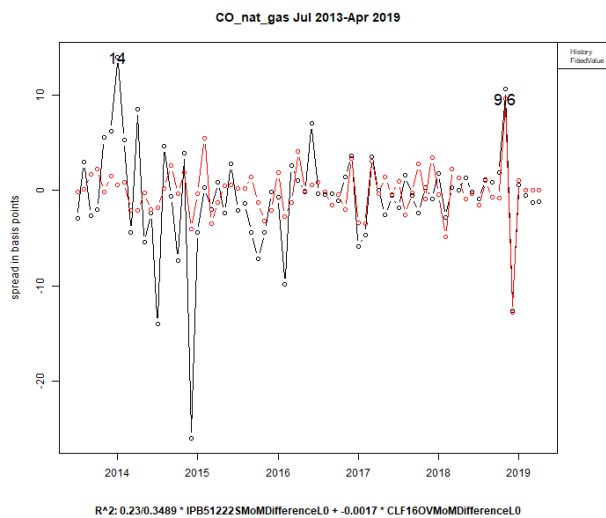
CO_nat_gas



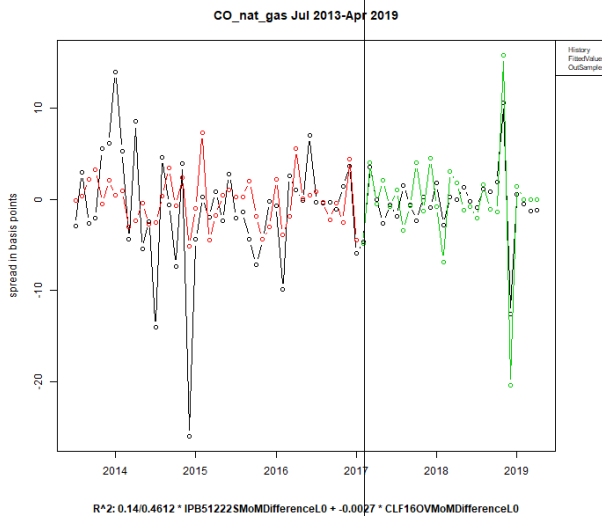
4.6.4 Model - 4

Backtesting on differenced values

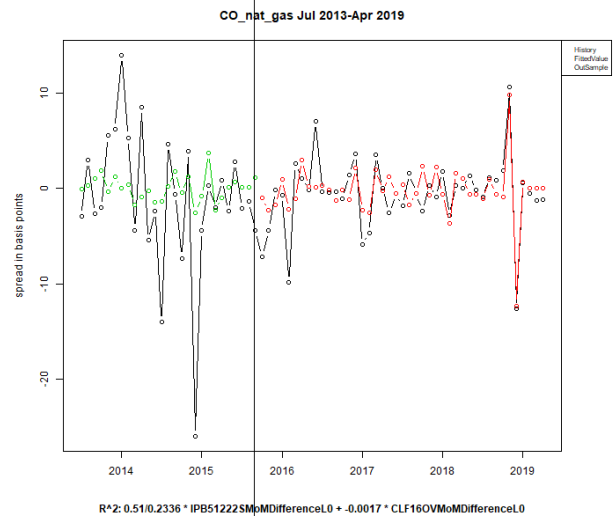
Backtesting on level values



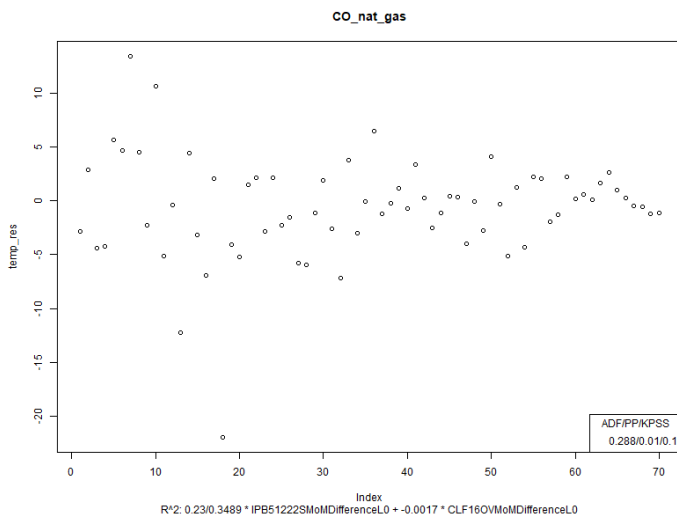
Out-of-Sample testing



Out-of-Sample testing



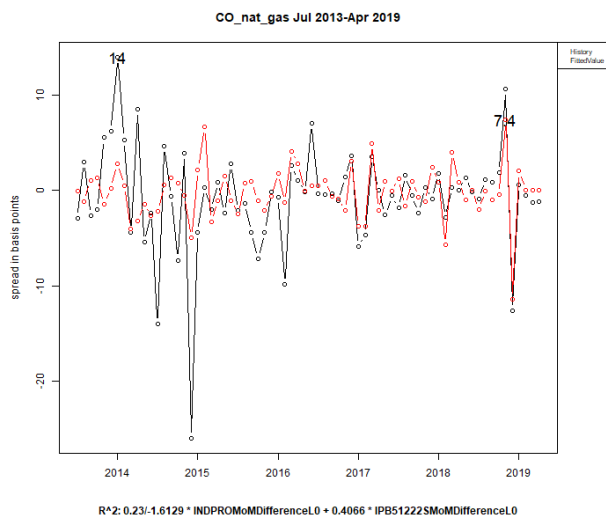
Stationarity of Residuals



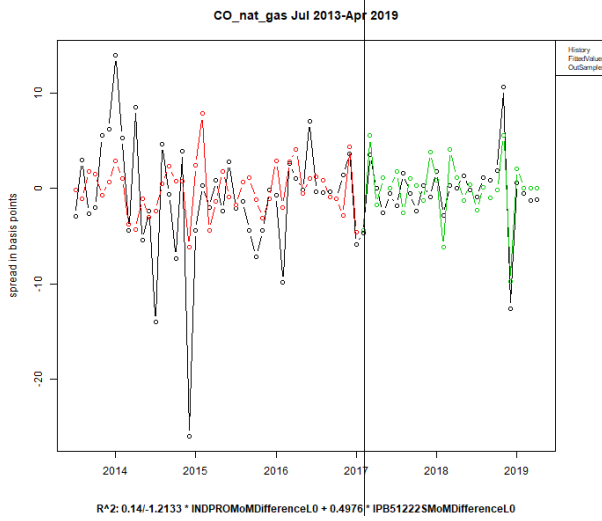
4.6.5 Model - 5

Backtesting on differenced values

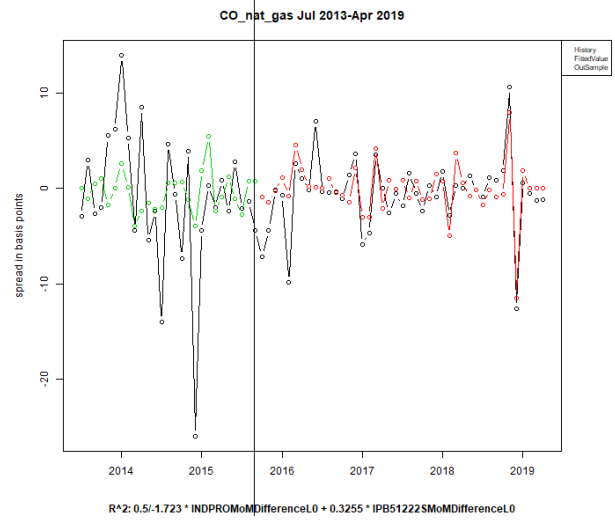
Backtesting on level values



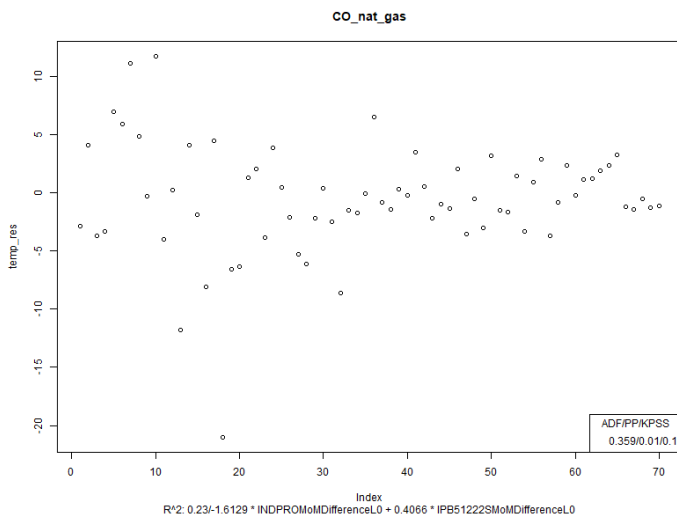
Out-of-Sample testing



Out-of-Sample testing



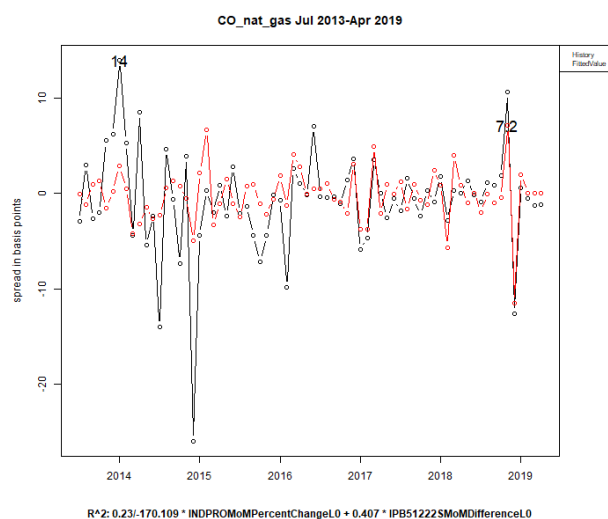
Stationarity of Residuals



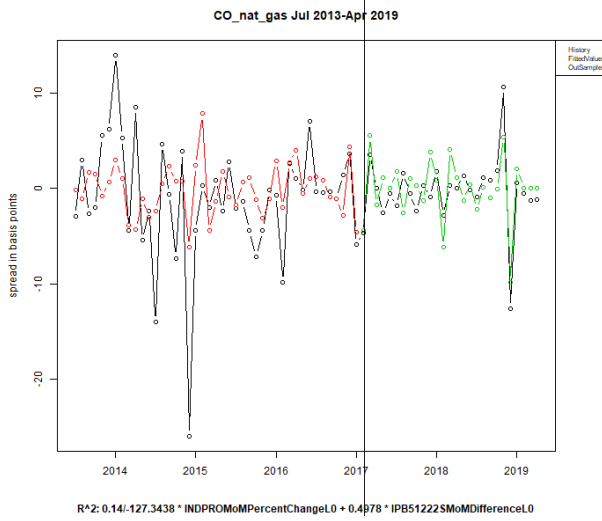
4.6.6 Model - 6

Backtesting on differenced values

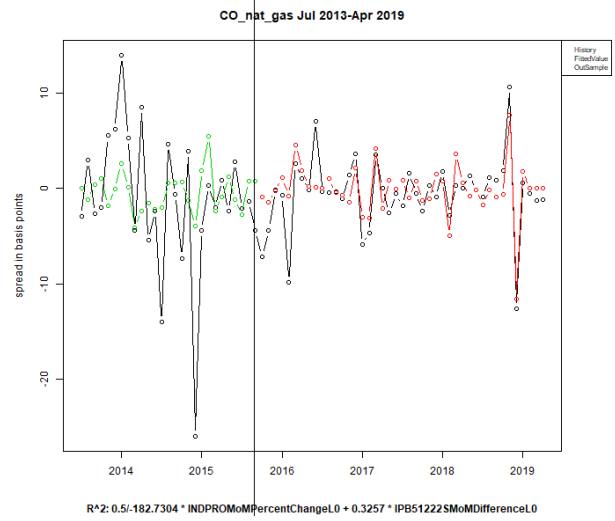
Backtesting on level values



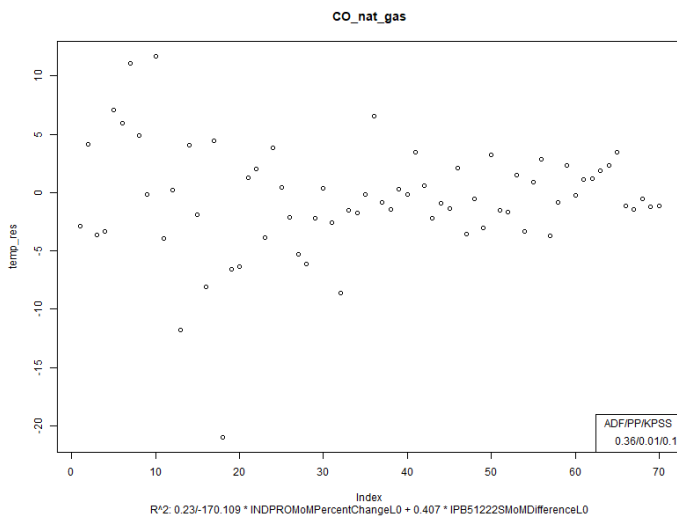
Out-of-Sample testing



Out-of-Sample testing



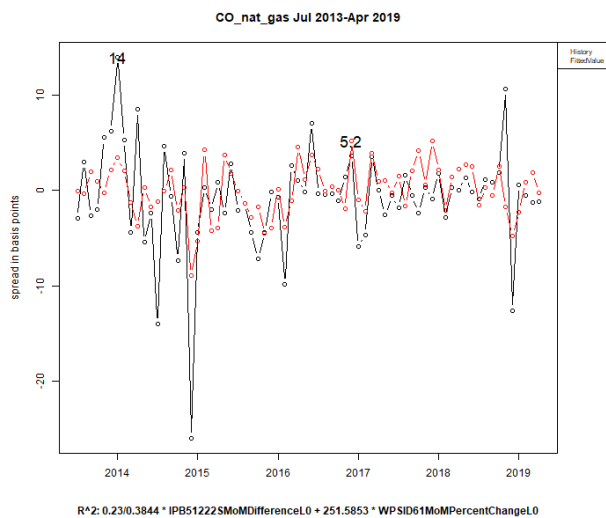
Stationarity of Residuals



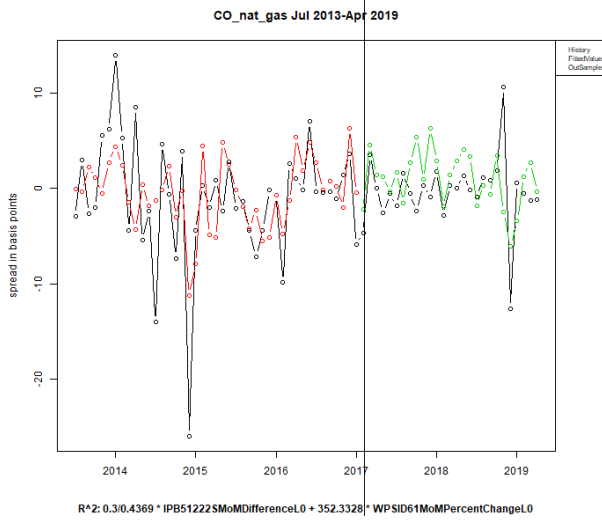
4.6.7 Model - 7

Backtesting on differenced values

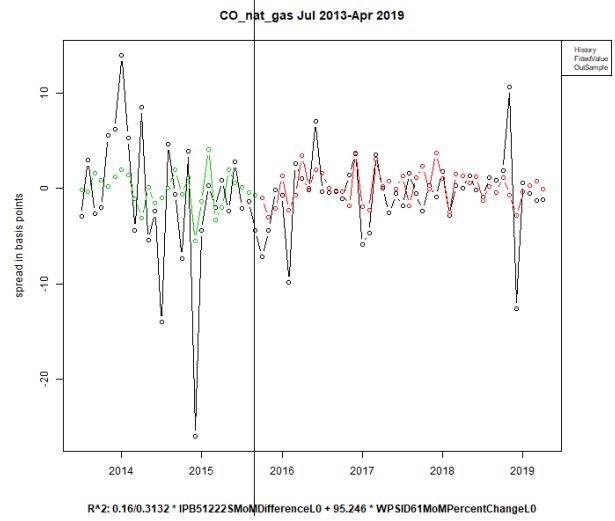
Backtesting on level values



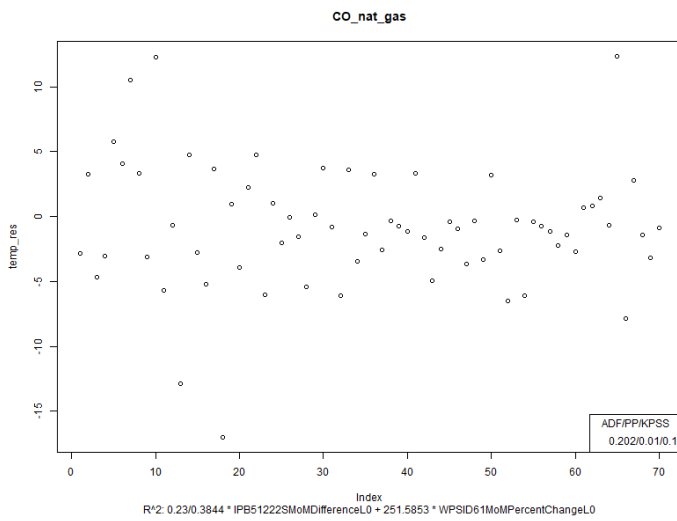
Out-of-Sample testing



Out-of-Sample testing



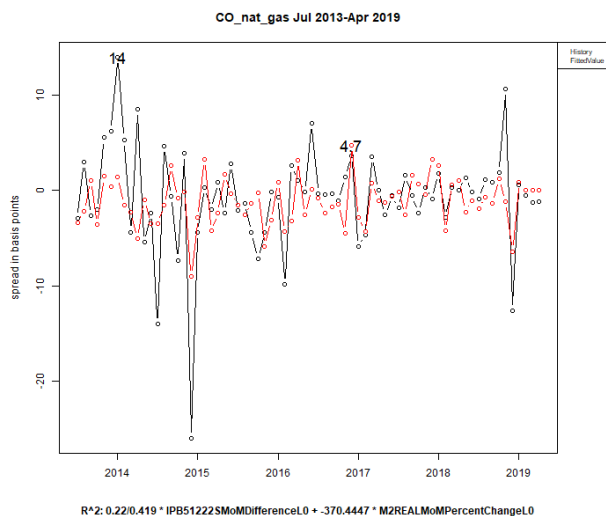
Stationarity of Residuals



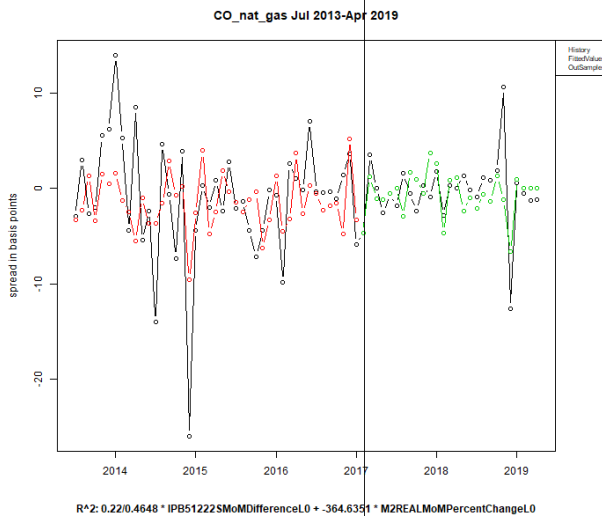
4.6.8 Model - 8

Backtesting on differenced values

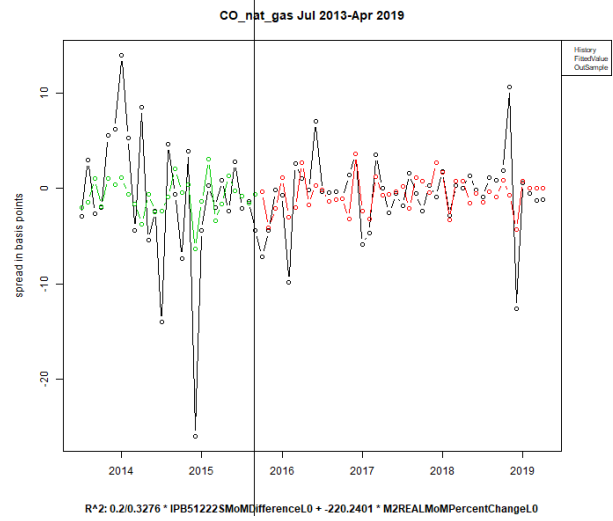
Backtesting on level values



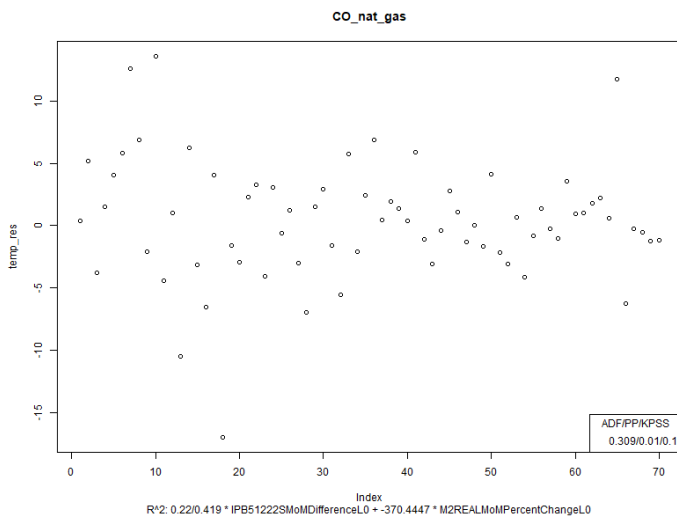
Out-of-Sample testing



Out-of-Sample testing



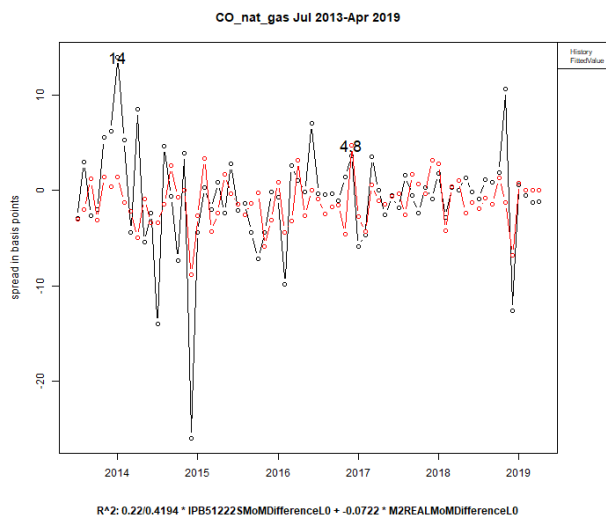
Stationarity of Residuals



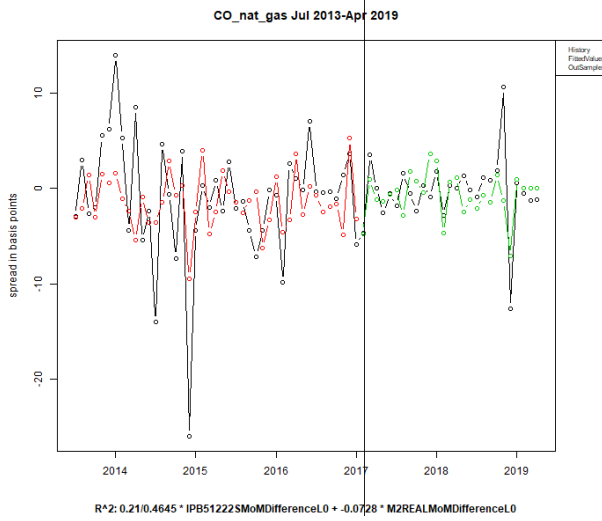
4.6.9 Model - 9

Backtesting on differenced values

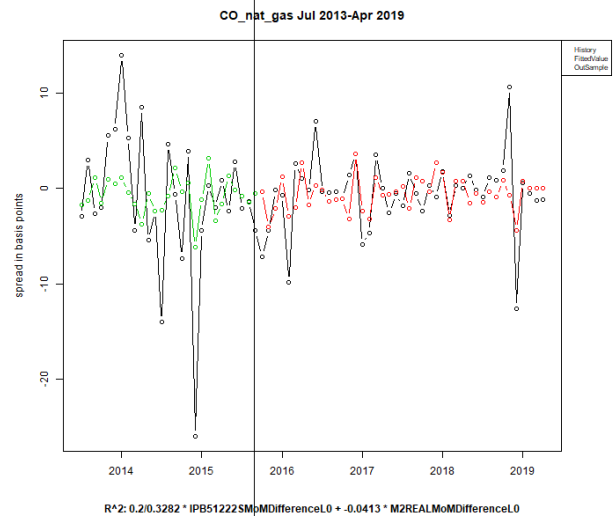
Backtesting on level values



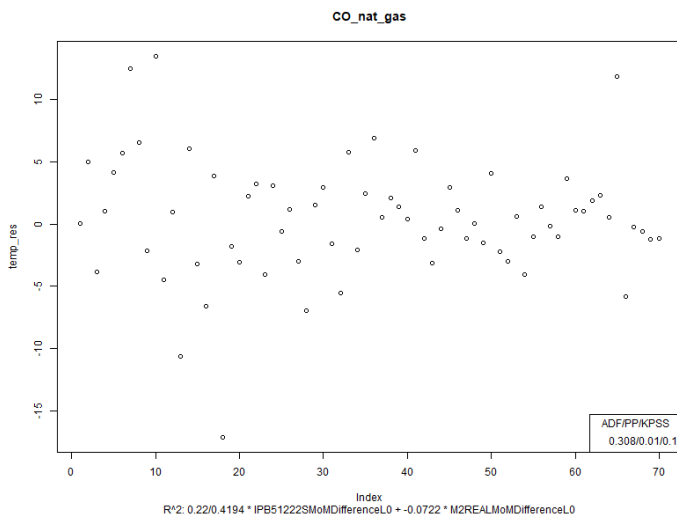
Out-of-Sample testing



Out-of-Sample testing



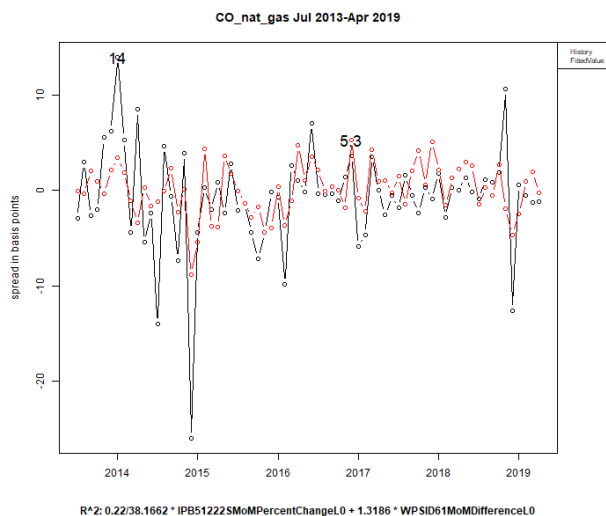
Stationarity of Residuals



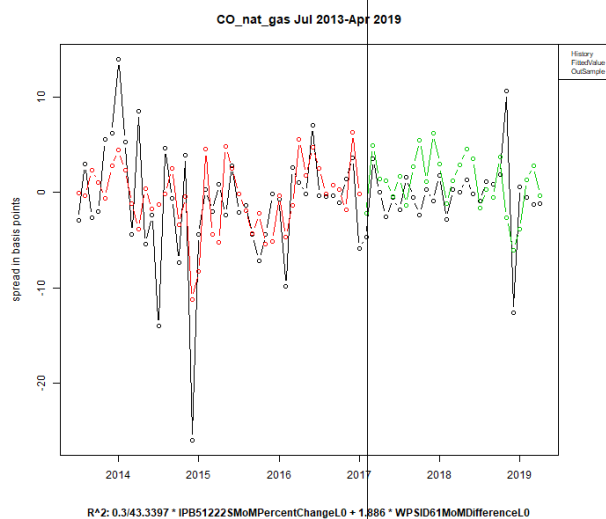
4.6.10 Model - 10

Backtesting on differenced values

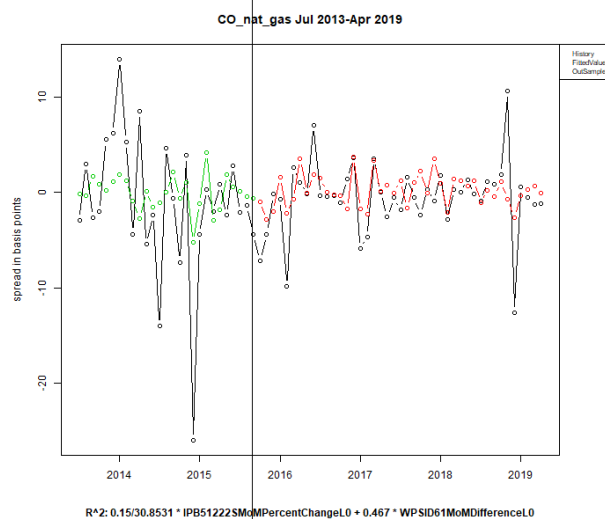
Backtesting on level values



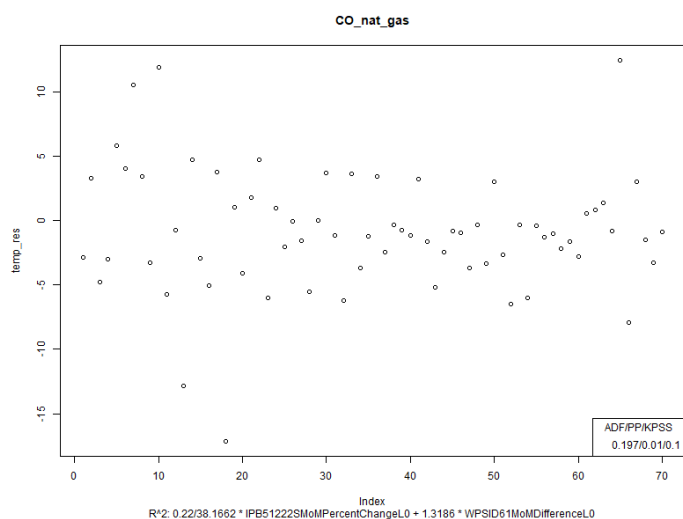
Out-of-Sample testing



Out-of-Sample testing



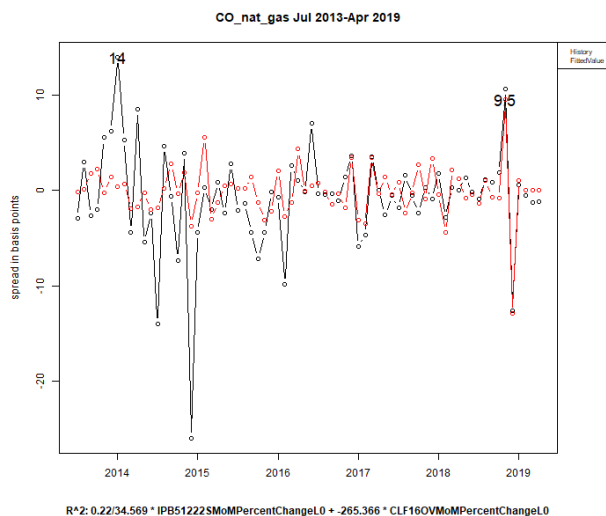
Stationarity of Residuals



4.6.11 Model - 11

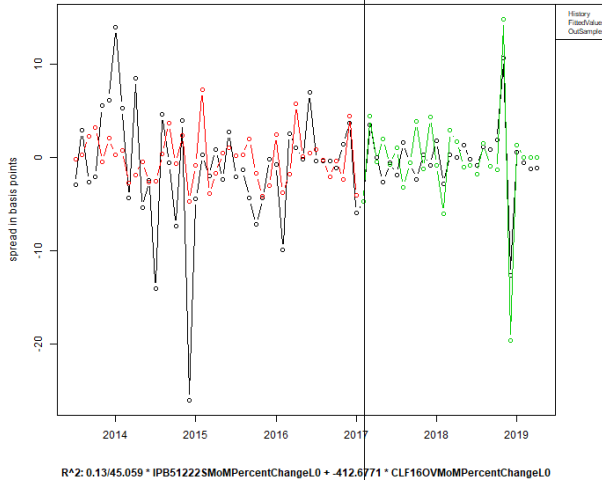
Backtesting on differenced values

Backtesting on level values



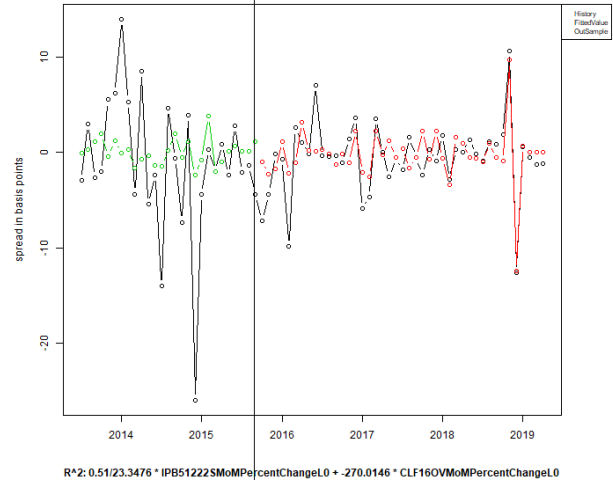
Out-of-Sample testing

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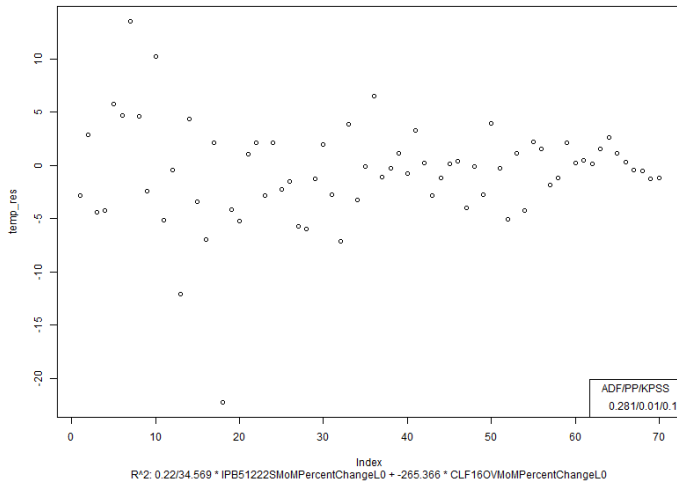
Out-of-Sample testing

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Stationarity of Residuals

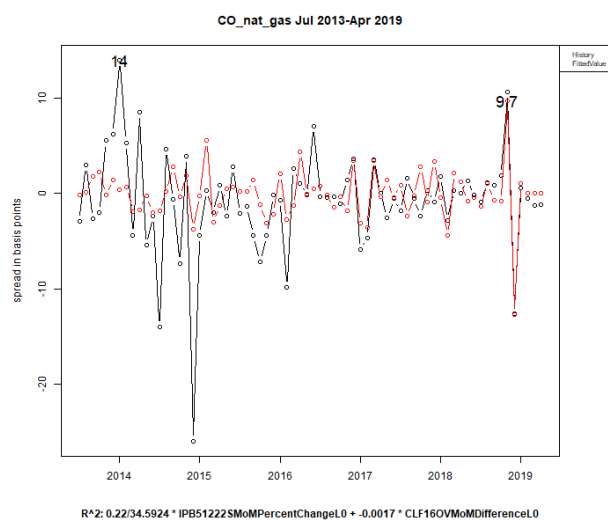
CO_nat_gas



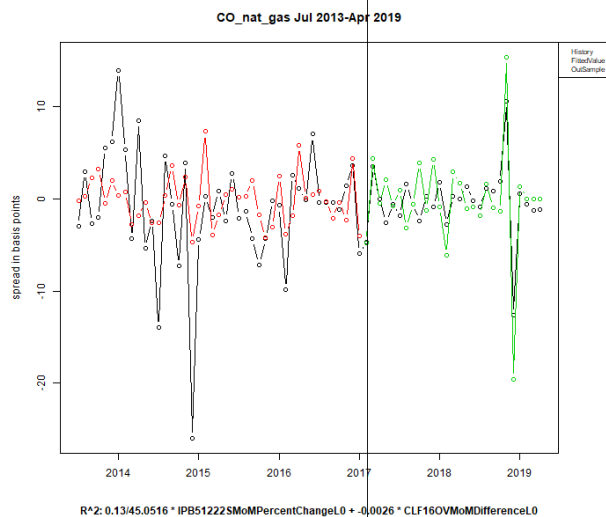
4.6.12 Model - 12

Backtesting on differenced values

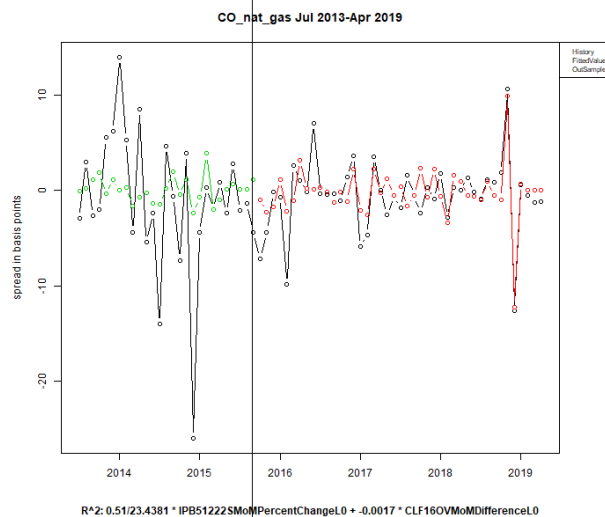
Backtesting on level values



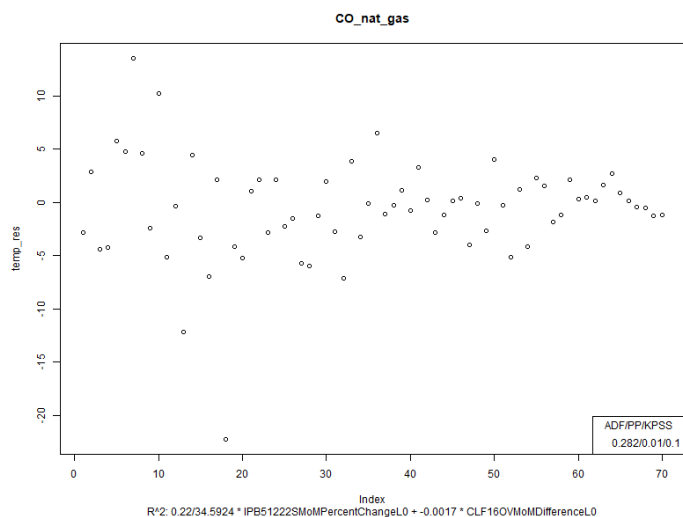
Out-of-Sample testing



Out-of-Sample testing



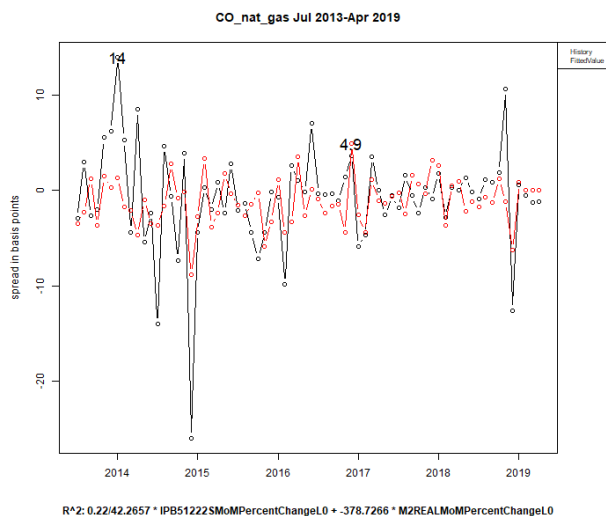
Stationarity of Residuals



4.6.13 Model - 13

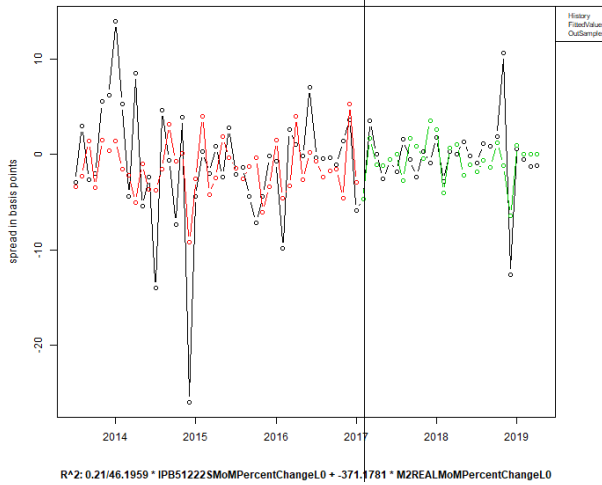
Backtesting on differenced values

Backtesting on level values



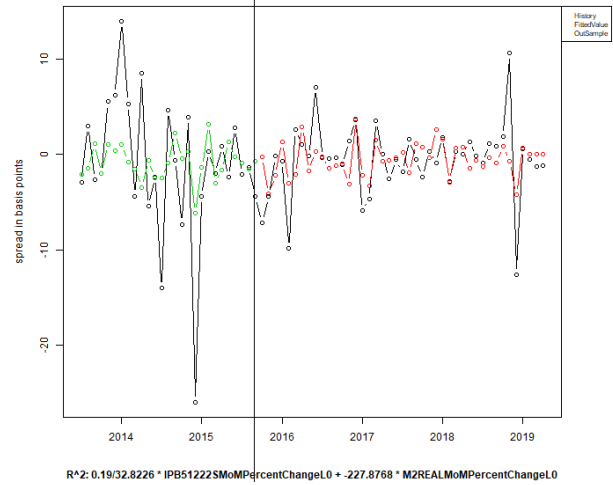
Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019



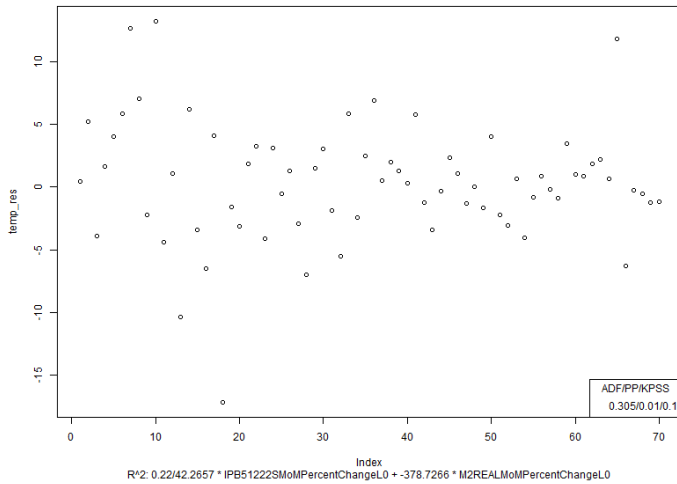
Out-of-Sample testing

CO_nat_gas Jul 2013-Apr 2019



Stationarity of Residuals

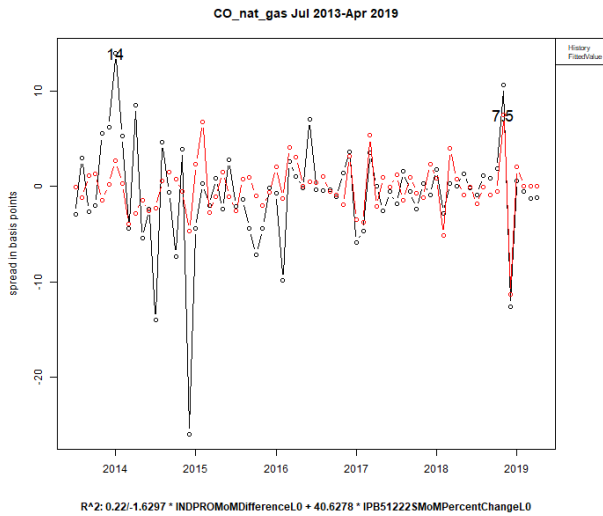
CO_nat_gas



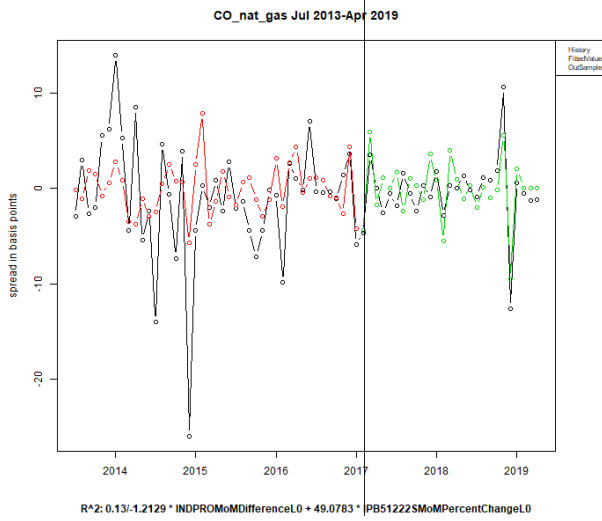
4.6.14 Model - 14

Backtesting on differenced values

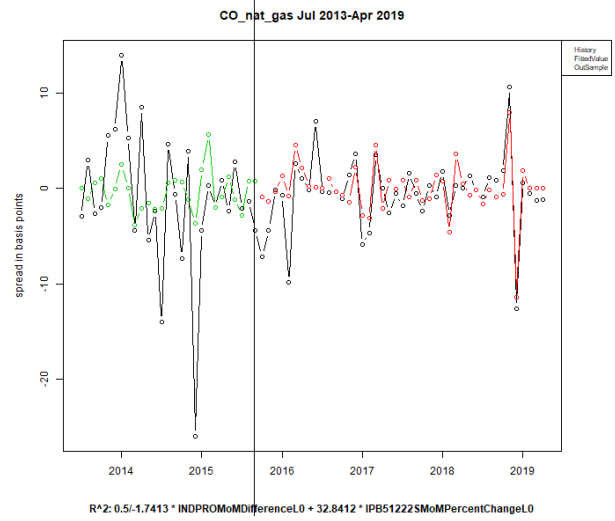
Backtesting on level values



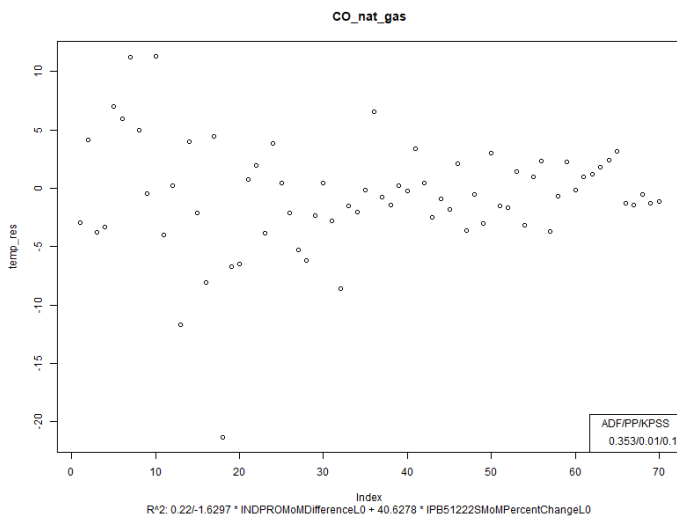
Out-of-Sample testing



Out-of-Sample testing



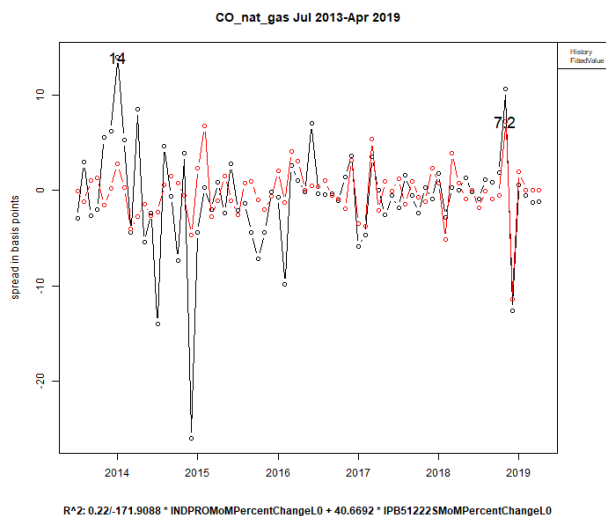
Stationarity of Residuals



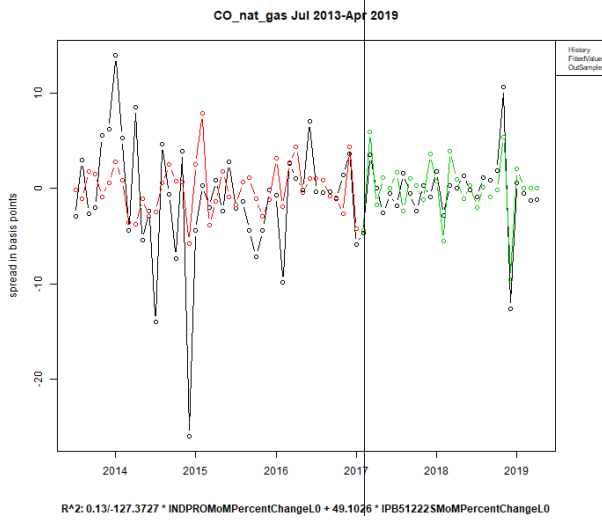
4.6.15 Model - 15

Backtesting on differenced values

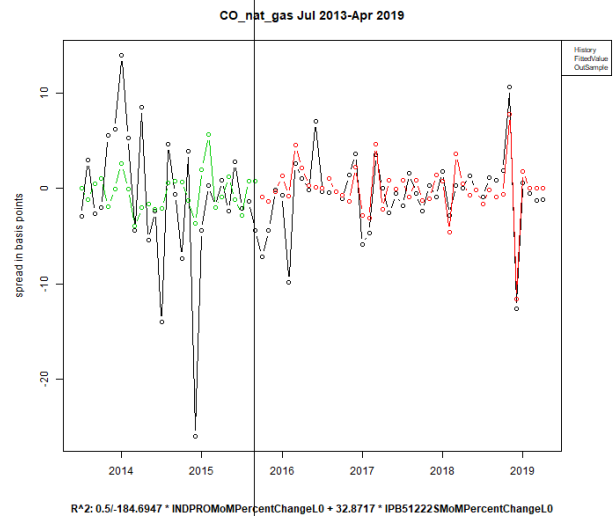
Backtesting on level values



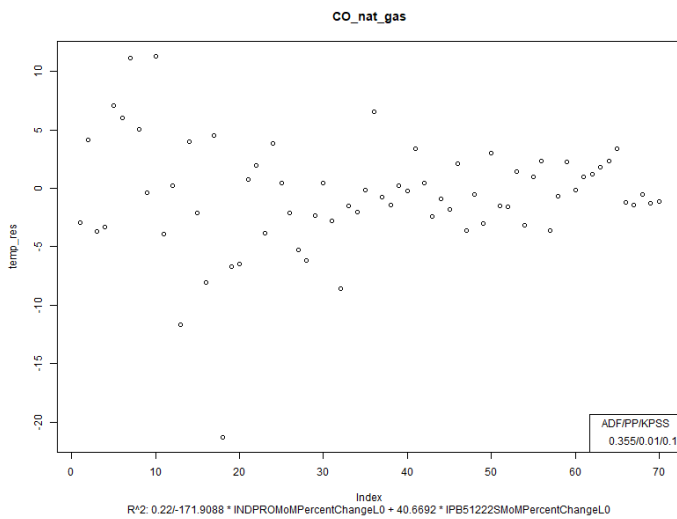
Out-of-Sample testing



Out-of-Sample testing



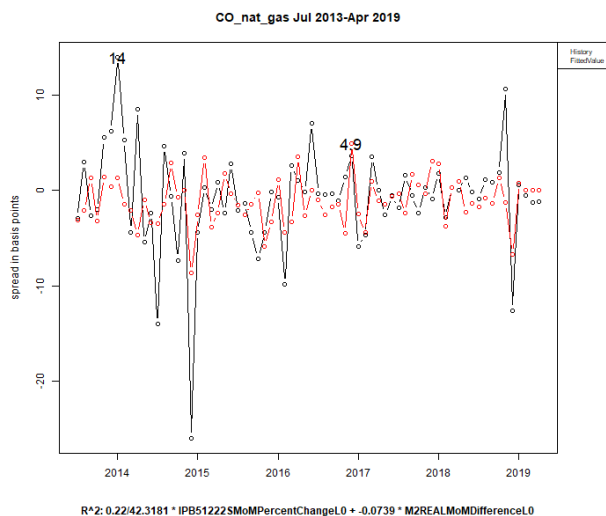
Stationarity of Residuals



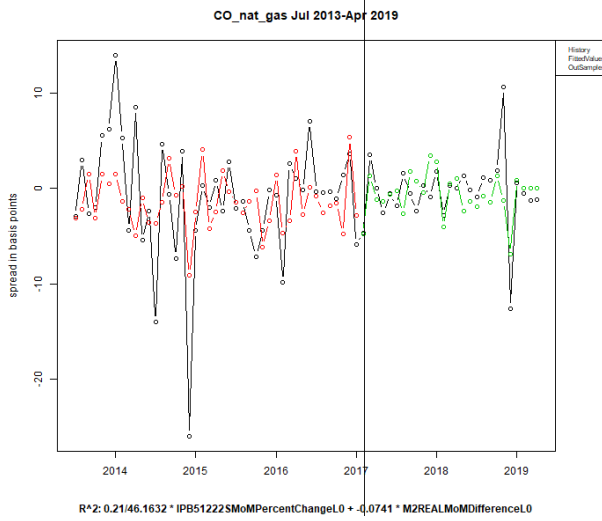
4.6.16 Model - 16

Backtesting on differenced values

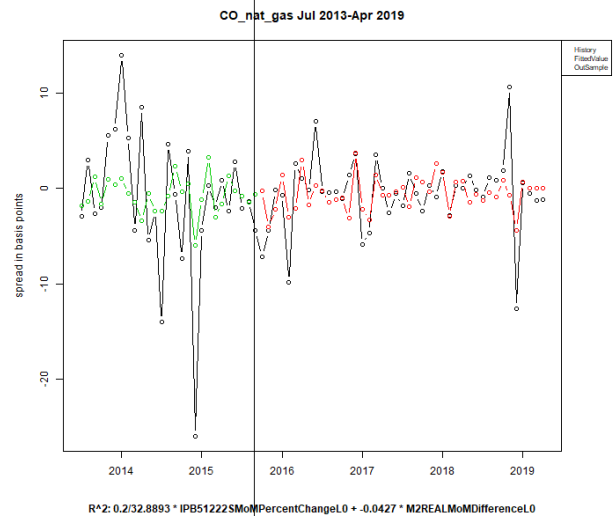
Backtesting on level values



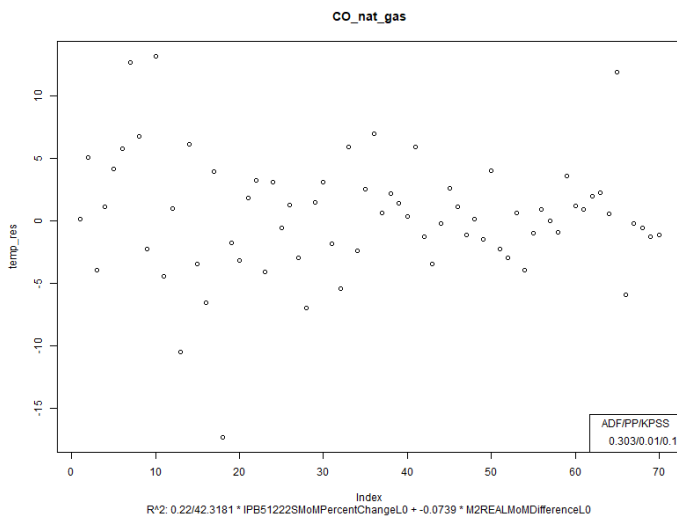
Out-of-Sample testing



Out-of-Sample testing



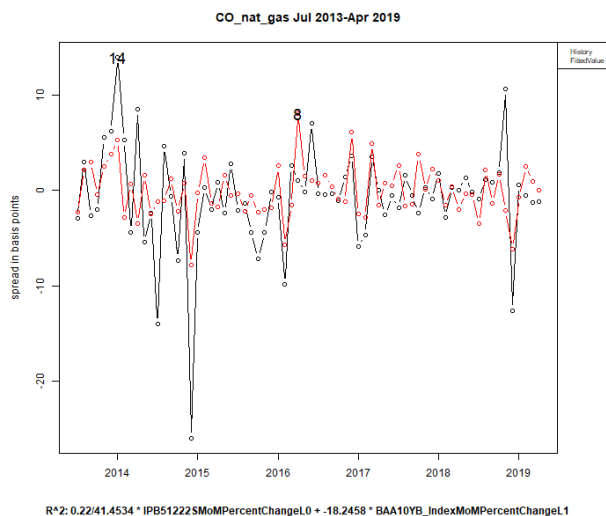
Stationarity of Residuals



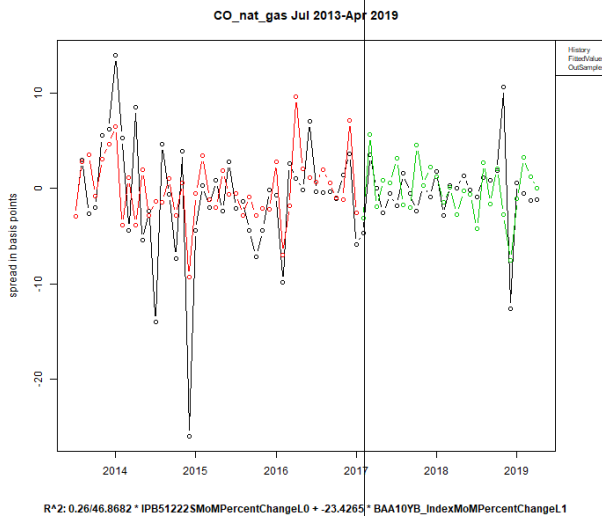
4.6.17 Model - 17

Backtesting on differenced values

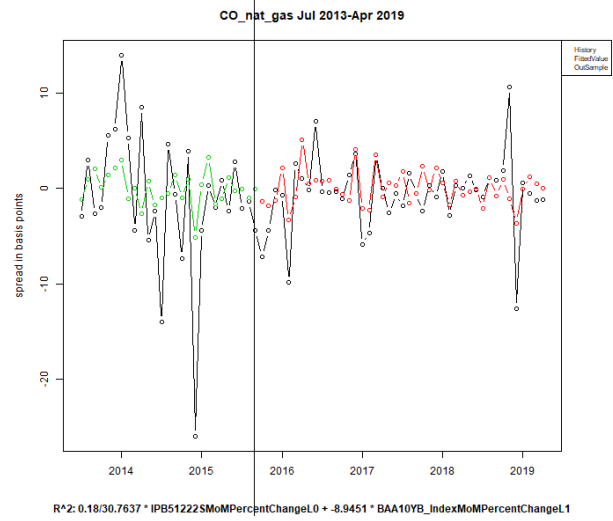
Backtesting on level values



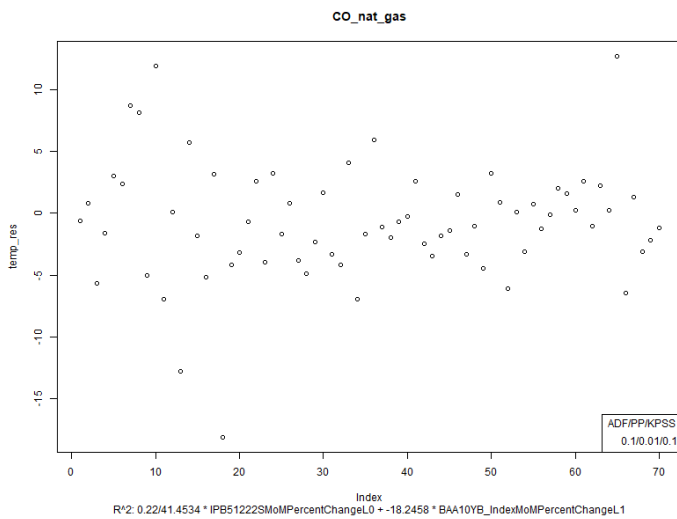
Out-of-Sample testing



Out-of-Sample testing



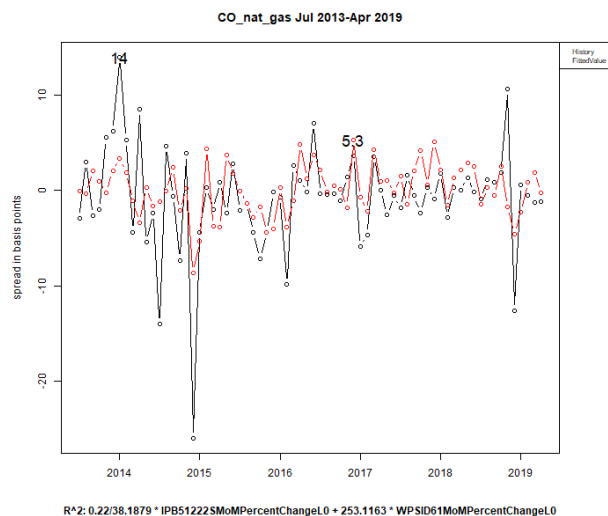
Stationarity of Residuals



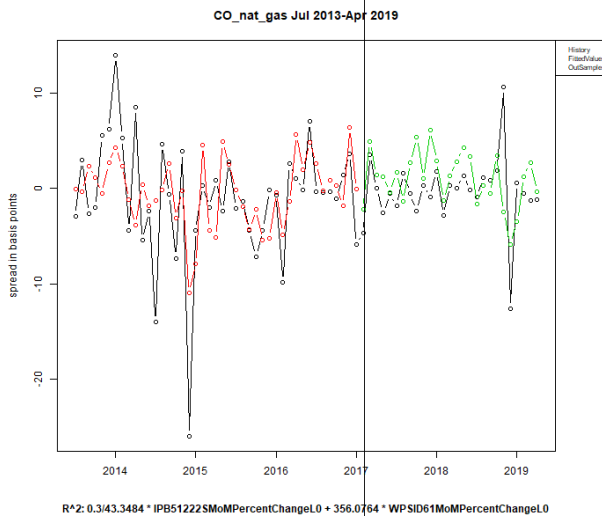
4.6.18 Model - 18

Backtesting on differenced values

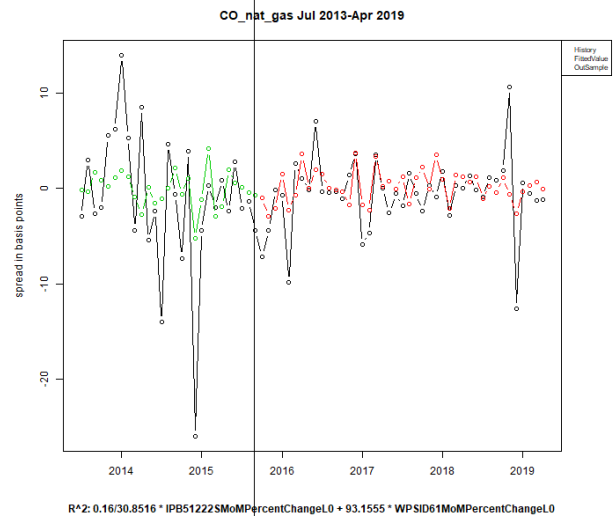
Backtesting on level values



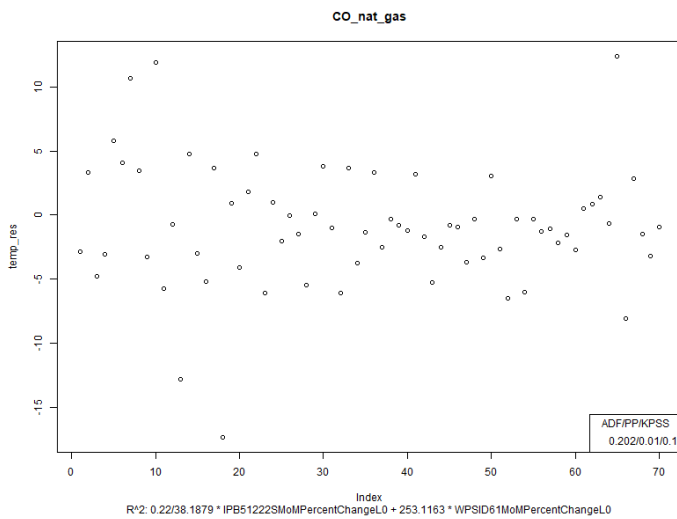
Out-of-Sample testing



Out-of-Sample testing



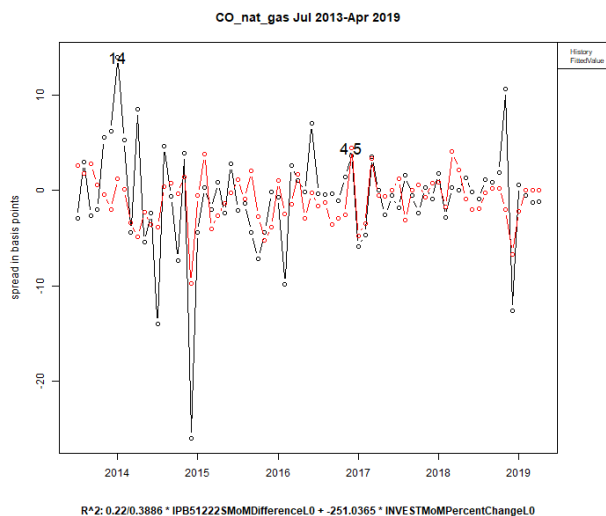
Stationarity of Residuals



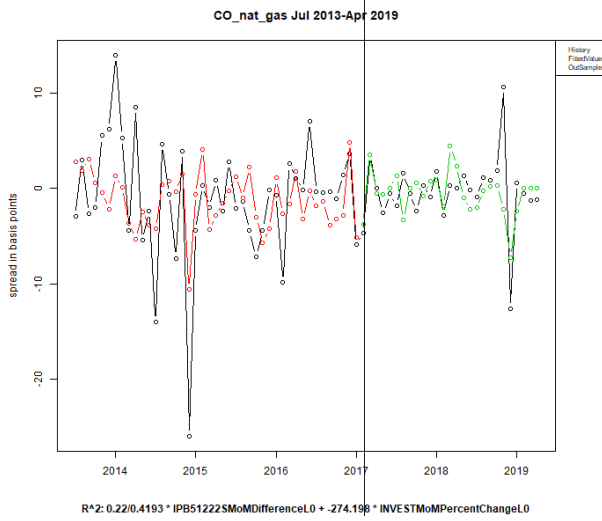
4.6.19 Model - 19

Backtesting on differenced values

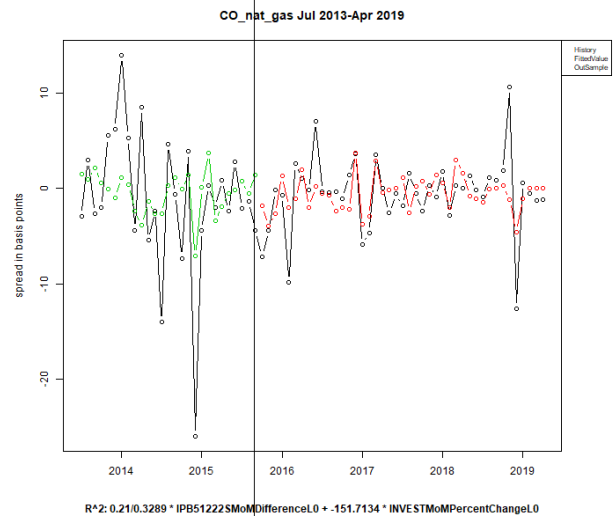
Backtesting on level values



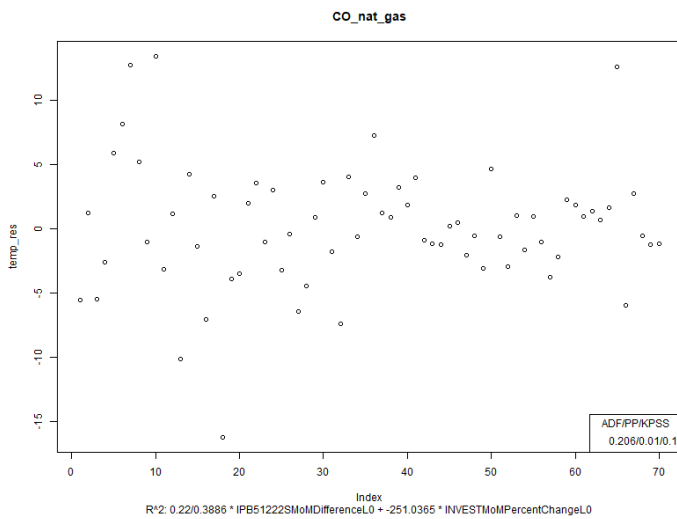
Out-of-Sample testing



Out-of-Sample testing



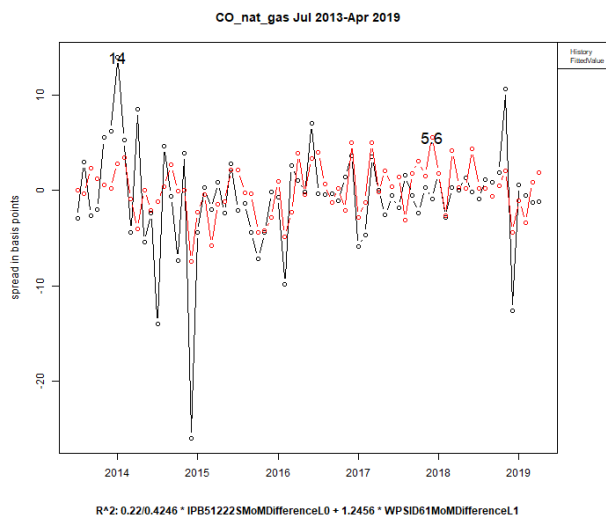
Stationarity of Residuals



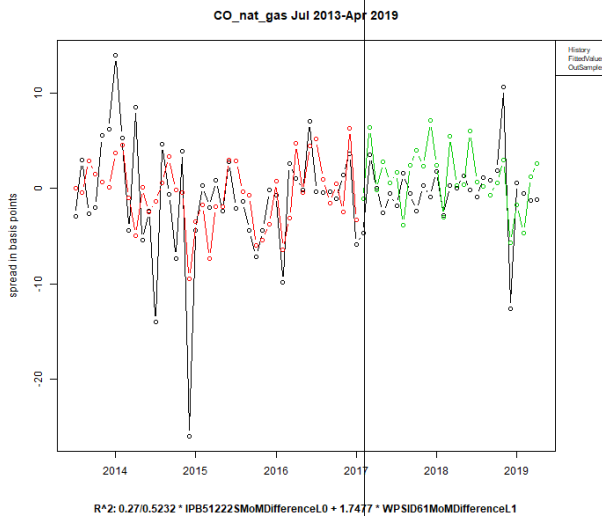
4.6.20 Model - 20

Backtesting on differenced values

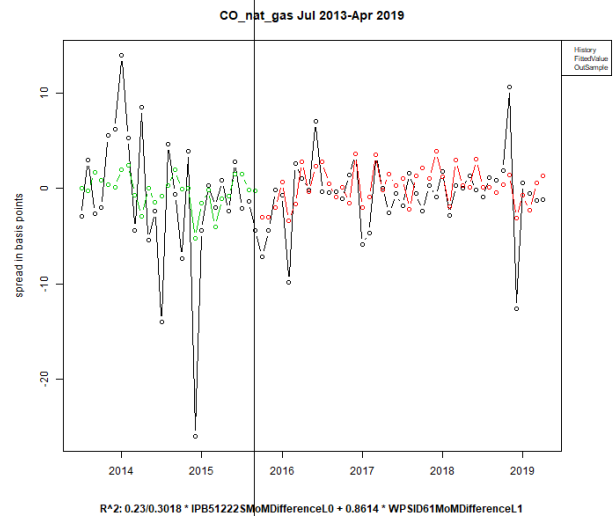
Backtesting on level values



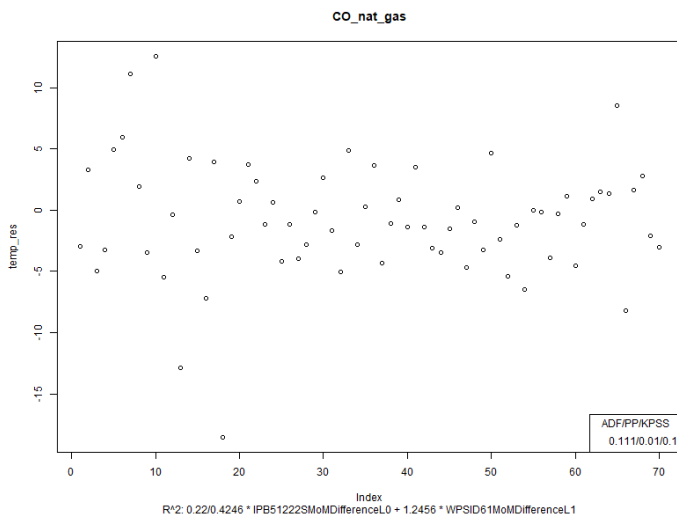
Out-of-Sample testing



Out-of-Sample testing



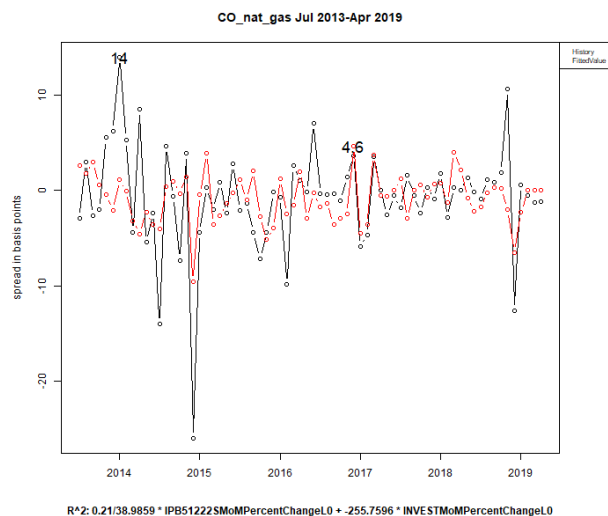
Stationarity of Residuals



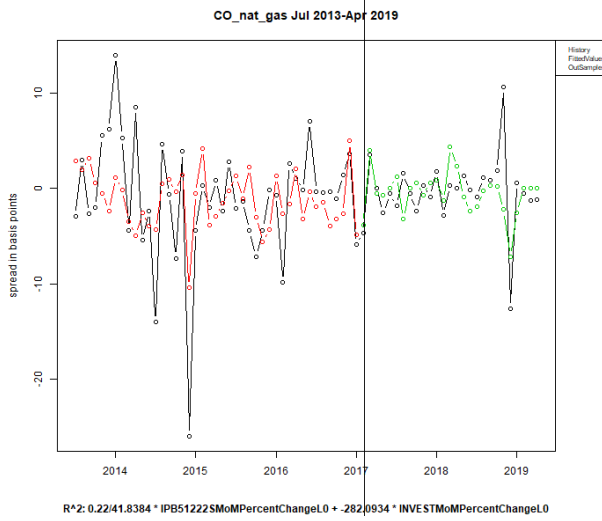
4.6.21 Model - 21

Backtesting on differenced values

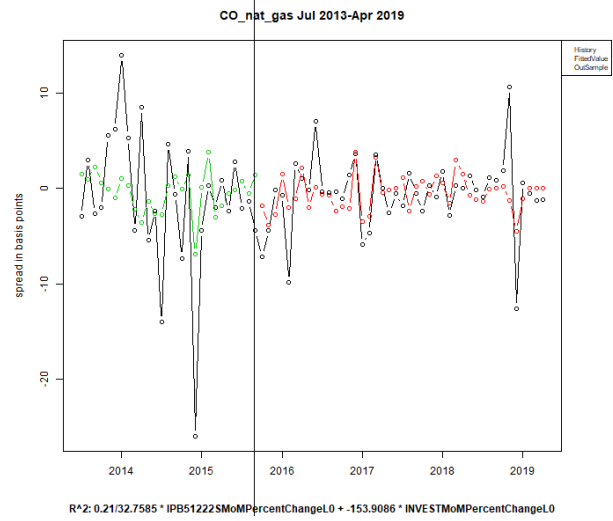
Backtesting on level values



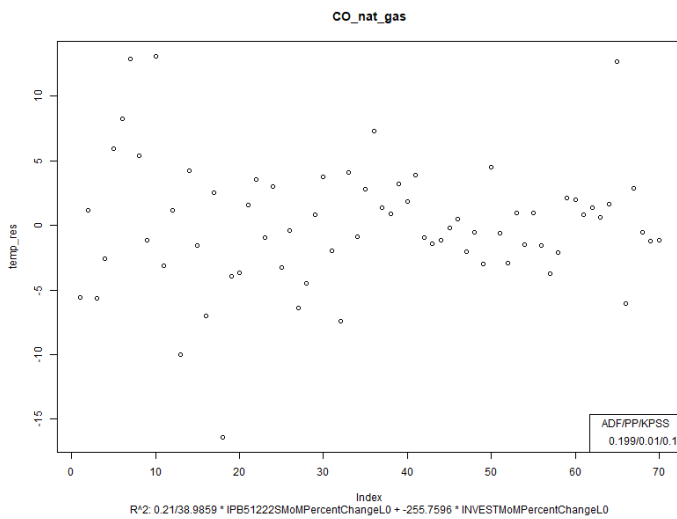
Out-of-Sample testing



Out-of-Sample testing



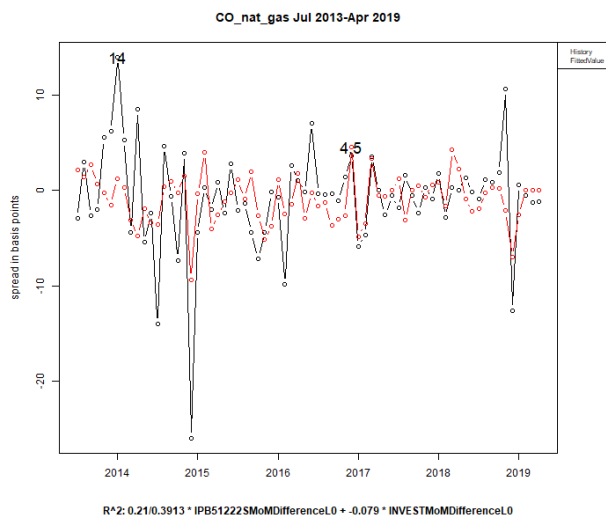
Stationarity of Residuals



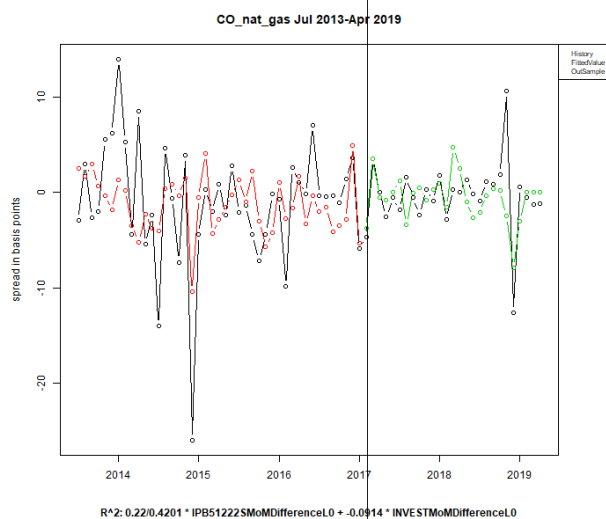
4.6.22 Model - 22

Backtesting on differenced values

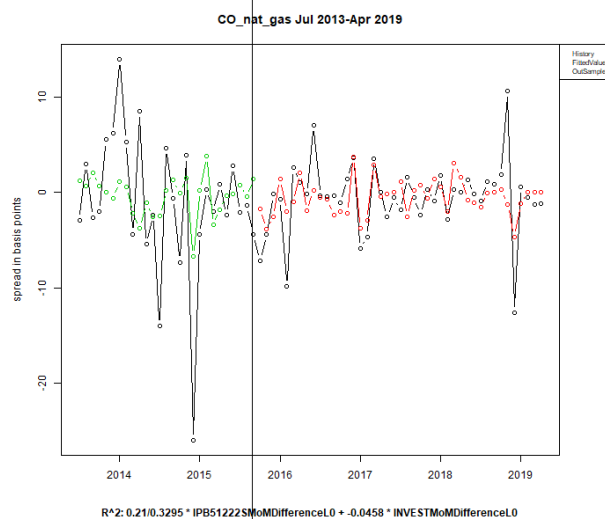
Backtesting on level values



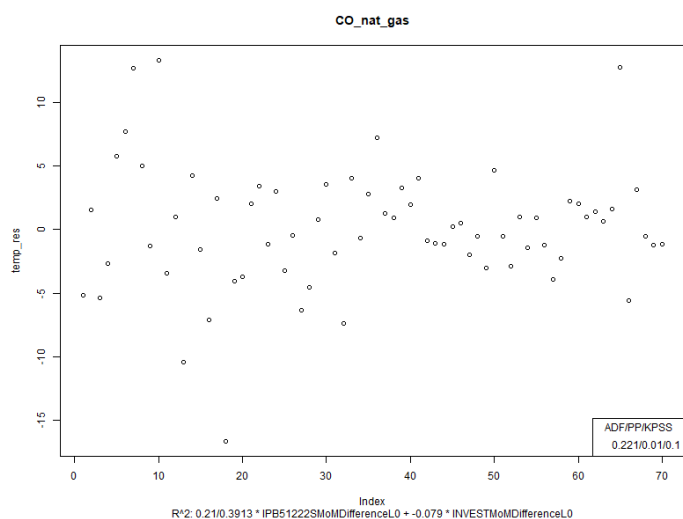
Out-of-Sample testing



Out-of-Sample testing



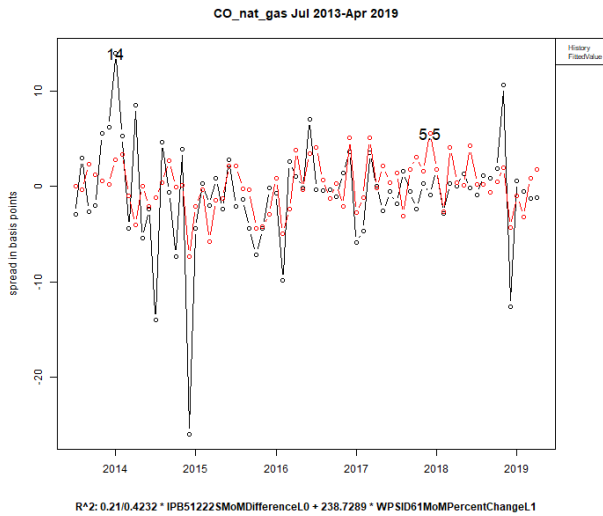
Stationarity of Residuals



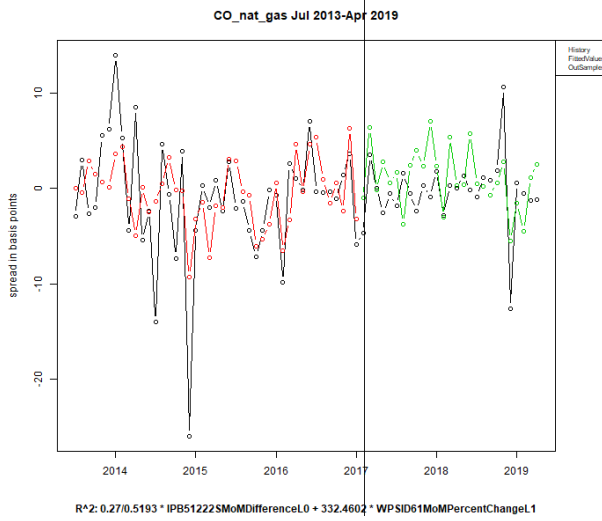
4.6.23 Model - 23

Backtesting on differenced values

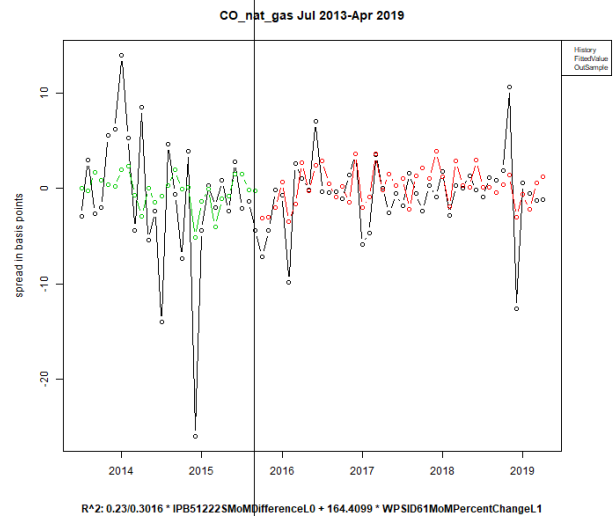
Backtesting on level values



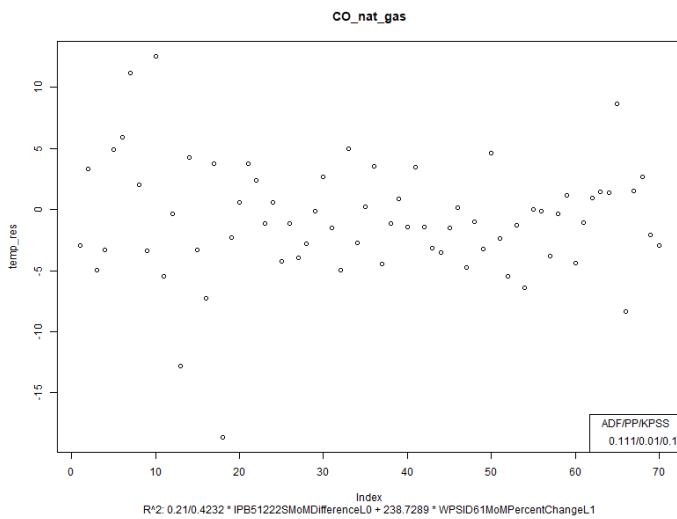
Out-of-Sample testing



Out-of-Sample testing



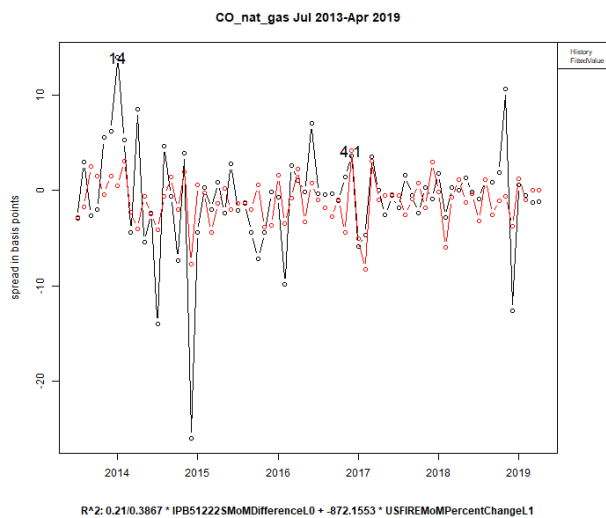
Stationarity of Residuals



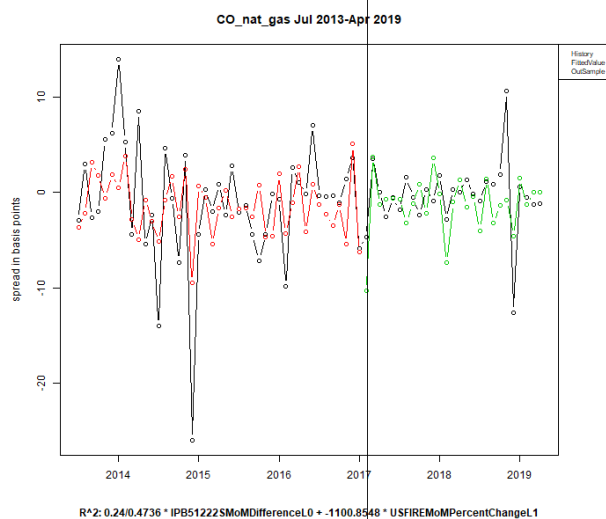
4.6.24 Model - 24

Backtesting on differenced values

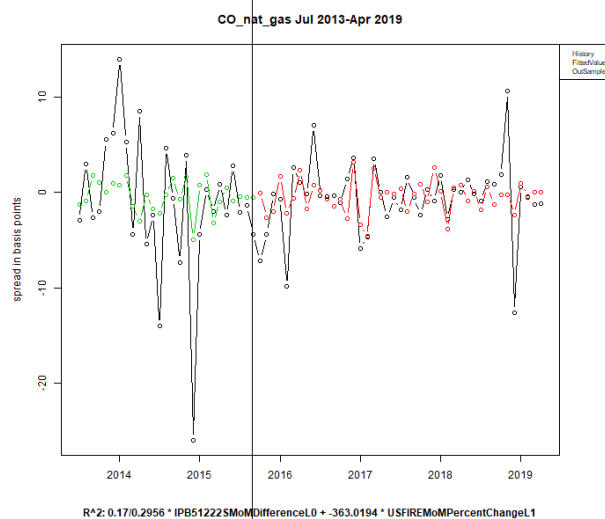
Backtesting on level values



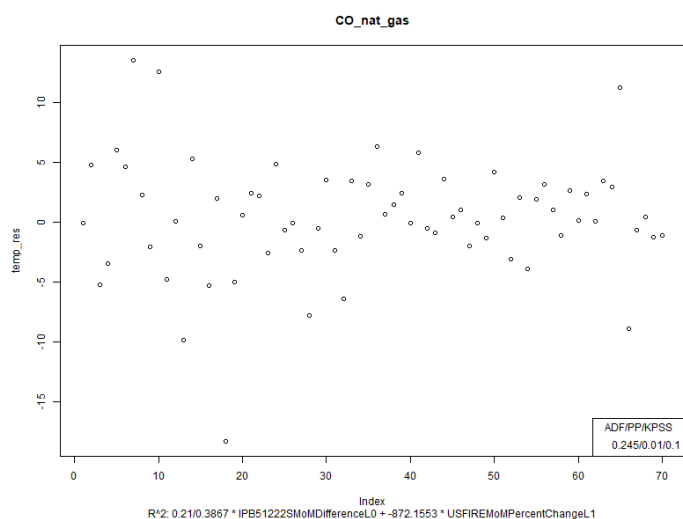
Out-of-Sample testing



Out-of-Sample testing



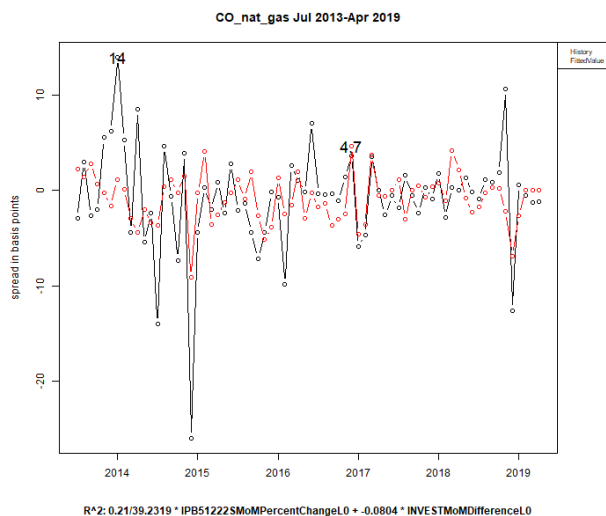
Stationarity of Residuals



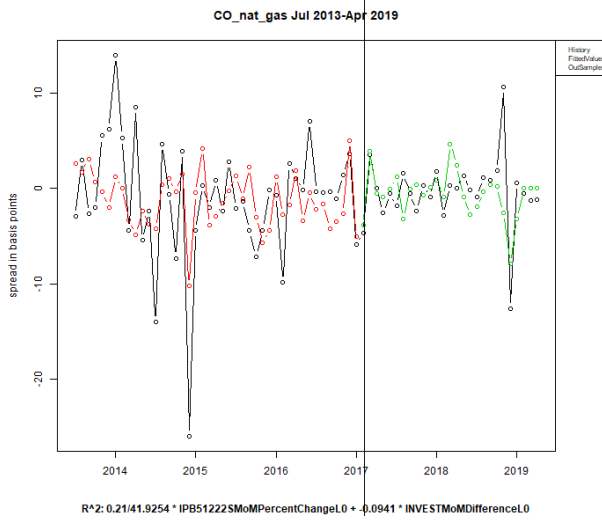
4.6.25 Model - 25

Backtesting on differenced values

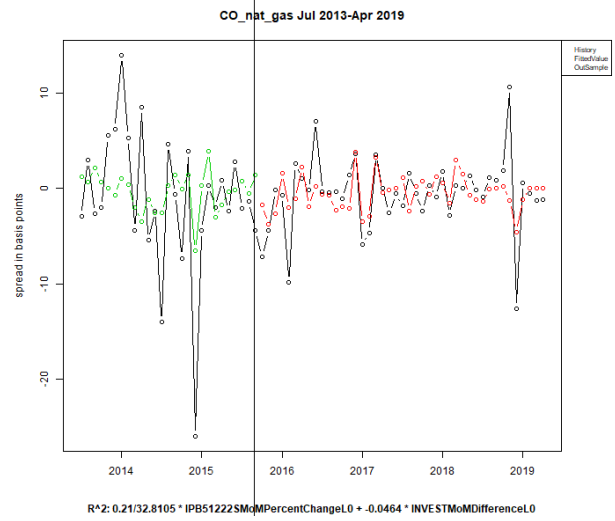
Backtesting on level values



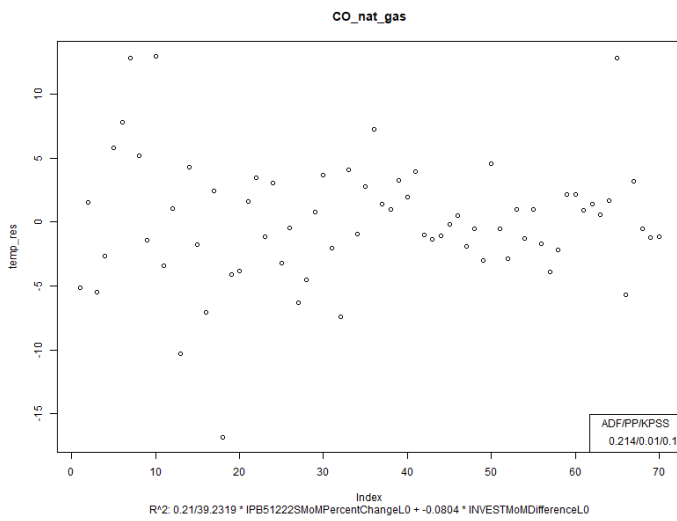
Out-of-Sample testing



Out-of-Sample testing



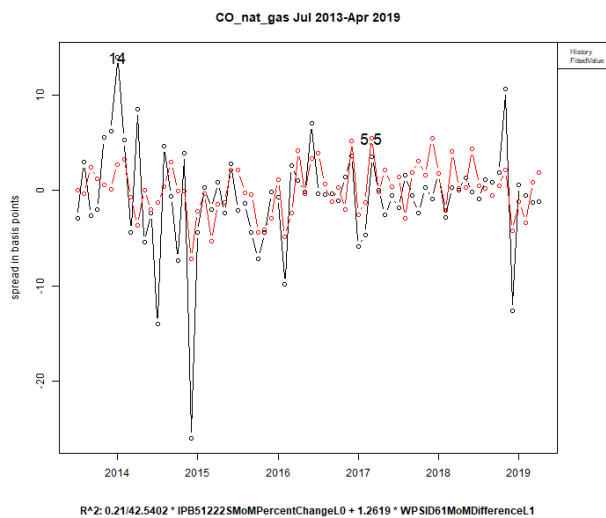
Stationarity of Residuals



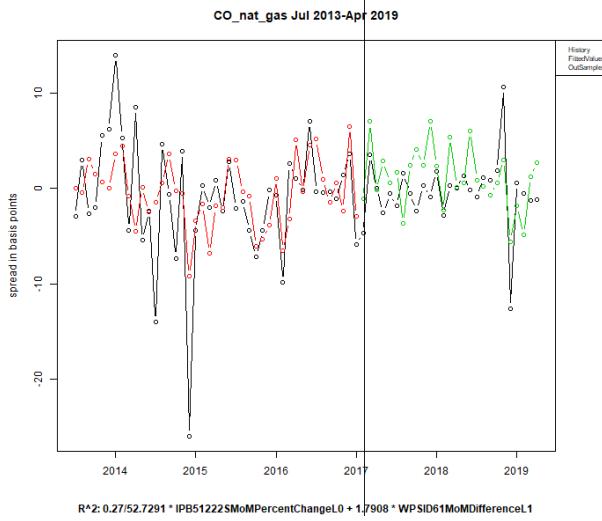
4.6.26 Model - 26

Backtesting on differenced values

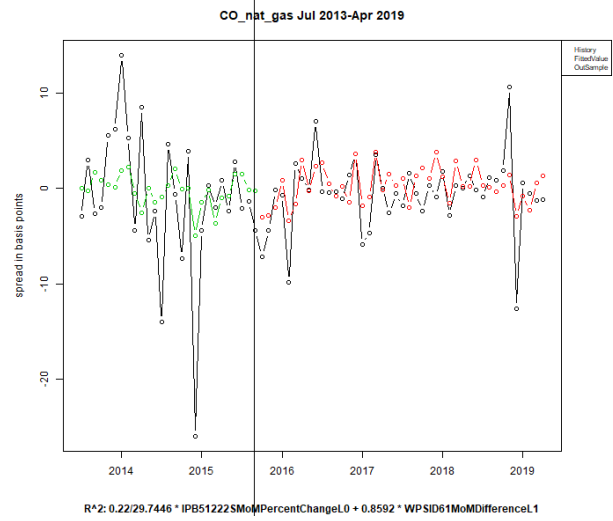
Backtesting on level values



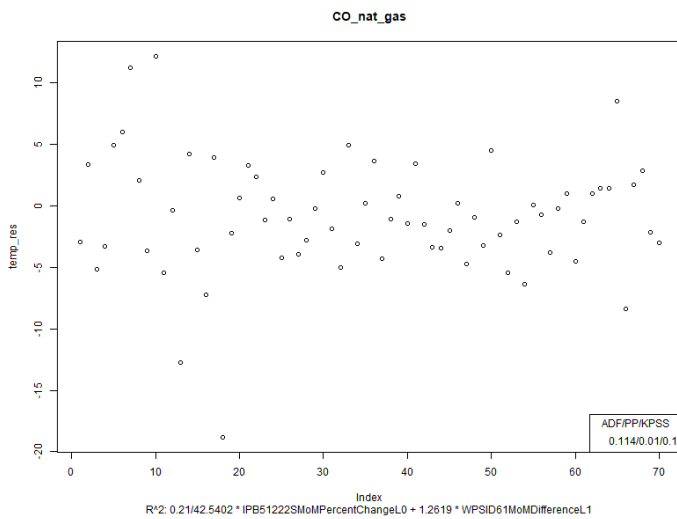
Out-of-Sample testing



Out-of-Sample testing



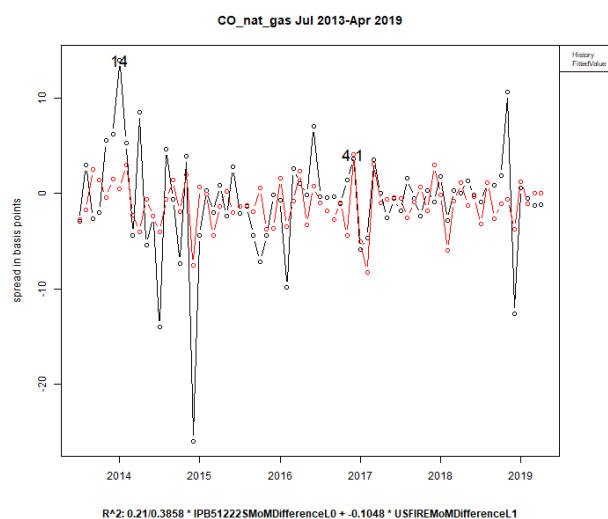
Stationarity of Residuals



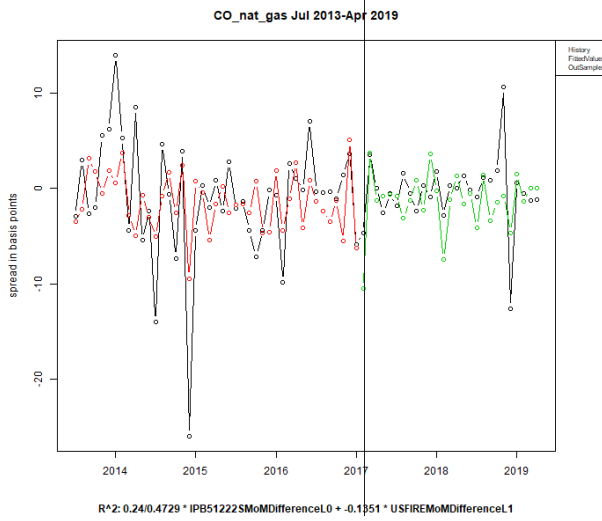
4.6.27 Model - 27

Backtesting on differenced values

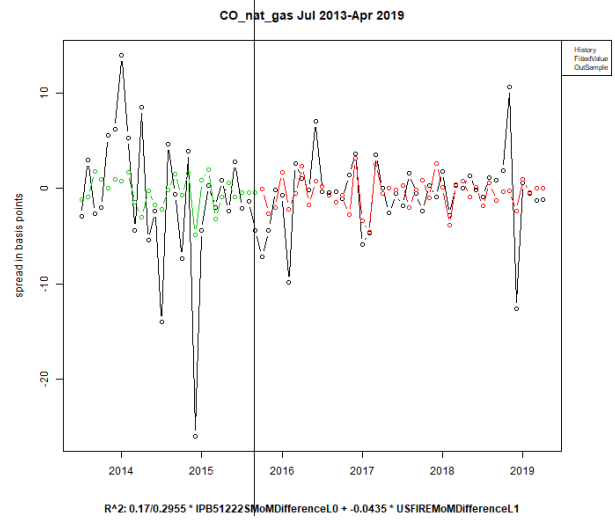
Backtesting on level values



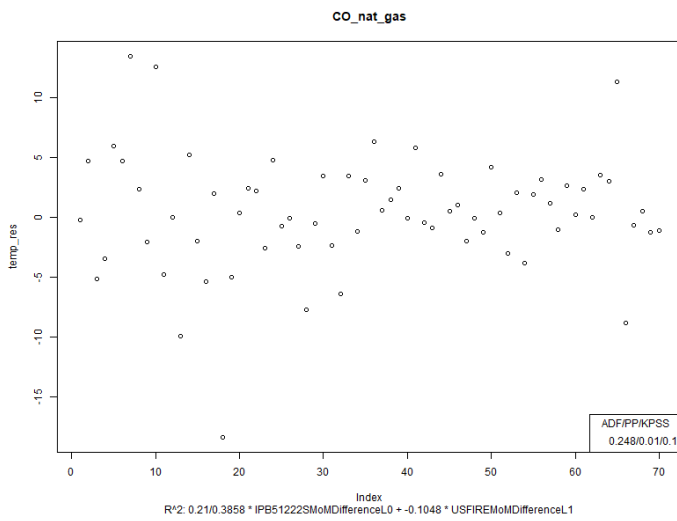
Out-of-Sample testing



Out-of-Sample testing



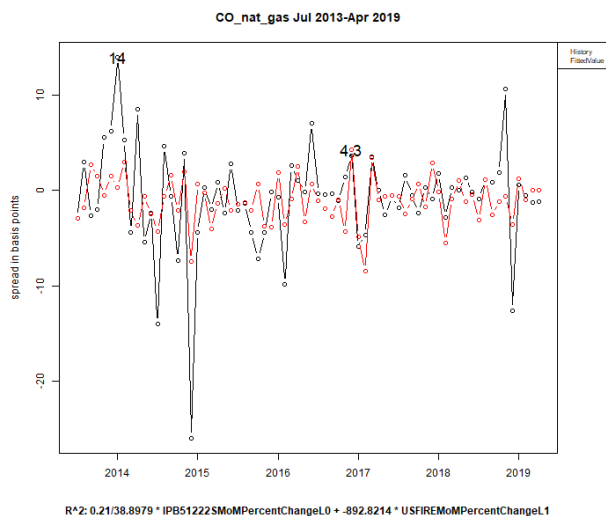
Stationarity of Residuals



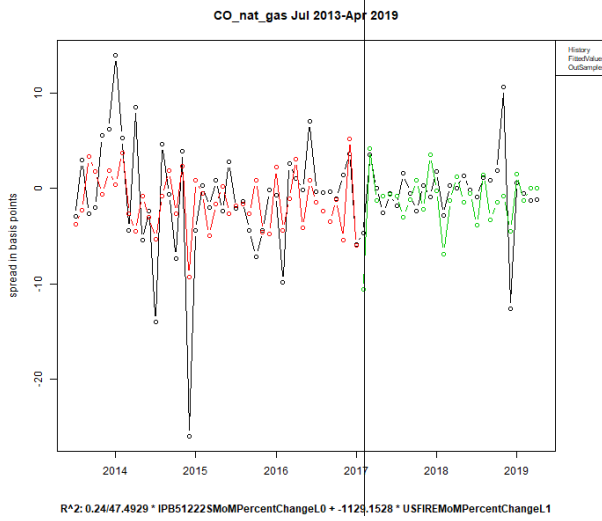
4.6.28 Model - 28

Backtesting on differenced values

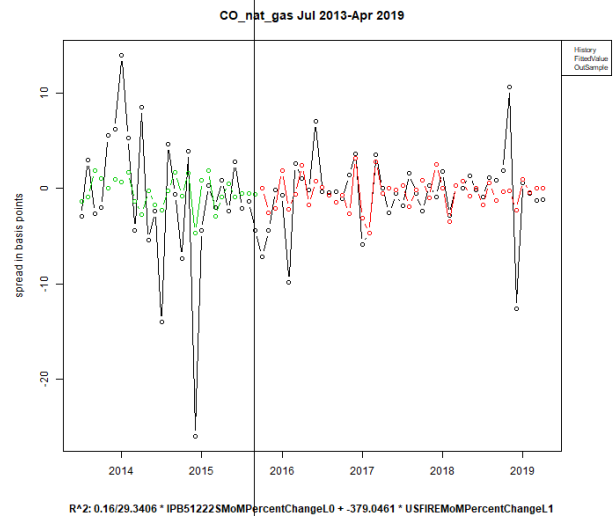
Backtesting on level values



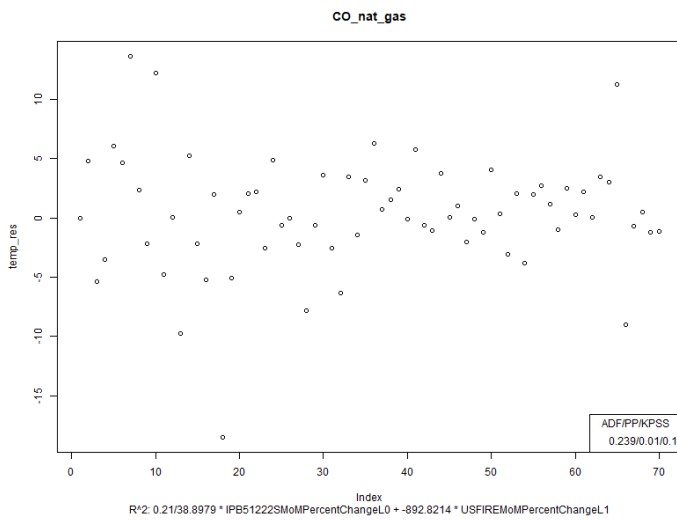
Out-of-Sample testing



Out-of-Sample testing



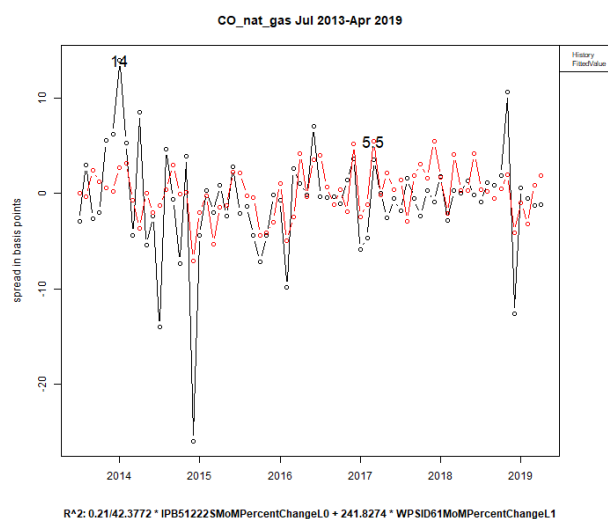
Stationarity of Residuals



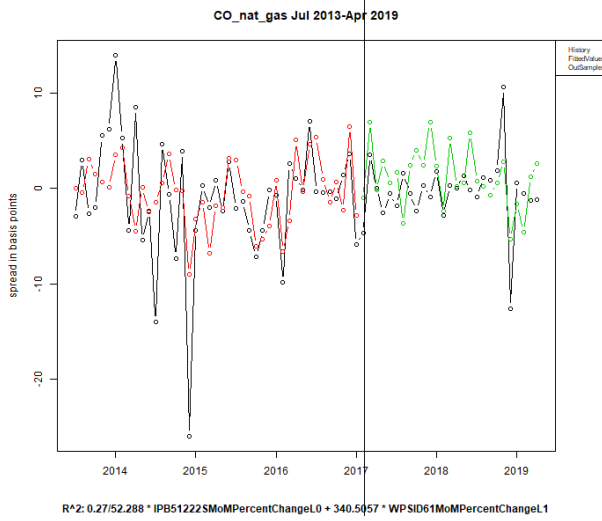
4.6.29 Model - 29

Backtesting on differenced values

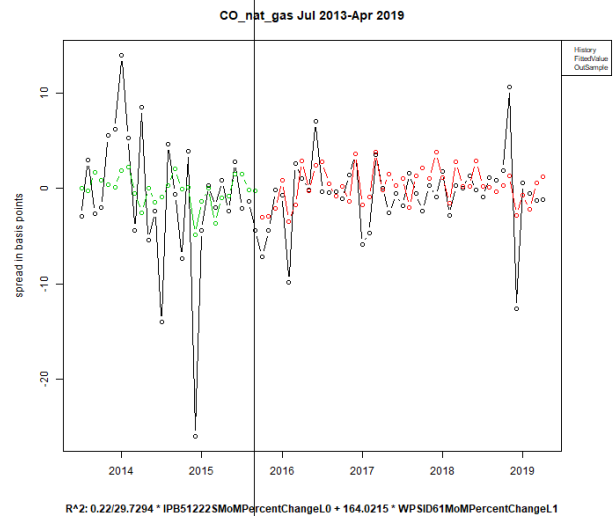
Backtesting on level values



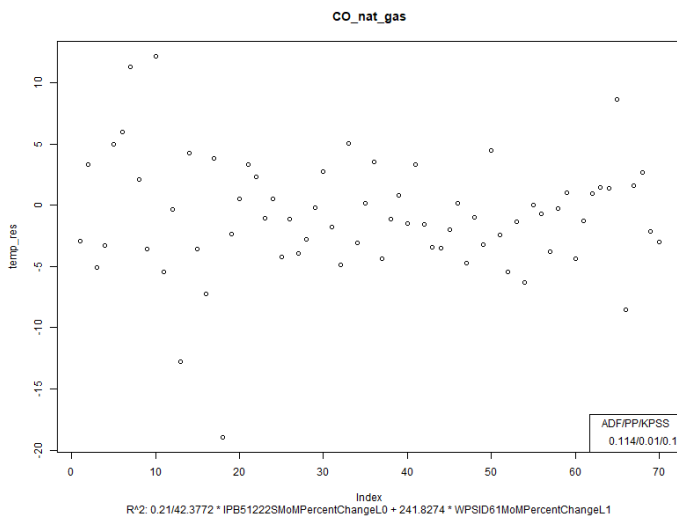
Out-of-Sample testing



Out-of-Sample testing



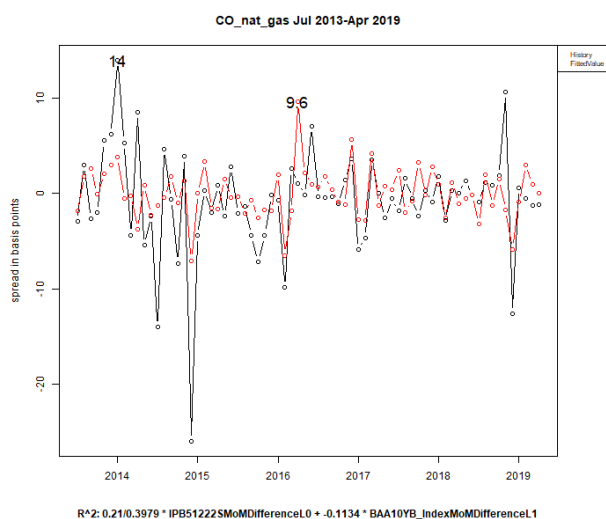
Stationarity of Residuals



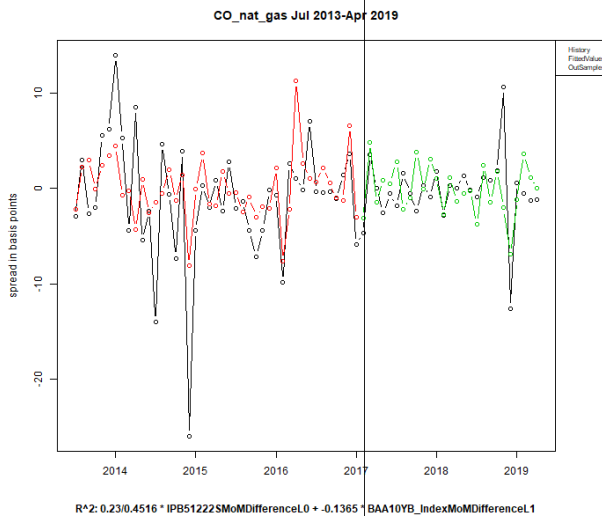
4.6.30 Model - 30

Backtesting on differenced values

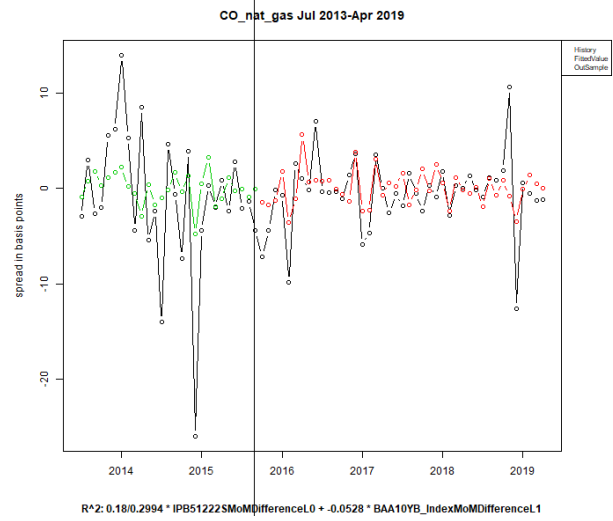
Backtesting on level values



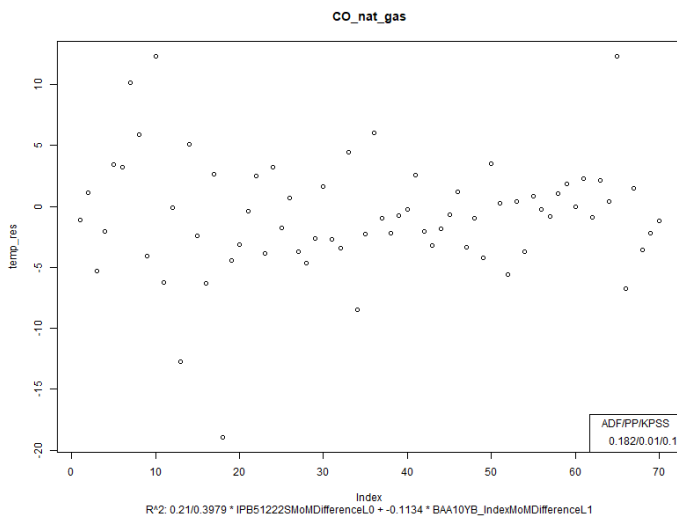
Out-of-Sample testing



Out-of-Sample testing



Stationarity of Residuals



4.7 Single Factor Model for CO₂gasoline

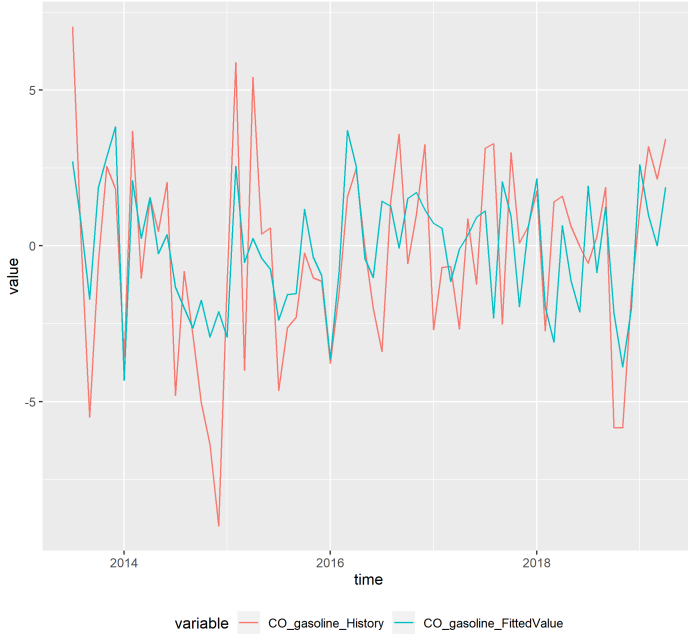
	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	BAA10YB_IndexMoMPercentChangeL0	0.37	326.29	-18.86	0.00	2.46			
2	BAA10YB_IndexMoMDifferenceL0	0.32	331.77	-0.12	0.00	2.26			
3	USGG30YR_IndexMoMDifferenceL0	0.14	348.13	7.11	0.00	2.83			
4	USGG30YR_IndexMoMPercentChangeL0	0.14	348.30	19.96	0.00	2.77			
5	USGG10YR_IndexMoMDifferenceL0	0.11	350.37	5.91	0.00	2.67			
6	USGG10YR_IndexMoMPercentChangeL0	0.11	350.63	12.70	0.00	2.60			
7	ISRATIOMoMPercentChangeL0	0.10	351.12	-146.83	0.01	2.57			
8	VIX_IndexMoMPercentChangeL0	0.10	351.16	-3.87	0.01	2.27			

Table 12: Single Factor Analysis-CO₂gasoline

4.7.1 Model - 1

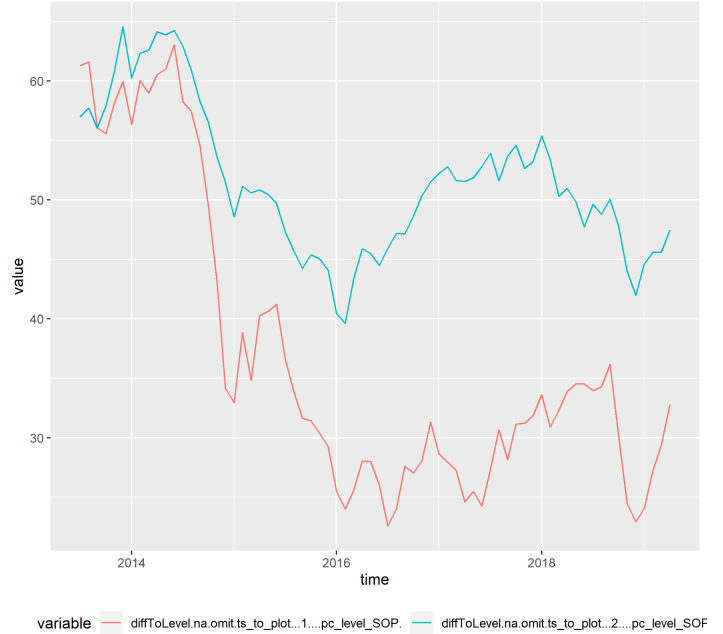
Backtesting on differenced values

CO_gasoline - Backtesting

 $R^2: 0.37 / -18.8629 * BAA10YB_IndexMoMPercentChangeL0$ 

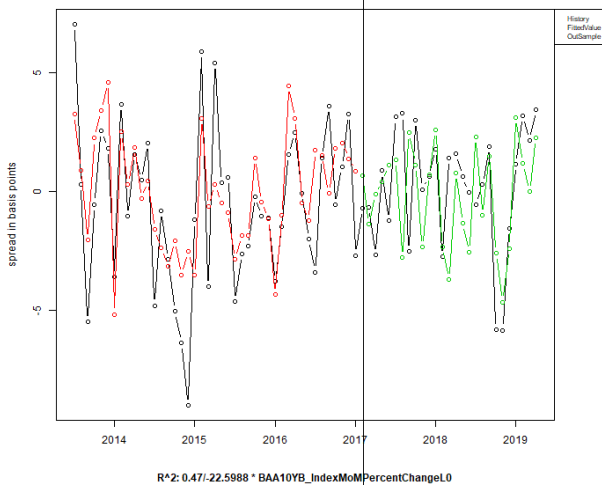
Backtesting on level values

CO_gasoline - Level Value Backtesting

 $R^2: 0.37 / -18.8629 * BAA10YB_IndexMoMPercentChangeL0$ 

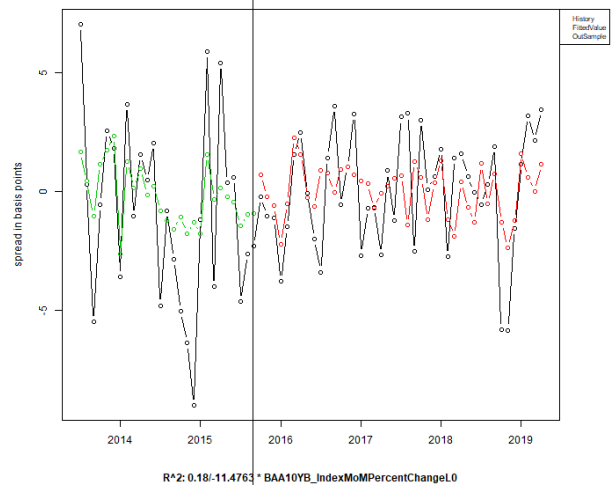
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

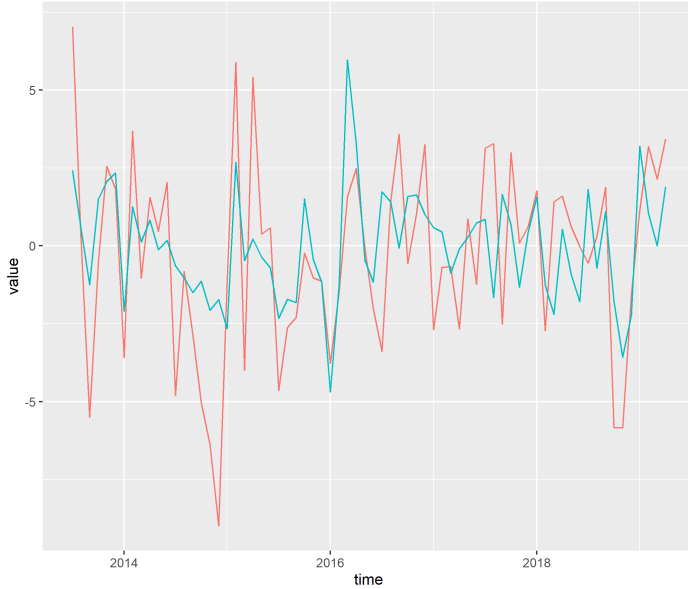


4.7.2 Model - 2

Backtesting on differenced values

CO_gasoline - Backtesting

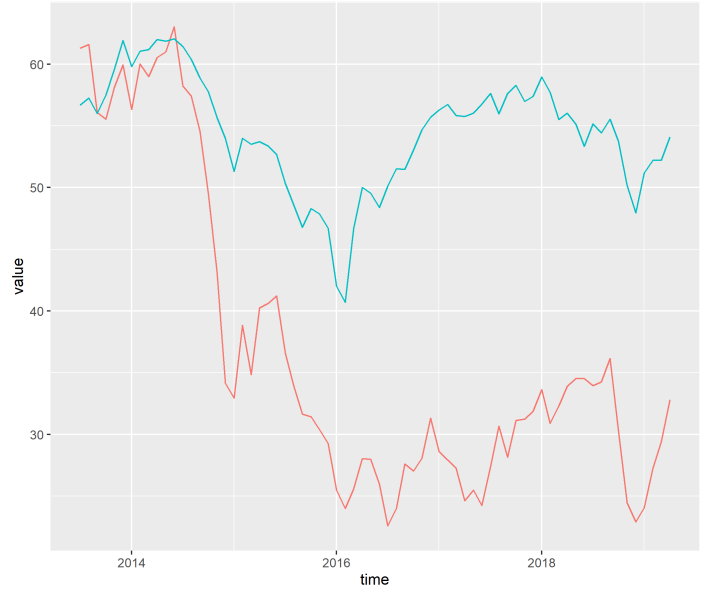
$R^2: 0.32 / -0.1208 * BAA10YB_IndexMoMDifferenceL0$



Backtesting on level values

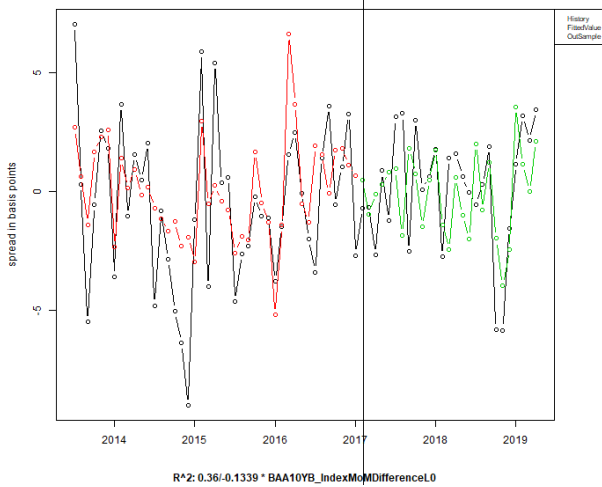
CO_gasoline - Level Value Backtesting

$R^2: 0.32 / -0.1208 * BAA10YB_IndexMoMDifferenceL0$



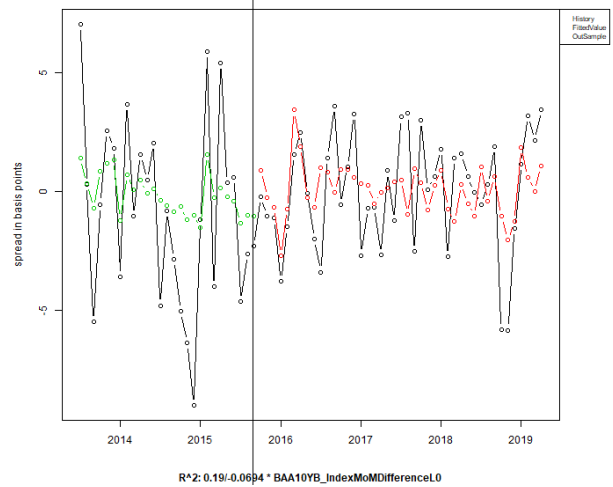
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



4.7.3 Model - 3

Backtesting on differenced values

CO_gasoline - Backtesting

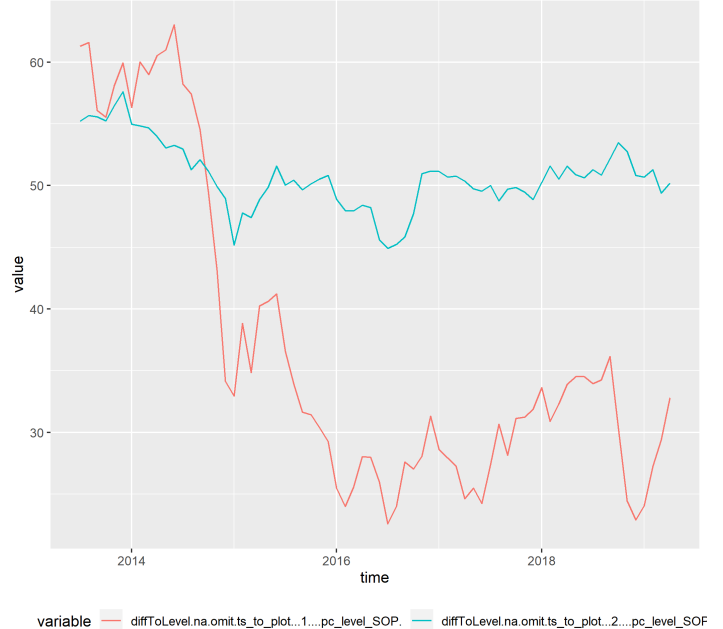
R²: 0.14/ 7.1116 * USGG30YR_IndexMoMDifferenceL0



Backtesting on level values

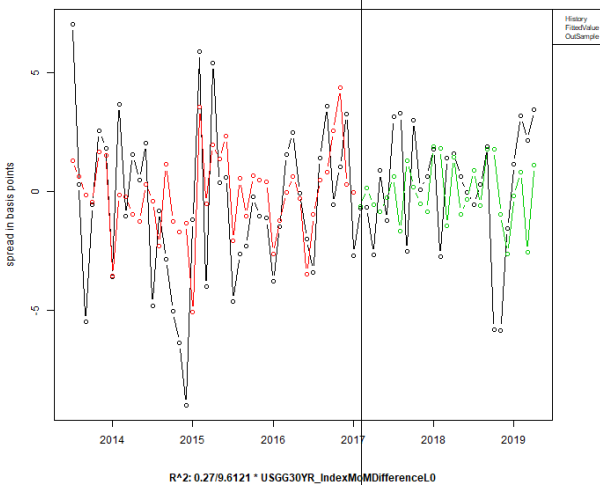
CO_gasoline - Level Value Backtesting

R²: 0.14/ 7.1116 * USGG30YR_IndexMoMDifferenceL0



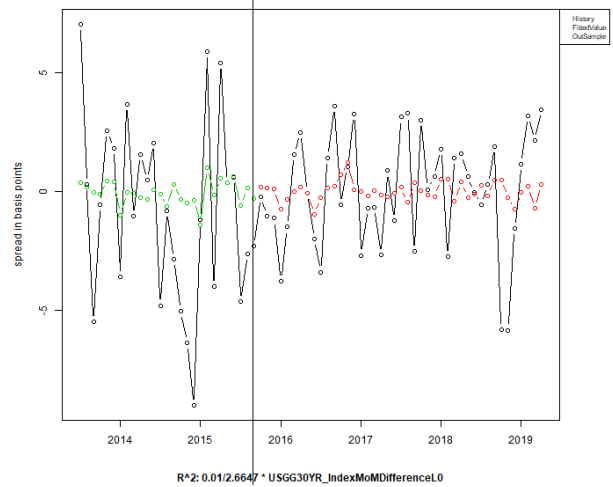
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

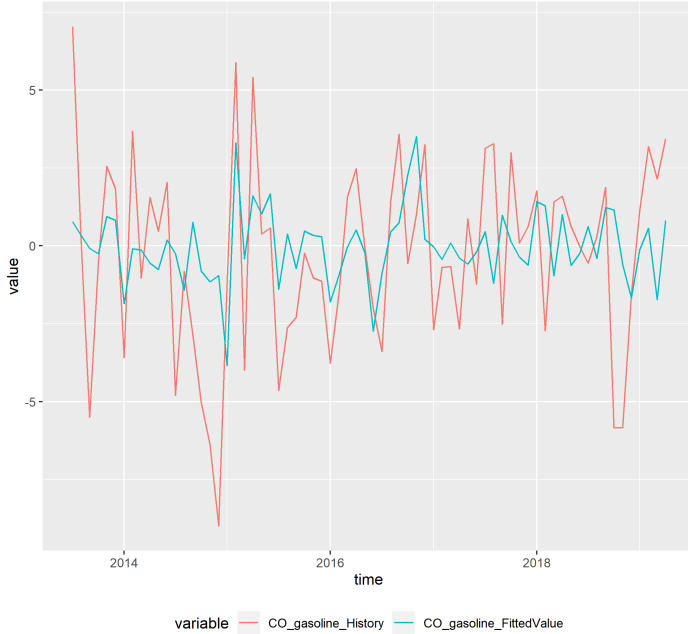


4.7.4 Model - 4

Backtesting on differenced values

CO_gasoline - Backtesting

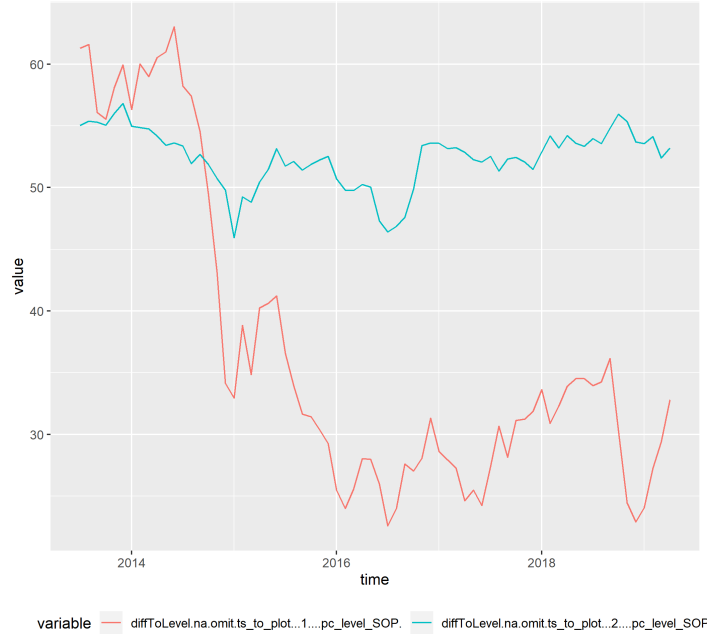
R²: 0.14/ 19.9606 * USGG30YR_IndexMoMPercentChangeL0



Backtesting on level values

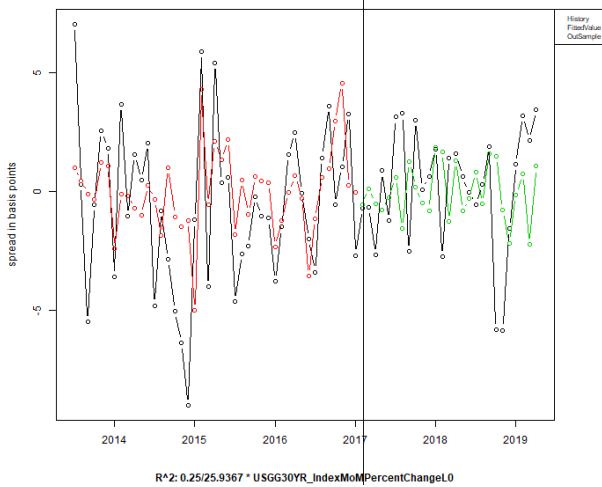
CO_gasoline - Level Value Backtesting

R²: 0.14/ 19.9606 * USGG30YR_IndexMoMPercentChangeL0



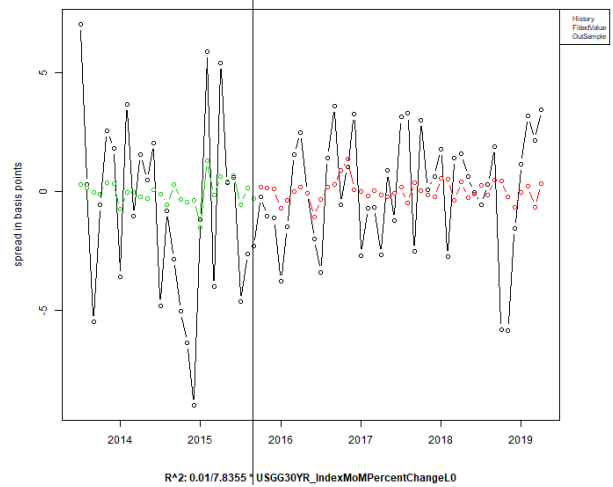
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

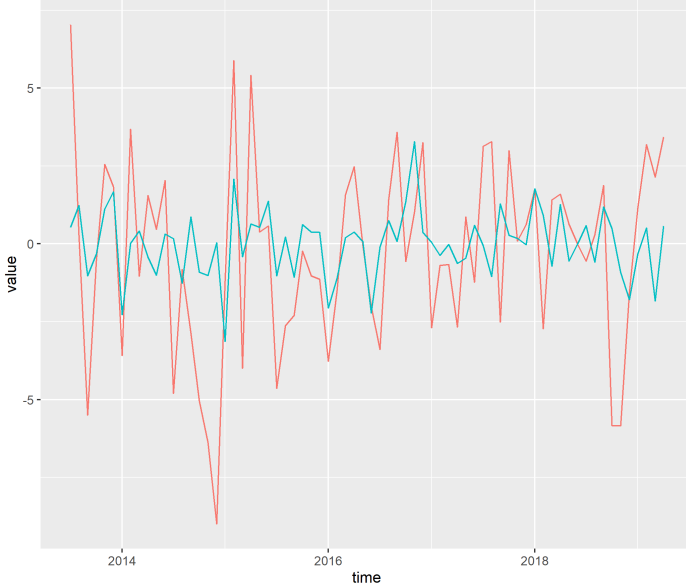


4.7.5 Model - 5

Backtesting on differenced values

CO_gasoline - Backtesting

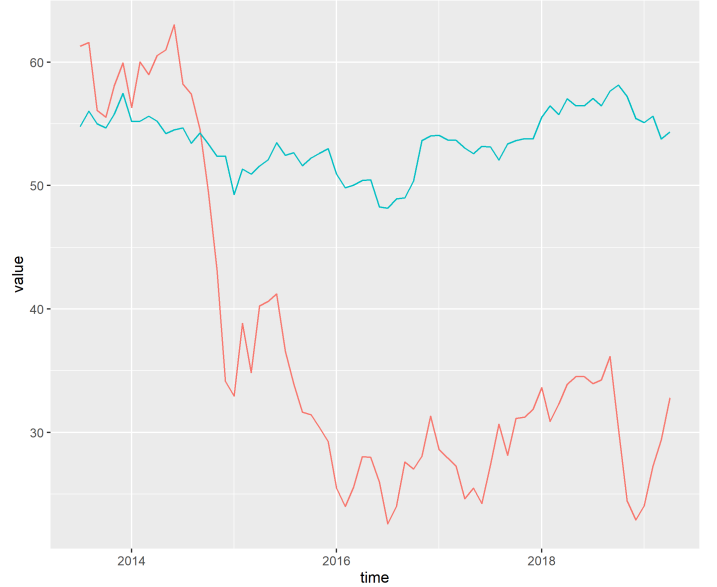
$R^2: 0.11 / 5.9053 * USGG10YR_IndexMoMDifferenceL0$



Backtesting on level values

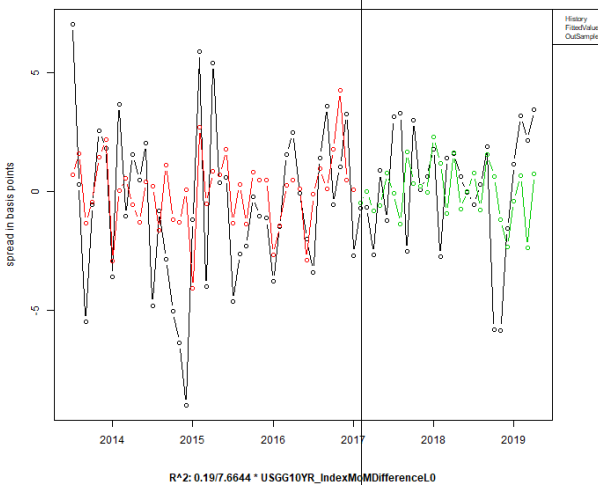
CO_gasoline - Level Value Backtesting

$R^2: 0.11 / 5.9053 * USGG10YR_IndexMoMDifferenceL0$



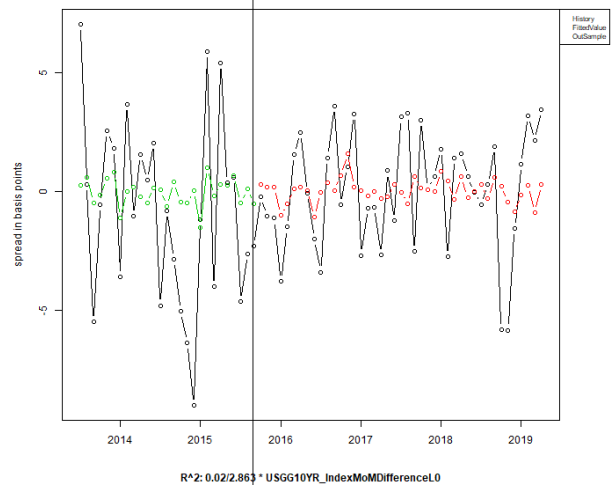
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

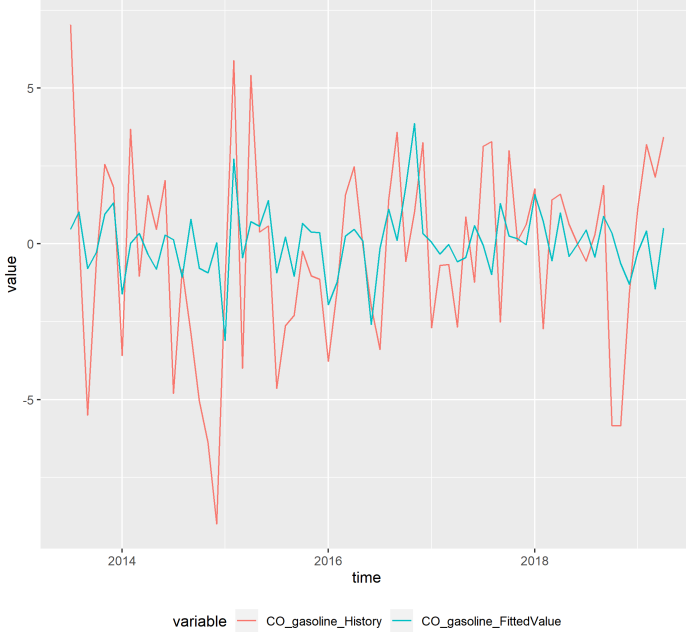


4.7.6 Model - 6

Backtesting on differenced values

CO_gasoline - Backtesting

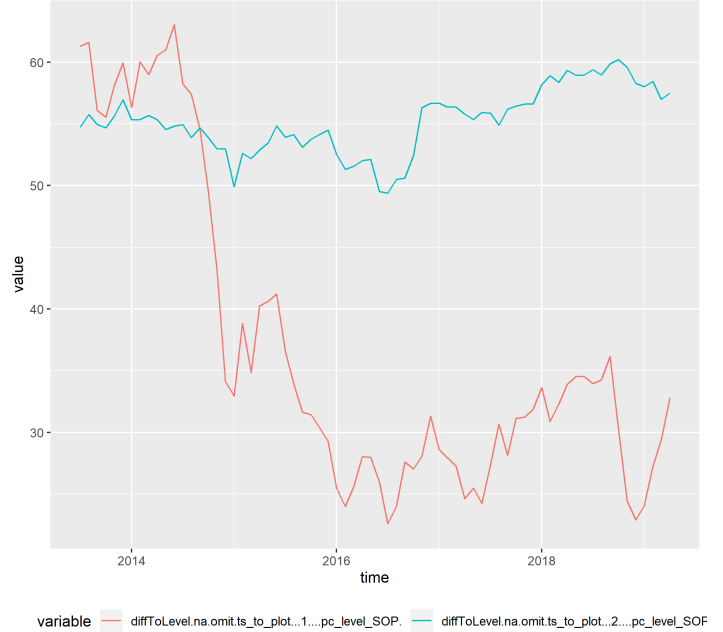
R²: 0.11/ 12.6966 * USGG10YR_IndexMoMPercentChangeL0



Backtesting on level values

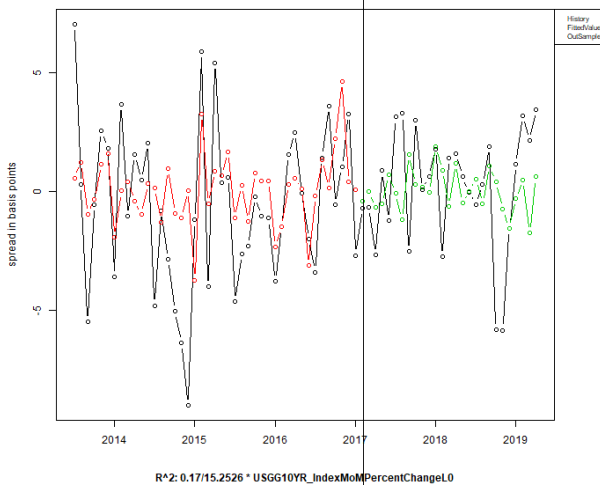
CO_gasoline - Level Value Backtesting

R²: 0.11/ 12.6966 * USGG10YR_IndexMoMPercentChangeL0



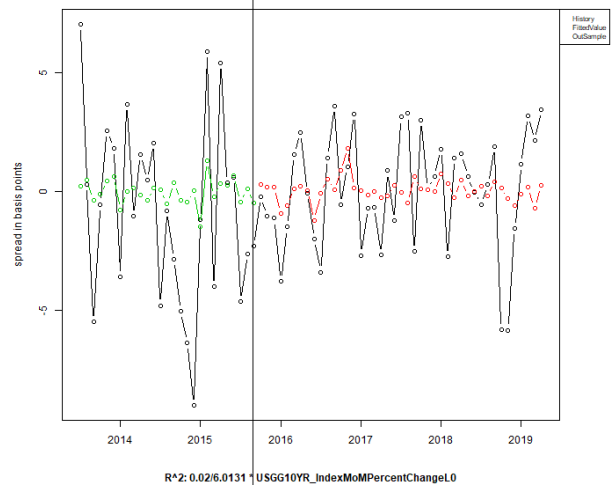
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

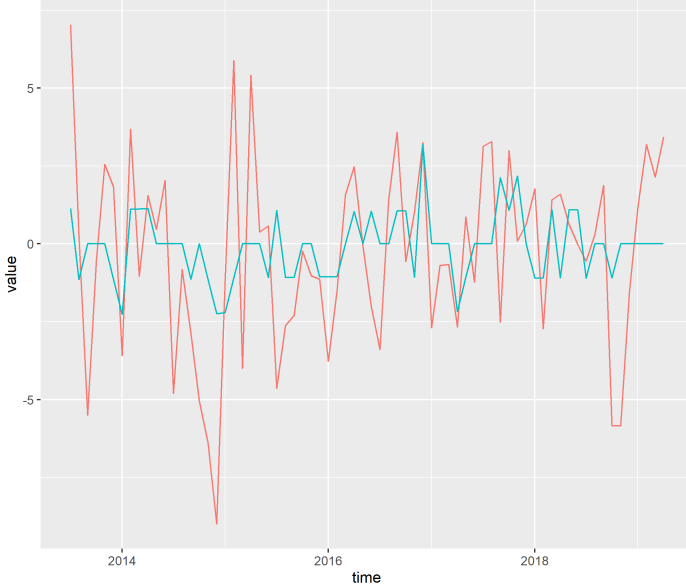


4.7.7 Model - 7

Backtesting on differenced values

CO_gasoline - Backtesting

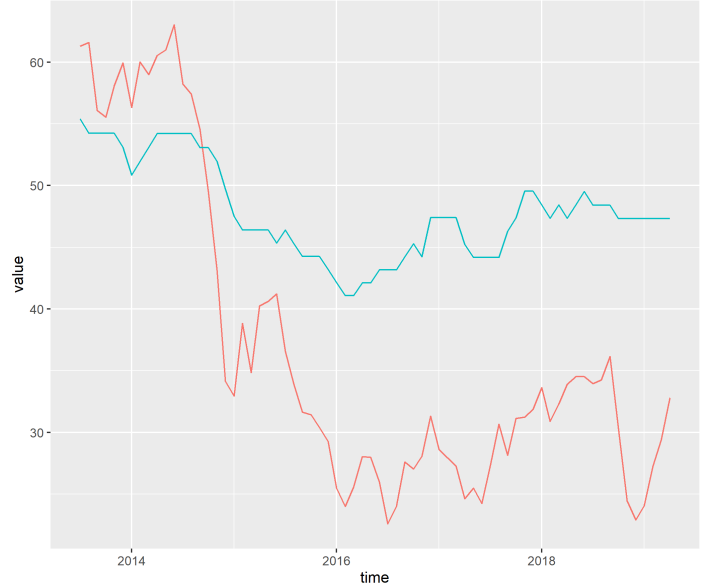
R²: 0.1/ -146.8314 * ISRATIOMoMPercentChangeL0



Backtesting on level values

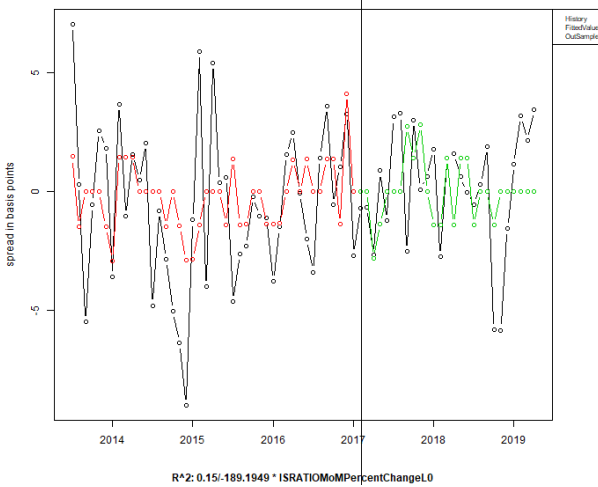
CO_gasoline - Level Value Backtesting

R²: 0.1/ -146.8314 * ISRATIOMoMPercentChangeL0



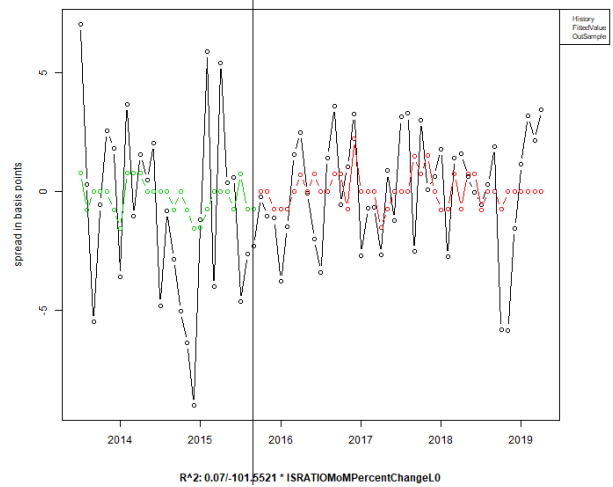
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019

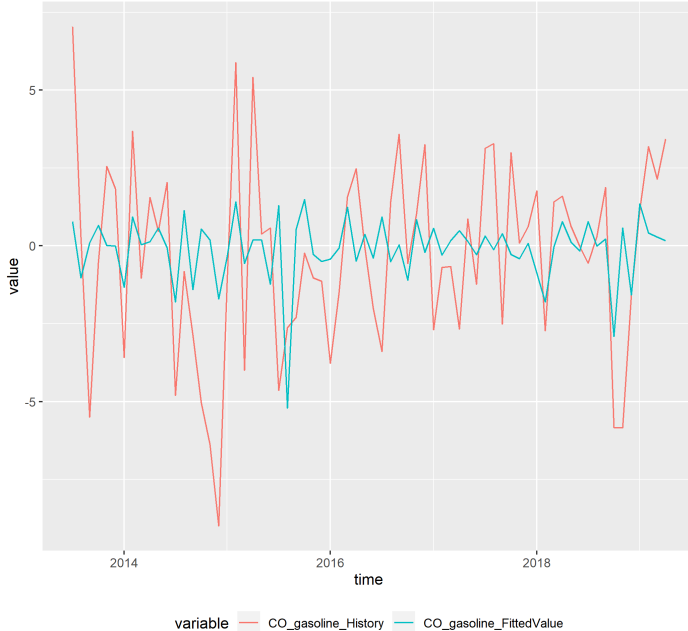


4.7.8 Model - 8

Backtesting on differenced values

CO_gasoline - Backtesting

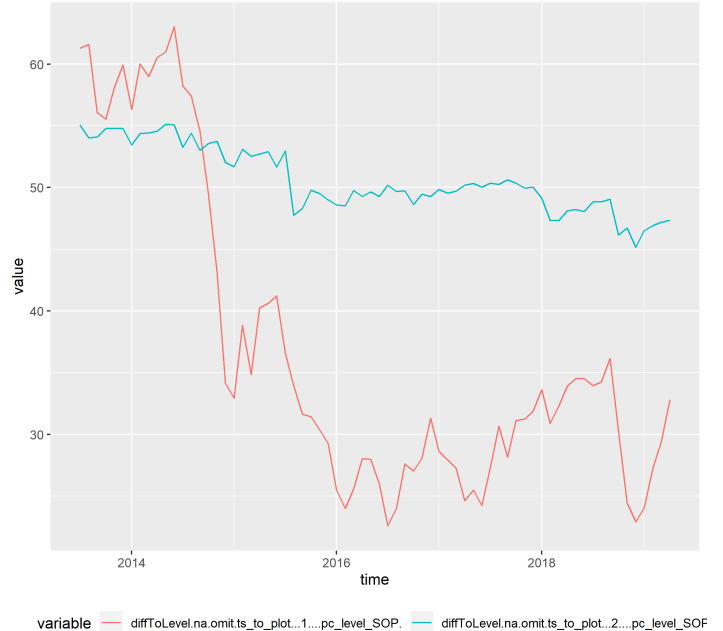
$R^2: 0.1 / -3.8705 * VIX_IndexMoMPercentChangeL0$



Backtesting on level values

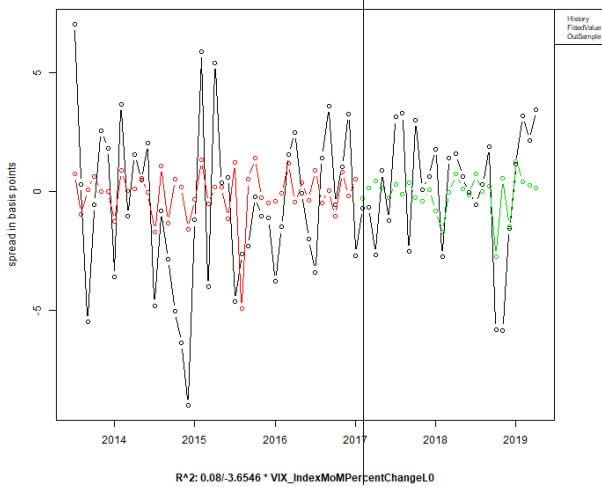
CO_gasoline - Level Value Backtesting

$R^2: 0.1 / -3.8705 * VIX_IndexMoMPercentChangeL0$



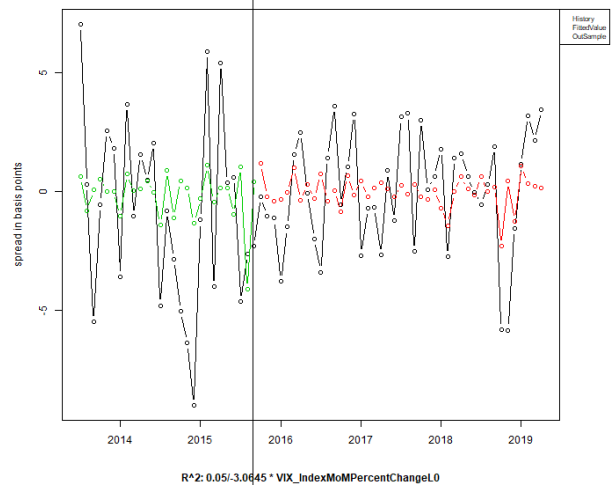
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



4.8 Two Factor Model for CO_gasoline

	V1	V2	r2adj	Var1	Var2	aic	RMSE	Criteria1	Criteria2	Criteria3
1	USGOVTMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.44	-0.03	-20.78	318.71	2.27			
2	USGOVTMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.44	-605.92	-20.76	318.74	2.27			
3	DTC THFNMMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.41	-31.41	-19.64	322.43	3.07			
4	DTC THFNMMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.41	-0.00	-19.65	322.47	3.08			
5	IPFINALMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.41	-76.23	-18.10	322.97	2.47			
6	IPFINALMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.41	-0.74	-18.10	323.06	2.47			
7	IPB51222SMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.40	-0.12	-19.30	323.60	2.67			
8	ISRATIOMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.40	-89.18	-17.38	323.60	2.30			
9	IPB51222SMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.40	-11.70	-19.27	323.63	2.66			
10	ISRATIOMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.40	-64.88	-17.42	323.79	2.30			
11	DTCOLNVHFNNMMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.40	-0.00	-19.44	324.18	2.73			
12	IPCONGDMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.40	-47.99	-18.33	324.24	2.44			
13	DTCOLNVHFNNMMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.40	-40.50	-19.41	324.29	2.72			
14	IPFPNSSMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.40	-57.43	-18.25	324.37	2.46			
15	IPCONGDMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.40	-0.47	-18.33	324.38	2.44			
16	INVESTMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.40	-80.57	-18.75	324.40	2.48			
17	IPFPNSSMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.39	-0.56	-18.25	324.52	2.46			
18	INVESTMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-0.02	-18.78	324.73	2.50			
19	HOUSTWMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.39	-0.01	-19.16	324.85	2.50			
20	IPMANICSMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-38.58	-17.96	324.97	2.42			
21	AWOTMANMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-3.93	-19.54	325.10	2.46			
22	DPCERA3M086SBEAMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-72.27	-19.29	325.10	2.47			
23	AWOTMANMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-17.13	-19.51	325.14	2.46			
24	AMBSLMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.39	0.01	-17.82	325.16	2.53		Fail	
25	IPMANICSMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-0.35	-18.02	325.18	2.43			
26	HOUSTWMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.39	-3.31	-18.99	325.18	2.49			
27	AAIIBEAR_IndexMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.39	-0.06	-19.02	325.21	2.48			
28	DPCERA3M086SBEAMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-0.59	-19.30	325.24	2.51			
29	IPB51222SMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.39	0.09	-18.66	325.38	2.51			
30	IPNCONGDMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.39	-45.97	-18.16	325.39	2.54			
31	AMBSLMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.39	21.90	-17.88	325.48	2.53		Fail	
32	CUMFNSMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-1.07	-18.89	325.52	2.42			
33	CUMFNSMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.39	-80.96	-18.89	325.56	2.42			
34	IPNCONGDMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.39	-0.45	-18.17	325.57	2.54			
35	IPB51222SMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.39	8.96	-18.70	325.60	2.50			
36	IPDCONGDMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.39	-0.16	-18.84	325.62	2.39			
37	IPDCONGDMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.38	-17.72	-18.80	325.68	2.40			
38	AAIIBEAR_IndexMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.38	-1.50	-19.15	325.81	2.45			
39	UEMPMEANMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.38	-0.44	-19.01	325.86	2.45			
40	UEMP5TO14MoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.38	0.00	-18.59	325.86	2.47			
41	USGOVTMoMDifferenceL0	BAA10YB_IndexMoMDifferenceL0	0.38	-0.03	-0.13	325.92	2.09			
42	USGOVTMoMPercentChangeL0	BAA10YB_IndexMoMDifferenceL0	0.38	-572.39	-0.13	325.96	2.09			
43	ism_man.pricesMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL0	0.38	0.09	-18.24	325.97	2.38			
44	CE16OVMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-0.00	-18.18	326.08	2.46			
45	HOUSTNEMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-1.22	-19.04	326.15	2.47			
46	CE16OVMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-176.93	-18.19	326.20	2.46			
47	AAIIBEAR_IndexMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0	0.38	0.06	-21.34	326.20	2.45			
48	USGG2YR_IndexMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.38	-2.84	-20.18	326.21	2.50			
49	UEMPMEANMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.38	-10.93	-19.04	326.28	2.48			
50	INDPROMoMPercentChangeL1	BAA10YB_IndexMoMPercentChangeL0	0.38	-37.33	-19.03	326.36	2.83			

Table 13: Two Factor Analysis-CO_gasolineTable 14: Relevant Variables: CO_gasoline

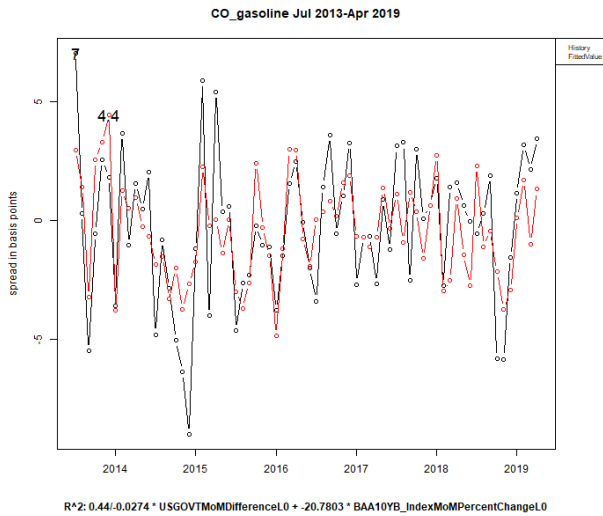
1	2	3
1 RPIMoMDifferenceL0	UEMP15T26MoMPercentChangeL0	BUSLOANSMoMDifferenceL0
2 RPIMoMDifferenceL1	UEMP15T26MoMPercentChangeL1	BUSLOANSMoMDifferenceL1
3 RPIMoMPercentChangeL0	UEMP27OVMoMDifferenceL0	BUSLOANSMoMPercentChangeL0
4 RPIMoMPercentChangeL1	UEMP27OVMoMDifferenceL1	BUSLOANSMoMPercentChangeL1
5 W875RX1MoMDifferenceL0	UEMP27OVMoMPercentChangeL0	REALLNMoMDifferenceL0
6 W875RX1MoMDifferenceL1	UEMP27OVMoMPercentChangeL1	REALLNMoMDifferenceL1
7 W875RX1MoMPercentChangeL0	PAYEMSMoMDifferenceL0	REALLNMoMPercentChangeL0
8 W875RX1MoMPercentChangeL1	PAYEMSMoMDifferenceL1	REALLNMoMPercentChangeL1
9 DPCERA3M086SBEAMoMDifferenceL0	PAYEMSMoMPercentChangeL0	WPSFD49207MoMDifferenceL0
10 DPCERA3M086SBEAMoMDifferenceL1	PAYEMSMoMPercentChangeL1	WPSFD49207MoMDifferenceL1
11 DPCERA3M086SBEAMoMPercentChangeL0	USGOODMoMDifferenceL0	WPSFD49207MoMPercentChangeL0
12 DPCERA3M086SBEAMoMPercentChangeL1	USGOODMoMDifferenceL1	WPSFD49207MoMPercentChangeL1
13 CMRMTSPLMoMDifferenceL0	USGOODMoMPercentChangeL0	WPSFD49502MoMDifferenceL0
14 CMRMTSPLMoMDifferenceL1	USGOODMoMPercentChangeL1	WPSFD49502MoMDifferenceL1
15 CMRMTSPLMoMPercentChangeL0	CES1021000001MoMDifferenceL0	WPSFD49502MoMPercentChangeL0
16 CMRMTSPLMoMPercentChangeL1	CES1021000001MoMDifferenceL1	WPSFD49502MoMPercentChangeL1
17 INDPROMoMDifferenceL0	CES1021000001MoMPercentChangeL0	WPSID61MoMDifferenceL0
18 INDPROMoMDifferenceL1	CES1021000001MoMPercentChangeL1	WPSID61MoMDifferenceL1
19 INDPROMoMPercentChangeL0	USCONSMoMDifferenceL0	WPSID61MoMPercentChangeL0
20 INDPROMoMPercentChangeL1	USCONSMoMDifferenceL1	WPSID61MoMPercentChangeL1
21 IPFPNSSMoMDifferenceL0	USCONSMoMPercentChangeL0	WPSID62MoMDifferenceL0
22 IPFPNSSMoMDifferenceL1	USCONSMoMPercentChangeL1	WPSID62MoMDifferenceL1
23 IPFPNSSMoMPercentChangeL0	MANEMPMoMDifferenceL0	WPSID62MoMPercentChangeL0
24 IPFPNSSMoMPercentChangeL1	MANEMPMoMDifferenceL1	WPSID62MoMPercentChangeL1
25 IPFINALMoMDifferenceL0	MANEMPMoMPercentChangeL0	MZMSLMoMDifferenceL0
26 IPFINALMoMDifferenceL1	MANEMPMoMPercentChangeL1	MZMSLMoMDifferenceL1
27 IPFINALMoMPercentChangeL0	DMANEMPMoMDifferenceL0	MZMSLMoMPercentChangeL0
28 IPFINALMoMPercentChangeL1	DMANEMPMoMDifferenceL1	MZMSLMoMPercentChangeL1
29 IPCONGDMoMDifferenceL0	DMANEMPMoMPercentChangeL0	DTCOLNVHFNMMDifferenceL0
30 IPCONGDMoMDifferenceL1	DMANEMPMoMPercentChangeL1	DTCOLNVHFNMMDifferenceL1
31 IPCONGDMoMPercentChangeL0	SRVPRDMoMDifferenceL0	DTCOLNVHFNMMDifferenceL0
32 IPCONGDMoMPercentChangeL1	SRVPRDMoMDifferenceL1	DTCOLNVHFNMMDifferenceL1
33 IPDCONGDMoMDifferenceL0	SRVPRDMoMPercentChangeL0	DTCTHFNMMDifferenceL0
34 IPDCONGDMoMDifferenceL1	SRVPRDMoMPercentChangeL1	DTCTHFNMMDifferenceL1
35 IPDCONGDMoMPercentChangeL0	USWTRADEMoMDifferenceL0	DTCTHFNMMDifferenceL0
36 IPDCONGDMoMPercentChangeL1	USWTRADEMoMDifferenceL1	DTCTHFNMMDifferenceL1
37 IPNCONGDMoMDifferenceL0	USWTRADEMoMPercentChangeL0	INVESTMoMDifferenceL0
38 IPNCONGDMoMDifferenceL1	USWTRADEMoMPercentChangeL1	INVESTMoMDifferenceL1
39 IPNCONGDMoMPercentChangeL0	USTRADEMoMDifferenceL0	INVESTMoMPercentChangeL0
40 IPNCONGDMoMPercentChangeL1	USTRADEMoMDifferenceL1	INVESTMoMPercentChangeL1
41 IPBUSEQMoMDifferenceL0	USTRADEMoMPercentChangeL0	ism_man_emplMoMDifferenceL0
42 IPBUSEQMoMDifferenceL1	USTRADEMoMPercentChangeL1	ism_man_emplMoMDifferenceL1
43 IPBUSEQMoMPercentChangeL0	USFIREMoMDifferenceL0	ism_man_emplMoMPercentChangeL0
44 IPBUSEQMoMPercentChangeL1	USFIREMoMDifferenceL1	ism_man_emplMoMPercentChangeL1
45 IPMATMoMDifferenceL0	USFIREMoMPercentChangeL0	ism_man_newordersMoMDifferenceL0
46 IPMATMoMDifferenceL1	USFIREMoMPercentChangeL1	ism_man_newordersMoMDifferenceL1
47 IPMATMoMPercentChangeL0	USGOVTMoMDifferenceL0	ism_man_newordersMoMPercentChangeL0

48	IPMATMoMPercentChangeL1	USGOVTMoMDifferenceL1	ism_man_newordersMoMPercentChangeL1
49	IPDMATMoMDifferenceL0	USGOVTMoMPercentChangeL0	ism_man_pmiMoMDifferenceL0
50	IPDMATMoMDifferenceL1	USGOVTMoMPercentChangeL1	ism_man_pmiMoMDifferenceL1
51	IPDMATMoMPercentChangeL0	CES0600000007MoMDifferenceL0	ism_man_pmiMoMPercentChangeL0
52	IPDMATMoMPercentChangeL1	CES0600000007MoMDifferenceL1	ism_man_pmiMoMPercentChangeL1
53	IPNMATMoMDifferenceL0	CES0600000007MoMPercentChangeL0	ism_man_pricesMoMDifferenceL0
54	IPNMATMoMDifferenceL1	CES0600000007MoMPercentChangeL1	ism_man_pricesMoMDifferenceL1
55	IPNMATMoMPercentChangeL0	AWOTMANMoMDifferenceL0	ism_man_pricesMoMPercentChangeL0
56	IPNMATMoMPercentChangeL1	AWOTMANMoMDifferenceL1	ism_man_pricesMoMPercentChangeL1
57	IPMANSICSMoMDifferenceL0	AWOTMANMoMPercentChangeL0	USGG2YR_IndexMoMDifferenceL0
58	IPMANSICSMoMDifferenceL1	AWOTMANMoMPercentChangeL1	USGG2YR_IndexMoMDifferenceL1
59	IPMANSICSMoMPercentChangeL0	AWHMANMoMDifferenceL0	USGG2YR_IndexMoMPercentChangeL0
60	IPMANSICSMoMPercentChangeL1	AWHMANMoMDifferenceL1	USGG2YR_IndexMoMPercentChangeL1
61	IPB51222SMoMDifferenceL0	AWHMANMoMPercentChangeL0	USGG10YR_IndexMoMDifferenceL0
62	IPB51222SMoMDifferenceL1	AWHMANMoMPercentChangeL1	USGG10YR_IndexMoMDifferenceL1
63	IPB51222SMoMPercentChangeL0	HOUSTMoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0
64	IPB51222SMoMPercentChangeL1	HOUSTMoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL1
65	IPFUELSMoMDifferenceL0	HOUSTMoMPercentChangeL0	USGG30YR_IndexMoMDifferenceL0
66	IPFUELSMoMDifferenceL1	HOUSTMoMPercentChangeL1	USGG30YR_IndexMoMDifferenceL1
67	IPFUELSMoMPercentChangeL0	HOUSTNEMoMDifferenceL0	USGG30YR_IndexMoMPercentChangeL0
68	IPFUELSMoMPercentChangeL1	HOUSTNEMoMDifferenceL1	USGG30YR_IndexMoMPercentChangeL1
69	CUMFNSMoMDifferenceL0	HOUSTNEMoMPercentChangeL0	VIX_IndexMoMDifferenceL0
70	CUMFNSMoMDifferenceL1	HOUSTNEMoMPercentChangeL1	VIX_IndexMoMDifferenceL1
71	CUMFNSMoMPercentChangeL0	HOUSTMWMoMDifferenceL0	VIX_IndexMoMPercentChangeL0
72	CUMFNSMoMPercentChangeL1	HOUSTMWMoMDifferenceL1	VIX_IndexMoMPercentChangeL1
73	CLF16OVMoMDifferenceL0	HOUSTMWMoMPercentChangeL0	MOVE_IndexMoMDifferenceL0
74	CLF16OVMoMDifferenceL1	HOUSTMWMoMPercentChangeL1	MOVE_IndexMoMDifferenceL1
75	CLF16OVMoMPercentChangeL0	HOUSTSMoMDifferenceL0	MOVE_IndexMoMPercentChangeL0
76	CLF16OVMoMPercentChangeL1	HOUSTSMoMDifferenceL1	MOVE_IndexMoMPercentChangeL1
77	CE16OVMoMDifferenceL0	HOUSTSMoMPercentChangeL0	AAIIBULL_IndexMoMDifferenceL0
78	CE16OVMoMDifferenceL1	HOUSTSMoMPercentChangeL1	AAIIBULL_IndexMoMDifferenceL1
79	CE16OVMoMPercentChangeL0	HOUSTWMoMDifferenceL0	AAIIBULL_IndexMoMPercentChangeL0
80	CE16OVMoMPercentChangeL1	HOUSTWMoMDifferenceL1	AAIIBULL_IndexMoMPercentChangeL1
81	UNRATEMoMDifferenceL0	HOUSTWMoMPercentChangeL0	AAIIBEAR_IndexMoMDifferenceL0
82	UNRATEMoMDifferenceL1	HOUSTWMoMPercentChangeL1	AAIIBEAR_IndexMoMDifferenceL1
83	UNRATEMoMPercentChangeL0	ISRATIOMoMDifferenceL0	AAIIBEAR_IndexMoMPercentChangeL0
84	UNRATEMoMPercentChangeL1	ISRATIOMoMDifferenceL1	AAIIBEAR_IndexMoMPercentChangeL1
85	UEMPMEANMoMDifferenceL0	ISRATIOMoMPercentChangeL0	SPX_Index_14DRSIMoMDifferenceL0
86	UEMPMEANMoMDifferenceL1	ISRATIOMoMPercentChangeL1	SPX_Index_14DRSIMoMDifferenceL1
87	UEMPMEANMoMPercentChangeL0	M1SLMoMDifferenceL0	SPX_Index_14DRSIMoMPercentChangeL0
88	UEMPMEANMoMPercentChangeL1	M1SLMoMDifferenceL1	SPX_Index_14DRSIMoMPercentChangeL1
89	UEMPLT5MoMDifferenceL0	M1SLMoMPercentChangeL0	SPX_Index_30DRSIMoMDifferenceL0
90	UEMPLT5MoMDifferenceL1	M1SLMoMPercentChangeL1	SPX_Index_30DRSIMoMDifferenceL1
91	UEMPLT5MoMPercentChangeL0	M2SLMoMDifferenceL0	SPX_Index_30DRSIMoMPercentChangeL0
92	UEMPLT5MoMPercentChangeL1	M2SLMoMDifferenceL1	SPX_Index_30DRSIMoMPercentChangeL1
93	UEMP5TO14MoMDifferenceL0	M2SLMoMPercentChangeL0	TED_G_IndexMoMDifferenceL0
94	UEMP5TO14MoMDifferenceL1	M2SLMoMPercentChangeL1	TED_G_IndexMoMDifferenceL1
95	UEMP5TO14MoMPercentChangeL0	M2REALMoMDifferenceL0	TED_G_IndexMoMPercentChangeL0
96	UEMP5TO14MoMPercentChangeL1	M2REALMoMDifferenceL1	TED_G_IndexMoMPercentChangeL1
97	UEMP15OVMoMDifferenceL0	M2REALMoMPercentChangeL0	BAA10YB_IndexMoMDifferenceL0
98	UEMP15OVMoMDifferenceL1	M2REALMoMPercentChangeL1	BAA10YB_IndexMoMDifferenceL1
99	UEMP15OVMoMPercentChangeL0	AMBSLMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0
100	UEMP15OVMoMPercentChangeL1	AMBSLMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL1
101	UEMP15T26MoMDifferenceL0	AMBSLMoMPercentChangeL0	
102	UEMP15T26MoMDifferenceL1	AMBSLMoMPercentChangeL1	

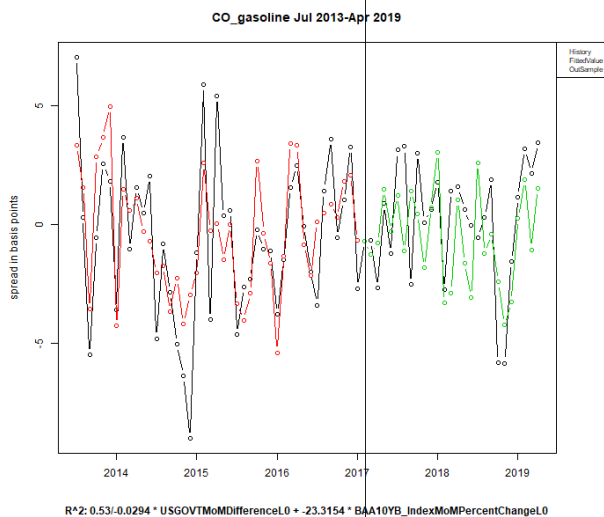
4.8.1 Model - 1

Backtesting on differenced values

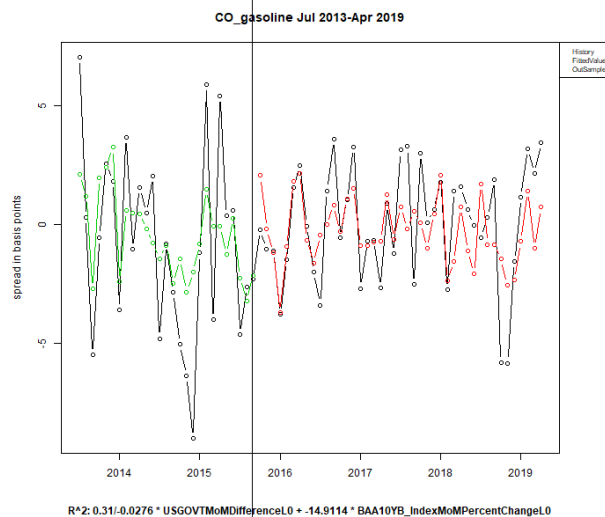
Backtesting on level values



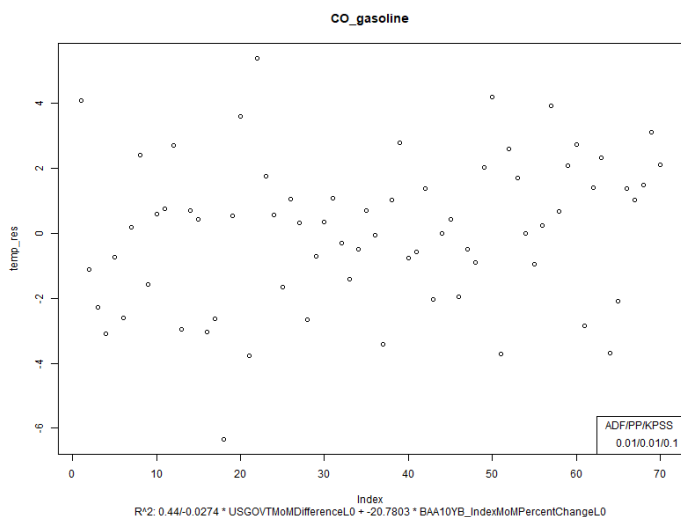
Out-of-Sample testing



Out-of-Sample testing



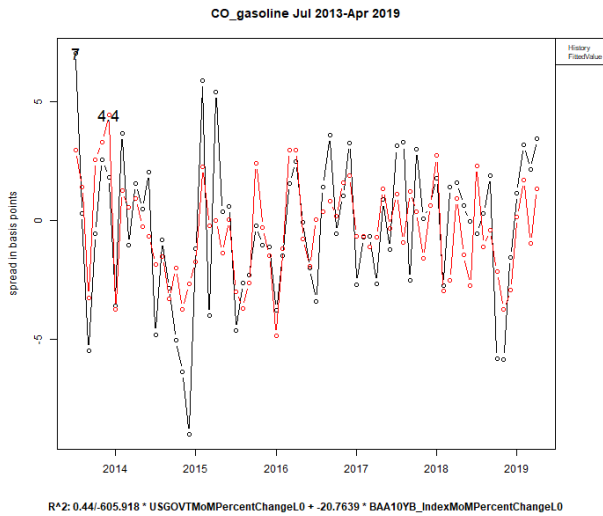
Stationarity of Residuals



4.8.2 Model - 2

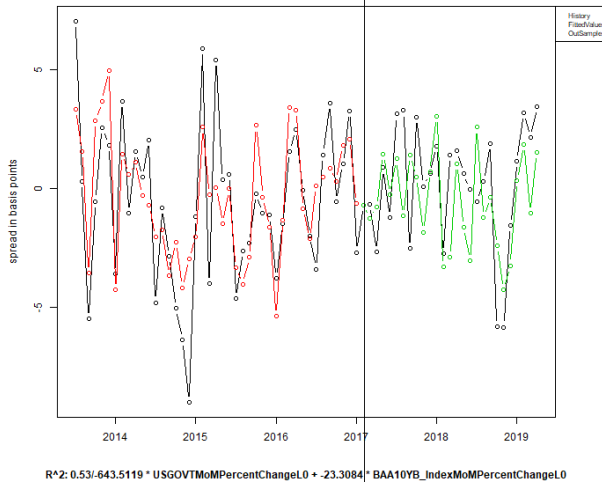
Backtesting on differenced values

Backtesting on level values



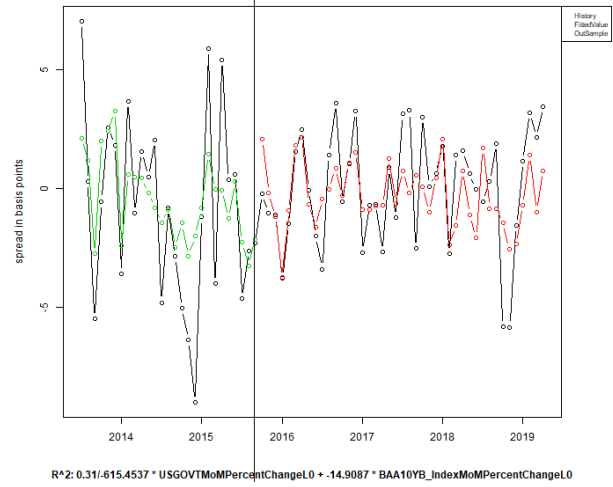
Out-of-Sample testing

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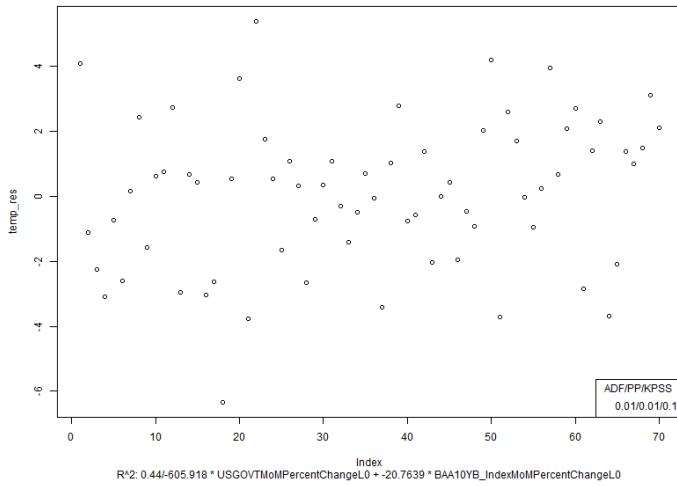
Out-of-Sample testing

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Stationarity of Residuals

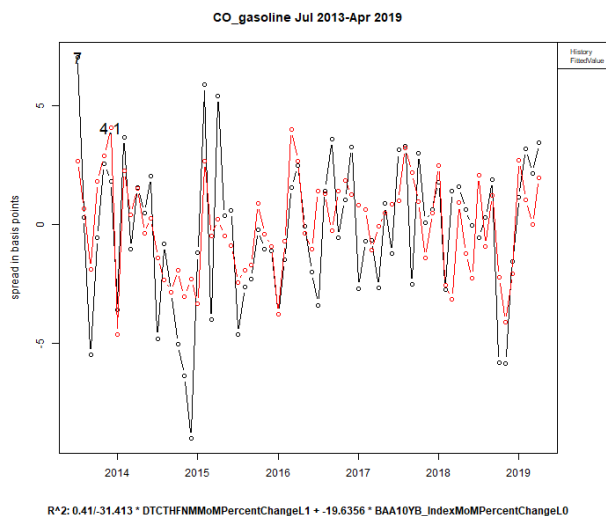
CO_gasoline



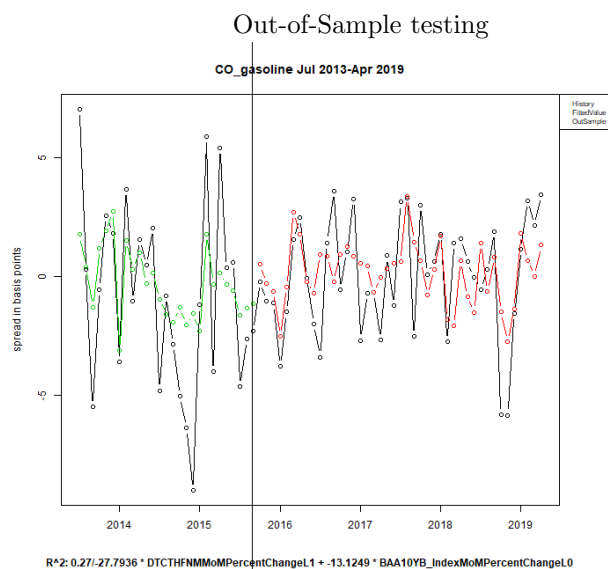
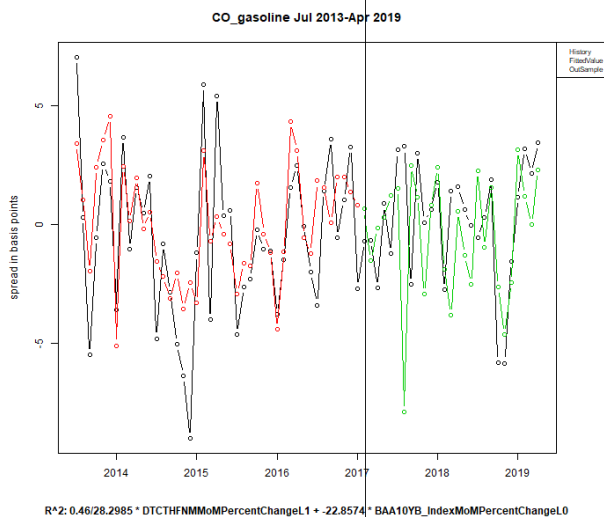
4.8.3 Model - 3

Backtesting on differenced values

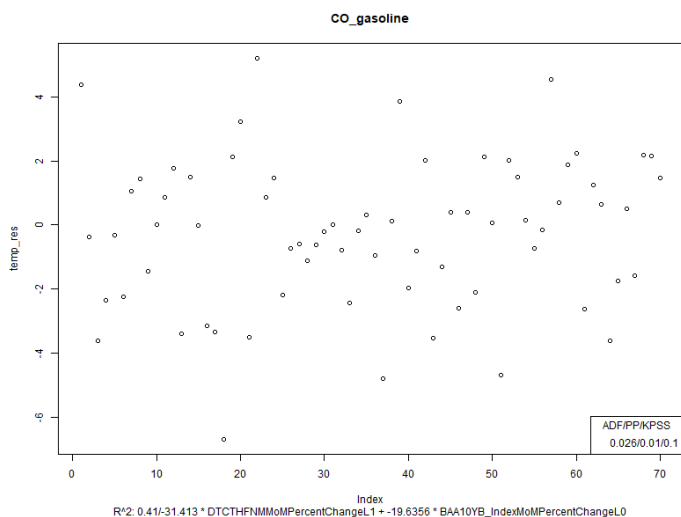
Backtesting on level values



Out-of-Sample testing



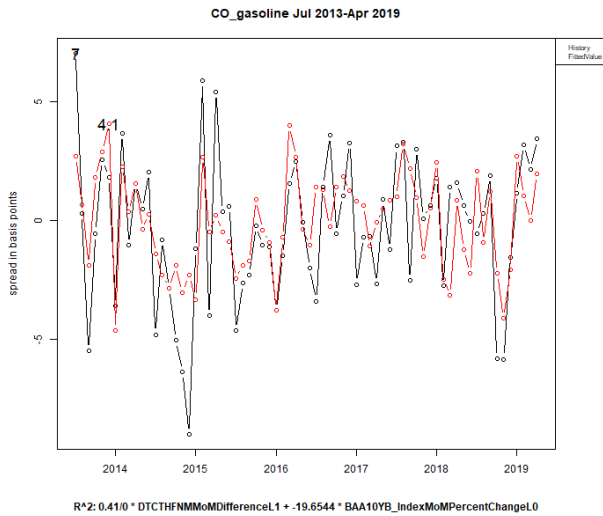
Stationarity of Residuals



4.8.4 Model - 4

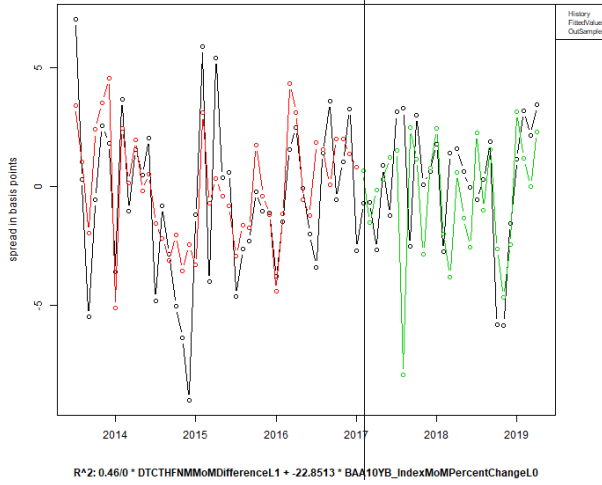
Backtesting on differenced values

Backtesting on level values



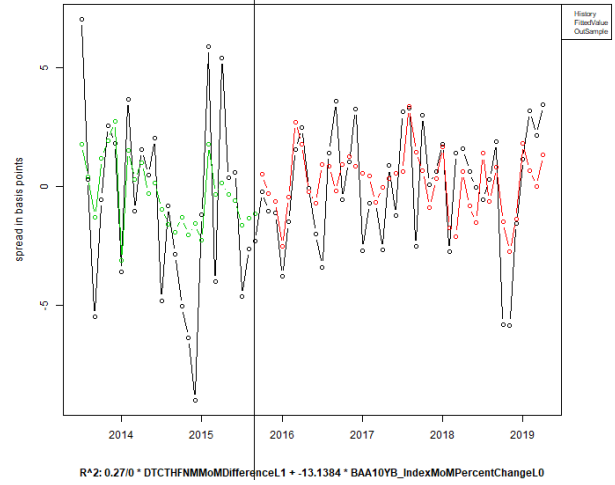
Out-of-Sample testing

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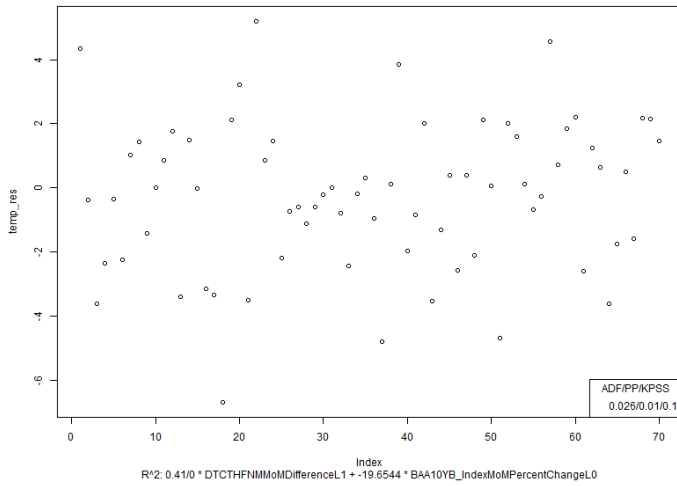
Out-of-Sample testing

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Stationarity of Residuals

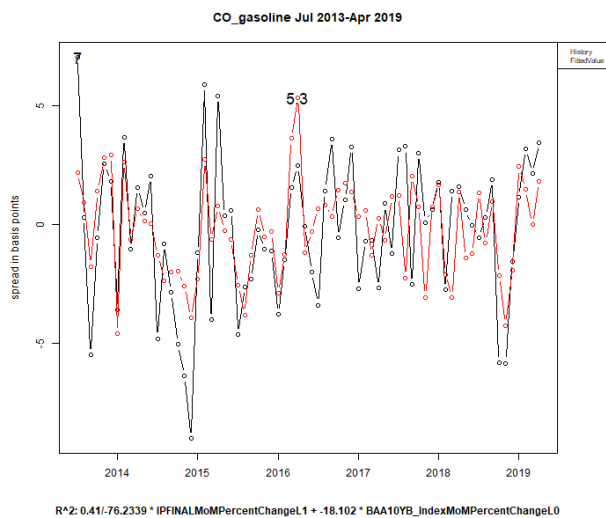
CO_gasoline



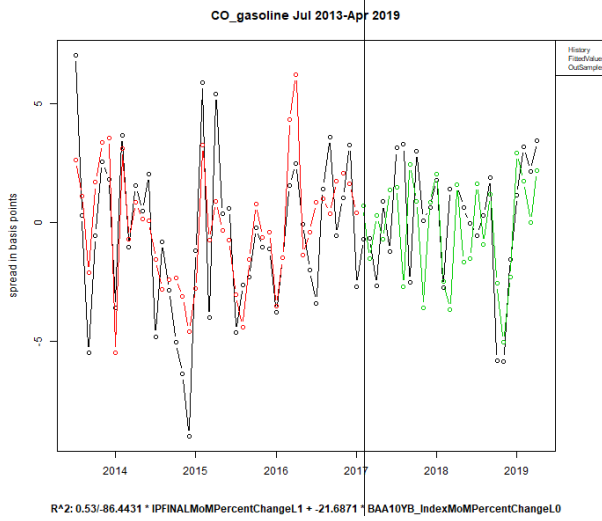
4.8.5 Model - 5

Backtesting on differenced values

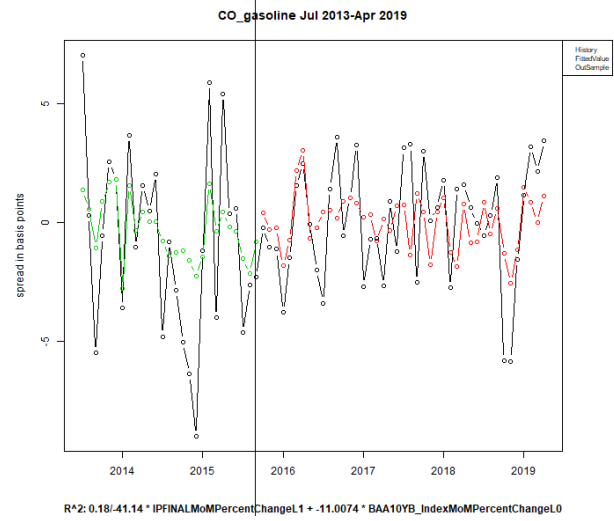
Backtesting on level values



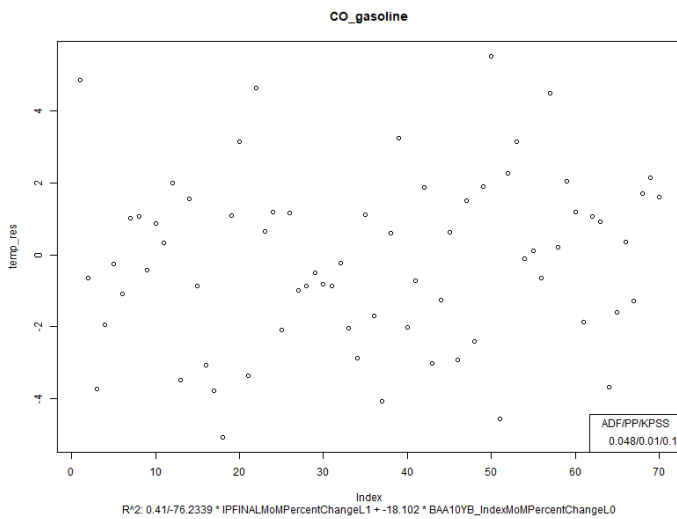
Out-of-Sample testing



Out-of-Sample testing



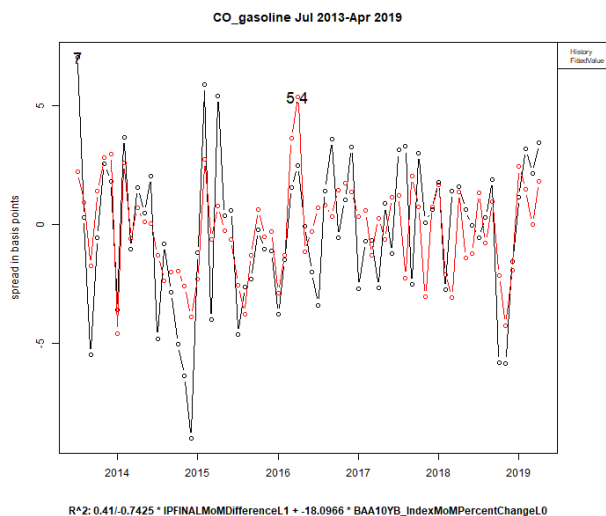
Stationarity of Residuals



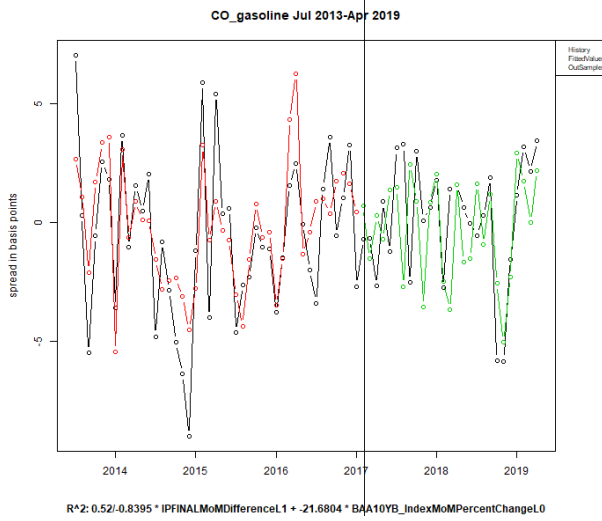
4.8.6 Model - 6

Backtesting on differenced values

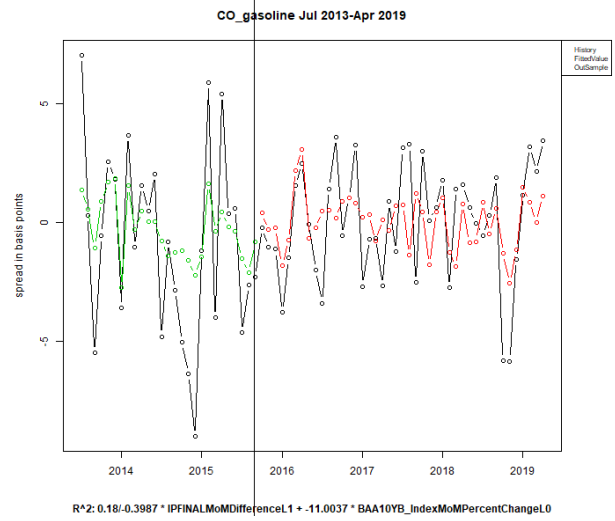
Backtesting on level values



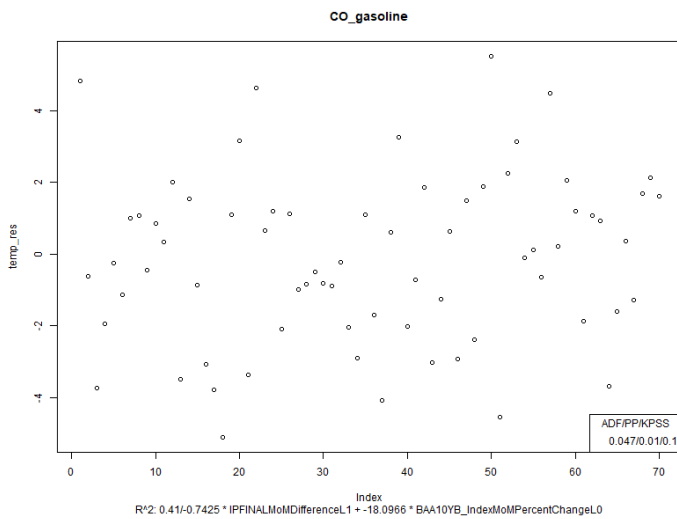
Out-of-Sample testing



Out-of-Sample testing



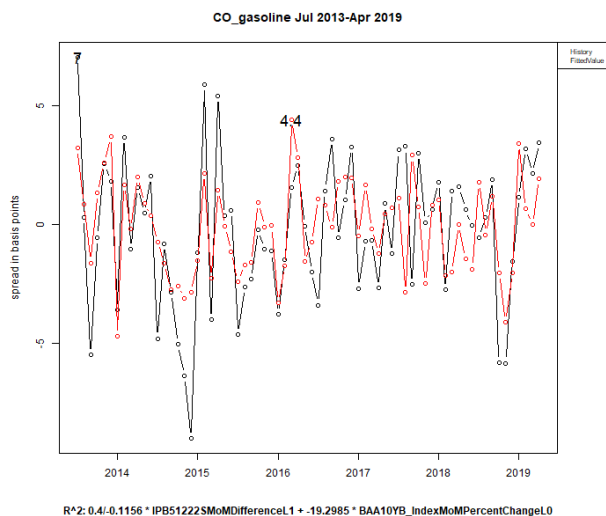
Stationarity of Residuals



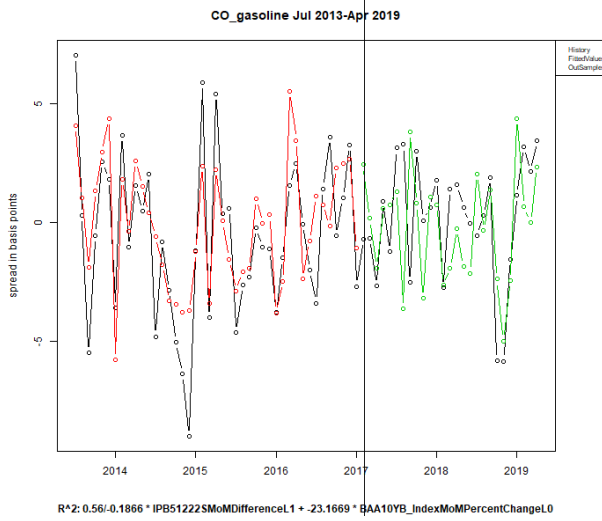
4.8.7 Model - 7

Backtesting on differenced values

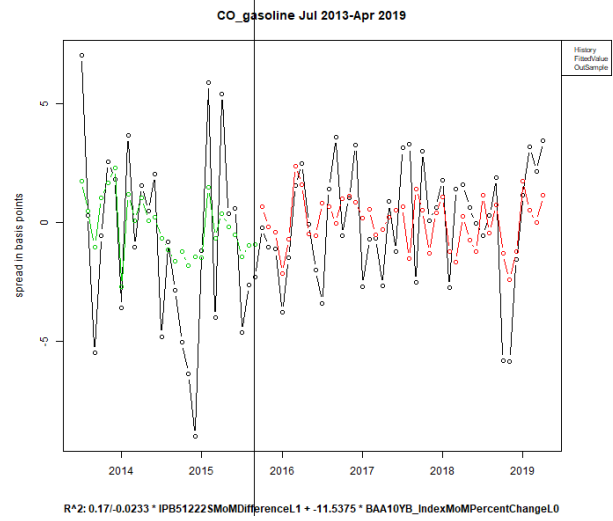
Backtesting on level values



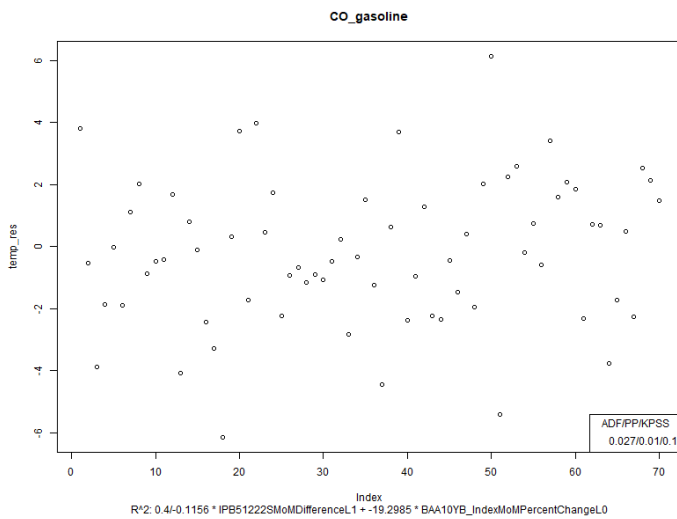
Out-of-Sample testing



Out-of-Sample testing



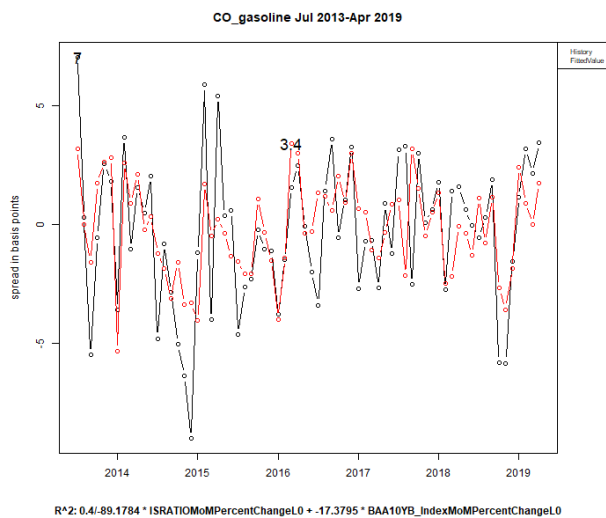
Stationarity of Residuals



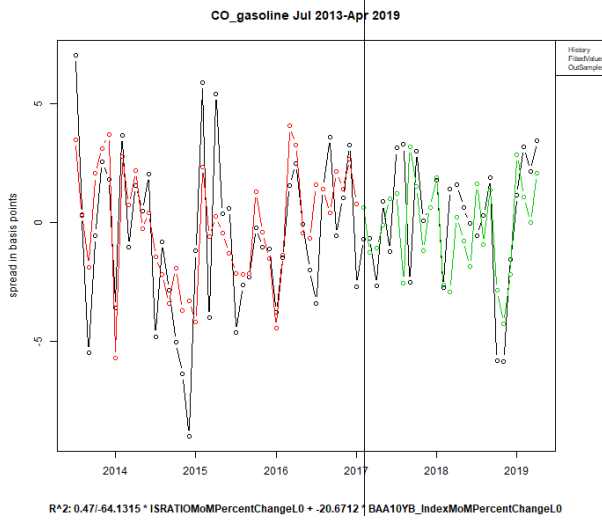
4.8.8 Model - 8

Backtesting on differenced values

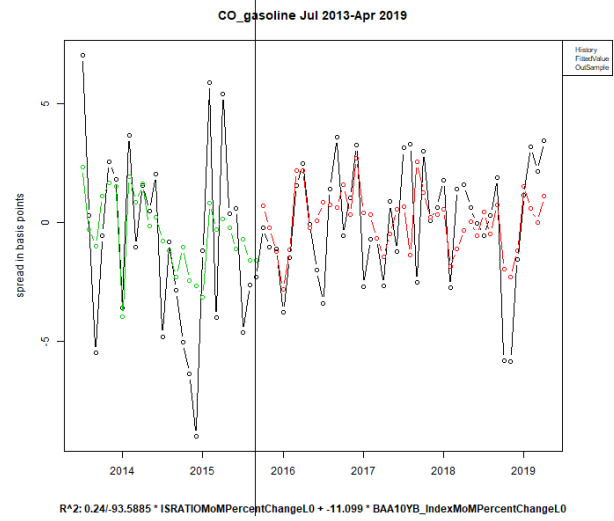
Backtesting on level values



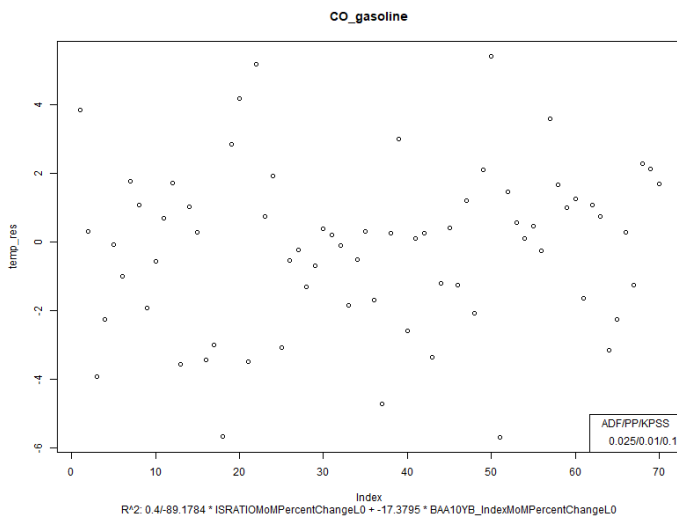
Out-of-Sample testing



Out-of-Sample testing



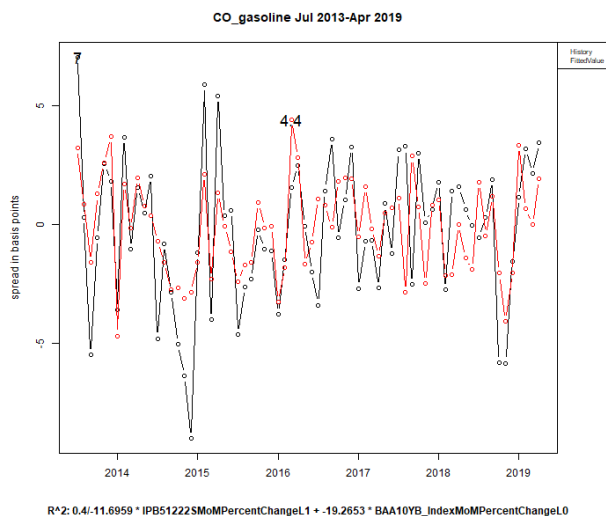
Stationarity of Residuals



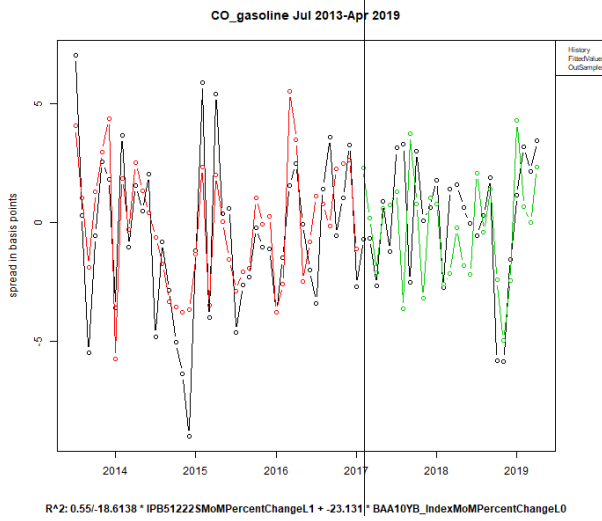
4.8.9 Model - 9

Backtesting on differenced values

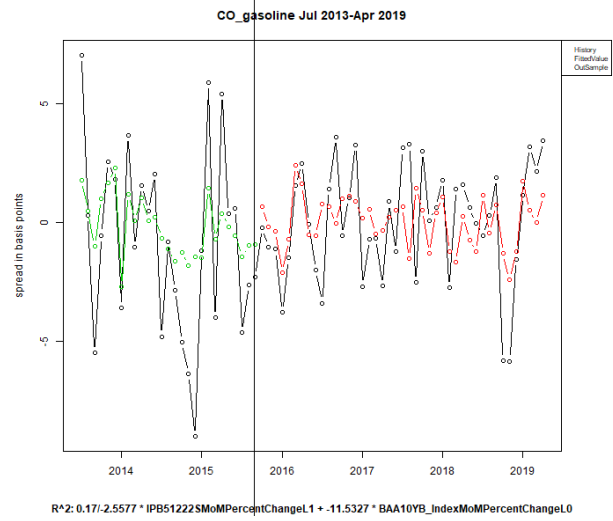
Backtesting on level values



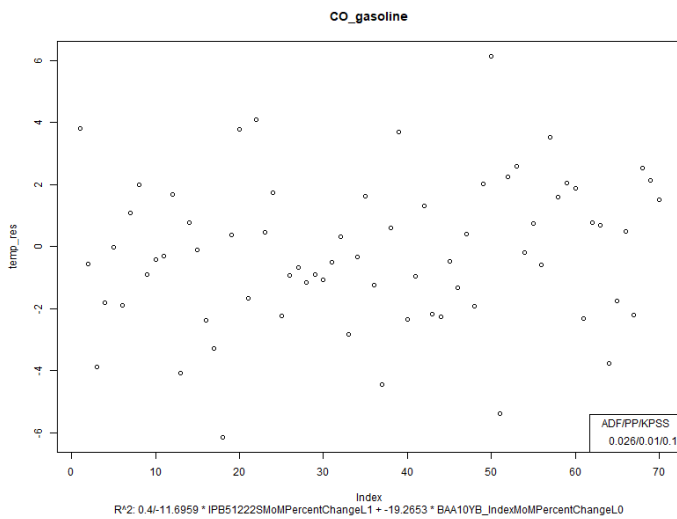
Out-of-Sample testing



Out-of-Sample testing



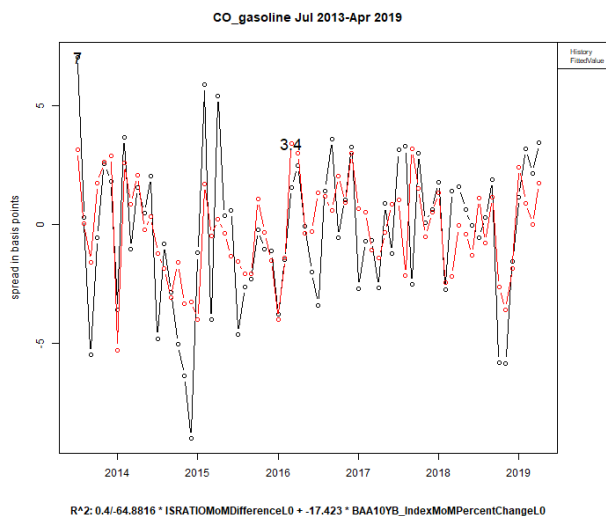
Stationarity of Residuals



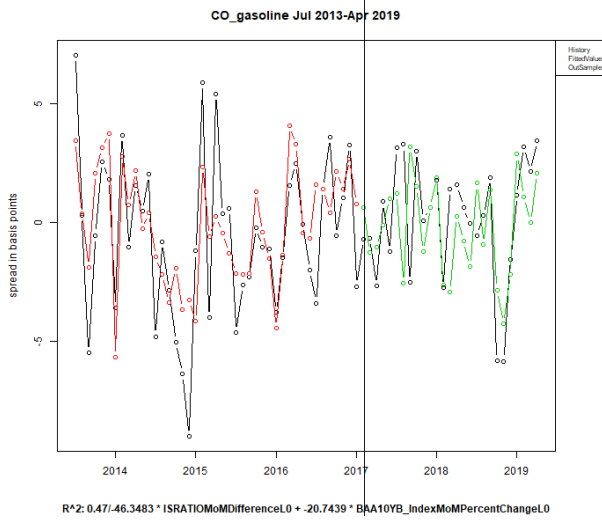
4.8.10 Model - 10

Backtesting on differenced values

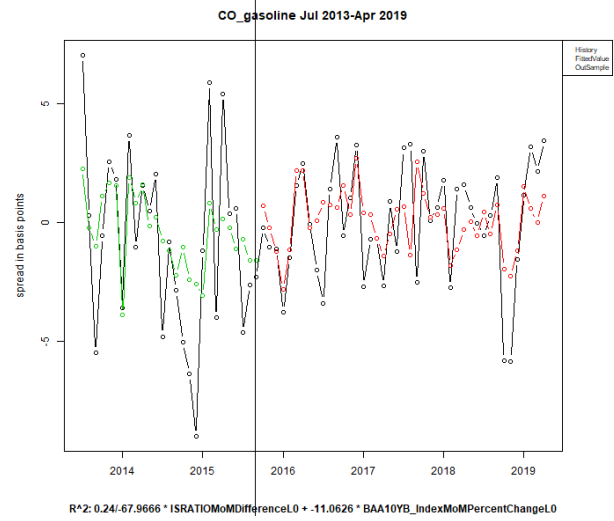
Backtesting on level values



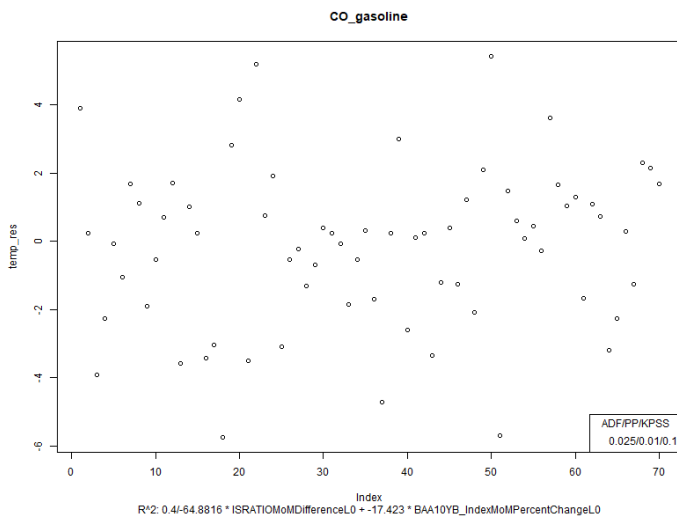
Out-of-Sample testing



Out-of-Sample testing



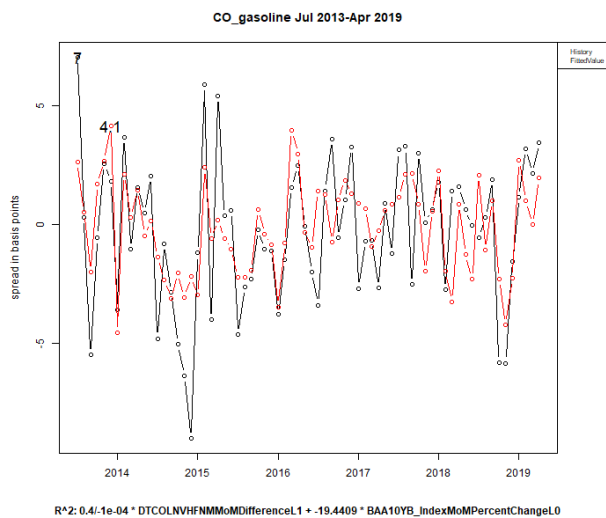
Stationarity of Residuals



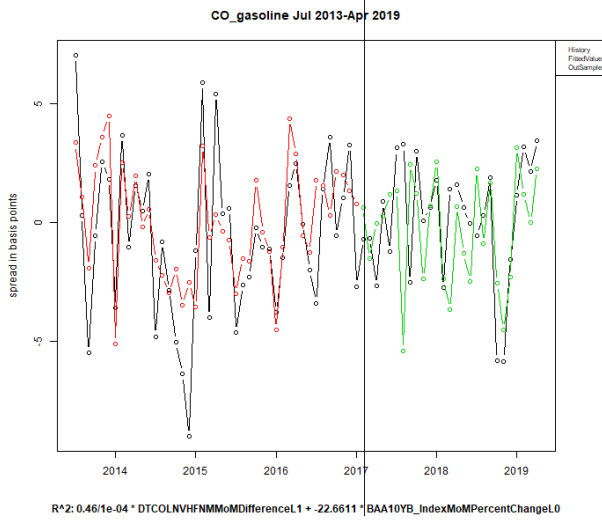
4.8.11 Model - 11

Backtesting on differenced values

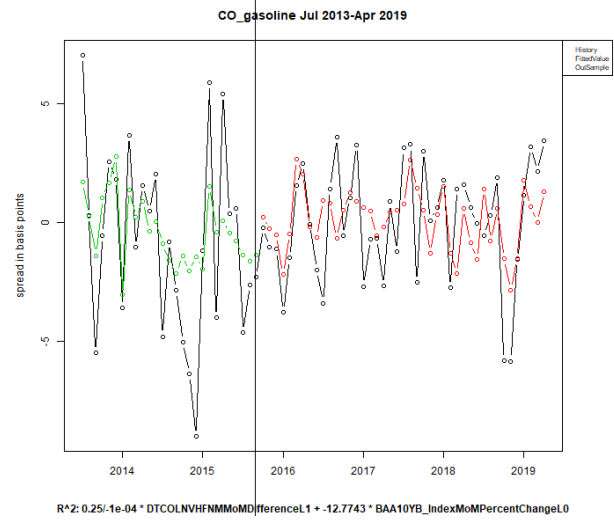
Backtesting on level values



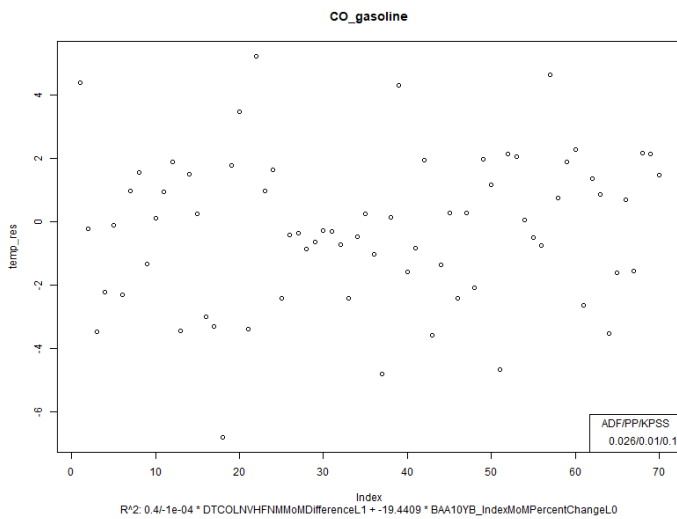
Out-of-Sample testing



Out-of-Sample testing



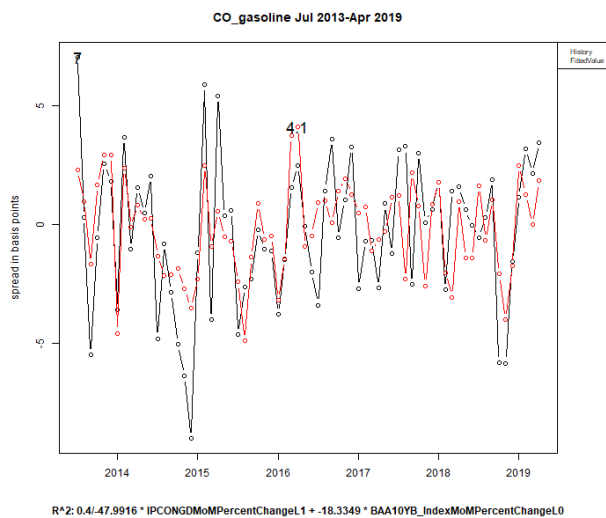
Stationarity of Residuals



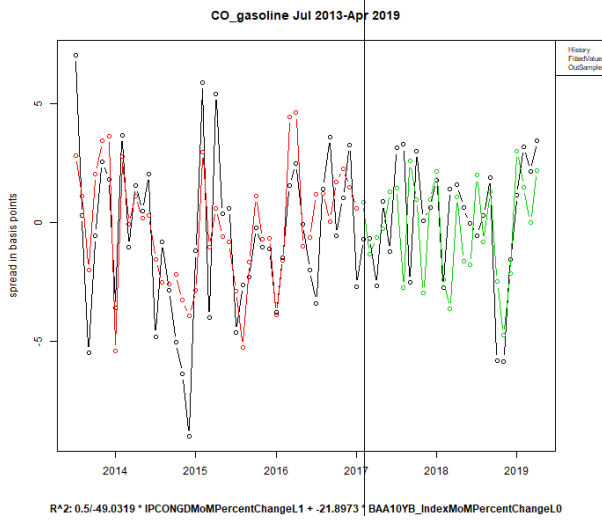
4.8.12 Model - 12

Backtesting on differenced values

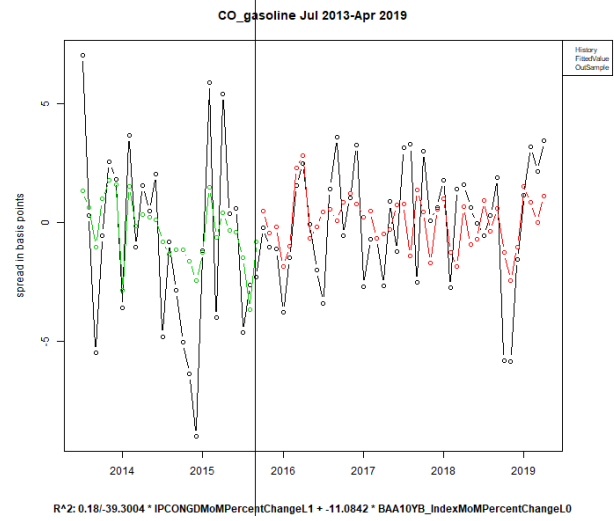
Backtesting on level values



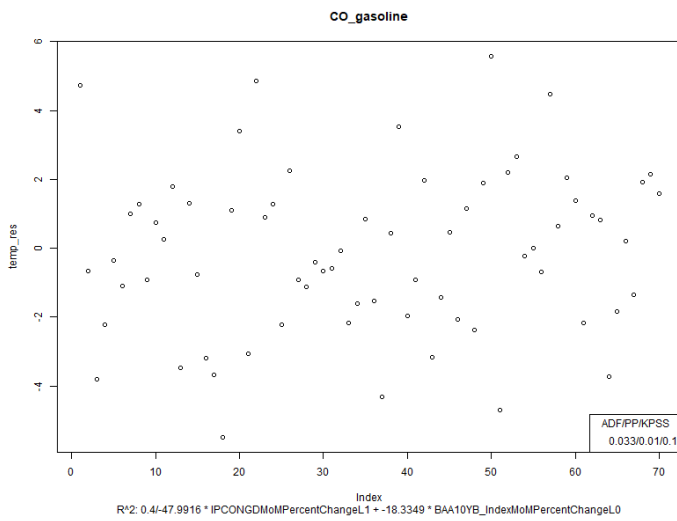
Out-of-Sample testing



Out-of-Sample testing



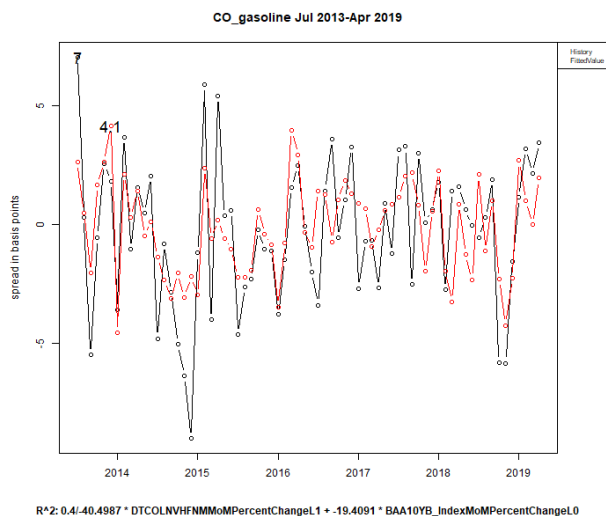
Stationarity of Residuals



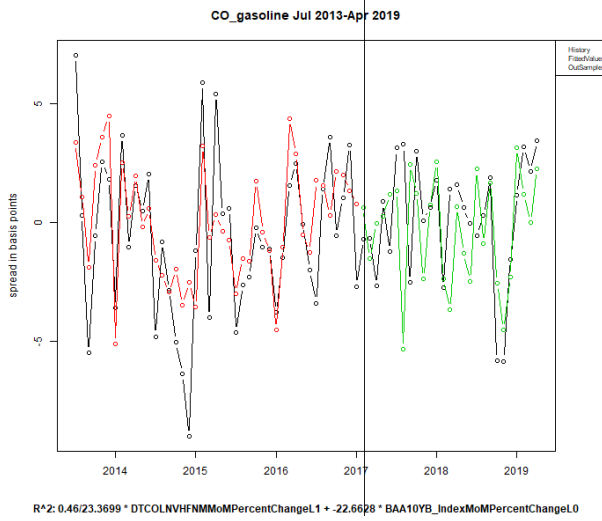
4.8.13 Model - 13

Backtesting on differenced values

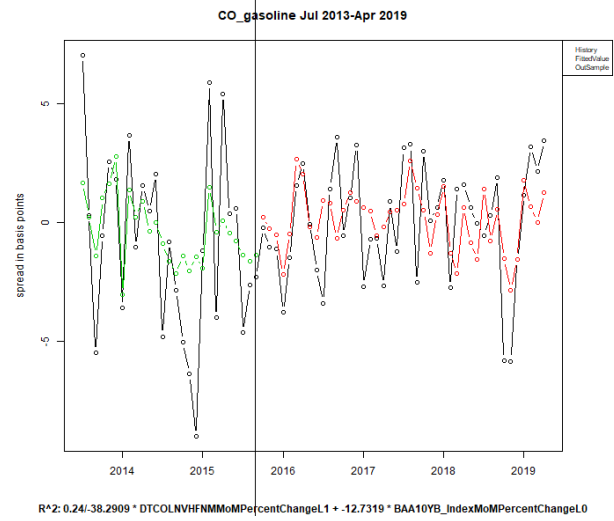
Backtesting on level values



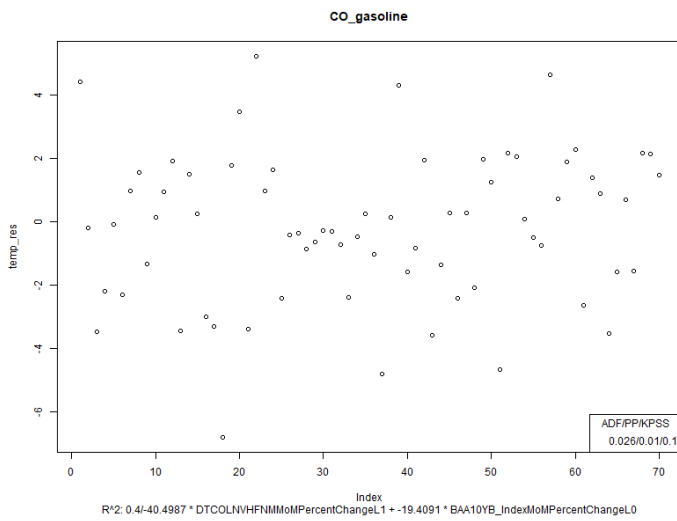
Out-of-Sample testing



Out-of-Sample testing



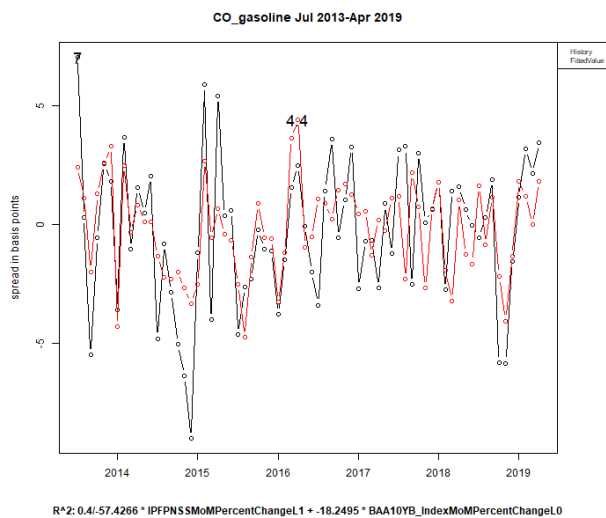
Stationarity of Residuals



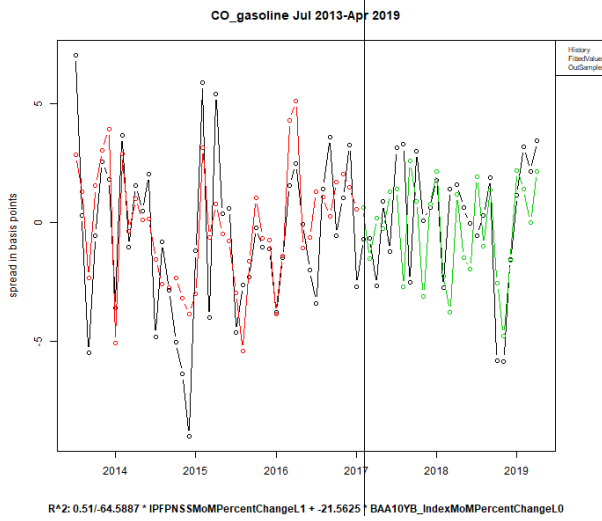
4.8.14 Model - 14

Backtesting on differenced values

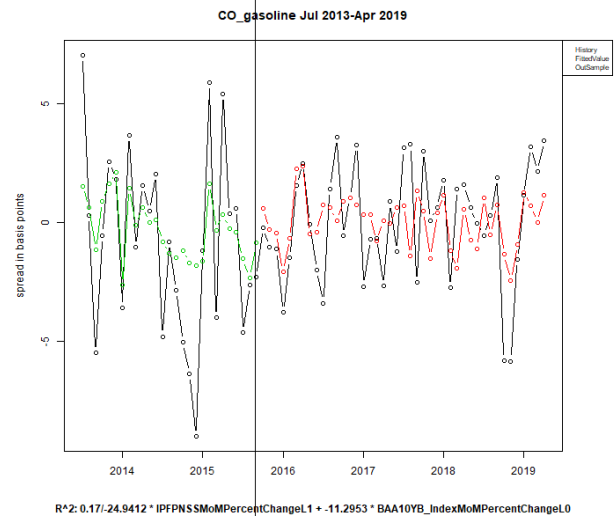
Backtesting on level values



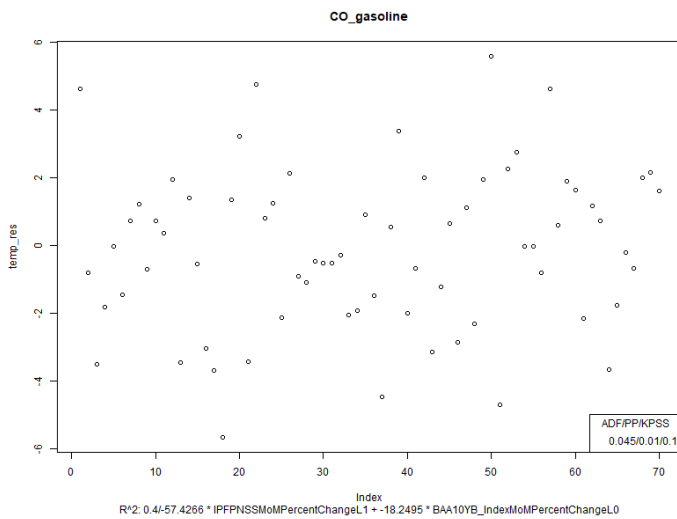
Out-of-Sample testing



Out-of-Sample testing



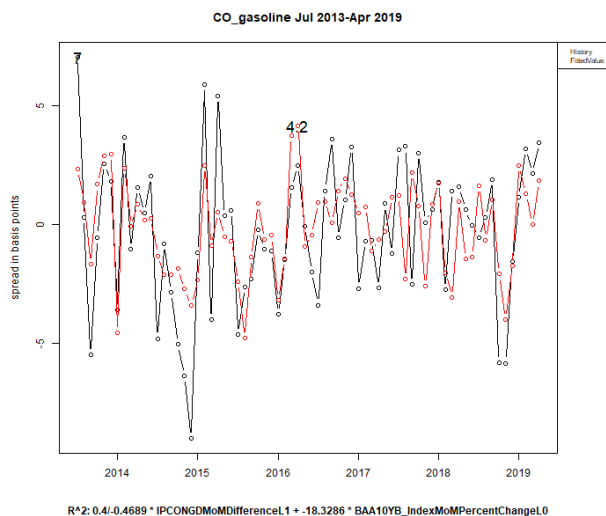
Stationarity of Residuals



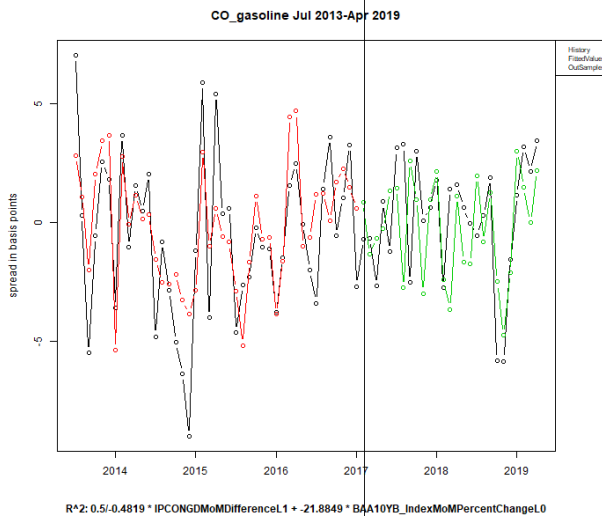
4.8.15 Model - 15

Backtesting on differenced values

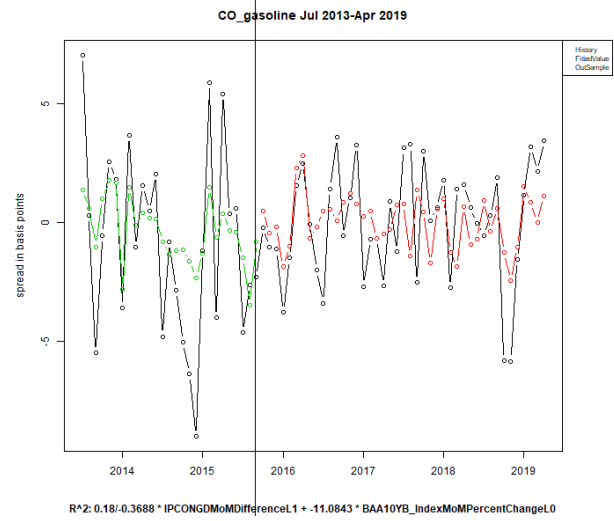
Backtesting on level values



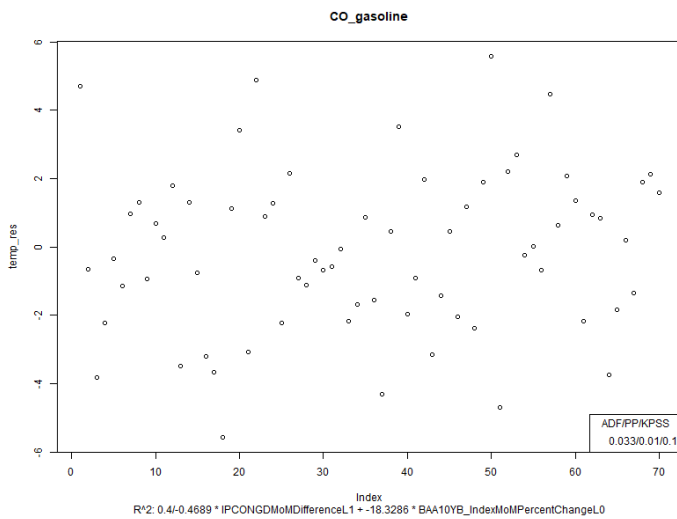
Out-of-Sample testing



Out-of-Sample testing



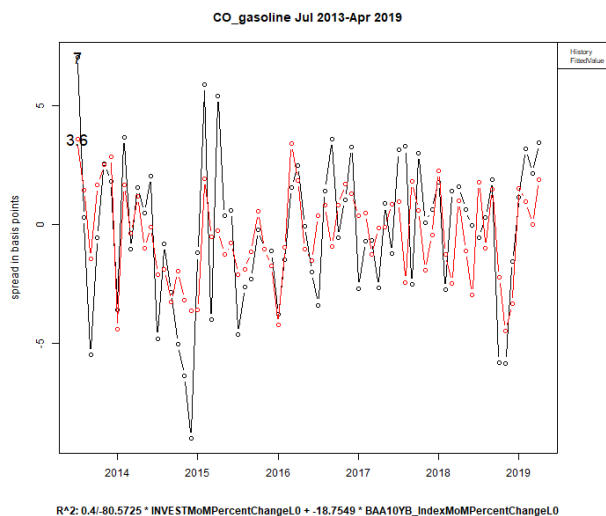
Stationarity of Residuals



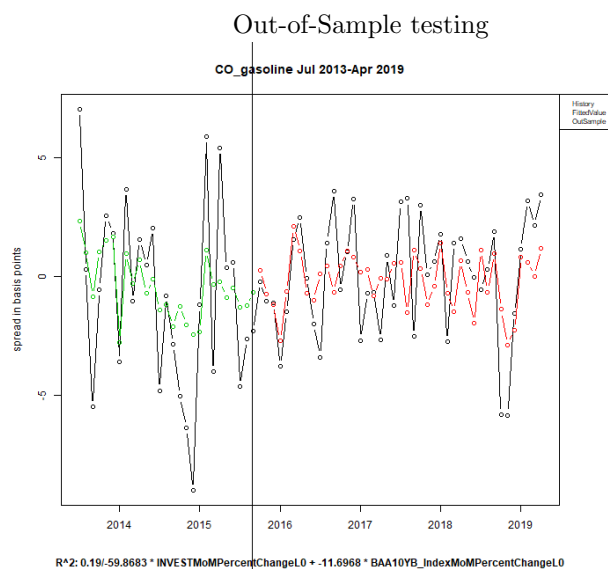
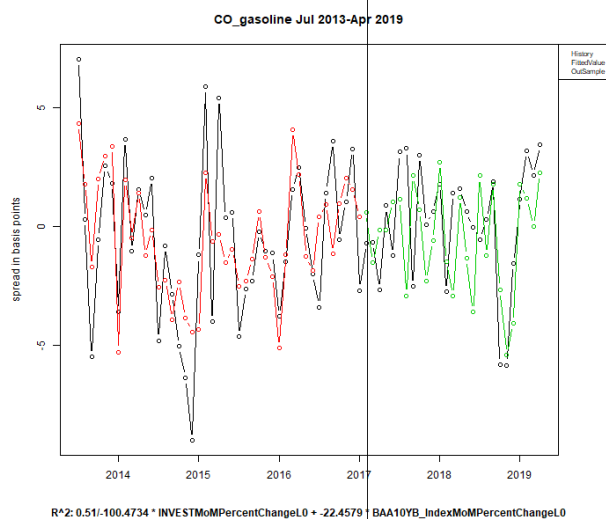
4.8.16 Model - 16

Backtesting on differenced values

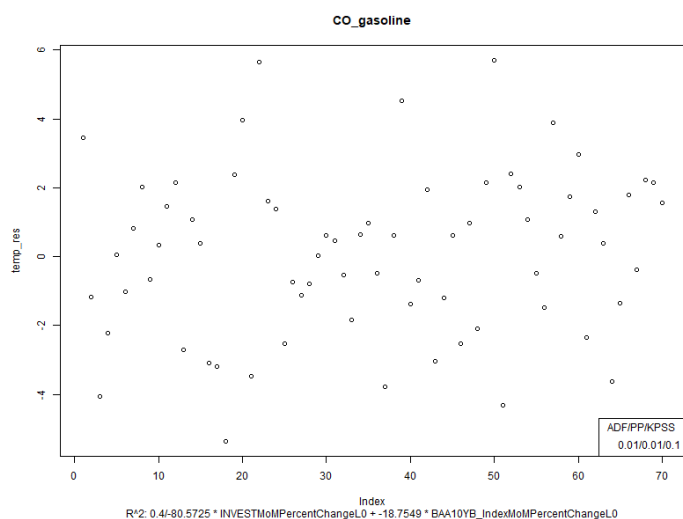
Backtesting on level values



Out-of-Sample testing



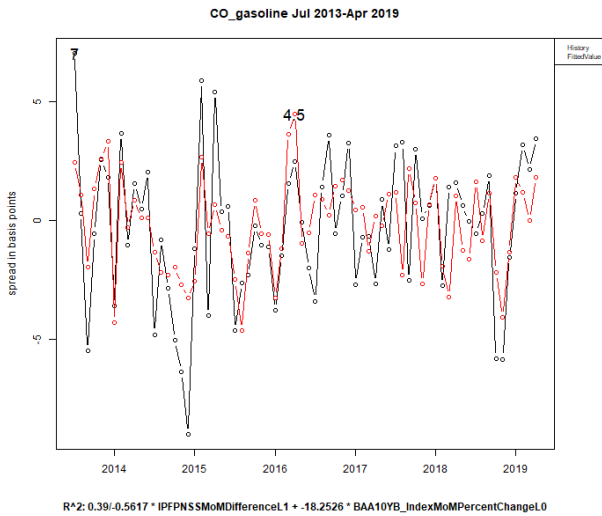
Stationarity of Residuals



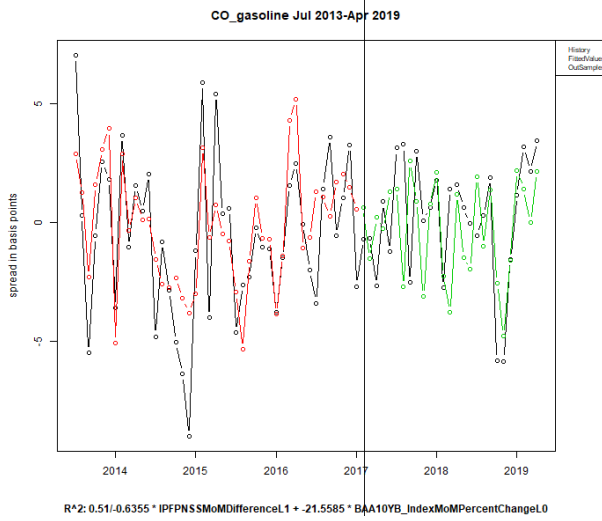
4.8.17 Model - 17

Backtesting on differenced values

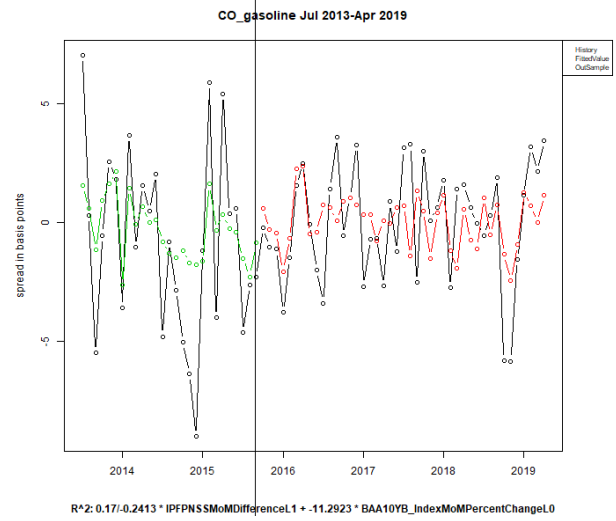
Backtesting on level values



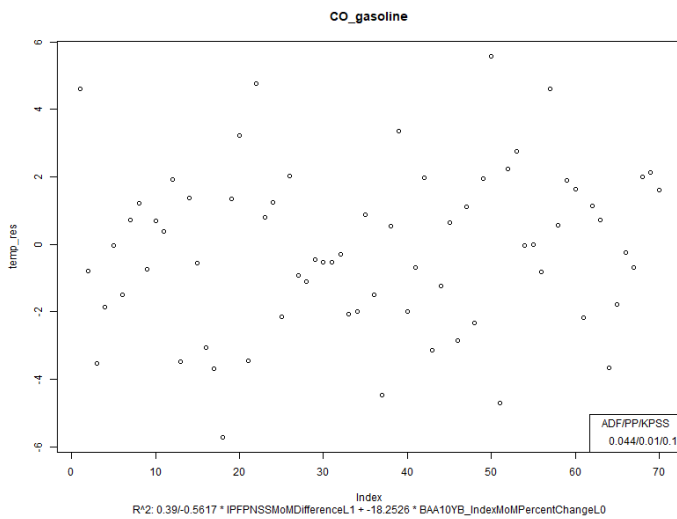
Out-of-Sample testing



Out-of-Sample testing



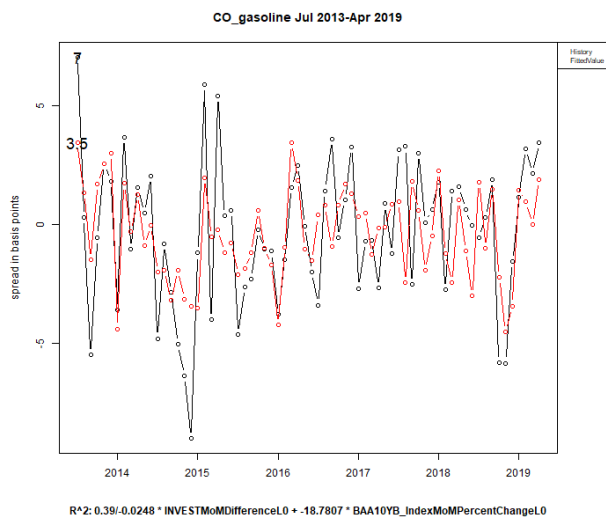
Stationarity of Residuals



4.8.18 Model - 18

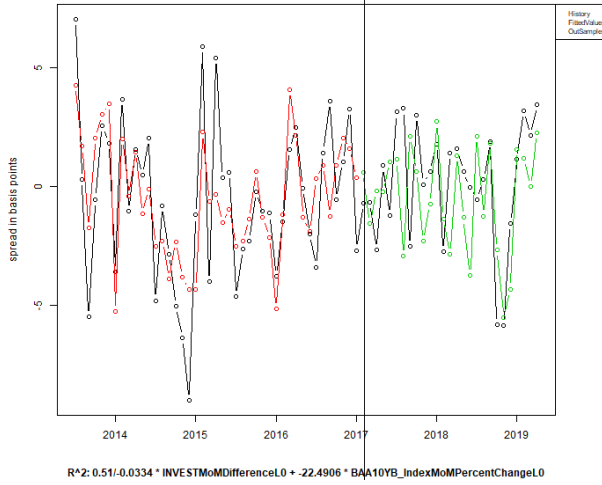
Backtesting on differenced values

Backtesting on level values



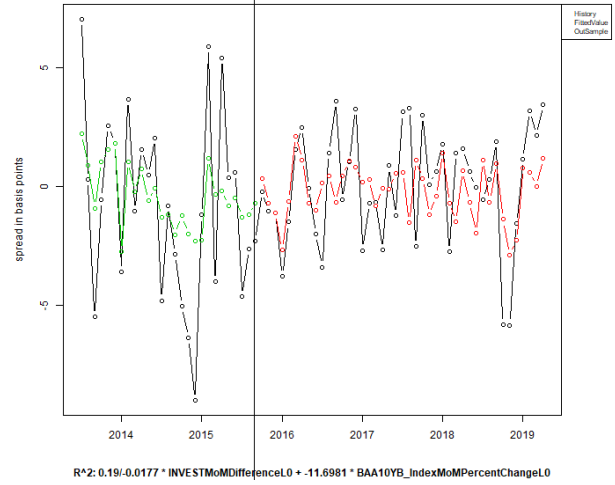
Out-of-Sample testing

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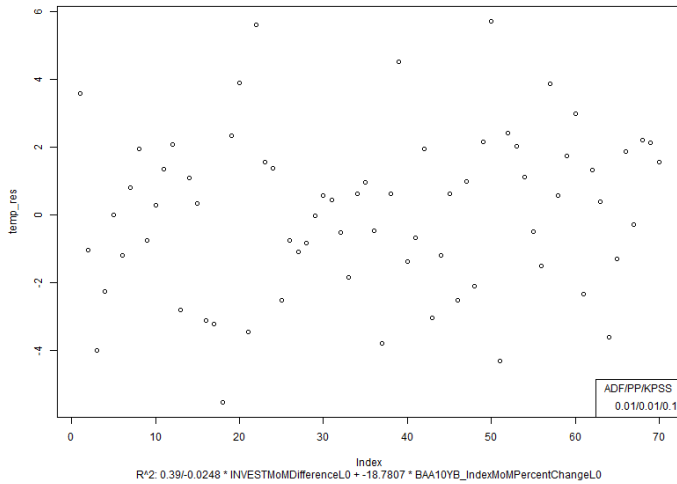
Out-of-Sample testing

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Stationarity of Residuals

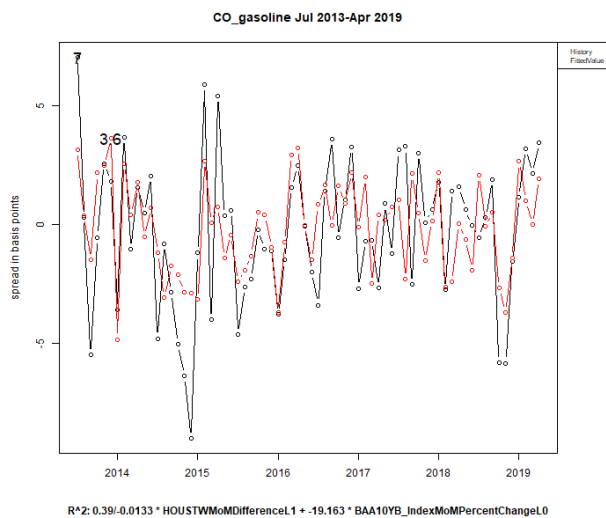
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4.8.19 Model - 19

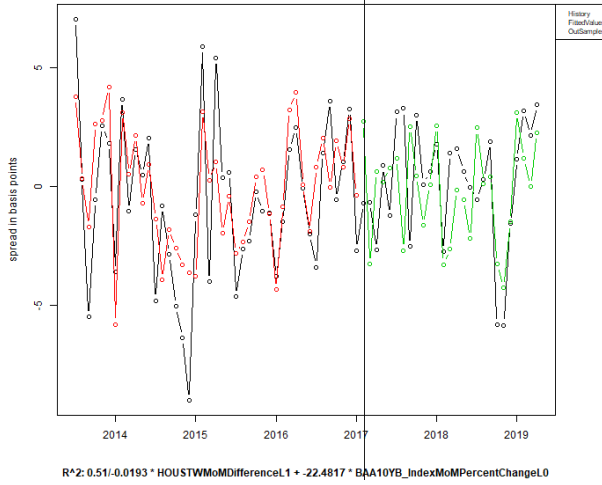
Backtesting on differenced values

Backtesting on level values



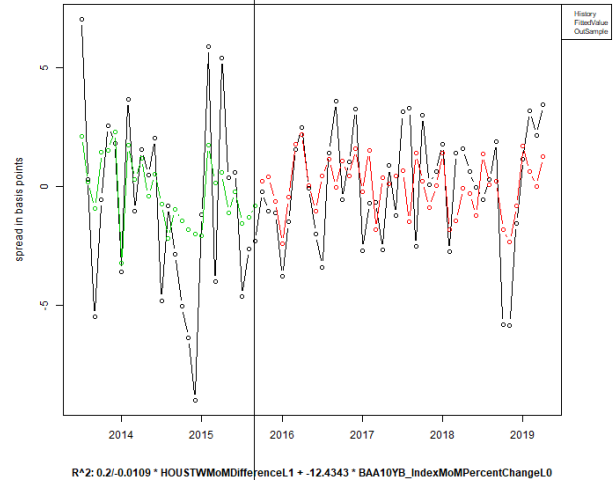
Out-of-Sample testing

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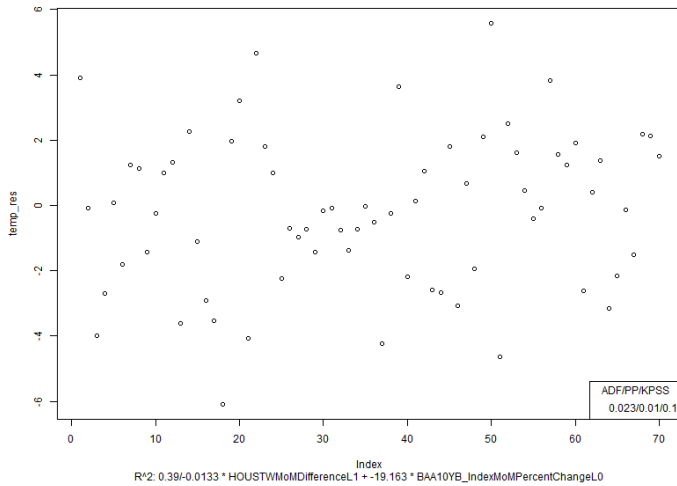
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Stationarity of Residuals

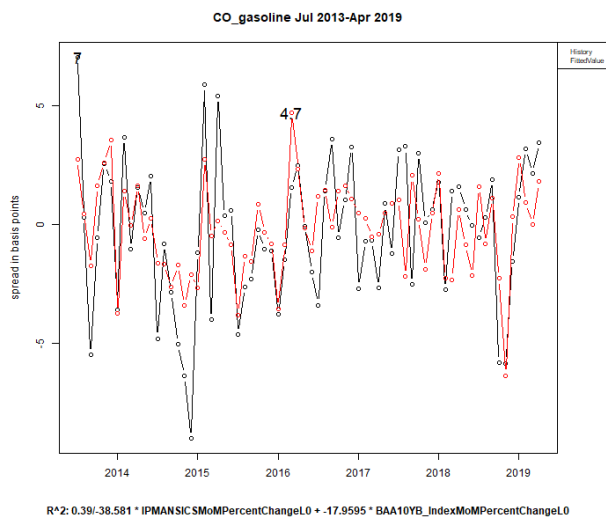
CO_gasoline



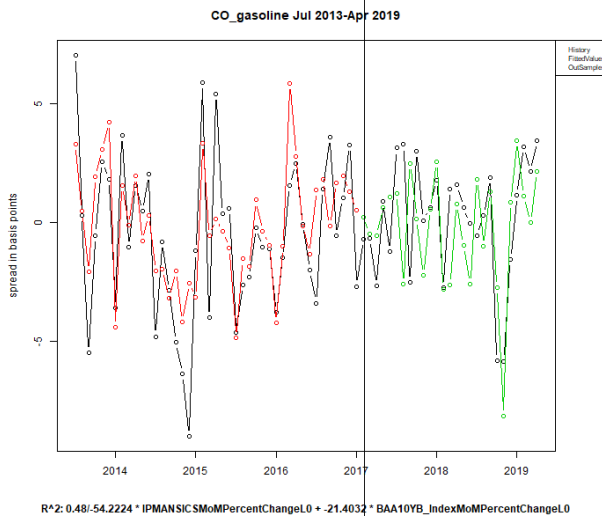
4.8.20 Model - 20

Backtesting on differenced values

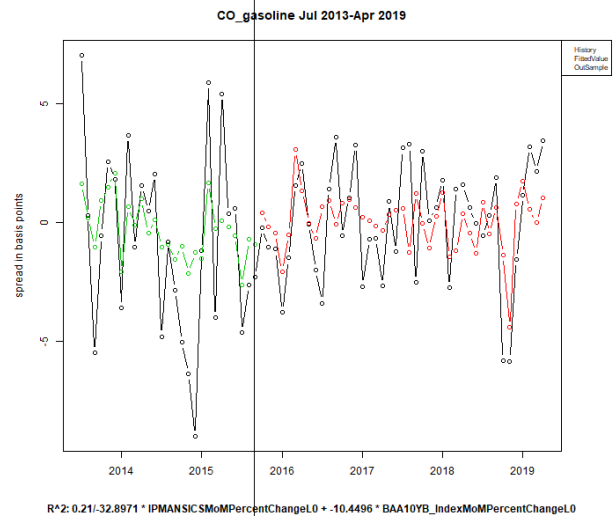
Backtesting on level values



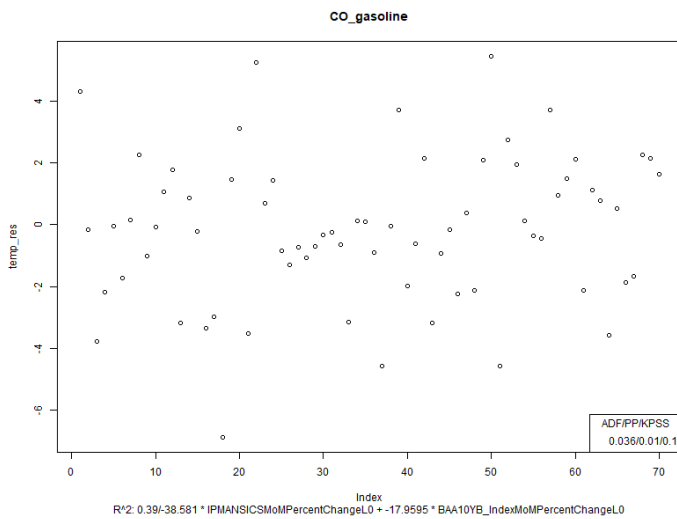
Out-of-Sample testing



Out-of-Sample testing



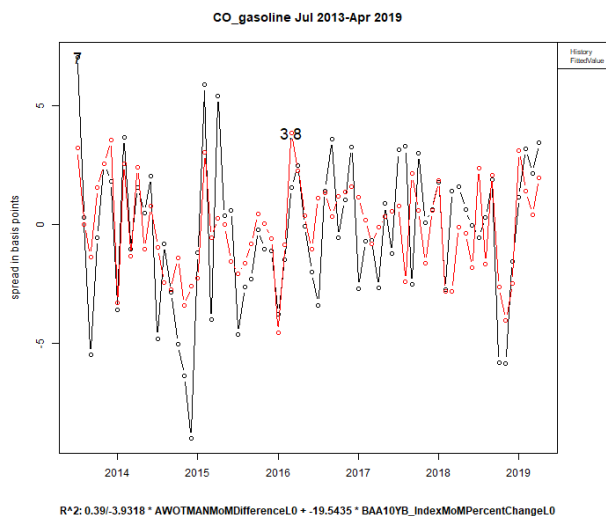
Stationarity of Residuals



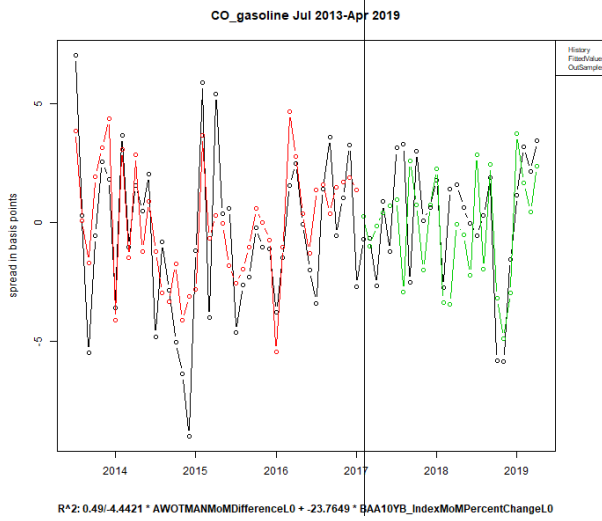
4.8.21 Model - 21

Backtesting on differenced values

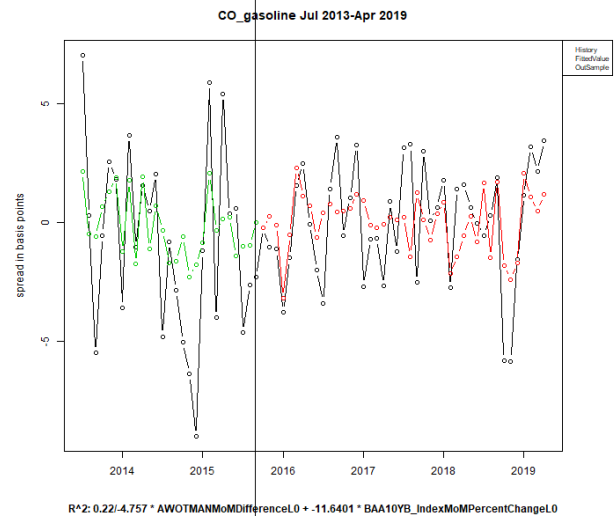
Backtesting on level values



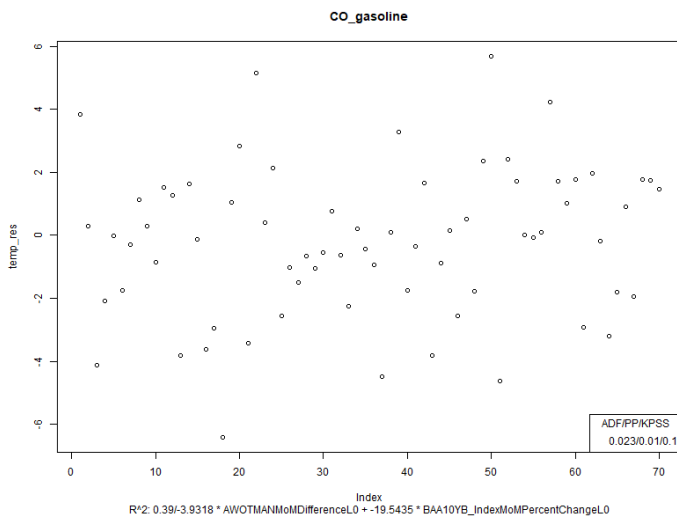
Out-of-Sample testing



Out-of-Sample testing



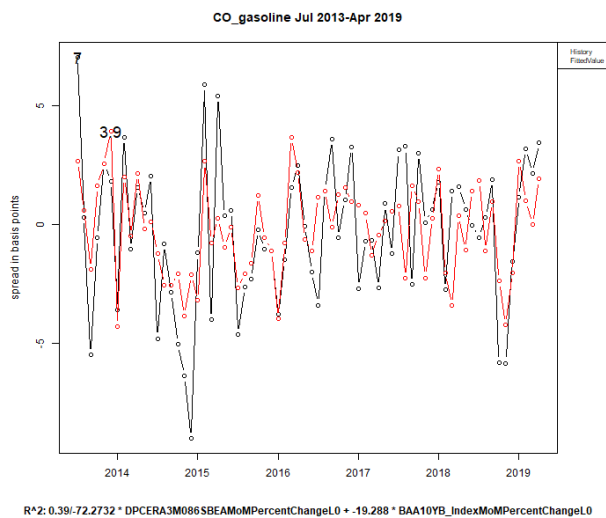
Stationarity of Residuals



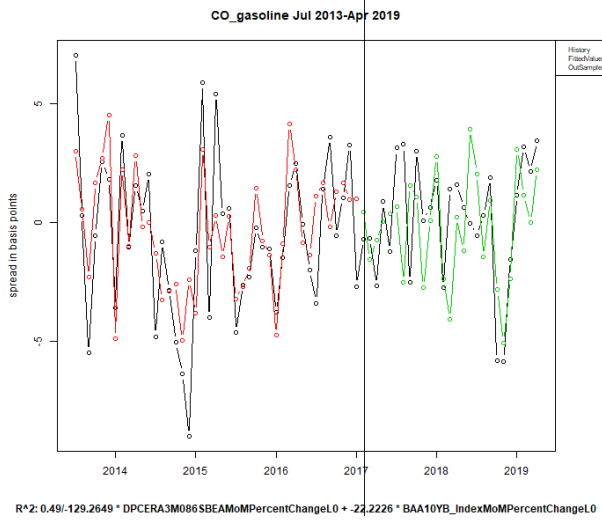
4.8.22 Model - 22

Backtesting on differenced values

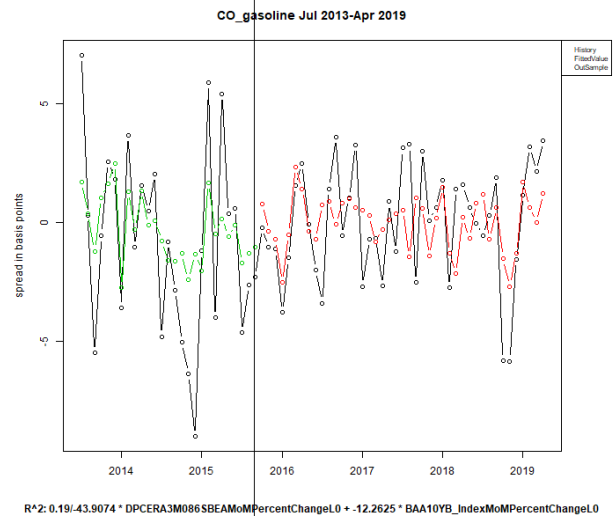
Backtesting on level values



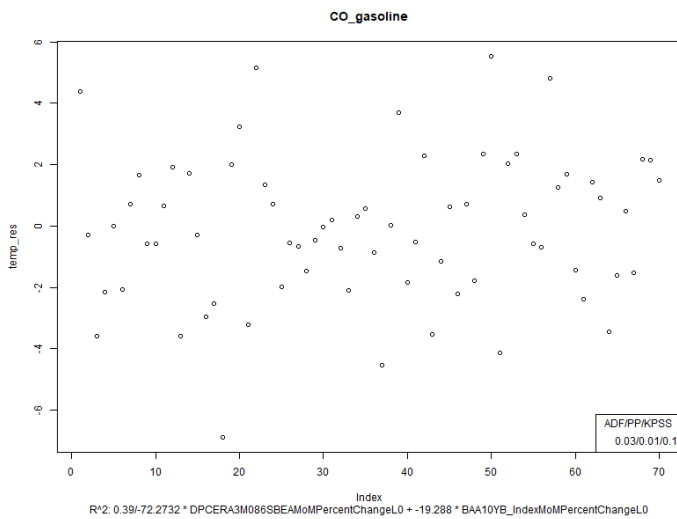
Out-of-Sample testing



Out-of-Sample testing



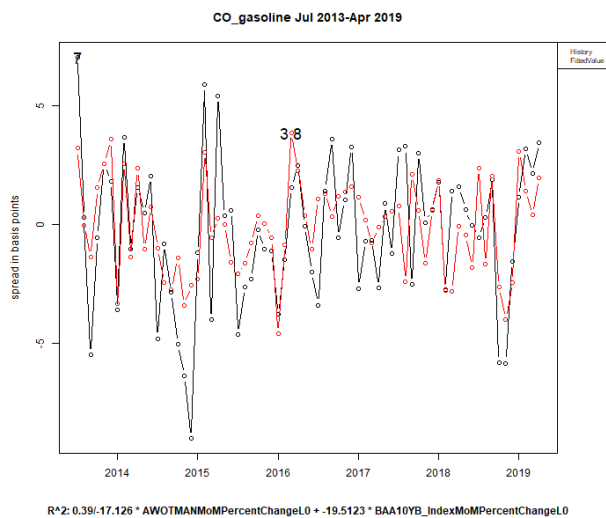
Stationarity of Residuals



4.8.23 Model - 23

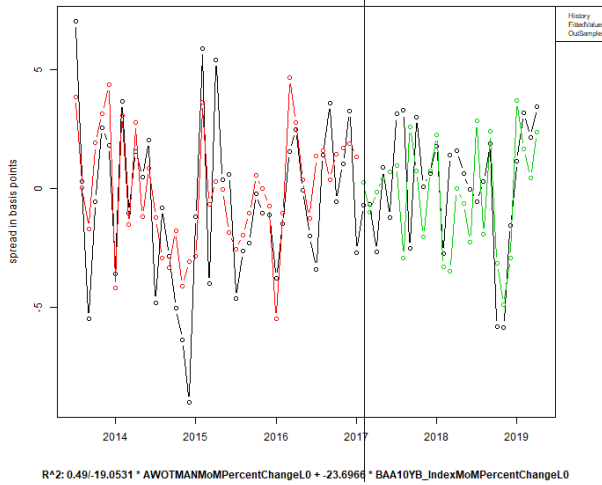
Backtesting on differenced values

Backtesting on level values



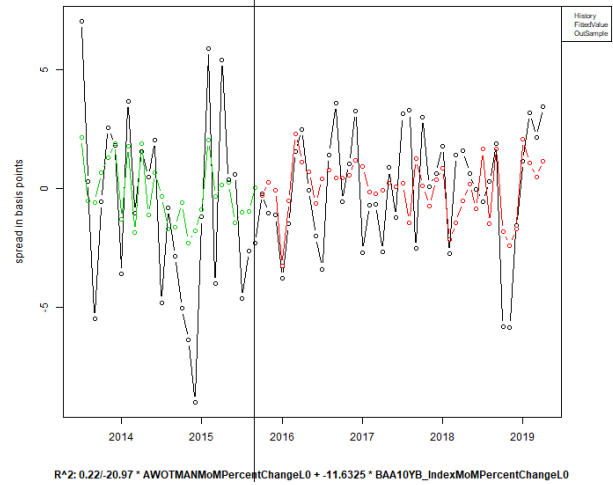
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



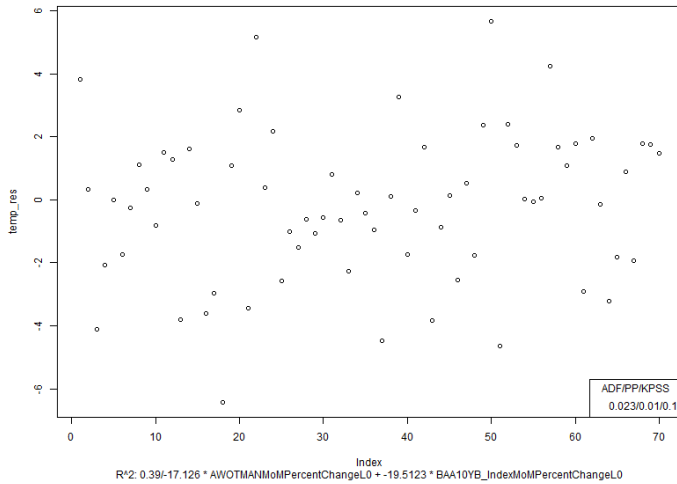
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Stationarity of Residuals

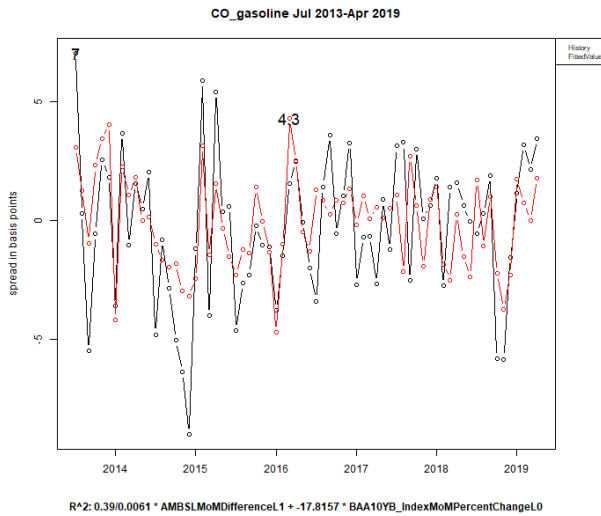
CO_gasoline



4.8.24 Model - 24

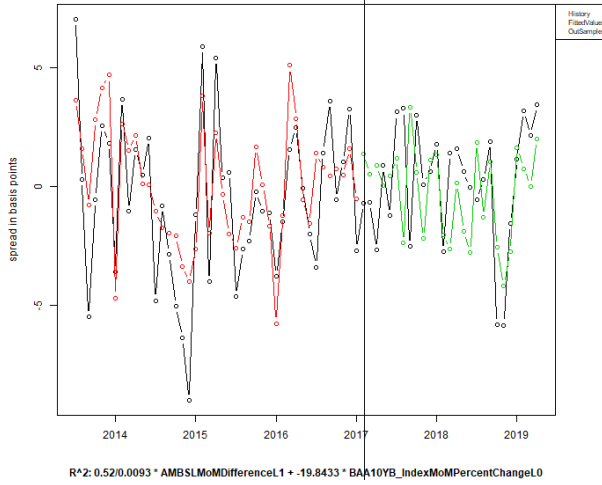
Backtesting on differenced values

Backtesting on level values



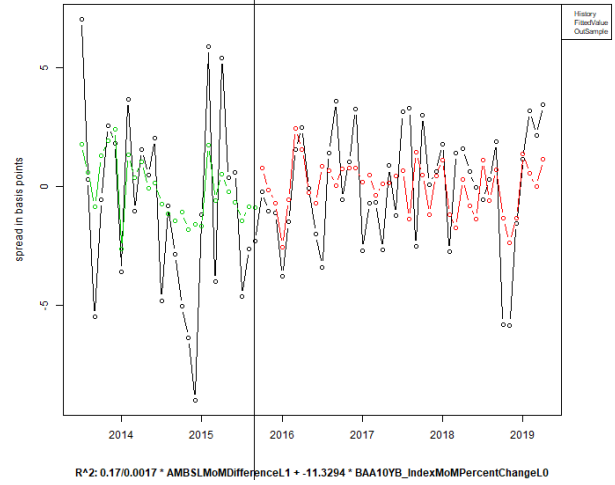
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



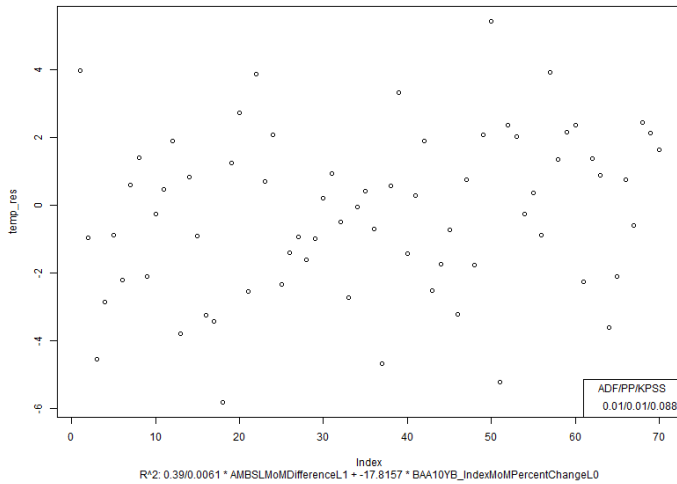
Out-of-Sample testing

CO_gasoline Jul 2013-Apr 2019



Stationarity of Residuals

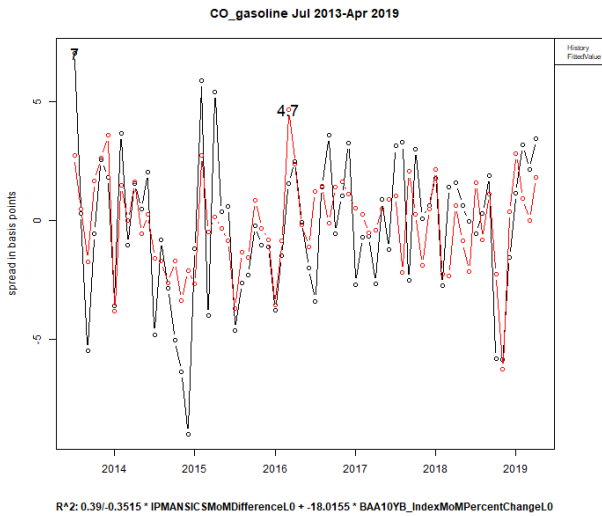
CO_gasoline



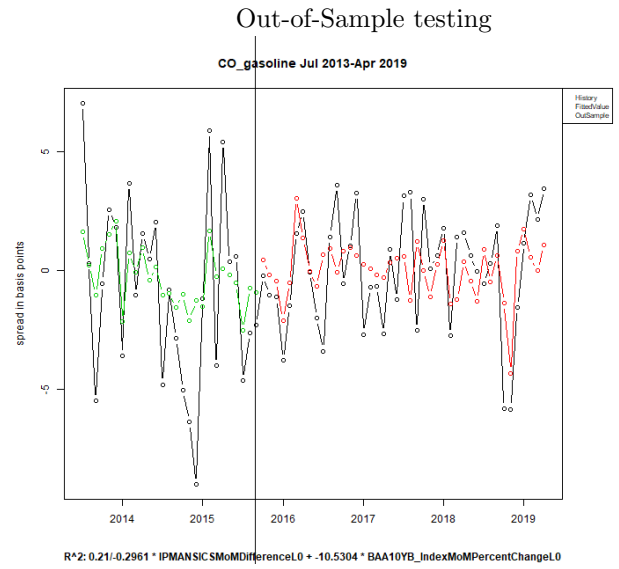
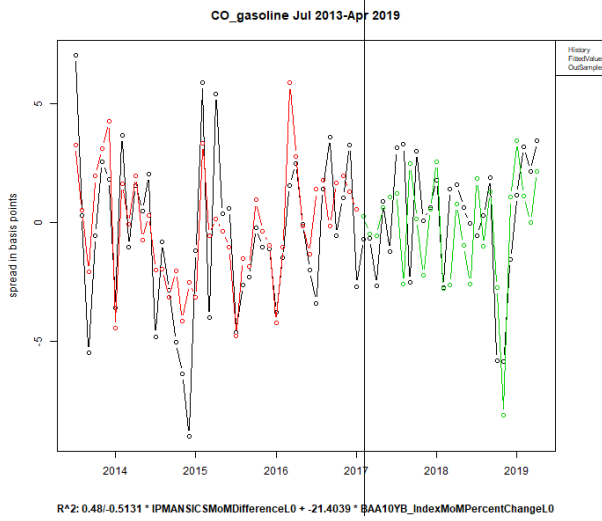
4.8.25 Model - 25

Backtesting on differenced values

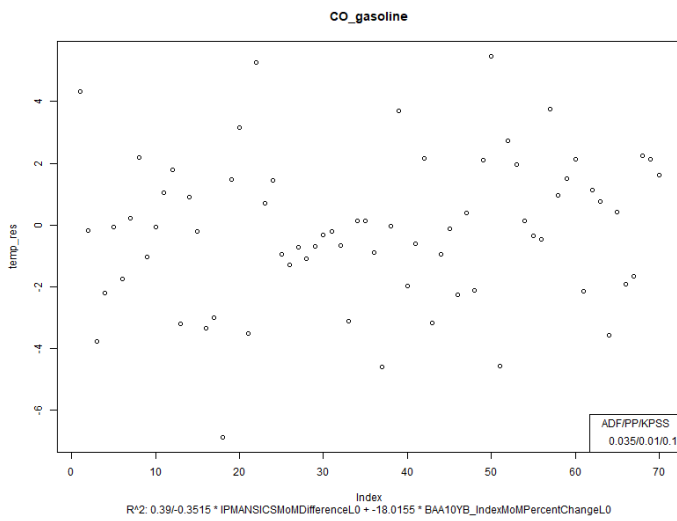
Backtesting on level values



Out-of-Sample testing



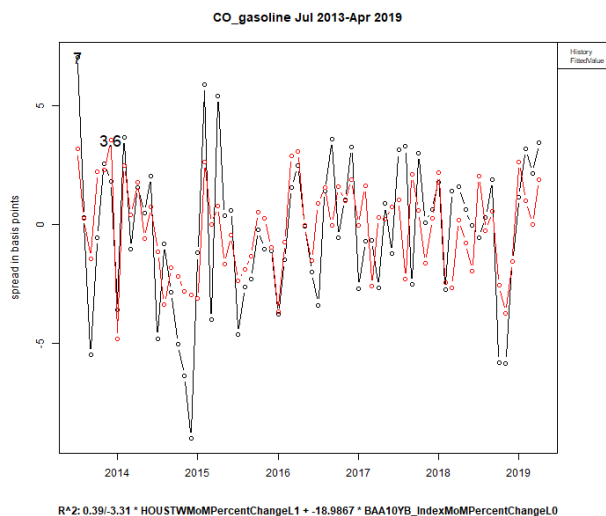
Stationarity of Residuals



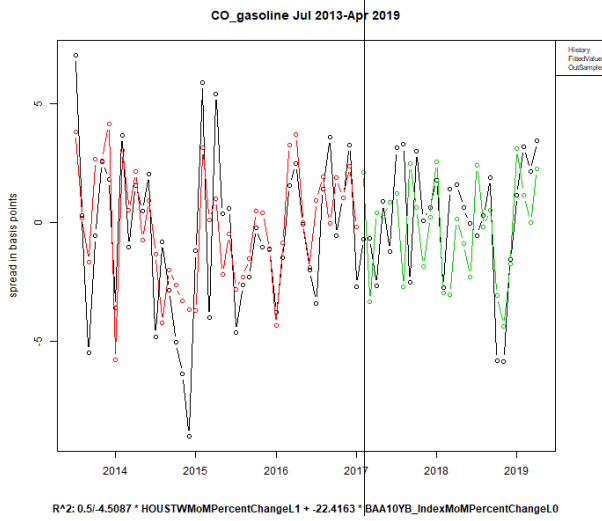
4.8.26 Model - 26

Backtesting on differenced values

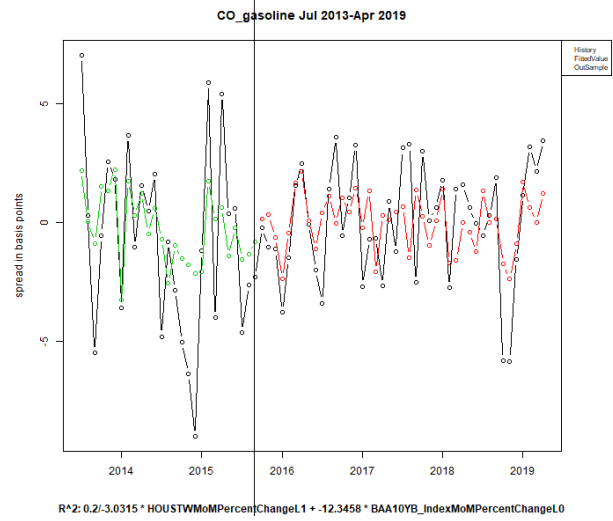
Backtesting on level values



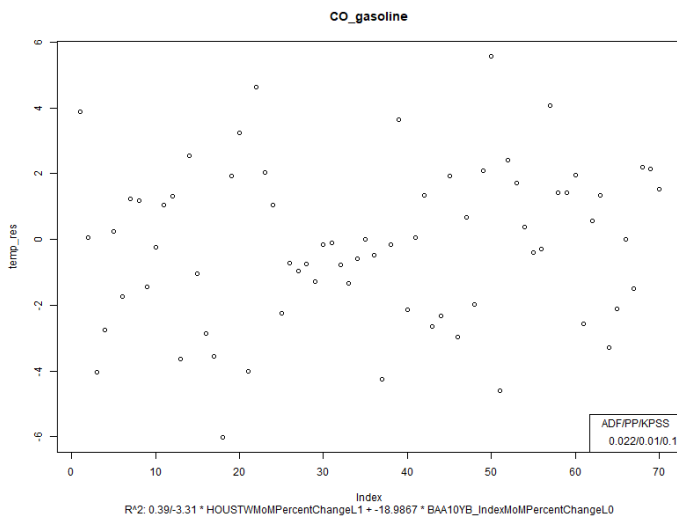
Out-of-Sample testing



Out-of-Sample testing



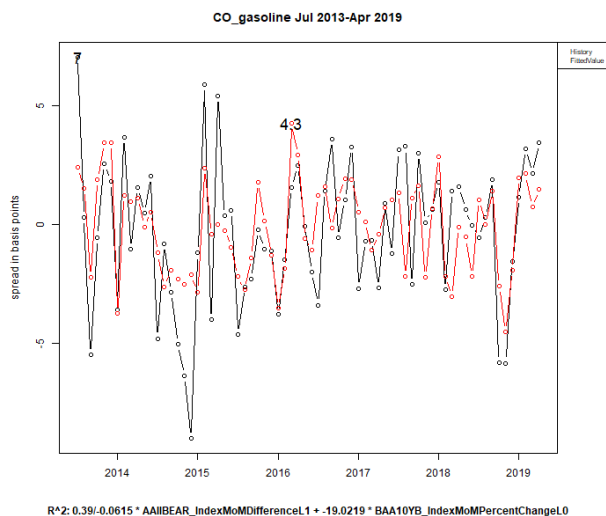
Stationarity of Residuals



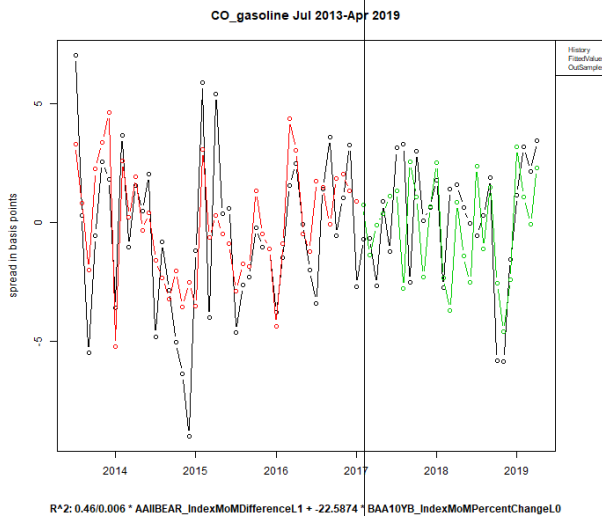
4.8.27 Model - 27

Backtesting on differenced values

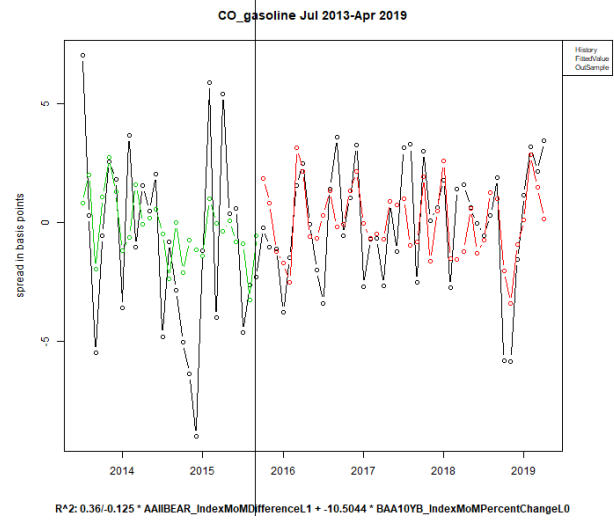
Backtesting on level values



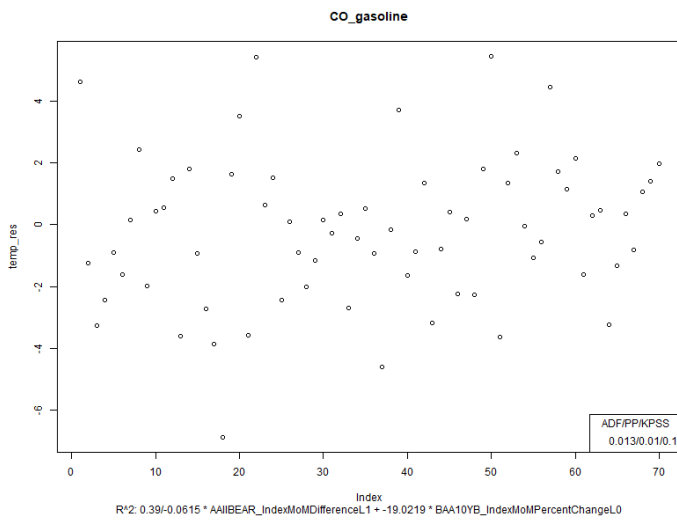
Out-of-Sample testing



Out-of-Sample testing



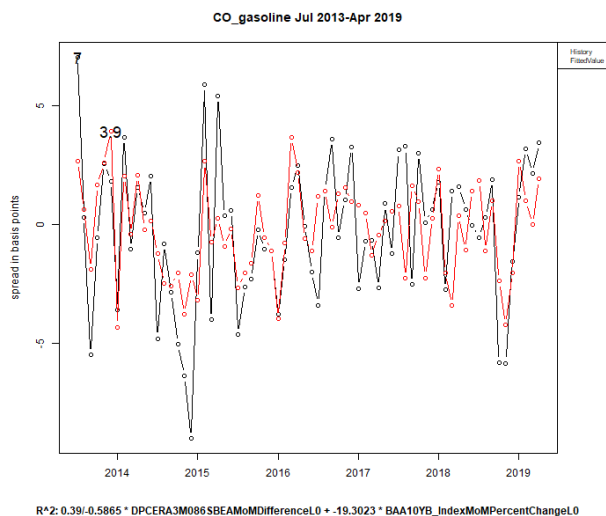
Stationarity of Residuals



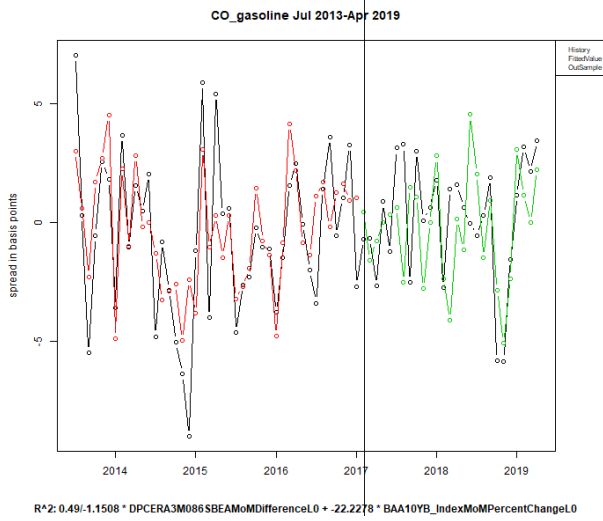
4.8.28 Model - 28

Backtesting on differenced values

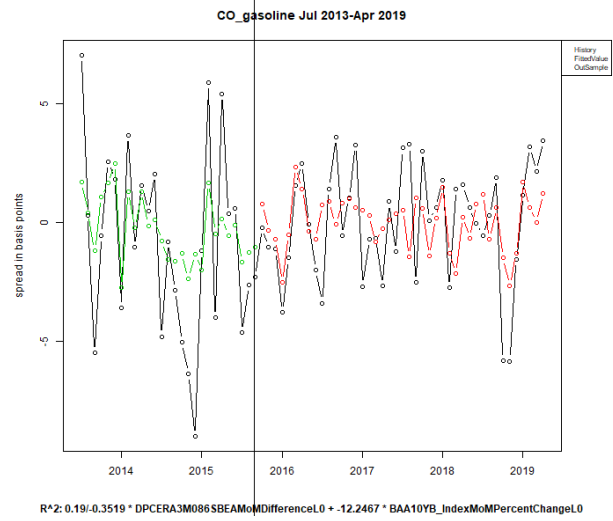
Backtesting on level values



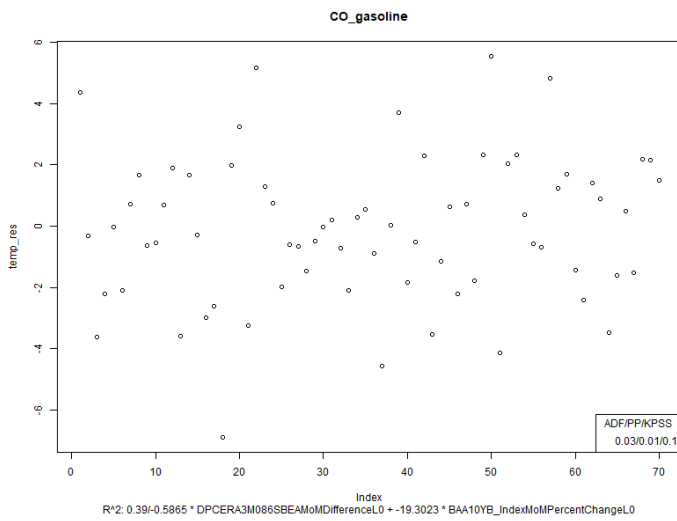
Out-of-Sample testing



Out-of-Sample testing



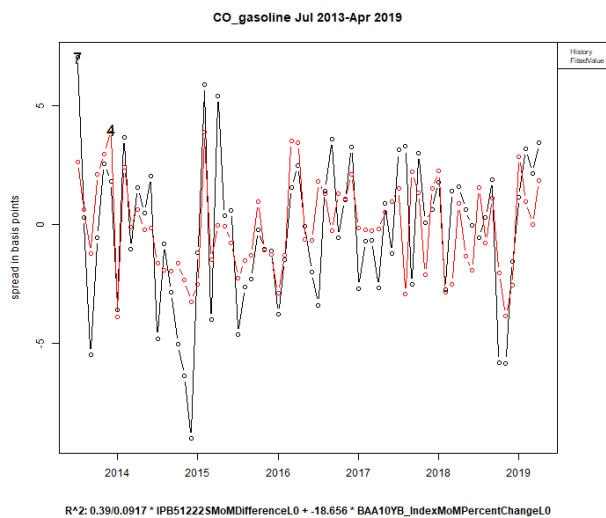
Stationarity of Residuals



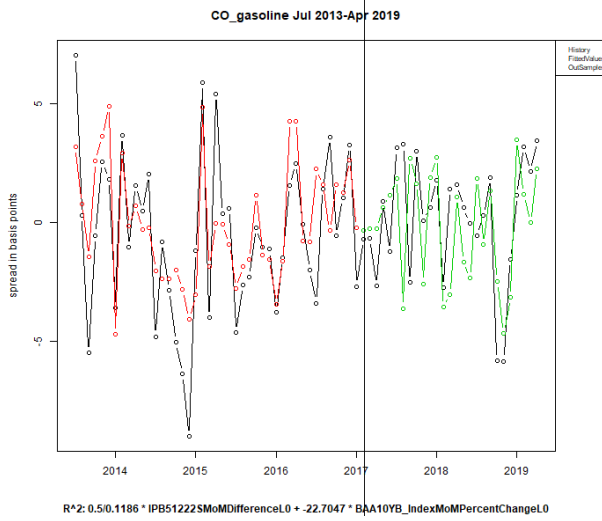
4.8.29 Model - 29

Backtesting on differenced values

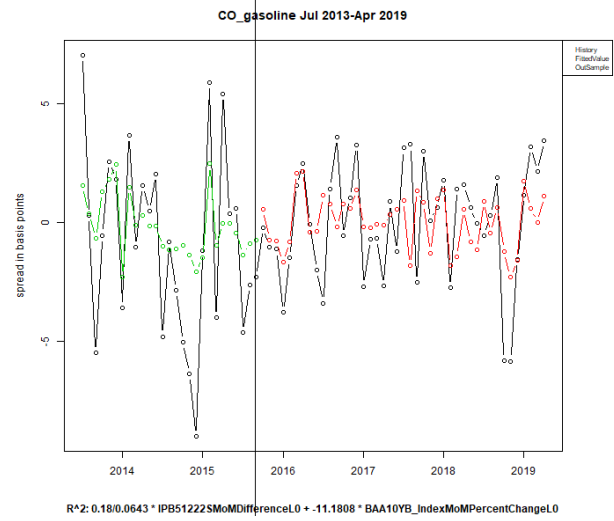
Backtesting on level values



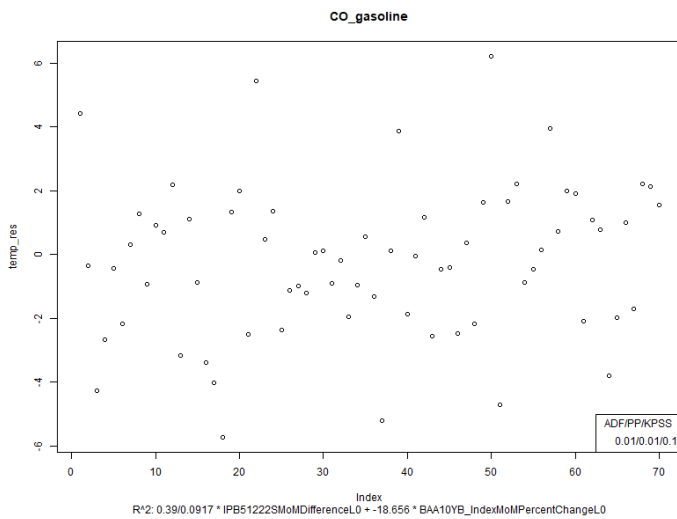
Out-of-Sample testing



Out-of-Sample testing



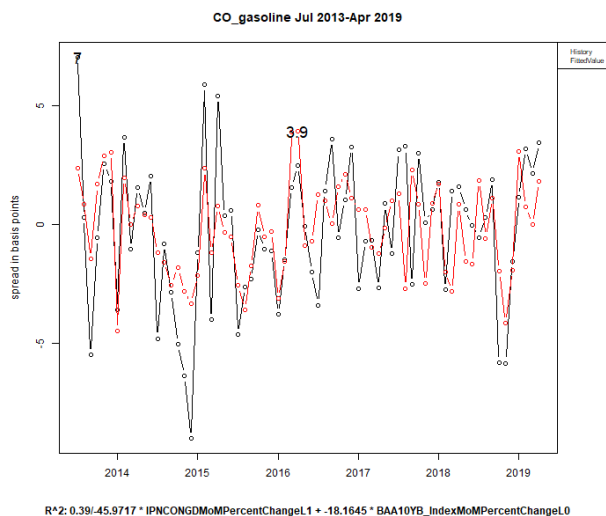
Stationarity of Residuals



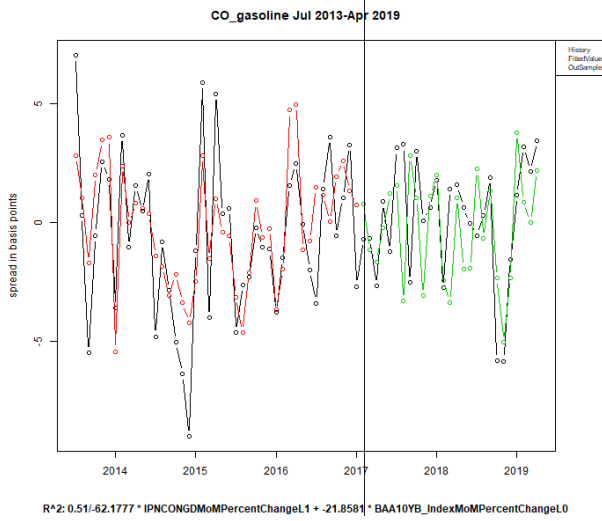
4.8.30 Model - 30

Backtesting on differenced values

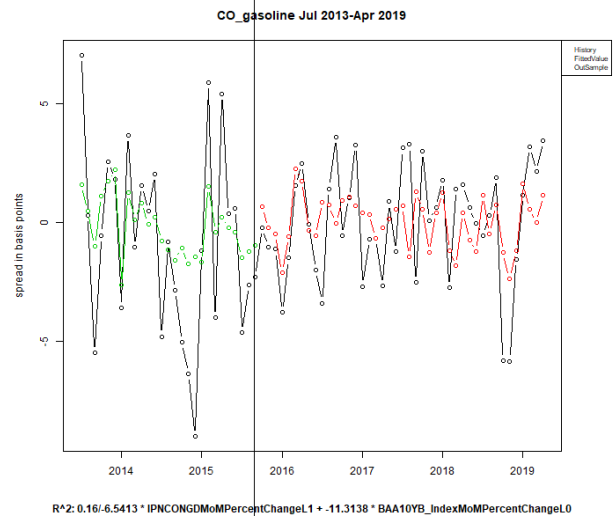
Backtesting on level values



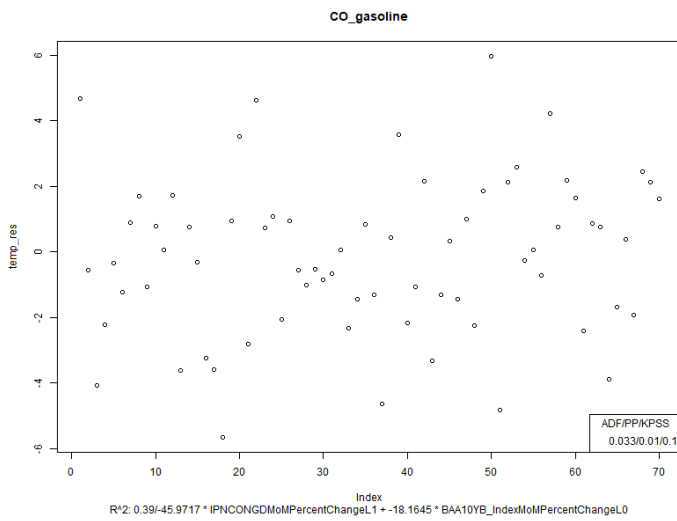
Out-of-Sample testing



Out-of-Sample testing



Stationarity of Residuals



4.9 Single Factor Model for CO_g*old*

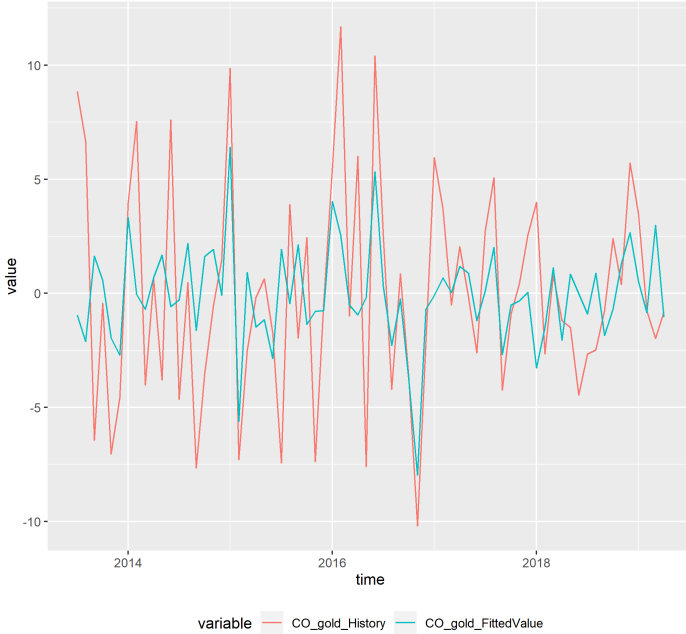
	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	USGG10YR_IndexMoMPercentChangeL0	0.21	401.44	-26.18	0.00	2.67			
2	USGG30YR_IndexMoMPercentChangeL0	0.18	404.50	-33.98	0.00	2.79			
3	USGG10YR_IndexMoMDifferenceL0	0.17	405.01	-10.86	0.00	2.76			
4	USGG30YR_IndexMoMDifferenceL0	0.16	406.16	-11.38	0.00	2.80			
5	USGG2YR_IndexMoMDifferenceL0	0.13	408.52	-14.40	0.00	3.50			
6	USGG2YR_IndexMoMPercentChangeL0	0.11	409.78	-10.74	0.01	2.66			

Table 15: Single Factor Analysis-CO_g*old*

4.9.1 Model - 1

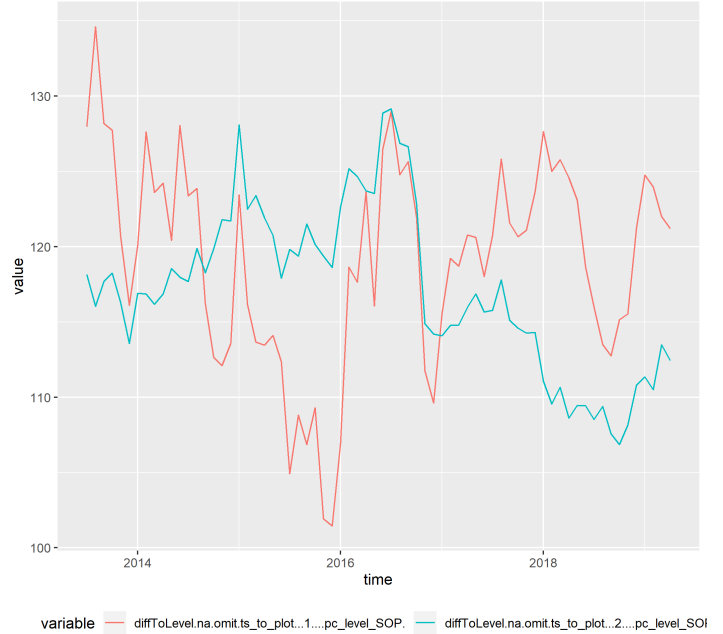
Backtesting on differenced values

CO_gold - Backtesting
R²: 0.21/-26.1767 * USGG10YR_IndexMoMPercentChangeL0



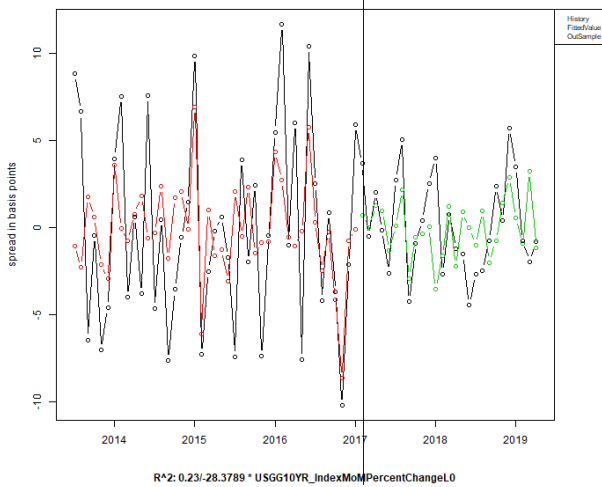
Backtesting on level values

CO_gold - Level Value Backtesting
R²: 0.21/-26.1767 * USGG10YR_IndexMoMPercentChangeL0



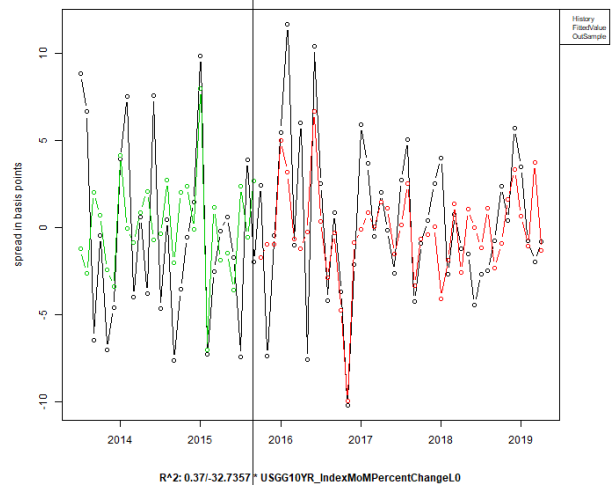
Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



Out-of-Sample testing

CO_gold Jul 2013-Apr 2019

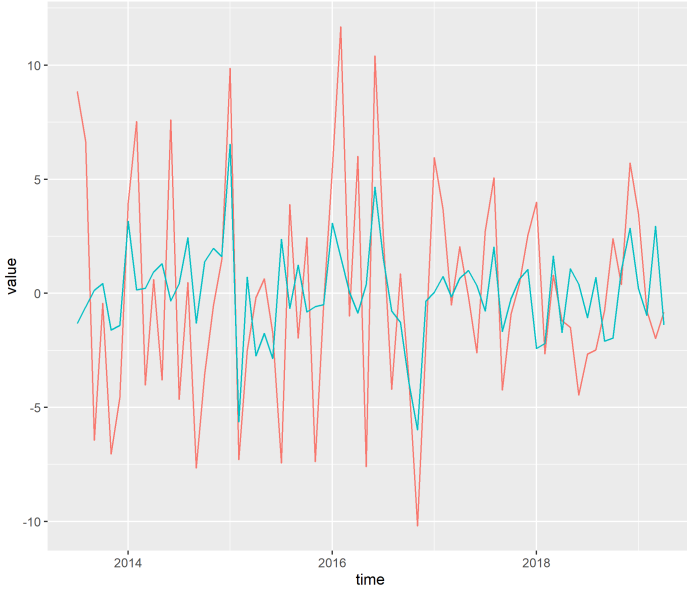


4.9.2 Model - 2

Backtesting on differenced values

CO_gold - Backtesting

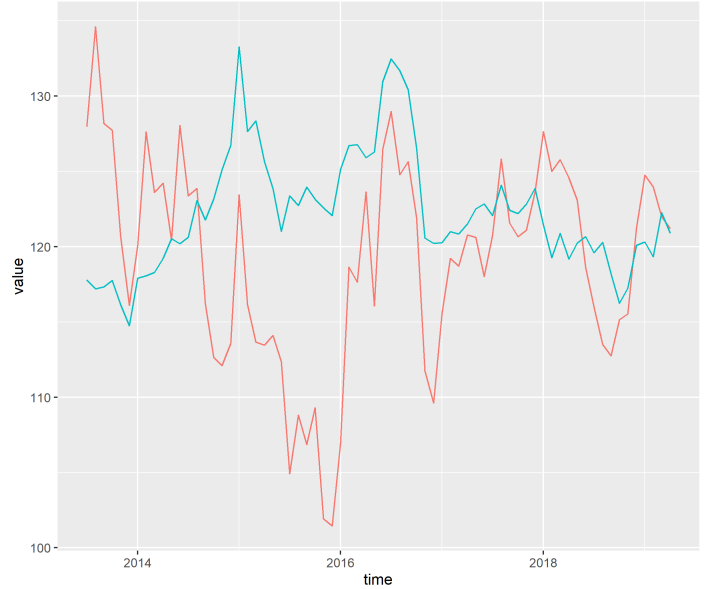
$R^2: 0.18 / -33.9817 * USGG30YR_IndexMoMPercentChangeL0$



Backtesting on level values

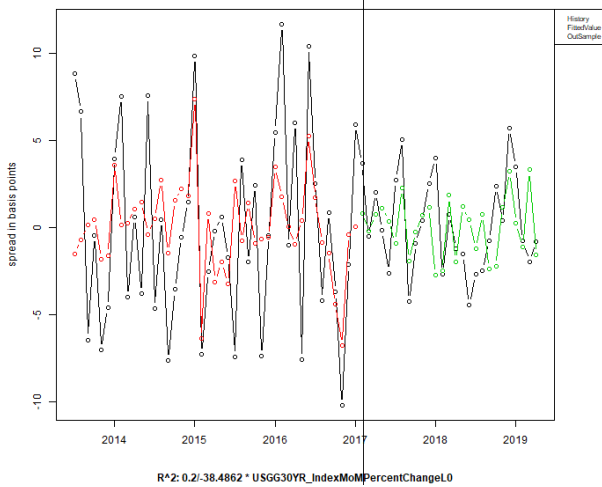
CO_gold - Level Value Backtesting

$R^2: 0.18 / -33.9817 * USGG30YR_IndexMoMPercentChangeL0$



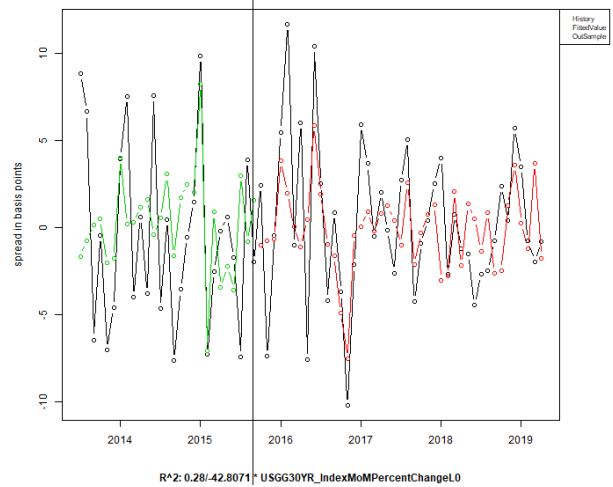
Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



4.9.3 Model - 3

Backtesting on differenced values

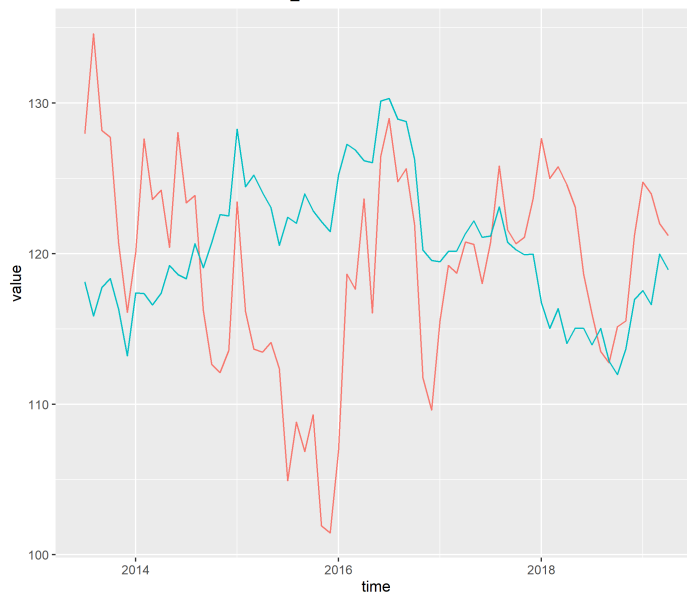
CO_gold - Backtesting

 $R^2: 0.17 / -10.8564 * USGG10YR_IndexMoMDifferenceL0$ 

variable — CO_gold_History — CO_gold_FittedValue

Backtesting on level values

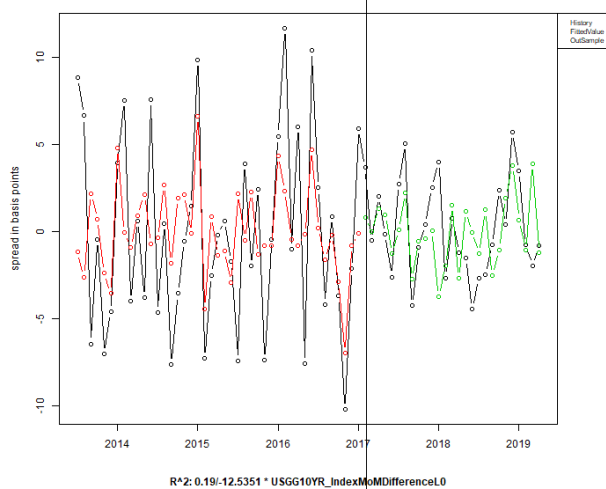
CO_gold - Level Value Backtesting

 $R^2: 0.17 / -10.8564 * USGG10YR_IndexMoMDifferenceL0$ 

variable — diffToLevel.na.omit.ts...1...pc_level_SOP — diffToLevel.na.omit.ts...2...pc_level_SOI

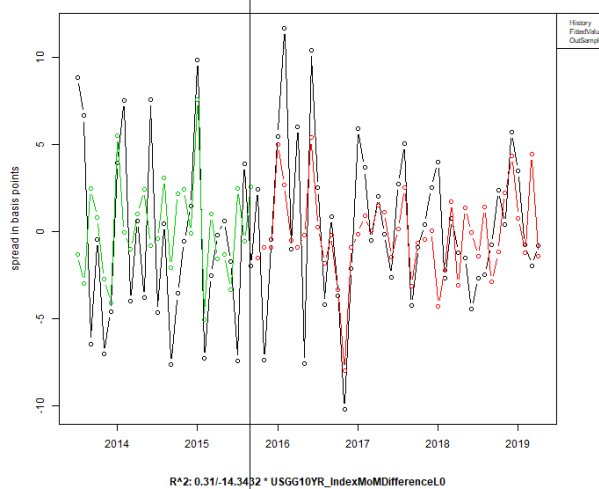
Out-of-Sample testing

CO_gold Jul 2013-Apr 2019

 $R^2: 0.19 / -12.5351 * USGG10YR_IndexMoMDifferenceL0$

Out-of-Sample testing

CO_gold Jul 2013-Apr 2019

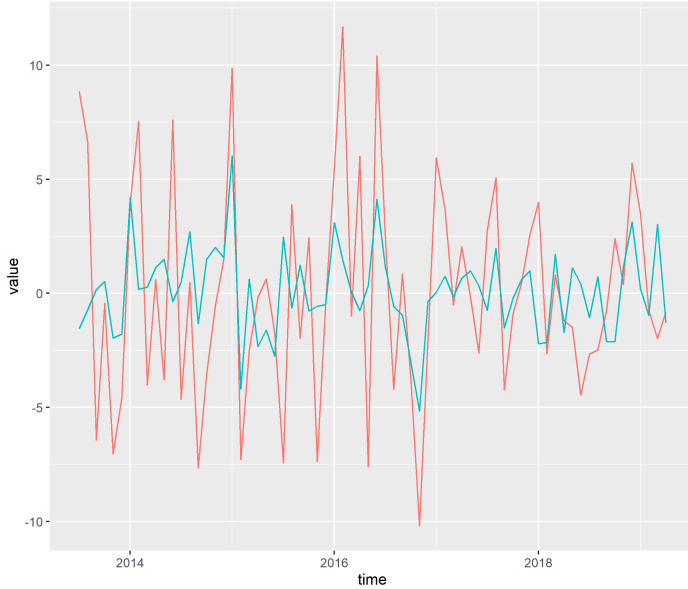
 $R^2: 0.31 / -14.3432 * USGG10YR_IndexMoMDifferenceL0$

4.9.4 Model - 4

Backtesting on differenced values

CO_gold - Backtesting

R²: 0.16/ -11.3799 * USGG30YR_IndexMoMDifferenceL0

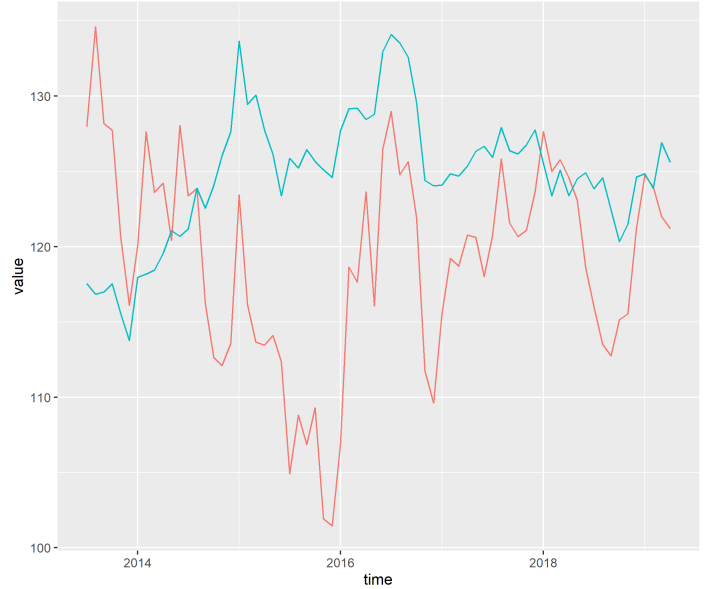


variable — CO_gold_History — CO_gold_FittedValue

Backtesting on level values

CO_gold - Level Value Backtesting

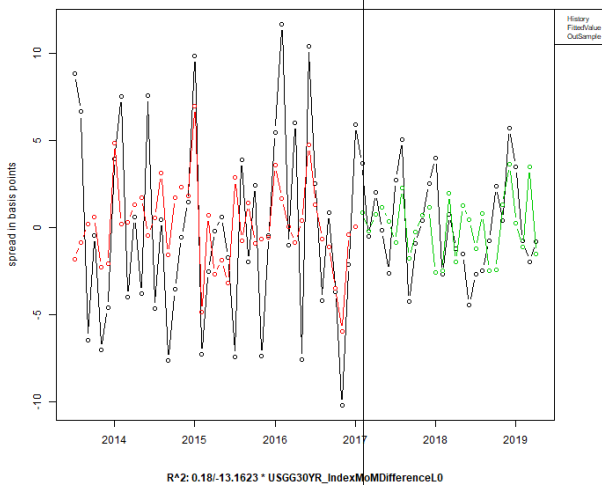
R²: 0.16/ -11.3799 * USGG30YR_IndexMoMDifferenceL0



variable — diffToLevel.na.omit.ts_to_plot...1...pc_level_SOP — diffToLevel.na.omit.ts_to_plot...2...pc_level_SOI

Out-of-Sample testing

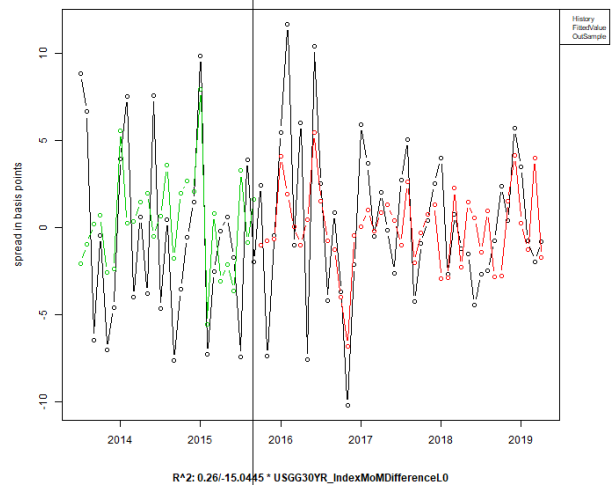
CO_gold Jul 2013-Apr 2019



R²: 0.18/-13.1623 * USGG30YR_IndexMoMDifferenceL0

Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



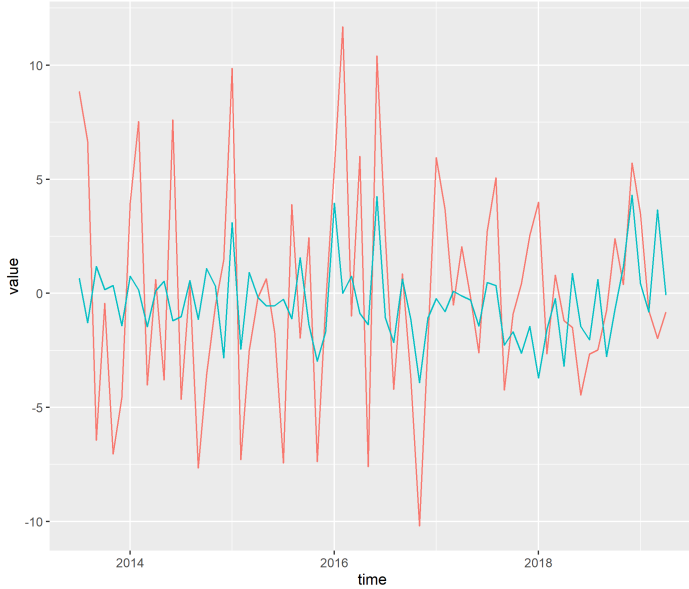
R²: 0.26/-15.0445 * USGG30YR_IndexMoMDifferenceL0

4.9.5 Model - 5

Backtesting on differenced values

CO_gold - Backtesting

$R^2: 0.13 / -14.3967 * USGG2YR_IndexMoMDifferenceL0$

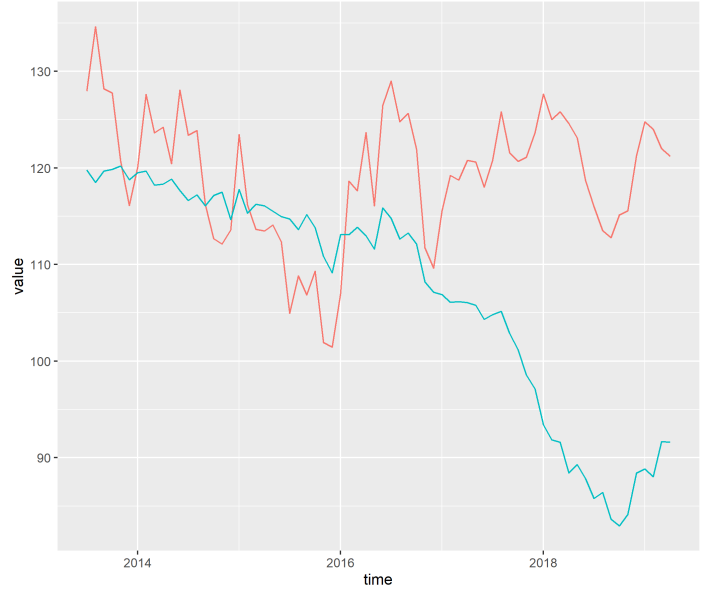


variable — CO_gold_History — CO_gold_FittedValue

Backtesting on level values

CO_gold - Level Value Backtesting

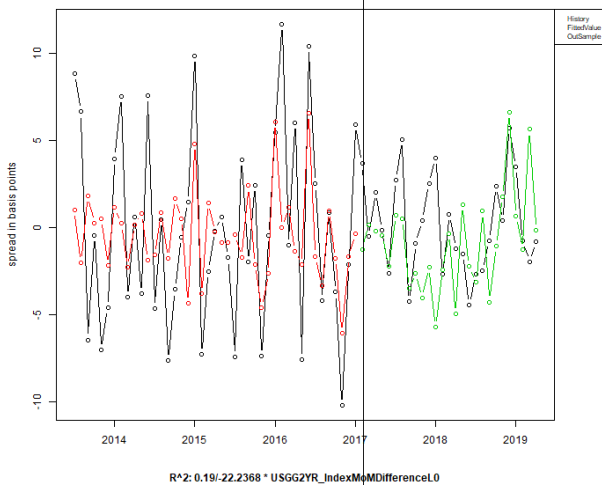
$R^2: 0.13 / -14.3967 * USGG2YR_IndexMoMDifferenceL0$



variable — diffToLevel.na.omit.ts...1....pc_level_SOP — diffToLevel.na.omit.ts...2....pc_level_SOI

Out-of-Sample testing

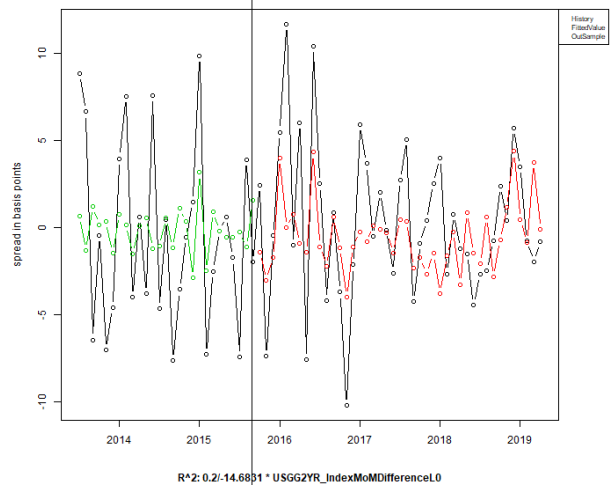
CO_gold Jul 2013-Apr 2019



$R^2: 0.19 / -22.2368 * USGG2YR_IndexMoMDifferenceL0$

Out-of-Sample testing

CO_gold Jul 2013-Apr 2019

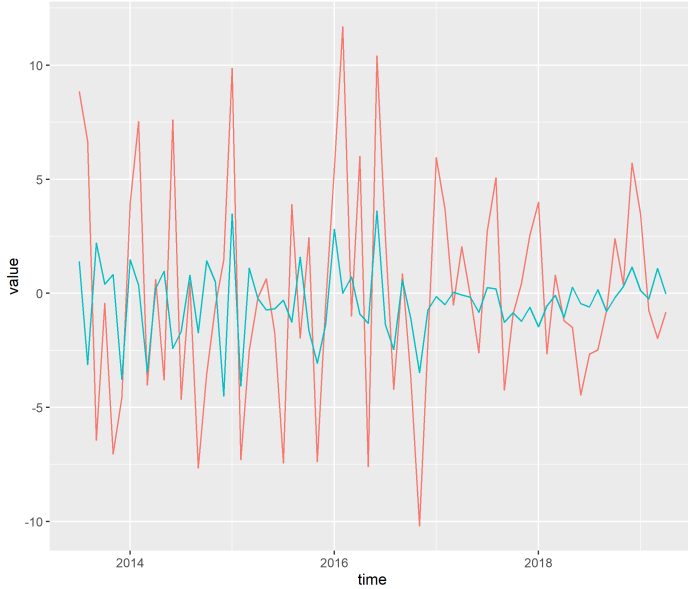


$R^2: 0.2 / -14.6811 * USGG2YR_IndexMoMDifferenceL0$

4.9.6 Model - 6

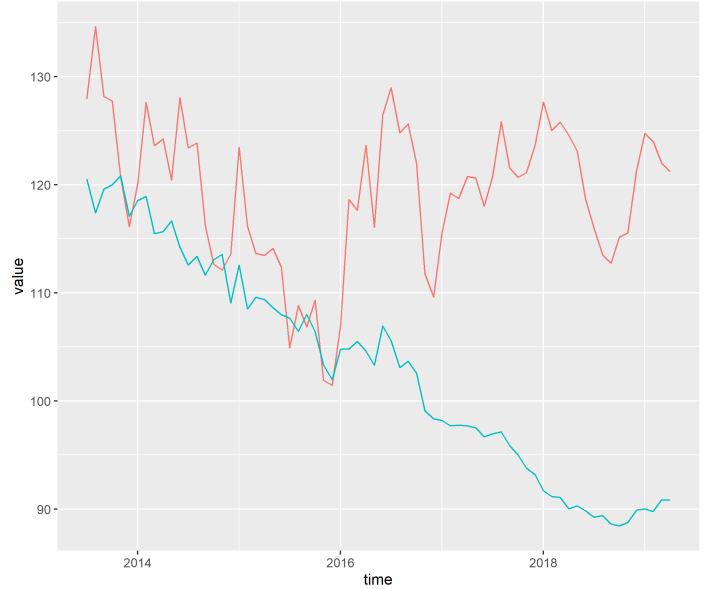
Backtesting on differenced values

CO_gold - Backtesting
R²: 0.11/-10.743 * USGG2YR_IndexMoMPercentChangeL0



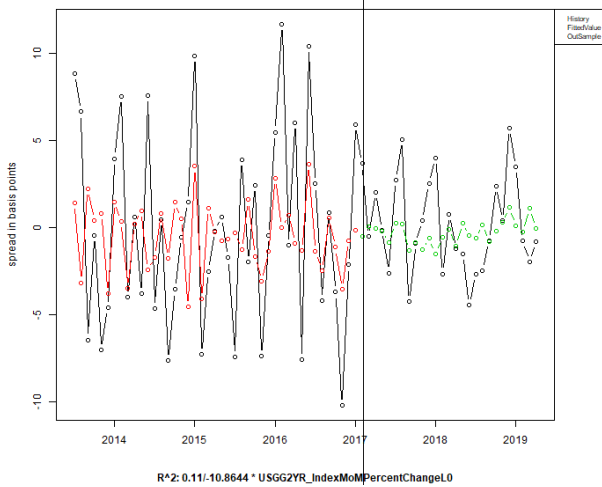
Backtesting on level values

CO_gold - Level Value Backtesting
R²: 0.11/-10.743 * USGG2YR_IndexMoMPercentChangeL0



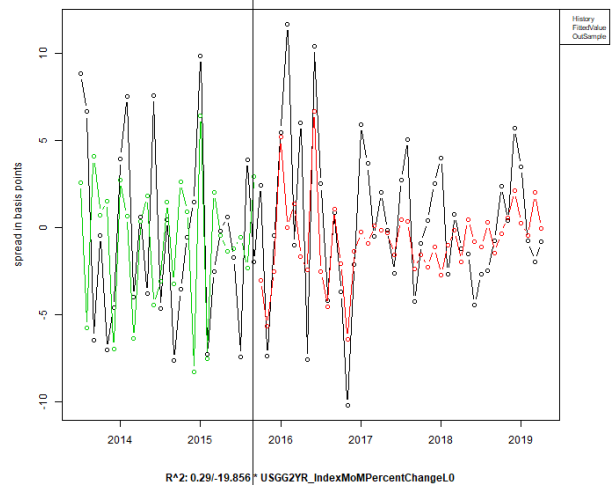
Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



4.10 Two Factor Model for CO₂old

	V1	V2	r2adj	Var1	Var2	aic	RMSE	Criteria1	Criteria2	Criteria3
1	CES0600000007MoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL0	0.28	-5.78	-27.78	395.95	3.05			
2	CES0600000007MoMPercentChangeL1	USGG10YR_IndexMoMPercentChangeL0	0.28	-236.93	-27.76	395.97	3.05			
3	CES0600000007MoMDifferenceL1	USGG30YR_IndexMoMPercentChangeL0	0.26	-6.36	-38.02	397.87	3.18			
4	CES0600000007MoMPercentChangeL1	USGG30YR_IndexMoMPercentChangeL0	0.26	-260.91	-37.97	397.91	3.17			
5	USGG10YR_IndexMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL0	0.25	-35.13	-13.14	398.50	2.91			
6	UEMP15T26MoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL0	0.25	-0.01	-23.10	398.74	2.64			
7	UEMP5TO14MoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL0	0.25	0.01	-28.03	398.86	2.54			
8	UEMP15T26MoMPercentChangeL1	USGG10YR_IndexMoMPercentChangeL0	0.24	-13.49	-23.73	399.33	2.80			
9	UEMP5TO14MoMPercentChangeL1	USGG10YR_IndexMoMPercentChangeL0	0.24	23.03	-27.68	399.38	2.53			
10	IPNMATMoMPercentChangeL0	USGG10YR_IndexMoMPercentChangeL0	0.24	-51.87	-26.59	399.42	2.83			
11	IPNMATMoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0	0.24	-0.53	-26.58	399.69	2.84			
12	USGG10YR_IndexMoMPercentChangeL0	USGG30YR_IndexMoMDifferenceL1	0.24	-26.58	-5.42	399.88	2.59			
13	CES0600000007MoMDifferenceL1	USGG10YR_IndexMoMDifferenceL0	0.24	-5.77	-11.66	399.94	3.15			
14	CES0600000007MoMPercentChangeL1	USGG10YR_IndexMoMDifferenceL0	0.24	-236.96	-11.65	399.95	3.14			
15	USGG10YR_IndexMoMPercentChangeL0	USGG30YR_IndexMoMPercentChangeL1	0.24	-26.38	-15.04	400.03	2.59			
16	CES0600000007MoMDifferenceL1	USGG30YR_IndexMoMDifferenceL0	0.24	-6.24	-12.74	400.10	3.18			
17	CES0600000007MoMPercentChangeL1	USGG30YR_IndexMoMDifferenceL0	0.24	-255.80	-12.73	400.12	3.17			
18	UEMP15T26MoMPercentChangeL0	USGG10YR_IndexMoMPercentChangeL0	0.24	11.92	-25.63	400.13	2.50			
19	USGG10YR_IndexMoMPercentChangeL0	BAA10YB_IndexMoMDifferenceL0	0.24	-33.33	-0.07	400.16	2.77			
20	USGG2YR_IndexMoMPercentChangeL1	USGG10YR_IndexMoMPercentChangeL0	0.23	-5.84	-29.10	400.26	2.78			
21	HOUSTMWMoMPercentChangeL1	USGG10YR_IndexMoMPercentChangeL0	0.23	-3.22	-24.35	400.56	2.55			
22	AMBSLMoMPercentChangeL0	USGG10YR_IndexMoMPercentChangeL0	0.23	38.03	-24.53	400.56	2.65			
23	USGG10YR_IndexMoMPercentChangeL0	AAIBEAR_IndexMoMPercentChangeL1	0.23	-26.63	2.74	400.59	2.81			
24	HOUSTMWMoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL0	0.23	-0.02	-24.33	400.63	2.59			
25	USCONSMoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0	0.23	0.02	-24.96	400.63	2.45			
26	USGG10YR_IndexMoMPercentChangeL0	AAIBULL_IndexMoMDifferenceL1	0.23	-26.06	-0.09	400.72	2.96			
27	IPDCONGDMoMPercentChangeL0	USGG10YR_IndexMoMPercentChangeL0	0.23	-30.91	-26.98	400.75	2.66			
28	AMBSLMoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0	0.23	0.01	-24.47	400.76	2.64			
29	USGG2YR_IndexMoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL0	0.23	-6.64	-27.45	400.77	3.01			
30	USGG10YR_IndexMoMPercentChangeL0	BAA10YB_IndexMoMPercentChangeL1	0.23	-24.79	8.01	400.80	2.71			
31	UEMP15OVMoMPercentChangeL0	USGG10YR_IndexMoMPercentChangeL0	0.23	19.74	-24.66	400.81	3.08			
32	HOUSTNEMoMPercentChangeL1	USGG10YR_IndexMoMPercentChangeL0	0.23	2.31	-27.09	400.81	2.56			
33	UEMP15T26MoMDifferenceL1	USGG30YR_IndexMoMPercentChangeL0	0.23	-0.01	-30.05	400.85	2.72			
34	AWHMANMoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL0	0.23	-4.68	-26.16	400.86	2.87			
35	AWHMANMoMPercentChangeL1	USGG10YR_IndexMoMPercentChangeL0	0.23	-195.51	-26.16	400.86	2.87			
36	USFIREMoMPercentChangeL0	USGG10YR_IndexMoMPercentChangeL0	0.23	412.11	-25.65	400.87	2.69			
37	USFIREMoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0	0.23	0.05	-25.69	400.89	2.70			
38	USGG10YR_IndexMoMPercentChangeL0	AAIBEAR_IndexMoMDifferenceL1	0.23	-26.27	0.10	400.93	2.83			
39	USGG10YR_IndexMoMPercentChangeL0	BAA10YB_IndexMoMDifferenceL1	0.23	-24.47	0.05	400.96	2.64			
40	USCONSMoMPercentChangeL0	USGG10YR_IndexMoMPercentChangeL0	0.23	125.24	-24.85	400.97	2.48			
41	USGG10YR_IndexMoMPercentChangeL0	AAIBEAR_IndexMoMPercentChangeL0	0.23	-28.88	-2.65	400.98	2.90			
42	UEMP5TO14MoMPercentChangeL0	USGG10YR_IndexMoMPercentChangeL0	0.23	18.37	-24.86	401.00	2.49			
43	IPDCONGDMoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0	0.23	-0.26	-27.01	401.03	2.67			
44	USGG10YR_IndexMoMPercentChangeL0	AAIBULL_IndexMoMPercentChangeL1	0.23	-26.00	-2.80	401.08	2.87			
45	UEMP15T26MoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0	0.22	0.01	-25.45	401.11	2.52			
46	UEMP5TO14MoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0	0.22	0.01	-24.96	401.16	2.51			
47	UEMP15OVMoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0	0.22	0.01	-24.75	401.16	2.71			
48	USGG10YR_IndexMoMPercentChangeL0	AAIBULL_IndexMoMPercentChangeL0	0.22	-27.60	2.79	401.16	2.82			
49	IPCONGDMoMPercentChangeL0	USGG10YR_IndexMoMPercentChangeL0	0.22	-61.78	-26.81	401.18	2.77			
50	UNRATEMoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL0	0.22	5.47	-28.67	401.20	2.71			

Table 16: Two Factor Analysis-CO₉oldTable 17: Relevant Variables: CO₉old

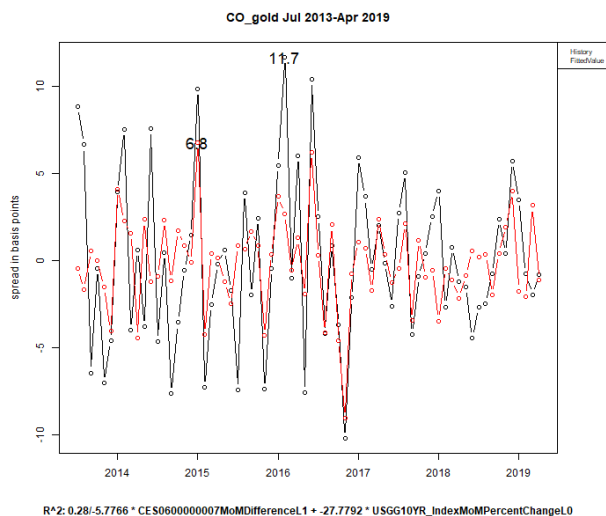
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3 RPIMoMPercentChangeL0	UEMP27OVMoMDifferenceL0	BUSLOANSMoMPercentChangeL0
4 RPIMoMPercentChangeL1	UEMP27OVMoMDifferenceL1	BUSLOANSMoMPercentChangeL1
5 W875RX1MoMDifferenceL0	UEMP27OVMoMPercentChangeL0	REALLNMoMDifferenceL0
6 W875RX1MoMDifferenceL1	UEMP27OVMoMPercentChangeL1	REALLNMoMDifferenceL1
7 W875RX1MoMPercentChangeL0	PAYEMSMoMDifferenceL0	REALLNMoMPercentChangeL0
8 W875RX1MoMPercentChangeL1	PAYEMSMoMDifferenceL1	REALLNMoMPercentChangeL1
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10 DPCERA3M086SBEAMoMDifferenceL1	PAYEMSMoMPercentChangeL1	WPSFD49207MoMDifferenceL1
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12 DPCERA3M086SBEAMoMPercentChangeL1	USGOODMoMDifferenceL1	WPSFD49207MoMPercentChangeL1
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14 CMRMTSPLMoMDifferenceL1	USGOODMoMPercentChangeL1	WPSFD49502MoMDifferenceL1
15 CMRMTSPLMoMPercentChangeL0	CES1021000001MoMDifferenceL0	WPSFD49502MoMPercentChangeL0
16 CMRMTSPLMoMPercentChangeL1	CES1021000001MoMDifferenceL1	WPSFD49502MoMPercentChangeL1
17 INDPROMoMDifferenceL0	CES1021000001MoMPercentChangeL0	WPSID61MoMDifferenceL0
18 INDPROMoMDifferenceL1	CES1021000001MoMPercentChangeL1	WPSID61MoMDifferenceL1
19 INDPROMoMPercentChangeL0	USCONSMoMDifferenceL0	WPSID61MoMPercentChangeL0
20 INDPROMoMPercentChangeL1	USCONSMoMDifferenceL1	WPSID61MoMPercentChangeL1
21 IPFPNSSMoMDifferenceL0	USCONSMoMPercentChangeL0	WPSID62MoMDifferenceL0
22 IPFPNSSMoMDifferenceL1	USCONSMoMPercentChangeL1	WPSID62MoMDifferenceL1
23 IPFPNSSMoMPercentChangeL0	MANEMPMoMDifferenceL0	WPSID62MoMPercentChangeL0
24 IPFPNSSMoMPercentChangeL1	MANEMPMoMDifferenceL1	WPSID62MoMPercentChangeL1
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26 IPFINALMoMDifferenceL1	MANEMPMoMPercentChangeL1	MZMSLMoMDifferenceL1
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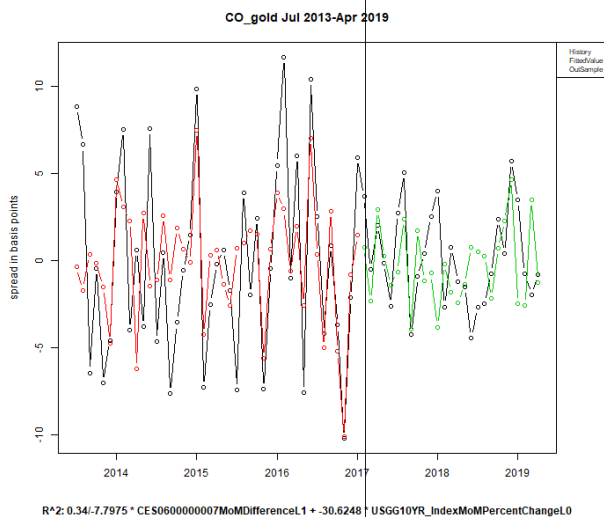
4.10.1 Model - 1

Backtesting on differenced values

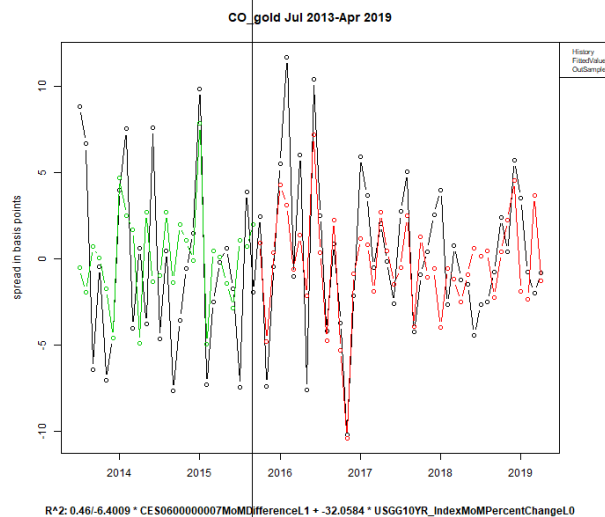
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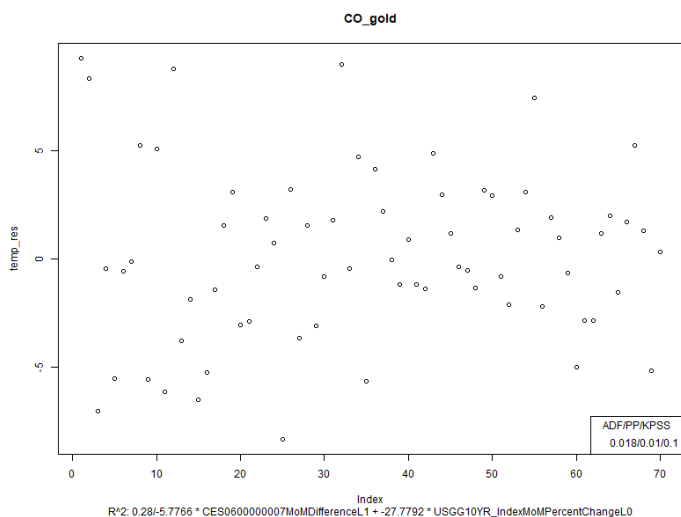
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Out-of-Sample testing



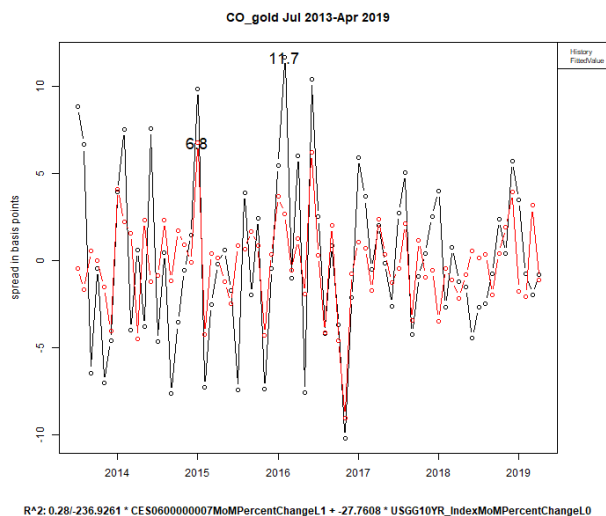
Stationarity of Residuals



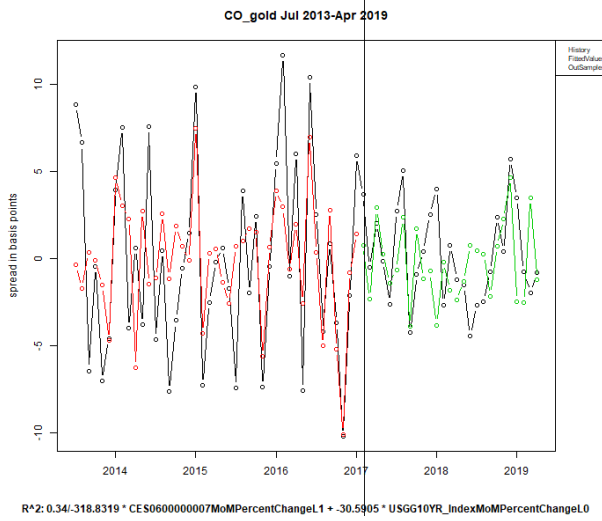
4.10.2 Model - 2

Backtesting on differenced values

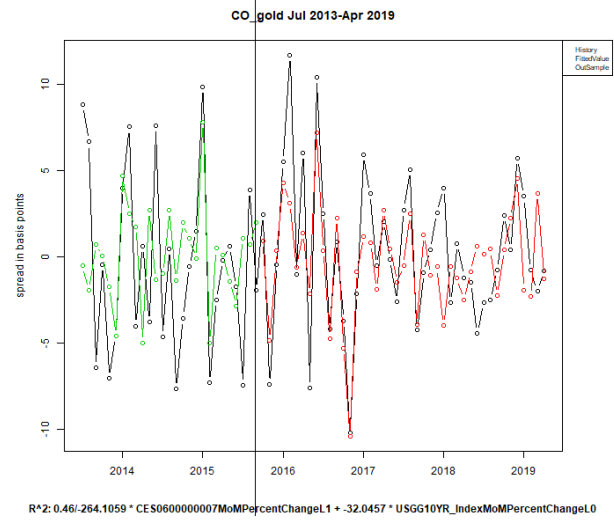
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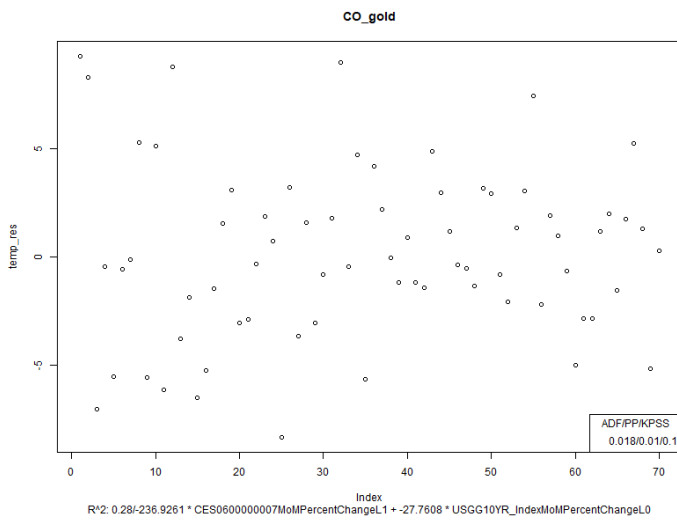
Out-of-Sample testing



Out-of-Sample testing



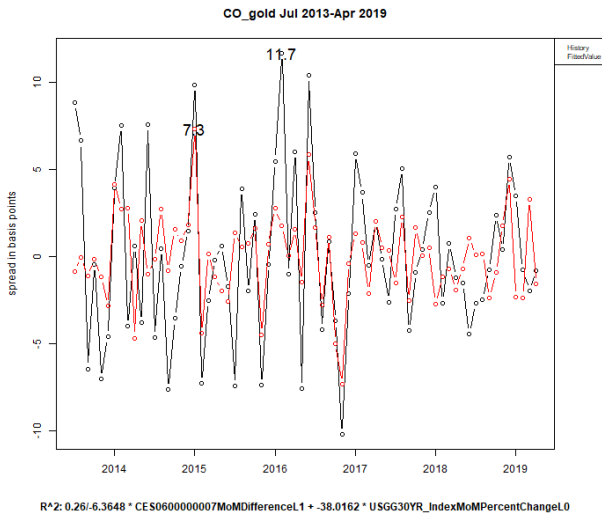
Stationarity of Residuals



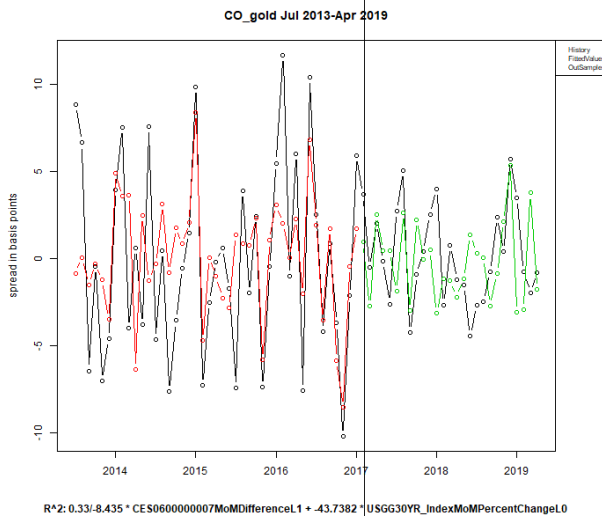
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Backtesting on differenced values

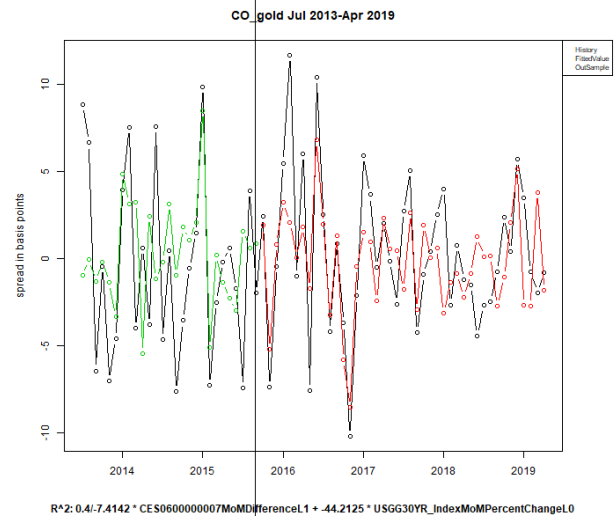
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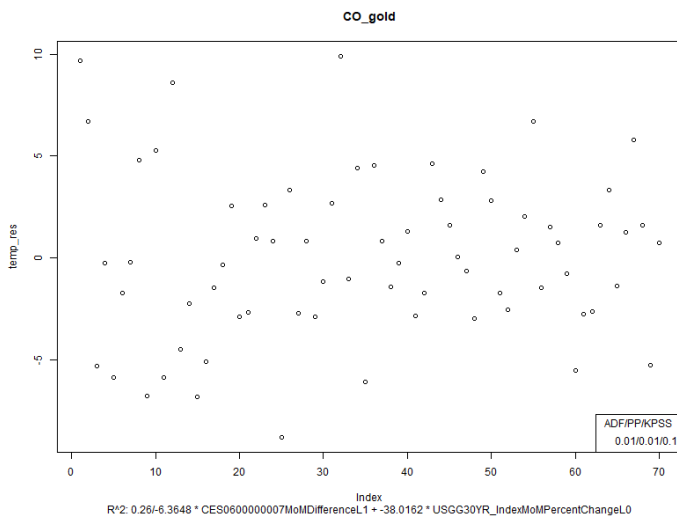
Out-of-Sample testing



Out-of-Sample testing



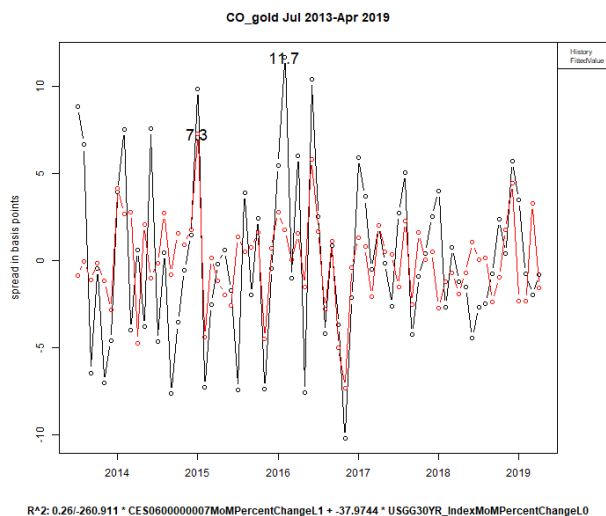
Stationarity of Residuals



4.10.4 Model - 4

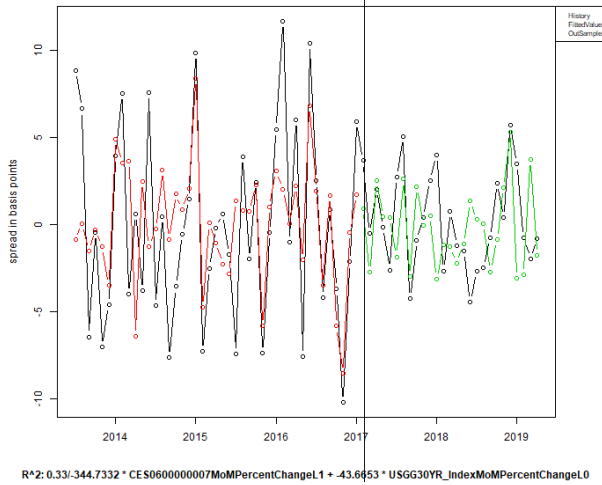
Backtesting on differenced values

Backtesting on level values



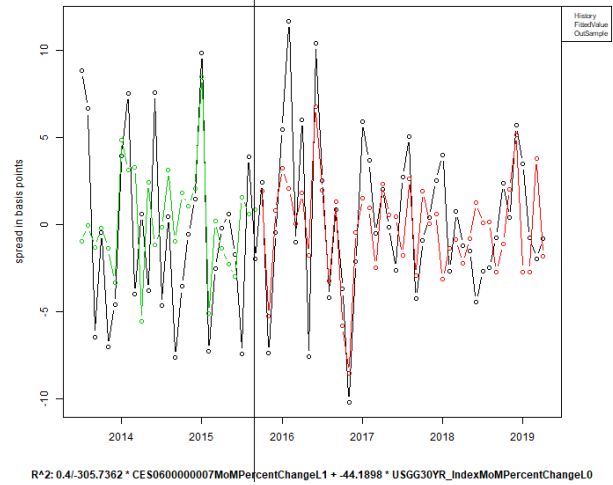
Out-of-Sample testing

CO_gold Jul 2013-Apr 2019



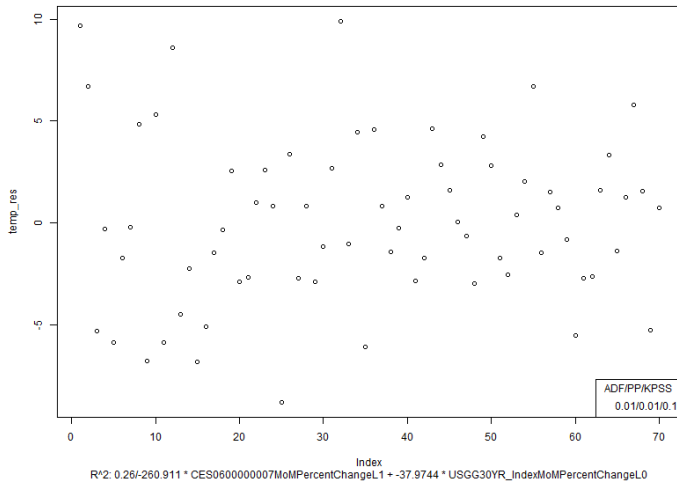
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CO_gold Jul 2013-Apr 2019



Stationarity of Residuals

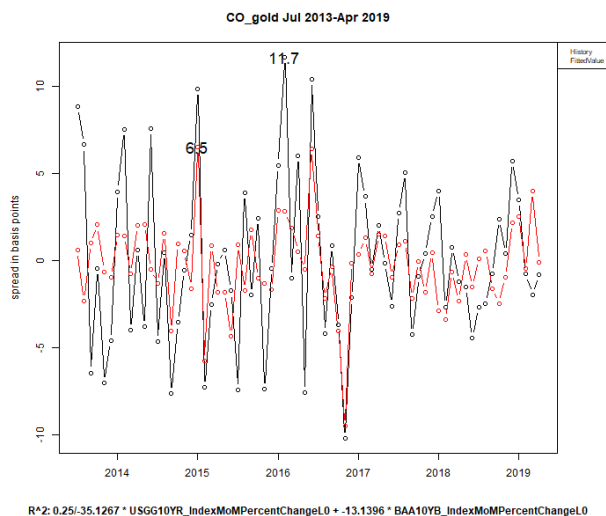
CO_gold



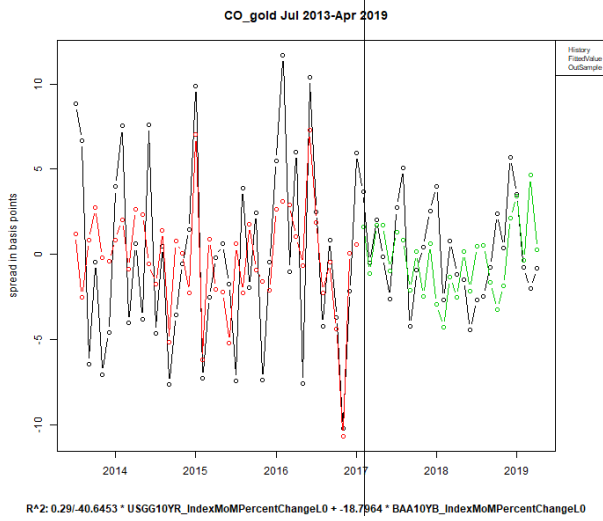
4.10.5 Model - 5

Backtesting on differenced values

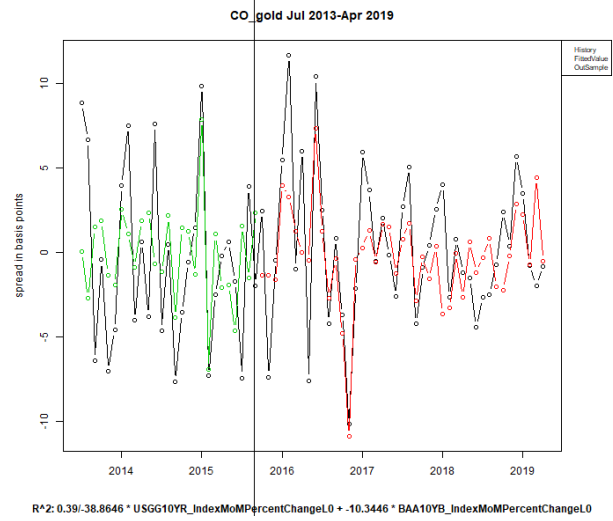
Backtesting on level values



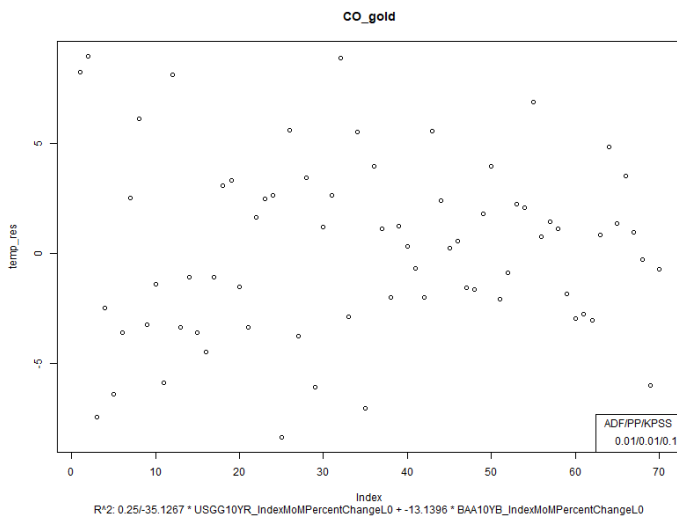
Out-of-Sample testing



Out-of-Sample testing



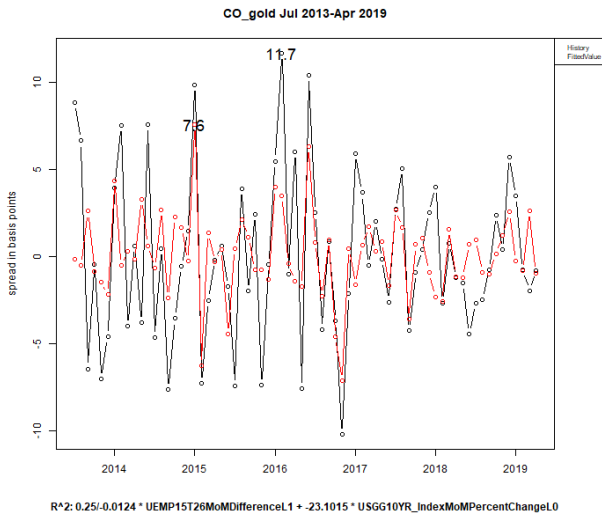
Stationarity of Residuals



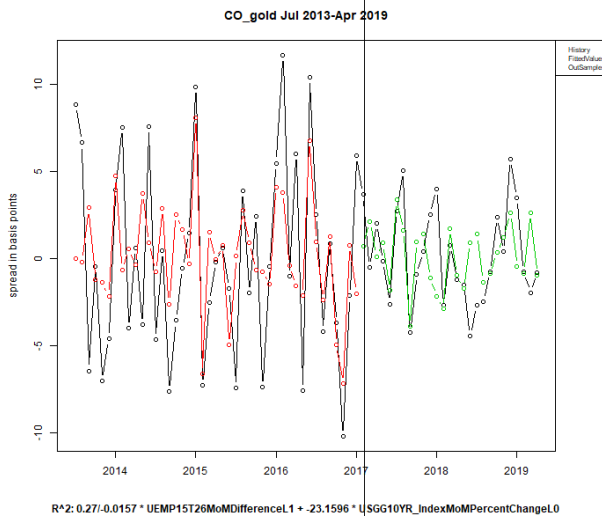
4.10.6 Model - 6

Backtesting on differenced values

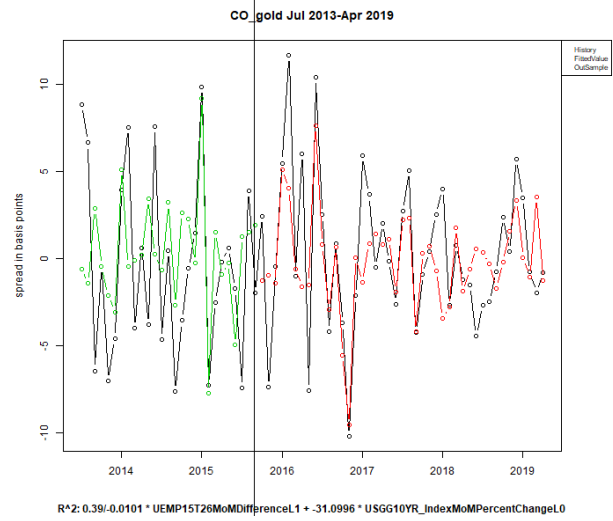
Backtesting on level values



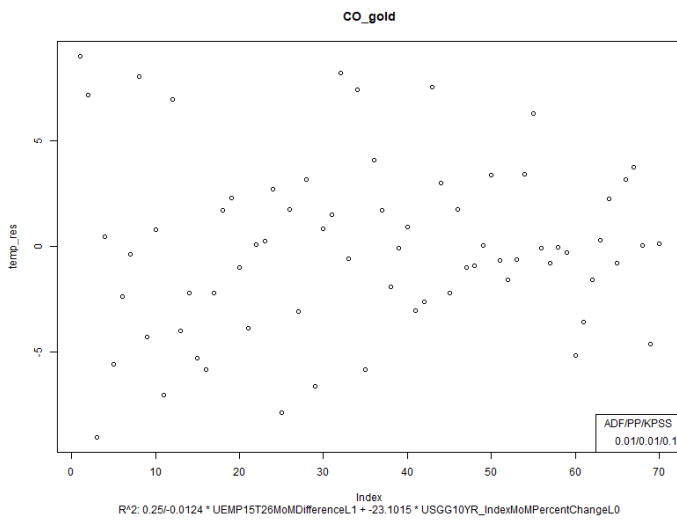
Out-of-Sample testing



Out-of-Sample testing



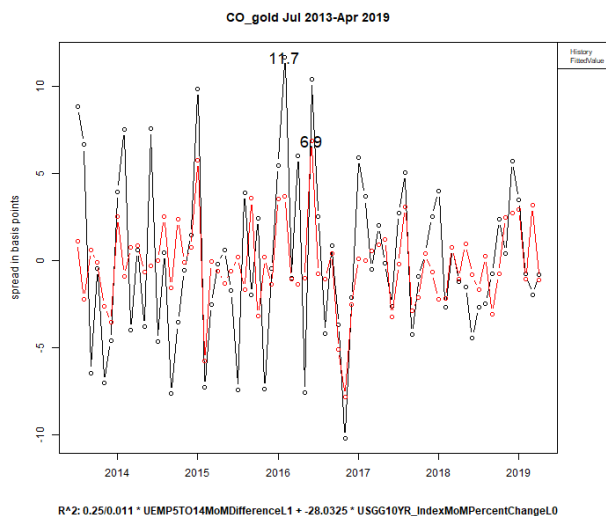
Stationarity of Residuals



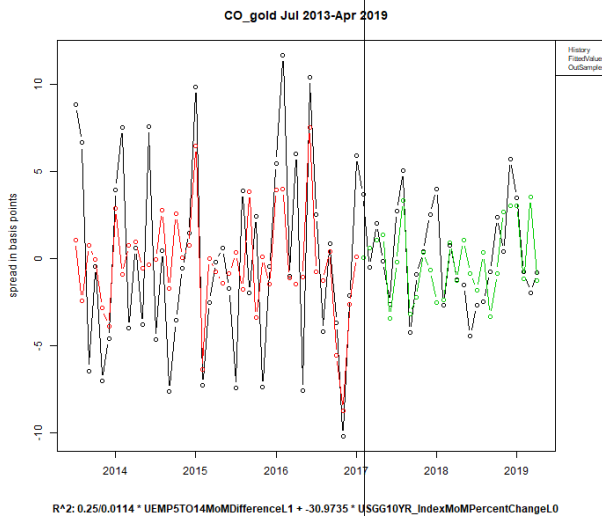
4.10.7 Model - 7

Backtesting on differenced values

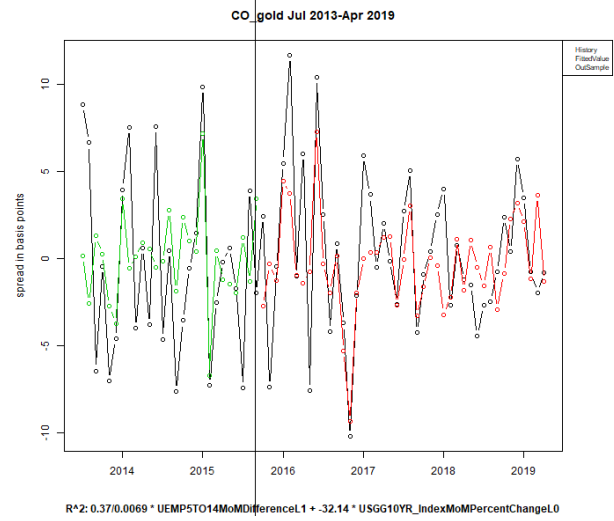
Backtesting on level values



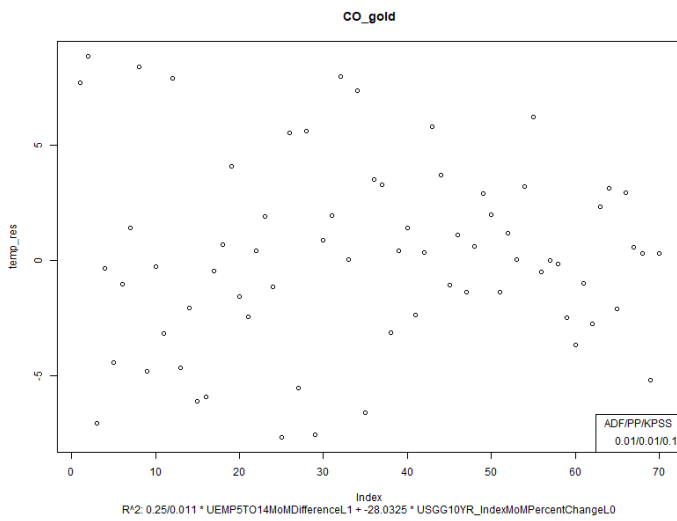
Out-of-Sample testing



Out-of-Sample testing



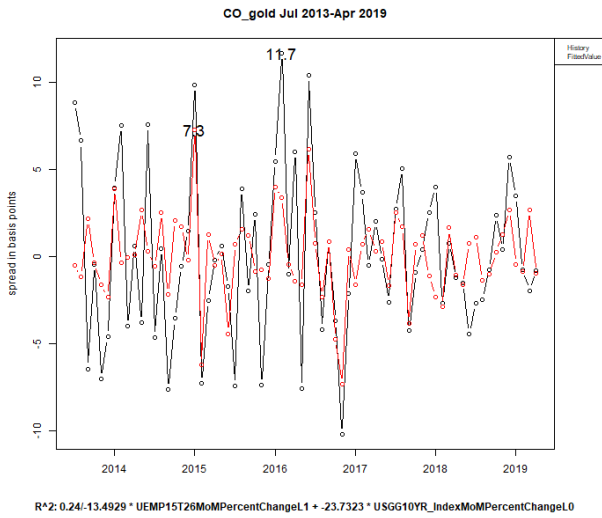
Stationarity of Residuals



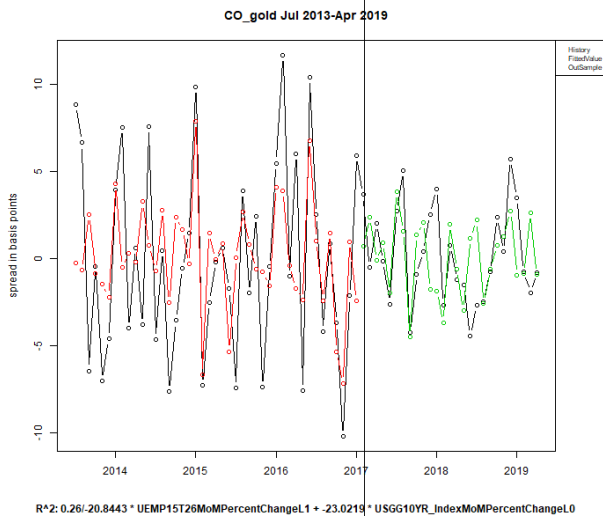
4.10.8 Model - 8

Backtesting on differenced values

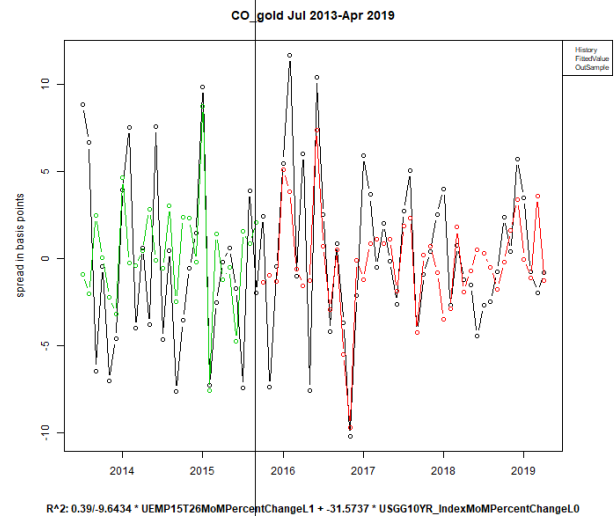
Backtesting on level values



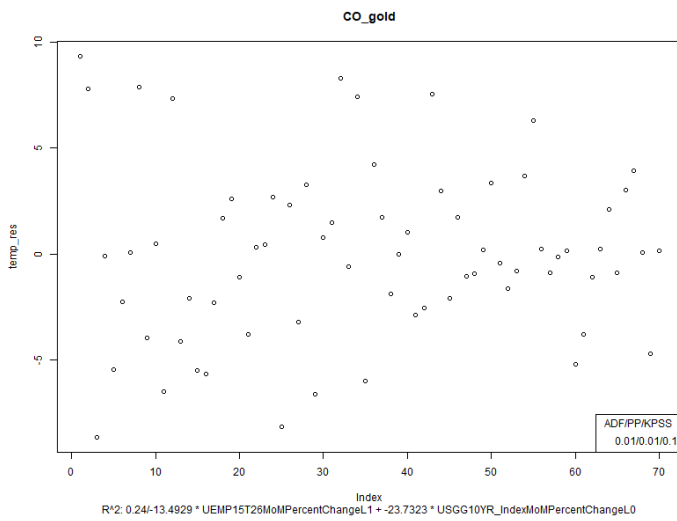
Out-of-Sample testing



Out-of-Sample testing



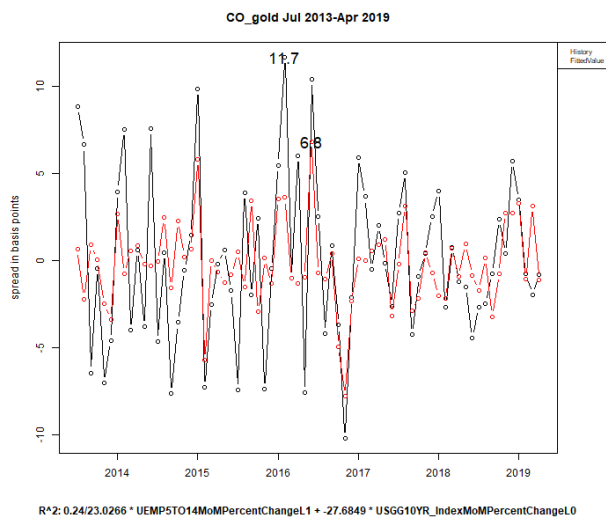
Stationarity of Residuals



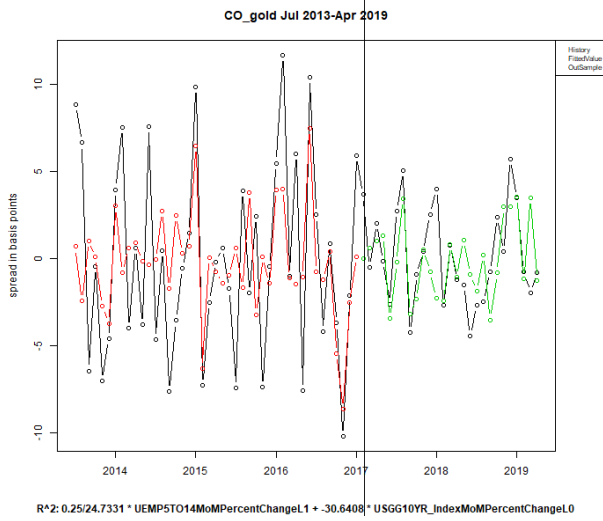
4.10.9 Model - 9

Backtesting on differenced values

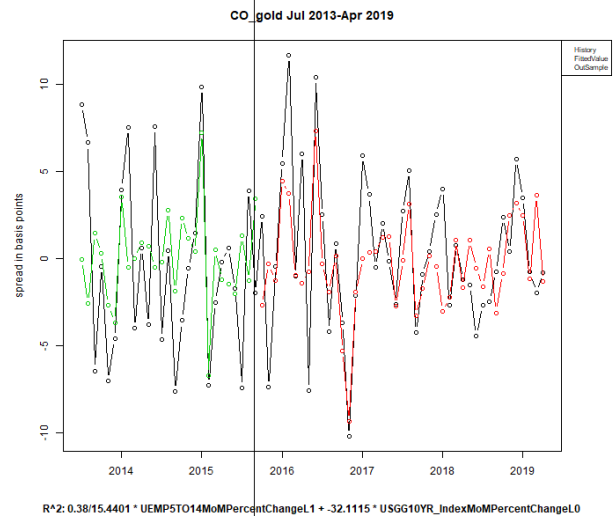
Backtesting on level values



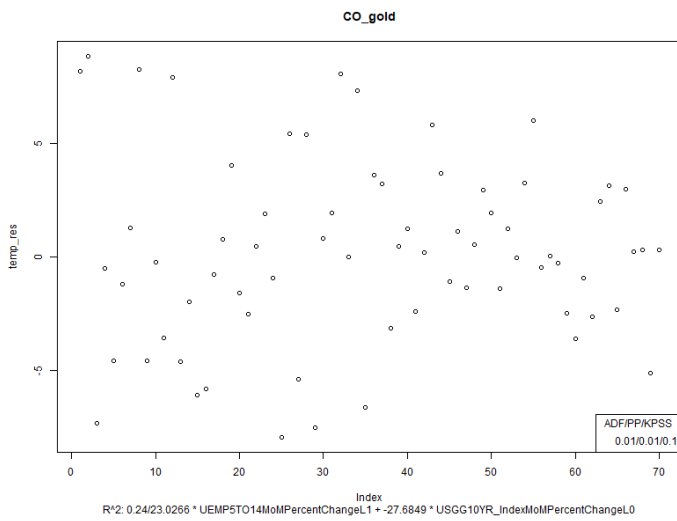
Out-of-Sample testing



Out-of-Sample testing



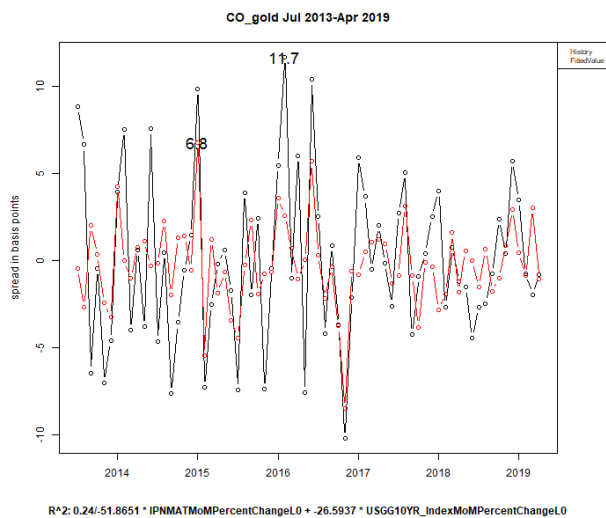
Stationarity of Residuals



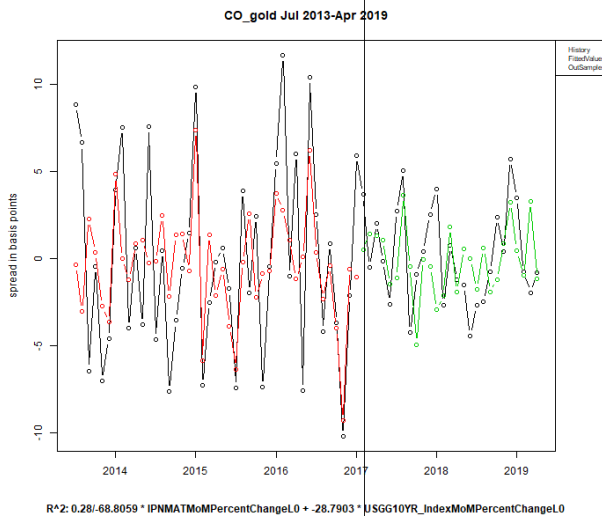
4.10.10 Model - 10

Backtesting on differenced values

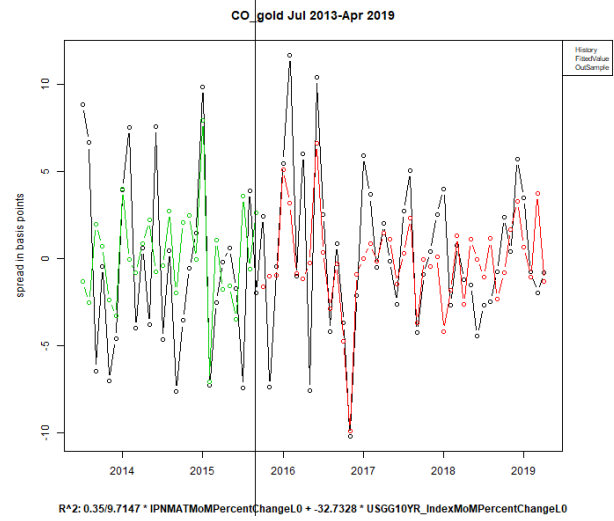
Backtesting on level values



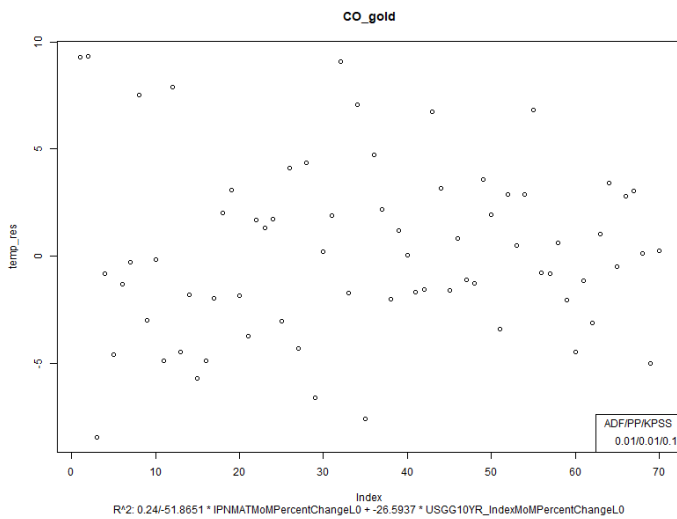
Out-of-Sample testing



Out-of-Sample testing



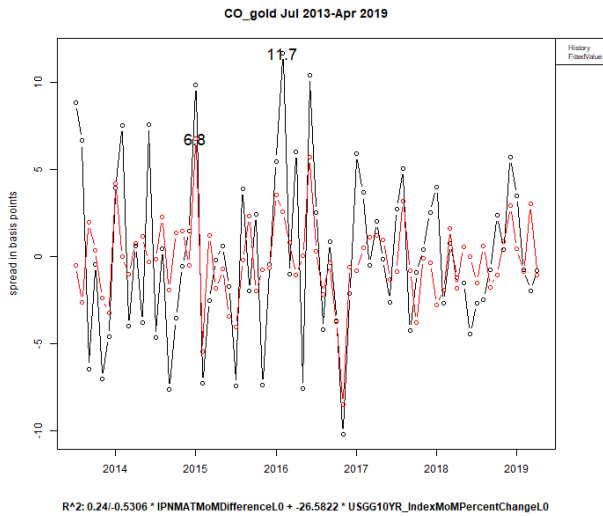
Stationarity of Residuals



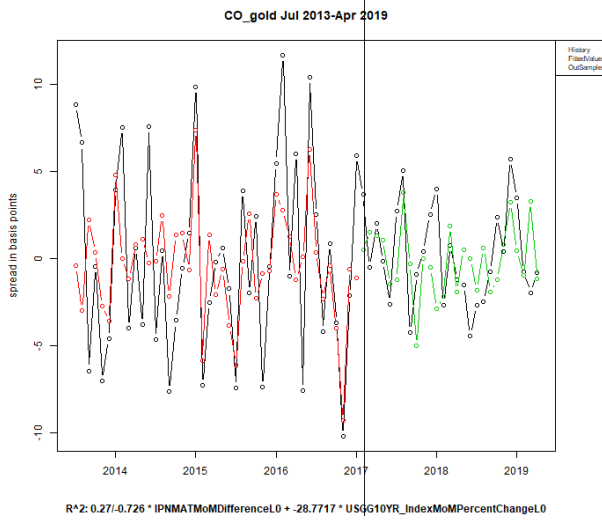
4.10.11 Model - 11

Backtesting on differenced values

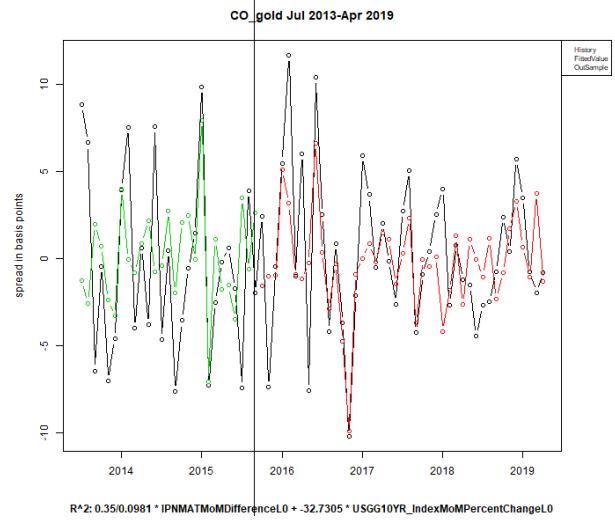
Backtesting on level values



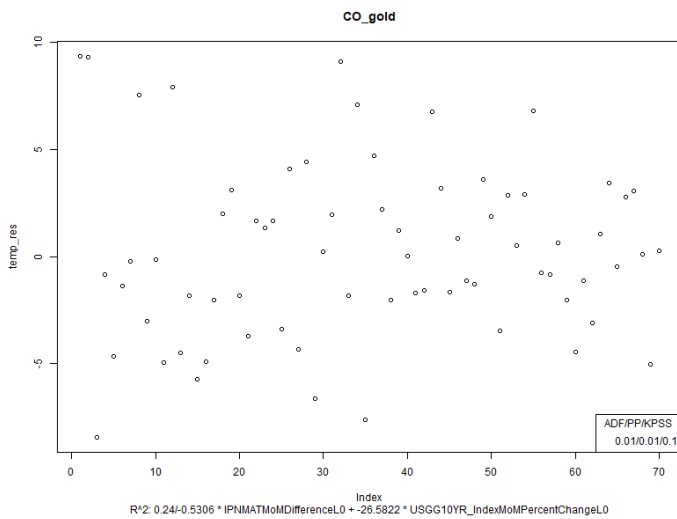
Out-of-Sample testing



Out-of-Sample testing



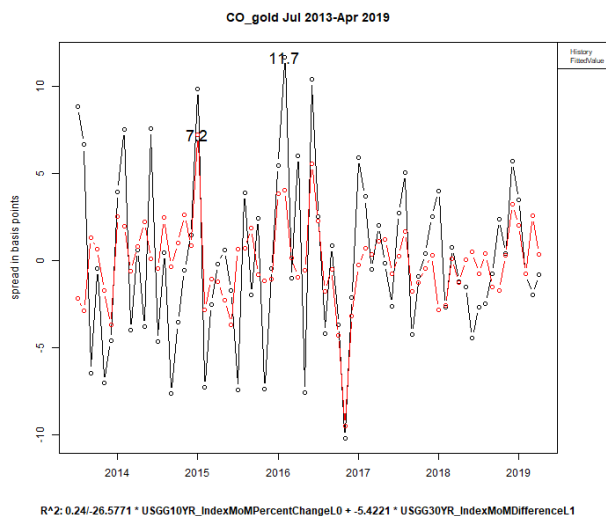
Stationarity of Residuals



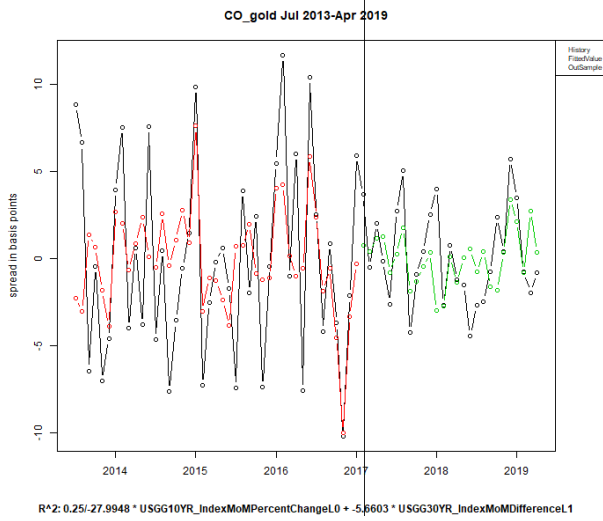
4.10.12 Model - 12

Backtesting on differenced values

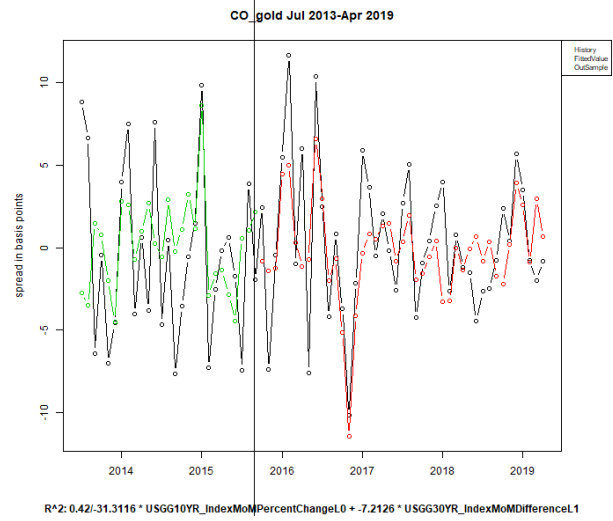
Backtesting on level values



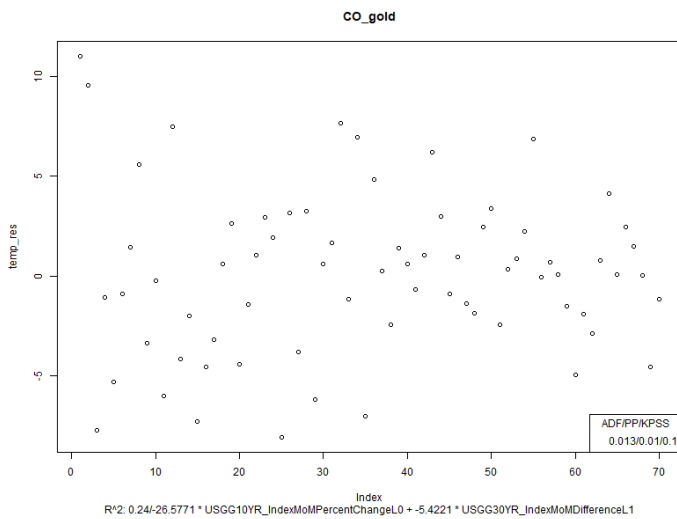
Out-of-Sample testing



Out-of-Sample testing



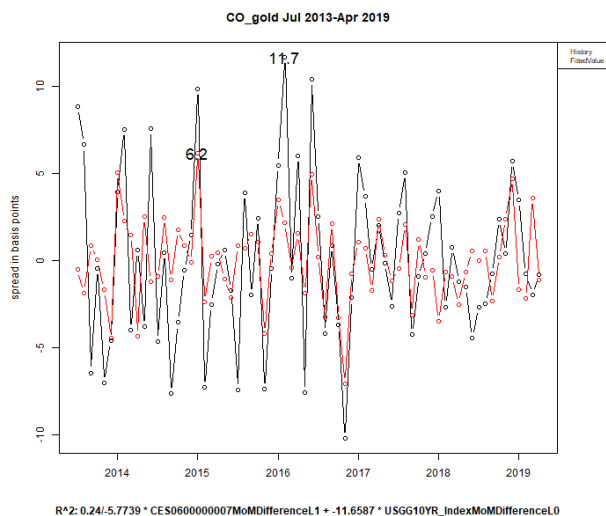
Stationarity of Residuals



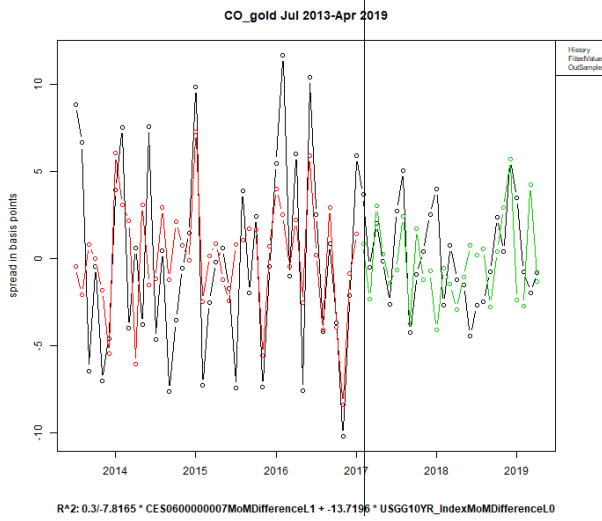
4.10.13 Model - 13

Backtesting on differenced values

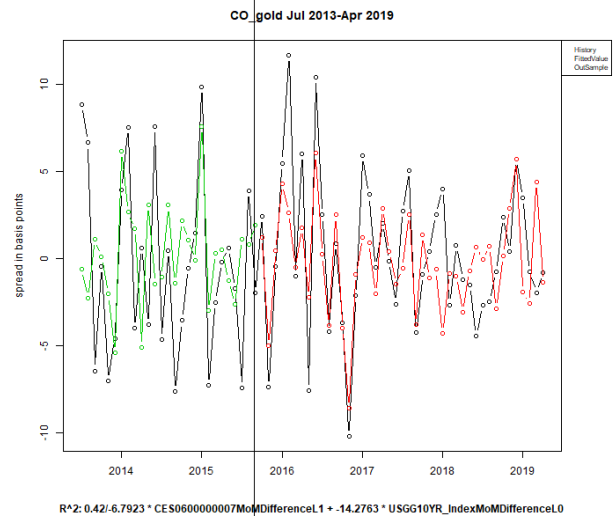
Backtesting on level values



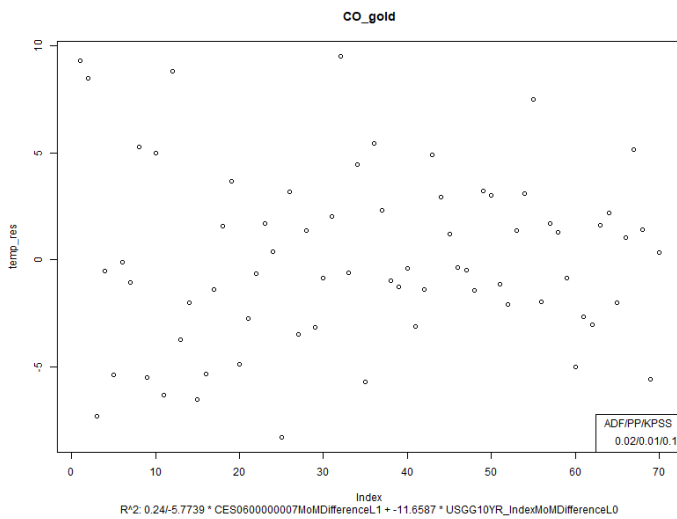
Out-of-Sample testing



Out-of-Sample testing



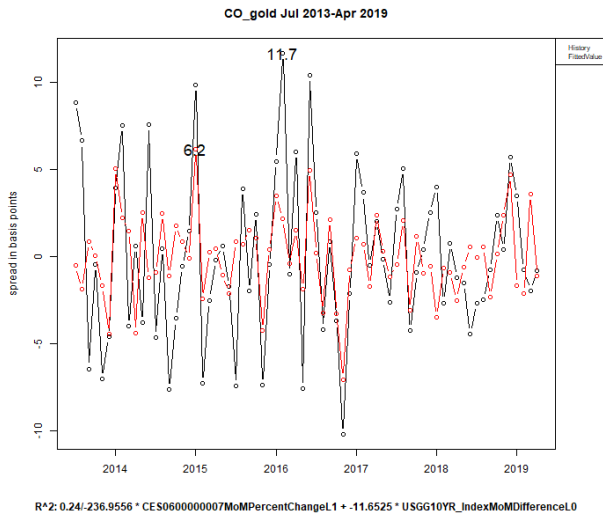
Stationarity of Residuals



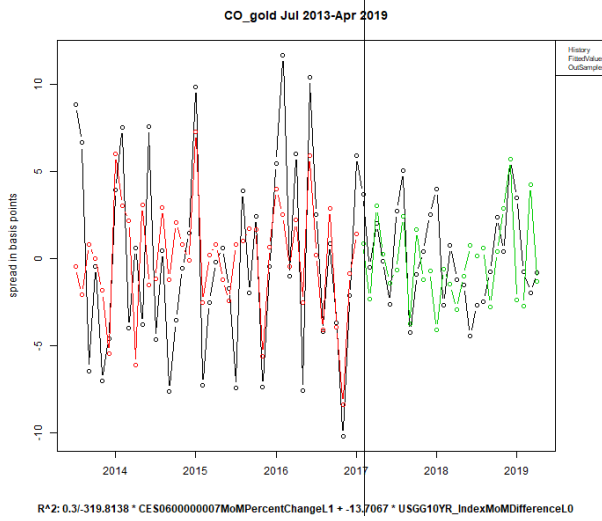
4.10.14 Model - 14

Backtesting on differenced values

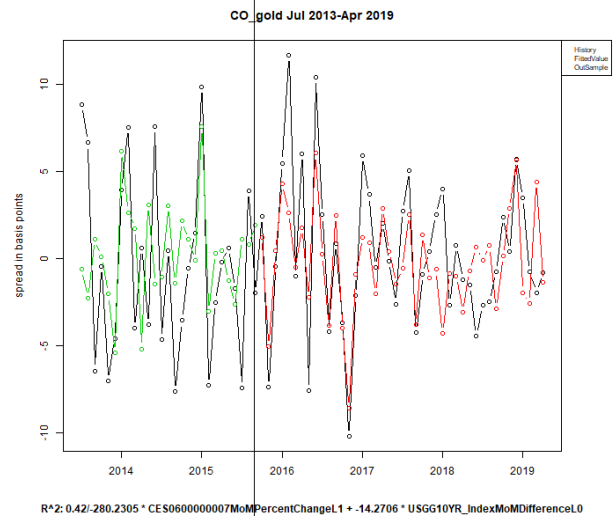
Backtesting on level values



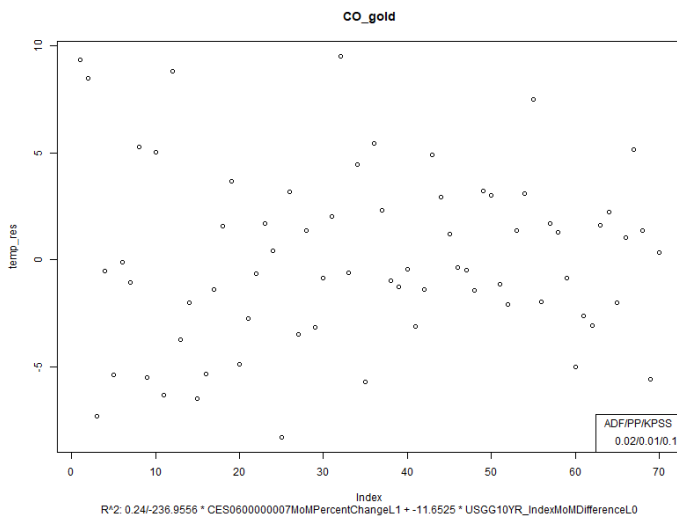
Out-of-Sample testing



Out-of-Sample testing



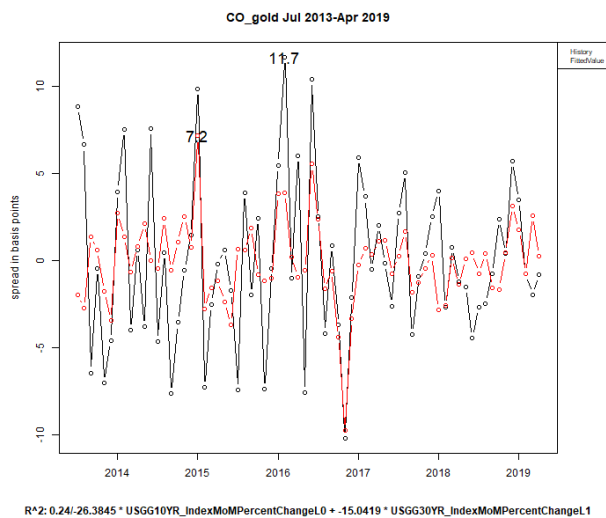
Stationarity of Residuals



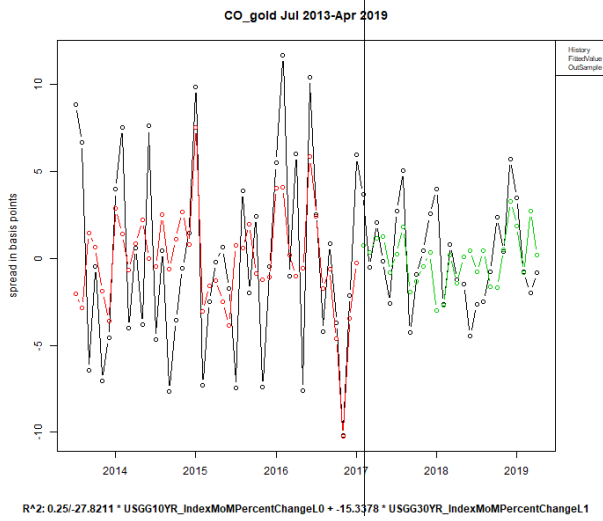
4.10.15 Model - 15

Backtesting on differenced values

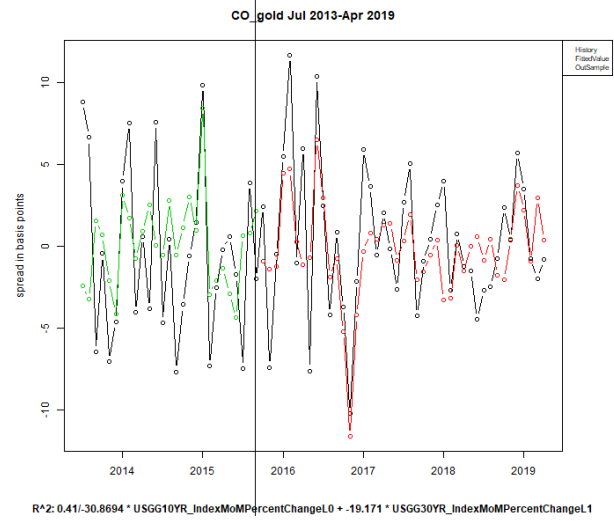
Backtesting on level values



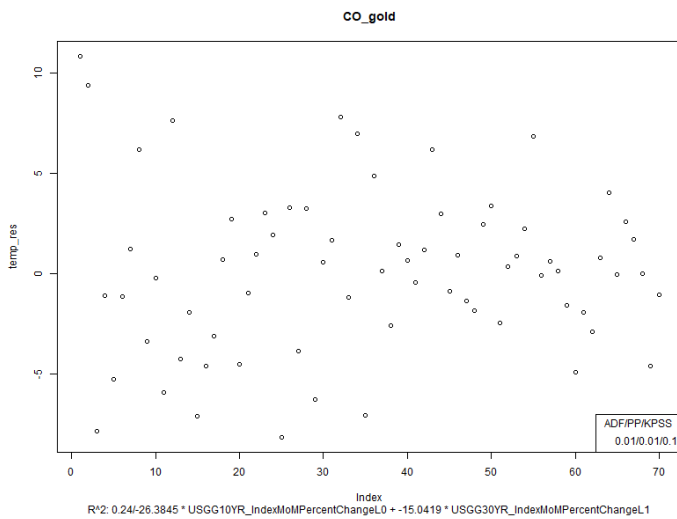
Out-of-Sample testing



Out-of-Sample testing



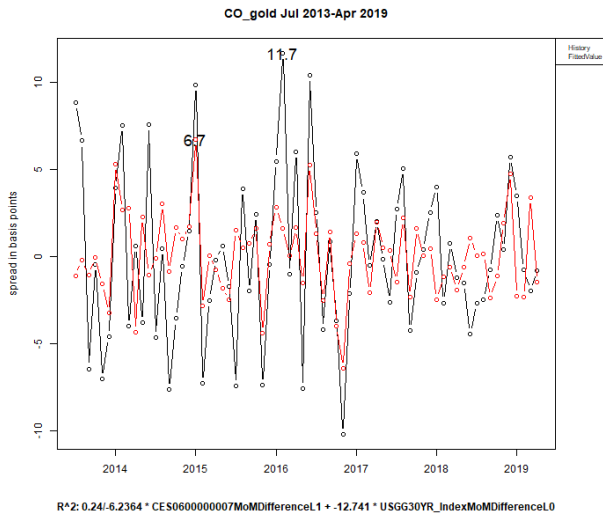
Stationarity of Residuals



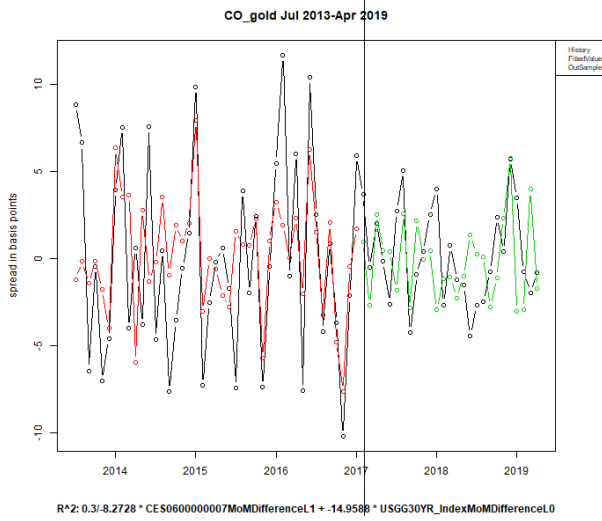
4.10.16 Model - 16

Backtesting on differenced values

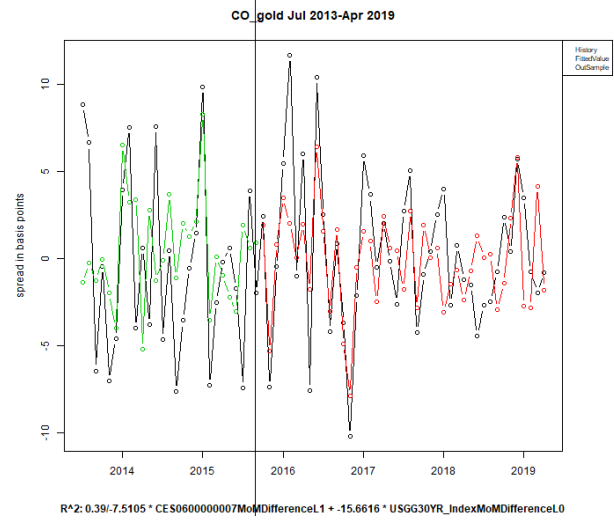
Backtesting on level values



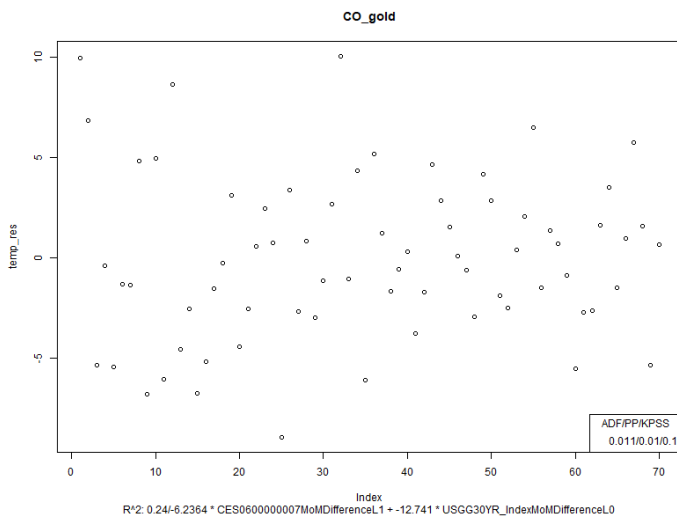
Out-of-Sample testing



Out-of-Sample testing



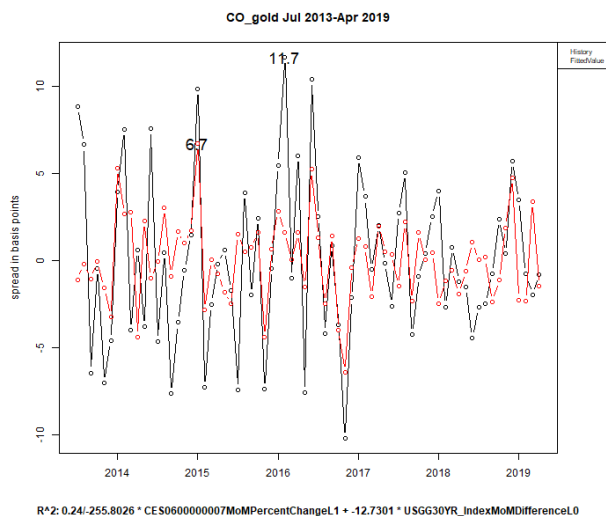
Stationarity of Residuals



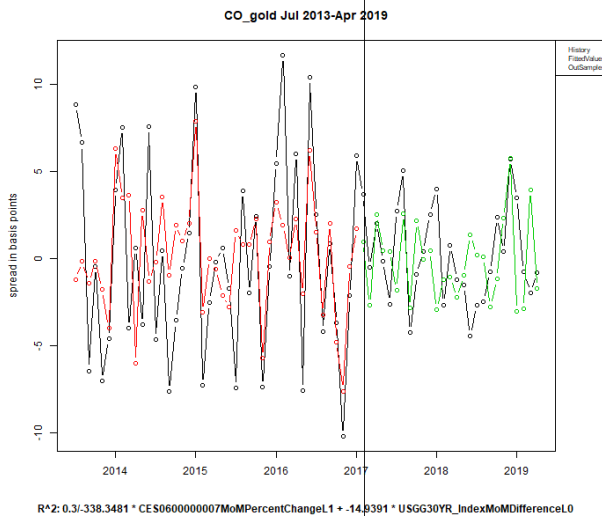
4.10.17 Model - 17

Backtesting on differenced values

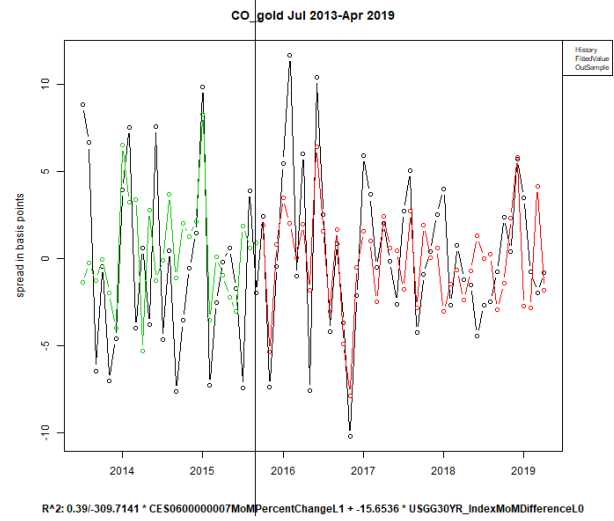
Backtesting on level values



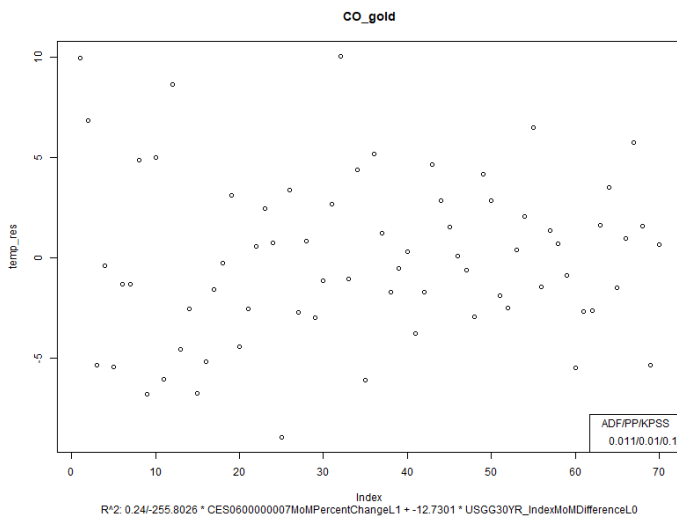
Out-of-Sample testing



Out-of-Sample testing



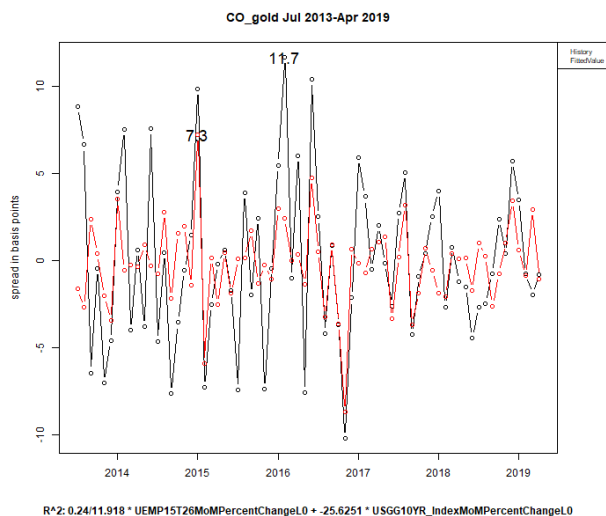
Stationarity of Residuals



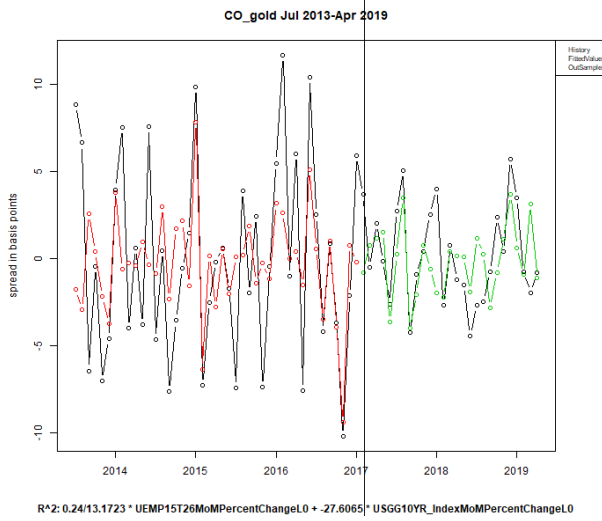
4.10.18 Model - 18

Backtesting on differenced values

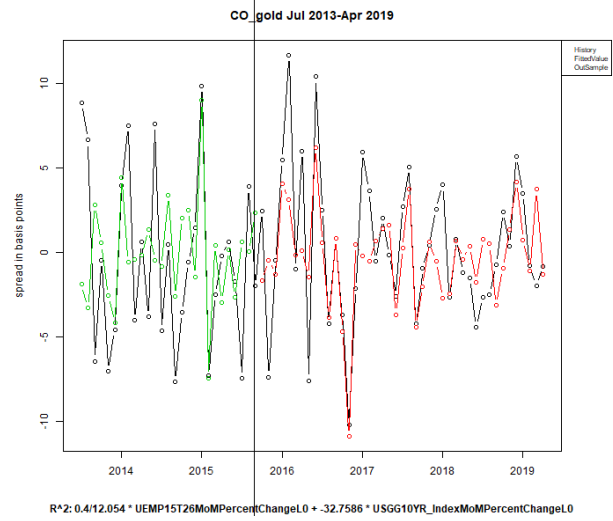
Backtesting on level values



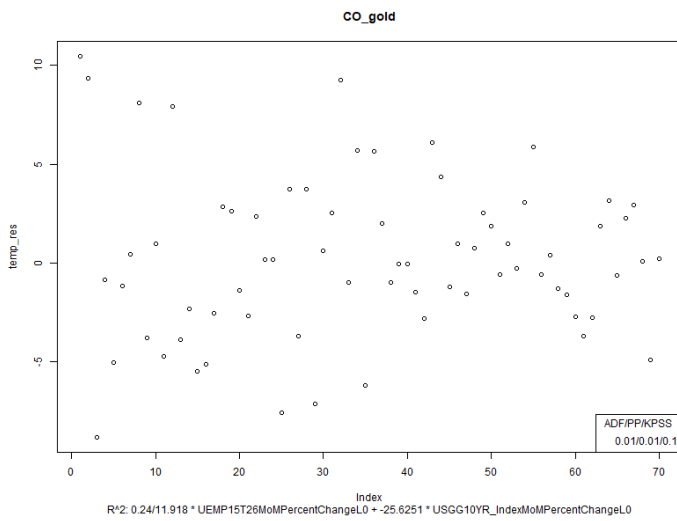
Out-of-Sample testing



Out-of-Sample testing



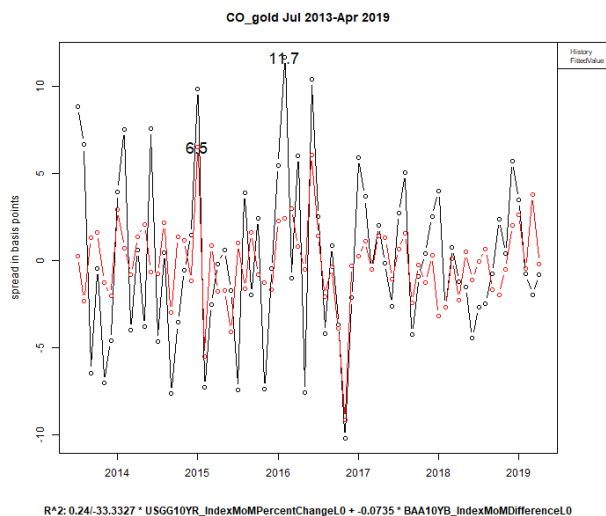
Stationarity of Residuals



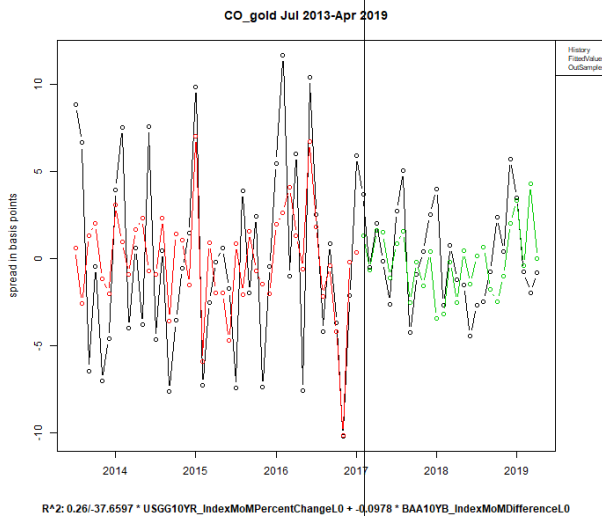
4.10.19 Model - 19

Backtesting on differenced values

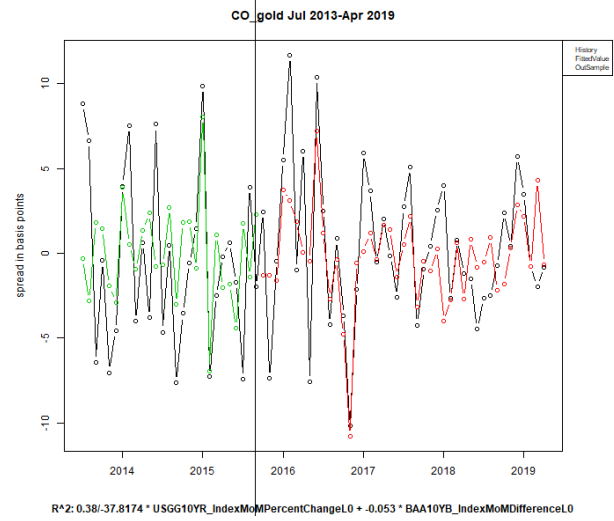
Backtesting on level values



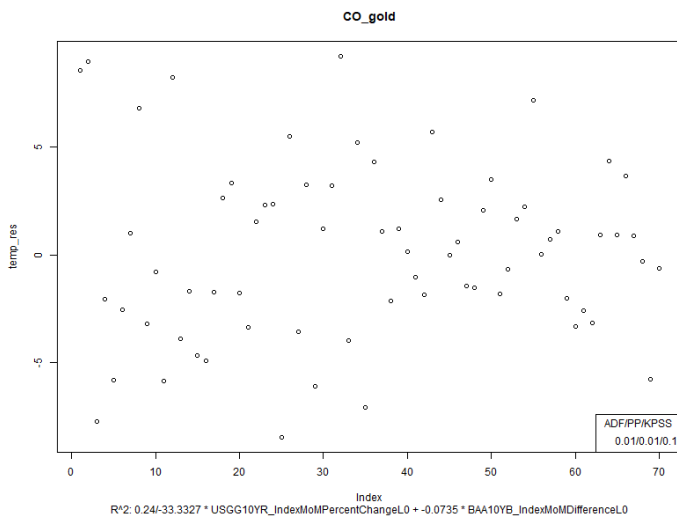
Out-of-Sample testing



Out-of-Sample testing



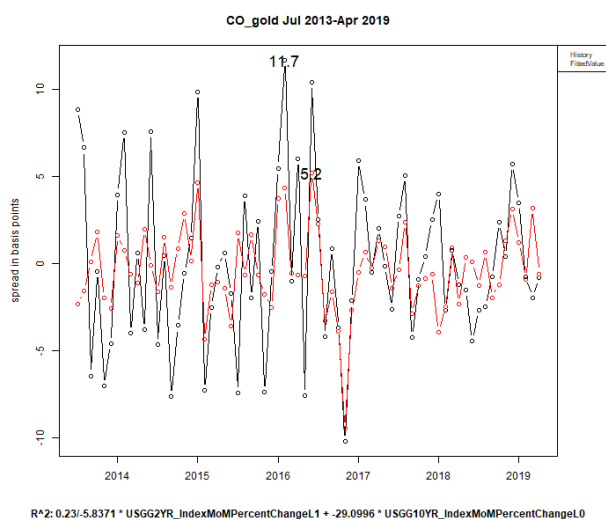
Stationarity of Residuals



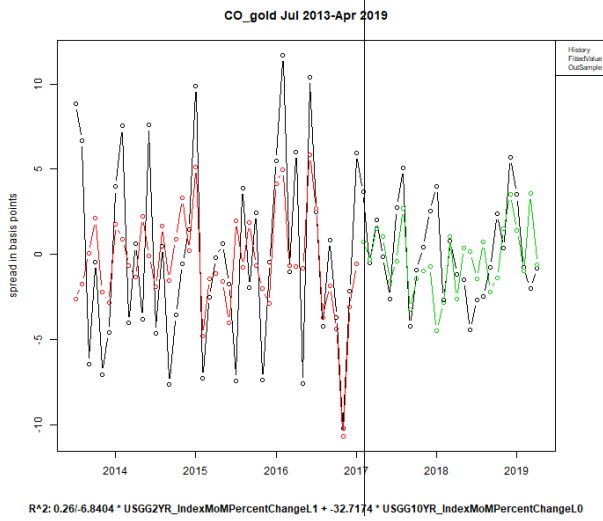
4.10.20 Model - 20

Backtesting on differenced values

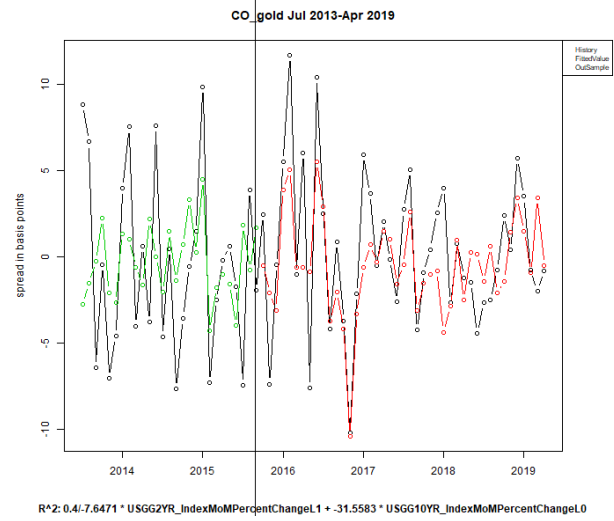
Backtesting on level values



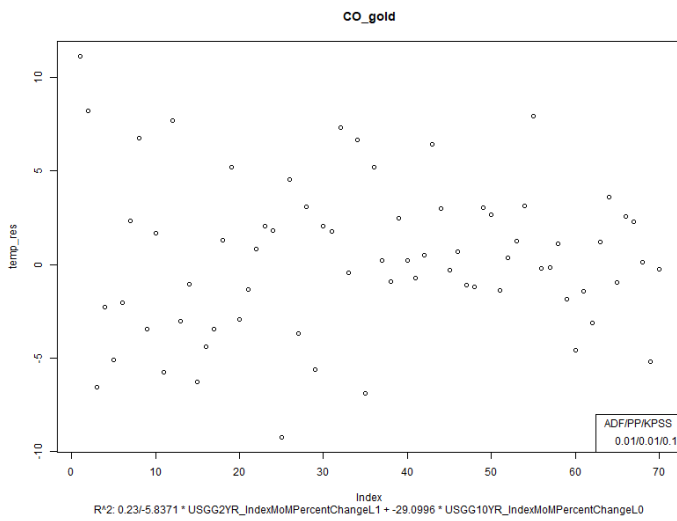
Out-of-Sample testing



Out-of-Sample testing



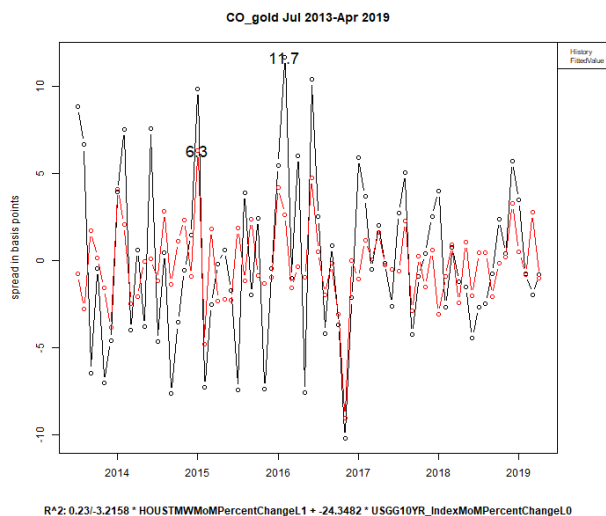
Stationarity of Residuals



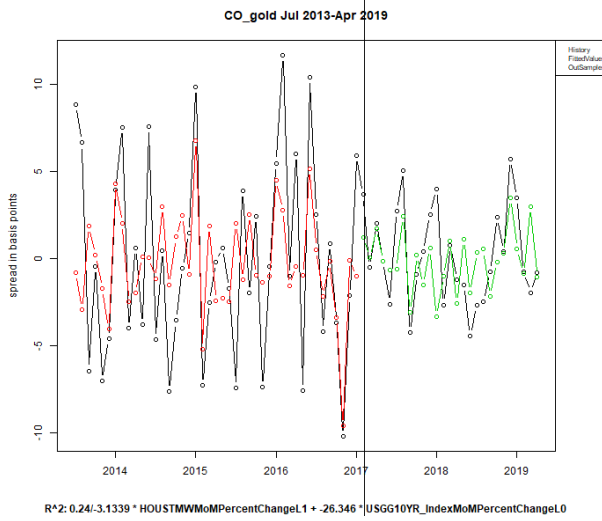
4.10.21 Model - 21

Backtesting on differenced values

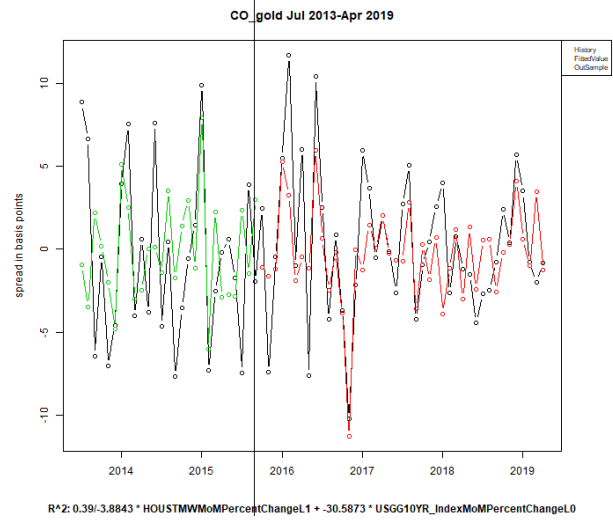
Backtesting on level values



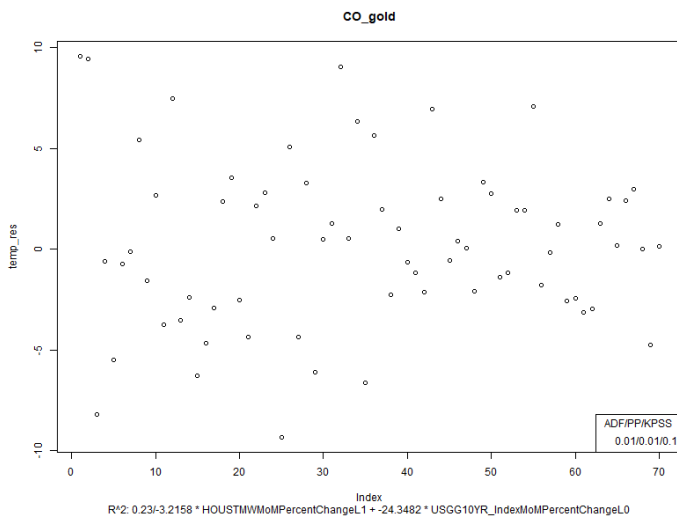
Out-of-Sample testing



Out-of-Sample testing



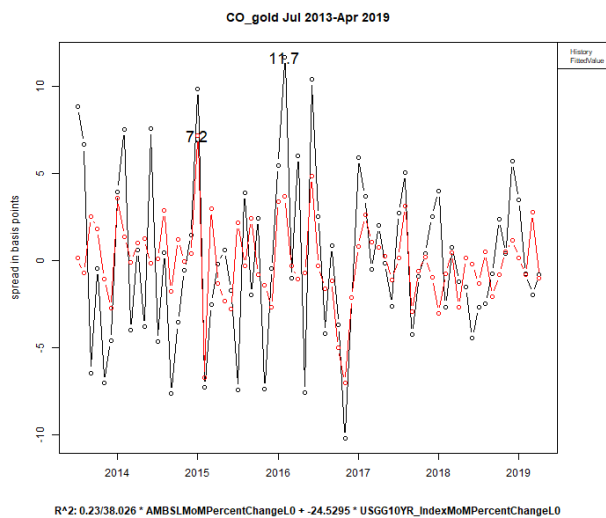
Stationarity of Residuals



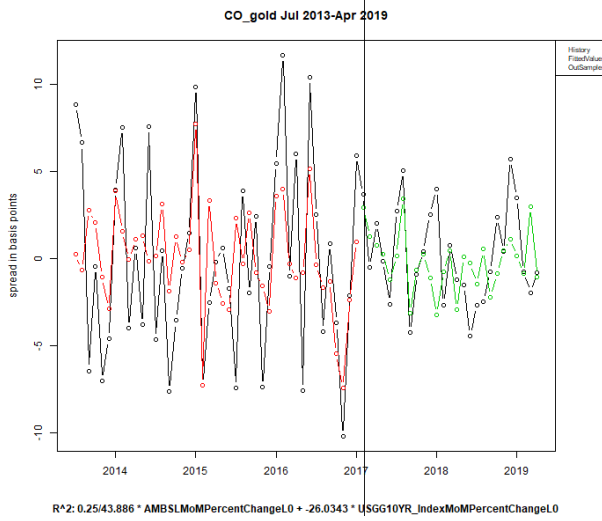
4.10.22 Model - 22

Backtesting on differenced values

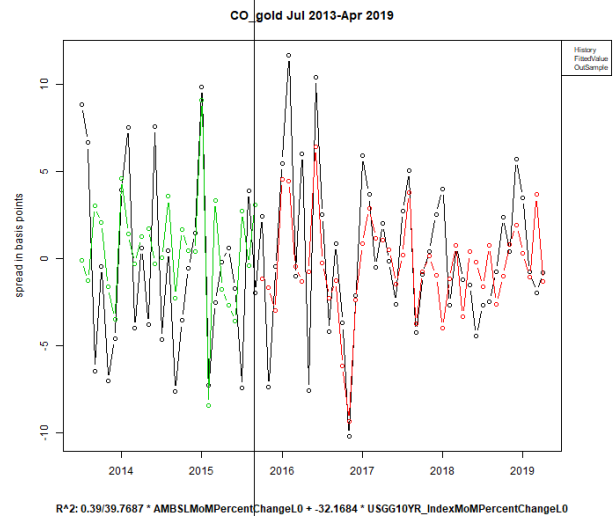
Backtesting on level values



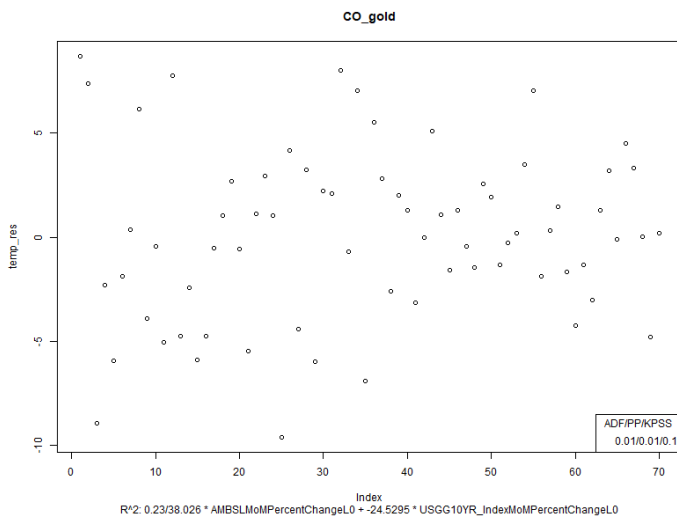
Out-of-Sample testing



Out-of-Sample testing



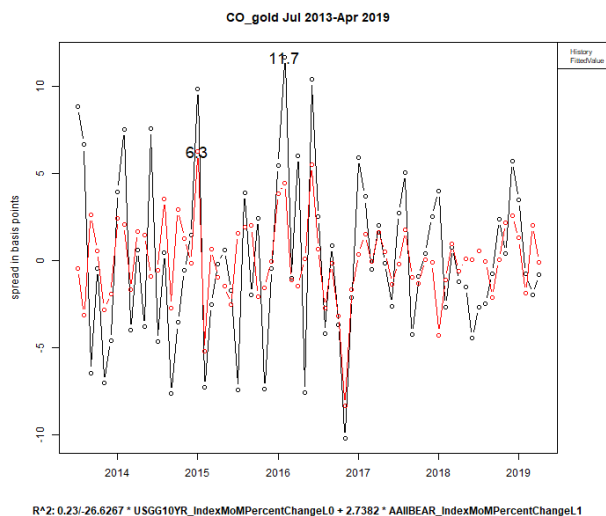
Stationarity of Residuals



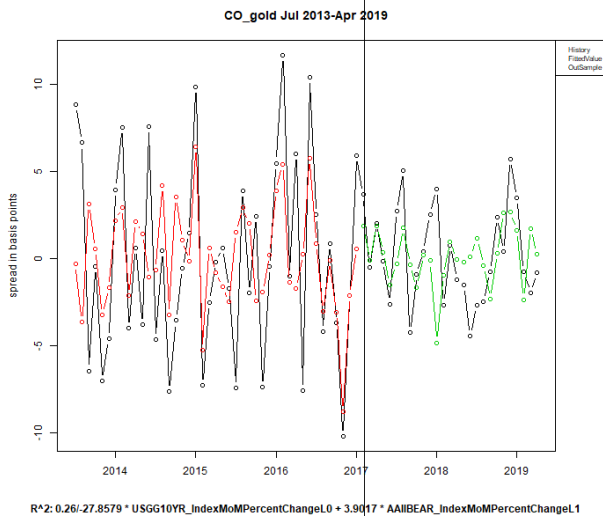
4.10.23 Model - 23

Backtesting on differenced values

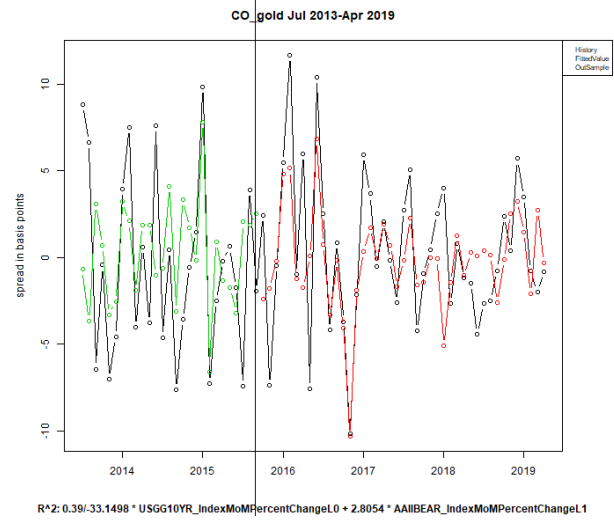
Backtesting on level values



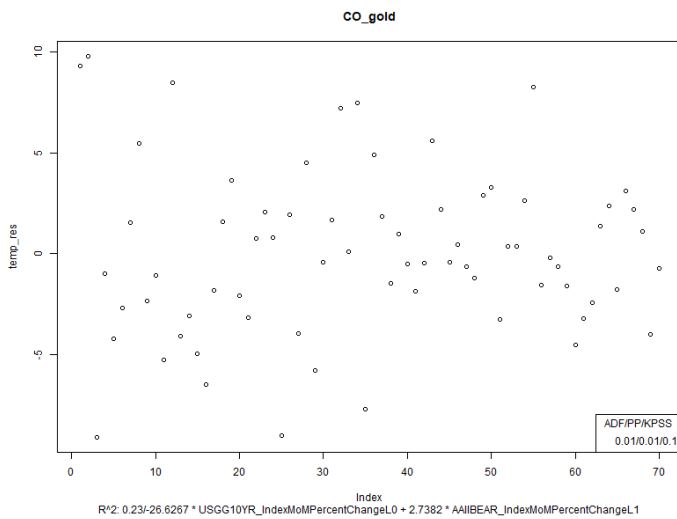
Out-of-Sample testing



Out-of-Sample testing



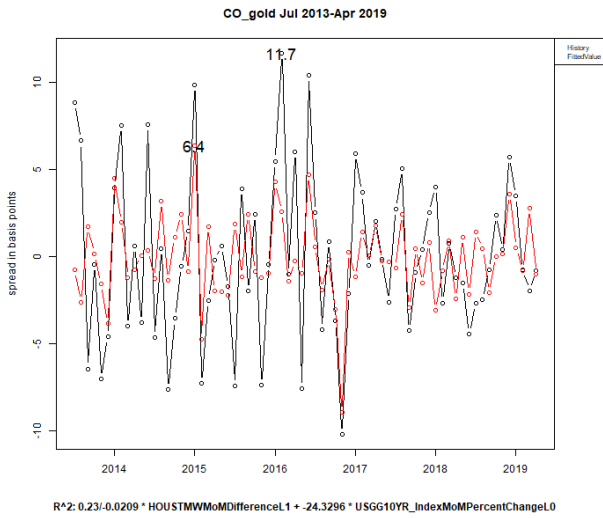
Stationarity of Residuals



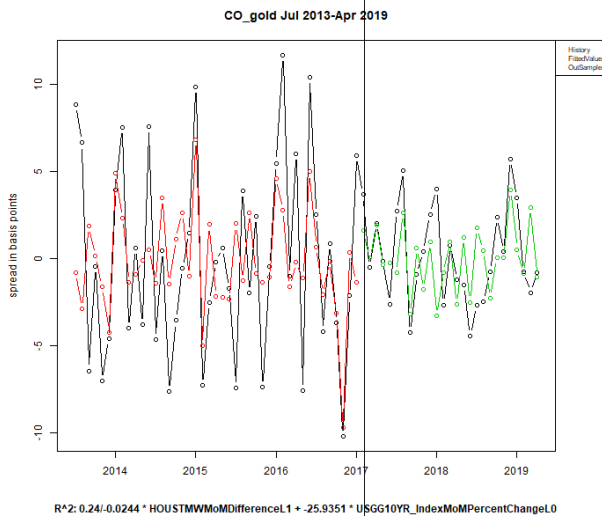
4.10.24 Model - 24

Backtesting on differenced values

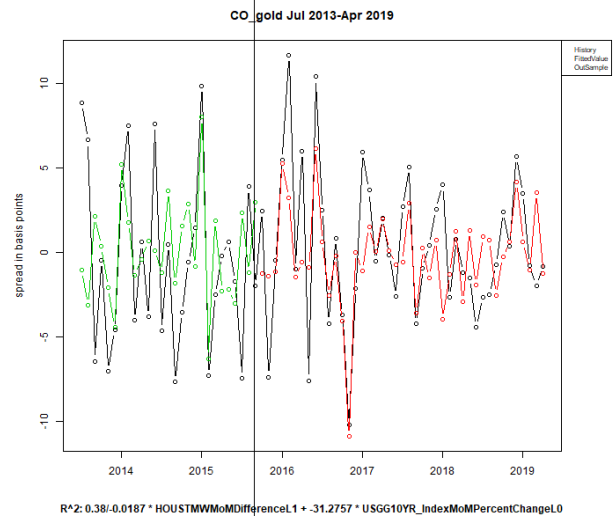
Backtesting on level values



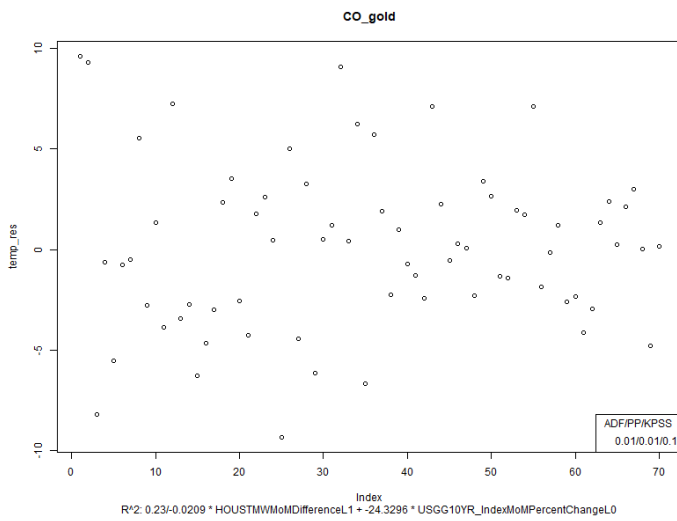
Out-of-Sample testing



Out-of-Sample testing



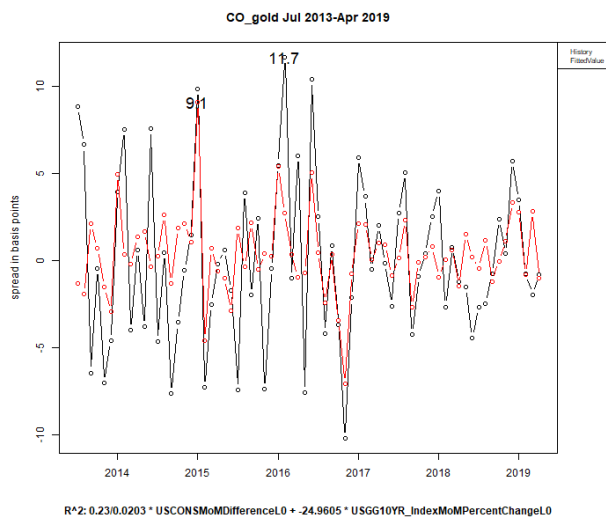
Stationarity of Residuals



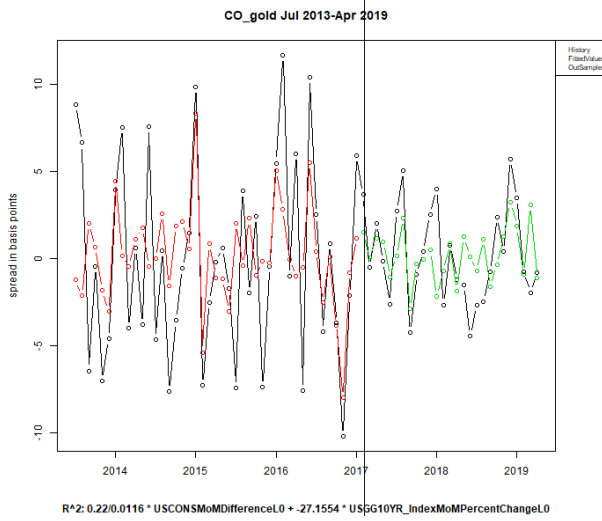
4.10.25 Model - 25

Backtesting on differenced values

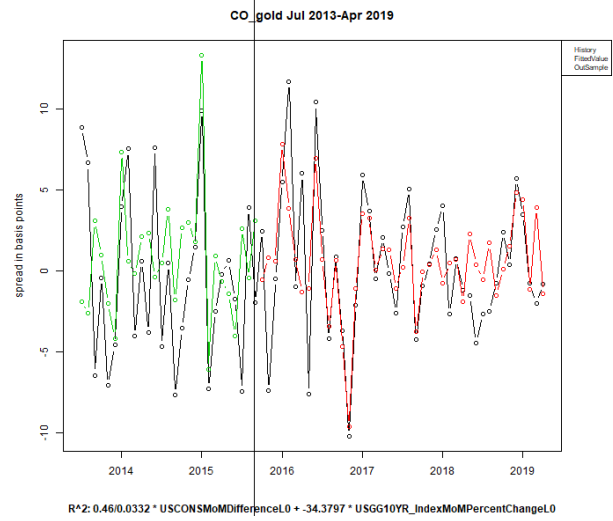
Backtesting on level values



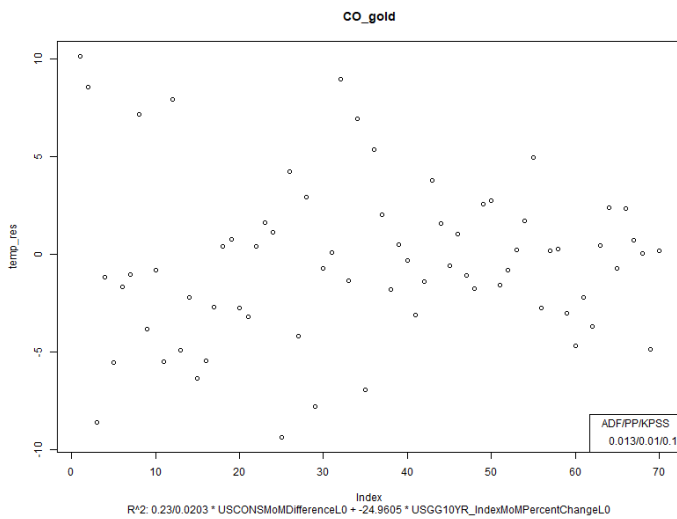
Out-of-Sample testing



Out-of-Sample testing



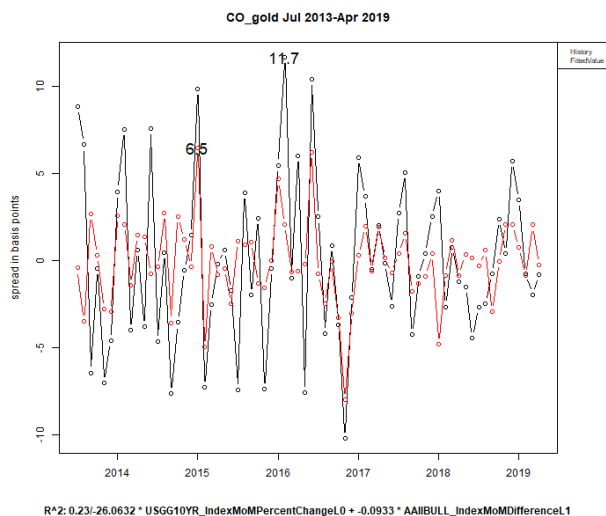
Stationarity of Residuals



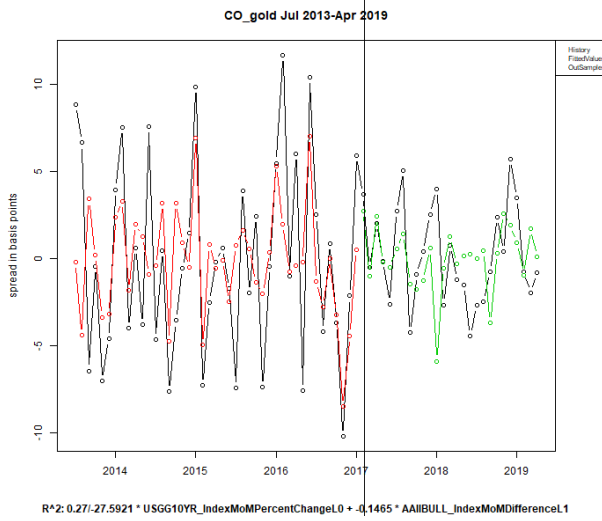
4.10.26 Model - 26

Backtesting on differenced values

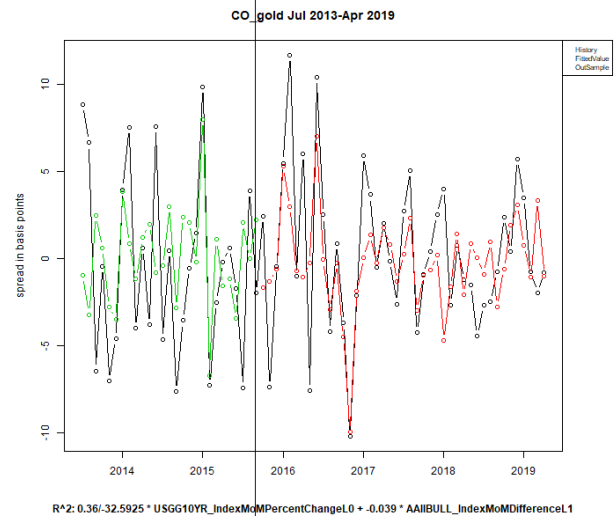
Backtesting on level values



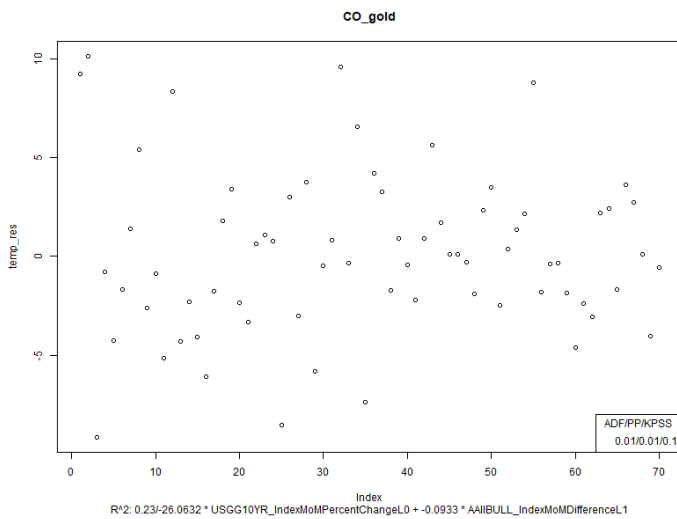
Out-of-Sample testing



Out-of-Sample testing



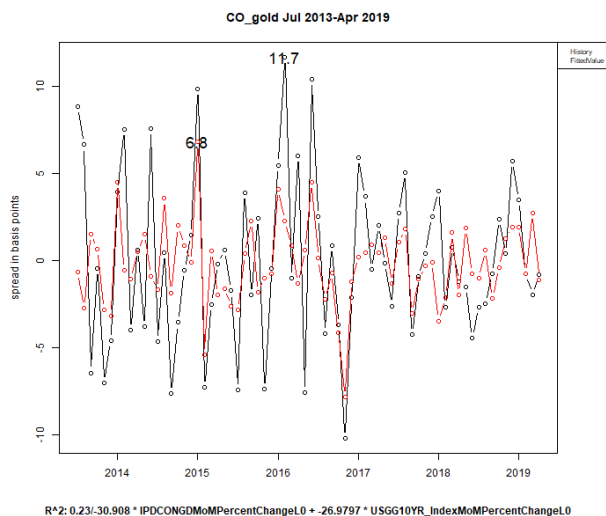
Stationarity of Residuals



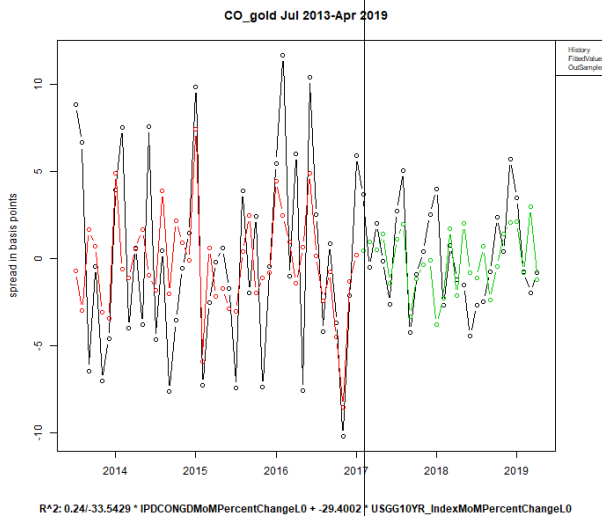
4.10.27 Model - 27

Backtesting on differenced values

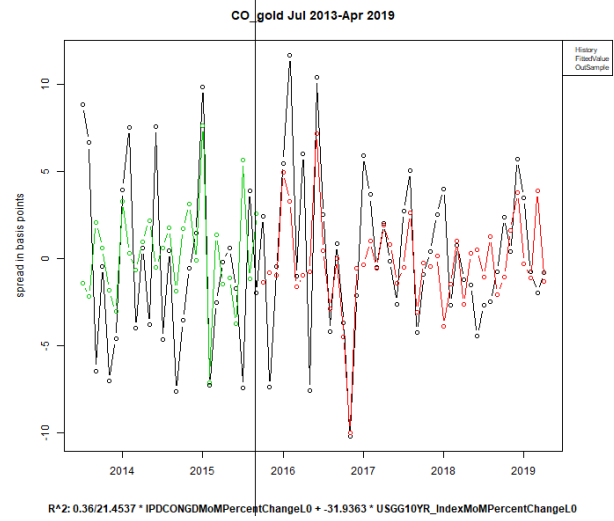
Backtesting on level values



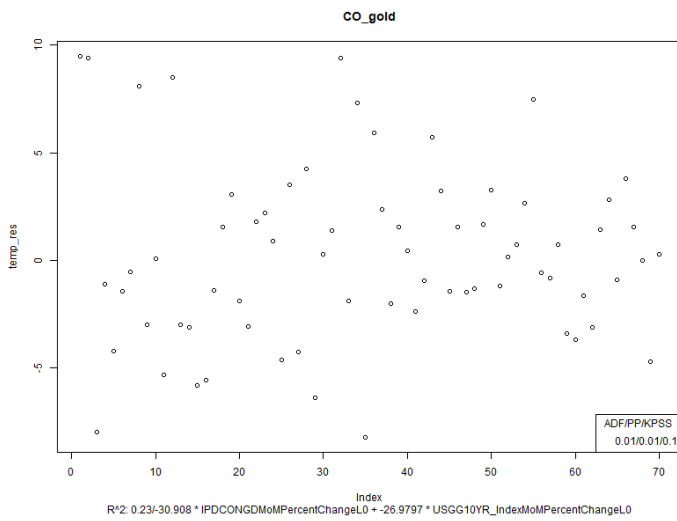
Out-of-Sample testing



Out-of-Sample testing



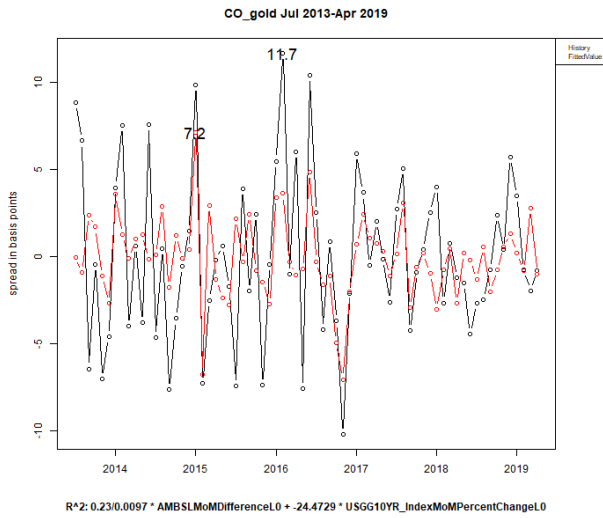
Stationarity of Residuals



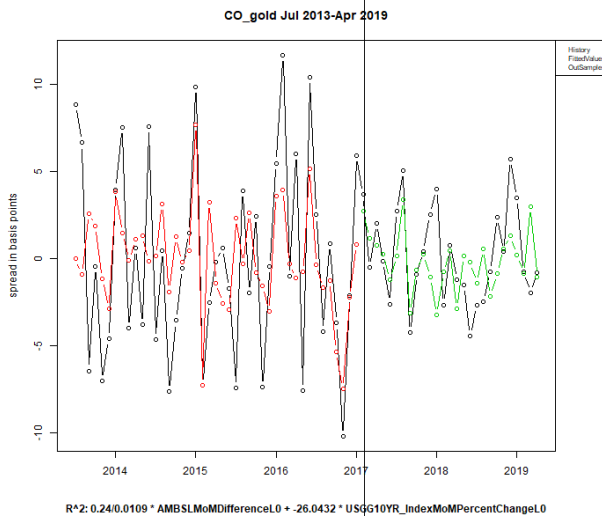
4.10.28 Model - 28

Backtesting on differenced values

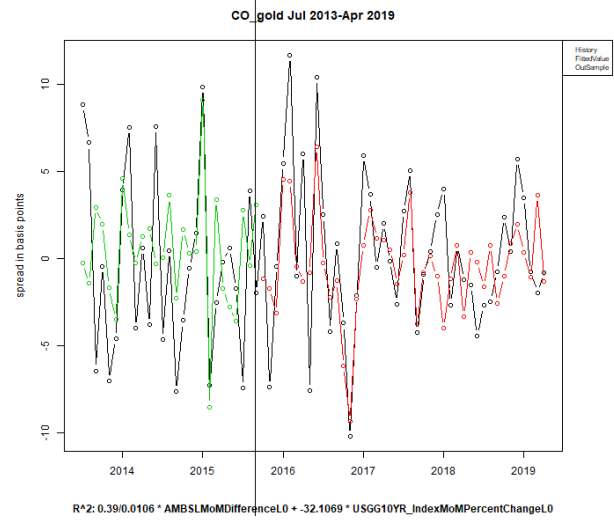
Backtesting on level values



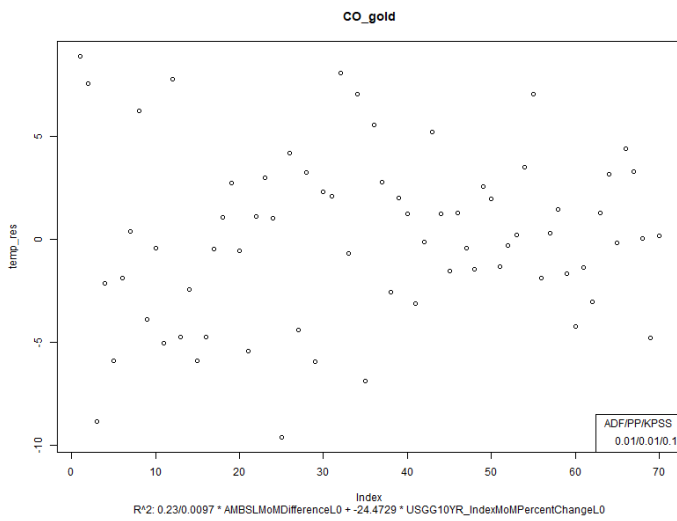
Out-of-Sample testing



Out-of-Sample testing



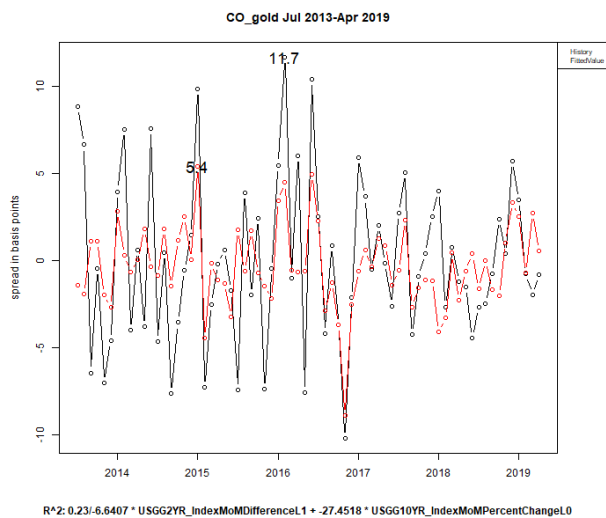
Stationarity of Residuals



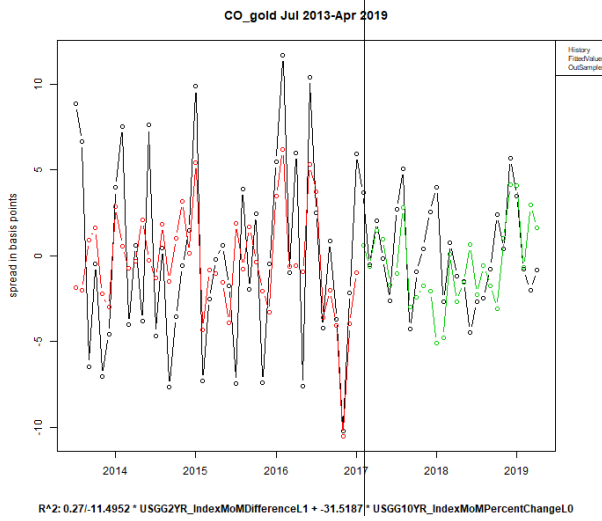
4.10.29 Model - 29

Backtesting on differenced values

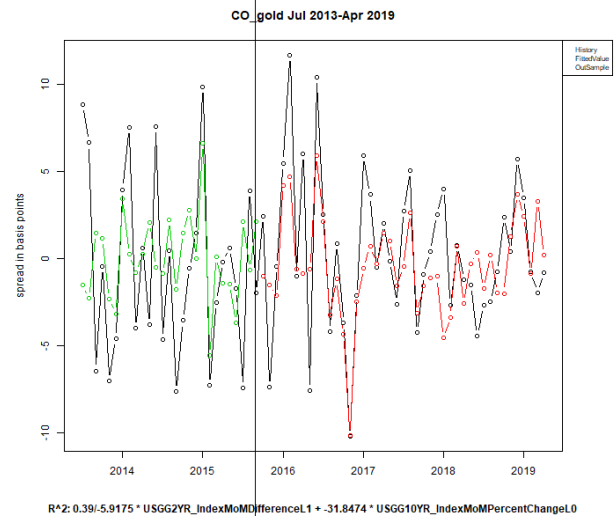
Backtesting on level values



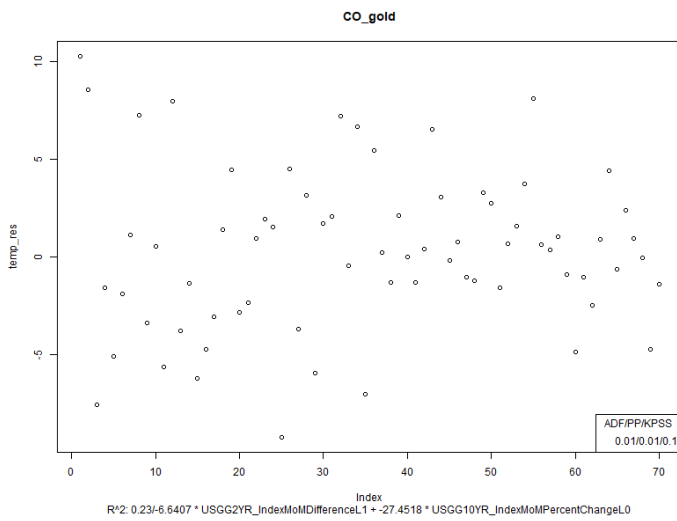
Out-of-Sample testing



Out-of-Sample testing



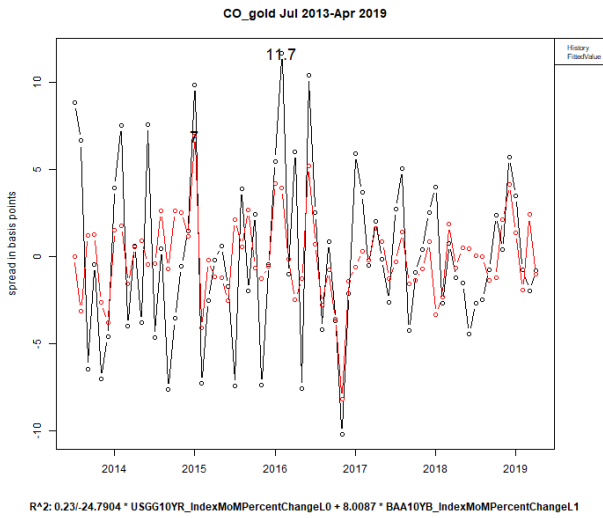
Stationarity of Residuals



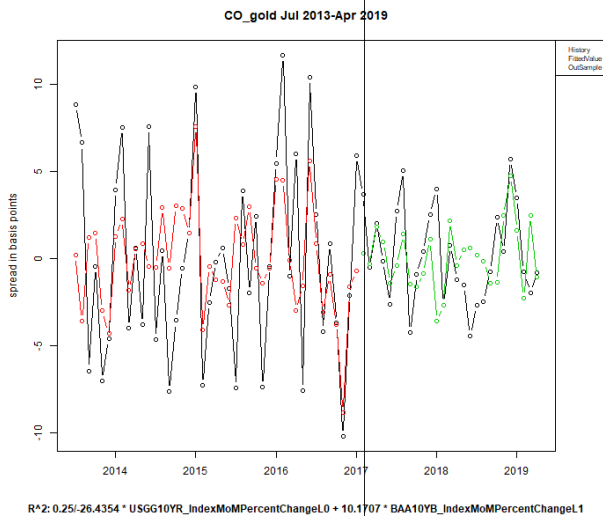
4.10.30 Model - 30

Backtesting on differenced values

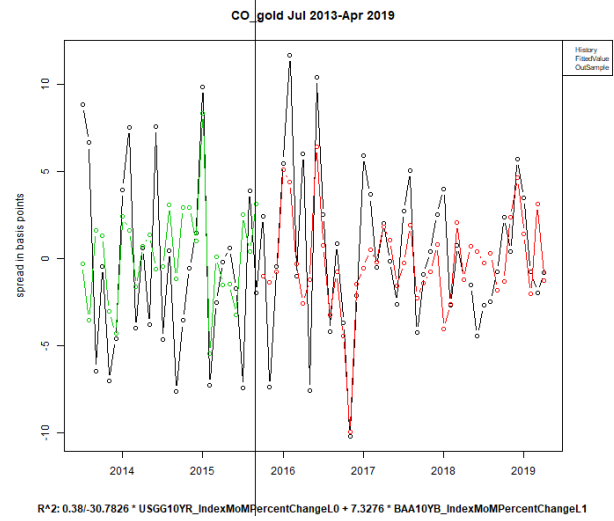
Backtesting on level values



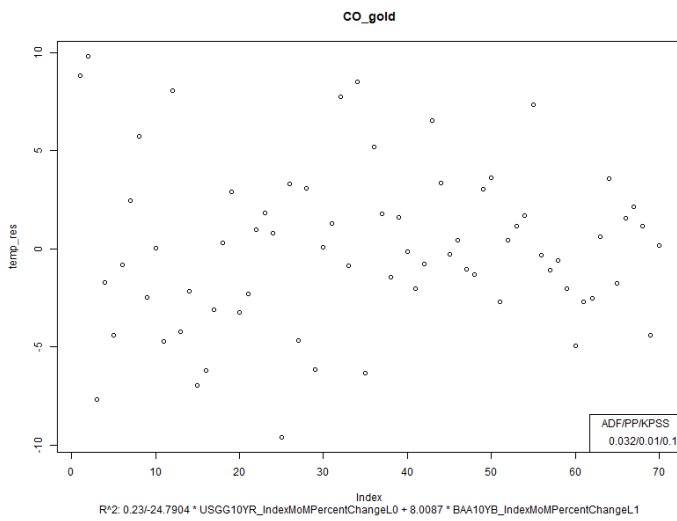
Out-of-Sample testing



Out-of-Sample testing



Stationarity of Residuals



4.11 Single Factor Model for CO_{silver}

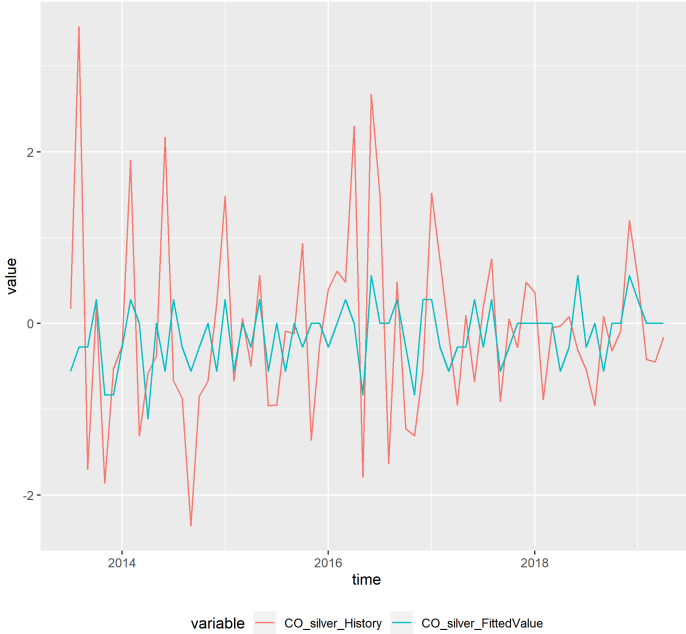
	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	UNRATEMoMDifferenceL0	0.12	202.69	2.78	0.00	0.57			
2	UNRATEMoMPercentChangeL0	0.12	202.77	14.45	0.00	0.66			

Table 18: Single Factor Analysis-CO_{silver}

4.11.1 Model - 1

Backtesting on differenced values

CO_silver - Backtesting
R²: 0.12/ 2.7831 * UNRATEMoMDifferenceL0



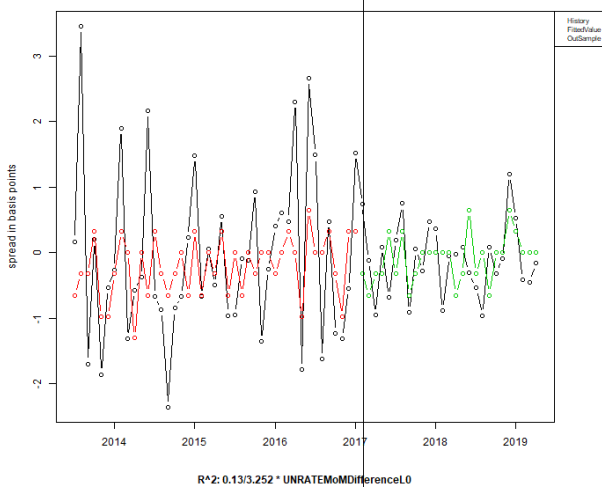
Backtesting on level values

CO_silver - Level Value Backtesting
R²: 0.12/ 2.7831 * UNRATEMoMDifferenceL0



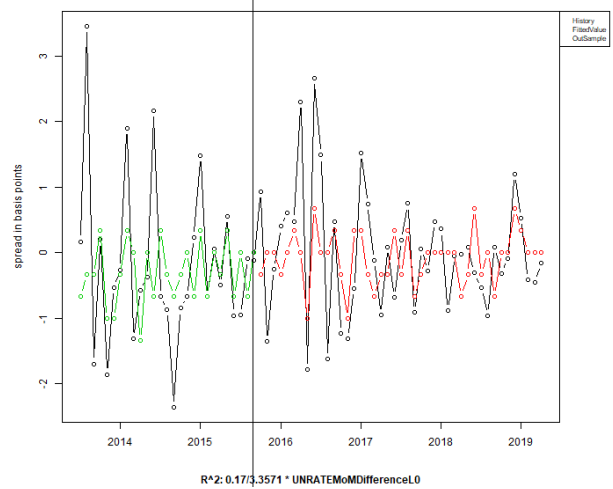
Out-of-Sample testing

CO_silver Jul 2013-Apr 2019



Out-of-Sample testing

CO_silver Jul 2013-Apr 2019



4.11.2 Model - 2

Backtesting on differenced values

CO_silver - Backtesting

R²: 0.12/ 14.4503 * UNRATEMoMPercentChangeL0



Backtesting on level values

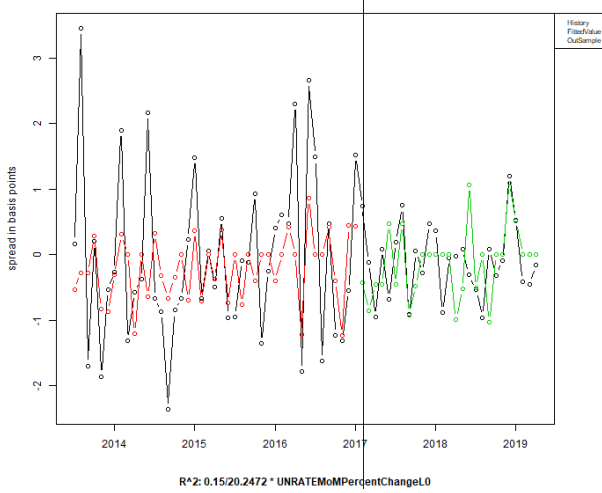
CO_silver - Level Value Backtesting

R²: 0.12/ 14.4503 * UNRATEMoMPercentChangeL0



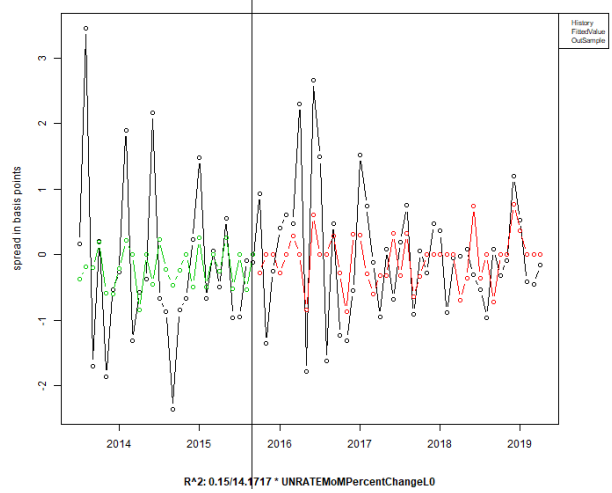
Out-of-Sample testing

CO_silver Jul 2013-Apr 2019



Out-of-Sample testing

CO_silver Jul 2013-Apr 2019



4.12 Two Factor Model for CO_{silver}

	V1	V2	r2adj	Var1	Var2	aic	RMSE	Criteria1	Criteria2	Criteria3
1										

Table 19: Two Factor Analysis-CO_{silver}

Table 20: Relevant Variables: CO_{silver}

1	2	3
1 RPIMoMDifferenceL0	UEMP15T26MoMPercentChangeL0	BUSLOANSMoMDifferenceL0
2 RPIMoMDifferenceL1	UEMP15T26MoMPercentChangeL1	BUSLOANSMoMDifferenceL1
3 RPIMoMPercentChangeL0	UEMP27OVMoMDifferenceL0	BUSLOANSMoMPercentChangeL0
4 RPIMoMPercentChangeL1	UEMP27OVMoMDifferenceL1	BUSLOANSMoMPercentChangeL1
5 W875RX1MoMDifferenceL0	UEMP27OVMoMPercentChangeL0	REALLNMoMDifferenceL0
6 W875RX1MoMDifferenceL1	UEMP27OVMoMPercentChangeL1	REALLNMoMDifferenceL1
7 W875RX1MoMPercentChangeL0	PAYEMSMoMDifferenceL0	REALLNMoMPercentChangeL0
8 W875RX1MoMPercentChangeL1	PAYEMSMoMDifferenceL1	REALLNMoMPercentChangeL1
9 DPCERA3M086SBEAMoMDifferenceL0	PAYEMSMoMPercentChangeL0	WPSFD49207MoMDifferenceL0
10 DPCERA3M086SBEAMoMDifferenceL1	PAYEMSMoMPercentChangeL1	WPSFD49207MoMDifferenceL1
11 DPCERA3M086SBEAMoMPercentChangeL0	USGOODMoMDifferenceL0	WPSFD49207MoMPercentChangeL0
12 DPCERA3M086SBEAMoMPercentChangeL1	USGOODMoMDifferenceL1	WPSFD49207MoMPercentChangeL1
13 CMRMTSPLMoMDifferenceL0	USGOODMoMPercentChangeL0	WPSFD49502MoMDifferenceL0
14 CMRMTSPLMoMDifferenceL1	USGOODMoMPercentChangeL1	WPSFD49502MoMDifferenceL1
15 CMRMTSPLMoMPercentChangeL0	CES1021000001MoMDifferenceL0	WPSFD49502MoMPercentChangeL0
16 CMRMTSPLMoMPercentChangeL1	CES1021000001MoMDifferenceL1	WPSFD49502MoMPercentChangeL1
17 INDPROMoMDifferenceL0	CES1021000001MoMPercentChangeL0	WPSID61MoMDifferenceL0
18 INDPROMoMDifferenceL1	CES1021000001MoMPercentChangeL1	WPSID61MoMDifferenceL1
19 INDPROMoMPercentChangeL0	USCONSMoMDifferenceL0	WPSID61MoMPercentChangeL0
20 INDPROMoMPercentChangeL1	USCONSMoMDifferenceL1	WPSID61MoMPercentChangeL1
21 IPFPNSSMoMDifferenceL0	USCONSMoMPercentChangeL0	WPSID62MoMDifferenceL0
22 IPFPNSSMoMDifferenceL1	USCONSMoMPercentChangeL1	WPSID62MoMDifferenceL1
23 IPFPNSSMoMPercentChangeL0	MANEMPMoMDifferenceL0	WPSID62MoMPercentChangeL0
24 IPFPNSSMoMPercentChangeL1	MANEMPMoMDifferenceL1	WPSID62MoMPercentChangeL1
25 IPFINALMoMDifferenceL0	MANEMPMoMPercentChangeL0	MZMSLMoMDifferenceL0
26 IPFINALMoMDifferenceL1	MANEMPMoMPercentChangeL1	MZMSLMoMDifferenceL1
27 IPFINALMoMPercentChangeL0	DMANEMPMoMDifferenceL0	MZMSLMoMPercentChangeL0
28 IPFINALMoMPercentChangeL1	DMANEMPMoMDifferenceL1	MZMSLMoMPercentChangeL1
29 IPCONGDMoMDifferenceL0	DMANEMPMoMPercentChangeL0	DTCOLNVHFNMMDifferenceL0
30 IPCONGDMoMDifferenceL1	DMANEMPMoMPercentChangeL1	DTCOLNVHFNMMDifferenceL1
31 IPCONGDMoMPercentChangeL0	SRVPRDMoMDifferenceL0	DTCOLNVHFNMMDifferenceL0
32 IPCONGDMoMPercentChangeL1	SRVPRDMoMDifferenceL1	DTCOLNVHFNMMDifferenceL1
33 IPDCONGDMoMDifferenceL0	SRVPRDMoMPercentChangeL0	DTCTHFNMMDifferenceL0
34 IPDCONGDMoMDifferenceL1	SRVPRDMoMPercentChangeL1	DTCTHFNMMDifferenceL1
35 IPDCONGDMoMPercentChangeL0	USWTRADEMoMDifferenceL0	DTCTHFNMMDifferenceL0
36 IPDCONGDMoMPercentChangeL1	USWTRADEMoMDifferenceL1	DTCTHFNMMDifferenceL1
37 IPNCONGDMoMDifferenceL0	USWTRADEMoMPercentChangeL0	INVESTMoMDifferenceL0
38 IPNCONGDMoMDifferenceL1	USWTRADEMoMPercentChangeL1	INVESTMoMDifferenceL1
39 IPNCONGDMoMPercentChangeL0	USTRADemoMDifferenceL0	INVESTMoMPercentChangeL0
40 IPNCONGDMoMPercentChangeL1	USTRADemoMDifferenceL1	INVESTMoMPercentChangeL1
41 IPBUSEQMoMDifferenceL0	USTRADemoMPercentChangeL0	ism_man_emplMoMDifferenceL0
42 IPBUSEQMoMDifferenceL1	USTRADemoMPercentChangeL1	ism_man_emplMoMDifferenceL1
43 IPBUSEQMoMPercentChangeL0	USFIREMoMDifferenceL0	ism_man_emplMoMPercentChangeL0

44	IPBUSEQMoMPercentChangeL1	USFIREMoMDifferenceL1	ism_man_emplMoMPercentChangeL1
45	IPMATMoMDifferenceL0	USFIREMoMPercentChangeL0	ism_man_newordersMoMDifferenceL0
46	IPMATMoMDifferenceL1	USFIREMoMPercentChangeL1	ism_man_newordersMoMDifferenceL1
47	IPMATMoMPercentChangeL0	USGOVTMoMDifferenceL0	ism_man_newordersMoMPercentChangeL0
48	IPMATMoMPercentChangeL1	USGOVTMoMDifferenceL1	ism_man_newordersMoMPercentChangeL1
49	IPDMATMoMDifferenceL0	USGOVTMoMPercentChangeL0	ism_man_pmiMoMDifferenceL0
50	IPDMATMoMDifferenceL1	USGOVTMoMPercentChangeL1	ism_man_pmiMoMDifferenceL1
51	IPDMATMoMPercentChangeL0	CES0600000007MoMDifferenceL0	ism_man_pmiMoMPercentChangeL0
52	IPDMATMoMPercentChangeL1	CES0600000007MoMDifferenceL1	ism_man_pmiMoMPercentChangeL1
53	IPNMATMoMDifferenceL0	CES0600000007MoMPercentChangeL0	ism_man_pricesMoMDifferenceL0
54	IPNMATMoMDifferenceL1	CES0600000007MoMPercentChangeL1	ism_man_pricesMoMDifferenceL1
55	IPNMATMoMPercentChangeL0	AWOTMANMoMDifferenceL0	ism_man_pricesMoMPercentChangeL0
56	IPNMATMoMPercentChangeL1	AWOTMANMoMDifferenceL1	ism_man_pricesMoMPercentChangeL1
57	IPMANSICSMoMDifferenceL0	AWOTMANMoMPercentChangeL0	USGG2YR_IndexMoMDifferenceL0
58	IPMANSICSMoMDifferenceL1	AWOTMANMoMPercentChangeL1	USGG2YR_IndexMoMDifferenceL1
59	IPMANSICSMoMPercentChangeL0	AWHMANMoMDifferenceL0	USGG2YR_IndexMoMPercentChangeL0
60	IPMANSICSMoMPercentChangeL1	AWHMANMoMDifferenceL1	USGG2YR_IndexMoMPercentChangeL1
61	IPB5122SMoMDifferenceL0	AWHMANMoMPercentChangeL0	USGG10YR_IndexMoMDifferenceL0
62	IPB5122SMoMDifferenceL1	AWHMANMoMPercentChangeL1	USGG10YR_IndexMoMDifferenceL1
63	IPB5122SMoMPercentChangeL0	HOUSTMoMDifferenceL0	USGG10YR_IndexMoMPercentChangeL0
64	IPB5122SMoMPercentChangeL1	HOUSTMoMDifferenceL1	USGG10YR_IndexMoMPercentChangeL1
65	IPFUELSMoMDifferenceL0	HOUSTMoMPercentChangeL0	USGG30YR_IndexMoMDifferenceL0
66	IPFUELSMoMDifferenceL1	HOUSTMoMPercentChangeL1	USGG30YR_IndexMoMDifferenceL1
67	IPFUELSMoMPercentChangeL0	HOUSTNEMoMDifferenceL0	USGG30YR_IndexMoMPercentChangeL0
68	IPFUELSMoMPercentChangeL1	HOUSTNEMoMDifferenceL1	USGG30YR_IndexMoMPercentChangeL1
69	CUMFNSMoMDifferenceL0	HOUSTNEMoMPercentChangeL0	VIX_IndexMoMDifferenceL0
70	CUMFNSMoMDifferenceL1	HOUSTNEMoMPercentChangeL1	VIX_IndexMoMDifferenceL1
71	CUMFNSMoMPercentChangeL0	HOUSTMWMoMDifferenceL0	VIX_IndexMoMPercentChangeL0
72	CUMFNSMoMPercentChangeL1	HOUSTMWMoMDifferenceL1	VIX_IndexMoMPercentChangeL1
73	CLF16OVMoMDifferenceL0	HOUSTMWMoMPercentChangeL0	MOVE_IndexMoMDifferenceL0
74	CLF16OVMoMDifferenceL1	HOUSTMWMoMPercentChangeL1	MOVE_IndexMoMDifferenceL1
75	CLF16OVMoMPercentChangeL0	HOUSTSMoMDifferenceL0	MOVE_IndexMoMPercentChangeL0
76	CLF16OVMoMPercentChangeL1	HOUSTSMoMDifferenceL1	MOVE_IndexMoMPercentChangeL1
77	CE16OVMoMDifferenceL0	HOUSTSMoMPercentChangeL0	AAIIBULL_IndexMoMDifferenceL0
78	CE16OVMoMDifferenceL1	HOUSTSMoMPercentChangeL1	AAIIBULL_IndexMoMDifferenceL1
79	CE16OVMoMPercentChangeL0	HOUSTWMoMDifferenceL0	AAIIBULL_IndexMoMPercentChangeL0
80	CE16OVMoMPercentChangeL1	HOUSTWMoMDifferenceL1	AAIIBULL_IndexMoMPercentChangeL1
81	UNRATEMoMDifferenceL0	HOUSTWMoMPercentChangeL0	AAIIBEAR_IndexMoMDifferenceL0
82	UNRATEMoMDifferenceL1	HOUSTWMoMPercentChangeL1	AAIIBEAR_IndexMoMDifferenceL1
83	UNRATEMoMPercentChangeL0	ISRATIOMoMDifferenceL0	AAIIBEAR_IndexMoMPercentChangeL0
84	UNRATEMoMPercentChangeL1	ISRATIOMoMDifferenceL1	AAIIBEAR_IndexMoMPercentChangeL1
85	UEMPMEANMoMDifferenceL0	ISRATIOMoMPercentChangeL0	SPX_Index_14DRSIMoMDifferenceL0
86	UEMPMEANMoMDifferenceL1	ISRATIOMoMPercentChangeL1	SPX_Index_14DRSIMoMDifferenceL1
87	UEMPMEANMoMPercentChangeL0	M1SLMoMDifferenceL0	SPX_Index_14DRSIMoMPercentChangeL0
88	UEMPMEANMoMPercentChangeL1	M1SLMoMDifferenceL1	SPX_Index_14DRSIMoMPercentChangeL1
89	UEMPLT5MoMDifferenceL0	M1SLMoMPercentChangeL0	SPX_Index_30DRSIMoMDifferenceL0
90	UEMPLT5MoMDifferenceL1	M1SLMoMPercentChangeL1	SPX_Index_30DRSIMoMDifferenceL1
91	UEMPLT5MoMPercentChangeL0	M2SLMoMDifferenceL0	SPX_Index_30DRSIMoMPercentChangeL0
92	UEMPLT5MoMPercentChangeL1	M2SLMoMDifferenceL1	SPX_Index_30DRSIMoMPercentChangeL1
93	UEMP5TO14MoMDifferenceL0	M2SLMoMPercentChangeL0	TED_G_IndexMoMDifferenceL0
94	UEMP5TO14MoMDifferenceL1	M2SLMoMPercentChangeL1	TED_G_IndexMoMDifferenceL1
95	UEMP5TO14MoMPercentChangeL0	M2REALMoMDifferenceL0	TED_G_IndexMoMPercentChangeL0
96	UEMP5TO14MoMPercentChangeL1	M2REALMoMDifferenceL1	TED_G_IndexMoMPercentChangeL1
97	UEMP15OVMoMDifferenceL0	M2REALMoMPercentChangeL0	BAA10YB_IndexMoMDifferenceL0
98	UEMP15OVMoMDifferenceL1	M2REALMoMPercentChangeL1	BAA10YB_IndexMoMDifferenceL1

99	UEMP15OVMoMPercentChangeL0	AMBSLMoMDifferenceL0	BAA10YB_IndexMoMPercentChangeL0
100	UEMP15OVMoMPercentChangeL1	AMBSLMoMDifferenceL1	BAA10YB_IndexMoMPercentChangeL1
101	UEMP15T26MoMDifferenceL0	AMBSLMoMPercentChangeL0	
102	UEMP15T26MoMDifferenceL1	AMBSLMoMPercentChangeL1	

4.13 Single Factor Model for CO_{grains}

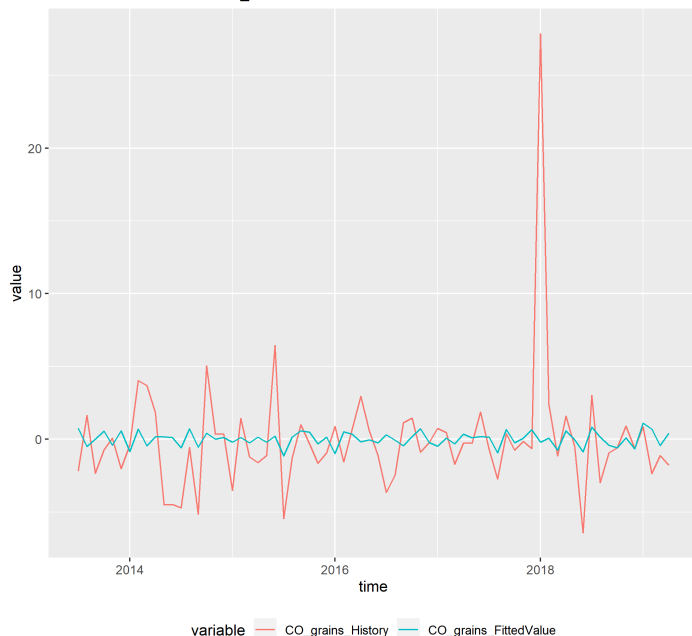
	Variable	adjR2	AIC	Coef	Robst p	RMSE_OS	Criteria1	Criteria2	Criteria3
1	AAIIBEAR_IndexMoMDifferenceL0	0.00	399.16	-0.06	0.19	5.67			

Table 21: Single Factor Analysis- CO_{grains}

4.13.1 Model - 1

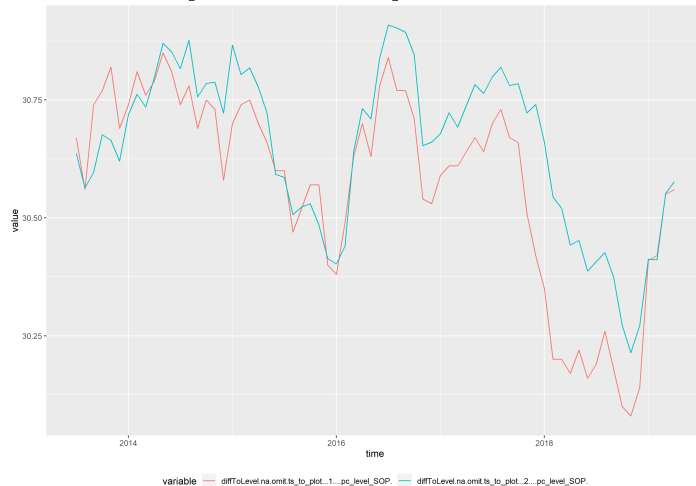
Backtesting on differenced values

CO_grains - Backtesting

 $R^2: 0 / -0.0601 * AAI\text{BEAR_IndexMoMDifferenceL0}$ 

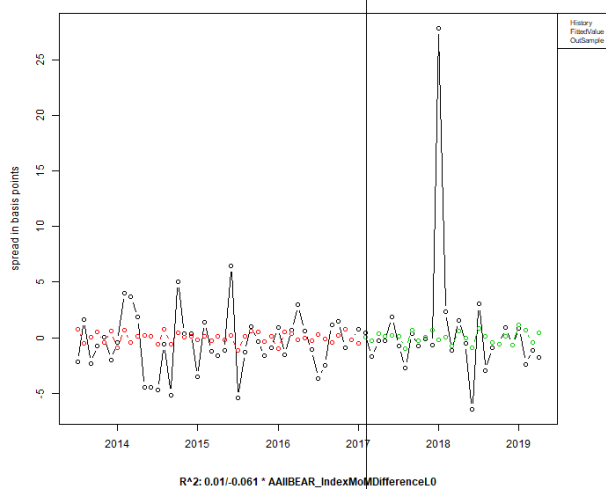
Backtesting on level values

FLIG_short - Level Value Backtesting

 $R^2: 0.68 / -0.4543 * USG10YR_IndexMoMDifferenceL0 + -0.0044 * BAA10YB_IndexMoMDifferenceL0$ 

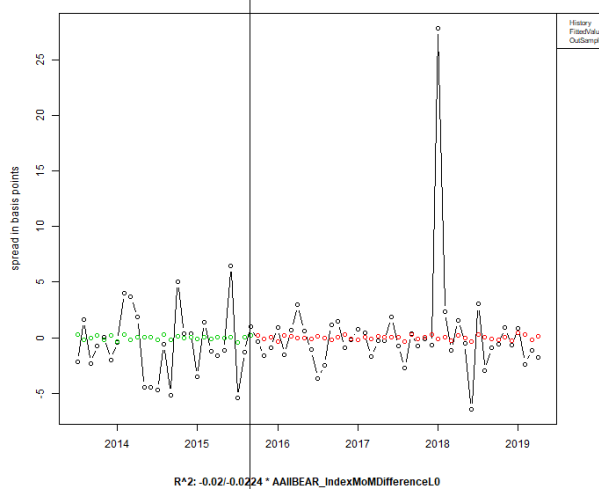
Out-of-Sample testing

CO_grains Jul 2013-Apr 2019



Out-of-Sample testing

CO_grains Jul 2013-Apr 2019



4.14 Two Factor Model for CO_grains

V1	V2	r2adj	Var1	Var2	aic	RMSE	Criteria1	Criteria2	Criteria3
1									

Table 22: Two Factor Analysis-CO_grains

Table 23: Relevant Variables: CO_grains

1	2	3
1 RPIMoMDifferenceL0	UEMP15T26MoMPercentChangeL0	BUSLOANSMoMDifferenceL0
2 RPIMoMDifferenceL1	UEMP15T26MoMPercentChangeL1	BUSLOANSMoMDifferenceL1
3 RPIMoMPercentChangeL0	UEMP27OVMoMDifferenceL0	BUSLOANSMoMPercentChangeL0
4 RPIMoMPercentChangeL1	UEMP27OVMoMDifferenceL1	BUSLOANSMoMPercentChangeL1
5 W875RX1MoMDifferenceL0	UEMP27OVMoMPercentChangeL0	REALLNMoMDifferenceL0
6 W875RX1MoMDifferenceL1	UEMP27OVMoMPercentChangeL1	REALLNMoMDifferenceL1
7 W875RX1MoMPercentChangeL0	PAYEMSMoMDifferenceL0	REALLNMoMPercentChangeL0
8 W875RX1MoMPercentChangeL1	PAYEMSMoMDifferenceL1	REALLNMoMPercentChangeL1
9 DPCERA3M086SBEAMoMDifferenceL0	PAYEMSMoMPercentChangeL0	WPSFD49207MoMDifferenceL0
10 DPCERA3M086SBEAMoMDifferenceL1	PAYEMSMoMPercentChangeL1	WPSFD49207MoMDifferenceL1
11 DPCERA3M086SBEAMoMPercentChangeL0	USGOODMoMDifferenceL0	WPSFD49207MoMPercentChangeL0
12 DPCERA3M086SBEAMoMPercentChangeL1	USGOODMoMDifferenceL1	WPSFD49207MoMPercentChangeL1
13 CMRMTSPLMoMDifferenceL0	USGOODMoMPercentChangeL0	WPSFD49502MoMDifferenceL0
14 CMRMTSPLMoMDifferenceL1	USGOODMoMPercentChangeL1	WPSFD49502MoMDifferenceL1
15 CMRMTSPLMoMPercentChangeL0	CES102100001MoMDifferenceL0	WPSFD49502MoMPercentChangeL0
16 CMRMTSPLMoMPercentChangeL1	CES102100001MoMDifferenceL1	WPSFD49502MoMPercentChangeL1
17 INDPROMoMDifferenceL0	CES102100001MoMPercentChangeL0	WPSID61MoMDifferenceL0
18 INDPROMoMDifferenceL1	CES102100001MoMPercentChangeL1	WPSID61MoMDifferenceL1
19 INDPROMoMPercentChangeL0	USCONSMoMDifferenceL0	WPSID61MoMPercentChangeL0
20 INDPROMoMPercentChangeL1	USCONSMoMDifferenceL1	WPSID61MoMPercentChangeL1
21 IPFPNSSMoMDifferenceL0	USCONSMoMPercentChangeL0	WPSID62MoMDifferenceL0
22 IPFPNSSMoMDifferenceL1	USCONSMoMPercentChangeL1	WPSID62MoMDifferenceL1
23 IPFPNSSMoMPercentChangeL0	MANEMPMoMDifferenceL0	WPSID62MoMPercentChangeL0
24 IPFPNSSMoMPercentChangeL1	MANEMPMoMDifferenceL1	WPSID62MoMPercentChangeL1
25 IPFINALMoMDifferenceL0	MANEMPMoMPercentChangeL0	MZMSLMoMDifferenceL0
26 IPFINALMoMDifferenceL1	MANEMPMoMPercentChangeL1	MZMSLMoMDifferenceL1
27 IPFINALMoMPercentChangeL0	DMANEMPMoMDifferenceL0	MZMSLMoMPercentChangeL0
28 IPFINALMoMPercentChangeL1	DMANEMPMoMDifferenceL1	MZMSLMoMPercentChangeL1
29 IPCONGDMoMDifferenceL0	DMANEMPMoMPercentChangeL0	DTCOLNVHFNMMDifferenceL0
30 IPCONGDMoMDifferenceL1	DMANEMPMoMPercentChangeL1	DTCOLNVHFNMMDifferenceL1
31 IPCONGDMoMPercentChangeL0	SRVPRDMoMDifferenceL0	DTCOLNVHFNMMDifferenceL0
32 IPCONGDMoMPercentChangeL1	SRVPRDMoMDifferenceL1	DTCOLNVHFNMMDifferenceL1
33 IPDCONGDMoMDifferenceL0	SRVPRDMoMPercentChangeL0	DTCTHFNMMDifferenceL0
34 IPDCONGDMoMDifferenceL1	SRVPRDMoMPercentChangeL1	DTCTHFNMMDifferenceL1
35 IPDCONGDMoMPercentChangeL0	USWTRADEMoMDifferenceL0	DTCTHFNMMDifferenceL0
36 IPDCONGDMoMPercentChangeL1	USWTRADEMoMDifferenceL1	DTCTHFNMMDifferenceL1
37 IPNCONGDMoMDifferenceL0	USWTRADEMoMPercentChangeL0	INVESTMoMDifferenceL0
38 IPNCONGDMoMDifferenceL1	USWTRADEMoMPercentChangeL1	INVESTMoMDifferenceL1
39 IPNCONGDMoMPercentChangeL0	USTRADEMoMDifferenceL0	INVESTMoMPercentChangeL0
40 IPNCONGDMoMPercentChangeL1	USTRADEMoMDifferenceL1	INVESTMoMPercentChangeL1
41 IPBUSEQMoMDifferenceL0	USTRADEMoMPercentChangeL0	ism_man_emplMoMDifferenceL0
42 IPBUSEQMoMDifferenceL1	USTRADEMoMPercentChangeL1	ism_man_emplMoMDifferenceL1
43 IPBUSEQMoMPercentChangeL0	USFIREMoMDifferenceL0	ism_man_emplMoMPercentChangeL0

44 IPBUSEQMoMPercentChangeL1
 45 IPMATMoMDifferenceL0
 46 IPMATMoMDifferenceL1
 47 IPMATMoMPercentChangeL0
 48 IPMATMoMPercentChangeL1
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 77 CE16OVMoMDifferenceL0
 78 CE16OVMoMDifferenceL1
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101	UEMP15T26MoMDifferenceL0	AMBSLMoMPercentChangeL0	
102	UEMP15T26MoMDifferenceL1	AMBSLMoMPercentChangeL1	